Bearing Housings

Split plummer block housings, Unsplit plummer block housings, Take-up housings, Flanged housings, Housings for specific industrial and railway applications

Catalogue

We pioneer motion

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1 Housing overview

Housing concept		Split plummer block housing					
Housing design		**SES**	**SAF**	**S30**	**SNS**	**RLE**	**KPG**
Details		**➤36 | 3**	**➤112 | 4**	**➤184 | 5**	**➤198 | 6**	**➤252 | 7**	**➤272 | 8**
d ≥ ...	**mm**	**20**	**–**	**110**	**115**	**180**	**470**
	inch	**$^{3}/_{4}$**	**1 $^{7}/_{16}$**	**–**	**4 $^{7}/_{16}$**	**–**	**–**
d ≤ ...	**mm**	**160**	**–**	**150**	**530**	**500**	**1250**
	inch	**5 $^{1}/_{2}$**	**10 $^{1}/_{2}$**	**–**	**19 $^{1}/_{2}$**	**–**	**–**
Bearing type	Spherical ball bearing	+	+	–	–	–	–
	Spherical roller bearing	+	+	+	+	+	+
	Barrel roller bearing	+	–	–	–	–	–
	Cylindrical roller bearing	–	–	–	–	–	–
	Deep groove ball bearing	+	–	–	–	–	–
	Angular contact ball bearing	–	–	–	–	–	–
	Spherical roller bearing, split	+	+	+	+	–	+
Installation method	On adapter sleeve	+	+	+	+	–	–
	On withdrawal sleeve	–	–	–	–	+	–
	On adjustment sleeve	–	–	–	–	–	+
	On cylindrical seat	+	+	+	+	+	–
Lubrication	Grease	+	+	+	+	+	+
	Oil	–	+	–	+	–	–
Seal	Double lip seal, split	✓	–	–	–	–	–
	Felt seal, split	✓	–	✓	–	–	–
	Labyrinth seal, unsplit	✓	✓	–	✓	✓	–
	Labyrinth seal, split	✓	✓	–	✓	–	–
	V ring seal, unsplit	✓	–	–	–	–	–
	Taconite seal, unsplit	✓	✓	–	✓	–	–
	Taconite seal, split	✓	✓	–	✓	–	–
	Bolt-on seal, unsplit	–	–	–	✓	–	–
	High-pressure packing, split	–	–	–	–	–	✓

\+ suitable
✓ available
– not available

		Unsplit plummer bearing housing		Take-up housing	Flange housing	
KPGZ	LOE	VRE3	BND	SPA	F112	F5
➤272 | 8	➤288 | 9	➤310 | 10	➤336 | 11	➤422 | 12	➤432 | 13	➤437 | 14
500	50	25	60	50	20	20
–	–	–	–	–	–	–
1320	240	120	420	400	60	100
–	–	–	–	–	–	–
–	–	–	–	–	+	+
+	+	–	+	+	–	+
–	–	–	–	–	–	+
–	–	+	–	–	–	–
–	–	+	–	–	–	–
–	–	+	–	–	–	–
+	–	–	–	–	–	–
–	+	–	+	+	–	+
–	–	–	–	–	–	–
–	–	–	–	–	–	–
+	+	+	+	–	+	–
+	–	+	+	+	+	+
–	+	–	–	–	–	–
–	–	–	–	–	–	–
–	–	✓	–	–	✓	✓
–	✓	–	✓	✓	–	–
–	–	–	–	–	–	–
–	–	✓	–	–	–	–
–	–	–	✓	✓	–	–
–	–	–	–	–	–	–
–	–	–	–	–	–	–
✓	–	–	–	–	–	–

2 Technical principles

2.1 Standard housings

2.1.1 Housing concepts

Bearing housings are subdivided by their basic design.

Split plummer block housing

For split plummer block housings, the housing body is split into an upper section and a lower section. This significantly simplifies assembly and maintenance. The two halves of a housing form a matching pair and cannot be swapped with parts from other housings.

1 Split plummer block housing

001C9E64

Unsplit plummer block housings

For unsplit plummer block housings, the housing body is a single piece, so the bearing seat has no joints. These housings are used where the bearings are subjected to very high loads.

Unsplit plummer block housings also include VRE3 plummer block housing units. These are available as fully assembled and greased bearing units comprising a housing, seals, bearings and shaft.

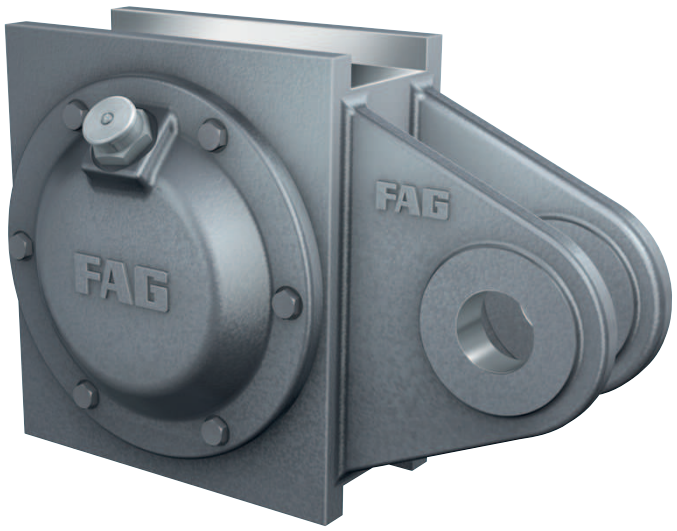
2 Unsplit plummer block housings

001C9E84

Take-up housing

SPA take-up housings were developed specifically for tensioner drum bearings in conveyor belt systems. The housings are unsplit. There is a fork-shaped drawbar eye to attach to the tensioning device.

3 Take-up housing



0001AD1C

Flange housing

Flange housings have a flange perpendicular to the shaft axis and are therefore the ideal connection structure for a wide range of machines and systems for which the use of plummer block housings would be too expensive.

4 Flange housing

001C9E44

2.1.2 Selection wizard for bearing housings

The selection wizard for bearing housings in medias provides extensive support for selecting housings and housing units. It takes account of the environmental conditions and the requirements placed on the bearing as well as the characteristics of the housings and the bearings suitable for the housings. Once the specified data have been entered, the software generates detailed lists of results that also contain information on other accessories.

medias | Engineering tool |
Housing selection wizard |
https://www.schaeffler.de/std/1D61

2.2 Design of the bearing arrangement

2.2.1 Locating and non-locating bearings

2 bearings installed at a specific distance from each other are required to guide and support a rotating shaft. There are various ways of achieving this. In many cases, the locating/non-locating bearing is used in which one bearing position has a locating bearing and the other has a non-locating bearing.

In addition, the fixed bearing and floating bearing are also often used.

2.2.2 Axial length compensation

On a shaft supported by two radial bearings, the distances between the bearing seats on the shaft and in the housing are often not identical because of manufacturing tolerances. The distances may also change as a result of temperature increases during operation. These differences in distance are compensated by the non-locating bearing.

There are bearing types in which the inner ring and outer ring cannot be displaced relative to each other due to their internal design. As a result, no axial length compensation between the shaft and housing can take place within the bearing. Examples include spherical roller bearings, barrel roller bearings and deep groove ball bearings. These bearings can only function as non-locating bearings if the inner ring or outer ring are aligned so that relative displacement is possible. For bearing housings, this means that it must be possible for the outer ring to move in the housing. The tolerance of the bearing seat in the housing is designed accordingly and is described in the information for the specific housing series under the sub-heading of accuracy.

2.2.3 Transmission of axial forces

A locating bearing is necessary for transmitting axial forces. The locating bearing also provides the shaft with axial guidance. To prevent axial stresses, shafts with more than 2 bearings have only one locating bearing.

2.2.4 Bearing concepts

One of the following concepts is used in each housing series to produce both locating and non-locating bearings:

- Housing in locating bearing design and non-locating bearing design
- Housing with locating rings

VR3unsplit plummer block housings, which have 2 bearings in a single housing, are a special case in this respect. It is possible to order complete VRE3 plummer block housing units which, depending on their design, feature a locating/non-locating bearing, an adjusted bearing arrangement or a floating bearing arrangement.

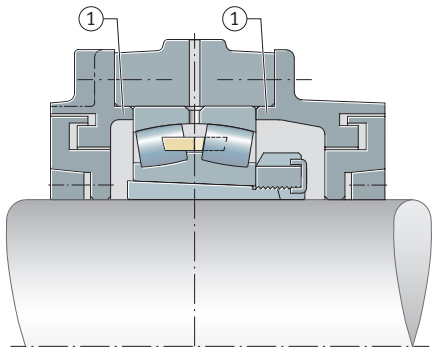
2.2.4.1 Housing in locating bearing design and non-locating bearing design

For this housing concept for achieving a locating or non-locating bearing arrangement, the housing must be ordered in a locating bearing design or a non-locating bearing design as required. This applies to the RLE, KPG, KPGZ, LOE, BND and SPA housings.

For the locating bearing version, the bearings are clamped axially between the covers on the housings. For the non-locating bearing version, the covers have shorter centring collars. As a result, the bearing can move axially.

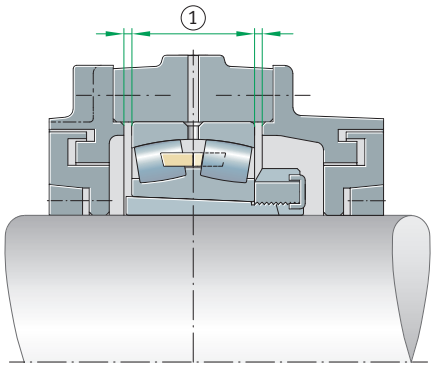
When placing the order, the designation indicates whether the housing should be supplied in a locating or non-locating bearing version.

5 Housing in locating bearing version



1	Centring collars on the covers secure the bearing in the axial direction

6 Housing in non-locating bearing version



1	Bearing can move in the axial direction

2.2.4.2 Housing with locating rings

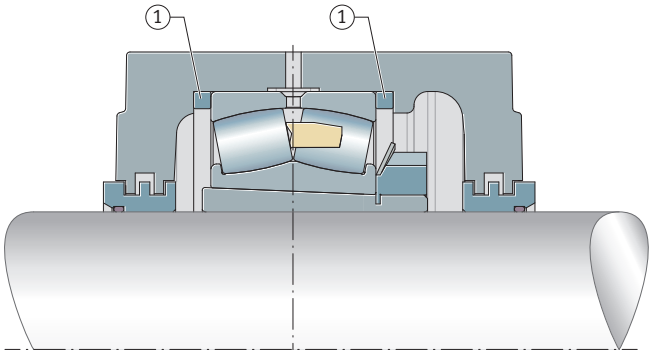
For this housing concept for achieving a locating or non-locating bearing arrangement, locating rings can be obtained as accessories and used to set the locating bearing function. This applies to the SES, S30, SNS and F5 housings.

For these housings, the bearing seats are designed such that the bearing can move in the axial direction and therefore acts as a non-locating bearing if locating rings are not inserted.

Once locating rings are inserted, the bearings are fixed axially. The locating rings are generally inserted in the housing on both sides of the bearing. Normally, an even number of locating rings is required for the concentric seating of the bearing in the housing. In some cases, a single locating ring is sufficient.

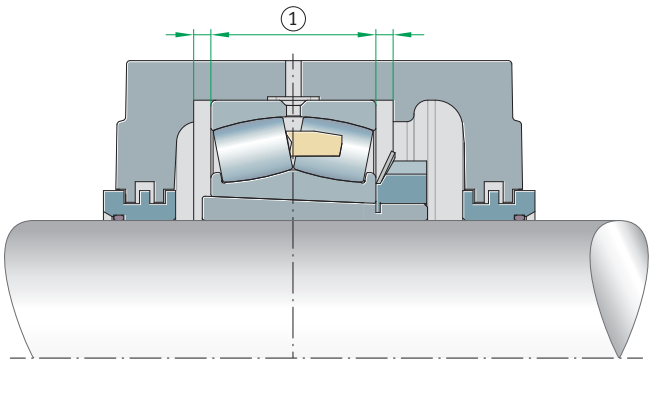
The number of locating rings required is shown in the product tables. Locating rings must be ordered separately.

7 Locating bearing arrangement with locating rings inserted



1	Locating rings secure the bearing in the axial direction

8 Non-locating bearing arrangement, no locating rings inserted



1	Bearing can move in the axial direction

2.3 Materials, corrosion protection

The standard material used for standard housings is gray cast iron, ductile cast iron or cast steel depending on the housing series.

The material is generally shown by adding a suffix to the housing designation. This suffix is omitted for housings which are only available in one material version.

1 Suffix for materials

Material	Suffix
Gray cast iron	L
Ductile cast iron	D
Ductile cast iron for low temperatures	LT
Steel or cast steel	S

Housings made from other materials are also available to order. The material should be selected on the basis of a careful analysis of the application and the associated requirements.

2.3.1 Gray cast iron

Cast iron to DIN EN 1561:2012 is generally used for standard housings made from gray cast iron. The graphite inclusions are in flake form in these cast materials with an iron/carbon/silicon base. When subject to tensile load, these act as internal notches. The tensile strength of gray cast iron is therefore limited and its ductility is comparatively low. The compressive strength of this alloy is, however, higher than its tensile strength by a factor of 4 and these alloys also have good inherent stability.

Gray cast iron is therefore used for housings with simple requirements. Furthermore, gray cast iron is the most economical version of the cast materials used in housings.

2.3.2 Ductile cast iron

Cast iron to DIN EN 1563:2019 is generally used for standard housings made from ductile cast iron. The graphite inclusions adopt a spheroidal form during casting due to the addition of magnesium or, more rarely, cerium or calcium. As a result, the material's tensile strength and ductility are higher than those of the grades with a flake graphite form.

Ductile cast iron is therefore used for housings with increased requirements. The costs of ductile cast iron are between those for gray cast iron and cast steel.

2.3.3 Cast steel

Steel to DIN EN 10293 is generally used for standard housings made from cast steel. Both unalloyed and alloyed steels are suitable for casting. The unalloyed cast steel used for housings combines moderate to high strength values with high elongation at fracture.

Cast steel is therefore used for housings with high requirements for mechanical properties and for high ductility.

The cost of cast steel is higher than the cost of gray cast iron or ductile cast iron. It is therefore recommended that checks be carried out to establish whether the material is really necessary to satisfy the requirements of the specific application.

2.3.4 Corrosion protection

All outer surfaces not machined by cutting have a universal coating. The universal coating complies with corrosivity class C2 to DIN EN ISO 12944.

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

If there are specific requirements relating to coatings and corrosion protection, special solutions are possible. The feasibility of special requirements must be assessed by Schaeffler.

The following finishes can be applied to the universal coating:

- synthetic resin
- polyurethane
- acrylic
- epoxy resin
- chlorinated rubber
- nitrocellulose
- acid-hardening hammer tone

2.4 Load capacity

Factors influencing the maximum load of plummer bearing housings:

- The strength of the housing
- The load capacity of the bearing
- The load direction
- For split plummer block housings, the strength of the connecting screws

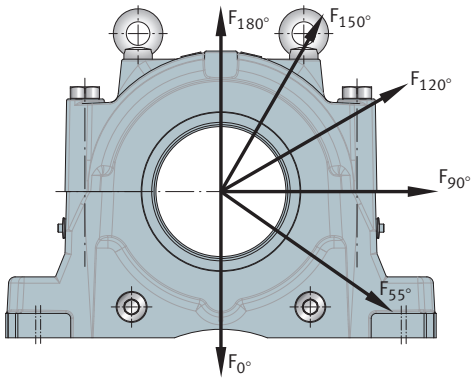
2.4.1 Rupture loads

Guide values are provided for the housing rupture load of some housing series. Where housings are available in different material versions, it must be noted that the rupture load depends on the housing material.

Guide values are given for the rupture load of plummer block housings and the maximum load capacity of connecting screws for the upper and lower housing sections. The guide values apply purely to static loads.

The guide values only apply if the flatness of the mounting surface to DIN EN ISO 1101 satisfies tolerance level IT7 to DIN EN ISO 286-1 (measured across the diagonal). The requirement for supporting loads is that the housing base surface has full, rigid support.

9 Load directions F for the guide values for the housing rupture load and the maximum load capacity of the connecting screws



2.4.2 Maximum load

The maximum load on the housing and the connecting screws and eye bolts must not be exceeded.

2.4.2.1 Safety factors

Safety factors must be taken into account when calculating the maximum load. A safety factor of 6 is normally applied to the housing rupture load in general mechanical engineering.

A safety factor of 5is sufficient for SNS split plummer block housings.

2.4.2.2 Axial load

If there is an axial load or a radial load that does not act vertically, we recommend that the housing should be secured horizontally by means of stops or pins, see ➤31|2.7.2.4.

If the housing is subjected to axial load, the maximum axial load of the fitted bearing must be taken into consideration. If the bearing is located on the shaft using an adapter sleeve, the axial retaining force of the bearing and adapter sleeve must also be taken into consideration.

2.4.2.3 Connecting screws

For split plummer block housings, the guide values for the maximum load capacity of the connecting screws must also be observed. Depending on the application, these may be the limiting factor for the load capacity of the housing.

2.4.2.4 Eye bolts

If the eye bolts provided are used to lift housings or housing components, their maximum load must not be exceeded, see ➤28|2.7.1.2.

2.5 Lubrication

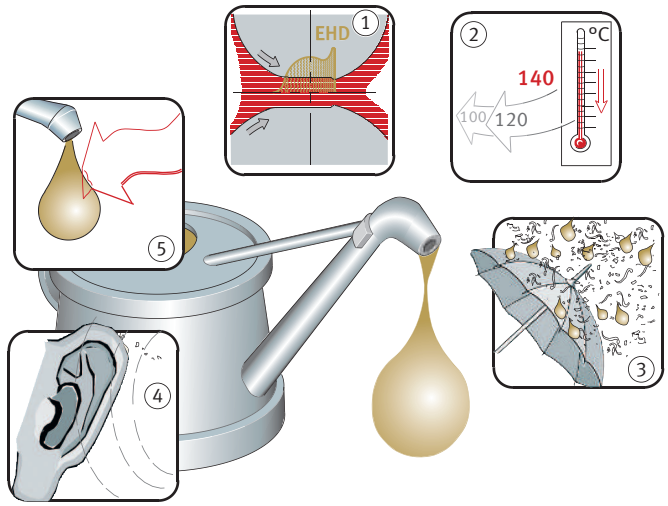
Lubrication is an essential requirement for the reliable function and long service life of roller bearings.

2.5.1 Functions of lubricant

The lubricant fulfils the following tasks:

- provides protection against corrosion
- dissipates heat from the bearing (in the case of oil lubrication)
- flushes out wear particles and contaminants (recirculating oil lubrication with filtration of the oil)
- supports the sealing effect of bearing seals (grease collar, pneumatic oil lubrication)

10 Functions of the lubricant



1	Formation of a lubricant film capable of supporting loads	2	Heat dissipation in the case of oil lubrication
3	Sealing of the bearing against external contaminants in the case of grease lubrication	4	Damping of running noise
5	Provide protection against corrosion		

2.5.2 Housing selection, type of lubrication

The design of a bearing housing determines how the bearing can be supplied with lubricant. This affects both the type of lubricant and the possibilities for replacing the lubricant after a certain service life.

Before selecting a bearing housing, the type of lubrication must be defined for the bearing, in other words whether the bearing is lubricated with grease or oil. The following criteria must be taken into consideration.

Criteria for grease lubrication

Grease lubrication requires less design work than oil lubrication. As a result, grease lubrication is normally preferred.

Advantages of grease lubrication:

- Less design work for feed systems than oil lubrication
- External sealing effect by using to a grease collar or grease-filled labyrinth seals
- Reservoir effect by storing grease in the housing
- Long service life with little maintenance work (lifetime lubrication possible in certain circumstances)

Criteria for oil lubrication

In contrast to grease lubrication, oil lubrication allows the direct dissipation of heat from the bearing in the lubricant, especially when high speeds and loads are involved. The best cooling effect is achieved by using recirculating oil lubrication together with an oil recooling system.

Advantages of oil lubrication:

- Heat dissipation from the bearing possible
- Removal of dirt and wear particles
- Good lubricant distribution and lubricant supply to contact areas
- Very low friction losses with minimal volume lubrication

Standard housings

Most standard housings are designed for grease lubrication, while some are designed for oil lubrication. There are also housing series that can be supplied in a design for either grease or oil lubrication.

2.5.3 Housings for grease lubrication

Grease lubrication is the most common type of lubrication for standard housings.

Where grease lubrication is used, bearings can be operated in many applications with lifetime lubrication, which means that the quantity of grease inserted during assembly is sufficient for the entire bearing service life if used with suitable seals. If subject to high stresses, the grease must be renewed by means of targeted relubrication.

2.5.3.1 Initial grease application

Adding grease to bearings and housings during assembly is known as the initial grease application.

Grease quantities

For initial application greasing, the rule of thumb is that the bearing should be filled with grease to 100 % and the empty volume of the housing should be filled to 60 %. The empty volume is the space that remains in the housing once the bearing, shaft and seals have been fitted. For housings with a labyrinth seal or Taconite seal, the labyrinth passages must also be filled with grease.

The exception to this rule of thumb is the recommendation for specific applications to fill the housing and seal cavities to 100 %. SES plummer block housings in applications with low speeds and a non-contact seal, where the grease must perform a sealing function, are one example of this.

For BND plummer housings and SPAtake-up housings, the general rule is that the bearing and the empty volume of the housing should be filled with grease to 100 %.

Tables are available for many housing series showing the recommended grease quantities for the initial grease application.

The recommended level of filling for greasing the bearing may differ from100 %, depending on the application and speed. A smaller grease volume is recommended for very high speeds, for example. This allows for the easier initial distribution of the grease as the bearing starts to move and restricts the increase in temperature caused by the process.

2.5.3.2 Relubrication

Relubrication involves inserting additional grease is into the housing at suitable positions. There must always be a method for ensuring that excess grease can exit the housing.

Positions for relubrication

Positions for relubrication are provided for each bearing housing based on the properties of the bearing and housing. Lubrication nipples are screwed in at these positions for relubrication.

For bearings with a lubrication groove and lubrication holes, the grease can be inserted centrally into the housing if the housing also features a suitable lubrication hole. The grease then travels straight to the bearing raceway.

Alternatively, the grease can be inserted into the side of the housing. The grease then travels past the end face to the bearing raceway.

There are separate positions for Taconite seals, and in some housings also for labyrinth seals, for relubricating the seals.

When relubricating bearings with a non-locating bearing function, there is a risk that the lubrication holes in the housing are not always aligned with the lubrication groove in the outer bearing ring and therefore no lubricant can enter the bearing. The risk becomes even greater as the width of the bearing sliding fit increases and the width of the lubrication groove in the bearing outer ring decreases. In these cases, the bearing should be relubricated at the side.

Escape of excess lubricant

If there is an open grease outlet hole or non-contact seal, there is no risk of overgreasing the bearing. If the temperature reaches a high level during relubrication due to the churning work of the grease, it will return to its initial value after it has been running for several hours once the excess grease has escaped.

Controlled metering must be ensured for the sake of the environment, . Any grease that escapes through the grease outlet holes must be collected by a suitable device.

Automatic lubricators

The use of automatic lubricators is recommended to provide bearings with a reliable supply of lubricant for relubrication. These allow a defined quantity of grease to be dispensed at the appropriate intervals. Schaeffler can supply suitable lubrication devices, for example the CONCEPT8 lubrication system.

TPI 252 | Lubricators |
https://www.schaeffler.de/std/1D4E

TPI 271 | Lubricator | OPTIME C1 |
https://www.schaeffler.de/std/1FC2

2.5.3.3 Grease types

Schaeffler recommended Arcanolroller bearing grease products to achieve a long service life and high reliability of the bearing. These have been specially designed and tested for bearing technology. Schaeffler can supply an extensive range of Arcanol grease products as part of its industrial service.

Further information

TPI 168 | Arcanol Rolling Bearing Greases |
https://www.schaeffler.de/std/1F66

TPI 176 | Lubrication of Rolling Bearings |
https://www.schaeffler.de/std/1F83

Arcanol lubricants | Safety data sheets and technical data sheets |
https://www.schaeffler.de/sds

2.5.4 Housings for oil lubrication

In addition to housings for grease lubrication, Schaeffler can also supply housings designed for oil lubrication.

For split housings, the joint between the upper and lower sections of the housing must be sealed with a thin layer of a commercial sealant (with permanent elasticity). The covers on the housing must also be sealed.

2.5.4.1 Lubrication method

Methods used for lubricating plummer block housings with oil:

- Oil bath lubrication
- Oil circulation lubrication

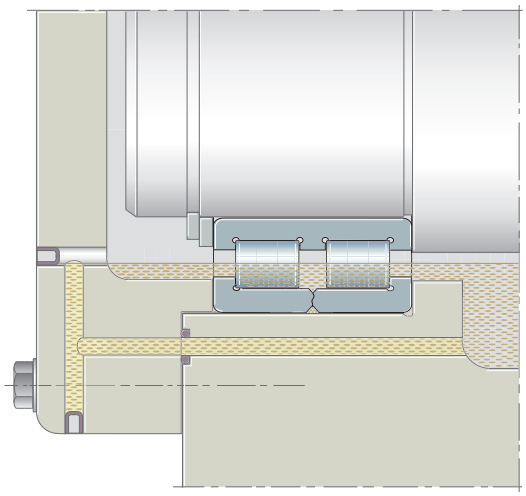
Oil bath lubrication

If oil bath lubrication is used, the oil level should be up to the centre of the lowest rolling element. If the oil level is higher than this, the bearing temperature may increase at high circumferential velocities due to splashing losses. The oil may also generate foam.

Oil bath lubrication system is suitable for speed characteristics up to $n \cdot d_M = 300000$ mm/min.

If $n \cdot d_M < 150000$ mm/min, the bearing may also be fully immersed.

11 Schematic diagram of oil bath lubrication



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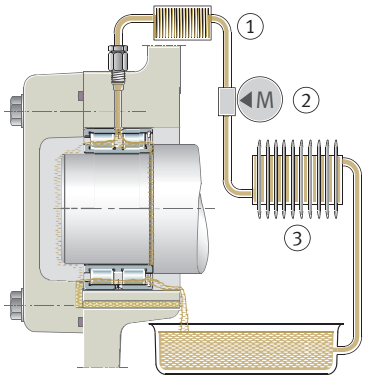
When using oil bath lubrication, venting the housing is essential. For example, the filling hole can be sealed using a bleed screw.

Oil circulation lubrication

The oil is cooled as it circulates in an oil circulation lubrication system. It therefore also dissipates heat from the bearing. In addition, dirt and wear particles can be flushed out of the bearing. We always recommend that the oil system has a filter.

12 Schematic diagram of an oil circulation lubrication system



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1	Filter	2	Pump
3	Cooling		

2.5.4.2 Oil selection

All mineral oils and synthetic oils are suitable for lubricating roller bearings.

Oils with a mineral oil base are most commonly used. They must fulfil at least the requirements of DIN 51517 or DIN 51524.

Special oils, often synthetic oils, are used for extreme operating conditions or if special requirements apply relating to oil resistance. In these cases, please consult the lubricant manufacturers or Schaeffler.

Further information

TPI 176 | Lubrication of Rolling Bearings |
https://www.schaeffler.de/std/1F83

2.6 Sealing

The seal has a considerable influence on the service life of a bearing. Its function is to keep the lubricant in the bearing and prevent the ingress of dirt into the bearing.

2 Dirt and its effect on the bearing

Dirt	Effect on the bearing
A large number of very small abrasive particles	• Generates wear in the bearing • An increase in clearance or noise indicates the end of the bearing's service life
Larger, hard particles over which the bearing travels	• Reduce fatigue service life • Pitting occurs at the pressure points at high bearing loads

2.6.1 Seals

Roller bearings, which do not have their own seals, are normally used in bearing housings, such as spherical roller bearings, barrel roller bearings and deep groove ball bearings. The bearing must therefore be sealed by the housing.

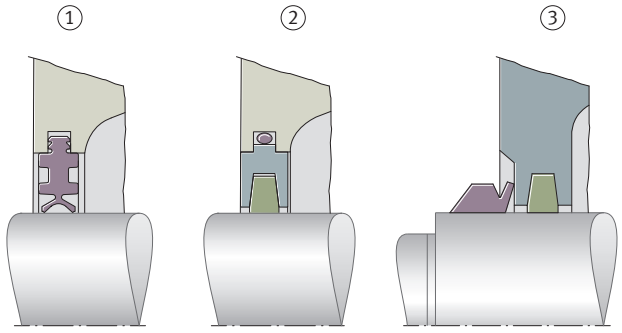
Contact seals, non-contact seals and combinations of these are available to seal the housing against the shaft, depending on the operating conditions.

2.6.1.1 Contact seals

Contact seals are normally in contact with the running surface due to radial contact force. The contact force should be kept low to keep frictional torque and temperature low as well. The frictional torque, temperature and seal wear are also affected by the seal lubrication on the running surface, the roughness of the running surface and the sliding velocity.

The contact seals used in bearing housings are usually double lip seals, felt seals and felt seals with a V ring.

13 Examples of contact seals



1	Double lip seal	2	Felt seal
3	Felt seal with V ring		

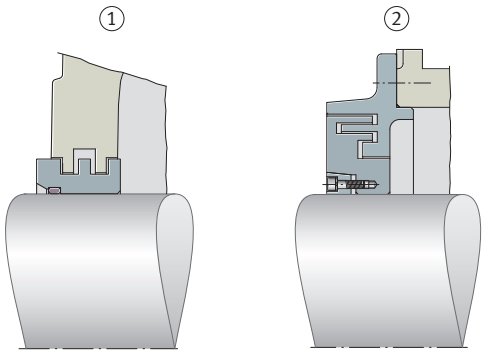
2.6.1.2 Non-contact seals

On non-contact seals, there is no friction in the lubricant gap other than lubricant friction. The seals do not suffer wear and remain functional for a long time. Since they do not generate heat, non-contact seals are also suitable for very high speeds.

The lubricant in the lubrication gap in these seals makes an important contribution to the sealing action. There is an option for separate relubrication on some of the seals.

The seals used in bearing housings are mainly radial labyrinths, axial labyrinths and combinations of them.

14 Examples of non-contact seals

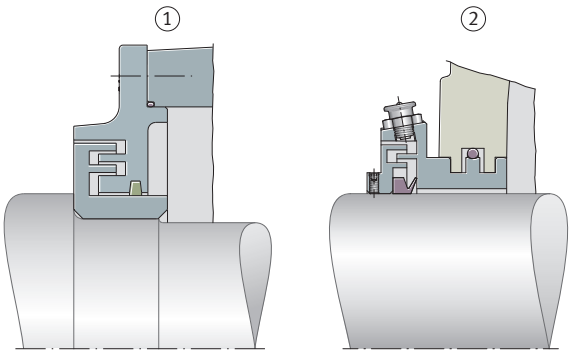


1	Radial labyrinth	2	Axial labyrinth

2.6.1.3 Combined seals

Combined seals comprising contact sealing elements and non-contact sealing elements are often used where there are high requirements for the sealing effect. The combination of a labyrinth seal that can be relubricated with a V ring is also known as a Taconite seal. This seal features a particularly good sealing effect against dust and abrasive particles.

15 Example of combined seals



1	Labyrinth and felt seal	2	Taconite seal

2.6.2 Cover

Where the shaft does not pass fully through the housing, one side of the housing is sealed by a cover. In split housings, the cover is generally inserted in the lower half of the housing and the upper half of the housing is fitted. In unsplit housings, the cover is flanged to the housing.

The maximum operating temperatures of the covers depend on the materials used for them. Various designed may be selected for some housings.

Many housings with a continuous shaft are also fitted with covers. These covers are not sealed and instead have a ring-shaped design and are sealed against the shaft.

2.7 Installation and removal

Schaeffler assembly manual MH 1 provides extensive information about the correct storage, assembly, removal and maintenance of rotary roller bearings. The assembly manual also provides information for the designer relating to the assembly, removal and maintenance of bearings which is required for designing the bearing.

Further information

MH 1 | Mounting Handbook |
https://www.schaeffler.de/std/1D53

HR 1 | Rolling Bearings |
https://www.schaeffler.de/std/1D3D

2.7.1 Basic specifications

The correct assembly of roller bearings and housings has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information.

The relevant legal regulations and other directives relating to environmental protection and occupational safety must be observed in your own interest.

2.7.1.1 Cleanliness

Cleanliness is a fundamental requirement for the successful assembly of roller bearings and housings.

Dirt can shorten the service life of roller bearings and must therefore be prevented

Measures for ensuring cleanliness include the following:

- Ensure that the assembly area is clean.
- Clean the mounting surface.
- Clean the housing, seals and cover, paying particular attention to any machining or moulding sand residues.
- Use only volatile solvents and lint-free cloths for cleaning.
- Do not remove roller bearings and adapter sleeves or withdrawal sleeves from their packaging until immediately before installing them. Do not remove the anti-corrosion protection from these components.

2.7.1.2 Eye bolts

On many housings, the housing body features 1 or 2 eye bolts to DIN 580. These are designed for use as anchoring points for the installation and removal of the housing. The load capacity of the eye bolts allows the housing to be lifted including, in many cases, the installed bearing, but excluding a shaft. Details of this are provided in the description of the relevant housing.

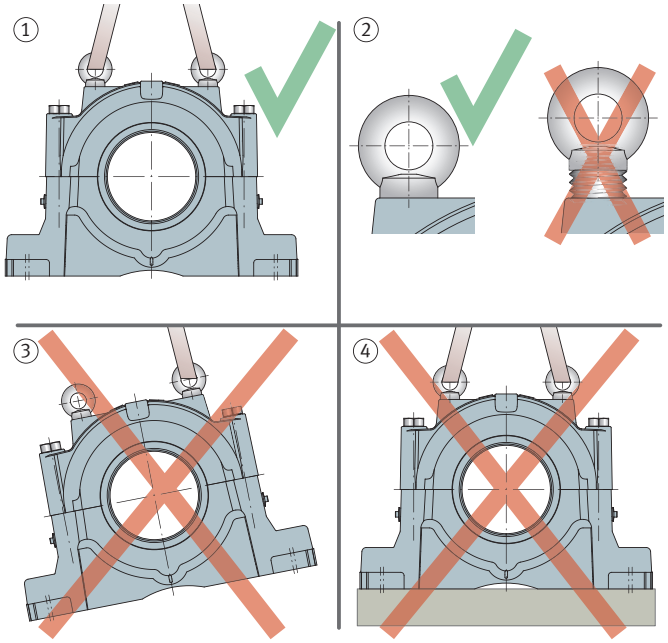
KPG and KPGZ split plummer block housings also feature eye bolts in the covers. These eye bolts must not be used for lifting the housing but only for lifting the covers.

Rupture of the eye bolts due to overload. Serious personal injury or property damage caused by large housing or housing parts falling.

Specifications for using eye bolts as locating points:

- Eye bolts must always be screwed fully into the housing.
- If there are several eye bolts on the housing body, all the eye bolts must be used simultaneously as anchoring points.
- Only use eye bolts for lifting the housing and, if permitted for this housing, the installed bearing. The eye bolts must not be subjected to any additional loads from parts attached to the housing.

16 Correct usage of eye bolts on the housing body



1	Correct usage of eye bolts as anchoring points	2	Screw in eye bolts fully
3	Always use all the eye bolts at the same time	4	No additional load caused by attached parts

2.7.2 Securing the housings

The main requirements for securing bearing housings are suitable mounting surfaces, the correct selection and use of base screws and, in certain cases, the additional horizontal anchoring of the housing.

2.7.2.1 Properties of the mounting surface

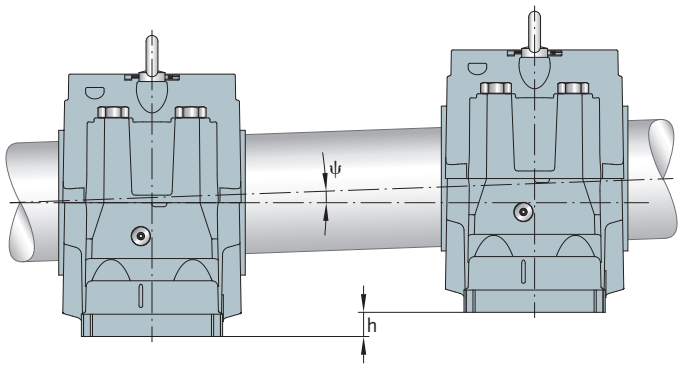
Requirements for the quality of the clamping surface:

- Sufficient strength to withstand the static and dynamic loads which occur during operation in the long term.
- Surface roughness: Ramax 12,5
- Flatness of the mounting surface to DIN EN ISO 1101 which corresponds to tolerance level IT7 to DIN EN ISO 286-1 (measured over the diagonals)
- No coatings

2.7.2.2 Level of mounting surfaces

A difference in level between the mounting surfaces of bearing housings will result in the misalignment of the shaft.

17 Misalignment of the shaft



ψ	Misalignment of the shaft	h	Difference in level between mounting surfaces

The maximum alignment error depends on the housing and seal versions. Differences in level must be compensated to ensure that the maximum misalignment is not exceeded. Levelling shims can be used for this purpose.

In addition, it must be ensured that the installed bearings can compensate for any misalignment.

2.7.2.3 Tightening torques for base fastening bolts

Base fastening screws are used for secure the housing on the mounting surface. They are not supplied with the housings.

The maximum tightening torques apply for 90 % use of the yield stress of the screw material 8.8 and for a friction factor of 0,14. We recommend that base fastening screws should be tightened to approx. 70 % of these values.

The following table lists the tightening torques for standard metric threads to DIN 13, DIN 962 and DIN ISO 965-2.

▦3 Tightening torques for screws with standard metric threads to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999

Thread	Tightening torque M_A	
	Max.	Recommended
	N m	N m
M6	11,3	8
M8	27,3	20
M10	54	35
M12	93	65
M16	230	160
M20	464	325
M24	798	550
M30	1597	1100
M36	2778	1950
M42	3991	2750
M48	6021	4250
M56	9650	6750
M64	14416	10000
M72	21081	14500
M80	29314	20500
M90	42525	29500
M100	59200	41000

The specified tightening torques apply to base fastening bolts with standard threads to ASTM A449 Grade 5 with a friction coefficient of 0.15. This friction coefficient can be assumed for steel bolts that are oiled to ASTM A449.

▦4 Tightening torques for base fastening screws to ASTM A449 Grade 5

Base fastening screws	Recommended tightening torque M_A	
	ft-lbs	Nm
$^1/_2$ " – 13 UNC	55	70
$^5/_8$ " – 11 UNC	105	140
$^3/_4$ " – 10 UNC	185	255
$^7/_8$ " – 9 UNC	300	405
1 " – 8 UNC	450	610
1 $^1/_4$ " – 7 UNC	785	1065
1 $^1/_2$ " – 6 UNC	1365	1850
1 $^5/_8$ " – 5 $^1/_2$ UNC	1290	1745
1 $^7/_8$ " – 5 UNC	2000	2710
2 " – 4 $^1/_2$ UNC	2405	3260
2 $^1/_4$ " – 4 $^1/_2$ UNC	3515	4770
2 $^1/_2$ " – 4 UNC	4810	6525
3 " – 4 UNC	8615	11680

2.7.2.4 Horizontal fastening

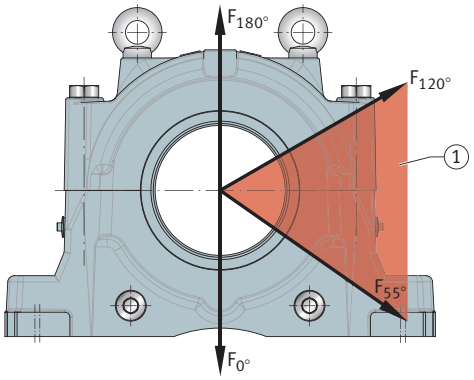
Plummer block housings may require the additional horizontal fastening of the housing as well as the base fastening screws.

Horizontal fastening is required in the following conditions:

- The load angle is between 55 ° and 120 °.
- An axial load is applied.

Depending on the housing, this fastening may take the form of stops in the load direction or pins.

18 Load directions on a plummer block housing



1	Load angle range within which the horizontal fastening of the housing is required

2.7.3 Fastening the bearings to the shaft

The full assembly of a bearing housing always includes the installation of the bearing.

2.7.3.1 Bearing seats on the shaft

The bearing seats on the shaft must be free from impact marks and burrs and must have adequate dimensional and geometrical accuracy.

The dimensional and geometrical accuracy requirements of the bearing seats with a tapered bearing bore and fastening by means of an adapter sleeve, withdrawal sleeve or adjustment sleeve are as follows:

- The diameter is at least in tolerance class h9 Ⓔ, ideally h8 Ⓔ
- Cylindricity to DIN EN ISO 1101 within IT5/2

The dimensional and geometrical accuracy requirements of the bearing seats with a cylindrical bearing bore and the bearing seated on the shaft depend on the operating conditions.

2.7.3.2 Hydraulic method

The installation and removal of large bearings requires high assembly forces and can be made easier by using the hydraulic method.

5 Suffix for sleeves

Sleeves	Suffix
Adapter sleeves with oil grooves on the tapered outer surface and a pump connection on the thread side	HG
Withdrawal sleeves with oil grooves on the tapered outer surface and 2 pump connections on the thread side offset from each other by 90 °	H

As from a shaft diameter of 150 mm, we recommend the hydraulic method to may assembly easier.

2.7.3.3 Location of bearings with a cylindrical bore on the shaft

Bearings with a cylindrical bore are installed in plummer block housings on a stepped shaft. First the bearing is slid onto the shaft and is then followed by a spacer sleeve. The outer diameter of the spacer sleeve must be identical to the larger diameter of the stepped shaft. The seal is slid onto the spacer sleeve. No spacer sleeve is used on a housing which is closed on one side.

Spacer sleeves are not supplied with the housings.

2.7.3.4 Spherical roller bearings with a tapered bore and adapter sleeve

Spherical roller bearings with a tapered bore that are located on the shaft using an adapter sleeve are often used in bearing housings. The adapter sleeve is slid onto the shaft first and the bearing is then slid onto the taper of the adapter sleeve. This therefore reduces the radial clearance of the bearing. At the same time it creates radial prestressing between the bearing inner ring, adapter sleeve and shaft, ensuring that the bearing is seated securely on the shaft.

Hydraulic nuts can be used to make pressing on the bearing easier.

The reduction in the radial clearance is found on spherical roller bearings by measuring the residual clearance using feeler gauges or measuring the sliding distance. The values given for the reduction in the radial clearance and the sliding distance on the taper ensure that the bearing is securely seated on the shaft.

6 Reduction in the radial clearance and sliding distance on the taper

Nominal bearing bore diameter		Reduction in radial clearance		Sliding distance on the taper 1:12				Sliding distance on the taper 1:30				Minimum radial clearance after installation, control value for		
d				Shaft		Adapter sleeve		Shaft		Adapter sleeve		Group N	Group 3	Group 4
mm		mm		mm		mm		mm		mm		mm	mm	mm
from	to	min.	Max.	min.	Max.	min.	Max.	min.	Max.	min.	Max.	min.	min.	min.
24	30	0,015	0,02	0,3	0,35	0,3	0,4	–	–	–	–	0,015	0,02	0,035
30	40	0,02	0,025	0,35	0,4	0,35	0,45	–	–	–	–	0,015	0,025	0,04
40	50	0,025	0,03	0,4	0,45	0,45	0,5	–	–	–	–	0,02	0,03	0,05
50	65	0,03	0,04	0,45	0,6	0,5	0,7	–	–	–	–	0,025	0,035	0,055
65	80	0,04	0,05	0,6	0,75	0,7	0,85	–	–	–	–	0,025	0,04	0,07
80	100	0,045	0,06	0,7	0,9	0,75	1	1,7	2,2	1,8	2,4	0,035	0,05	0,08
100	120	0,05	0,07	0,7	1,1	0,8	1,2	1,9	2,7	2	2,8	0,05	0,065	0,1
120	140	0,065	0,09	1,1	1,4	1,2	1,5	2,7	3,5	2,8	3,6	0,055	0,08	0,11
140	160	0,075	0,1	1,2	1,6	1,3	1,7	3	4	3,1	4,2	0,055	0,09	0,13
160	180	0,08	0,11	1,3	1,7	1,4	1,9	3,2	4,2	3,3	4,6	0,06	0,1	0,15
180	200	0,09	0,13	1,4	2	1,5	2,2	3,5	4,5	3,6	5	0,07	0,1	0,16
200	225	0,1	0,14	1,6	2,2	1,7	2,4	4	5,5	4,2	5,7	0,08	0,12	0,18
225	250	0,11	0,15	1,7	2,4	1,8	2,6	4,2	6	4,6	6,2	0,09	0,13	0,2
250	280	0,12	0,17	1,9	2,6	2	2,9	4,7	6,7	4,8	6,9	0,1	0,14	0,22
280	315	0,13	0,19	2	3	2,2	3,2	5	7,5	5,2	7,7	0,11	0,15	0,24
315	355	0,15	0,21	2,4	3,4	2,6	3,6	6	8,2	6,2	8,4	0,12	0,17	0,26
355	400	0,17	0,23	2,6	3,6	2,9	3,9	6,5	9	5,8	9,2	0,13	0,19	0,29
400	450	0,2	0,26	3,1	4,1	3,4	4,4	7,7	10	8	10,4	0,13	0,2	0,31
450	500	0,21	0,28	3,3	4,4	3,6	4,8	8,2	11	8,4	11,2	0,16	0,23	0,35
500	560	0,24	0,32	3,7	5	4,1	5,4	9,2	12,5	9,6	12,8	0,17	0,25	0,36
560	630	0,26	0,35	4	5,4	4,4	5,9	10	13,5	10,4	14	0,2	0,29	0,41
630	710	0,3	0,4	4,6	6,2	5,1	6,8	11,5	15,5	12	16	0,21	0,31	0,45
710	800	0,34	0,45	5,3	7	5,8	7,6	13,3	17,5	13,6	18	0,23	0,35	0,51
800	900	0,37	0,5	5,7	7,8	6,3	8,5	14,3	19,5	14,8	20	0,27	0,39	0,57

Nominal bearing bore diameter		Reduction in radial clearance		Sliding distance on the taper 1:12				Sliding distance on the taper 1:30				Minimum radial clearance after installation, control value for		
d				Shaft		Adapter sleeve		Shaft		Adapter sleeve		Group N	Group 3	Group 4
mm		mm		mm		mm		mm		mm		mm	mm	mm
from	to	min.	Max.	min.	Max.	min.	Max.	min.	Max.	min.	Max.	min.	min.	min.
900	1000	0,41	0,55	6,3	8,5	7	9,4	15,8	21	16,4	22	0,3	0,43	0,64
1000	1120	0,45	0,6	6,8	9	7,6	10,2	17	23	18	24	0,32	0,48	0,7
1120	1250	0,49	0,65	7,4	9,8	8,3	11	18,5	25	19,6	26	0,34	0,54	0,77
1250	1400	0,55	0,72	8,3	10,8	9,3	12,1	21	27	22,2	28,3	0,36	0,59	0,84

2.7.4 Maintenance

Careful maintenance of a housing and monitoring the operating state of the bearing make a significant contribution towards achieving a long service life and reliable operation.

Regular maintenance must be carried out and the intervals for the various maintenance operations depend on the ambient and operating conditions.

Only carry out maintenance work when the machine is at a standstill and make preparations to prevent the machine from starting up unintentionally.

2.7.4.1 Regular maintenance work

Regular maintenance work on bearing housings are as follows:

- Checking the alignment of the housing.
- Retightening of the base fastening screws and connecting screws. This should be carried out initially more frequently after commissioning and at longer intervals later.
- Inspect the housing for signs of damage. Any indications such as loud noises or unusual grease escape must also be checked.
- Relubrication of the seals. Some have lubrication nipples for this purpose. If the ambient air contains high levels of dust, the seals must be relubricated more frequently.
- Relubrication of the installed bearings. The grease is inserted into the housing through lubrication nipples and then gets to the bearing. During relubrication, it must be possible for excess grease to escape through non-contact seals or opened grease outlet holes.
- Cleaning the housing to remove baked-on material and other coarse dirt.

If the housing is not cleaned correctly, dirt or moisture may get into the housing. In particular, do not use compressed air, steam cleaners or similar cleaning methods around seals

2.8 Special solutions for standard housings

In addition to standard Schaeffler housings, other versions of these housings can be supplied or the package extended.

Basis for special solutions:

- Satisfaction of special technical requirements
- Compliance with special customer-specific requirements

The special solutions and additional services described below are available on request.

Bearing seat tolerance class

Bearing seat tolerance class according to customer specification.

High temperature seal

The felt seal can be replaced by a high-temperature seal to extend application limits.

Properties of the high-temperature seal:

- Temperature resistance up to +300 °C
- Circumferential speed up to 20 m/s

Lubrication nipples

Lubrication nipples or fastening parts for the relubrication holes that deviate from the standard package are available.

Long term packaging

Long term packaging of housings allows a storage time from 3 a to 10 a, up to , depending on the preservative and climatic conditions. The storage regulations must be observed.

Long term packaging is available for housings with a mass of 30 kg or greater.

Acceptance inspection of products

The acceptance inspection of products by the customer can be carried out on site. The acceptance documents and inspection certificates can be prepared to Schaeffler specifications or customer specifications.

Acceptance inspection certificate

Preparation of an acceptance test certificate 3.1 to DIN EN 10204.

3 SES split plummer block housing

medias | Product catalogue |
SES split plummer block housing |
https://www.schaeffler.de/std/21F9

3.1 Housing design

Split plummer block housings and the associated bearings form bearing arrangement units that can be matched, through the appropriate selection of accessories, to a wide range of applications. Applications include agricultural machinery, the paper processing industry, mining, materials processing, the steel industry and power stations.

Split plummer block housings SES are available in the following design variants:

- standard design with 2 extended slots for foot screws
- special design with 4 holes for foot screws

Housings of size SES524-620 and larger have 1 eye bolt.

Bearing housings, standard design

Housings in the standard design have 2 extended slots for location with 2 foot screws.

19 Split plummer block housings SES, standard design

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Bearing housings, special design with 4 holes for foot screws

Housings with 4 holes for foot screws are available as a special design by agreement. These are indicated by the suffix G944DB in the ordering designation.

Hole dimensions ➤57|3.9.1

20 Split plummer block housings SES, special design with 4 holes for foot screws

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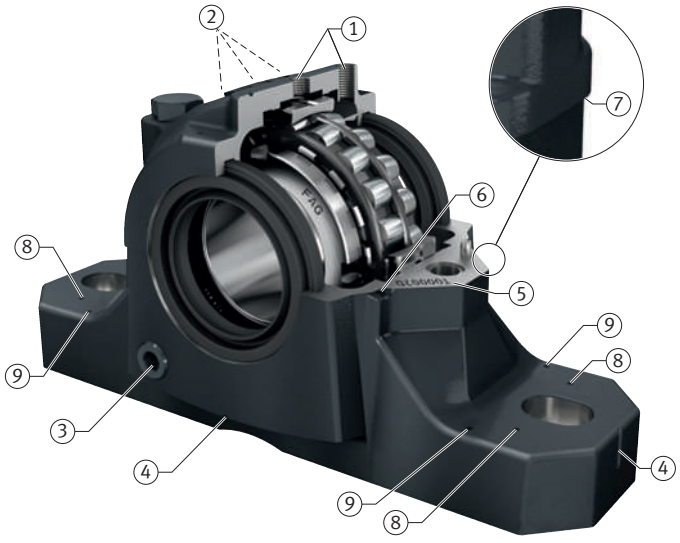
3.1.1 Advantages of the detailed design

The details in the housing design are systematically matched to practical application and make the SES plummer block housings particularly flexible and economical to use.

21 Advantages of the detailed design



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1	Lubrication holes for central and lateral lubricant feed	2	Marked connection points for condition monitoring
3	Grease outlet hole	4	Indentations indicating the housing centre
5	Serial numbering of housing halves	6	Dismounting notches
7	Projection of upper housing section	8	Marked positions for additional fixing holes
9	Marked positions for pins		

3.1.1.1 Lubrication holes

2 lubrication holes M10×1 in the upper housing section enable central and lateral lubricant feed ➤37|21, (1). The central lubrication hole is connected to a long axial lubrication groove in the upper housing section and thus ensures supply of lubricant irrespective of bearing position to all bearing series that can be fitted. 1 button head lubrication nipple and 1 taper type lubrication nipple are included with the housing.

The lubrication holes also enable the connection of automatic lubricators ➤50|3.5.4.

3.1.1.2 Marked connection points for condition monitoring

Cast-in pilot holes in the upper housing section indicate suitable connection points for condition monitoring ➤37|21, (2). Schaeffler recommends the following online measuring systems for monitoring the condition of machines:

- Schaeffler OPTIME: wireless, cost-effective, easy to use
- Schaeffler SmartCheck: modular construction, ideal for widely fluctuating operating conditions such as speed or load
- other sensors for vibration monitoring

The online measuring systems must be ordered separately.

22 Examples of condition monitoring and lubricant supply systems

1	Condition monitoring system Schaeffler OPTIME	2	Lubricator OPTIME C1

Further information

Service Info | What is OPTIME and how does it work? |
https://www.schaeffler.de/std/1FF0

TPI 214 | Schaeffler SmartCheck |
https://www.schaeffler.de/std/1B6C

3.1.1.3 Grease outlet hole

A grease outlet hole in the lower housing section allows the escape of used and superfluous grease during relubrication ➤37|21, (3).

3.1.1.4 Indentations to mark the housing centre

Indentations on the end and side surfaces of the lower housing section mark the housing centre. The marks ensure rapid alignment and reduce assembly time.

3.1.1.5 Serial numbering of housing halves

The upper housing section and lower housing section are matched to each other and must not be transposed with other housing halves. Serial numbers marked on the housing cap and housing base ensure that the upper and lower housing sections are correctly matched during assembly. Furthermore, the serial number ensures that the product can be clearly identified and is beneficial for traceability and documentation.

3.1.1.6 Dismounting notches

Recessed dismounting notches on the interface of the housing halves make it easier to remove the upper housing section.

3.1.1.7 Projection of upper housing section

The projection of the upper housing ensures the following:

- It prevents the collection of fluids and contaminants at the interface of the two housing halves
- It reduces the risk of dirt entering the housing
- It reduces the risk of corrosion

3.1.1.8 Marked positions for additional fixing holes

4 additional fixing holes allow the housing to be mounted on T-beams. Cast-in pilot holes in the lower housing section indicate the positions for the additional fixing holes ➤37|21, (8).

3.1.1.9 Marked positions for pins

The use of pins to fix the housing is also available as an option. The optimum positions for the holes for the pins are marked on the lower housing section ➤37|21, (9) ➤57|3.9.1.

3.1.1.10 Improved material

An improved material increases the rigidity of the housing. As a result, the robust housing is even more suitable for aggressive environmental conditions.

3.1.2 Modular concept

The housings are designed in accordance with a modular concept. Each housing can be fitted with rolling bearings of various diameter series and width series if they have the correct outside diameter for the housing.

Depending on their design, the bearings can be located either directly on the shaft or by means of an adapter sleeve. This gives different shaft diameters for the same bearing size. The spacings between the shaft and housing body are compensated by means of appropriately matched seals.

3.1.3 Locating and non-locating bearings

The bearing seats in the housing are machined so that the bearings can move and can act as non-locating bearings.

Locating bearings can be achieved by inserting 1 FRM locating ring on each side of the bearing outer ring. The bearing is then seated in the centre of the housing.

2 FRM locating rings must be ordered separately for each locating bearing.

Locating rings must be ordered separately.

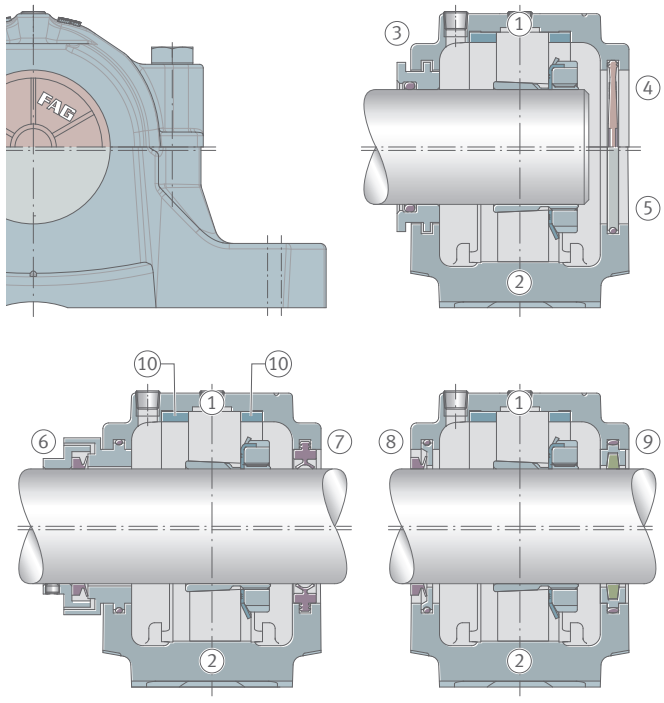
3.1.4 Housing configurations

The modular design of plummer block housings allows a wide range of possible combinations.

Starting with the standard components, the following features can be varied in the housing configuration:

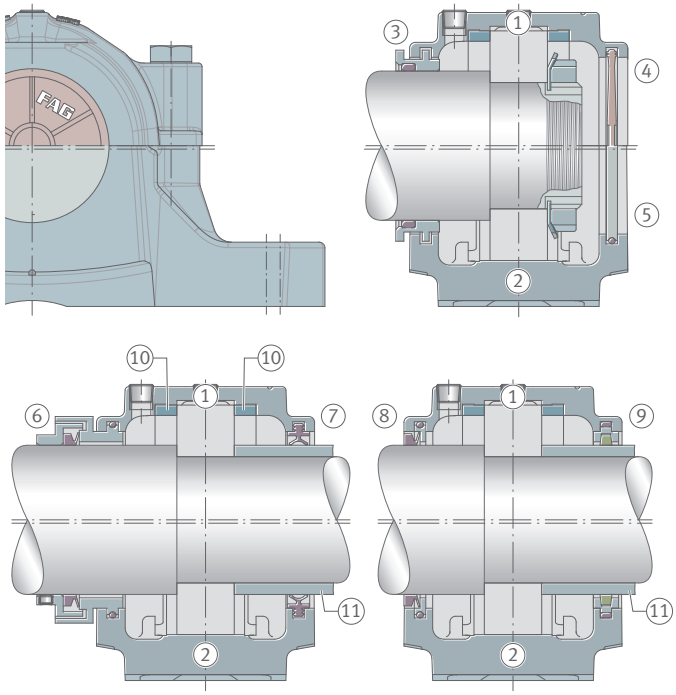
- Fastening: bearings with a tapered bore by an adapter sleeve on a shaft with a constant diameter or bearings with a cylindrical bore on a stepped shaft
- Housing sealing using a double lip seal, felt seal, labyrinth seal, V ring seal orTaconite seal
- Continuous shaft with an open housing on both sides or non-continuous shaft with housing closed on one side
- Plastic EDK cover or steel EDKT cover
- Bearing in the form of a non-locating bearing or a locating bearing by the additional insertion of 2 FRM locating rings
- Various bearing types

23 Plummer block housings SES for bearings with tapered bore and adapter sleeve, overview of combinations



1	Locating bearing	2	Non-locating bearing
3	Labyrinth seal ETS	4	Cover EDK
5	Cover EDKT	6	Taconite seal ETC
7	Double lip seal EDH	8	V ring seal EDV
9	Felt seal EFS	10	Locating ring FRM

24 Plummer block housings SES for bearings with cylindrical bore, overview of combinations

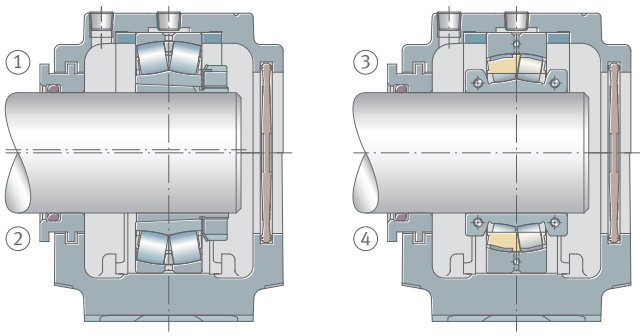


1	Locating bearing	2	Non-locating bearing
3	Labyrinth seal ETS	4	Cover EDK
5	Cover EDKT	6	Taconite seal ETC
7	Double lip seal EDH	8	V ring seal EDV
9	Felt seal EFS	10	Locating ring FRM
11	Spacer sleeve, not included in the scope of delivery		

Installation of split spherical roller bearing

For SES plummer block housings, an unsplit spherical roller bearing with a tapered bore and adapter sleeve can be replaced by a split spherical roller bearing.

25 Plummer block housings SES with split and unsplit spherical roller bearing



1	Locating bearing with unsplit bearing	2	Non-locating bearing with unsplit bearing
3	Locating bearing with split bearing	4	Non-locating bearing with split bearing

3.1.5 Accessories

Accessories must be ordered separately.

Seals

Detailed information about ➤50|3.6

26 EDH double lip seals

0001AD20

27 EFS felt seals

0007EEB2

28 ETS labyrinth seals

0007EEBA

29 EDV V ring seals

0001AD21

30 Taconite ETC seals

0001AD2B

Cover

Detailed information about ➤50|3.6

31 Plastic EDK covers

0001AD6F

32 Steel and FKM EDKT covers

0001AD23

Locating rings

33 FRM locating rings

0001AD2A

3.2 Suitable bearings

SES split plummer block housings are suitable for the installation of spherical roller bearings, barrel roller bearings and spherical ball bearings with a tapered or cylindrical bore as well as deep groove ball bearings. The shaft diameters are 20 mm to 160 mm and $^{3}/_{4}$ inch to 5 $^{1}/_{2}$ inch.

7 Suitable bearings for SESsplit plummer block housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	21306..-K	21322..-K
	22205..-K	22232..-K
	22308..-K	22332..-K
	23218..-K	23232..-K
Spherical roller bearings with a cylindrical bore	21304	21322
	22205	22232
	22308	22332
	23218	23232
Sealed spherical roller bearings with a cylindrical bore [1)]	WS22205-E1-XL-2RSR	22232-E1A-XL-2RSR-M
Sealed spherical roller bearings with a tapered bore and adapter sleeve [1)]	WS22205-E1-XL-K-2RSR	22232-E1A-XL-K-2RSR-M
Split spherical roller bearings with a cylindrical bore	222SM55-TVPA	222SM140-TVPA
	222S.203	222S.508
Spherical ball bearings with a tapered bore and adapter sleeve	1205-K	1222-K
	1305-K	1322-K
	2205-K	2222-K
	2305-K	2322-K
Spherical ball bearings with a cylindrical bore	1205	1226
	1304	1322
	2205	2222
	2304	2322
Barrel roller bearings with a tapered bore and adapter sleeve	20205-K	20232-K
	20305-K	20324-K
Barrel roller bearings with a cylindrical bore	20205	20232
	20304	20330
Deep groove ball bearings with a cylindrical bore	6205	6232
	6304	6324

1) When using sealed spherical roller bearings with a spherical bore and adapter sleeve, the possible non-locating bearing displacement is reduced. To ensure that the correct combination of housing and bearing is selected, please contact Schaeffler.

Split spherical roller bearings

In bearing replacement, unsplit spherical roller bearings with an adapter sleeve can be replaced by split spherical roller bearings. In many cases, the use of split spherical roller bearings significantly reduces the work associated with bearing replacement.

In order to ensure selection of the correct combination of housing and bearing, please contact us.

Further information

TPI 250 | Split spherical roller bearings |
https://www.schaeffler.de/std/1F81

3.3 Materials, corrosion protection

8 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron GJL	L	✓	RAL 7016, anthracite grey
Ductile cast iron GJS	D	o	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

3.4 Load carrying capacity

The guide values for the housing rupture load apply to the standard housing material gray cast iron. 1,6 times this value applies for ductile cast iron.

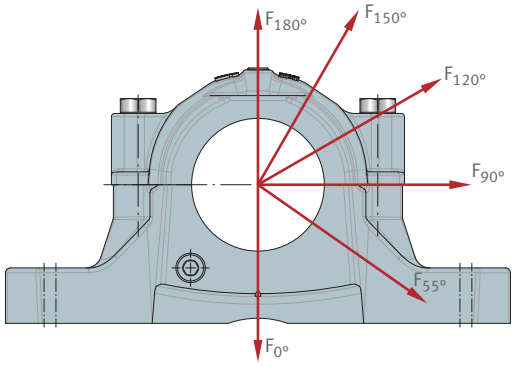
When determining the permissible static load, safety factors must be applied:

- safety factor of 6 relative to the guide values: housing rupture load for gray cast iron, ductile cast iron
- safety factor of 3 relative to the maximum load carrying capacity of the connecting screws and foot screws

Additional factors influencing load carrying capacity:

- If the housing is subjected to axial load, the permissible axial load of the bearing fitted must be taken into consideration.
- If an adapter sleeve is used to locate the bearing on the shaft, the axial retaining force of the bearing and adapter sleeve must also be taken into consideration.

34 Load directions F



9 Guide values for housing rupture load and maximum load capacity of the connecting screws, tightening torques

Housing		Housing rupture load in load direction F [1]							Connecting screws				
									G	M_A [2]	Max. load capacity of 2 screws in load direction F [3]		
		0 °	55 °	90 °	120 °	150 °	180 °	Axial			120 °	150 °	180 °
		kN	kN	kN	kN	kN	kN	kN	–	N m	kN	kN	kN
SES505	SES205	100	155	95	70	60	80	52	M10	51	60	35	30
SES506-605	SES206-305	130	170	100	80	65	85	55	M10	51	60	35	30
SES507-606	SES207	140	190	115	85	80	95	60	M10	51	60	35	30

Housing		Housing rupture load in load direction F [1]							Connecting screws				
									G	M_A [2]	Max. load capacity of 2 screws in load direction F [3]		
		0 °	55 °	90 °	120 °	150 °	180 °	Axial			120 °	150 °	180 °
		kN	kN	kN	kN	kN	kN	kN	–	N m	kN	kN	kN
SES508-607	SES208-307	150	215	130	95	85	110	70	M10	51	60	35	30
SES509	SES209	160	230	140	100	90	115	75	M10	51	60	35	30
SES510-608	SES210	170	265	155	120	110	130	85	M10	51	60	35	30
SES511-609	SES211	270	290	215	160	130	140	90	M12	87	80	45	40
SES512-610	SES212	210	300	180	130	120	150	100	M12	87	80	45	40
SES513-611	SES213	270	340	205	150	130	170	110	M12	87	80	45	40
–	SES214	270	340	205	150	130	170	110	M12	87	80	45	40
SES515-612	SES215	290	410	250	185	160	205	135	M12	87	80	45	40
SES516-613	SES216	350	430	260	190	175	215	140	M12	87	80	45	40
SES517	SES217	500	525	420	250	205	240	190	M12	87	80	45	40
SES518-615	SES218	430	550	340	250	215	275	180	M16	215	180	100	90
SES519-616	–	870	670	450	270	230	290	220	M16	215	180	100	90
SES520-617	–	500	740	660	325	270	310	310	M20	430	260	150	130
SES618-318	–	550	630	375	280	250	300	200	M20	430	260	150	130
SES522-619	–	600	680	410	310	275	340	220	M20	430	260	150	130
SES524-620	–	800	790	470	350	320	400	260	M20	430	260	150	130
SES526	–	900	900	540	410	360	450	295	M24	740	360	210	180
SES622-322	–	950	1000	600	450	400	500	320	M24	740	360	210	180
SES528	–	1165	1160	895	530	435	530	530	M24	740	360	210	180
SES624-324	–	1050	1180	710	530	475	600	360	M24	740	360	210	180
SES530	–	1100	1200	730	540	480	600	390	M24	740	360	210	180
SES626-326	–	1170	1320	750	600	530	630	400	M24	740	360	210	180
SES532	–	1300	1450	860	640	570	720	470	M24	740	360	210	180
SES628-328	–	1350	1500	900	670	600	750	500	M24	740	360	210	180
SES630-330	–	1500	1700	1000	750	670	850	550	M24	740	360	210	180
SES632-332	–	1700	1900	1120	850	750	950	620	M30	1450	640	370	320

1) For housings made from flake graphite cast iron. Housings made from spheroidal graphite cast iron have a housing rupture load that is 1,6 times higher than that of housings made from flake graphite cast iron.

2) The tightening torques are maximum values when utilising 90 % of the yield stress of the screw material and with a friction factor of 0,14. Schaeffler recommends that the screws should be tightened to 70 % of these values.

3) Maximum load carrying capacity in load direction F for maintaining contact at the parting surface between the upper and lower housing sections

G	–	Thread
M_A	Nm	Tightening torque

3.5 Lubrication

Split plummer block housings SES are principally intended for grease lubrication.

3.5.1 Greases

In order to achieve a long operating life and high operational security of the bearing arrangement, Schaeffler recommends the use of Arcanol rolling bearing greases. These rolling bearing greases have been designed and tested for bearing arrangement engineering.

Further information

TPI 168 | Arcanol Rolling Bearing Greases |
https://www.schaeffler.de/std/1F66

TPI 176 | Lubrication of Rolling Bearings |
https://www.schaeffler.de/std/1F83

3.5.2 Grease quantities for initial greasing

For initial greasing, the basic rule is that the bearing should be filled with grease to 100 % and the free volume of the housing to 60 %. The free volume is the space that remains in the housing once the bearing, adapter sleeve, shaft and seals have been fitted.

The recommended grease quantities specified in the following table take account of the following criteria:

- filling of the free volume to 60 %
- grease with a density of 0,9 g/cm³

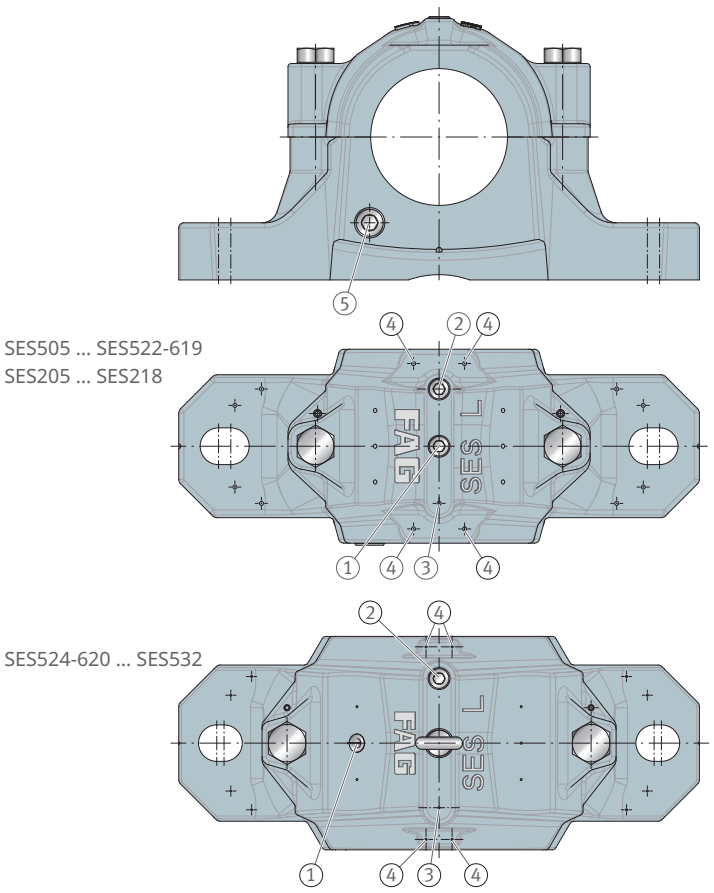
10 Recommended grease quantity for initial grease application

Housing			Recommended grease quantity	
Designation		Free volume	Volume	Mass
		cm³	≈ cm³	≈ g
SES505	SES205	72	43	39
SES506-605	SES206-305	118	71	64
SES507-606	SES207	145	87	78
SES508-607	SES208-307	164	98	89
SES509	SES209	184	110	99
SES510-608	SES210	211	127	114
SES511-609	SES211	263	158	142
SES512-610	SES212	388	233	210
SES513-611	SES213	474	284	256
–	SES214	537	322	290
SES515-612	SES215	605	363	327
SES516-613	SES216	816	490	441
SES517	SES217	961	577	519
SES518-615	SES218	1250	750	675
SES519-616	–	1421	853	767
SES520-617	–	1855	1113	1002
SES618-318	–	1759	1055	950
SES522-619	–	2513	1508	1357
SES524-620	–	2974	1784	1606
SES526	–	3289	1973	1776
SES622-322	–	3148	1889	1700
SES528	–	4211	2527	2274
SES624-324	–	3704	2222	2000
SES530	–	5132	3079	2771
SES626-326	–	4815	2890	2600
SES532	–	6053	3632	3268
SES628-328	–	5740	3445	3100
SES630-330	–	6850	4110	3700
SES632-332	–	8330	5000	4500

3.5.3 Relubrication

3.5.3.1 Lubrication holes

35 Positions for relubrication



1	Central lubrication hole, for bearings with lubrication groove	2	Lateral lubrication hole, for bearings without lubrication groove
3	Marked positions for further lubrication holes for lubrication of the bearing	4	Marked positions for further lubrication holes for lubrication of seals
5	Grease outlet hole		

For relubrication on bearings with a circumferential lubrication groove, the grease is inserted into the housing through the **central lubrication hole** ➤48|35, (1). For this type of relubrication, the grease acts on the actual bearing raceway.

For relubrication on bearings without a lubrication groove, the grease must be inserted into the housing through the **lateral lubrication hole** ➤48|35, (2). In this case, the housing cavities on the side with the lubrication nipple must be completely filled with grease so that the relubrication grease can immediately act on the bearing.

As an alternative to the lubrication holes already in the housing, lubrication holes may be drilled at the other positions indicated by the cast pilot holes ➤48|35, (3).

Lubrication holes may be drilled at other marked points for relubricating labyrinth seals ➤48|35, (4).

Each lubrication hole is sealed with 1 screw plug, which must be removed and permanently replaced with one of the supplied lubrication nipples.

Supplied lubrication nipples with dust cap:

- 1 button head lubricating nipple DIN 3404-A M10×1 St
- 1 taper type lubrication nipple DIN 71412-A M10×1

3.5.3.2 Grease outlet hole

The grease outlet hole is sealed with 1 screw plug to DIN 906.

In order to prevent overgreasing, the screw plug in the grease outlet hole in the lower housing section must be removed while relubrication is being carried out ➤48|35, (5). This allows superfluous grease to escape. When using double lip seals EDH in particular, the superfluous grease must be able to escape. Otherwise, there is a risk that the double lip seal will be pressed out of the housing.

The grease outlet hole must then be closed off again using the screw plug.

In adverse environmental conditions, there is a risk of dirt penetrating the interior of the housing when the grease outlet holes are opened.

3.5.3.3 Dimensions

11 Dimensions of the grease outlet hole and lubrication holes

Housing		Thread for	
		Grease outlet hole	Lubrication hole
SES505	SES205	M10×1	M10×1
SES506-605	SES206-305		
SES507-606	SES207		
SES508-607	SES208-307		
SES509	SES209		
SES510-608	SES210		
SES511-609	SES211	M14×1,5	M10×1
SES512-610	SES212		
SES513-611	SES213		
–	SES214		
SES515-612	SES215	M20×1,5	M10×1
SES516-613	SES216		
SES517	SES217		
SES518-615	SES218		
SES519-616	–		
SES520-617	–		
SES618-318	–		
SES522-619	–		
SES524-620	–		
SES526	–		
SES622-322	–		
SES528	–		
SES624-324	–		
SES530	–		
SES626-326	–		
SES532	–		
SES628-328	–		
SES630-330	–		
SES632-332	–		

3.5.4 Automatic lubricators

The housings can be directly connected to automatic lubricators, such as OPTIME C1 or CONCEPT1 from Schaeffler.

36 Plummer block housing SES with automatic lubricator CONCEPT1

001935AE

We recommend that the grease outlet bore is left open when using an automatic lubricator to allow the escape of used grease.

Further information

TPI 252 | Lubricators |
https://www.schaeffler.de/std/1D4E

3.6 Sealing

Seals

For the sealing of the bearing housings, the standard seals available are the double lip seal, the felt seal, the labyrinth seal, the V ring seal and the Taconite seal. These seals are matched to the rectangular section annular slots on both sides of the housings and are principally suitable for grease lubrication.

The seals must be ordered separately. They are supplied separately. If a continuous shaft is present, 2 seals must be ordered.

Special seals, such as split labyrinth seals, are also available from Schaeffler by agreement.

Covers

Covers are used with housings closed on one side. The covers fit in the rectangular section annular slots in the housings. The covers must be ordered separately.

3.6.1 Double lip seals EDH

The seal lips of the double lip seal EDH slide on the rotating shaft. The outer seal lip prevents ingress of contamination into the bearing. This effect is supported by the grease inserted between the seal lips on assembly. The inner lip prevents lubricant from escaping from the housing.

The seal is made from acrylonitrile butadiene rubber NBR and is suitable for temperatures from −30 °C up to +100 °C.

The seal is suitable for circumferential velocities of up to 13 m/s. For continuous operation, Schaeffler recommends a maximum circumferential velocity of 6 m/s.

The permissible shaft misalignment is 0,5° in both directions.

The double lip seal is a split seal. The seal halves are easily inserted in the annual slots of the housing parts.

3.6.2 EFS felt seals

EFS felt seals are ideal for grease lubrication. The felt seals are made up of 1 split adapter with an inserted, oil-impregnated felt strip. The adapter is secured to prevent it rotating by 1 O-Ring aus FKM O-ring made fromFKM in the annular slot of the housing.

Felt seals can be used in temperatures of up to +100 °C. To order, aramide packings are available for higher temperatures (suffixG600).

The seal is suitable for circumferential velocities of up to 5 m/s, and, after the running-in process, up to 15 m/s.

The permissible shaft misalignment is 0,5° in both directions.

The felt seal is a split seal. The seal halves are easy to insert in the annular slots of the housing parts.

3.6.3 Labyrinth seals ETS

Unsplit labyrinth seals ETS allow non-contact sealing, which makes them suitable for high circumferential velocities. The O-ring made from fluoro rubber FKM, which is pressed between the labyrinth ring and shaft, is suitable for temperatures of up to +200 °C.

The permissible shaft misalignment is 0,5° in both directions.

If necessary, the labyrinth can be relubricated. For relubrication, 1 lubrication hole must be made in the upper housing section for each labyrinth seal. Cast-in pilot holes in the housing indicate the optimum positions for the lubrication holes ➤48|35, (4).

Split labyrinth seals are available by agreement.

3.6.4 V ring seals EDV

In the case of V ring seals EDV, the seal lip is in axial contact with the sliding surface which is located in the rectangular section annular slot of the housing. The V ring is made from NBR and is suitable for temperatures of up to +100 °C.

The seal is suitable, when using grease lubrication, for circumferential velocities of up to 12 m/s. If circumferential velocities of more than 8 m/s are present, axial location is necessary.

The permissible shaft misalignment is 0,5° in both directions.

3.6.5 ETC Taconite seals

Taconite ETC seals are combination seals comprising a labyrinth seal and a V ring. Taconite seals are suitable for extreme operating conditions in terms of dirt and dust. The V ring is made of NBR and is suitable for temperatures of up to +100 °C.

With grease lubrication, the seal is suitable for circumferential velocities of up to 12 m/s. Axial location is required for circumferential velocities of over 8 m/s.

The permissible shaft misalignment is 0,5° in both directions.

Taconite seals suitable for relubrication are available on request.

ETC split Taconite seals are available on request.

3.6.6 Covers EDK

Covers EDK are made from plastic and are suitable for long term temperatures of up to +120 °C.

3.6.7 Covers EDKT

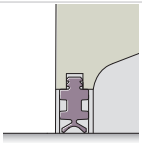
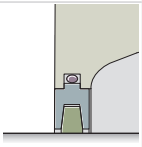
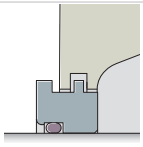
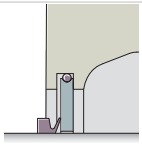
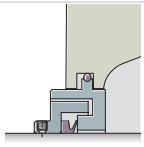
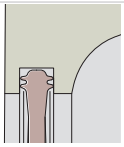
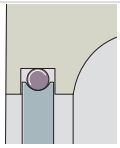
Covers EDKT are made from steel and FKM and are suitable for temperatures of up to +200 °C. These covers should always be used in overhead and side mounting arrangements, regardless of the temperature.

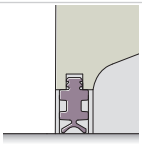
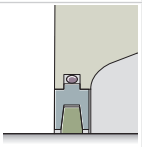
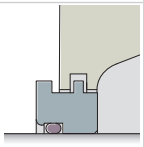
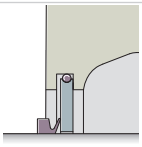
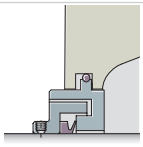
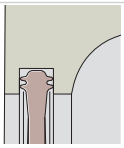
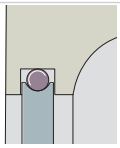
Covers are available by agreement.

3.6.8 Overview of characteristics and operating ranges

The overview shows the characteristics of the seals and covers, their suitability and the demands on the shaft.

12 Standard seals and covers for plummer block housings SES

Design features and suitability		Seals					Covers	
		Double lip seal	Felt seal	Labyrinth seal	V ring seal	Taconite seal	Made from plastic	Made from steel
								
Designation		EDH	EFS	ETS	EDV	ETC	EDK	EDKT
Material		NBR	Steel, felt, FKM	Steel, FKM	Steel, NBR	Steel, NBR	Plastic	Steel, FKM
Split seal		Yes	Yes	No	No	No	Not applicable	Not applicable
Pieces per pack [1]		2	2	1	2	1	1	1
Suitability for sealing against								
Dust		++	+	+	+	++	+	++
Fine solid particles		++	–	+	+	++	+	++
Coarse solid particles		+	o	+	–	++	++	++
Slivers		+	++	++	–	++	++	++
Spray liquids		+	–	–	+	++	+	++
Operating limits								
Long term temperature	°C	–30 ... +100	–30 ... +100	–20 ... +200	–30 ... +100	–30 ... +100	–40 ... +120	–20 ... +200
	°F	–22 ... +210	–22 ... +210	–4 ... +390	–22 ... +210	–22 ... +210	–40 ... +250	–4 ... +390
		Restricted due to NBR		Restricted due to FKM	Restricted due to NBR	Restricted due to NBR		Restricted due to FKM

Design features and suitability		Seals					Covers	
		Double lip seal	Felt seal	Labyrinth seal	V ring seal	Taconite seal	Made from plastic	Made from steel
								
Circumferential velocity	m/s	≤ 13 Permanently ≤ 6	≤ 5 After running-in ≤ 15	No restriction	≤ 12	≤ 12	Not applicable	Not applicable
Misalignment	°	≤ 0,5	≤ 0,5	≤ 0,5	≤ 0,5	≤ 0,5	Not applicable	Not applicable
Low friction		+	+ After running-in	++	+	+	Not applicable	Not applicable
Axial shaft displacement (suitability as non-locating bearing)		++	++	+	o	+	Not applicable	Not applicable
Vertical arrangement		+	–	–	o	o	–	+
Grease retention during relubrication		++ [2]	o	o	o	+	+	++
Oil lubrication		o	–	–	–	–	o	++
Direct sunlight		+	++	++	–	++	o	++
Shaft requirements								
Tolerance class [3]		h8 (h9)	h8 (h9)	h8 (h9)	h8 (h9)	h8 (h9)	Not applicable	Not applicable
Max. roughness	µm	Ra 3,2	Ra 3,2	Ra 3,2	Ra 3,2	Ra 3,2	Not applicable	Not applicable

++ highly suitable
+ suitable
o suitable with restrictions
– not suitable

1) Only multiples of 2 are possible as an order quantity, as there are 2 items per pack

2) Precondition: the grease outlet hole is open during relubrication.

3) The envelope condition Ⓔ applies here.

3.7 Dimensions, tolerances

3.7.1 Dimensions

The dimensions of housings SES correspond to ISO 113 and DIN 736 to DIN 739. Housings SES are interchangeable with housings SN, SNE and SNV.

3.7.2 Tolerances for the bearing seat

The bearing seat in the housings is machined to the tolerance class G7 to DIN EN ISO 286-1. The tolerance indicated applies to the condition on delivery, in other words before the screws connecting the upper and lower sections are loosened.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

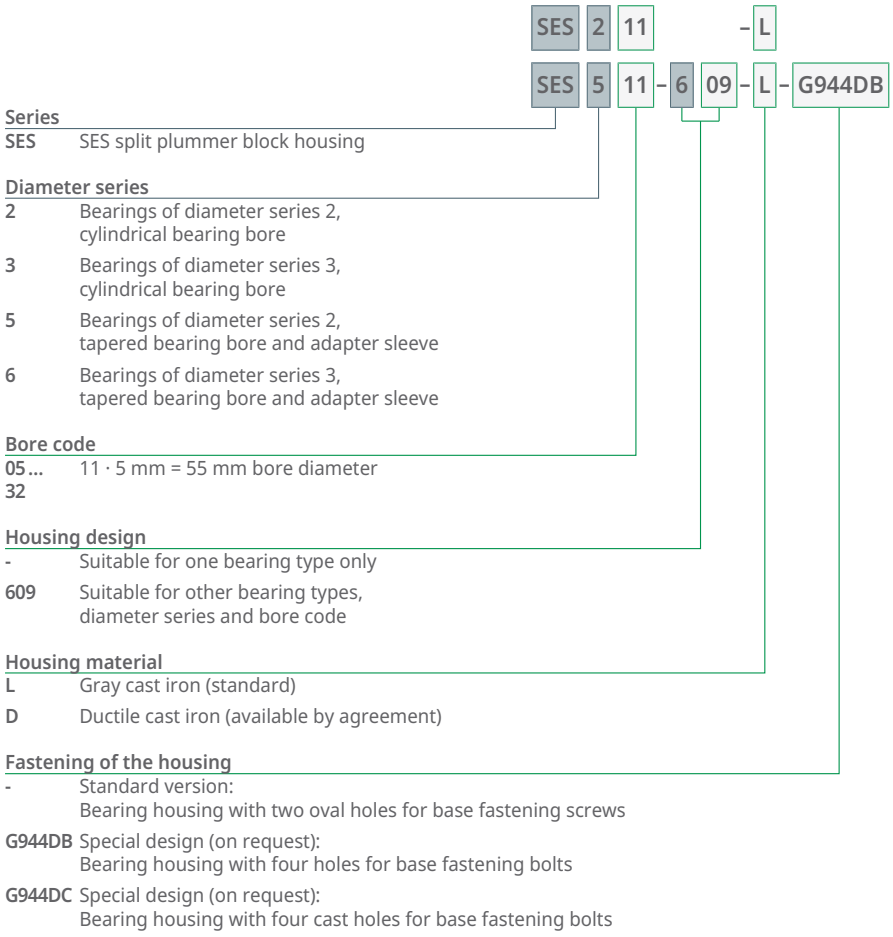
3.8 Structure of the ordering designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

Bearing housing

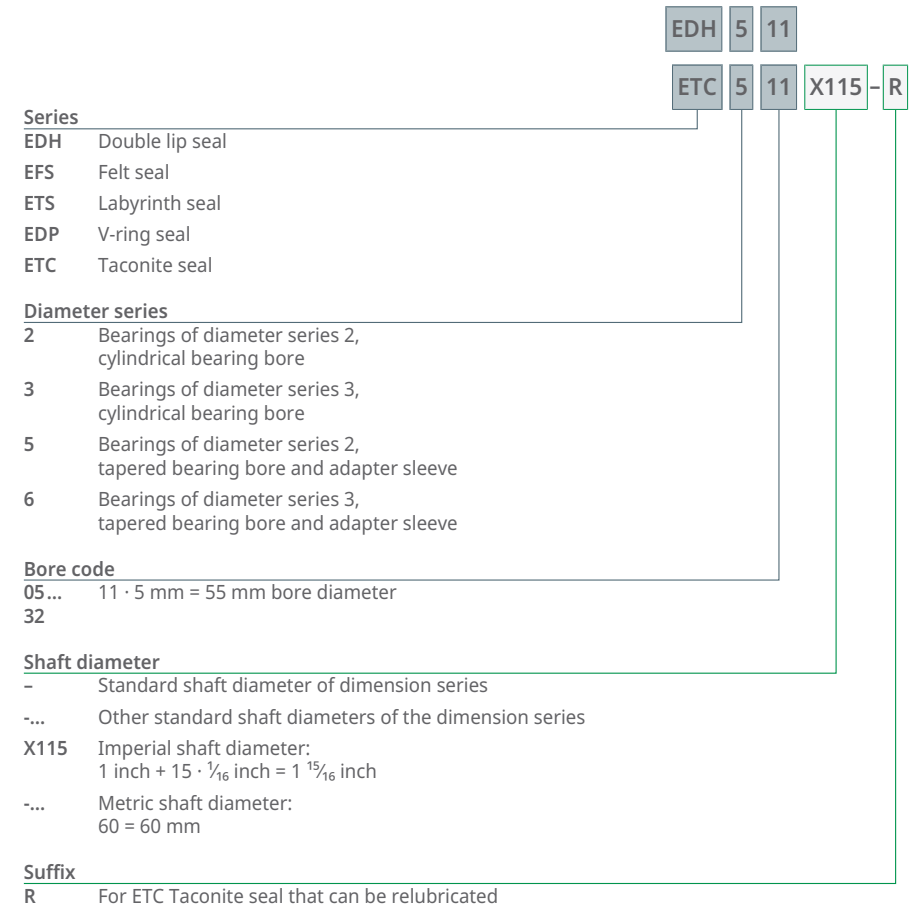
The designation structure of split plummer block housings SES is based on DIN 736 to DIN 739. Due to the modular concept, the same bearings can be installed in a single SES housing as in several of the SN housings that are also referred to in these standards.

37 Structure of the designations of SES plummer block housings



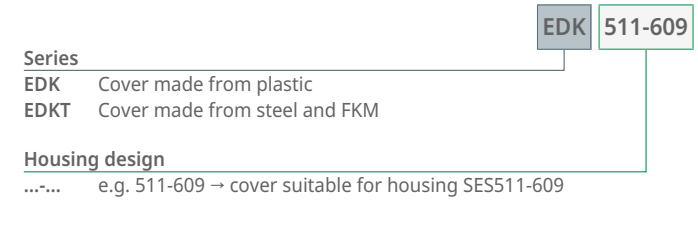
Seals

38 Structure of the designations of standard seals



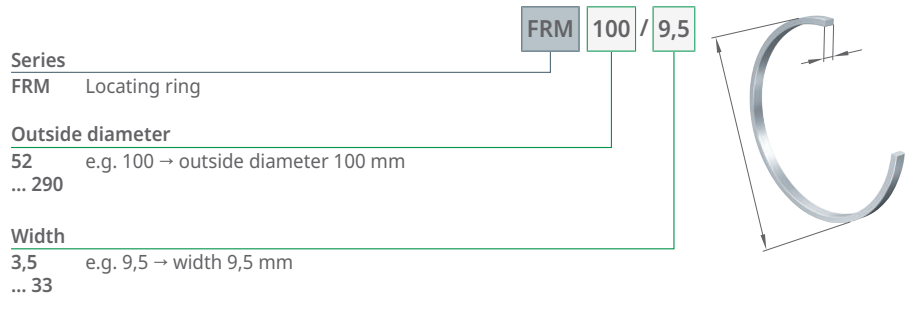
Cover

39 Structure of the designations of covers



Locating rings

40 Structure of the designations of locating rings



3.8.1 Ordering examples

When a split plummer block housing SES is ordered, the housing designation only describes the housing body. The other components such as seals, covers or locating rings must be ordered separately in the specific design required. The rolling bearing and, if necessary, the adapter sleeve must also be ordered separately.

Housings with a bearing fitted give non-locating bearing arrangements. Locating bearing arrangements can be achieved by the insertion of 1 locating ring FRM on each side of the bearing outer ring.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

3.8.1.1 Example 1

Example of a housing configuration, characteristics:

- Housing from standard series made from gray cast iron, closed on one side
- Shaft with a diameter of 50 mm, location by means of adapter sleeve
- 22211-E1-XL-K spherical roller bearing in the form of a locating bearing
- Double lip seal

13 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	SES511-609-L
1	Spherical roller bearing	22211-E1-XL-K
1	Adapter sleeve	H311
2	Locating ring	FRM100/9,5
1	Cover	EDK511-609
2 [1)]	Double lip seal (2 pieces per pack)	EDH511

1) Only multiples of 2 are possible as an order quantity, as there are 2 items per pack

3.8.1.2 Example 2

Example of a housing configuration, characteristics:

- Housing from standard series made from gray cast iron, open on both sides
- Continuous shaft, diameter 70 mm
- 222SM70-TVPA split spherical roller bearing in the form of a locating bearing
- Felt seals

14 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	SES516-613-L
1	Split spherical roller bearing	222SM70-TVPA
2	Locating ring	FRM140/12,5
2	Felt seal (2 pieces per pack)	EFS516

3.8.1.3 Example 3

Example of a housing configuration, characteristics:

- Housing made from ductile cast iron, closed on one side, 4 holes for base fastening screws
- Stepped shaft, diameters 90 mm and 100 mm
- 23218-E1-XL-TVPB spherical roller bearing in the form of a non-locating bearing
- Taconite seals

15 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	SES218-D-G944DB
1	Spherical roller bearing	23218-E1-XL-TVPB
1	Shaft nut	KM18
1	Tab washer	MB18
1	Cover	EDK218
1	Taconite seal	ETC218

3.9 Design of the adjacent construction

3.9.1 Additional holes for foot screws and pins

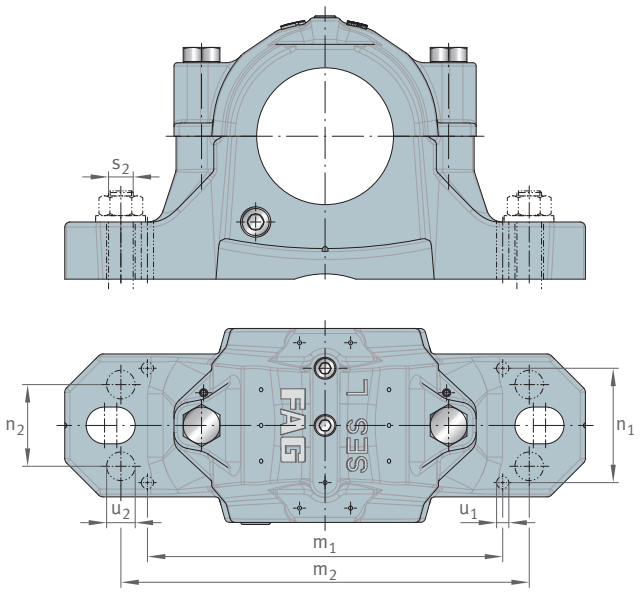
For many applications, it is sufficient to fix the housing to the adjacent construction with 2 foot screws. The housing base in the standard design has 2 extended slots, to enable good alignment during assembly.

For location on T profiles, 4 foot screws are required. Cast-in markings indicate the locations at which 4 additional holes can be produced for fixing purposes.

Housings with 4 holes for foot screws are available as a special design by agreement. These are indicated by the suffix G944DB in the ordering designation.

Further markings indicate the locations at which holes can be produced for pins to secure the position.

41 Additional holes for foot screws and pins

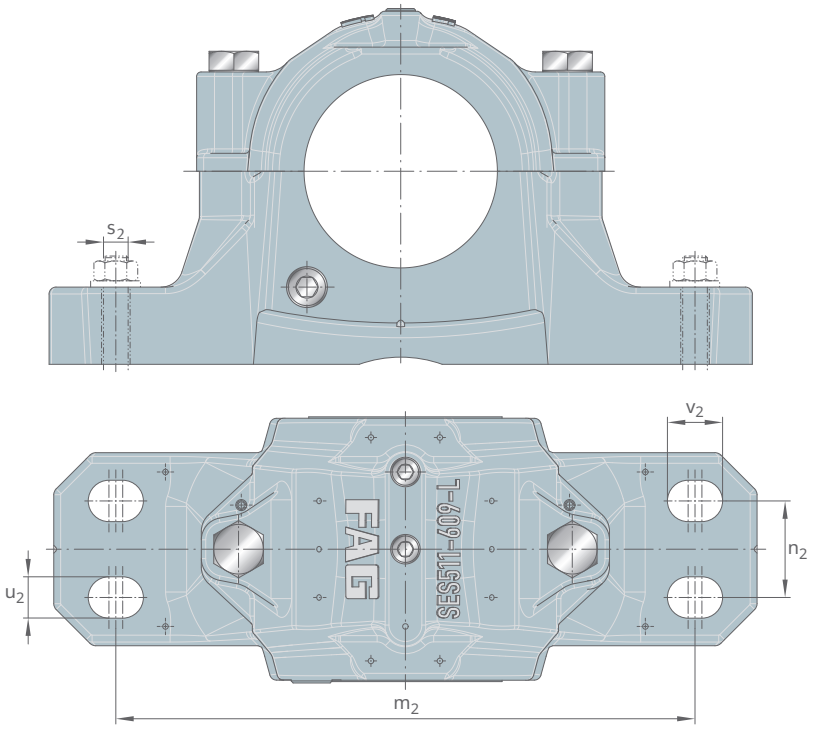


16 Dimensions of additional holes for base fastening screws and pins

Housing		Pins			Base fastening screws			
		m_1	n_1	u_1	m_2	n_2	u_2	s_2
		mm	mm	mm	mm	mm	mm	–
SES505	SES205	100	34	4	114	25	9	M8
SES506-605	SES206-305	119	39	4	130	29	9	M8
SES507-606	SES207	124	40	4	138	29	9	M8
SES508-607	SES208-307	137	43	4	160	34	11	M10
SES509	SES209	138	45	4	160	34	11	M10
SES510-608	SES210	140	46	6	160	34	11	M10
SES511-609	SES211	174	56	6	200	40	14	M12
SES512-610	SES212	174	56	6	200	40	14	M12
SES513-611	SES213	194	61	6	220	48	14	M12
–	SES214	190	52	4	220	48	14	M12
SES515-612	SES215	194	60	8	220	48	14	M12
SES516-613	SES216	218	67	8	252	52	18	M16
SES517	SES217	220	75	8	252	52	18	M16
SES518-615	SES218	250	80	8	280	58	18	M16
SES519-616	–	250	82	8	280	58	18	M16
SES520-617	–	270	90	8	300	66	18	M16
SES618-318	–	276	84	8	300	66	18	M16
SES522-619	–	296	92	8	320	74	18	M16
SES524-620	–	300	91	8	330	74	18	M16
SES526	–	328	97	10	370	80	22	M20
SES622-322	–	338	96	10	370	80	22	M20
SES528	–	360	128	12	400	92	26	M24
SES624-324	–	378	120	12	430	100	26	M24
SES530	–	374	116	12	430	100	26	M24
SES626-326	–	392	114	12	450	100	26	M24
SES532	–	394	120	12	450	100	26	M24
SES628-328	–	442	123	12	500	100	26	M24
SES630-330	–	482	130	12	540	100	26	M24
SES632-332	–	489	138	12	570	100	33	M30

Housings with 4 cast oval holes for base fastening screws are available as a special design on request. These are identified by the suffix G944DC in the order designation.

42 4 cast oval holes for base fastening screws



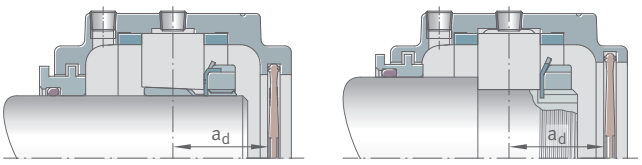
17 Dimensions of the cast holes for base fastening screws

Housing		Base fastening screws				
		m_2	n_2	u_2	v_2	s_2
		mm	mm	mm	mm	–
SES511-609	SES211	210	35	15	20	M12
SES512-610	SES212	210	35	15	24	M12
SES513-611	SES213	230	40	15	24	M12
–	SES214	230	40	15	24	M12
SES515-612	SES215	230	40	15	20	M12
SES516-613	SES216	260	50	18	24	M16
SES517	SES217	260	50	18	24	M16
SES518-615	SES218	290	50	18	24	M16
SES519-616	–	290	50	18	24	M16
SES520-617	–	320	60	18	24	M16
SES618-318	–	320	60	18	24	M16
SES522-619	–	350	70	18	24	M16
SES524-620	–	350	70	18	24	M16
SES526	–	380	70	22	28	M20
SES622-322	–	385	70	22	28	M20
SES528	–	420	80	26	32	M24
SES624-324	–	450	90	26	32	M24
SES530	–	450	90	26	32	M24
SES626-326	–	450	90	26	32	M24
SES532	–	450	90	26	32	M24
SES628-328	–	500	90	26	32	M24
SES630-330	–	550	90	26	32	M24
SES632-332	–	570	90	35	42	M30

3.9.2 Permissible shaft end designs

In the case of a housing closed on one side, the length of the shaft in the housing must be restricted such that no contact occurs between the shaft and the cover. In this case, the displacement facility of the bearing in a non-locating bearing arrangement must be taken into account. The basis for determining the permissible length of the shaft is the distance between the centre of the bearing seat and the cover ➤60|43 ➤60|18.

43 Determining the permissible length of the shaft



a_d	Spacing between centre of bearing seat and cover

18 Spacing between centre of the bearing seat and cover

Housing		Spacing a_d
		mm
SES505	SES205	26
SES506-605	SES206-305	31
SES507-606	SES207	33
SES508-607	SES208-307	34,5
SES509	SES209	33,5
SES510-608	SES210	36
SES511-609	SES211	38,5
SES512-610	SES212	43,5
SES513-611	SES213	46
–	SES214	42,3
SES515-612	SES215	48,5
SES516-613	SES216	51
SES517	SES217	53,5
SES518-615	SES218	61
SES519-616	–	62,5
SES520-617	–	70
SES618-318	–	67,3
SES522-619	–	77,5
SES524-620	–	81,5
SES526	–	84
SES622-322	–	80,3
SES528	–	91,5
SES624-324	–	84
SES530	–	99
SES626-326	–	91,5
SES532	–	106,5
SES628-328	–	96,5
SES630-330	–	101,5
SES632-332	–	106,5

3.9.3 Shaft design

Requirements governing the design of the shaft can be found in the publication for the bearing used, e.g. the respective bearing chapter in the main catalogue:

HR 1 | Rolling Bearings |
https://www.schaeffler.de/std/1D3D

The minimum requirements for the shaft must be observed for the running surfaces of the seals ➤52|▦12.

3.10 Fitting and dismantling

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

Further information

MON 92 | Split plummer block housings | SES |
https://www.schaeffler.de/std/1FD9

3.10.1 Eye bolts

There is 1 eye bolt to DIN 580:2018 in each housing upper section as from housing size SES524-620. This eye bolt is provided as a locating point for the installation and removal of the housing. The load capacity of the eye bolt allows the housing, including the bearing fitted in the housing, to be lifted.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

3.10.2 Foot screws

Foot screws are used for screw mounting of the housings to the locating surface. Foot screws are not included in the scope of delivery of the housings.

The appropriate screw size for each housing is given in the product tables.

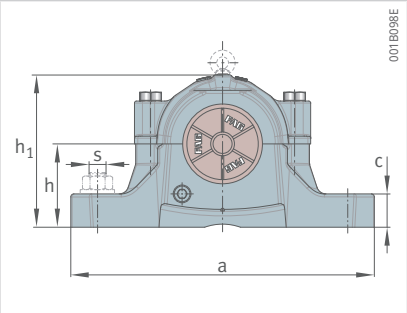
The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

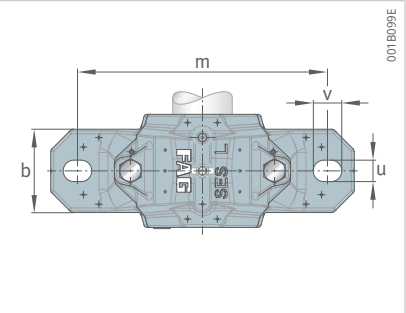
3.11 Product tables

3.11.1 Explanations of the product tables

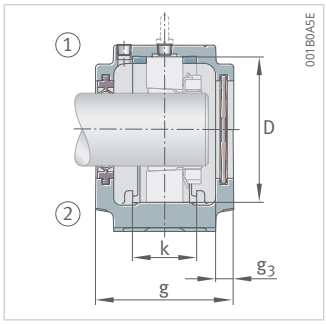
(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
b	mm	Base width
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
d_2	mm	Shaft shoulder diameter
g	mm	Housing body width
g_3	mm	Housing body wall thickness
g_L	mm	Total width, with 2 labyrinth seals
g_T	mm	Total width, with 2 Taconite seals
g_V	mm	Total width, with 2 V ring seals
h	mm	Distance to shaft axis
h_1	mm	Overall height
k	mm	Bearing seat width
m	mm	Fixing hole spacing
m	kg	Mass
s	inch or mm	Screw size
u	mm	Slot width
v	mm	Slot length

3.11.2 Split plummer block housings SES, dimensions



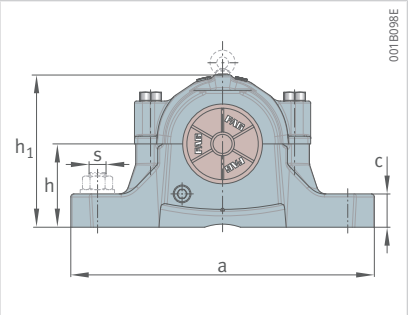


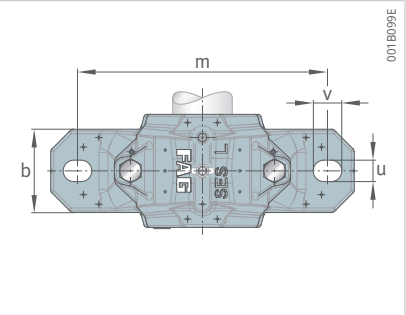
Housing	D	h	h_1	g	b	c	a
	mm	mm	mm	mm	mm	mm	mm
SES205-L	**52**	40	74	67	46	19	165
SES505-L	**52**	40	74	67	46	19	165
SES206-305-L	**62**	50	89	77	52	22	185
SES506-605-L	**62**	50	89	77	52	22	185
SES207-L	**72**	50	93	82	52	22	185
SES507-606-L	**72**	50	93	82	52	22	185
SES208-307-L	**80**	60	108	85	60	25	205
SES508-607-L	**80**	60	108	85	60	25	205
SES209-L	**85**	60	109	85	60	25	205
SES509-L	**85**	60	109	85	60	25	205
SES210-L	**90**	60	113	90	60	25	205
SES510-608-L	**90**	60	113	90	60	25	205
SES211-L	**100**	70	128	95	70	28	255
SES511-609-L	**100**	70	128	95	70	28	255
SES212-L	**110**	70	134	105	70	30	255
SES512-610-L	**110**	70	134	105	70	30	255
SES213-L	**120**	80	150	110	80	30	275
SES513-611-L	**120**	80	150	110	80	30	275
SES214-L	**125**	80	158	105	80	30	275
SES215-L	**130**	80	156	115	80	30	280
SES515-612-L	**130**	80	156	115	80	30	280
SES216-L	**140**	95	177	120	90	32	315
SES516-613-L	**140**	95	177	120	90	32	315
SES217-L	**150**	95	183	125	90	32	320
SES517-L	**150**	95	183	125	90	32	320
SES218-L	**160**	100	194	140	100	35	345
SES518-615-L	**160**	100	194	140	100	35	345
SES519-616-L	**170**	112	212	145	100	35	345
SES520-617-L	**180**	112	218	160	110	40	380
SES618-318-L	**190**	112	229	155	110	40	380
SES522-619-L	**200**	125	242	175	120	45	410
SES524-620-L	**215**	140	271	185	120	45	410
SES526-L	**230**	150	290	190	130	50	445
SES622-322-L	**240**	150	298	185	130	50	450
SES528-L	**250**	150	302	205	150	50	500
SES624-324-L	**260**	160	321	190	160	60	530
SES530-L	**270**	160	323	220	160	60	530
SES626-326-L	**280**	160	344	205	160	60	550



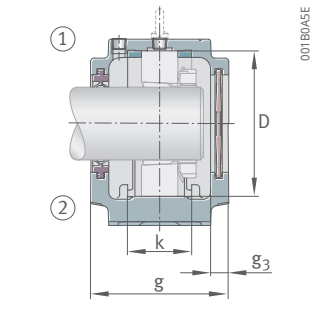
m	v	u	s		k	g3	m
mm	mm	mm	–	inch	mm	mm	≈ kg
130	20	15	M12	1/2	25	10	1,6
130	20	15	M12	1/2	25	10	1,45
150	20	15	M12	1/2	32	10	2,4
150	20	15	M12	1/2	32	10	2,12
150	20	15	M12	1/2	34	11	2,1
150	20	15	M12	1/2	34	11	2,27
170	20	15	M12	1/2	39	11	3,53
170	20	15	M12	1/2	39	11	2,95
170	20	15	M12	1/2	30	12	3,63
170	20	15	M12	1/2	30	12	3,1
170	20	15	M12	1/2	41	12	3,81
170	20	15	M12	1/2	41	12	3,27
210	24	18	M16	5/8	44	12	4,18
210	24	18	M16	5/8	44	12	4,41
210	24	18	M16	5/8	48	12	4,82
210	24	18	M16	5/8	48	12	5,2
230	24	18	M16	5/8	51	13	6,2
230	24	18	M16	5/8	51	13	6,5
230	24	18	M16	5/8	39	15	6,3
230	24	18	M16	5/8	56	13	6,6
230	24	18	M16	5/8	56	13	7
260	28	22	M20	3/4	58	13	8,8
260	28	22	M20	3/4	58	13	9,3
260	28	22	M20	3/4	61	13	9,4
260	28	22	M20	3/4	61	13	9,8
290	28	22	M20	3/4	65	13	11,8
290	28	22	M20	3/4	65	13	12,9
290	28	22	M20	3/4	68	14	13,8
320	32	26	M24	7/8	70	14	17,1
320	32	26	M24	7/8	74	15	19,1
350	32	26	M24	7/8	80	14	22,8
350	32	26	M24	7/8	86	15	26,2
380	35	28	M24	1	90	15	33,1
390	35	28	M24	1	90	18	36,1
420	42	35	M30	1 1/4	98	15	40,3
450	42	35	M30	1 1/4	96	18	47,5
450	42	35	M30	1 1/4	106	15	48,5
450	42	35	M30	1 1/4	103	18	55,6

3.11.2 Split plummer block housings SES, dimensions





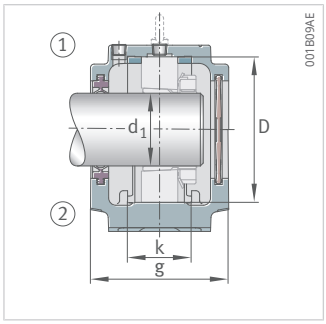
Housing	D	h	h_1	g	b	c	a
	mm	mm	mm	mm	mm	mm	mm
SES532-L	**290**	170	344	235	160	60	550
SES628-328-L	**300**	170	366	215	170	65	620
SES630-330-L	**320**	190	386	225	180	65	650
SES632-332-L	**340**	200	406	235	190	70	680



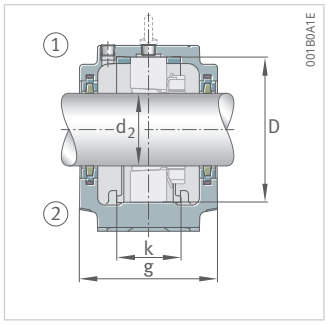
m	v	u	s		k	g_3	m
mm	mm	mm	–	inch	mm	mm	≈ kg
470	42	35	M30	1 1/4	114	15	56,7
520	42	35	M30	1 1/4	112	18	68,7
560	42	35	M30	1 1/4	118	18	77,9
580	50	42	M36	1 7/16	118	18	92,5

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

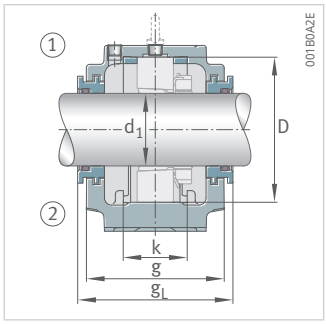


Double lip seal EDH, cover

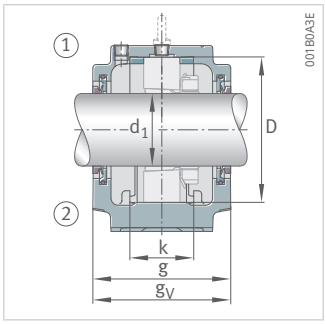


Felt seal EFS

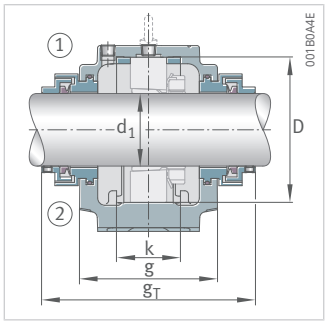
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES505-L	**19,05**	$^3/_4$	1205-K	H205×012	FRM52/5	EDK505
SES505-L	**19,05**	$^3/_4$	20205-K	H205×012	FRM52/5	EDK505
SES505-L	**19,05**	$^3/_4$	2205-K	H305×012	FRM52/3,5	EDK505
SES505-L	**19,05**	$^3/_4$	22205..-K	H305×012	FRM52/3,5	EDK505
SES506-605-L	**19,05**	$^3/_4$	1305-K	H305×012	FRM62/7,5	EDK506-605
SES506-605-L	**19,05**	$^3/_4$	20305..-K	H305×012	FRM62/7,5	EDK506-605
SES506-605-L	**19,05**	$^3/_4$	2305-K	H2305×012	FRM62/4	EDK506-605
SES505-L	**20**	–	1205-K	H205	FRM52/5	EDK505
SES505-L	**20**	–	2205-K	H305	FRM52/3,5	EDK505
SES505-L	**20**	–	22205..-K	H305	FRM52/3,5	EDK505
SES505-L	**20**	–	WS22205-E1-XL-K-2RSR	H2205-T-WS	FRM52/1	EDK505
SES506-605-L	**20**	–	1305-K	H305	FRM62/7,5	EDK506-605
SES506-605-L	**20**	–	20305..-K	H305	FRM62/7,5	EDK506-605
SES506-605-L	**20**	–	2305-K	H2305	FRM62/4	EDK506-605
SES506-605-L	**22,225**	$^7/_8$	1206-K	H206×014	FRM62/8	EDK506-605
SES506-605-L	**22,225**	$^7/_8$	20206..K	H206×014	FRM62/8	EDK506-605
SES506-605-L	**22,225**	$^7/_8$	2206-K	H306×014	FRM62/6	EDK506-605
SES506-605-L	**22,225**	$^7/_8$	22206..-K	H306×014	FRM62/6	EDK506-605
SES506-605-L	**23,813**	$^{15}/_{16}$	1206-K	H206×015	FRM62/8	EDK506-605
SES506-605-L	**23,813**	$^{15}/_{16}$	20206-K	H206×015	FRM62/8	EDK506-605
SES506-605-L	**23,813**	$^{15}/_{16}$	2206-K	H306×015	FRM62/6	EDK506-605
SES506-605-L	**23,813**	$^{15}/_{16}$	22206..-K	H306×015	FRM62/6	EDK506-605
SES507-606-L	**23,813**	$^{15}/_{16}$	1306-K	H306×015	FRM72/7,5	EDK507-606
SES507-606-L	**23,813**	$^{15}/_{16}$	20306..-K	H306×015	FRM72/7,5	EDK507-606
SES507-606-L	**23,813**	$^{15}/_{16}$	21306..-K	H306×015	FRM72/7,5	EDK507-606
SES507-606-L	**23,813**	$^{15}/_{16}$	2306-K	H2306×015	FRM72/3,5	EDK507-606
SES506-605-L	**25**	–	1206-K	H206	FRM62/8	EDK506-605
SES506-605-L	**25**	–	20206..-K	H206	FRM62/8	EDK506-605
SES506-605-L	**25**	–	2206-K	H306	FRM62/6	EDK506-605
SES506-605-L	**25**	–	22206..-K	H306	FRM62/6	EDK506-605
SES506-605-L	**25**	–	WS22206-E1-XL-K-2RSR	H2206-T-WS	FRM62/3,5	EDK506-605
SES507-606-L	**25**	–	1306-K	H306	FRM72/7,5	EDK507-606
SES507-606-L	**25**	–	20306..-K	H306	FRM72/7,5	EDK507-606
SES507-606-L	**25**	–	21306..-K	H306	FRM72/7,5	EDK507-606
SES507-606-L	**25**	–	2306-K	H2306	FRM72/3,5	EDK507-606
SES506-605-L	**25,4**	**1**	1206-K	H206×100	FRM62/8	EDK506-605
SES506-605-L	**25,4**	**1**	20206..-K	H206×100	FRM62/8	EDK506-605
SES506-605-L	**25,4**	**1**	2206-K	H306×100	FRM62/6	EDK506-605
SES506-605-L	**25,4**	**1**	22206..-K	H306×100	FRM62/6	EDK506-605
SES507-606-L	**25,4**	**1**	1306-K	H306×100	FRM72/7,5	EDK507-606



Labyrinth seal ETS



V ring seal EDV

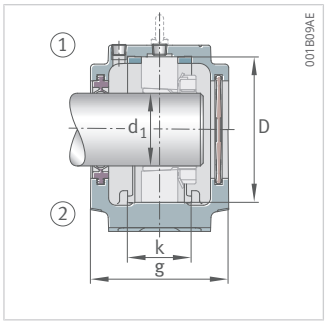


Taconite seal ETC

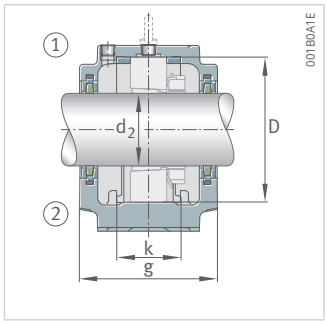
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
–	EFS505×012	ETS505×012	80	EDV505×012	67	ETC505×012	125
–	EFS505×012	ETS505×012	80	EDV505×012	67	ETC505×012	125
–	EFS505×012	ETS505×012	80	EDV505×012	67	ETC505×012	125
–	EFS505×012	ETS505×012	80	EDV505×012	67	ETC505×012	125
EDH605×012	EFS605×012	ETS605×012	89	EDV605×012	77	ETC605×012	135
EDH605×012	EFS605×012	ETS605×012	89	EDV605×012	77	ETC605×012	135
EDH605×012	EFS605×012	ETS605×012	89	EDV605×012	77	ETC605×012	135
EDH505	EFS505	ETS505	80	EDV505	67	ETC505	125
EDH505	EFS505	ETS505	80	EDV505	67	ETC505	125
EDH505	EFS505	ETS505	80	EDV505	67	ETC505	125
EDH505	EFS505	ETS505	80	EDV505	67	ETC505	125
EDH605	EFS605	ETS605	89	EDV605	77	ETC605	135
EDH605	EFS605	ETS605	89	EDV605	77	ETC605	135
EDH605	EFS605	ETS605	89	EDV605	77	ETC605	135
–	EFS506×014	ETS506×014	89	EDV506×014	77	ETC506×014	135
–	EFS506×014	ETS506×014	89	EDV506×014	77	ETC506×014	135
–	EFS506×014	ETS506×014	89	EDV506×014	77	ETC506×014	135
–	EFS506×014	ETS506×014	89	EDV506×014	77	ETC506×014	135
–	EFS506×015	ETS506×015	89	EDV506×015	77	ETC506×015	135
–	EFS506×015	ETS506×015	89	EDV506×015	77	ETC506×015	135
–	EFS506×015	ETS506×015	89	EDV506×015	77	ETC506×015	135
–	EFS506×015	ETS506×015	89	EDV506×015	77	ETC506×015	135
–	EFS606×015	ETS606×015	94	EDV606×015	82	ETC606×015	140
–	EFS606×015	ETS606×015	94	EDV606×015	82	ETC606×015	140
–	EFS606×015	ETS606×015	94	EDV606×015	82	ETC606×015	140
–	EFS606×015	ETS606×015	94	EDV606×015	82	ETC606×015	140
EDH506	EFS506	ETS506	89	EDV506	77	ETC506	135
EDH506	EFS506	ETS506	89	EDV506	77	ETC506	135
EDH506	EFS506	ETS506	89	EDV506	77	ETC506	135
EDH506	EFS506	ETS506	89	EDV506	77	ETC506	135
EDH506	EFS506	ETS506	89	EDV506	77	ETC506	135
EDH606	EFS606	ETS606	94	EDV606	82	ETC606	140
EDH606	EFS606	ETS606	94	EDV606	82	ETC606	140
EDH606	EFS606	ETS606	94	EDV606	82	ETC606	140
EDH606	EFS606	ETS606	94	EDV606	82	ETC606	140
EDH506	EFS506	ETS506×100	89	EDV506	77	ETC506×100	135
EDH506	EFS506	ETS506×100	89	EDV506	77	ETC506×100	135
EDH506	EFS506	ETS506×100	89	EDV506	77	ETC506×100	135
EDH506	EFS506	ETS506×100	89	EDV506	77	ETC506×100	135
EDH606	EFS606	ETS606×100	94	EDV606	82	ETC606	140

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

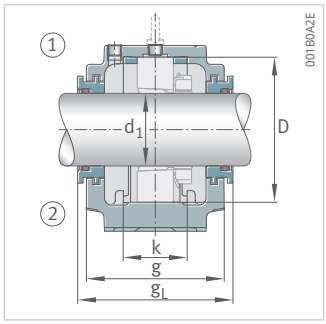


Double lip seal EDH, cover

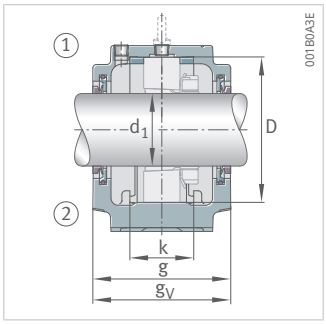


Felt seal EFS

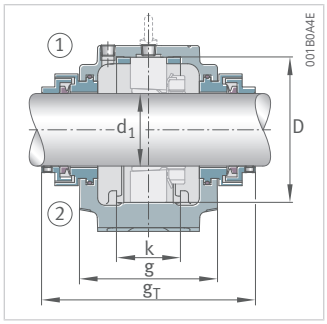
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES507-606-L	**25,4**	**1**	20306..-K	H2306×100	FRM72/7,5	EDK507-606
SES507-606-L	**25,4**	**1**	21306..-K	H2306×100	FRM72/7,5	EDK507-606
SES507-606-L	**25,4**	**1**	2306-K	H2306×100	FRM72/3,5	EDK507-606
SES508-607-L	**28,575**	**1 1/8**	1307-K	H307×102	FRM80/9	EDK508-607
SES508-607-L	**28,575**	**1 1/8**	20307-K	H307×102	FRM80/9	EDK508-607
SES508-607-L	**28,575**	**1 1/8**	21307..-K	H307×102	FRM80/9	EDK508-607
SES508-607-L	**28,575**	**1 1/8**	2307-K	H2307×102	FRM80/4	EDK508-607
SES507-606-L	**30**	**–**	1207-K	H207	FRM72/8,5	EDK507-606
SES507-606-L	**30**	**–**	20207..-K	H207	FRM72/8,5	EDK507-606
SES507-606-L	**30**	**–**	2207-K	H307	FRM72/5,5	EDK507-606
SES507-606-L	**30**	**–**	22207..-K	H307	FRM72/5,5	EDK507-606
SES507-606-L	**30**	**–**	WS22207-E1-XL-K-2RSR	H2207-T-WS	FRM72/3	EDK507-606
SES508-607-L	**30**	**–**	1307-K	H307	FRM80/9	EDK508-607
SES508-607-L	**30**	**–**	20307-K	H307	FRM80/9	EDK508-607
SES508-607-L	**30**	**–**	21307..-K	H307	FRM80/9	EDK508-607
SES508-607-L	**30**	**–**	2307-K	H2307	FRM80/4	EDK508-607
SES507-606-L	**30,163**	**1 3/16**	1207-K	H207×103	FRM72/8,5	EDK507-606
SES507-606-L	**30,163**	**1 3/16**	20207-K	H207×103	FRM72/8,5	EDK507-606
SES507-606-L	**30,163**	**1 3/16**	2207-K	H307×103	FRM72/5,5	EDK507-606
SES507-606-L	**30,163**	**1 3/16**	22207..-K	H307×103	FRM72/5,5	EDK507-606
SES508-607-L	**30,163**	**1 3/16**	1307-K	H307×103	FRM80/9	EDK508-607
SES508-607-L	**30,163**	**1 3/16**	20307..-K	H307×103	FRM80/9	EDK508-607
SES508-607-L	**30,163**	**1 3/16**	21307..-K	H307×103	FRM80/9	EDK508-607
SES508-607-L	**30,163**	**1 3/16**	2307-K	H2307×103	FRM80/4	EDK508-607
SES508-607-L	**31,75**	**1 1/4**	1208-K	H208×104	FRM80/10,5	EDK508-607
SES508-607-L	**31,75**	**1 1/4**	20208-K	H208×104	FRM80/10,5	EDK508-607
SES508-607-L	**31,75**	**1 1/4**	2208-K	H308×104	FRM80/8	EDK508-607
SES508-607-L	**31,75**	**1 1/4**	22208..-K	H308×104	FRM80/8	EDK508-607
SES510-608-L	**31,75**	**1 1/4**	1308-K	H308×104	FRM90/9	EDK510-608
SES510-608-L	**31,75**	**1 1/4**	20308-K	H308×104	FRM90/9	EDK510-608
SES510-608-L	**31,75**	**1 1/4**	21308..-K	H308×104	FRM90/9	EDK510-608
SES510-608-L	**31,75**	**1 1/4**	22308..-K	H2308×104	FRM90/4	EDK510-608
SES510-608-L	**31,75**	**1 1/4**	2308-K	H2308×104	FRM90/4	EDK510-608
SES510-608-L	**33,338**	**1 5/16**	1308-K	H308×105	FRM90/9	EDK510-608
SES510-608-L	**33,338**	**1 5/16**	20308-K	H308×105	FRM90/9	EDK510-608
SES510-608-L	**33,338**	**1 5/16**	21308..-K	H308×105	FRM90/9	EDK510-608
SES510-608-L	**33,338**	**1 5/16**	22308..-K	H2308×105	FRM90/4	EDK510-608
SES510-608-L	**33,338**	**1 5/16**	2308-K	H2308×105	FRM90/4	EDK510-608
SES508-607-L	**34,925**	**1 3/8**	1208-K	H208×106	FRM80/10,5	EDK508-607
SES508-607-L	**34,925**	**1 3/8**	20208-K	H208×106	FRM80/10,5	EDK508-607



Labyrinth seal ETS



V ring seal EDV

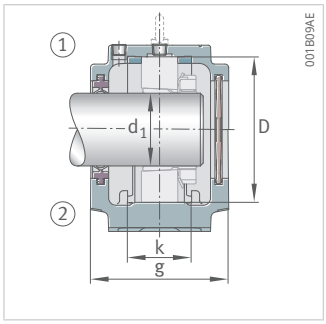


Taconite seal ETC

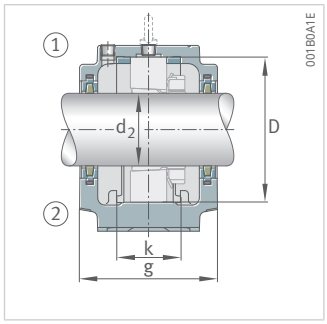
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH606	EFS606	ETS606×100	94	EDV606	82	ETC606	140
EDH606	EFS606	ETS606×100	94	EDV606	82	ETC606	140
EDH606	EFS606	ETS606×100	94	EDV606	82	ETC606	140
EDH607×102	EFS607×102	ETS607×102	97	EDV607×102	85	–	–
EDH607×102	EFS607×102	ETS607×102	97	EDV607×102	85	–	–
EDH607×102	EFS607×102	ETS607×102	97	EDV607×102	85	–	–
EDH607×102	EFS607×102	ETS607×102	97	EDV607×102	85	–	–
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH507-305	EFS507-305	ETS507-305	94	EDV507-305	81	ETC507-305	144
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
EDH607	EFS607	ETS607	97	EDV607	85	ETC607	145
–	EFS508×104	ETS508×104	97	EDV508×104	85	ETC508×104	150
–	EFS508×104	ETS508×104	97	EDV508×104	85	ETC508×104	150
–	EFS508×104	ETS508×104	97	EDV508×104	85	ETC508×104	150
–	EFS508×104	ETS508×104	97	EDV508×104	85	ETC508×104	150
EDH608×104	EFS608×104	ETS608×104	102	EDV608×104	90	ETC608×104	150
EDH608×104	EFS608×104	ETS608×104	102	EDV608×104	90	ETC608×104	150
EDH608×104	EFS608×104	ETS608×104	102	EDV608×104	90	ETC608×104	150
EDH608×104	EFS608×104	ETS608×104	102	EDV608×104	90	ETC608×104	150
EDH608×104	EFS608×104	ETS608×104	102	EDV608×104	90	ETC608×104	150
EDH608×104	EFS608×105	ETS608×105	102	EDV608×105	90	ETC608×105	150
EDH608×104	EFS608×105	ETS608×105	102	EDV608×105	90	ETC608×105	150
EDH608×104	EFS608×105	ETS608×105	102	EDV608×105	90	ETC608×105	150
EDH608×104	EFS608×105	ETS608×105	102	EDV608×105	90	ETC608×105	150
EDH608×104	EFS608×105	ETS608×105	102	EDV608×105	90	ETC608×105	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

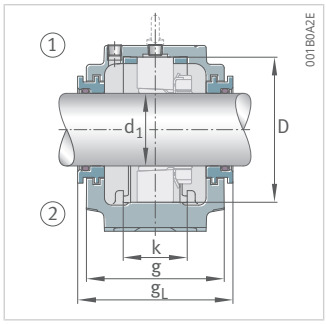


Double lip seal EDH, cover

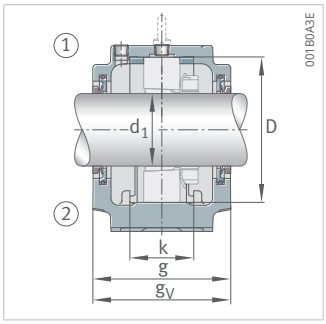


Felt seal EFS

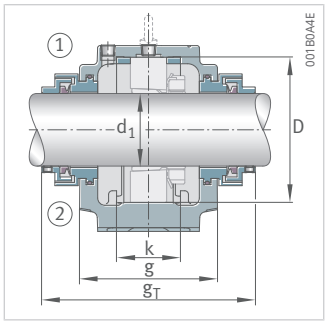
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES508-607-L	**34,925**	**1 3/8**	2208-K	H308×106	FRM80/8	EDK508-607
SES508-607-L	**34,925**	**1 3/8**	22208..-K	H308×106	FRM80/8	EDK508-607
SES510-608-L	**34,925**	**1 3/8**	1308-K	H308×106	FRM90/9	EDK510-608
SES510-608-L	**34,925**	**1 3/8**	20308-K	H308×106	FRM90/9	EDK510-608
SES510-608-L	**34,925**	**1 3/8**	21308..-K	H308×106	FRM90/9	EDK510-608
SES510-608-L	**34,925**	**1 3/8**	22308..-K	H2308×106	FRM90/4	EDK510-608
SES510-608-L	**34,925**	**1 3/8**	2308-K	H2308×106	FRM90/4	EDK510-608
SES508-607-L	**35**	**–**	1208-K	H208	FRM80/10,5	EDK508-607
SES508-607-L	**35**	**–**	20208-K	H208	FRM80/10,5	EDK508-607
SES508-607-L	**35**	**–**	2208-K	H308	FRM80/8	EDK508-607
SES508-607-L	**35**	**–**	22208..-K	H308	FRM80/8	EDK508-607
SES508-607-L	**35**	**–**	WS22208-E1-XL-K-2RSR	H2208-T-WS	FRM80/5,5	EDK508-607
SES510-608-L	**35**	**–**	1308-K	H308	FRM90/9	EDK510-608
SES510-608-L	**35**	**–**	20308-K	H308	FRM90/9	EDK510-608
SES510-608-L	**35**	**–**	21308..-K	H308	FRM90/9	EDK510-608
SES510-608-L	**35**	**–**	22308..-K	H2308	FRM90/4	EDK510-608
SES510-608-L	**35**	**–**	2308-K	H2308	FRM90/4	EDK510-608
SES510-608-L	**35**	**–**	WS22308-E1-XL-K-2RSR	H3308-T-WS	FRM90/1,5	EDK510-608
SES509-L	**36,513**	**1 7/16**	1209-K	H209×107	FRM85/5,5	EDK509
SES509-L	**36,513**	**1 7/16**	20209-K	H209×107	FRM85/5,5	EDK509
SES509-L	**36,513**	**1 7/16**	2209-K	H309×107	FRM85/3,5	EDK509
SES509-L	**36,513**	**1 7/16**	22209..-K	H309×107	FRM85/3,5	EDK509
SES511-609-L	**36,513**	**1 7/16**	1309-K	H309×107	FRM100/9,5	EDK511-609
SES511-609-L	**36,513**	**1 7/16**	20309-K	H309×107	FRM100/9,5	EDK511-609
SES511-609-L	**36,513**	**1 7/16**	21309..-K	H309×107	FRM100/9,5	EDK511-609
SES511-609-L	**36,513**	**1 7/16**	22309..-K	H2309×107	FRM100/4	EDK511-609
SES511-609-L	**36,513**	**1 7/16**	2309-K	H2309×107	FRM100/4	EDK511-609
SES509-L	**38,1**	**1 1/2**	1209-K	H209×108	FRM85/5,5	EDK509
SES509-L	**38,1**	**1 1/2**	20209-K	H209×108	FRM85/5,5	EDK509
SES509-L	**38,1**	**1 1/2**	2209-K	H309×108	FRM85/3,5	EDK509
SES509-L	**38,1**	**1 1/2**	22209..-K	H309×108	FRM85/3,5	EDK509
SES511-609-L	**38,1**	**1 1/2**	1309-K	H309×108	FRM100/9,5	EDK511-609
SES511-609-L	**38,1**	**1 1/2**	20309-K	H309×108	FRM100/9,5	EDK511-609
SES511-609-L	**38,1**	**1 1/2**	21309..-K	H309×108	FRM100/9,5	EDK511-609
SES511-609-L	**38,1**	**1 1/2**	22309..-K	H2309×108	FRM100/4	EDK511-609
SES511-609-L	**38,1**	**1 1/2**	2309-K	H2309×108	FRM100/4	EDK511-609
SES511-609-L	**39,688**	**1 9/16**	1309-K	H309×109	FRM100/9,5	EDK511-609
SES511-609-L	**39,688**	**1 9/16**	20309-K	H309×109	FRM100/9,5	EDK511-609
SES511-609-L	**39,688**	**1 9/16**	21309..-K	H309×109	FRM100/9,5	EDK511-609
SES511-609-L	**39,688**	**1 9/16**	22309..-K	H2309×109	FRM100/4	EDK511-609



Labyrinth seal ETS



V ring seal EDV

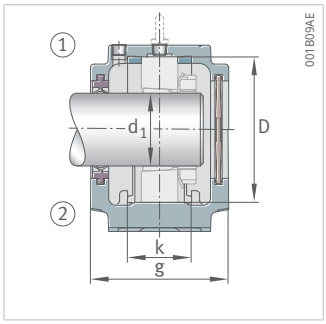


Taconite seal ETC

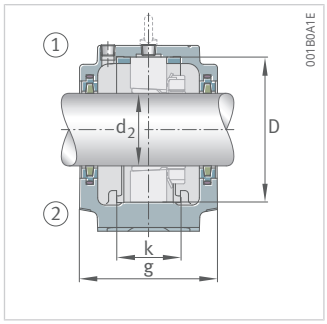
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH508	EFS508	ETS508	97	EDV508	85	ETC508	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH608	EFS608	ETS608	102	EDV608	90	ETC608	150
EDH509×107	EFS509×107	ETS509×107	97	EDV509×107	85	ETC509×107	150
EDH509×107	EFS509×107	ETS509×107	97	EDV509×107	85	ETC509×107	150
EDH509×107	EFS509×107	ETS509×107	97	EDV509×107	85	ETC509×107	150
EDH509×107	EFS509×107	ETS509×107	97	EDV509×107	85	ETC509×107	150
EDH609×107	EFS609×107	ETS609×107	107	EDV609×107	95	ETC609×107	155
EDH609×107	EFS609×107	ETS609×107	107	EDV609×107	95	ETC609×107	155
EDH609×107	EFS609×107	ETS609×107	107	EDV609×107	95	ETC609×107	155
EDH609×107	EFS609×107	ETS609×107	107	EDV609×107	95	ETC609×107	155
EDH609×107	EFS609×107	ETS609×107	107	EDV609×107	95	ETC609×107	155
–	EFS509×108	ETS509×108	97	EDV509×108	85	ETC509×108	150
–	EFS509×108	ETS509×108	97	EDV509×108	85	ETC509×108	150
–	EFS509×108	ETS509×108	97	EDV509×108	85	ETC509×108	150
–	EFS509×108	ETS509×108	97	EDV509×108	85	ETC509×108	150
EDH609×107	EFS609×108	ETS609×108	107	EDV609×108	95	ETC609×108	155
EDH609×107	EFS609×108	ETS609×108	107	EDV609×108	95	ETC609×108	155
EDH609×107	EFS609×108	ETS609×108	107	EDV609×108	95	ETC609×108	155
EDH609×107	EFS609×108	ETS609×108	107	EDV609×108	95	ETC609×108	155
EDH609×107	EFS609×108	ETS609×108	107	EDV609×108	95	ETC609×108	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

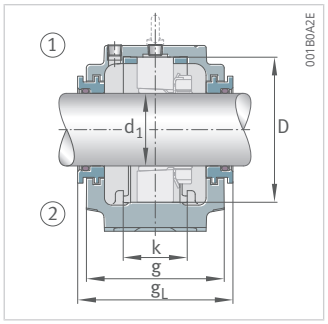


Double lip seal EDH, cover

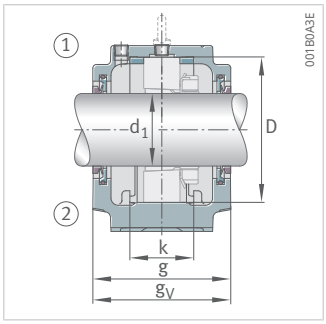


Felt seal EFS

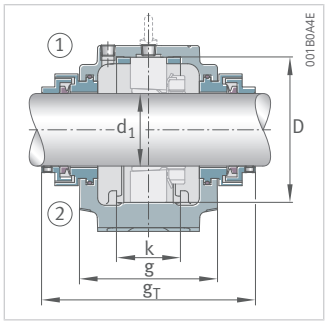
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES511-609-L	**39,688**	**1 9/16**	2309-K	H2309×109	FRM100/4	EDK511-609
SES509-L	**40**	**–**	1209-K	H209	FRM85/5,5	EDK509
SES509-L	**40**	**–**	20209-K	H209	FRM85/5,5	EDK509
SES509-L	**40**	**–**	2209-K	H309	FRM85/3,5	EDK509
SES509-L	**40**	**–**	22209..-K	H309	FRM85/3,5	EDK509
SES509-L	**40**	**–**	WS22209-E1-XL-K-2RSR	H2209-T-WS	FRM85/1	EDK509
SES511-609-L	**40**	**–**	1309-K	H309	FRM100/9,5	EDK511-609
SES511-609-L	**40**	**–**	20309-K	H309	FRM100/9,5	EDK511-609
SES511-609-L	**40**	**–**	21309..-K	H309	FRM100/9,5	EDK511-609
SES511-609-L	**40**	**–**	22309..-K	H2309	FRM100/4	EDK511-609
SES511-609-L	**40**	**–**	2309-K	H2309	FRM100/4	EDK511-609
SES511-609-L	**40**	**–**	WS22309-E1-XL-K-2RSR	H3309-T-WS	FRM100/1	EDK511-609
SES512-610-L	**41,275**	**1 5/8**	1310-K	H310×110	FRM110/10,5	EDK512-610
SES512-610-L	**41,275**	**1 5/8**	20310-K	H310×110	FRM110/10,5	EDK512-610
SES512-610-L	**41,275**	**1 5/8**	21310..-K	H310×110	FRM110/10,5	EDK512-610
SES512-610-L	**41,275**	**1 5/8**	22310..-K	H2310×110	FRM110/4	EDK512-610
SES512-610-L	**41,275**	**1 5/8**	2310-K	H2310×110	FRM110/4	EDK512-610
SES510-608-L	**42,863**	**1 11/16**	1210-K	H210×111	FRM90/10,5	EDK510-608
SES510-608-L	**42,863**	**1 11/16**	20210-K	H210×111	FRM90/10,5	EDK510-608
SES510-608-L	**42,863**	**1 11/16**	2210-K	H310×111	FRM90/9	EDK510-608
SES510-608-L	**42,863**	**1 11/16**	22210..-K	H310×111	FRM90/9	EDK510-608
SES512-610-L	**42,863**	**1 11/16**	1310-K	H310×111	FRM110/10,5	EDK512-610
SES512-610-L	**42,863**	**1 11/16**	20310-K	H310×111	FRM110/10,5	EDK512-610
SES512-610-L	**42,863**	**1 11/16**	21310..-K	H310×111	FRM110/10,5	EDK512-610
SES512-610-L	**42,863**	**1 11/16**	22310..-K	H2310×111	FRM110/4	EDK512-610
SES512-610-L	**42,863**	**1 11/16**	2310-K	H2310×111	FRM110/4	EDK512-610
SES510-608-L	**44,45**	**1 3/4**	1210-K	H210×112	FRM90/10,5	EDK510-608
SES510-608-L	**44,45**	**1 3/4**	20210-K	H210×112	FRM90/10,5	EDK510-608
SES510-608-L	**44,45**	**1 3/4**	2210-K	H310×112	FRM90/9	EDK510-608
SES510-608-L	**44,45**	**1 3/4**	22210..-K	H310×112	FRM90/9	EDK510-608
SES512-610-L	**44,45**	**1 3/4**	1310-K	H310×112	FRM110/10,5	EDK512-610
SES512-610-L	**44,45**	**1 3/4**	20310-K	H310×112	FRM110/10,5	EDK512-610
SES512-610-L	**44,45**	**1 3/4**	21310..-K	H310×112	FRM110/10,5	EDK512-610
SES512-610-L	**44,45**	**1 3/4**	22310..-K	H2310×112	FRM110/4	EDK512-610
SES512-610-L	**44,45**	**1 3/4**	2310-K	H2310×112	FRM110/4	EDK512-610
SES510-608-L	**45**	**–**	1210-K	H210	FRM90/10,5	EDK510-608
SES510-608-L	**45**	**–**	20210-K	H210	FRM90/10,5	EDK510-608
SES510-608-L	**45**	**–**	2210-K	H310	FRM90/9	EDK510-608
SES510-608-L	**45**	**–**	22210..-K	H310	FRM90/9	EDK510-608
SES510-608-L	**45**	**–**	WS22210-E1-XL-K-2RSR	H2210-T-WS	FRM90/6,5	EDK510-608



Labyrinth seal ETS



V ring seal EDV

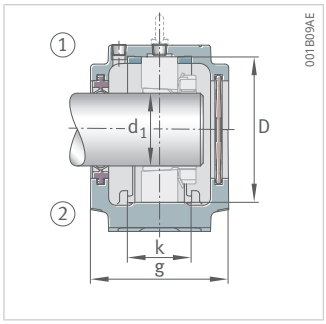


Taconite seal ETC

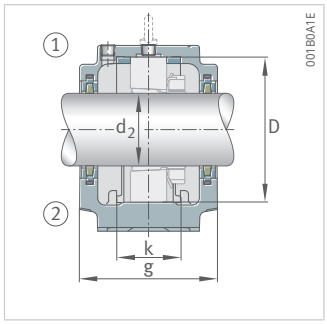
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH509	EFS509	ETS509	97	EDV509	85	ETC509	150
EDH509	EFS509	ETS509	97	EDV509	85	ETC509	150
EDH509	EFS509	ETS509	97	EDV509	85	ETC509	150
EDH509	EFS509	ETS509	97	EDV509	85	ETC509	150
EDH509	EFS509	ETS509	97	EDV509	85	ETC509	150
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH609	EFS609	ETS609	107	EDV609	95	ETC609	155
EDH610×110	EFS610×110	ETS610×110	117	EDV610×110	105	ETC610×110	165
EDH610×110	EFS610×110	ETS610×110	117	EDV610×110	105	ETC610×110	165
EDH610×110	EFS610×110	ETS610×110	117	EDV610×110	105	ETC610×110	165
EDH610×110	EFS610×110	ETS610×110	117	EDV610×110	105	ETC610×110	165
EDH610×110	EFS610×110	ETS610×110	117	EDV610×110	105	ETC610×110	165
–	EFS510×111	ETS510×111	102	EDV510×111	90	ETC510×111	155
–	EFS510×111	ETS510×111	102	EDV510×111	90	ETC510×111	155
–	EFS510×111	ETS510×111	102	EDV510×111	90	ETC510×111	155
–	EFS510×111	ETS510×111	102	EDV510×111	90	ETC510×111	155
EDH610×110	EFS610×111	ETS610×111	117	EDV610×111	105	ETC610×111	165
EDH610×110	EFS610×111	ETS610×111	117	EDV610×111	105	ETC610×111	165
EDH610×110	EFS610×111	ETS610×111	117	EDV610×111	105	ETC610×111	165
EDH610×110	EFS610×111	ETS610×111	117	EDV610×111	105	ETC610×111	165
EDH610×110	EFS610×111	ETS610×111	117	EDV610×111	105	ETC610×111	165
EDH510-307	EFS510-307	ETS510×112	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510×112	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510×112	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510×112	102	EDV510-307	88	ETC510-307	148
EDH610	EFS610	ETS610×112	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610×112	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610×112	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610×112	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610×112	117	EDV610	105	ETC610	165
EDH510-307	EFS510-307	ETS510-307	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510-307	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510-307	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510-307	102	EDV510-307	88	ETC510-307	148
EDH510-307	EFS510-307	ETS510-307	102	EDV510-307	88	ETC510-307	148

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

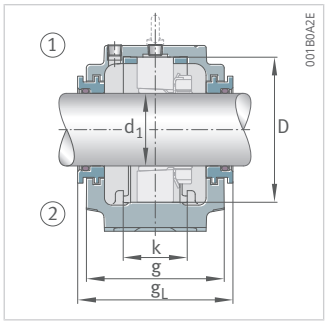


Double lip seal EDH, cover

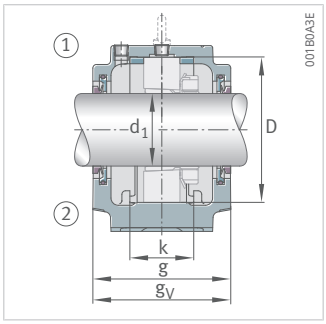


Felt seal EFS

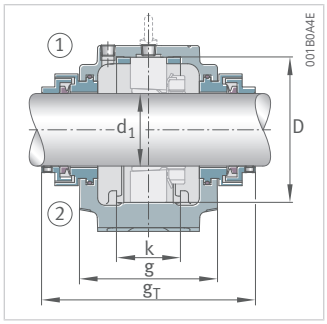
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES512-610-L	**45**	**–**	1310-K	H310	FRM110/10,5	EDK512-610
SES512-610-L	**45**	**–**	20310-K	H310	FRM110/10,5	EDK512-610
SES512-610-L	**45**	**–**	21310..-K	H310	FRM110/10,5	EDK512-610
SES512-610-L	**45**	**–**	22310..-K	H2310	FRM110/4	EDK512-610
SES512-610-L	**45**	**–**	2310-K	H2310	FRM110/4	EDK512-610
SES513-611-L	**47,625**	**1 7/8**	1311-K	H311×114	FRM120/11	EDK513-611
SES513-611-L	**47,625**	**1 7/8**	20311-K	H311×114	FRM120/11	EDK513-611
SES513-611-L	**47,625**	**1 7/8**	21311..-K	H311×114	FRM120/11	EDK513-611
SES513-611-L	**47,625**	**1 7/8**	22311..-K	H2311×114	FRM120/4	EDK513-611
SES513-611-L	**47,625**	**1 7/8**	2311-K	H2311×114	FRM120/4	EDK513-611
SES511-609-L	**49,213**	**1 15/16**	1211-K	H211×115	FRM100/11,5	EDK511-609
SES511-609-L	**49,213**	**1 15/16**	20211-K	H211×115	FRM100/11,5	EDK511-609
SES511-609-L	**49,213**	**1 15/16**	2211-K	H311×115	FRM100/9,5	EDK511-609
SES511-609-L	**49,213**	**1 15/16**	22211..-K	H311×115	FRM100/9,5	EDK511-609
SES513-611-L	**49,213**	**1 15/16**	1311-K	H311×115	FRM120/11	EDK513-611
SES513-611-L	**49,213**	**1 15/16**	20311-K	H311×115	FRM120/11	EDK513-611
SES513-611-L	**49,213**	**1 15/16**	21311..-K	H311×115	FRM120/11	EDK513-611
SES513-611-L	**49,213**	**1 15/16**	22311..-K	H2311×115	FRM120/4	EDK513-611
SES513-611-L	**49,213**	**1 15/16**	2311-K	H2311×115	FRM120/4	EDK513-611
SES511-609-L	**50**	**–**	1211-K	H211	FRM100/11,5	EDK511-609
SES511-609-L	**50**	**–**	20211-K	H211	FRM100/11,5	EDK511-609
SES511-609-L	**50**	**–**	2211-K	H311	FRM100/9,5	EDK511-609
SES511-609-L	**50**	**–**	22211..-K	H311	FRM100/9,5	EDK511-609
SES511-609-L	**50**	**–**	WS22211-E1-XL-K-2RSR	H2211-T-WS	FRM100/6,5	EDK511-609
SES513-611-L	**50**	**–**	1311-K	H311	FRM120/11	EDK513-611
SES513-611-L	**50**	**–**	20311-K	H311	FRM120/11	EDK513-611
SES513-611-L	**50**	**–**	21311..-K	H311	FRM120/11	EDK513-611
SES513-611-L	**50**	**–**	22311..-K	H2311	FRM120/4	EDK513-611
SES513-611-L	**50**	**–**	2311-K	H2311	FRM120/4	EDK513-611
SES511-609-L	**50,8**	**2**	1211-K	H211×200	FRM100/11,5	EDK511-609
SES511-609-L	**50,8**	**2**	20211-K	H211×200	FRM100/11,5	EDK511-609
SES511-609-L	**50,8**	**2**	2211-K	H311×200	FRM100/9,5	EDK511-609
SES511-609-L	**50,8**	**2**	22211..-K	H311×200	FRM100/9,5	EDK511-609
SES513-611-L	**50,8**	**2**	1311-K	H311×200	FRM120/11	EDK513-611
SES513-611-L	**50,8**	**2**	20311-K	H311×200	FRM120/11	EDK513-611
SES513-611-L	**50,8**	**2**	21311..-K	H311×200	FRM120/11	EDK513-611
SES513-611-L	**50,8**	**2**	22311..-K	H2311×200	FRM120/4	EDK513-611
SES513-611-L	**50,8**	**2**	2311-K	H2311×200	FRM120/4	EDK513-611
SES512-610-L	**55**	**–**	1212-K	H212	FRM110/13	EDK512-610
SES512-610-L	**55**	**–**	20212-K	H212	FRM110/13	EDK512-610



Labyrinth seal ETS



V ring seal EDV

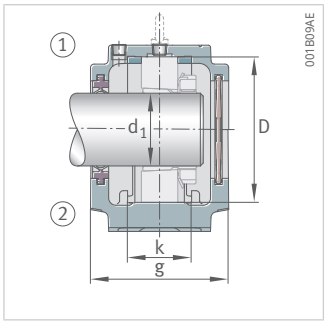


Taconite seal ETC

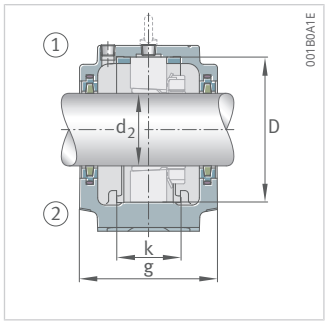
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH610	EFS610	ETS610	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610	117	EDV610	105	ETC610	165
EDH610	EFS610	ETS610	117	EDV610	105	ETC610	165
EDH611×114	EFS611×114	ETS611×114	122	EDV611×114	110	ETC611×114	170
EDH611×114	EFS611×114	ETS611×114	122	EDV611×114	110	ETC611×114	170
EDH611×114	EFS611×114	ETS611×114	122	EDV611×114	110	ETC611×114	170
EDH611×114	EFS611×114	ETS611×114	122	EDV611×114	110	ETC611×114	170
EDH611×114	EFS611×114	ETS611×114	122	EDV611×114	110	ETC611×114	170
EDH511	EFS511×115	ETS511×115	107	EDV511×115	95	ETC511×115	165
EDH511	EFS511×115	ETS511×115	107	EDV511×115	95	ETC511×115	165
EDH511	EFS511×115	ETS511×115	107	EDV511×115	95	ETC511×115	165
EDH511	EFS511×115	ETS511×115	107	EDV511×115	95	ETC511×115	165
EDH611	EFS611×115	ETS611×115	122	EDV611×115	110	ETC611×115	170
EDH611	EFS611×115	ETS611×115	122	EDV611×115	110	ETC611×115	170
EDH611	EFS611×115	ETS611×115	122	EDV611×115	110	ETC611×115	170
EDH611	EFS611×115	ETS611×115	122	EDV611×115	110	ETC611×115	170
EDH611	EFS611×115	ETS611×115	122	EDV611×115	110	ETC611×115	170
EDH511	EFS511	ETS511	107	EDV511	95	ETC511	165
EDH511	EFS511	ETS511	107	EDV511	95	ETC511	165
EDH511	EFS511	ETS511	107	EDV511	95	ETC511	165
EDH511	EFS511	ETS511	107	EDV511	95	ETC511	165
EDH511	EFS511	ETS511	107	EDV511	95	ETC511	165
EDH611	EFS611	ETS611	122	EDV611	110	ETC611	170
EDH611	EFS611	ETS611	122	EDV611	110	ETC611	170
EDH611	EFS611	ETS611	122	EDV611	110	ETC611	170
EDH611	EFS611	ETS611	122	EDV611	110	ETC611	170
EDH611	EFS611	ETS611	122	EDV611	110	ETC611	170
EDH511	EFS511	ETS511×200	107	EDV511	95	ETC511×200	165
EDH511	EFS511	ETS511×200	107	EDV511	95	ETC511×200	165
EDH511	EFS511	ETS511×200	107	EDV511	95	ETC511×200	165
EDH511	EFS511	ETS511×200	107	EDV511	95	ETC511×200	165
EDH611	EFS611	ETS611×200	122	EDV611	110	ETC611×200	170
EDH611	EFS611	ETS611×200	122	EDV611	110	ETC611×200	170
EDH611	EFS611	ETS611×200	122	EDV611	110	ETC611×200	170
EDH611	EFS611	ETS611×200	122	EDV611	110	ETC611×200	170
EDH611	EFS611	ETS611×200	122	EDV611	110	ETC611×200	170
EDH512	EFS512	ETS512	117	EDV512	105	ETC512	175
EDH512	EFS512	ETS512	117	EDV512	105	ETC512	175

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

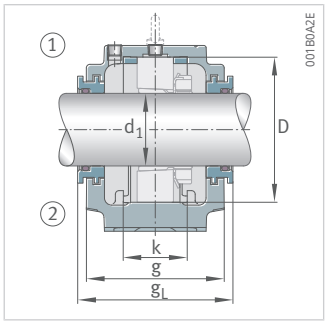


Double lip seal EDH, cover

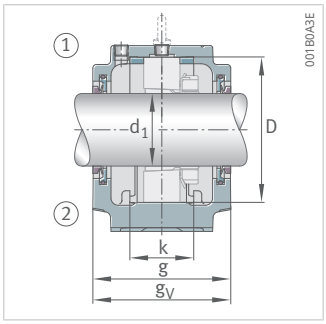


Felt seal EFS

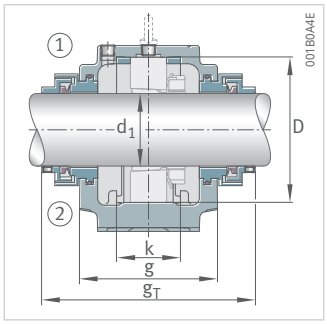
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES512-610-L	**55**	–	2212-K	H312	FRM110/10	EDK512-610
SES512-610-L	**55**	–	22212..-K	H312	FRM110/10	EDK512-610
SES512-610-L	**55**	–	222SM55-TVPA	–	FRM110/10	EDK512-610
SES512-610-L	**55**	–	WS22212-E1-XL-K-2RSR	H2212-T-WS	FRM110/7	EDK512-610
SES515-612-L	**55**	–	1312-K	H312	FRM130/12,5	EDK515-612
SES515-612-L	**55**	–	20312-K	H312	FRM130/12,5	EDK515-612
SES515-612-L	**55**	–	21312..-K	H312	FRM130/12,5	EDK515-612
SES515-612-L	**55**	–	22312..-K	H2312	FRM130/5	EDK515-612
SES515-612-L	**55**	–	2312-K	H2312	FRM130/5	EDK515-612
SES515-612-L	**55**	–	WS22312-E1-XL-K-2RSR	H3312-T-WS	FRM130/1,5	EDK515-612
SES513-611-L	**55,563**	**2 3/16**	1213-K	H213×203	FRM120/14	EDK513-611
SES513-611-L	**55,563**	**2 3/16**	20213-K	H213×203	FRM120/14	EDK513-611
SES513-611-L	**55,563**	**2 3/16**	2213-K	H313×203	FRM120/10	EDK513-611
SES513-611-L	**55,563**	**2 3/16**	22213..-K	H313×203	FRM120/10	EDK513-611
SES513-611-L	**55,563**	**2 3/16**	222S.203	–	FRM120/10	EDK513-611
SES516-613-L	**55,563**	**2 3/16**	1313-K	H313×203	FRM140/12,5	EDK516-613
SES516-613-L	**55,563**	**2 3/16**	20313-K	H313×203	FRM140/12,5	EDK516-613
SES516-613-L	**55,563**	**2 3/16**	21313..-K	H313×203	FRM140/12,5	EDK516-613
SES516-613-L	**55,563**	**2 3/16**	22313..-K	H2313×203	FRM140/5	EDK516-613
SES516-613-L	**55,563**	**2 3/16**	2313-K	H2313×203	FRM140/5	EDK516-613
SES513-611-L	**57,15**	**2 1/4**	1213-K	H213×204	FRM120/14	EDK513-611
SES513-611-L	**57,15**	**2 1/4**	20213-K	H213×204	FRM120/14	EDK513-611
SES513-611-L	**57,15**	**2 1/4**	2213-K	H313×204	FRM120/10	EDK513-611
SES513-611-L	**57,15**	**2 1/4**	22213..-K	H313×204	FRM120/10	EDK513-611
SES513-611-L	**57,15**	**2 1/4**	222S.204	–	FRM120/10	EDK513-611
SES516-613-L	**57,15**	**2 1/4**	1313-K	H313×204	FRM140/12,5	EDK516-613
SES516-613-L	**57,15**	**2 1/4**	20313-K	H313×204	FRM140/12,5	EDK516-613
SES516-613-L	**57,15**	**2 1/4**	21313..-K	H313×204	FRM140/12,5	EDK516-613
SES516-613-L	**57,15**	**2 1/4**	22313..-K	H2313×204	FRM140/5	EDK516-613
SES516-613-L	**57,15**	**2 1/4**	2313-K	H2313×204	FRM140/5	EDK516-613
SES513-611-L	**60**	–	1213-K	H213	FRM120/14	EDK513-611
SES513-611-L	**60**	–	20213-K	H213	FRM120/14	EDK513-611
SES513-611-L	**60**	–	2213-K	H313	FRM120/10	EDK513-611
SES513-611-L	**60**	–	22213..-K	H313	FRM120/10	EDK513-611
SES513-611-L	**60**	–	222SM60-TVPA	–	FRM120/10	EDK513-611
SES513-611-L	**60**	–	WS22213-E1-XL-K-2RSR	H2213-T-WS	FRM120/6,5	EDK513-611
SES513-611-L	**60**	–	WS22311-E1-XL-K-2RSR	H3311-T-WS	FRM120/1	EDK513-611
SES516-613-L	**60**	–	1313-K	H313	FRM140/12,5	EDK516-613
SES516-613-L	**60**	–	20313-K	H313	FRM140/12,5	EDK516-613
SES516-613-L	**60**	–	21313..-K	H313	FRM140/12,5	EDK516-613



Labyrinth seal ETS



V ring seal EDV



Taconite seal ETC

Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH512	EFS512	ETS512	117	EDV512	105	ETC512	175
EDH512	EFS512	ETS512	117	EDV512	105	ETC512	175
EDH512	EFS512	ETS512	117	EDV512	105	ETC512	175
EDH512	EFS512	ETS512	117	EDV512	105	ETC512	175
EDH612	EFS612	ETS612	127	EDV612	115	ETC612	175
EDH612	EFS612	ETS612	127	EDV612	115	ETC612	175
EDH612	EFS612	ETS612	127	EDV612	115	ETC612	175
EDH612	EFS612	ETS612	127	EDV612	115	ETC612	175
EDH612	EFS612	ETS612	127	EDV612	115	ETC612	175
EDH612	EFS612	ETS612	127	EDV612	115	ETC612	175
EDH513×203	EFS513×203	ETS513×203	122	EDV513×203	110	ETC513×203	180
EDH513×203	EFS513×203	ETS513×203	122	EDV513×203	110	ETC513×203	180
EDH513×203	EFS513×203	ETS513×203	122	EDV513×203	110	ETC513×203	180
EDH513×203	EFS513×203	ETS513×203	122	EDV513×203	110	ETC513×203	180
EDH513×203	EFS513×203	ETS513×203	122	EDV513×203	110	ETC513×203	180
–	EFS613×203	ETS613×203	134	EDV613×203	120	ETC613×203	180
–	EFS613×203	ETS613×203	134	EDV613×203	120	ETC613×203	180
–	EFS613×203	ETS613×203	134	EDV613×203	120	ETC613×203	180
–	EFS613×203	ETS613×203	134	EDV613×203	120	ETC613×203	180
–	EFS613×203	ETS613×203	134	EDV613×203	120	ETC613×203	180
–	EFS513×204	ETS513×204	122	EDV513×204	110	ETC513×204	180
–	EFS513×204	ETS513×204	122	EDV513×204	110	ETC513×204	180
–	EFS513×204	ETS513×204	122	EDV513×204	110	ETC513×204	180
–	EFS513×204	ETS513×204	122	EDV513×204	110	ETC513×204	180
–	EFS513×204	ETS513×204	122	EDV513×204	110	ETC513×204	180
–	EFS613×204	ETS613×204	134	EDV613×204	120	ETC613×204	180
–	EFS613×204	ETS613×204	134	EDV613×204	120	ETC613×204	180
–	EFS613×204	ETS613×204	134	EDV613×204	120	ETC613×204	180
–	EFS613×204	ETS613×204	134	EDV613×204	120	ETC613×204	180
–	EFS613×204	ETS613×204	134	EDV613×204	120	ETC613×204	180
EDH513	EFS513	ETS513	122	EDV513	110	ETC513	180
EDH513	EFS513	ETS513	122	EDV513	110	ETC513	180
EDH513	EFS513	ETS513	122	EDV513	110	ETC513	180
EDH513	EFS513	ETS513	122	EDV513	110	ETC513	180
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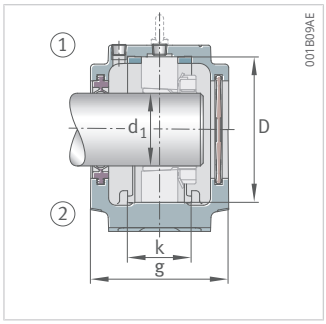
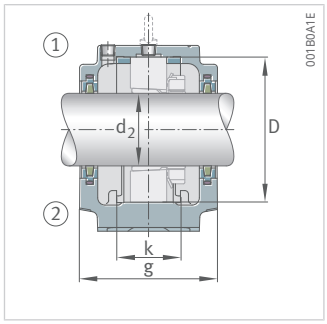
EDH513	EFS513	ETS513	122	EDV513	110	ETC513	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

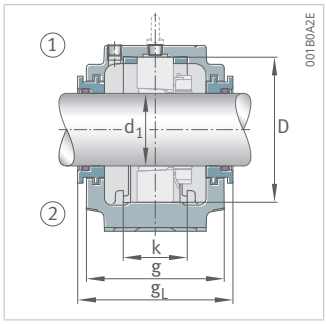


Double lip seal EDH, cover

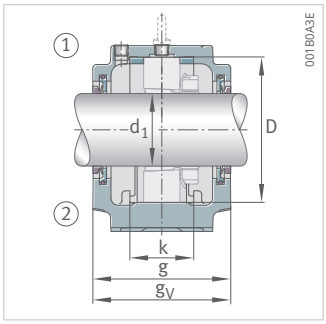


Felt seal EFS

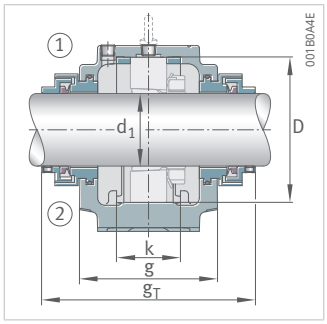
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES516-613-L	**60**	–	22313..-K	H2313	FRM140/5	EDK516-613
SES516-613-L	**60**	–	2313-K	H2313	FRM140/5	EDK516-613
SES516-613-L	**60,325**	**2 3/8**	1313-K	H313×206	FRM140/12,5	EDK516-613
SES516-613-L	**60,325**	**2 3/8**	20313-K	H313×206	FRM140/12,5	EDK516-613
SES516-613-L	**60,325**	**2 3/8**	21313..-K	H313×206	FRM140/12,5	EDK516-613
SES516-613-L	**60,325**	**2 3/8**	22313..-K	H2313×206	FRM140/5	EDK516-613
SES516-613-L	**60,325**	**2 3/8**	2313..-K	H2313×206	FRM140/5	EDK516-613
SES515-612-L	**61,913**	**2 7/16**	1215-K	H215×207	FRM130/15,5	EDK515-612
SES515-612-L	**61,913**	**2 7/16**	20215-K	H215×207	FRM130/15,5	EDK515-612
SES515-612-L	**61,913**	**2 7/16**	2215-K	H315×207	FRM130/12,5	EDK515-612
SES515-612-L	**61,913**	**2 7/16**	22215..-K	H315×207	FRM130/12,5	EDK515-612
SES515-612-L	**61,913**	**2 7/16**	222S.207	–	FRM130/12,5	EDK515-612
SES518-615-L	**61,913**	**2 7/16**	1315-K	H315×207	FRM160/14	EDK518-615
SES518-615-L	**61,913**	**2 7/16**	20315-K	H315×207	FRM160/14	EDK518-615
SES518-615-L	**61,913**	**2 7/16**	21315..-K	H315×207	FRM160/14	EDK518-615
SES518-615-L	**61,913**	**2 7/16**	22315..-K	H2315×207	FRM160/5	EDK518-615
SES518-615-L	**61,913**	**2 7/16**	2315-K	H2315×207	FRM160/5	EDK518-615
SES515-612-L	**63,5**	**2 1/2**	1215-K	H215×208	FRM130/15,5	EDK515-612
SES515-612-L	**63,5**	**2 1/2**	20215-K	H215×208	FRM130/15,5	EDK515-612
SES515-612-L	**63,5**	**2 1/2**	2215-K	H315×208	FRM130/12,5	EDK515-612
SES515-612-L	**63,5**	**2 1/2**	22215..-K	H315×208	FRM130/12,5	EDK515-612
SES515-612-L	**63,5**	**2 1/2**	222S.208	–	FRM130/12,5	EDK515-612
SES518-615-L	**63,5**	**2 1/2**	1315-K	H315×208	FRM160/14	EDK518-615
SES518-615-L	**63,5**	**2 1/2**	20315-K	H315×208	FRM160/14	EDK518-615
SES518-615-L	**63,5**	**2 1/2**	21315..-K	H315×208	FRM160/14	EDK518-615
SES518-615-L	**63,5**	**2 1/2**	22315..-K	H2315×208	FRM160/5	EDK518-615
SES518-615-L	**63,5**	**2 1/2**	2315-K	H2315×208	FRM160/5	EDK518-615
SES515-612-L	**65**	–	1215-K	H215	FRM130/15,5	EDK515-612
SES515-612-L	**65**	–	20215-K	H215	FRM130/15,5	EDK515-612
SES515-612-L	**65**	–	2215-K	H315	FRM130/12,5	EDK515-612
SES515-612-L	**65**	–	22215..-K	H315	FRM130/12,5	EDK515-612
SES515-612-L	**65**	–	222SM65-TVPA	–	FRM130/12,5	EDK515-612
SES515-612-L	**65**	–	WS22215-E1-XL-K-2RSR	H2215-T-WS	FRM130/9	EDK515-612
SES518-615-L	**65**	–	1315-K	H315	FRM160/14	EDK518-615
SES518-615-L	**65**	–	20315-K	H315	FRM160/14	EDK518-615
SES518-615-L	**65**	–	21315..-K	H315	FRM160/14	EDK518-615
SES518-615-L	**65**	–	22315..-K	H2315	FRM160/5	EDK518-615
SES518-615-L	**65**	–	2315-K	H2315	FRM160/5	EDK518-615
SES518-615-L	**65**	–	WS22315-E1-XL-K-2RSR	H3315-T-WS	FRM160/1	EDK518-615
SES518-615-L	**66,675**	**2 5/8**	1315-K	H315×210	FRM160/14	EDK518-615



Labyrinth seal ETS



V ring seal EDV

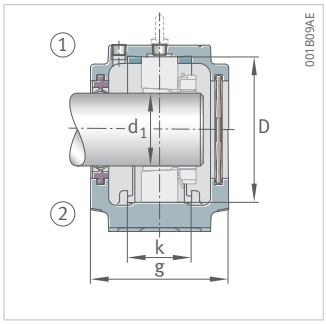


Taconite seal ETC

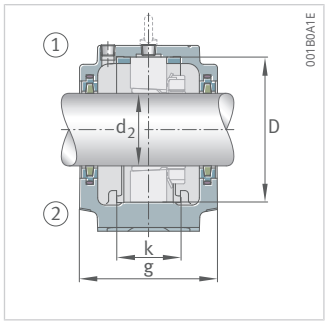
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH613	EFS613	ETS613	134	EDV613	120	ETC613	180
EDH515×207	EFS515×207	ETS515×207	127	EDV515×207	115	ETC515×207	175
EDH515×207	EFS515×207	ETS515×207	127	EDV515×207	115	ETC515×207	175
EDH515×207	EFS515×207	ETS515×207	127	EDV515×207	115	ETC515×207	175
EDH515×207	EFS515×207	ETS515×207	127	EDV515×207	115	ETC515×207	175
EDH515×207	EFS515×207	ETS515×207	127	EDV515×207	115	ETC515×207	175
EDH615×207	EFS615×207	ETS615×207	154	EDV615×207	140	ETC615×207	200
EDH615×207	EFS615×207	ETS615×207	154	EDV615×207	140	ETC615×207	200
EDH615×207	EFS615×207	ETS615×207	154	EDV615×207	140	ETC615×207	200
EDH615×207	EFS615×207	ETS615×207	154	EDV615×207	140	ETC615×207	200
EDH615×207	EFS615×207	ETS615×207	154	EDV615×207	140	ETC615×207	200
EDH515×207	EFS515×208	ETS515×208	127	EDV515×208	115	ETC515×208	175
EDH515×207	EFS515×208	ETS515×208	127	EDV515×208	115	ETC515×208	175
EDH515×207	EFS515×208	ETS515×208	127	EDV515×208	115	ETC515×208	175
EDH515×207	EFS515×208	ETS515×208	127	EDV515×208	115	ETC515×208	175
EDH515×207	EFS515×208	ETS515×208	127	EDV515×208	115	ETC515×208	175
EDH615×207	EFS615×208	ETS615×208	154	EDV615×208	140	ETC615×208	200
EDH615×207	EFS615×208	ETS615×208	154	EDV615×208	140	ETC615×208	200
EDH615×207	EFS615×208	ETS615×208	154	EDV615×208	140	ETC615×208	200
EDH615×207	EFS615×208	ETS615×208	154	EDV615×208	140	ETC615×208	200
EDH615×207	EFS615×208	ETS615×208	154	EDV615×208	140	ETC615×208	200
EDH515	EFS515	ETS515	127	EDV515	115	ETC515	175
EDH515	EFS515	ETS515	127	EDV515	115	ETC515	175
EDH515	EFS515	ETS515	127	EDV515	115	ETC515	175
EDH515	EFS515	ETS515	127	EDV515	115	ETC515	175
EDH515	EFS515	ETS515	127	EDV515	115	ETC515	175
EDH515	EFS515	ETS515	127	EDV515	115	ETC515	175
EDH615	EFS615	ETS615	154	EDV615	140	ETC615	200
EDH615	EFS615	ETS615	154	EDV615	140	ETC615	200
EDH615	EFS615	ETS615	154	EDV615	140	ETC615	200
EDH615	EFS615	ETS615	154	EDV615	140	ETC615	200
EDH615	EFS615	ETS615	154	EDV615	140	ETC615	200
EDH615	EFS615	ETS615	154	EDV615	140	ETC615	200
EDH615	EFS615×210	ETS615×210	154	EDV615×210	140	ETC615×210	200

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

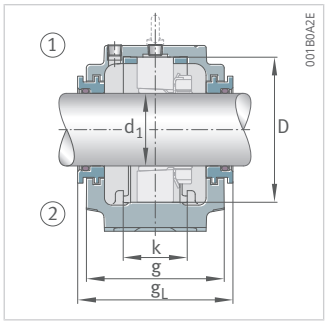


Double lip seal EDH, cover

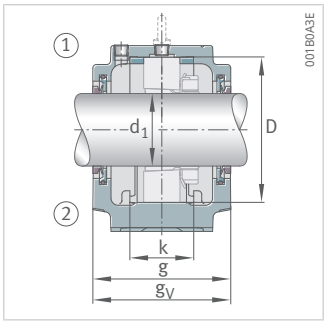


Felt seal EFS

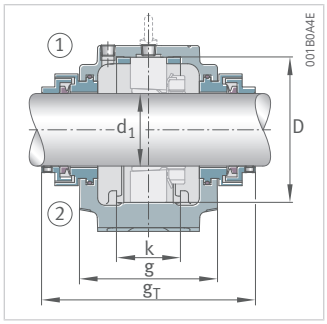
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES518-615-L	**66,675**	**2 5/8**	20315-K	H315×210	FRM160/14	EDK518-615
SES518-615-L	**66,675**	**2 5/8**	21315..-K	H315×210	FRM160/14	EDK518-615
SES518-615-L	**66,675**	**2 5/8**	22315..-K	H2315×210	FRM160/5	EDK518-615
SES518-615-L	**66,675**	**2 5/8**	2315-K	H2315×210	FRM160/5	EDK518-615
SES516-613-L	**68,263**	**2 11/16**	1216-K	H216×211	FRM140/16	EDK516-613
SES516-613-L	**68,263**	**2 11/16**	20216-K	H216×211	FRM140/16	EDK516-613
SES516-613-L	**68,263**	**2 11/16**	2216-K	H316×211	FRM140/12,5	EDK516-613
SES516-613-L	**68,263**	**2 11/16**	22216..-K	H316×211	FRM140/12,5	EDK516-613
SES516-613-L	**68,263**	**2 11/16**	222S.211	–	FRM140/12,5	EDK516-613
SES519-616-L	**68,263**	**2 11/16**	1316-K	H316×211	FRM170/14,5	EDK519-616
SES519-616-L	**68,263**	**2 11/16**	20316-K	H316×211	FRM170/14,5	EDK519-616
SES519-616-L	**68,263**	**2 11/16**	21316..-K	H316×211	FRM170/14,5	EDK519-616
SES519-616-L	**68,263**	**2 11/16**	22316..-K	H2316×211	FRM170/5	EDK519-616
SES519-616-L	**68,263**	**2 11/16**	2316-K	H2316×211	FRM170/5	EDK519-616
SES516-613-L	**69,85**	**2 3/4**	1216-K	H216×212	FRM140/16	EDK516-613
SES516-613-L	**69,85**	**2 3/4**	20216-K	H216×212	FRM140/16	EDK516-613
SES516-613-L	**69,85**	**2 3/4**	2216-K	H316×212	FRM140/12,5	EDK516-613
SES516-613-L	**69,85**	**2 3/4**	22216..-K	H316×212	FRM140/12,5	EDK516-613
SES519-616-L	**69,85**	**2 3/4**	1316-K	H316×212	FRM170/14,5	EDK519-616
SES519-616-L	**69,85**	**2 3/4**	20316-K	H316×212	FRM170/14,5	EDK519-616
SES519-616-L	**69,85**	**2 3/4**	21316..-K	H316×212	FRM170/14,5	EDK519-616
SES519-616-L	**69,85**	**2 3/4**	22316..-K	H2316×212	FRM170/5	EDK519-616
SES519-616-L	**69,85**	**2 3/4**	2316-K	H2316×212	FRM170/5	EDK519-616
SES516-613-L	**70**	**–**	1216-K	H216	FRM140/16	EDK516-613
SES516-613-L	**70**	**–**	20216..-K	H216	FRM140/16	EDK516-613
SES516-613-L	**70**	**–**	2216-K	H316	FRM140/12,5	EDK516-613
SES516-613-L	**70**	**–**	22216..-K	H316	FRM140/12,5	EDK516-613
SES516-613-L	**70**	**–**	222SM70-TVPA	–	FRM140/12,5	EDK516-613
SES516-613-L	**70**	**–**	WS22216-E1-XL-K-2RSR	H2216-T-WS	FRM140/9	EDK516-613
SES519-616-L	**70**	**–**	1316-K	H316	FRM170/14,5	EDK519-616
SES519-616-L	**70**	**–**	20316-K	H316	FRM170/14,5	EDK519-616
SES519-616-L	**70**	**–**	21316..-K	H316	FRM170/14,5	EDK519-616
SES519-616-L	**70**	**–**	22316..-K	H2316	FRM170/5	EDK519-616
SES519-616-L	**70**	**–**	2316-K	H2316	FRM170/5	EDK519-616
SES519-616-L	**73,025**	**2 7/8**	1316-K	H316×214	FRM170/14,5	EDK519-616
SES519-616-L	**73,025**	**2 7/8**	20316-K	H316×214	FRM170/14,5	EDK519-616
SES519-616-L	**73,025**	**2 7/8**	21316..-K	H316×214	FRM170/14,5	EDK519-616
SES519-616-L	**73,025**	**2 7/8**	22316..-K	H2316×214	FRM170/5	EDK519-616
SES519-616-L	**73,025**	**2 7/8**	2316-K	H2316×214	FRM170/5	EDK519-616
SES517-L	**74,613**	**2 15/16**	1217-K	H217×215	FRM150/16,5	EDK517



Labyrinth seal ETS



V ring seal EDV

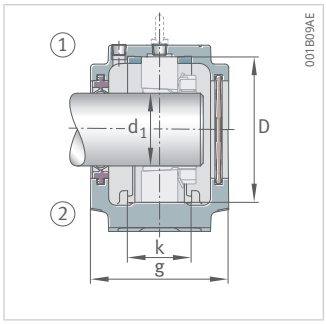


Taconite seal ETC

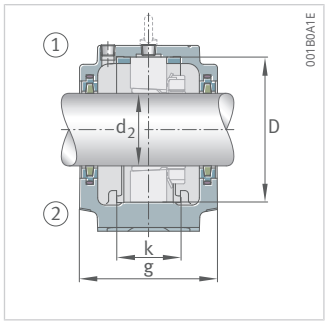
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH615	EFS615×210	ETS615×210	154	EDV615×210	140	ETC615×210	200
EDH615	EFS615×210	ETS615×210	154	EDV615×210	140	ETC615×210	200
EDH615	EFS615×210	ETS615×210	154	EDV615×210	140	ETC615×210	200
EDH615	EFS615×210	ETS615×210	154	EDV615×210	140	ETC615×210	200
EDH516×211	EFS516×211	ETS516×211	134	EDV516×211	120	ETC516×211	205
EDH516×211	EFS516×211	ETS516×211	134	EDV516×211	120	ETC516×211	205
EDH516×211	EFS516×211	ETS516×211	134	EDV516×211	120	ETC516×211	205
EDH516×211	EFS516×211	ETS516×211	134	EDV516×211	120	ETC516×211	205
EDH516×211	EFS516×211	ETS516×211	134	EDV516×211	120	ETC516×211	205
EDH616×211	EFS616×211	ETS616×211	161	EDV616×211	145	ETC616×211	205
EDH616×211	EFS616×211	ETS616×211	161	EDV616×211	145	ETC616×211	205
EDH616×211	EFS616×211	ETS616×211	161	EDV616×211	145	ETC616×211	205
EDH616×211	EFS616×211	ETS616×211	161	EDV616×211	145	ETC616×211	205
EDH616×211	EFS616×211	ETS616×211	161	EDV616×211	145	ETC616×211	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH516	EFS516	ETS516	134	EDV516	120	ETC516	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616	EFS616	ETS616	161	EDV616	145	ETC616	205
EDH616×214	EFS616×214	ETS616×214	161	EDV616×214	145	ETC616×214	205
EDH616×214	EFS616×214	ETS616×214	161	EDV616×214	145	ETC616×214	205
EDH616×214	EFS616×214	ETS616×214	161	EDV616×214	145	ETC616×214	205
EDH616×214	EFS616×214	ETS616×214	161	EDV616×214	145	ETC616×214	205
EDH616×214	EFS616×214	ETS616×214	161	EDV616×214	145	ETC616×214	205
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

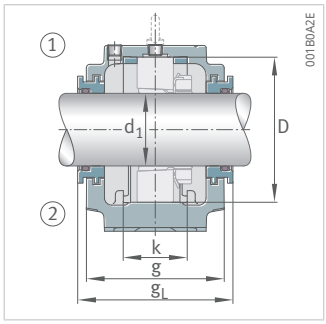


Double lip seal EDH, cover

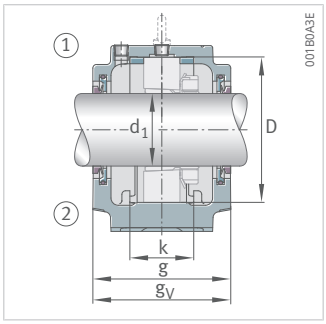


Felt seal EFS

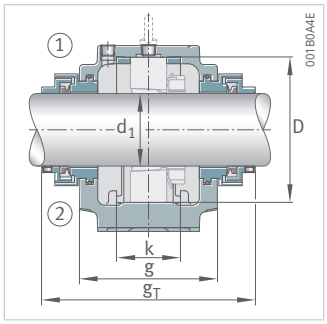
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES517-L	**74,613**	**2 15/16**	20217-K	H217×215	FRM150/16,5	EDK517
SES517-L	**74,613**	**2 15/16**	2217-K	H317×215	FRM150/12,5	EDK517
SES517-L	**74,613**	**2 15/16**	22217..-K	H317×215	FRM150/12,5	EDK517
SES517-L	**74,613**	**2 15/16**	222S.215	–	FRM150/12,5	EDK517
SES520-617-L	**74,613**	**2 15/16**	1317-K	H317×215	FRM180/14,5	EDK520-617
SES520-617-L	**74,613**	**2 15/16**	20317-K	H317×215	FRM180/14,5	EDK520-617
SES520-617-L	**74,613**	**2 15/16**	20317-K	H2317×215	FRM180/5	EDK520-617
SES520-617-L	**74,613**	**2 15/16**	21317..-K	H317×215	FRM180/14,5	EDK520-617
SES520-617-L	**74,613**	**2 15/16**	22317..-K	H2317×215	FRM180/5	EDK520-617
SES520-617-L	**74,613**	**2 15/16**	2317-K	H2317×215	FRM180/5	EDK520-617
SES517-L	**75**	**–**	1217-K	H217	FRM150/16,5	EDK517
SES517-L	**75**	**–**	20217-K	H217	FRM150/16,5	EDK517
SES517-L	**75**	**–**	2217-K	H317	FRM150/12,5	EDK517
SES517-L	**75**	**–**	22217..-K	H317	FRM150/12,5	EDK517
SES517-L	**75**	**–**	222SM75-TVPA	–	FRM150/12,5	EDK517
SES517-L	**75**	**–**	WS22217-E1-XL-K-2RSR	H2217-T-WS	FRM150/8,5	EDK517
SES520-617-L	**75**	**–**	1317-K	H317	FRM180/14,5	EDK520-617
SES520-617-L	**75**	**–**	21317..-K	H317	FRM180/14,5	EDK520-617
SES520-617-L	**75**	**–**	22317..-K	H2317	FRM180/5	EDK520-617
SES520-617-L	**75**	**–**	2317-K	H2317	FRM180/5	EDK520-617
SES517-L	**76,2**	**3**	1217-K	H217×300	FRM150/16,5	EDK517
SES517-L	**76,2**	**3**	20217-K	H217×300	FRM150/16,5	EDK517
SES517-L	**76,2**	**3**	2217-K	H317×300	FRM150/12,5	EDK517
SES517-L	**76,2**	**3**	22217..-K	H317×300	FRM150/12,5	EDK517
SES517-L	**76,2**	**3**	222S.300	–	FRM150/12,5	EDK517
SES520-617-L	**76,2**	**3**	1317-K	H317×300	FRM180/14,5	EDK520-617
SES520-617-L	**76,2**	**3**	20317-K	H317×300	FRM180/14,5	EDK520-617
SES520-617-L	**76,2**	**3**	21317..-K	H317×300	FRM180/14,5	EDK520-617
SES520-617-L	**76,2**	**3**	22317..-K	H2317×300	FRM180/5	EDK520-617
SES520-617-L	**76,2**	**3**	2317-K	H2317×300	FRM180/5	EDK520-617
SES518-615-L	**79,375**	**3 1/8**	1218-K	H218×302	FRM160/17,5	EDK518-615
SES518-615-L	**79,375**	**3 1/8**	20218-K	H218×302	FRM160/17,5	EDK518-615
SES518-615-L	**79,375**	**3 1/8**	2218-K	H318×302	FRM160/12,5	EDK518-615
SES518-615-L	**79,375**	**3 1/8**	22218..-K	H318×302	FRM160/12,5	EDK518-615
SES518-615-L	**79,375**	**3 1/8**	23218..-K	H2318×302	FRM160/6,25	EDK518-615
SES618-318-L	**79,375**	**3 1/8**	1318-K	H318×302	FRM190/15,5	EDK218
SES618-318-L	**79,375**	**3 1/8**	20318-K	H318×302	FRM190/15,5	EDK218
SES618-318-L	**79,375**	**3 1/8**	21318..-K	H318×302	FRM190/15,5	EDK218
SES618-318-L	**79,375**	**3 1/8**	22318..-K	H2318×302	FRM190/5	EDK218
SES618-318-L	**79,375**	**3 1/8**	2318-K	H2318×302	FRM190/5	EDK218



Labyrinth seal ETS



V ring seal EDV

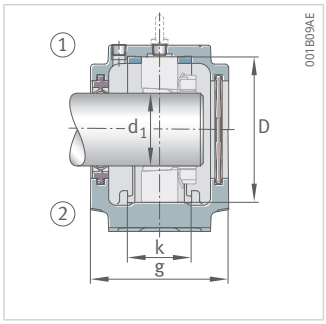


Taconite seal ETC

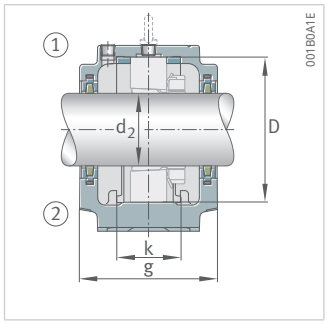
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH517	EFS517	ETS517	139	EDV517	125	ETC517	210
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH617	EFS617	ETS617	176	EDV617	160	ETC617	220
EDH517	EFS517×300	ETS517×300	139	EDV517×300	125	ETC517×300	210
EDH517	EFS517×300	ETS517×300	139	EDV517×300	125	ETC517×300	210
EDH517	EFS517×300	ETS517×300	139	EDV517×300	125	ETC517×300	210
EDH517	EFS517×300	ETS517×300	139	EDV517×300	125	ETC517×300	210
EDH517	EFS517×300	ETS517×300	139	EDV517×300	125	ETC517×300	210
EDH617	EFS617×300	ETS617×300	176	EDV617×300	160	ETC617×300	220
EDH617	EFS617×300	ETS617×300	176	EDV617×300	160	ETC617×300	220
EDH617	EFS617×300	ETS617×300	176	EDV617×300	160	ETC617×300	220
EDH617	EFS617×300	ETS617×300	176	EDV617×300	160	ETC617×300	220
EDH617	EFS617×300	ETS617×300	176	EDV617×300	160	ETC617×300	220
EDH518	EFS518	–	–	EDV518	140	–	–
EDH518	EFS518	–	–	EDV518	140	–	–
EDH518	EFS518	–	–	EDV518	140	–	–
EDH518	EFS518	–	–	EDV518	140	–	–
EDH518	EFS518	–	–	EDV518	140	–	–
EDH618	EFS618	ETS618×302	170,3	EDV618	163	ETC618×302	196
EDH618	EFS618	ETS618×302	170,3	EDV618	163	ETC618×302	196
EDH618	EFS618	ETS618×302	170,3	EDV618	163	ETC618×302	196
EDH618	EFS618	ETS618×302	170,3	EDV618	163	ETC618×302	196
EDH618	EFS618	ETS618×302	170,3	EDV618	163	ETC618×302	196

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

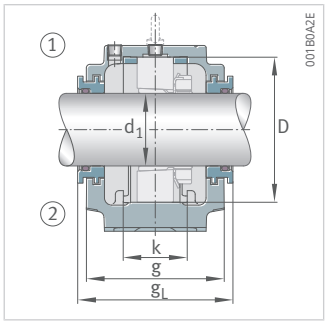


Double lip seal EDH, cover

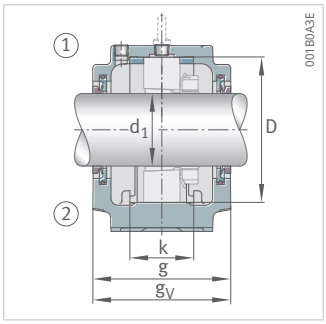


Felt seal EFS

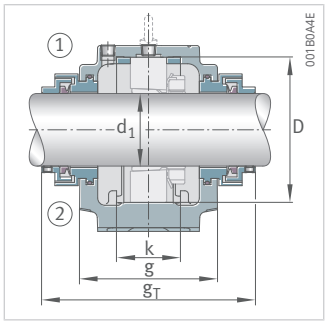
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES518-615-L	**80**	–	1218-K	H218	FRM160/17,5	EDK518-615
SES518-615-L	**80**	–	20218-K	H218	FRM160/17,5	EDK518-615
SES518-615-L	**80**	–	2218-K	H318	FRM160/12,5	EDK518-615
SES518-615-L	**80**	–	22218..-K	H318	FRM160/12,5	EDK518-615
SES518-615-L	**80**	–	222SM80-TVPA	–	FRM160/12,5	EDK518-615
SES518-615-L	**80**	–	23218..-K	H2318	FRM160/6,25	EDK518-615
SES518-615-L	**80**	–	WS22218-E1-XL-K-2RSR	H2218-T-WS	FRM160/8,5	EDK518-615
SES618-318-L	**80**	–	1318-K	H318	FRM190/15,5	EDK218
SES618-318-L	**80**	–	20318-K	H318	FRM190/15,5	EDK218
SES618-318-L	**80**	–	21318..-K	H318	FRM190/15,5	EDK218
SES618-318-L	**80**	–	22318..-K	H2318	FRM190/5	EDK218
SES618-318-L	**80**	–	2318-K	H2318	FRM190/5	EDK218
SES518-615-L	**80,963**	**3 3/16**	1218-K	H218×303	FRM160/17,5	EDK518-615
SES518-615-L	**80,963**	**3 3/16**	20218-K	H218×303	FRM160/17,5	EDK518-615
SES518-615-L	**80,963**	**3 3/16**	2218-K	H318×303	FRM160/12,5	EDK518-615
SES518-615-L	**80,963**	**3 3/16**	22218..-K	H318×303	FRM160/12,5	EDK518-615
SES518-615-L	**80,963**	**3 3/16**	222S.303	–	FRM160/12,5	EDK518-615
SES518-615-L	**80,963**	**3 3/16**	23218..-K	H2318×303	FRM160/6,25	EDK518-615
SES618-318-L	**80,963**	**3 3/16**	1318-K	H318×303	FRM190/15,5	EDK218
SES618-318-L	**80,963**	**3 3/16**	20318-K	H318×303	FRM190/15,5	EDK218
SES618-318-L	**80,963**	**3 3/16**	21318..-K	H318×303	FRM190/15,5	EDK218
SES618-318-L	**80,963**	**3 3/16**	22318..-K	H2318×303	FRM190/5	EDK218
SES618-318-L	**80,963**	**3 3/16**	2318-K	H2318×303	FRM190/5	EDK218
SES518-615-L	**82,55**	**3 1/4**	1218-K	H218×304	FRM160/17,5	EDK518-615
SES518-615-L	**82,55**	**3 1/4**	20218-K	H218×304	FRM160/17,5	EDK518-615
SES518-615-L	**82,55**	**3 1/4**	2218-K	H318×304	FRM160/12,5	EDK518-615
SES518-615-L	**82,55**	**3 1/4**	22218..-K	H318×304	FRM160/12,5	EDK518-615
SES518-615-L	**82,55**	**3 1/4**	222S.304	–	FRM160/12,5	EDK518-615
SES518-615-L	**82,55**	**3 1/4**	23218..-K	H2318×304	FRM160/6,25	EDK518-615
SES522-619-L	**82,55**	**3 1/4**	1319-K	H319×304	FRM200/17,5	EDK522-619
SES522-619-L	**82,55**	**3 1/4**	20319-K	H319×304	FRM200/17,5	EDK522-619
SES522-619-L	**82,55**	**3 1/4**	21319..-K	H319×304	FRM200/17,5	EDK522-619
SES522-619-L	**82,55**	**3 1/4**	22319..-K	H2319×304	FRM200/6,5	EDK522-619
SES522-619-L	**82,55**	**3 1/4**	2319-K	H2319×304	FRM200/6,5	EDK522-619
SES618-318-L	**82,55**	**3 1/4**	1318-K	H318×304	FRM190/15,5	EDK218
SES618-318-L	**82,55**	**3 1/4**	20318-K	H318×304	FRM190/15,5	EDK218
SES618-318-L	**82,55**	**3 1/4**	21318..-K	H318×304	FRM190/15,5	EDK218
SES618-318-L	**82,55**	**3 1/4**	22318..-K	H2318×304	FRM190/5	EDK218
SES618-318-L	**82,55**	**3 1/4**	2318-K	H2318×304	FRM190/5	EDK218
SES519-616-L	**85**	–	1219-K	H219	FRM170/18	EDK519-616



Labyrinth seal ETS



V ring seal EDV

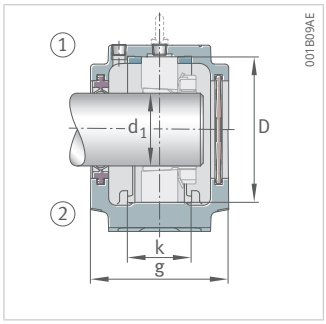


Taconite seal ETC

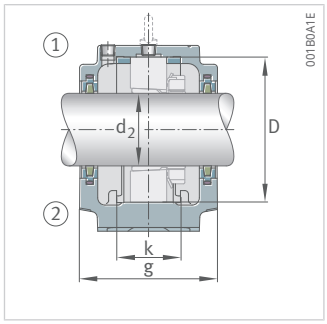
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH518	EFS518	ETS518	154	EDV518	140	ETC518	225
EDH618	EFS618	ETS618	170,3	EDV618	162,5	ETC618	178
EDH618	EFS618	ETS618	170,3	EDV618	162,5	ETC618	178
EDH618	EFS618	ETS618	170,3	EDV618	162,5	ETC618	178
EDH618	EFS618	ETS618	170,3	EDV618	162,5	ETC618	178
EDH618	EFS618	ETS618	170,3	EDV618	162,5	ETC618	178
EDH518	EFS518×303	ETS518×303	154	EDV518×303	140	ETC518×303	225
EDH518	EFS518×303	ETS518×303	154	EDV518×303	140	ETC518×303	225
EDH518	EFS518×303	ETS518×303	154	EDV518×303	140	ETC518×303	225
EDH518	EFS518×303	ETS518×303	154	EDV518×303	140	ETC518×303	225
EDH518	EFS518×303	ETS518×303	154	EDV518×303	140	ETC518×303	225
EDH518	EFS518×303	ETS518×303	154	EDV518×303	140	ETC518×303	225
EDH618	EFS618×303	ETS618×303	170,3	EDV618×303	163	ETC618×303	196
EDH618	EFS618×303	ETS618×303	170,3	EDV618×303	163	ETC618×303	196
EDH618	EFS618×303	ETS618×303	170,3	EDV618×303	163	ETC618×303	196
EDH618	EFS618×303	ETS618×303	170,3	EDV618×303	163	ETC618×303	196
EDH618	EFS618×303	ETS618×303	170,3	EDV618×303	163	ETC618×303	196
–	EFS518×304	ETS518×304	154	EDV518×304	140	ETC518×304	225
–	EFS518×304	ETS518×304	154	EDV518×304	140	ETC518×304	225
–	EFS518×304	ETS518×304	154	EDV518×304	140	ETC518×304	225
–	EFS518×304	ETS518×304	154	EDV518×304	140	ETC518×304	225
–	EFS518×304	ETS518×304	154	EDV518×304	140	ETC518×304	225
–	EFS518×304	ETS518×304	154	EDV518×304	140	ETC518×304	225
–	EFS619×304	ETS619×304	191	EDV619×304	175	ETC619×304	235
–	EFS619×304	ETS619×304	191	EDV619×304	175	ETC619×304	235
–	EFS619×304	ETS619×304	191	EDV619×304	175	ETC619×304	235
–	EFS619×304	ETS619×304	191	EDV619×304	175	ETC619×304	235
–	EFS619×304	ETS619×304	191	EDV619×304	175	ETC619×304	235
EDH618×304	EFS618×304	ETS618×304	170,3	EDV618×304	163	ETC618×304	196
EDH618×304	EFS618×304	ETS618×304	170,3	EDV618×304	163	ETC618×304	196
EDH618×304	EFS618×304	ETS618×304	170,3	EDV618×304	163	ETC618×304	196
EDH618×304	EFS618×304	ETS618×304	170,3	EDV618×304	163	ETC618×304	196
EDH618×304	EFS618×304	ETS618×304	170,3	EDV618×304	163	ETC618×304	196
EDH519	EFS519	ETS519	161	EDV519	145	ETC519	220

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

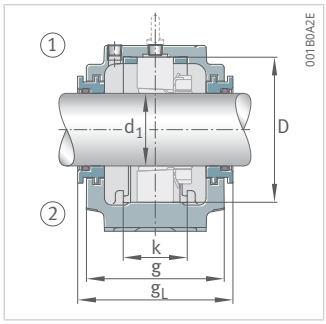


Double lip seal EDH, cover

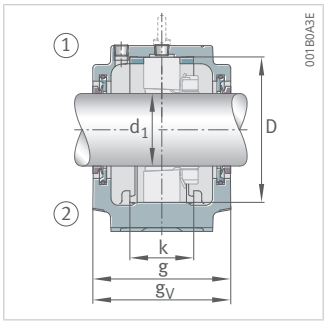


Felt seal EFS

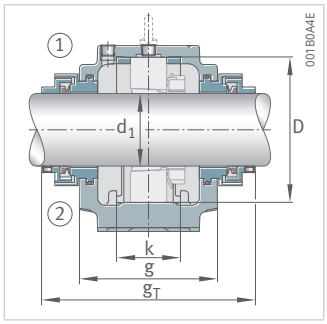
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES519-616-L	**85**	–	20219-K	H219	FRM170/18	EDK519-616
SES519-616-L	**85**	–	2219-K	H319	FRM170/12,5	EDK519-616
SES519-616-L	**85**	–	22219..-K	H319	FRM170/12,5	EDK519-616
SES519-616-L	**85**	–	222SM85-TVPA	–	FRM170/12,5	EDK519-616
SES522-619-L	**85**	–	1319-K	H319	FRM200/17,5	EDK522-619
SES522-619-L	**85**	–	20319-K	H319	FRM200/17,5	EDK522-619
SES522-619-L	**85**	–	21319..-K	H319	FRM200/17,5	EDK522-619
SES522-619-L	**85**	–	22319..-K	H2319	FRM200/6,5	EDK522-619
SES522-619-L	**85**	–	2319-K	H2319	FRM200/6,5	EDK522-619
SES519-616-L	**85,725**	**3 3/8**	1219-K	H219×306	FRM170/18	EDK519-616
SES519-616-L	**85,725**	**3 3/8**	20219-K	H219×306	FRM170/18	EDK519-616
SES519-616-L	**85,725**	**3 3/8**	2219-K	H319×306	FRM170/12,5	EDK519-616
SES519-616-L	**85,725**	**3 3/8**	22219..-K	H319×306	FRM170/12,5	EDK519-616
SES522-619-L	**85,725**	**3 3/8**	1319-K	H319×306	FRM200/17,5	EDK522-619
SES522-619-L	**85,725**	**3 3/8**	20319-K	H319×306	FRM200/17,5	EDK522-619
SES522-619-L	**85,725**	**3 3/8**	21319..-K	H319×306	FRM200/17,5	EDK522-619
SES522-619-L	**85,725**	**3 3/8**	22319..-K	H2319×306	FRM200/6,5	EDK522-619
SES522-619-L	**85,725**	**3 3/8**	2319-K	H2319×306	FRM200/6,5	EDK522-619
SES520-617-L	**87,313**	**3 7/16**	1220-K	H220×307	FRM180/18	EDK520-617
SES520-617-L	**87,313**	**3 7/16**	20220-K	H220×307	FRM180/18	EDK520-617
SES520-617-L	**87,313**	**3 7/16**	2220-K	H320×307	FRM180/12	EDK520-617
SES520-617-L	**87,313**	**3 7/16**	22220..-K	H320×307	FRM180/12	EDK520-617
SES520-617-L	**87,313**	**3 7/16**	222S.307	–	FRM180/18	EDK520-617
SES520-617-L	**87,313**	**3 7/16**	23220..-K	H2320×307	FRM180/4,85	EDK520-617
SES524-620-L	**87,313**	**3 7/16**	1320-K	H320×307	FRM215/19,5	EDK524-620
SES524-620-L	**87,313**	**3 7/16**	20320-K	H320×307	FRM215/19,5	EDK524-620
SES524-620-L	**87,313**	**3 7/16**	21320..-K	H320×307	FRM215/19,5	EDK524-620
SES524-620-L	**87,313**	**3 7/16**	22320..-K	H2320×307	FRM215/6,5	EDK524-620
SES524-620-L	**87,313**	**3 7/16**	2320-K	H2320×307	FRM215/6,5	EDK524-620
SES520-617-L	**88,9**	**3 1/2**	1220-K	H220×308	FRM180/18	EDK520-617
SES520-617-L	**88,9**	**3 1/2**	20220-K	H220×308	FRM180/18	EDK520-617
SES520-617-L	**88,9**	**3 1/2**	2220-K	H320×308	FRM180/12	EDK520-617
SES520-617-L	**88,9**	**3 1/2**	22220..-K	H320×308	FRM180/12	EDK520-617
SES520-617-L	**88,9**	**3 1/2**	222S.308	–	FRM180/12	EDK520-617
SES520-617-L	**88,9**	**3 1/2**	23220..-K	H2320×308	FRM180/4,85	EDK520-617
SES524-620-L	**88,9**	**3 1/2**	1320-K	H320×308	FRM215/19,5	EDK524-620
SES524-620-L	**88,9**	**3 1/2**	20320-K	H320×308	FRM215/19,5	EDK524-620
SES524-620-L	**88,9**	**3 1/2**	21320..-K	H320×308	FRM215/19,5	EDK524-620
SES524-620-L	**88,9**	**3 1/2**	22320..-K	H2320×308	FRM215/6,5	EDK524-620
SES524-620-L	**88,9**	**3 1/2**	2320-K	H2320×308	FRM215/6,5	EDK524-620



Labyrinth seal ETS



V ring seal EDV

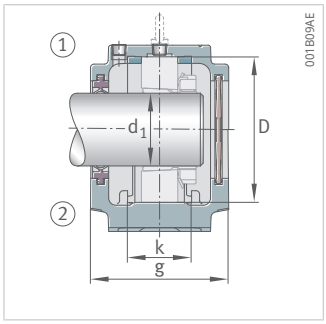


Taconite seal ETC

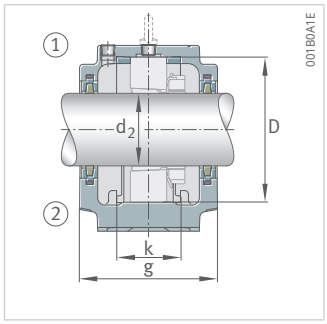
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH519	EFS519	ETS519	161	EDV519	145	ETC519	220
EDH519	EFS519	ETS519	161	EDV519	145	ETC519	220
EDH519	EFS519	ETS519	161	EDV519	145	ETC519	220
EDH519	EFS519	ETS519	161	EDV519	145	ETC519	220
EDH619	EFS619	ETS619	191	EDV619	175	ETC619	235
EDH619	EFS619	ETS619	191	EDV619	175	ETC619	235
EDH619	EFS619	ETS619	191	EDV619	175	ETC619	235
EDH619	EFS619	ETS619	191	EDV619	175	ETC619	235
EDH619	EFS619	ETS619	191	EDV619	175	ETC619	235
EDH519	EFS519	ETS519×306	161	EDV519	145	ETC519×306	220
EDH519	EFS519	ETS519×306	161	EDV519	145	ETC519×306	220
EDH519	EFS519	ETS519×306	161	EDV519	145	ETC519×306	220
EDH519	EFS519	ETS519×306	161	EDV519	145	ETC519×306	220
EDH619	EFS619	ETS619×306	191	EDV619	175	ETC619×306	235
EDH619	EFS619	ETS619×306	191	EDV619	175	ETC619×306	235
EDH619	EFS619	ETS619×306	191	EDV619	175	ETC619×306	235
EDH619	EFS619	ETS619×306	191	EDV619	175	ETC619×306	235
EDH619	EFS619	ETS619×306	191	EDV619	175	ETC619×306	235
EDH520×307	EFS520×307	ETS520×307	176	EDV520×307	160	ETC520×307	230
EDH520×307	EFS520×307	ETS520×307	176	EDV520×307	160	ETC520×307	230
EDH520×307	EFS520×307	ETS520×307	176	EDV520×307	160	ETC520×307	230
EDH520×307	EFS520×307	ETS520×307	176	EDV520×307	160	ETC520×307	230
EDH520×307	EFS520×307	ETS520×307	176	EDV520×307	160	ETC520×307	230
EDH520×307	EFS520×307	ETS520×307	176	EDV520×307	160	ETC520×307	230
EDH620×307	EFS620×307	ETS620×307	199	EDV620×307	185	ETC620×307	240
EDH620×307	EFS620×307	ETS620×307	199	EDV620×307	185	ETC620×307	240
EDH620×307	EFS620×307	ETS620×307	199	EDV620×307	185	ETC620×307	240
EDH620×307	EFS620×307	ETS620×307	199	EDV620×307	185	ETC620×307	240
EDH620×307	EFS620×307	ETS620×307	199	EDV620×307	185	ETC620×307	240
EDH520	EFS520×308	ETS520×308	176	EDV520	160	ETC520×308	230
EDH520	EFS520×308	ETS520×308	176	EDV520	160	ETC520×308	230
EDH520	EFS520×308	ETS520×308	176	EDV520	160	ETC520×308	230
EDH520	EFS520×308	ETS520×308	176	EDV520	160	ETC520×308	230
EDH520	EFS520×308	ETS520×308	176	EDV520	160	ETC520×308	230
EDH520	EFS520×308	ETS520×308	176	EDV520	160	ETC520×308	230
EDH620	EFS620×308	ETS620×308	199	EDV620	185	ETC620×308	240
EDH620	EFS620×308	ETS620×308	199	EDV620	185	ETC620×308	240
EDH620	EFS620×308	ETS620×308	199	EDV620	185	ETC620×308	240
EDH620	EFS620×308	ETS620×308	199	EDV620	185	ETC620×308	240
EDH620	EFS620×308	ETS620×308	199	EDV620	185	ETC620×308	240

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

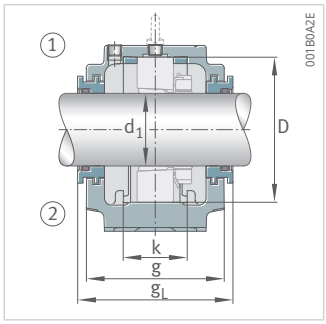


Double lip seal EDH, cover

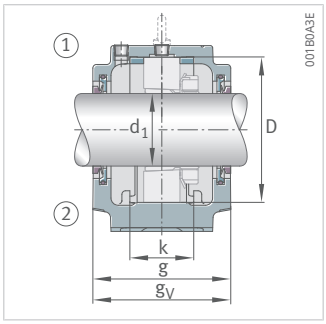


Felt seal EFS

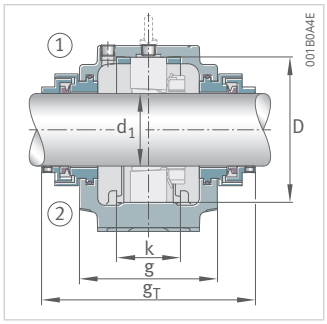
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES520-617-L	**90**	–	1220-K	H220	FRM180/18	EDK520-617
SES520-617-L	**90**	–	20220-K	H220	FRM180/18	EDK520-617
SES520-617-L	**90**	–	2220-K	H320	FRM180/12	EDK520-617
SES520-617-L	**90**	–	22220..-K	H320	FRM180/12	EDK520-617
SES520-617-L	**90**	–	222SM90-TVPA	–	FRM180/12	EDK520-617
SES520-617-L	**90**	–	23220..-K	H2320	FRM180/4,85	EDK520-617
SES520-617-L	**90**	–	WS22220-E1-XL-K-2RSR	H2220-T-WS	FRM180/7,5	EDK520-617
SES524-620-L	**90**	–	1320-K	H320	FRM215/19,5	EDK524-620
SES524-620-L	**90**	–	20320-K	H320	FRM215/19,5	EDK524-620
SES524-620-L	**90**	–	21320..-K	H320	FRM215/19,5	EDK524-620
SES524-620-L	**90**	–	22320..-K	H2320	FRM215/6,5	EDK524-620
SES524-620-L	**90**	–	2320-K	H2320	FRM215/6,5	EDK524-620
SES520-617-L	**92,075**	**3 5/8**	1220-K	H220×310	FRM180/18	EDK520-617
SES520-617-L	**92,075**	**3 5/8**	20220-K	H220×310	FRM180/18	EDK520-617
SES520-617-L	**92,075**	**3 5/8**	2220-K	H320×310	FRM180/12	EDK520-617
SES520-617-L	**92,075**	**3 5/8**	22220..-K	H320×310	FRM180/12	EDK520-617
SES520-617-L	**92,075**	**3 5/8**	23220..-K	H2320×310	FRM180/4,85	EDK520-617
SES524-620-L	**92,075**	**3 5/8**	1320-K	H320×310	FRM215/19,5	EDK524-620
SES524-620-L	**92,075**	**3 5/8**	20320-K	H320×310	FRM215/19,5	EDK524-620
SES524-620-L	**92,075**	**3 5/8**	21320..-K	H320×310	FRM215/19,5	EDK524-620
SES524-620-L	**92,075**	**3 5/8**	22320..-K	H2320×310	FRM215/6,5	EDK524-620
SES524-620-L	**92,075**	**3 5/8**	2320-K	H2320×310	FRM215/6,5	EDK524-620
SES520-617-L	**93,663**	**3 11/16**	1220-K	H220×311	FRM180/18	EDK520-617
SES520-617-L	**93,663**	**3 11/16**	20220-K	H220×311	FRM180/18	EDK520-617
SES520-617-L	**93,663**	**3 11/16**	2220-K	H320×311	FRM180/12	EDK520-617
SES520-617-L	**93,663**	**3 11/16**	22220..-K	H320×311	FRM180/12	EDK520-617
SES520-617-L	**93,663**	**3 11/16**	23220..-K	H2320×311	FRM180/4,85	EDK520-617
SES524-620-L	**93,663**	**3 11/16**	1320-K	H320×311	FRM215/19,5	EDK524-620
SES524-620-L	**93,663**	**3 11/16**	20320-K	H320×311	FRM215/19,5	EDK524-620
SES524-620-L	**93,663**	**3 11/16**	21320..-K	H320×311	FRM215/19,5	EDK524-620
SES524-620-L	**93,663**	**3 11/16**	22320..-K	H2320×311	FRM215/6,5	EDK524-620
SES524-620-L	**93,663**	**3 11/16**	2320-K	H2320×311	FRM215/6,5	EDK524-620
SES522-619-L	**95,25**	**3 3/4**	1222-K	H222×312	FRM200/21	EDK522-619
SES522-619-L	**95,25**	**3 3/4**	20222-K	H222×312	FRM200/21	EDK522-619
SES522-619-L	**95,25**	**3 3/4**	22222-K	H322×312	FRM200/13,5	EDK522-619
SES522-619-L	**95,25**	**3 3/4**	23222-K	H2322×312	FRM200/5,1	EDK522-619
SES622-322-L	**95,25**	**3 3/4**	1322-K	H322×312	FRM240/20	EDK522-619
SES622-322-L	**95,25**	**3 3/4**	21322..-K	H322×312	FRM240/20	EDK522-619
SES622-322-L	**95,25**	**3 3/4**	22322..-K	H2322×312	FRM240/5	EDK522-619
SES622-322-L	**95,25**	**3 3/4**	2322-K	H2322×312	FRM240/5	EDK522-619



Labyrinth seal ETS



V ring seal EDV

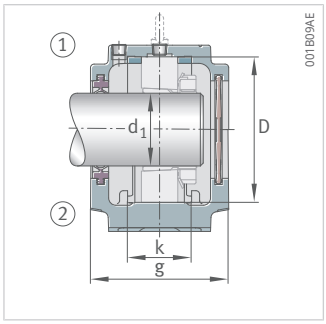


Taconite seal ETC

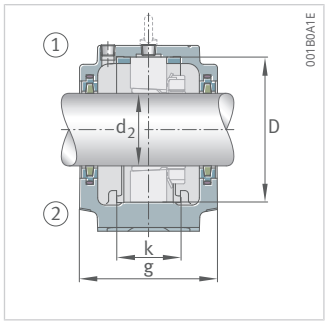
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH520	EFS520	ETS520	176	EDV520	160	ETC520	230
EDH620	EFS620	ETS620	199	EDV620	185	ETC620	240
EDH620	EFS620	ETS620	199	EDV620	185	ETC620	240
EDH620	EFS620	ETS620	199	EDV620	185	ETC620	240
EDH620	EFS620	ETS620	199	EDV620	185	ETC620	240
EDH620	EFS620	ETS620	199	EDV620	185	ETC620	240
EDH520×310	EFS520×310	ETS520×310	176	EDV520×310	160	ETC520×310	230
EDH520×310	EFS520×310	ETS520×310	176	EDV520×310	160	ETC520×310	230
EDH520×310	EFS520×310	ETS520×310	176	EDV520×310	160	ETC520×310	230
EDH520×310	EFS520×310	ETS520×310	176	EDV520×310	160	ETC520×310	230
EDH520×310	EFS520×310	ETS520×310	176	EDV520×310	160	ETC520×310	230
EDH620×310	EFS620×310	ETS620×310	199	EDV620×310	185	ETC620×310	240
EDH620×310	EFS620×310	ETS620×310	199	EDV620×310	185	ETC620×310	240
EDH620×310	EFS620×310	ETS620×310	199	EDV620×310	185	ETC620×310	240
EDH620×310	EFS620×310	ETS620×310	199	EDV620×310	185	ETC620×310	240
EDH620×310	EFS620×310	ETS620×310	199	EDV620×310	185	ETC620×310	240
EDH520×310	EFS520×311	ETS520×311	176	EDV520×311	160	ETC520×311	230
EDH520×310	EFS520×311	ETS520×311	176	EDV520×311	160	ETC520×311	230
EDH520×310	EFS520×311	ETS520×311	176	EDV520×311	160	ETC520×311	230
EDH520×310	EFS520×311	ETS520×311	176	EDV520×311	160	ETC520×311	230
EDH520×310	EFS520×311	ETS520×311	176	EDV520×311	160	ETC520×311	230
EDH620×310	EFS620×311	ETS620×311	199	EDV620×311	185	ETC620×311	240
EDH620×310	EFS620×311	ETS620×311	199	EDV620×311	185	ETC620×311	240
EDH620×310	EFS620×311	ETS620×311	199	EDV620×311	185	ETC620×311	240
EDH620×310	EFS620×311	ETS620×311	199	EDV620×311	185	ETC620×311	240
EDH620×310	EFS620×311	ETS620×311	199	EDV620×311	185	ETC620×311	240
EDH522×312	EFS522×312	ETS522×312	191	EDV522×312	175	ETC522×312	250
EDH522×312	EFS522×312	ETS522×312	191	EDV522×312	175	ETC522×312	250
EDH522×312	EFS522×312	ETS522×312	191	EDV522×312	175	ETC522×312	250
EDH522×312	EFS522×312	ETS522×312	191	EDV522×312	175	ETC522×312	250
EDH522×312	EFS522×312	ETS522×312	196,5	EDV522×312	180,5	ETC522×312	255,5
EDH522×312	EFS522×312	ETS522×312	196,5	EDV522×312	180,5	ETC522×312	255,5
EDH522×312	EFS522×312	ETS522×312	196,5	EDV522×312	180,5	ETC522×312	255,5
EDH522×312	EFS522×312	ETS522×312	196,5	EDV522×312	180,5	ETC522×312	255,5

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

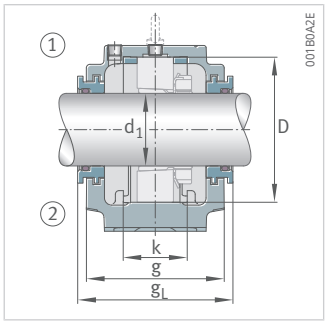


Double lip seal EDH, cover

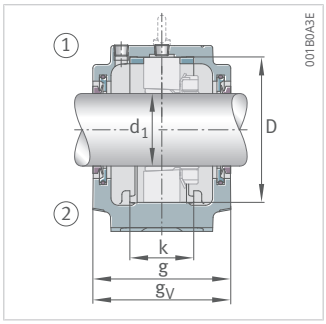


Felt seal EFS

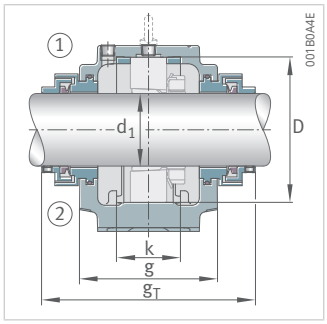
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES522-619-L	**98,425**	**3 7/8**	1222-K	H222×314	FRM200/21	EDK522-619
SES522-619-L	**98,425**	**3 7/8**	20222-K	H222×314	FRM200/21	EDK522-619
SES522-619-L	**98,425**	**3 7/8**	22222-K	H322×314	FRM200/13,5	EDK522-619
SES522-619-L	**98,425**	**3 7/8**	23222-K	H2322×314	FRM200/5,1	EDK522-619
SES622-322-L	**98,425**	**3 7/8**	1322-K	H322×314	FRM240/20	EDK522-619
SES622-322-L	**98,425**	**3 7/8**	21322..-K	H322×314	FRM240/20	EDK522-619
SES622-322-L	**98,425**	**3 7/8**	22322..-K	H2322×314	FRM240/5	EDK522-619
SES622-322-L	**98,425**	**3 7/8**	2322-K	H2322×314	FRM240/5	EDK522-619
SES522-619-L	**100**	**–**	1222-K	H222	FRM200/21	EDK522-619
SES522-619-L	**100**	**–**	20222-K	H222	FRM200/21	EDK522-619
SES522-619-L	**100**	**–**	22222..-K	H322	FRM200/13,5	EDK522-619
SES522-619-L	**100**	**–**	2222-K	H322	FRM200/13,5	EDK522-619
SES522-619-L	**100**	**–**	222SM100-TVPA	–	FRM200/13,5	EDK522-619
SES522-619-L	**100**	**–**	23222..-K	H2322	FRM200/5,1	EDK522-619
SES522-619-L	**100**	**–**	WS22222-E1-XL-K-2RSR	H2222-T-WS	FRM200/8,5	EDK522-619
SES622-322-L	**100**	**–**	1322-K	H322	FRM240/20	EDK522-619
SES622-322-L	**100**	**–**	21322..-K	H322	FRM240/20	EDK522-619
SES622-322-L	**100**	**–**	22322..-K	H2322	FRM240/5	EDK522-619
SES622-322-L	**100**	**–**	2322-K	H2322	FRM240/5	EDK522-619
SES522-619-L	**100,013**	**3 15/16**	1222-K	H222×315	FRM200/21	EDK522-619
SES522-619-L	**100,013**	**3 15/16**	20222-K	H322×315	FRM200/13,5	EDK522-619
SES522-619-L	**100,013**	**3 15/16**	22222..-K	H322×315	FRM200/13,5	EDK522-619
SES522-619-L	**100,013**	**3 15/16**	2222-K	H322×315	FRM200/13,5	EDK522-619
SES522-619-L	**100,013**	**3 15/16**	222S.315	–	FRM200/13,5	EDK522-619
SES522-619-L	**100,013**	**3 15/16**	23222..-K	H2322×315	FRM200/5,1	EDK522-619
SES622-322-L	**100,013**	**3 15/16**	1322-K	H322×315	FRM240/20	EDK522-619
SES622-322-L	**100,013**	**3 15/16**	21322..-K	H322×315	FRM240/20	EDK522-619
SES622-322-L	**100,013**	**3 15/16**	22322..-K	H2322×315	FRM240/5	EDK522-619
SES622-322-L	**100,013**	**3 15/16**	2322-K	H2322×315	FRM240/5	EDK522-619
SES522-619-L	**101,6**	**4**	1222-K	H222×400	FRM200/21	EDK522-619
SES522-619-L	**101,6**	**4**	20222-K	H322×400	FRM200/13,5	EDK522-619
SES522-619-L	**101,6**	**4**	22222..-K	H322×400	FRM200/13,5	EDK522-619
SES522-619-L	**101,6**	**4**	2222-K	H322×400	FRM200/13,5	EDK522-619
SES522-619-L	**101,6**	**4**	222S.400	–	FRM200/13,5	EDK522-619
SES522-619-L	**101,6**	**4**	23222..-K	H2322×400	FRM200/5,1	EDK522-619
SES622-322-L	**101,6**	**4**	1322-K	H322×400	FRM240/20	EDK522-619
SES622-322-L	**101,6**	**4**	21322..-K	H322×400	FRM240/20	EDK522-619
SES622-322-L	**101,6**	**4**	22322..-K	H2322×400	FRM240/5	EDK522-619
SES622-322-L	**101,6**	**4**	2322-K	H2322×400	FRM240/5	EDK522-619
SES524-620-L	**106,363**	**4 3/16**	20224..-K	H3024×403	FRM215/23	EDK524-620



Labyrinth seal ETS



V ring seal EDV

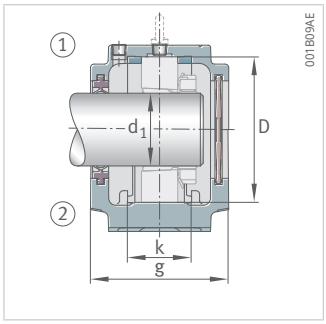


Taconite seal ETC

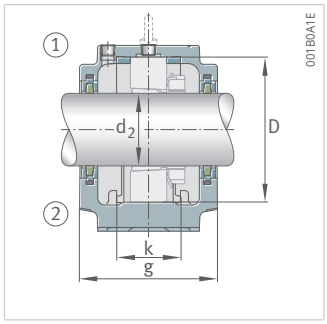
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH522×314	EFS522×314	ETS522×314	191	EDV522×314	175	ETC522×314	250
EDH522×314	EFS522×314	ETS522×314	191	EDV522×314	175	ETC522×314	250
EDH522×314	EFS522×314	ETS522×314	191	EDV522×314	175	ETC522×314	250
EDH522×314	EFS522×314	ETS522×314	191	EDV522×314	175	ETC522×314	250
EDH522×314	EFS522×314	ETS522×314	196,5	EDV522×314	180,5	ETC522×314	255,5
EDH522×314	EFS522×314	ETS522×314	196,5	EDV522×314	180,5	ETC522×314	255,5
EDH522×314	EFS522×314	ETS522×314	196,5	EDV522×314	180,5	ETC522×314	255,5
EDH522×314	EFS522×314	ETS522×314	196,5	EDV522×314	180,5	ETC522×314	255,5
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	191	EDV522-622	175	ETC522-622	250
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522-622	ETS522-622	196,5	EDV522-622	180,5	ETC522-622	255,5
EDH522-622	EFS522×400	ETS522×400	191	EDV522×400	175	ETC522×400	250
EDH522-622	EFS522×400	ETS522×400	191	EDV522×400	175	ETC522×400	250
EDH522-622	EFS522×400	ETS522×400	191	EDV522×400	175	ETC522×400	250
EDH522-622	EFS522×400	ETS522×400	191	EDV522×400	175	ETC522×400	250
EDH522-622	EFS522×400	ETS522×400	191	EDV522×400	175	ETC522×400	250
EDH522-622	EFS522×400	ETS522×400	191	EDV522×400	175	ETC522×400	250
EDH522-622	EFS522×400	ETS522×400	196,5	EDV522×400	180,5	ETC522×400	255,5
EDH522-622	EFS522×400	ETS522×400	196,5	EDV522×400	180,5	ETC522×400	255,5
EDH522-622	EFS522×400	ETS522×400	196,5	EDV522×400	180,5	ETC522×400	255,5
EDH522-622	EFS522×400	ETS522×400	196,5	EDV522×400	180,5	ETC522×400	255,5
EDH524×403	EFS524×403	ETS524×403	199	EDV524×403	185	ETC524×403	260

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

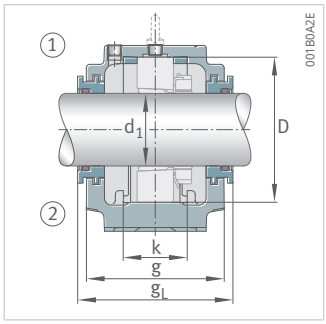


Double lip seal EDH, cover

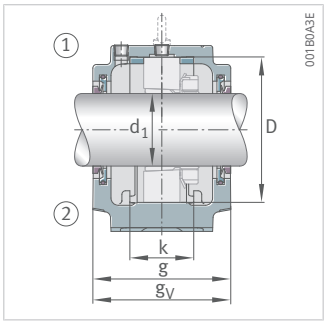


Felt seal EFS

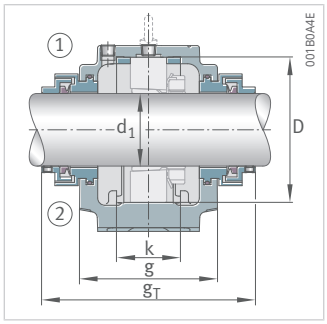
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES524-620-L	**106,363**	**4 3/16**	22224..-K	H3124×403	FRM215/14	EDK524-620
SES524-620-L	**106,363**	**4 3/16**	222S.403	–	FRM215/14	EDK524-620
SES524-620-L	**106,363**	**4 3/16**	23224..-K	H2324×403	FRM215/5	EDK524-620
SES624-324-L	**106,363**	**4 3/16**	20324..-K	H3124×403	FRM260/20,5	EDK524-620
SES624-324-L	**106,363**	**4 3/16**	22324..-K	H2324×403	FRM260/5	EDK524-620
SES524-620-L	**107,95**	**4 1/4**	20224..-K	H3024×404	FRM215/23	EDK524-620
SES524-620-L	**107,95**	**4 1/4**	22224..-K	H3124×404	FRM215/14	EDK524-620
SES524-620-L	**107,95**	**4 1/4**	23224..-K	H2324×404	FRM215/5	EDK524-620
SES624-324-L	**107,95**	**4 1/4**	20324..-K	H3124×404	FRM260/20,5	EDK524-620
SES624-324-L	**107,95**	**4 1/4**	22324..-K	H2324×404	FRM260/5	EDK524-620
SES524-620-L	**110**	**–**	20224..-K	H3024	FRM215/23	EDK524-620
SES524-620-L	**110**	**–**	22224..-K	H3124	FRM215/14	EDK524-620
SES524-620-L	**110**	**–**	222SM110-TVPA	–	FRM215/14	EDK524-620
SES524-620-L	**110**	**–**	23224..-K	H2324	FRM215/5	EDK524-620
SES524-620-L	**110**	**–**	WS22224-E1-XL-K-2RSR	H2224-T-WS	FRM215/8,5	EDK524-620
SES624-324-L	**110**	**–**	20324..-K	H3124	FRM260/20,5	EDK524-620
SES624-324-L	**110**	**–**	22324..-K	H2324	FRM260/5	EDK524-620
SES526-L	**112,713**	**4 7/16**	20226..-K	H3026×407	FRM230/25	EDK526
SES526-L	**112,713**	**4 7/16**	22226..-K	H3126×407	FRM230/13	EDK526
SES526-L	**112,713**	**4 7/16**	222S.407	–	FRM230/13	EDK526
SES526-L	**112,713**	**4 7/16**	23226..-K	H2326×407	FRM230/5	EDK526
SES626-326-L	**112,713**	**4 7/16**	22326..-K	H2326×407	FRM280/5	EDK526
SES526-L	**114,3**	**4 1/2**	20226..-K	H3026×408	FRM230/25	EDK526
SES526-L	**114,3**	**4 1/2**	22226..-K	H3126×408	FRM230/13	EDK526
SES526-L	**114,3**	**4 1/2**	222S.408	–	FRM230/13	EDK526
SES526-L	**114,3**	**4 1/2**	23226..-K	H2326×408	FRM230/5	EDK526
SES626-326-L	**114,3**	**4 1/2**	22326..-K	H2326×408	FRM280/5	EDK526
SES526-L	**115**	**–**	20226..-K	H3026	FRM230/25	EDK526
SES526-L	**115**	**–**	22226..-K	H3126	FRM230/13	EDK526
SES526-L	**115**	**–**	222SM115-TVPA	–	FRM230/25	EDK526
SES526-L	**115**	**–**	23226..-K	H2326	FRM230/5	EDK526
SES526-L	**115**	**–**	WS22226-E1-XL-K-2RSR	H2226-T-WS	FRM230/7,5	EDK526
SES526-L	**120,65**	**4 3/4**	20226..-K	H3026×412	FRM230/25	EDK526
SES526-L	**120,65**	**4 3/4**	22226..-K	H3126×412	FRM230/13	EDK526
SES526-L	**120,65**	**4 3/4**	23226..-K	H2326×412	FRM230/5	EDK526
SES626-326-L	**120,65**	**4 3/4**	22326..-K	H2326×412	FRM280/5	EDK526
SES528-L	**125**	**–**	20228..-K	H3028	FRM250/28	EDK528
SES528-L	**125**	**–**	22228..-K	H3128	FRM250/15	EDK528
SES528-L	**125**	**–**	22228-E1A-XL-K-2RSR-M	H3128	FRM250/15	EDK528
SES528-L	**125**	**–**	222SM125-TVPA	–	FRM250/15	EDK528



Labyrinth seal ETS



V ring seal EDV

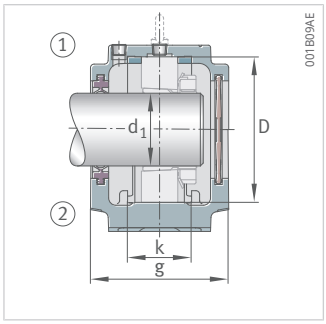


Taconite seal ETC

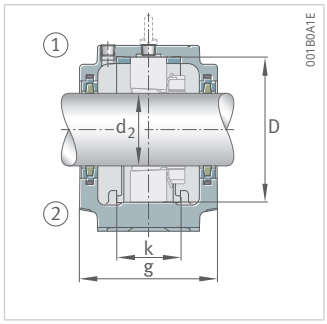
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH524×403	EFS524×403	ETS524×403	199	EDV524×403	185	ETC524×403	260
EDH524×403	EFS524×403	ETS524×403	199	EDV524×403	185	ETC524×403	260
EDH524×403	EFS524×403	ETS524×403	199	EDV524×403	185	ETC524×403	260
EDH524×403	EFS524×403	ETS524×403	201,5	EDV524×403	185	ETC524×403	262,5
EDH524×403	EFS524×403	ETS524×403	201,5	EDV524×403	185	ETC524×403	262,5
EDH524×403	EFS524×404	ETS524×404	199	EDV524×404	185	ETC524×404	260
EDH524×403	EFS524×404	ETS524×404	199	EDV524×404	185	ETC524×404	260
EDH524×403	EFS524×404	ETS524×404	199	EDV524×404	185	ETC524×404	260
EDH524×403	EFS524×404	ETS524×404	201,5	EDV524×404	185	ETC524×404	262,5
EDH524×403	EFS524×404	ETS524×404	201,5	EDV524×404	185	ETC524×404	262,5
EDH524-624	EFS524-624	ETS524-624	199	EDV524-624	185	ETC524-624	260
EDH524-624	EFS524-624	ETS524-624	199	EDV524-624	185	ETC524-624	260
EDH524-624	EFS524-624	ETS524-624	199	EDV524-624	185	ETC524-624	260
EDH524-624	EFS524-624	ETS524-624	199	EDV524-624	185	ETC524-624	260
EDH524-624	EFS524-624	ETS524-624	199	EDV524-624	185	ETC524-624	260
EDH524-624	EFS524-624	ETS524-624	201,5	EDV524-624	187,5	ETC524-624	262,5
EDH524-624	EFS524-624	ETS524-624	201,5	EDV524-624	187,5	ETC524-624	262,5
EDH526×407	EFS526×407	ETS526×407	207	EDV526×407	190	ETC526×407	265
EDH526×407	EFS526×407	ETS526×407	207	EDV526×407	190	ETC526×407	265
EDH526×407	EFS526×407	ETS526×407	207	EDV526×407	190	ETC526×407	265
EDH526×407	EFS526×407	ETS526×407	207	EDV526×407	190	ETC526×407	265
EDH526×407	EFS526×407	ETS526×407	222	EDV526×407	205	ETC526×407	280
EDH526	EFS526	ETS526×408	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526×408	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526×408	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526×408	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526×408	222	EDV526×408	205	ETC526×408	280
EDH526	EFS526	ETS526	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526	207	EDV526	190	ETC526	265
EDH526	EFS526	ETS526	207	EDV526	190	ETC526	265
EDH526×412	EFS526×412	ETS526×412	207	EDV526×412	190	ETC526×412	265
EDH526×412	EFS526×412	ETS526×412	207	EDV526×412	190	ETC526×412	265
EDH526×412	EFS526×412	ETS526×412	207	EDV526×412	190	ETC526×412	265
EDH526×412	EFS526×412	ETS526×412	222	EDV526×412	205	ETC526×412	280
EDH528	EFS528	ETS528	222	EDV528	205	ETC528	285
EDH528	EFS528	ETS528	222	EDV528	205	ETC528	285
EDH528	EFS528	ETS528	222	EDV528	205	ETC528	285
EDH528	EFS528	ETS528	222	EDV528	205	ETC528	285

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

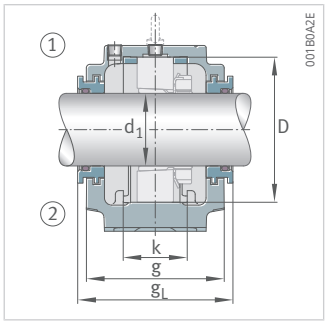


Double lip seal EDH, cover

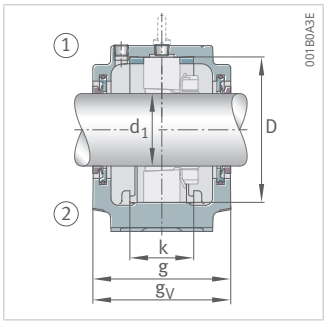


Felt seal EFS

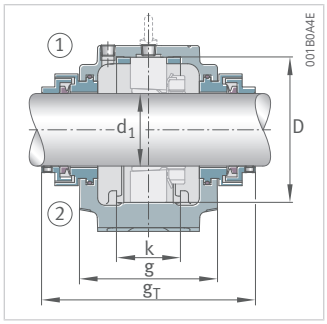
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES528-L	**125**	–	23228..-K	H2328	FRM250/5	EDK528
SES628-328-L	**125**	–	22328..-K	H2328	FRM300/5	EDK528
SES528-L	**125,413**	$4\,^{15}/_{16}$	20228..-K	H3028×415	FRM250/28	EDK528
SES528-L	**125,413**	$4\,^{15}/_{16}$	22228..-K	H3128×415	FRM250/15	EDK528
SES528-L	**125,413**	$4\,^{15}/_{16}$	222S.415	–	FRM250/15	EDK528
SES528-L	**125,413**	$4\,^{15}/_{16}$	23228..-K	H2328×415	FRM250/5	EDK528
SES628-328-L	**125,413**	$4\,^{15}/_{16}$	22328..-K	H2328×415	FRM300/5	EDK528
SES528-L	**127**	**5**	20228..-K	H3028×500	FRM250/28	EDK528
SES528-L	**127**	**5**	22228..-K	H3128×500	FRM250/15	EDK528
SES528-L	**127**	**5**	222S.500	–	FRM250/15	EDK528
SES528-L	**127**	**5**	23228..-K	H2328×500	FRM250/5	EDK528
SES628-328-L	**127**	**5**	22328..-K	H2328×500	FRM300/5	EDK528
SES626-326-L	**130**	–	22326..-K	H2326	FRM280/5	EDK526
SES530-L	**131,763**	$5\,^{3}/_{16}$	20230..-K	H3030×503	FRM270/30,5	EDK530
SES530-L	**131,763**	$5\,^{3}/_{16}$	22230..-K	H3130×503	FRM270/16,5	EDK530
SES530-L	**131,763**	$5\,^{3}/_{16}$	222S.503	–	FRM270/16,5	EDK530
SES530-L	**131,763**	$5\,^{3}/_{16}$	23230..-K	H2330×503	FRM270/5	EDK530
SES630-330-L	**131,763**	$5\,^{3}/_{16}$	22330..-K	H2330×503	FRM320/5	EDK530
SES530-L	**133,35**	$5\,^{1}/_{4}$	20230..-K	H3030×504	FRM270/30,5	EDK530
SES530-L	**133,35**	$5\,^{1}/_{4}$	22230..-K	H3130×504	FRM270/16,5	EDK530
SES530-L	**133,35**	$5\,^{1}/_{4}$	23230..-K	H2330×504	FRM270/5	EDK530
SES630-330-L	**133,35**	$5\,^{1}/_{4}$	22330..-K	H2330×504	FRM320/5	EDK530
SES530-L	**135**	–	20230..-K	H3030	FRM270/30,5	EDK530
SES530-L	**135**	–	22230..-K	H3130	FRM270/16,5	EDK530
SES530-L	**135**	–	222SM135-TVPA	–	FRM270/16,5	EDK530
SES530-L	**135**	–	23230..-K	H2330	FRM270/5	EDK530
SES630-330-L	**135**	–	22330..-K	H2330	FRM320/5	EDK530
SES532-L	**138,113**	$5\,^{7}/_{16}$	20232..-K	H3032×507	FRM290/33	EDK532
SES532-L	**138,113**	$5\,^{7}/_{16}$	22232..-K	H3132×507	FRM290/17	EDK532
SES532-L	**138,113**	$5\,^{7}/_{16}$	222S.507	–	FRM290/17	EDK532
SES532-L	**138,113**	$5\,^{7}/_{16}$	23232..-K	H2332×507	FRM290/5	EDK532
SES632-332-L	**138,113**	$5\,^{7}/_{16}$	22332..-K	H2332×507	FRM340/5	EDK532
SES532-L	**139,7**	$5\,^{1}/_{2}$	20232..-K	H3032×508	FRM290/33	EDK532
SES532-L	**139,7**	$5\,^{1}/_{2}$	22232..-K	H3132×508	FRM290/17	EDK532
SES532-L	**139,7**	$5\,^{1}/_{2}$	222S.508	–	FRM290/17	EDK532
SES532-L	**139,7**	$5\,^{1}/_{2}$	23232..-K	H2332×508	FRM290/5	EDK532
SES632-332-L	**139,7**	$5\,^{1}/_{2}$	22332..-K	H2332×508	FRM340/5	EDK532
SES532-L	**140**	–	20232..-K	H3032(-HG)	FRM290/33	EDK532
SES532-L	**140**	–	22232..-K	H3132(-HG)	FRM290/17	EDK532
SES532-L	**140**	–	22232-E1A-XL-K-2RSR-M	H3132(-HG)	FRM290/17	EDK532



Labyrinth seal ETS



V ring seal EDV

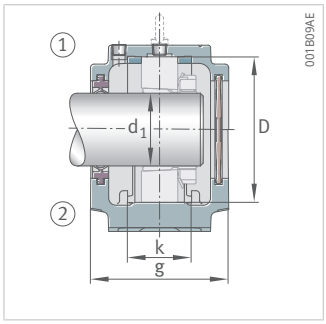


Taconite seal ETC

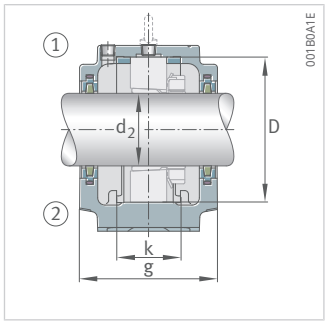
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH528	EFS528	ETS528	222	EDV528	205	ETC528	285
EDH528	EFS528	ETS528	232	EDV528	215	ETC528	295
EDH528	EFS528	ETS528×415	222	EDV528	205	ETC528×415	285
EDH528	EFS528	ETS528×415	222	EDV528	205	ETC528×415	285
EDH528	EFS528	ETS528×415	222	EDV528	205	ETC528×415	285
EDH528	EFS528	ETS528×415	222	EDV528	205	ETC528×415	285
EDH528	EFS528	ETS528×415	232	EDV528×415	215	ETC528×415	295
EDH528	EFS528×500	ETS528×500	222	EDV528×500	205	ETC528×500	285
EDH528	EFS528×500	ETS528×500	222	EDV528×500	205	ETC528×500	285
EDH528	EFS528×500	ETS528×500	222	EDV528×500	205	ETC528×500	285
EDH528	EFS528×500	ETS528×500	222	EDV528×500	205	ETC528×500	285
EDH528×500	EFS528×500	ETS528×500	232	EDV528×500	215	ETC528×500	295
EDH526	EFS526	ETS526	222	EDV526	205	ETC526	280
EDH530×503	EFS530×503	ETS530×503	237	EDV530×503	220	ETC530×503	295
EDH530×503	EFS530×503	ETS530×503	237	EDV530×503	220	ETC530×503	295
EDH530×503	EFS530×503	ETS530×503	237	EDV530×503	220	ETC530×503	295
EDH530×503	EFS530×503	ETS530×503	237	EDV530×503	220	ETC530×503	295
EDH530×503	EFS530×503	ETS530×503	242	EDV530×503	225	ETC530×503	300
EDH530×504	EFS530×504	ETS530×504	237	EDV530×504	220	ETC530×504	295
EDH530×504	EFS530×504	ETS530×504	237	EDV530×504	220	ETC530×504	295
EDH530×504	EFS530×504	ETS530×504	237	EDV530×504	220	ETC530×504	295
EDH530×504	EFS530×504	ETS530×504	242	EDV530×504	225	ETC530×504	300
EDH530	EFS530	ETS530	237	EDV530	220	ETC530	295
EDH530	EFS530	ETS530	237	EDV530	220	ETC530	295
EDH530	EFS530	ETS530	237	EDV530	220	ETC530	295
EDH530	EFS530	ETS530	237	EDV530	220	ETC530	295
EDH530	EFS530	ETS530	242	EDV530	225	ETC530	300
EDH532×507	EFS532×507	ETS532×507	252	EDV532×507	235	ETC532×507	315
EDH532×507	EFS532×507	ETS532×507	252	EDV532×507	235	ETC532×507	315
EDH532×507	EFS532×507	ETS532×507	252	EDV532×507	235	ETC532×507	315
EDH532×507	EFS532×507	ETS532×507	252	EDV532×507	235	ETC532×507	315
EDH532×507	EFS532×507	ETS532×507	258	EDV532×507	235	ETC532×507	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	258	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315

3.11.3 Split plummer block housings SES, accessories, bearings with tapered bore and adapter sleeve

For metric or inch size shaft

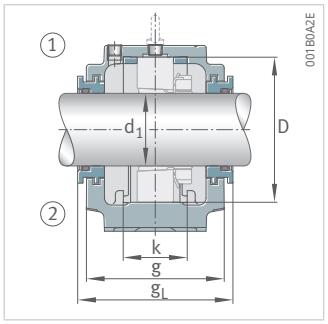


Double lip seal EDH, cover

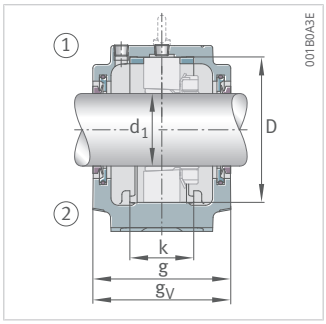


Felt seal EFS

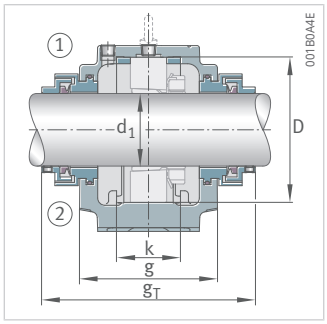
Housing	d_1		Bearing	Adapter sleeve	Locating ring	Cover
	mm	inch	–	–	–	–
SES532-L	**140**	**–**	222SM140-TVPA	–	FRM290/17	EDK532
SES532-L	**140**	**–**	23232..-K	H2332(-HG)	FRM290/5	EDK532
SES632-332-L	**140**	**–**	22332..-K	H2332(-HG)	FRM340/5	EDK532



Labyrinth seal ETS



V ring seal EDV

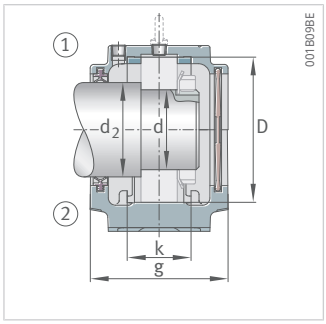


Taconite seal ETC

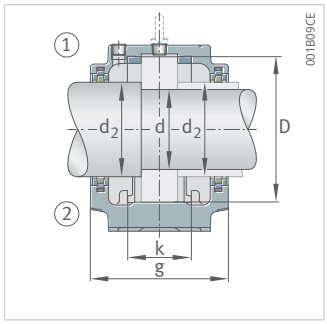
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	252	EDV532	235	ETC532	315
EDH532	EFS532	ETS532	258	EDV532	235	ETC532	315

3.11.4 Split plummer block housings SES, accessories, bearings with cylindrical bore

For metric shaft

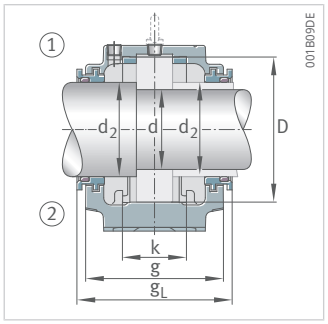


Double lip seal EDH, cover

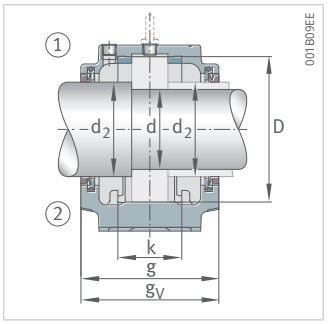


Felt seal EFS

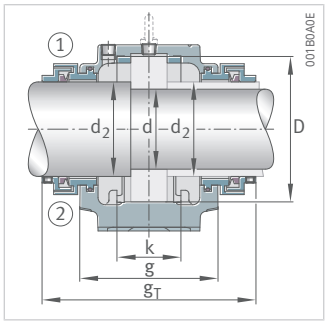
Housing	d	d_2	Bearing	Locating ring	Cover
	mm	mm	–	–	–
SES205-L	**25**	30	1205	FRM52/5	EDK506-605
SES205-L	**25**	30	20205	FRM52/5	EDK506-605
SES205-L	**25**	30	2205	FRM52/3,5	EDK506-605
SES205-L	**25**	30	22205	FRM52/3,5	EDK506-605
SES205-L	**25**	30	6205	FRM52/5	EDK506-605
SES205-L	**25**	30	WS22205-E1-XL-2RSR	FRM52/1	EDK506-605
SES206-305-L	**25**	30	1305	FRM62/7,5	EDK507-606
SES206-305-L	**25**	30	20305	FRM62/7,5	EDK507-606
SES206-305-L	**25**	30	21305	FRM62/7,5	EDK507-606
SES206-305-L	**25**	30	2305	FRM62/4	EDK507-606
SES206-305-L	**25**	30	6305	FRM62/7,5	EDK507-606
SES206-305-L	**30**	35	1206	FRM62/8	EDK507-606
SES206-305-L	**30**	35	20206	FRM62/8	EDK507-606
SES206-305-L	**30**	35	2206	FRM62/6	EDK507-606
SES206-305-L	**30**	35	22206	FRM62/6	EDK507-606
SES206-305-L	**30**	35	6206	FRM62/8	EDK507-606
SES206-305-L	**30**	35	WS22206-E1-XL-2RSR	FRM62/3,5	EDK507-606
SES507-606-L	**30**	35	1306	FRM72/7,5	EDK507-606
SES507-606-L	**30**	35	20306	FRM72/7,5	EDK507-606
SES507-606-L	**30**	35	21306	FRM72/7,5	EDK507-606
SES507-606-L	**30**	35	2306	FRM72/3,5	EDK507-606
SES507-606-L	**30**	35	6306	FRM72/7,5	EDK507-606
SES207-L	**35**	45	1207	FRM72/8,5	EDK509
SES207-L	**35**	45	20207	FRM72/8,5	EDK509
SES207-L	**35**	45	2207	FRM72/5,5	EDK509
SES207-L	**35**	45	22207	FRM72/5,5	EDK509
SES207-L	**35**	45	6207	FRM72/8,5	EDK509
SES207-L	**35**	45	WS22207-E1-XL-2RSR	FRM72/3	EDK509
SES208-307-L	**35**	45	1307	FRM80/9	EDK510-608
SES208-307-L	**35**	45	20307	FRM80/9	EDK510-608
SES208-307-L	**35**	45	21307	FRM80/9	EDK510-608
SES208-307-L	**35**	45	2307	FRM80/4	EDK510-608
SES208-307-L	**35**	45	6307	FRM80/9	EDK510-608
SES208-307-L	**40**	50	1208	FRM80/10,5	EDK510-608
SES208-307-L	**40**	50	20208	FRM80/10,5	EDK510-608
SES208-307-L	**40**	50	2208	FRM80/8	EDK510-608
SES208-307-L	**40**	50	22208	FRM80/8	EDK510-608
SES208-307-L	**40**	50	6208	FRM80/10,5	EDK510-608
SES208-307-L	**40**	50	WS22208-E1-XL-2RSR	FRM80/5,5	EDK510-608
SES510-608-L	**40**	50	1308	FRM90/9	EDK510-608



Labyrinth seal ETS



V ring seal EDV

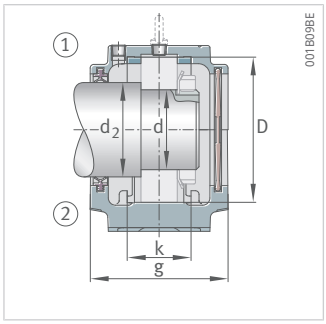


Taconite seal ETC

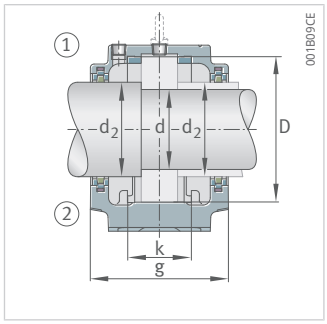
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH205	–	ETS205	90	–	–	ETC205	140
EDH205	–	ETS205	90	–	–	ETC205	140
EDH205	–	ETS205	90	–	–	ETC205	140
EDH205	–	ETS205	90	–	–	ETC205	140
EDH205	–	ETS205	90	–	–	ETC205	140
EDH205	–	ETS205	90	–	–	ETC205	140
EDH507-305	EFS507-305	ETS507-305	90	EDV507-305	77	ETC507-305	140
EDH507-305	EFS507-305	ETS507-305	90	EDV507-305	77	ETC507-305	140
EDH507-305	EFS507-305	ETS507-305	90	EDV507-305	77	ETC507-305	140
EDH507-305	EFS507-305	ETS507-305	90	EDV507-305	77	ETC507-305	140
EDH507-305	EFS507-305	ETS507-305	90	EDV507-305	77	ETC507-305	140
EDH206-306	–	ETS206-306	89	EDV206-306	77	ETC206-306	150
EDH206-306	–	ETS206-306	89	EDV206-306	77	ETC206-306	150
EDH206-306	–	ETS206-306	89	EDV206-306	77	ETC206-306	150
EDH206-306	–	ETS206-306	89	EDV206-306	77	ETC206-306	150
EDH206-306	–	ETS206-306	89	EDV206-306	77	ETC206-306	150
EDH206-306	–	ETS206-306	89	EDV206-306	77	ETC206-306	150
EDH206-306	–	ETS206-306	93	EDV206-306	81	ETC206-306	154
EDH206-306	–	ETS206-306	93	EDV206-306	81	ETC206-306	154
EDH206-306	–	ETS206-306	93	EDV206-306	81	ETC206-306	154
EDH206-306	–	ETS206-306	93	EDV206-306	81	ETC206-306	154
EDH206-306	–	ETS206-306	93	EDV206-306	81	ETC206-306	154
EDH207	–	ETS207	96	EDV207	82	ETC207	160
EDH207	–	ETS207	96	EDV207	82	ETC207	160
EDH207	–	ETS207	96	EDV207	82	ETC207	160
EDH207	–	ETS207	96	EDV207	82	ETC207	160
EDH207	–	ETS207	96	EDV207	82	ETC207	160
EDH207	–	ETS207	96	EDV207	82	ETC207	160
EDH510-307	EFS510-307	ETS510-307	99	EDV510-307	85	ETC510-307	145
EDH510-307	EFS510-307	ETS510-307	99	EDV510-307	85	ETC510-307	145
EDH510-307	EFS510-307	ETS510-307	99	EDV510-307	85	ETC510-307	145
EDH510-307	EFS510-307	ETS510-307	99	EDV510-307	85	ETC510-307	145
EDH510-307	EFS510-307	ETS510-307	99	EDV510-307	85	ETC510-307	145
EDH208-308	EFS208-308	ETS208-308	98	EDV208-308	85	ETC208-308	160
EDH208-308	EFS208-308	ETS208-308	98	EDV208-308	85	ETC208-308	160
EDH208-308	EFS208-308	ETS208-308	98	EDV208-308	85	ETC208-308	160
EDH208-308	EFS208-308	ETS208-308	98	EDV208-308	85	ETC208-308	160
EDH208-308	EFS208-308	ETS208-308	98	EDV208-308	85	ETC208-308	160
EDH208-308	EFS208-308	ETS208-308	98	EDV208-308	85	ETC208-308	160
EDH208-308	EFS208-308	ETS208-308	101	EDV208-308	88	ETC208-308	163

3.11.4 Split plummer block housings SES, accessories, bearings with cylindrical bore

For metric shaft

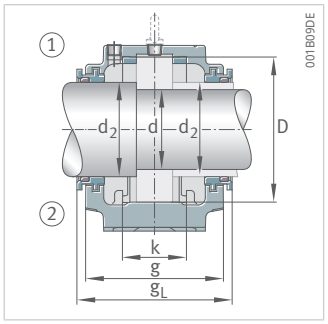


Double lip seal EDH, cover

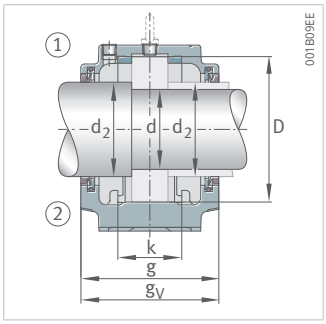


Felt seal EFS

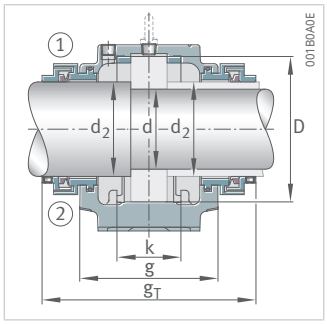
Housing	d	d_2	Bearing	Locating ring	Cover
	mm	mm	–	–	–
SES510-608-L	**40**	50	21308	FRM90/9	EDK510-608
SES510-608-L	**40**	50	22308	FRM90/4	EDK510-608
SES510-608-L	**40**	50	2308	FRM90/4	EDK510-608
SES510-608-L	**40**	50	WS22308-E1-XL-2RSR	FRM90/1,5	EDK510-608
SES209-L	**45**	55	1209	FRM85/5,5	EDK511-609
SES209-L	**45**	55	20209	FRM85/5,5	EDK511-609
SES209-L	**45**	55	2209	FRM85/3,5	EDK511-609
SES209-L	**45**	55	22209	FRM85/3,5	EDK511-609
SES209-L	**45**	55	6209	FRM85/5,5	EDK511-609
SES209-L	**45**	55	WS22209-E1-XL-2RSR	FRM85/1	EDK511-609
SES511-609-L	**45**	55	1309	FRM100/9,5	EDK511-609
SES511-609-L	**45**	55	20309	FRM100/9,5	EDK511-609
SES511-609-L	**45**	55	21309	FRM100/9,5	EDK511-609
SES511-609-L	**45**	55	22309	FRM100/4	EDK511-609
SES511-609-L	**45**	55	2309	FRM100/4	EDK511-609
SES511-609-L	**45**	55	6309	FRM100/9,5	EDK511-609
SES210-L	**50**	60	1210	FRM90/10,5	EDK512-610
SES210-L	**50**	60	20210	FRM90/10,5	EDK512-610
SES210-L	**50**	60	2210	FRM90/9	EDK512-610
SES210-L	**50**	60	22210	FRM90/9	EDK512-610
SES210-L	**50**	60	6210	FRM90/10,5	EDK512-610
SES210-L	**50**	60	WS22210-E1-XL-2RSR	FRM90/6,5	EDK512-610
SES512-610-L	**50**	60	1310	FRM110/10,5	EDK512-610
SES512-610-L	**50**	60	20310	FRM110/10,5	EDK512-610
SES512-610-L	**50**	60	21310	FRM110/10,5	EDK512-610
SES512-610-L	**50**	60	22310	FRM110/4	EDK512-610
SES512-610-L	**50**	60	2310	FRM110/4	EDK512-610
SES512-610-L	**50**	60	6310	FRM110/10,5	EDK512-610
SES211-L	**55**	65	1211	FRM100/11,5	EDK513-611
SES211-L	**55**	65	20211	FRM100/11,5	EDK513-611
SES211-L	**55**	65	2211	FRM100/9,5	EDK513-611
SES211-L	**55**	65	22211	FRM100/9,5	EDK513-611
SES211-L	**55**	65	6211	FRM100/11,5	EDK513-611
SES211-L	**55**	65	WS22211-E1-XL-2RSR	FRM100/6,5	EDK513-611
SES513-611-L	**55**	65	1311	FRM120/11	EDK513-611
SES513-611-L	**55**	65	20311	FRM120/11	EDK513-611
SES513-611-L	**55**	65	21311	FRM120/11	EDK513-611
SES513-611-L	**55**	65	22311	FRM120/4	EDK513-611
SES513-611-L	**55**	65	2311	FRM120/4	EDK513-611
SES513-611-L	**55**	65	6311	FRM120/11	EDK513-611



Labyrinth seal ETS



V ring seal EDV

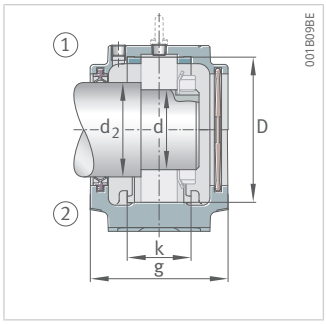


Taconite seal ETC

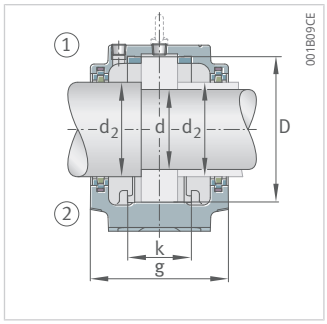
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH208-308	EFS208-308	ETS208-308	101	EDV208-308	88	ETC208-308	163
EDH208-308	EFS208-308	ETS208-308	101	EDV208-308	88	ETC208-308	163
EDH208-308	EFS208-308	ETS208-308	101	EDV208-308	88	ETC208-308	163
EDH208-308	EFS208-308	ETS208-308	101	EDV208-308	88	ETC208-308	163
EDH209-309	EFS209-309	ETS209-309	97	EDV209-309	85	ETC209-309	160
EDH209-309	EFS209-309	ETS209-309	97	EDV209-309	85	ETC209-309	160
EDH209-309	EFS209-309	ETS209-309	97	EDV209-309	85	ETC209-309	160
EDH209-309	EFS209-309	ETS209-309	97	EDV209-309	85	ETC209-309	160
EDH209-309	EFS209-309	ETS209-309	97	EDV209-309	85	ETC209-309	160
EDH209-309	EFS209-309	ETS209-309	97	EDV209-309	85	ETC209-309	160
EDH209-309	EFS209-309	ETS209-309	107	EDV209-309	95	ETC209-309	170
EDH209-309	EFS209-309	ETS209-309	107	EDV209-309	95	ETC209-309	170
EDH209-309	EFS209-309	ETS209-309	107	EDV209-309	95	ETC209-309	170
EDH209-309	EFS209-309	ETS209-309	107	EDV209-309	95	ETC209-309	170
EDH209-309	EFS209-309	ETS209-309	107	EDV209-309	95	ETC209-309	170
EDH209-309	EFS209-309	ETS209-309	107	EDV209-309	95	ETC209-309	170
EDH210-310	EFS210-310	ETS210-310	102	EDV210-310	90	ETC210-310	165
EDH210-310	EFS210-310	ETS210-310	102	EDV210-310	90	ETC210-310	165
EDH210-310	EFS210-310	ETS210-310	102	EDV210-310	90	ETC210-310	165
EDH210-310	EFS210-310	ETS210-310	102	EDV210-310	90	ETC210-310	165
EDH210-310	EFS210-310	ETS210-310	102	EDV210-310	90	ETC210-310	165
EDH210-310	EFS210-310	ETS210-310	102	EDV210-310	90	ETC210-310	165
EDH210-310	EFS210-310	ETS210-310	117	EDV210-310	105	ETC210-310	180
EDH210-310	EFS210-310	ETS210-310	117	EDV210-310	105	ETC210-310	180
EDH210-310	EFS210-310	ETS210-310	117	EDV210-310	105	ETC210-310	180
EDH210-310	EFS210-310	ETS210-310	117	EDV210-310	105	ETC210-310	180
EDH210-310	EFS210-310	ETS210-310	117	EDV210-310	105	ETC210-310	180
EDH210-310	EFS210-310	ETS210-310	117	EDV210-310	105	ETC210-310	180
EDH211-311	EFS211-311	ETS211-311	107	EDV211-311	95	ETC211-311	170
EDH211-311	EFS211-311	ETS211-311	107	EDV211-311	95	ETC211-311	170
EDH211-311	EFS211-311	ETS211-311	107	EDV211-311	95	ETC211-311	170
EDH211-311	EFS211-311	ETS211-311	107	EDV211-311	95	ETC211-311	170
EDH211-311	EFS211-311	ETS211-311	107	EDV211-311	95	ETC211-311	170
EDH211-311	EFS211-311	ETS211-311	107	EDV211-311	95	ETC211-311	170
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185

3.11.4 Split plummer block housings SES, accessories, bearings with cylindrical bore

For metric shaft

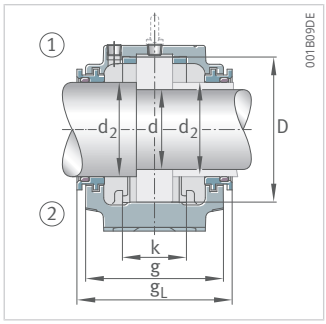


Double lip seal EDH, cover

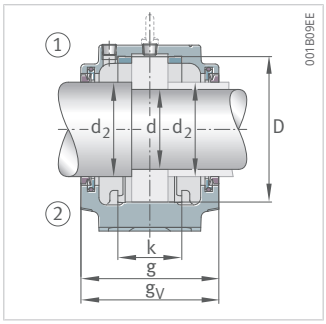


Felt seal EFS

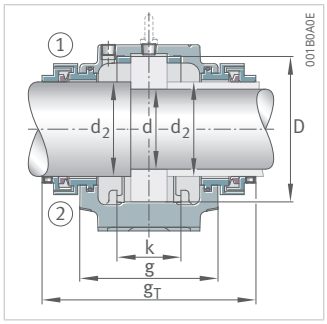
Housing	d	d_2	Bearing	Locating ring	Cover
	mm	mm	–	–	–
SES513-611-L	**55**	65	WS22311-E1-XL-2RSR	FRM120/1	EDK513-611
SES212-L	**60**	70	1212	FRM110/13	EDK515-612
SES212-L	**60**	70	20212	FRM110/13	EDK515-612
SES212-L	**60**	70	2212	FRM110/10	EDK515-612
SES212-L	**60**	70	22212	FRM110/10	EDK515-612
SES212-L	**60**	70	6212	FRM110/13	EDK515-612
SES212-L	**60**	70	WS22212-E1-XL-2RSR	FRM110/7	EDK515-612
SES515-612-L	**60**	70	1312	FRM130/12,5	EDK515-612
SES515-612-L	**60**	70	20312	FRM130/12,5	EDK515-612
SES515-612-L	**60**	70	21312	FRM130/12,5	EDK515-612
SES515-612-L	**60**	70	22312	FRM130/5	EDK515-612
SES515-612-L	**60**	70	2312	FRM130/5	EDK515-612
SES515-612-L	**60**	70	6312	FRM130/12,5	EDK515-612
SES515-612-L	**60**	70	WS22312-E1-XL-2RSR	FRM130/1,5	EDK515-612
SES213-L	**65**	75	1213	FRM120/14	EDK516-613
SES213-L	**65**	75	20213	FRM120/14	EDK516-613
SES213-L	**65**	75	2213	FRM120/10	EDK516-613
SES213-L	**65**	75	22213	FRM120/10	EDK516-613
SES213-L	**65**	75	6213	FRM120/14	EDK516-613
SES213-L	**65**	75	WS22213-E1-XL-2RSR	FRM120/6,5	EDK516-613
SES516-613-L	**65**	75	1313	FRM140/12,5	EDK516-613
SES516-613-L	**65**	75	20313	FRM140/12,5	EDK516-613
SES516-613-L	**65**	75	21313	FRM140/12,5	EDK516-613
SES516-613-L	**65**	75	22313	FRM140/5	EDK516-613
SES516-613-L	**65**	75	2313	FRM140/5	EDK516-613
SES516-613-L	**65**	75	6313	FRM140/12,5	EDK516-613
SES214-L	**70**	80	1214	FRM125/7,5	EDK217
SES214-L	**70**	80	20214	FRM125/7,5	EDK217
SES214-L	**70**	80	2214	FRM125/4	EDK217
SES214-L	**70**	80	22214	FRM125/4	EDK217
SES214-L	**70**	80	6214	FRM125/7,5	EDK217
SES214-L	**70**	80	WS22214-E1-XL-2RSR	FRM125/1	EDK217
SES517-L	**70**	80	1314	FRM150/13	EDK517
SES517-L	**70**	80	20314	FRM150/13	EDK517
SES517-L	**70**	80	21314	FRM150/13	EDK517
SES517-L	**70**	80	22314	FRM150/5	EDK517
SES517-L	**70**	80	2314	FRM150/5	EDK517
SES517-L	**70**	80	6314	FRM150/13	EDK517
SES215-L	**75**	85	1215	FRM130/15,5	EDK518-615
SES215-L	**75**	85	20215	FRM130/15,5	EDK518-615



Labyrinth seal ETS



V ring seal EDV

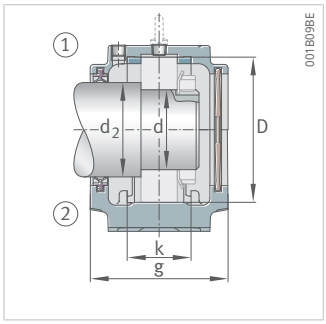


Taconite seal ETC

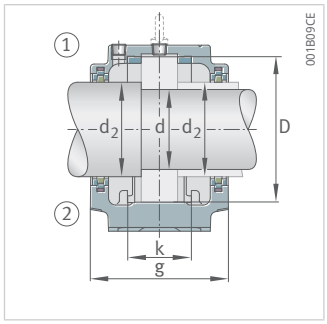
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH211-311	EFS211-311	ETS211-311	122	EDV211-311	110	ETC211-311	185
EDH212-312	–	ETS212-312	117	EDV212-312	105	ETC212-312	185
EDH212-312	–	ETS212-312	117	EDV212-312	105	ETC212-312	185
EDH212-312	–	ETS212-312	117	EDV212-312	105	ETC212-312	185
EDH212-312	–	ETS212-312	117	EDV212-312	105	ETC212-312	185
EDH212-312	–	ETS212-312	117	EDV212-312	105	ETC212-312	185
EDH212-312	–	ETS212-312	117	EDV212-312	105	ETC212-312	185
EDH212-312	–	ETS212-312	127	EDV212-312	115	ETC212-312	195
EDH212-312	–	ETS212-312	127	EDV212-312	115	ETC212-312	195
EDH212-312	–	ETS212-312	127	EDV212-312	115	ETC212-312	195
EDH212-312	–	ETS212-312	127	EDV212-312	115	ETC212-312	195
EDH212-312	–	ETS212-312	127	EDV212-312	115	ETC212-312	195
EDH212-312	–	ETS212-312	127	EDV212-312	115	ETC212-312	195
EDH213-313	–	ETS213-313	124	EDV213-313	110	ETC213-313	190
EDH213-313	–	ETS213-313	124	EDV213-313	110	ETC213-313	190
EDH213-313	–	ETS213-313	124	EDV213-313	110	ETC213-313	190
EDH213-313	–	ETS213-313	124	EDV213-313	110	ETC213-313	190
EDH213-313	–	ETS213-313	124	EDV213-313	110	ETC213-313	190
EDH213-313	–	ETS213-313	124	EDV213-313	110	ETC213-313	190
EDH213-313	–	ETS213-313	134	EDV213-313	120	ETC213-313	200
EDH213-313	–	ETS213-313	134	EDV213-313	120	ETC213-313	200
EDH213-313	–	ETS213-313	134	EDV213-313	120	ETC213-313	200
EDH213-313	–	ETS213-313	134	EDV213-313	120	ETC213-313	200
EDH213-313	–	ETS213-313	134	EDV213-313	120	ETC213-313	200
EDH214	EFS214	ETS214	120,25	EDV214	110	ETC214	143
EDH214	EFS214	ETS214	120,25	EDV214	110	ETC214	143
EDH214	EFS214	ETS214	120,25	EDV214	110	ETC214	143
EDH214	EFS214	ETS214	120,25	EDV214	110	ETC214	143
EDH214	EFS214	ETS214	120,25	EDV214	110	ETC214	143
EDH214	EFS214	ETS214	120,25	EDV214	110	ETC214	143
EDH314	–	ETS314	139	EDV314	125	ETC314	205
EDH314	–	ETS314	139	EDV314	125	ETC314	205
EDH314	–	ETS314	139	EDV314	125	ETC314	205
EDH314	–	ETS314	139	EDV314	125	ETC314	205
EDH314	–	ETS314	139	EDV314	125	ETC314	205
EDH314	–	ETS314	139	EDV314	125	ETC314	205
EDH215-315	EFS215-315	ETS215-315	129	EDV215-315	115	ETC215-315	195
EDH215-315	EFS215-315	ETS215-315	129	EDV215-315	115	ETC215-315	195

3.11.4 Split plummer block housings SES, accessories, bearings with cylindrical bore

For metric shaft

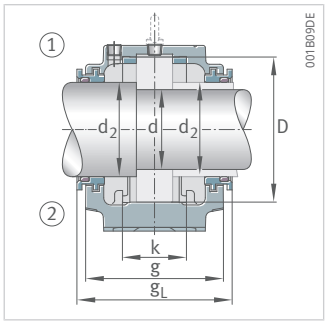


Double lip seal EDH, cover

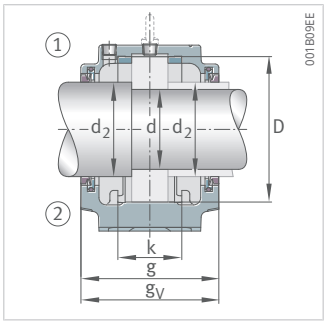


Felt seal EFS

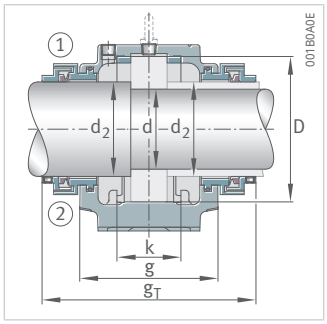
Housing	d	d_2	Bearing	Locating ring	Cover
	mm	mm	–	–	–
SES215-L	**75**	85	2215	FRM130/12,5	EDK518-615
SES215-L	**75**	85	22215	FRM130/12,5	EDK518-615
SES215-L	**75**	85	6215	FRM130/15,5	EDK518-615
SES215-L	**75**	85	WS22215-E1-XL-2RSR	FRM130/9	EDK518-615
SES518-615-L	**75**	85	1315	FRM160/14	EDK518-615
SES518-615-L	**75**	85	20315	FRM160/14	EDK518-615
SES518-615-L	**75**	85	21315	FRM160/14	EDK518-615
SES518-615-L	**75**	85	22315	FRM160/5	EDK518-615
SES518-615-L	**75**	85	2315	FRM160/5	EDK518-615
SES518-615-L	**75**	85	6315	FRM160/14	EDK518-615
SES216-L	**80**	90	1216	FRM140/16	EDK216
SES216-L	**80**	90	20216	FRM140/16	EDK216
SES216-L	**80**	90	2216	FRM140/12,5	EDK216
SES216-L	**80**	90	22216	FRM140/12,5	EDK216
SES216-L	**80**	90	6216	FRM140/16	EDK216
SES216-L	**80**	90	WS22216-E1-XL-2RSR	FRM140/9	EDK216
SES519-616-L	**80**	90	1316	FRM170/14,5	EDK519-616
SES519-616-L	**80**	90	20316	FRM170/14,5	EDK519-616
SES519-616-L	**80**	90	21316	FRM170/14,5	EDK519-616
SES519-616-L	**80**	90	22316	FRM170/5	EDK519-616
SES519-616-L	**80**	90	2316	FRM170/5	EDK519-616
SES519-616-L	**80**	90	6316	FRM170/14,5	EDK519-616
SES217-L	**85**	95	1217	FRM150/16,5	EDK217
SES217-L	**85**	95	20217	FRM150/16,5	EDK217
SES217-L	**85**	95	2217	FRM150/12,5	EDK217
SES217-L	**85**	95	22217	FRM150/12,5	EDK217
SES217-L	**85**	95	6217	FRM150/16,5	EDK217
SES217-L	**85**	95	WS22217-E1-XL-2RSR	FRM150/8,5	EDK217
SES520-617-L	**85**	95	1317	FRM180/14,5	EDK520-617
SES520-617-L	**85**	95	20317	FRM180/14,5	EDK520-617
SES520-617-L	**85**	95	21317	FRM180/14,5	EDK520-617
SES520-617-L	**85**	95	22317	FRM180/5	EDK520-617
SES520-617-L	**85**	95	2317	FRM180/5	EDK520-617
SES520-617-L	**85**	95	6317	FRM180/14,5	EDK520-617
SES218-L	**90**	100	1218	FRM160/17,5	EDK218
SES218-L	**90**	100	20218	FRM160/17,5	EDK218
SES218-L	**90**	100	2218	FRM160/12,5	EDK218
SES218-L	**90**	100	22218	FRM160/12,5	EDK218
SES218-L	**90**	100	23218	FRM160/6,25	EDK218
SES218-L	**90**	100	6218	FRM160/17,5	EDK218



Labyrinth seal ETS



V ring seal EDV

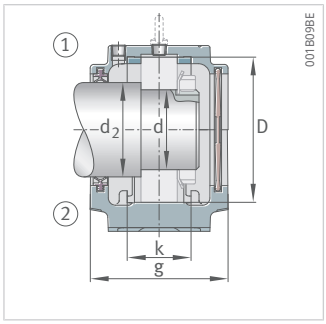


Taconite seal ETC

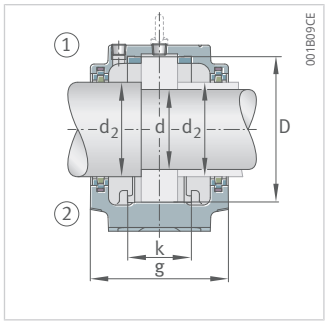
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH215-315	EFS215-315	ETS215-315	129	EDV215-315	115	ETC215-315	195
EDH215-315	EFS215-315	ETS215-315	129	EDV215-315	115	ETC215-315	195
EDH215-315	EFS215-315	ETS215-315	129	EDV215-315	115	ETC215-315	195
EDH215-315	EFS215-315	ETS215-315	129	EDV215-315	115	ETC215-315	195
EDH215-315	EFS215-315	ETS215-315	154	EDV215-315	140	ETC215-315	220
EDH215-315	EFS215-315	ETS215-315	154	EDV215-315	140	ETC215-315	220
EDH215-315	EFS215-315	ETS215-315	154	EDV215-315	140	ETC215-315	220
EDH215-315	EFS215-315	ETS215-315	154	EDV215-315	140	ETC215-315	220
EDH215-315	EFS215-315	ETS215-315	154	EDV215-315	140	ETC215-315	220
EDH215-315	EFS215-315	ETS215-315	154	EDV215-315	140	ETC215-315	220
EDH216	EFS216	ETS216	134	EDV216	120	ETC216	200
EDH216	EFS216	ETS216	134	EDV216	120	ETC216	200
EDH216	EFS216	ETS216	134	EDV216	120	ETC216	200
EDH216	EFS216	ETS216	134	EDV216	120	ETC216	200
EDH216	EFS216	ETS216	134	EDV216	120	ETC216	200
EDH216	EFS216	ETS216	134	EDV216	120	ETC216	200
EDH316	EFS316	ETS316	161	EDV316	145	ETC316	218
EDH316	EFS316	ETS316	161	EDV316	145	ETC316	218
EDH316	EFS316	ETS316	161	EDV316	145	ETC316	218
EDH316	EFS316	ETS316	161	EDV316	145	ETC316	218
EDH316	EFS316	ETS316	161	EDV316	145	ETC316	218
EDH316	EFS316	ETS316	161	EDV316	145	ETC316	218
EDH217	–	ETS217	139	EDV217	123	ETC217	205
EDH217	–	ETS217	139	EDV217	123	ETC217	205
EDH217	–	ETS217	139	EDV217	123	ETC217	205
EDH217	–	ETS217	139	EDV217	123	ETC217	205
EDH217	–	ETS217	139	EDV217	123	ETC217	205
EDH217	–	ETS217	139	EDV217	123	ETC217	205
EDH317	EFS317	ETS317	176	EDV317	160	ETC317	238
EDH317	EFS317	ETS317	176	EDV317	160	ETC317	238
EDH317	EFS317	ETS317	176	EDV317	160	ETC317	238
EDH317	EFS317	ETS317	176	EDV317	160	ETC317	238
EDH317	EFS317	ETS317	176	EDV317	160	ETC317	238
EDH317	EFS317	ETS317	176	EDV317	160	ETC317	238
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220

3.11.4 Split plummer block housings SES, accessories, bearings with cylindrical bore

For metric shaft

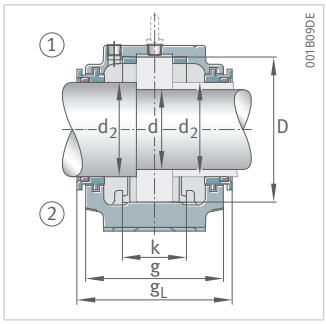


Double lip seal EDH, cover

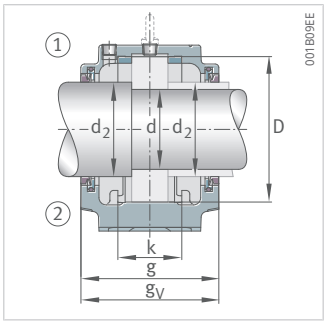


Felt seal EFS

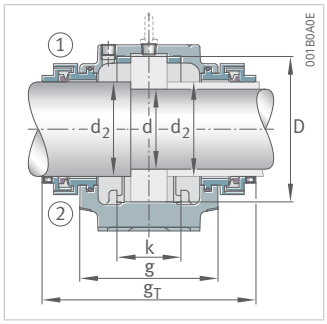
Housing	d	d_2	Bearing	Locating ring	Cover
	mm	mm	–	–	–
SES218-L	**90**	100	WS22218-E1-XL-2RSR	FRM160/8,5	EDK218
SES618-318-L	**90**	105	1318	FRM190/15,5	EDK218
SES618-318-L	**90**	105	20318	FRM190/15,5	EDK218
SES618-318-L	**90**	105	21318	FRM190/15,5	EDK218
SES618-318-L	**90**	105	22318	FRM190/5	EDK218
SES618-318-L	**90**	105	2318	FRM190/5	EDK218
SES618-318-L	**90**	105	6318	FRM190/15,5	EDK218
SES519-616-L	**95**	110	1219	FRM170/18	EDK519-616
SES519-616-L	**95**	110	20219	FRM170/18	EDK519-616
SES519-616-L	**95**	110	2219	FRM170/12,5	EDK519-616
SES519-616-L	**95**	110	22219	FRM170/12,5	EDK519-616
SES519-616-L	**95**	110	6219	FRM170/18	EDK519-616
SES522-619-L	**95**	110	1319	FRM200/17,5	EDK522-619
SES522-619-L	**95**	110	20319	FRM200/17,5	EDK522-619
SES522-619-L	**95**	110	21319	FRM200/17,5	EDK522-619
SES522-619-L	**95**	110	22319	FRM200/6,5	EDK522-619
SES522-619-L	**95**	110	2319	FRM200/6,5	EDK522-619
SES522-619-L	**95**	110	6319	FRM200/17,5	EDK522-619
SES520-617-L	**100**	115	1220	FRM180/18	EDK520-617
SES520-617-L	**100**	115	20220	FRM180/18	EDK520-617
SES520-617-L	**100**	115	2220	FRM180/12	EDK520-617
SES520-617-L	**100**	115	22220	FRM180/12	EDK520-617
SES520-617-L	**100**	115	23220	FRM180/4,85	EDK520-617
SES520-617-L	**100**	115	6220	FRM180/18	EDK520-617
SES520-617-L	**100**	115	WS22220-E1-XL-2RSR	FRM180/7,5	EDK520-617
SES524-620-L	**100**	115	1320	FRM215/19,5	EDK524-620
SES524-620-L	**100**	115	20320	FRM215/19,5	EDK524-620
SES524-620-L	**100**	115	21320	FRM215/19,5	EDK524-620
SES524-620-L	**100**	115	22320	FRM215/6,5	EDK524-620
SES524-620-L	**100**	115	2320	FRM215/6,5	EDK524-620
SES524-620-L	**100**	115	6320	FRM215/19,5	EDK524-620
SES522-619-L	**110**	125	1222	FRM200/21	EDK522-619
SES522-619-L	**110**	125	20222	FRM200/21	EDK522-619
SES522-619-L	**110**	125	2222	FRM200/13,5	EDK522-619
SES522-619-L	**110**	125	22222	FRM200/13,5	EDK522-619
SES522-619-L	**110**	125	23222	FRM200/5,1	EDK522-619
SES522-619-L	**110**	125	6222	FRM200/21	EDK522-619
SES522-619-L	**110**	125	WS22222-E1-XL-2RSR	FRM200/8,5	EDK522-619
SES622-322-L	**110**	125	1322	FRM240/20	EDK522-619
SES622-322-L	**110**	125	20322	FRM240/20	EDK522-619



Labyrinth seal ETS



V ring seal EDV

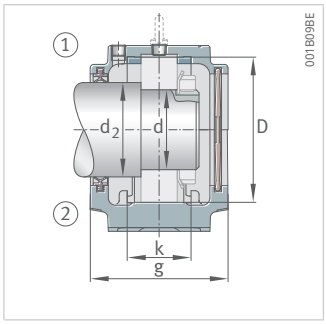


Taconite seal ETC

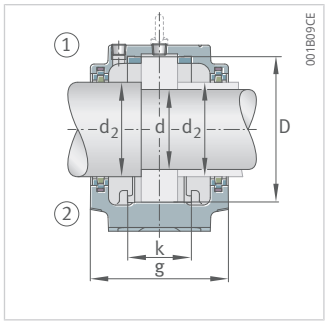
Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH218	EFS218	ETS218	154	EDV218	140	ETC218	220
EDH318	EFS318	ETS318	170,3	EDV318	162,5	–	–
EDH318	EFS318	ETS318	170,3	EDV318	162,5	–	–
EDH318	EFS318	ETS318	170,3	EDV318	162,5	–	–
EDH318	EFS318	ETS318	170,3	EDV318	162,5	–	–
EDH318	EFS318	ETS318	170,3	EDV318	162,5	–	–
EDH318	EFS318	ETS318	170,3	EDV318	162,5	–	–
EDH219	EFS219	ETS219	161	EDV219	158	–	–
EDH219	EFS219	ETS219	161	EDV219	158	–	–
EDH219	EFS219	ETS219	161	EDV219	158	–	–
EDH219	EFS219	ETS219	161	EDV219	158	–	–
EDH219	EFS219	ETS219	161	EDV219	158	–	–
EDH319	EFS319	ETS319	191	EDV319	175	ETC319	253
EDH319	EFS319	ETS319	191	EDV319	175	ETC319	253
EDH319	EFS319	ETS319	191	EDV319	175	ETC319	253
EDH319	EFS319	ETS319	191	EDV319	175	ETC319	253
EDH319	EFS319	ETS319	191	EDV319	175	ETC319	253
EDH319	EFS319	ETS319	191	EDV319	175	ETC319	253
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH220	EFS220	ETS220	176	EDV220	160	ETC220	245
EDH320	EFS320	ETS320	199	EDV320	185	ETC320	260
EDH320	EFS320	ETS320	199	EDV320	185	ETC320	260
EDH320	EFS320	ETS320	199	EDV320	185	ETC320	260
EDH320	EFS320	ETS320	199	EDV320	185	ETC320	260
EDH320	EFS320	ETS320	199	EDV320	185	ETC320	260
EDH320	EFS320	ETS320	199	EDV320	185	ETC320	260
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	191	EDV222-322	175	ETC222-322	255
EDH222-322	EFS222-322	ETS222-322	196,5	EDV222-322	180,5	ETC222-322	260,5
EDH222-322	EFS222-322	ETS222-322	196,5	EDV222-322	180,5	ETC222-322	260,5

3.11.4 Split plummer block housings SES, accessories, bearings with cylindrical bore

For metric shaft

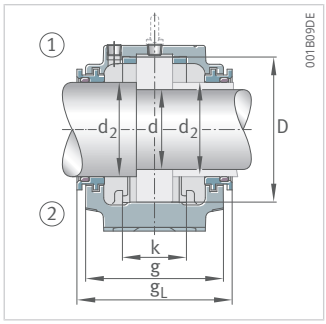


Double lip seal EDH, cover

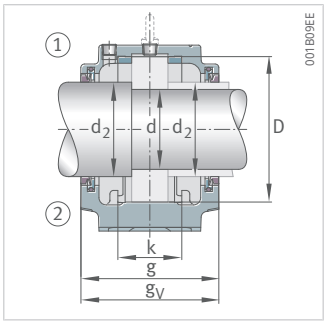


Felt seal EFS

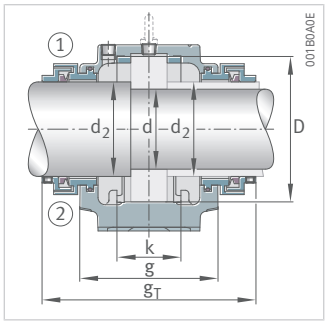
Housing	d	d_2	Bearing	Locating ring	Cover
	mm	mm	–	–	–
SES622-322-L	**110**	125	21322	FRM240/20	EDK522-619
SES622-322-L	**110**	125	22322	FRM240/5	EDK522-619
SES622-322-L	**110**	125	2322	FRM240/5	EDK522-619
SES622-322-L	**110**	125	6322	FRM240/20	EDK522-619
SES626-326-L	**115**	150	22326	FRM280/5	EDK526
SES524-620-L	**120**	135	1224	FRM215/22	EDK524-620
SES524-620-L	**120**	135	20224	FRM215/23	EDK524-620
SES524-620-L	**120**	135	22224	FRM215/14	EDK524-620
SES524-620-L	**120**	135	23224	FRM215/5	EDK524-620
SES524-620-L	**120**	135	6224	FRM215/23	EDK524-620
SES524-620-L	**120**	135	WS22224-E1-XL-2RSR	FRM215/8,5	EDK524-620
SES624-324-L	**120**	135	20324	FRM260/20,5	EDK524-620
SES624-324-L	**120**	135	22324	FRM260/5	EDK524-620
SES624-324-L	**120**	135	6324	FRM260/20,5	EDK524-620
SES526-L	**130**	145	1226	FRM230/22	EDK526
SES526-L	**130**	145	20226	FRM230/25	EDK526
SES526-L	**130**	145	22226	FRM230/13	EDK526
SES526-L	**130**	145	23226	FRM230/5	EDK526
SES526-L	**130**	145	6226	FRM230/25	EDK526
SES526-L	**130**	145	WS22226-E1-XL-2RSR	FRM230/7,5	EDK526
SES626-326-L	**130**	150	20326	FRM280/22,5	EDK526
SES528-L	**140**	155	20228	FRM250/28	EDK528
SES528-L	**140**	155	22228	FRM250/15	EDK528
SES528-L	**140**	155	22228-E1A-XL-2RSR-M	FRM250/15	EDK528
SES528-L	**140**	155	23228	FRM250/5	EDK528
SES528-L	**140**	155	6228	FRM250/28	EDK528
SES628-328-L	**140**	160	20328	FRM300/25	EDK528
SES628-328-L	**140**	160	22328	FRM300/5	EDK528
SES530-L	**150**	165	20230	FRM270/30,5	EDK530
SES530-L	**150**	165	22230	FRM270/16,5	EDK530
SES530-L	**150**	165	23230	FRM270/5	EDK530
SES530-L	**150**	165	6230	FRM270/30,5	EDK530
SES630-330-L	**150**	170	20330	FRM320/26,5	EDK530
SES630-330-L	**150**	170	22330	FRM320/5	EDK530
SES532-L	**160**	175	20232	FRM290/33	EDK532
SES532-L	**160**	175	22232	FRM290/17	EDK532
SES532-L	**160**	175	22232-E1A-XL-2RSR-M	FRM290/17	EDK532
SES532-L	**160**	175	23232	FRM290/5	EDK532
SES532-L	**160**	175	6232	FRM290/33	EDK532
SES632-332-L	**160**	180	22332	FRM340/5	EDK532



Labyrinth seal ETS



V ring seal EDV



Taconite seal ETC

Double lip seal	Felt seal	Labyrinth seal	g_L	V-ring seal	g_V	Taconite seal	g_T
–	–	–	mm	–	mm	–	mm
EDH222-322	EFS222-322	ETS222-322	196,5	EDV222-322	180,5	ETC222-322	260,5
EDH222-322	EFS222-322	ETS222-322	196,5	EDV222-322	180,5	ETC222-322	260,5
EDH222-322	EFS222-322	ETS222-322	196,5	EDV222-322	180,5	ETC222-322	260,5
EDH222-322	EFS222-322	ETS222-322	196,5	EDV222-322	180,5	ETC222-322	260,5
EDH326	EFS326	ETS326	222	EDV326	205	-	–
EDH224-324	EFS224-324	ETS224-324	199	EDV224-324	185	ETC224-324	270
EDH224-324	EFS224-324	ETS224-324	199	EDV224-324	185	ETC224-324	270
EDH224-324	EFS224-324	ETS224-324	199	EDV224-324	185	ETC224-324	270
EDH224-324	EFS224-324	ETS224-324	199	EDV224-324	185	ETC224-324	270
EDH224-324	EFS224-324	ETS224-324	199	EDV224-324	185	ETC224-324	270
EDH224-324	EFS224-324	ETS224-324	199	EDV224-324	185	ETC224-324	270
EDH224-324	EFS224-324	ETS224-324	201,5	EDV224-324	187,5	ETC224-324	272,5
EDH224-324	EFS224-324	ETS224-324	201,5	EDV224-324	187,5	ETC224-324	272,5
EDH224-324	EFS224-324	ETS224-324	201,5	EDV224-324	187,5	ETC224-324	272,5
EDH226	EFS226	ETS226	207	EDV226	190	ETC226	275
EDH226	EFS226	ETS226	207	EDV226	190	ETC226	275
EDH226	EFS226	ETS226	207	EDV226	190	ETC226	275
EDH226	EFS226	ETS226	207	EDV226	190	ETC226	275
EDH226	EFS226	ETS226	207	EDV226	190	ETC226	275
EDH226	EFS526	ETS226	207	EDV226	190	ETC226	275
EDH326	EFS326	ETS326	222	EDV326	205	-	–
EDH228	EFS228	ETS228	222	EDV228	205	ETC228	290
EDH228	EFS228	ETS228	222	EDV228	205	ETC228	290
EDH228	EFS228	ETS228	222	EDV228	205	ETC228	290
EDH228	EFS228	ETS228	222	EDV228	205	ETC228	290
EDH228	EFS228	ETS228	222	EDV228	205	ETC228	290
EDH328	EFS328	ETS328	232	EDV328	215	–	–
EDH328	EFS328	ETS328	232	EDV328	215	–	–
EDH230	EFS230	ETS230	237	EDV230	220	ETC230	310
EDH230	EFS230	ETS230	237	EDV230	220	ETC230	310
EDH230	EFS230	ETS230	237	EDV230	220	ETC230	310
EDH230	EFS230	ETS230	237	EDV230	220	ETC230	310
EDH330	EFS330	ETS330	242	EDV330	225	–	–
EDH330	EFS330	ETS330	242	EDV330	225	–	–
EDH232	–	ETS232	252	EDV232	235	ETC232	325
EDH232	–	ETS232	252	EDV232	235	ETC232	325
EDH232	–	ETS232	252	EDV232	235	ETC232	325
EDH232	–	ETS232	252	EDV232	235	ETC232	325
EDH232	–	ETS232	252	EDV232	235	ETC232	325
EDH332	EFS332	ETS332	258	EDV332	235	–	–

4 SAF split plummer block housings

medias | Product catalogue |
SAF split plummer block housings |
https://www.schaeffler.de/std/21FA

Bearing units consist of an SAF split plummer block housing and the appropriate bearing. If bearing units are combined with the appropriate accessories, the bearing units can be adapted to a wide range of applications. The housings are specifically designed for imperial size shaft centre heights.

4.1 Housing design

Bearing housing

44 SAF and FSAF split plummer block housings made of gray cast iron

0019A26F

45 SAFD and FSAFD split plummer block housings made of ductile cast iron

0019A267

If a housing in the SAF..U universal design is ordered, it will be supplied together with the following accessories:

- 2 labyrinth seals, suitable for the standard imperial shaft diameter depending on the housing version
- 1 locating ring or 2 locating rings, suitable for the relevant spherical roller bearing, depending on the housing version
- 1 cover
- Selection of lubrication nipples

The supplied accessories enable the following arrangements to be achieved in any combination:

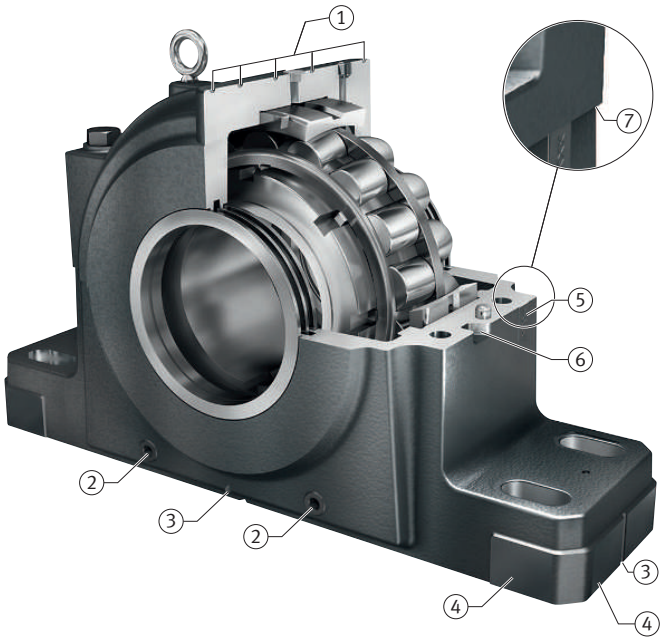
- Assembly in the form of locating or non-locating bearing: Additional insertion of one SR locating ring or several SR locating rings enables the conversion of a non-locating bearing into a locating bearing.
- Continuous shaft with an open housing on both sides or non-continuous shaft with housing closed on one side: Insertion of a EC or ECR cover closes the housing on one side.

The supplied accessories enable various assemblies in any combination ➤116|4.1.3.

4.1.1 Advantages of the detailed design

The new generation of SAF split plummer block housings has a range of design details that are beneficial in terms of assembly, operation and maintenance.

46 Overview of the detailed design



0001A76D

1	Marked connection points for lubricant supply and condition monitoring	2	Grease outlet holes
3	Indentations to mark the housing centre	4	Machined locating surfaces
5	Serial numbering of housing halves	6	Dismantling notches
7	Projection of upper housing section		

4.1.1.1 Precision

The bearing seat in SAF split plummer block housings is machined to G7.

4.1.1.2 Marked connection points for lubricant supply and condition monitoring

Certain positions on the upper part of the housing are ideal for connecting the lubricant supply and for attaching the sensors for condition monitoring. Cast marking points and threaded holes with sealing caps indicate these positions. The marked connection points facilitate the connection of the lubricant supply and the sensors for condition monitoring and ensure correct positioning.

A connection to the following systems is possible for condition monitoring:

- Schaeffler SmartCheck, the modular online measuring system for machine monitoring
- Schaeffler OPTIME
- Other sensors for vibration monitoring

Order the monitoring systems and lubricators separately.

47 Examples of condition monitoring and lubricant supply systems

1	Condition monitoring system Schaeffler OPTIME	2	Lubricator OPTIME C1

Further information

Further information can be found in the following publications:

TPI 214 | Schaeffler SmartCheck |
https://www.schaeffler.de/std/1B6C

TPI 252 | Lubricators |
https://www.schaeffler.de/std/1D4E

TPI 271 | Lubricator | OPTIME C1 |
https://www.schaeffler.de/std/1FC2

SDO | OPTIME

4.1.1.3 Grease outlet holes

A total of 4 grease outlet holes, 2 on each side of the lower housing section, allow used and excess grease to escape during relubrication. During installation, ensure that at least one of the grease outlet holes remains accessible.

4.1.1.4 Indentations to mark the housing centre

Indentations on the end and side surfaces of the lower housing section mark the housing centre. The marks ensure rapid alignment and reduce assembly time.

4.1.1.5 Machined locating surfaces

Machined locating surfaces on the lower housing section allow the precise alignment of the housing.

4.1.1.6 Serial numbering of housing halves

Serial numbers marked on the upper and lower housing sections are correctly ensure that the upper and lower housing sections are correctly paired during assembly. The serial number allows the product to be uniquely identified.

4.1.1.7 Dismounting notches

Recessed dismounting notches on the interface of the housing halves make it easier to remove the upper housing section.

4.1.1.8 Projection of upper housing section

The projection of the upper housing ensures the following:

- It prevents the collection of fluids and contaminants at the interface of the two housing halves
- It reduces the risk of dirt entering the housing
- It reduces the risk of corrosion

4.1.2 Locating and non-locating bearings

The bearing seats in the housing are machined such that the bearings can move so that they can act as non-locating bearings. Locating bearings can be achieved by inserting one or more SR locating rings.

4.1.3 Housing configurations

The modular design of plummer block housings allows a wide range of possible combinations.

Possible configurations with housings of universal design

The following configurations are possible based on the components that are supplied in an SAF..U universal housing package:

- Location of bearings with a tapered bore by means of an adapter sleeve on a shaft of constant diameter or of bearings with a cylindrical bore on a stepped shaft
- For standard imperial shaft diameters: Housing sealed with labyrinth seal
- Assembly in the form of locating or non-locating bearing: Additional insertion of one SR locating ring or several SR locating rings enables the conversion of a non-locating bearing into a locating bearing.
- Continuous shaft with housing open on both sides
- Non-continuous shaft with housing closed on one side by inserting a EC or ECR cover

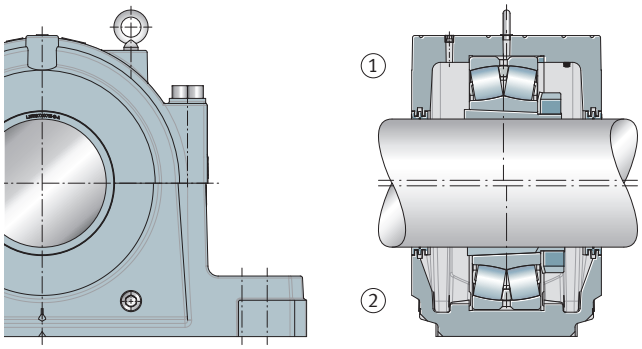
Various housing configurations are possible for a universal housing.

Example 1

Characteristics:

- Spherical roller bearings with a tapered bore and adapter sleeve
- Continuous shaft
- Locating or non-locating bearings

48 SAF500 and SAFD0 for bearings with a tapered bore and adapter sleeve, 2 LER labyrinth seals



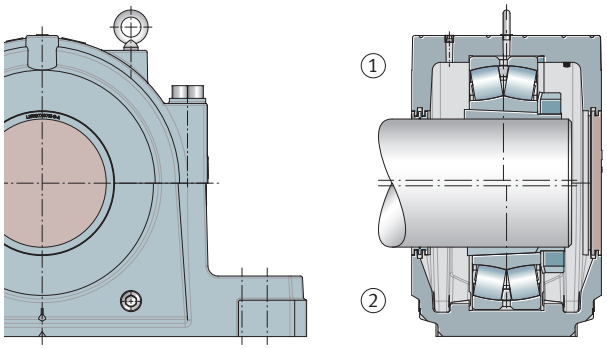
1	Locating bearing	2	Non-locating bearing

Example 2

Characteristics:

- Spherical roller bearings with a tapered bore and adapter sleeve
- Non-continuous shaft
- Locating or non-locating bearings

49 SAF500 and SAFD0for bearings with a tapered bore and adapter sleeve LER labyrinth seal and EC or ECR cover



1	Locating bearing	2	Non-locating bearing

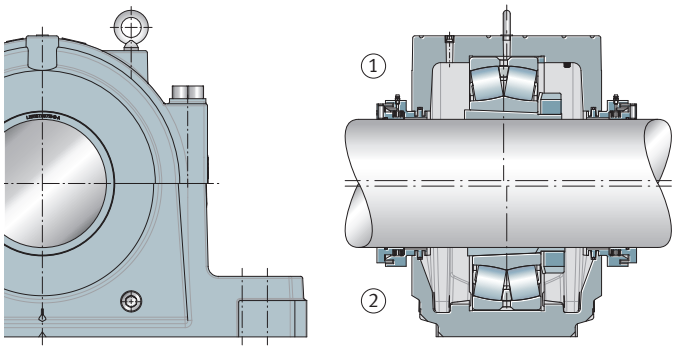
Possible combinations with other accessories

Additional seal versions are available.

- Taconite seals for the following shaft versions:
  - Shafts with a standard shaft diameter
  - Shafts with non-standard diameters
  - Imperial and metric shafts
- Labyrinth seals for the following shaft versions:
  - Shafts with non-standard diameters
  - Imperial and metric shafts

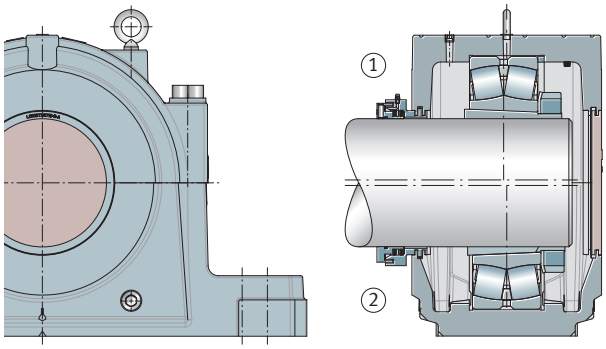
Configurations for SAF500 and SAFD0 plummer block housings

50 SAF500 and SAFD0 plummer block housings for bearings with a tapered bore and adapter sleeve, 2 Taconite seals SuperTAC II TA



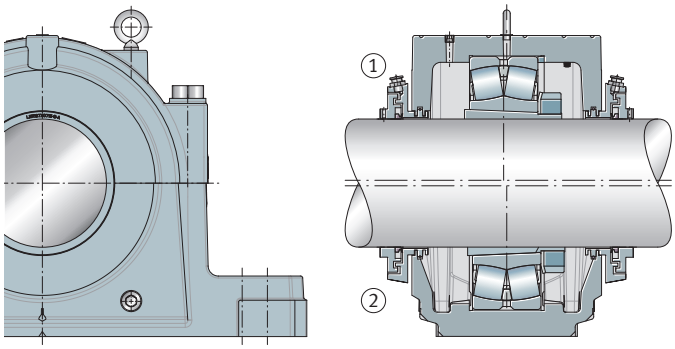
1	Locating bearing	2	Non-locating bearing

51 SAF500 and SAFD0 for bearings with a tapered bore and adapter sleeve, Taconite seal SuperTAC II TA and EC or ECR cover



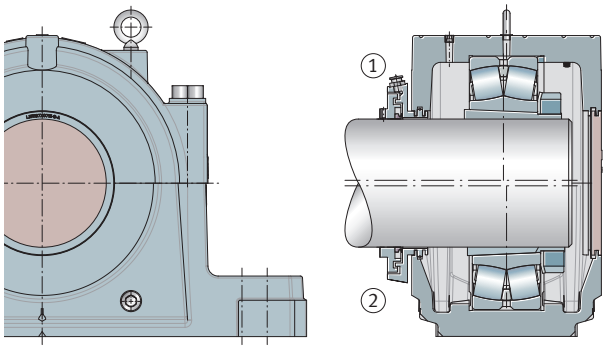
1	Locating bearing	2	Non-locating bearing

52 SAF500 and SAFD0 plummer block housings for bearings with a tapered bore and adapter sleeve, 2 Taconite seals MaxTAC NTC



1	Locating bearing	2	Non-locating bearing

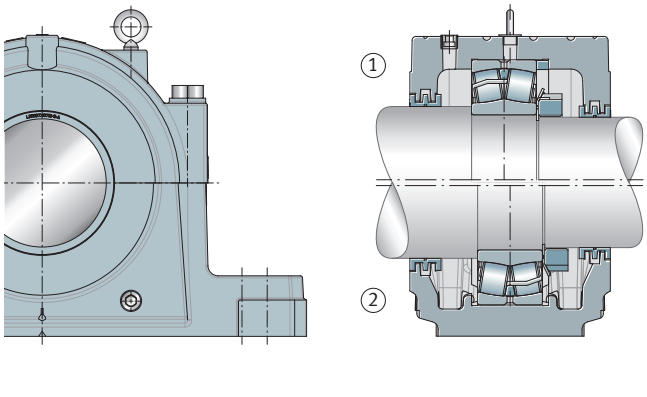
53 SAF500 and SAFD0 for bearings with a tapered bore and adapter sleeve, Taconite seal MaxTAC NTC and EC or ECR cover



1	Locating bearing	2	Non-locating bearing

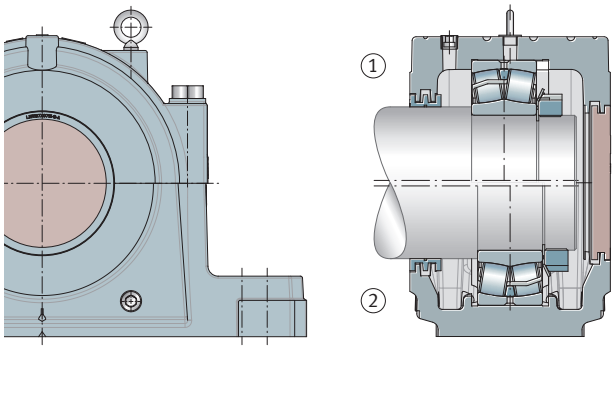
Configurations for SAFD200 plummer block housings

54 SAFD200 for bearings with a cylindrical bore and stepped shaft, 2 LER labyrinth seals



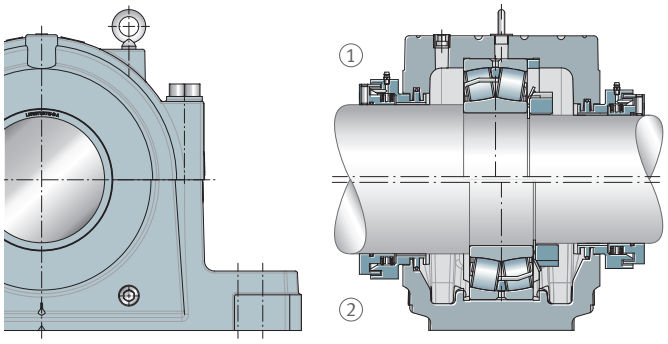
1	Locating bearing	2	Non-locating bearing

55 SAFD200 for bearings with a cylindrical bore and stepped shaft, LER labyrinth seal and EC cover



1	Locating bearing	2	Non-locating bearing

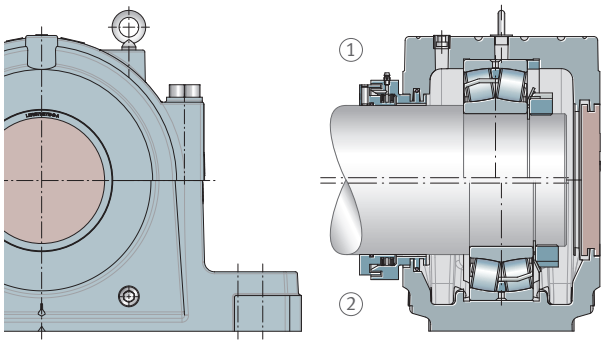
56 SAFD200 for bearings with a cylindrical bore and stepped shaft, 2 Taconite seals SuperTAC II TA



001AC77B

1	Locating bearing	2	Non-locating bearing

57 SAFD200 for bearings with a cylindrical bore and stepped shaft, Taconite seal SuperTAC II TA and EC cover

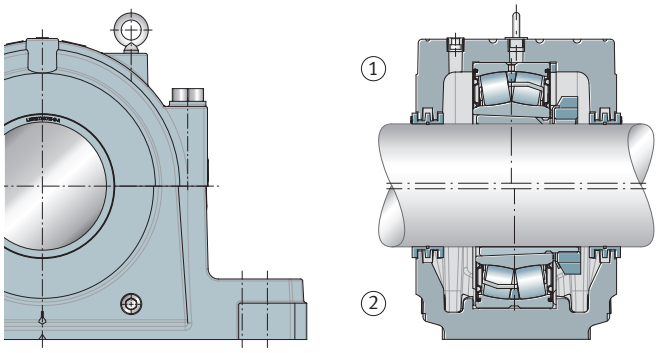


001AC78B

1	Locating bearing	2	Non-locating bearing

Configurations for SAFD500-WS plummer block housings

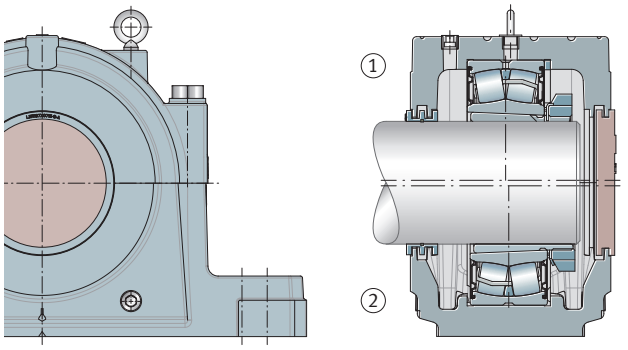
58 SAFD500-WS for bearings with a tapered bore and adapter sleeve, 2 LER labyrinth seals



001A152F

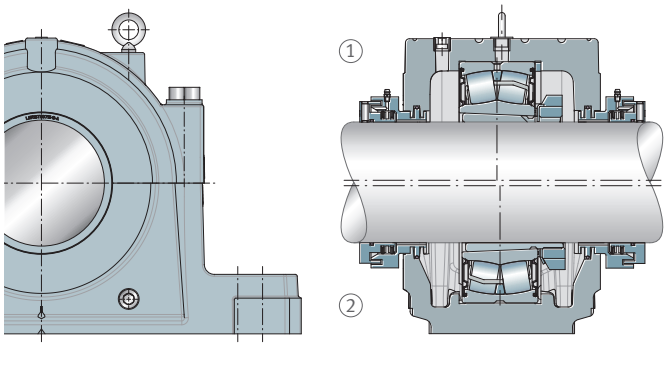
1	Locating bearing	2	Non-locating bearing

59 SAFD500-WS for bearings with a tapered bore and adapter sleeve, LER seal and EC cover



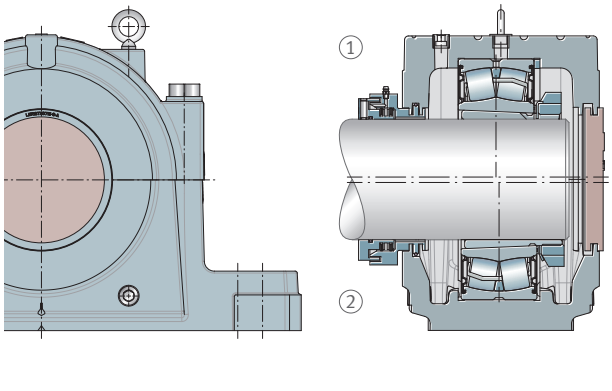
1	Locating bearing	2	Non-locating bearing

60 SAFD500-WS for bearings with a tapered bore and adapter sleeve, 2 Taconite seals SuperTAC II TA



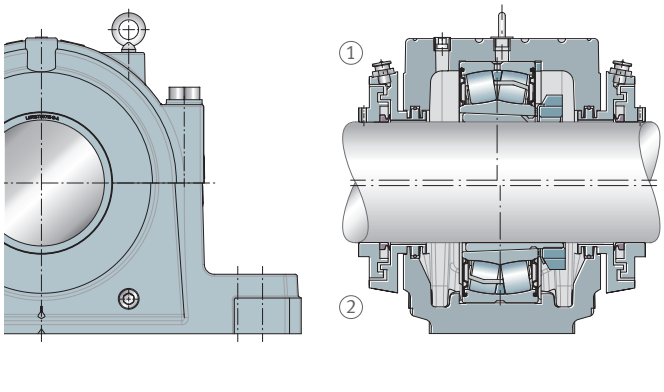
1	Locating bearing	2	Non-locating bearing

61 SAFD500-WS for bearings with a tapered bore and adapter sleeve, Taconite seal SuperTAC II TA and EC or ECR cover



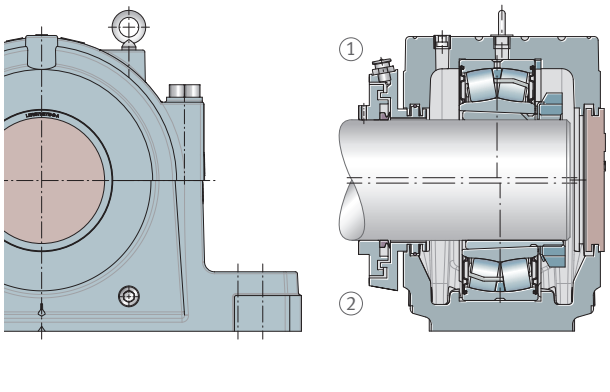
1	Locating bearing	2	Non-locating bearing

62 SAFD500-WS for bearings with a tapered bore and adapter sleeve, 2 Taconite seals MaxTAC NTC



1	Locating bearing	2	Non-locating bearing

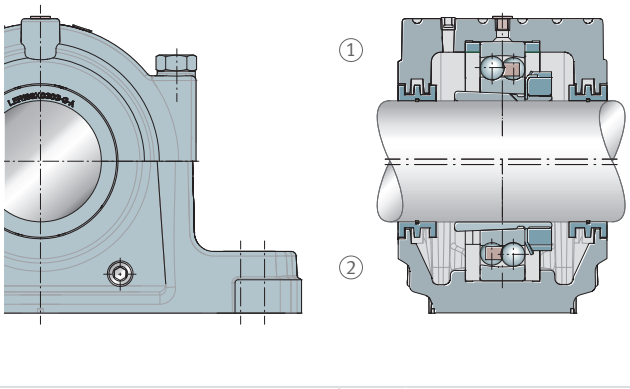
63 SAFD500-WS for bearings with a tapered bore and adapter sleeve, Taconite seal MaxTAC NTC and EC cover



1	Locating bearing	2	Non-locating bearing

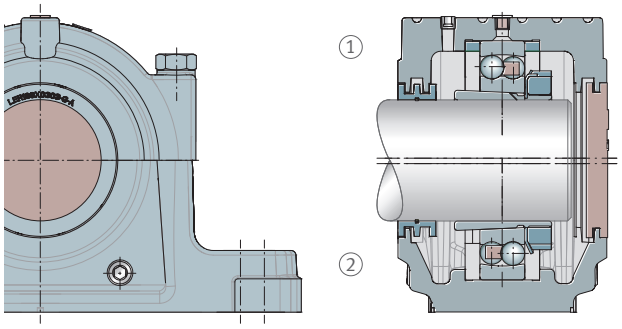
Configurations for SAFD1500 plummer block housings

64 SAFD1500 for bearings with a tapered bore and adapter sleeve, 2 LER labyrinth seals



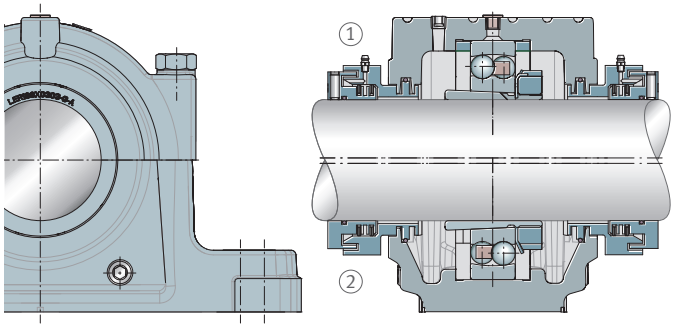
1	Locating bearing	2	Non-locating bearing

65 SAFD1500 for bearings with a tapered bore and adapter sleeve, LER seal and EC cover



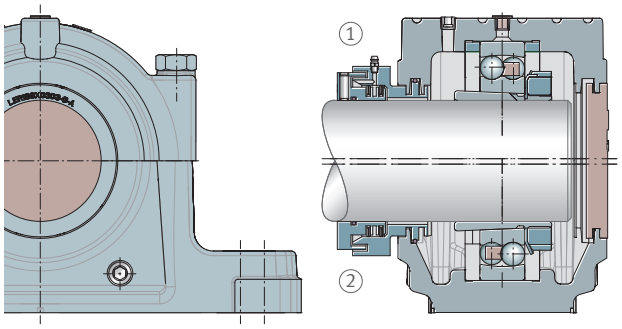
1	Locating bearing	2	Non-locating bearing

66 SAFD1500 for bearings with a tapered bore and adapter sleeve, 2 Taconite seals SuperTAC II TA



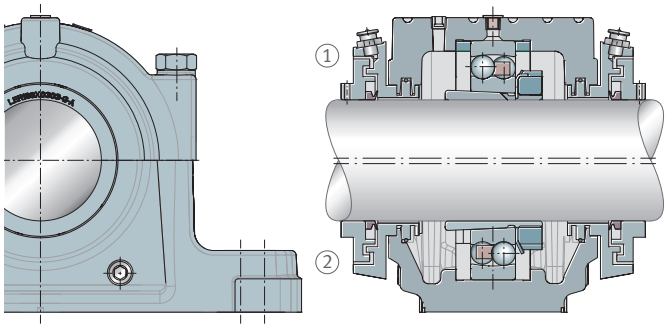
1	Locating bearing	2	Non-locating bearing

67 SAFD1500 for bearings with a tapered bore and adapter sleeve, Taconite seal SuperTAC II TA and EC cover



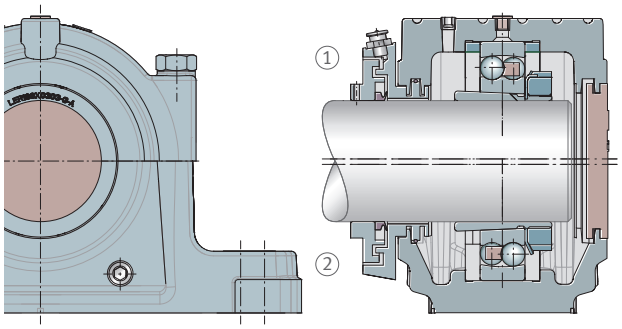
1	Locating bearing	2	Non-locating bearing

68 SAFD1500 for bearings with a tapered bore and adapter sleeve, 2 Taconite seals MaxTAC NTC



1	Locating bearing	2	Non-locating bearing

69 SAFD1500 for bearings with a tapered bore and adapter sleeve, Taconite seal MaxTAC NTC and EC cover

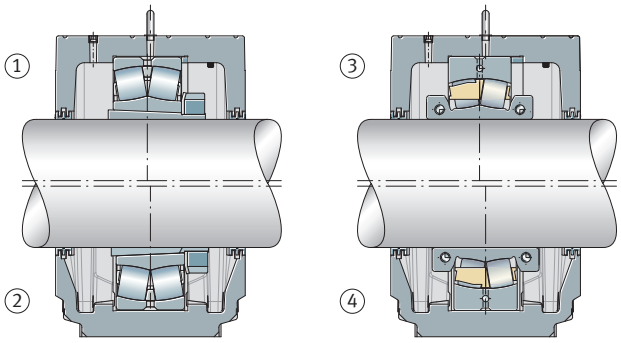


1	Locating bearing	2	Non-locating bearing

Installation of split spherical roller bearing

For SAF plummer block housings, a 222S or 230S split spherical roller bearing can replace an unsplit spherical roller bearing with a tapered bore and adapter sleeve ➤124|70.

70 SAF plummer block housings with split and unsplit spherical roller bearing



1	Locating bearing with unsplit bearing	2	Non-locating bearing with unsplit bearing
3	Locating bearing with split bearing	4	Non-locating bearing with split bearing

4.1.4 Size-dependent housing features

The design configuration of the housings varies in some respects depending on the housing size.

19 Housing sizes and features SAF500

Housing	Oval holes for base fastening screws in the lower housing	Connecting screws in the upper housing	Eye bolts in the upper housing
	Quantity	Quantity	Quantity
SAF509	2 (SAF)	2	–
SAF510	2 (SAF)	2	–
SAF511	2 (SAF)	2	–
SAF513	2 (SAF)	2	–
SAF515	2 (SAF) or 4 (FSAF)	2	–
SAF516	2 (SAF) or 4 (FSAF)	2	–
SAF517	2 (SAF) or 4 (FSAF)	4	–
SAF518	2 (SAF) or 4 (FSAF)	4	–
SAF520	2 (SAF) or 4 (FSAF)	4	–
SAF522	4 (SAF)	4	2
SAF524	4 (SAF)	4	2
SAF526	4 (SAF)	4	2
SAF528	4 (SAF)	4	2
SAF530	4 (SAF)	4	2
SAF532	4 (SAF)	4	2
SAF534	4 (SAF)	4	2
SAF536	4 (SAF)	4	2
SAF538	4 (SAF)	4	2
SAF540	4 (SAF)	4	2
SAF544	4 (SAF)	4	2

20 Housing sizes and features SAFD200

Housing	Oval holes for base fastening screws in the lower housing	Connecting screws in the upper housing	Eye bolts in the upper housing
	Quantity	Quantity	Quantity
SAFD216	2 (SAF) or 4 (FSAF)	2	–
SAFD217	2 (SAF) or 4 (FSAF)	4	–
SAFD218	2 (SAF) or 4 (FSAF)	4	–
SAFD220	2 (SAF) or 4 (FSAF)	4	–
SAFD222	4 (SAF)	4	2
SAFD224	4 (SAF)	4	2
SAFD226	4 (SAF)	4	2
SAFD228	4 (SAF)	4	2
SAFD230	4 (SAF)	4	2
SAFD232	4 (SAF)	4	2
SAFD234	4 (SAF)	4	2
SAFD236	4 (SAF)	4	2
SAFD238	4 (SAF)	4	2
SAFD240	4 (SAF)	4	2
SAFD244	4 (SAF)	4	2

21 Housing sizes and features SAFD500-WS

Housing	Oval holes for base fastening screws in the lower housing	Connecting screws in the upper housing	Eye bolts in the upper housing
	Quantity	Quantity	Quantity
SAFD513-WS	2 (SAF)	2	–
SAFD515-WS	2 (SAF) or 4 (FSAF)	2	–
SAFD516-WS	2 (SAF) or 4 (FSAF)	2	–
SAFD517-WS	2 (SAF) or 4 (FSAF)	4	–
SAFD518-WS	2 (SAF) or 4 (FSAF)	4	–
SAFD520-WS	2 (SAF) or 4 (FSAF)	4	–
SAFD522-WS	4 (SAF)	4	2
SAFD524-WS	4 (SAF)	4	2
SAFD526-WS	4 (SAF)	4	2

22 Housing sizes and features SAFD1500

Housing	Oval holes for base fastening screws in the lower housing	Connecting screws in the upper housing	Eye bolts in the upper housing
	Quantity	Quantity	Quantity
SAFD1509	2 (SAF)	2	–
SAFD1510	2 (SAF)	2	–
SAFD1511	2 (SAF)	2	–
SAFD1513	2 (SAF)	2	–
SAFD1515	2 (SAF) or 4 (FSAF)	4	–
SAFD1516	2 (SAF) or 4 (FSAF)	4	–
SAFD1517	2 (SAF) or 4 (FSAF)	4	–
SAFD1518	2 (SAF) or 4 (FSAF)	4	–
SAFD1520	2 (SAF) or 4 (FSAF)	4	–
SAFD1522	4 (SAF)	4	2

23 Housing sizes and features SAFD0

Housing	Oval holes for base fastening screws in the lower housing	Connecting screws in the upper housing	Eye bolts in the upper housing
	Quantity	Quantity	Quantity
SAFD024	2 (SAF) or 4 (FSAF)	4	–
SAFD026	4 (SAF)	4	2
SAFD028	4 (SAF)	4	2
SAFD030	4 (SAF)	4	2
SAFD032	4 (SAF)	4	2
SAFD034	4 (SAF)	4	2
SAFD036	4 (SAF)	4	2
SAFD038	4 (SAF)	4	2
SAFD040	4 (SAF)	4	2
SAFD044	4 (SAF)	4	2
SAFD048	4 (SAF)	4	2
SAFD052	4 (SAF)	4	2
SAFD056	4 (SAF)	4	2

4.1.5 Accessories

Accessories must be ordered separately.

Seals

71 LER labyrinth seals

0019A1DE

72 Taconite TA (SuperTAC II TA) seals

0019A1FE

73 Taconite NTC (MaxTAC NTC) seals

0019A1EE

Cover

74 EC and ECR covers

0019A1CE

Locating rings

75 SR locating rings

0019A1BE

4.2 Suitable bearings

The dimensions of SAF split plummer block housings are tailored to spherical roller bearings and spherical ball bearings. The shaft diameters are 1 $^{3}/_{8}$ inch to 10 $^{1}/_{2}$ inch.

24 Suitable bearings for SAF split plummer block housings

Bearing type	Housing	Size	
		from	to
Spherical roller bearings with a tapered bore and adapter sleeve	SAF500	22209..-K	22244..-K
	SAFD0	23024..-K	23056..-K
Spherical roller bearings with a cylindrical bore	SAFD200	22216	22244
	SAFD0	22216	22244
Split spherical roller bearings	SAF500	222S	230S
Spherical ball bearings with a tapered bore and adapter sleeve	SAFD1500	1209..-K	1222..-K

Further information

HR 1 | Rolling Bearings |
https://www.schaeffler.de/std/1D3D

TPI 250 | Split spherical roller bearings |
https://www.schaeffler.de/std/1F81

MON 102 | Split plummer block housing | SAF, SDAF |
https://www.schaeffler.de/std/1FDA

4.3 Materials, corrosion protection

25 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron to ASTM A48 Class 35	L	✓	RAL 7016, anthracite grey
Ductile cast iron to ASTM A536 Grade 65-45-12	D	o	RAL 3020, traffic red

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

4.4 Load capacity

Safety factors must be taken into consideration for defining the maximum static load. A safety factor of 5 is normally applied to the housing rupture load in general mechanical engineering.

A safety factor is not required for the values given for the load capacity of the connecting screws.

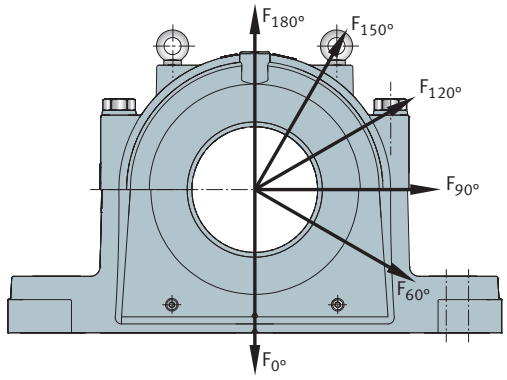
Additional factors influencing load carrying capacity:

- If the housing is subjected to axial load, the permissible axial load of the bearing fitted must be taken into consideration.
- If an adapter sleeve is used to locate the bearing on the shaft, the axial retaining force of the bearing and adapter sleeve must also be taken into consideration.

The housing must be secured by stoppers in the load direction for a load direction between 60 ° and 120 ° or for an axial load.

The specified tightening torques apply to connecting screws to SAE J 429 and ASTM A449 Grade 5.

76 Load directions F



26 Guide values for housing rupture load for gray cast iron housing for spherical roller bearings 222..-K

Housing	Housing rupture load in load direction F						
	0 °	60 °	90 °	120 °	150 °	180 °	Axial
	kN	kN	kN	kN	kN	kN	kN
SAF515	824	277	211	152	140	168	101
FSAF515	824	576	436	216	171	171	101
FSAF516	855	521	402	179	159	189	132
SAF517	956	312	229	168	154	185	132
FSAF517	956	612	395	236	189	189	132
SAF518	1047	342	251	184	169	202	146
FSAF518	1047	694	527	258	207	262	146
SAF520	1561	509	374	274	252	301	176
FSAF520	1561	1034	785	387	307	309	176
SAF522	1636	1086	823	405	322	325	203
SAF524	2090	1386	1052	518	414	414	203
SAF526	2882	1911	1450	714	569	572	247
SAF528	2432	1612	1223	603	480	563	292
SAF532	2818	1868	1419	698	556	558	343
SAF534	2762	1833	1390	685	545	547	426

Housing	Housing rupture load in load direction F						
	0 °	60 °	90 °	120 °	150 °	180 °	Axial
	kN	kN	kN	kN	kN	kN	kN
SAF536	3013	1999	1517	747	596	596	490
SAF538	2931	1944	1475	727	578	580	533
SAF544	4863	3225	2449	1208	961	963	709

27 Guide values for housing rupture load for ductile cast iron housing for spherical roller bearings 222 and 222..-K

Housing		Housing rupture load in load direction F						
		0 °	60 °	90 °	120 °	150 °	180 °	Axial
		kN	kN	kN	kN	kN	kN	kN
SAFD509	–	614	373	238	104	83	136	87
SAFD510	–	728	350	217	132	144	172	87
SAFD511	–	681	380	252	153	167	200	118
SAFD513	–	1256	396	278	169	185	221	154
SAFD515	–	1256	423	322	232	214	256	154
FSAFD515	–	1256	878	665	329	261	261	154
SAFD516	SAFD216	1337	427	325	235	216	258	206
FSAFD516	FSAFD216	1337	815	629	280	249	296	206
SAFD517	SAFD217	1495	488	358	262	241	289	206
FSAFD517	FSAFD217	1495	957	618	369	296	296	206
SAFD518	SAFD218	1654	540	396	290	267	319	230
FSAFD518	FSAFD218	1654	1096	832	407	327	414	230
SAFD520	SAFD220	2511	819	601	441	405	485	283
FSAFD520	FSAFD220	2511	1663	1263	623	494	497	283
SAFD522	SAFD222	2669	1772	1342	660	526	530	332
SAFD524	SAFD224	3410	2260	1716	845	675	675	332
SAFD526	SAFD226	4791	3176	2411	1187	947	950	411
SAFD528	SAFD228	4109	2724	2067	1018	812	951	493
SAFD530	SAFD230	4352	2887	2189	1077	861	861	550
SAFD532	SAFD232	4836	3207	2435	1199	954	958	589
SAFD534	SAFD234	5101	3385	2567	1265	1006	1011	786
SAFD536	SAFD236	5354	3553	2695	1328	1059	1059	871
SAFD538	SAFD238	5251	3482	2641	1303	1036	1040	955
SAFD540	SAFD240	5991	3973	3016	1488	1186	1484	1060
SAFD544	SAFD244	8955	5939	4510	2224	1769	1774	1306

28 Tightening torques and maximum carrying capacity of connecting screws

Housing			Connecting screws	Tightening torque M_A		Maximum load capacity in load direction F [1]			
						90 °	120 °	150 °	180 °
SAF500, FSAF500	SAFD200, FSAFD200	SAFD0, FSAFD0		ft-lbs	Nm	kN	kN	kN	kN
SAF509	–	–	$^{7}/_{16}$ " – 14 UNC	35	45	345	181	147	173
SAF510	–	–	$^{1}/_{2}$ " – 13 UNC	55	70	464	244	197	232
SAF511	–	–	$^{1}/_{2}$ " – 13 UNC	55	70	464	244	197	232
SAF513	SAFD213	–	$^{1}/_{2}$ " – 13 UNC	55	70	464	244	197	232
SAF513	FSAFD213	–	$^{1}/_{2}$ " – 13 UNC	55	70	464	244	197	232
SAF515	SAFD215	–	$^{1}/_{2}$ " – 13 UNC	55	70	464	244	197	232
FSAF515	FSAFD215	–	$^{1}/_{2}$ " – 13 UNC	55	70	464	244	197	232
SAF516	SAFD516	–	$^{5}/_{8}$ " – 11 UNC	105	140	464	244	197	232
FSAF516	FSAFD216	–	$^{5}/_{8}$ " – 11 UNC	105	140	464	244	197	232
SAF517	SAFD517	–	$^{5}/_{8}$ " – 11 UNC	105	140	464	244	197	232
FSAF517	FSAFD217	–	$^{5}/_{8}$ " – 11 UNC	105	140	464	244	197	232
SAF518	SAFD518	–	$^{5}/_{8}$ " – 11 UNC	105	140	725	260	230	240
FSAF518	FSAFD218	–	$^{5}/_{8}$ " – 11 UNC	105	140	725	260	230	240
SAF520	SAFD220	SAFD024	$^{5}/_{8}$ " – 11 UNC	105	140	725	260	230	240
FSAF520	FSAFD220	FSAFD024	$^{5}/_{8}$ " – 11 UNC	105	140	725	260	230	240
SAF522	SAFD222	SAFD026	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF524	SAFD224	SAFD028	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF526	SAFD226	–	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
–	–	SAFD030	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
–	–	SAFD032	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF528	SAFD228	SAFD232	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF530	SAFD230	–	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF532	SAFD232	SAFD036	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
–	–	SAFD038	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF534	SAFD234	SAFD040	$^{3}/_{4}$ " – 10 UNC	185	255	1094	574	465	547
SAF536	SAFD236	–	1 " – 8 UNC	450	610	1983	1041	843	992
SAF538	SAFD238	SAFD044	1 " – 8 UNC	450	610	1983	1041	843	992
SAF540	SAFD240	SAFD048	1 " – 8 UNC	450	610	1983	1041	843	992
SAF544	SAFD244	SAFD052	1 $^{1}/_{4}$ " – 7 UNC	785	1065	2793	1466	1187	1396
–	–	SAFD056	1 $^{1}/_{4}$ " – 7 UNC	785	1065	2793	1466	1187	1396

[1] Maximum load carrying capacity in load direction F for maintaining contact at the parting surface between the upper and lower housing sections

4.5 Lubrication

SAF split plummer block housings are mainly designed for grease lubrication. In certain conditions, oil bath lubrication is possible for LER labyrinth seals. Contact Schaeffler Group Industrial before using oil bath lubrication..

4.5.1 Grease types

In order to achieve a long operating life and high operational security of the bearing arrangement, Schaeffler recommends the use of Arcanol rolling bearing greases. These rolling bearing greases have been designed and tested for bearing arrangement engineering.

4.5.2 Grease quantities

For initial application greasing, the rule of thumb is that the bearing should be filled with grease to 100 % and the empty volume of the housing should be filled to 60 %. The recommended grease quantities are based on this rule of thumb. The free volume is the space that remains in the housing once the bearing, adapter sleeve, shaft and seals have been installed.

29 Recommended grease quantity for bearings at fill level 100 %

Housing	**SAFD200**	**SAF500**	**SAFD0**
Spherical roller bearing	**222**	**222..-K**	**230..-K**
Bore code	**Recommended grease quantity**		
	≈ g		
09	–	20	–
10	–	20	–
11	–	30	–
13	–	45	–
15	–	50	–
16	75	75	–
17	80	80	–
18	105	105	–
20	150	150	–
22	260	260	–
24	265	265	150
26	340	340	190
28	410	410	230
30	525	525	250
32	655	655	270
34	880	880	380
36	890	890	510
38	1070	1070	460
40	1220	1220	670
44	1695	1695	880
48	–	–	610
52	–	–	1240
56	–	–	1500

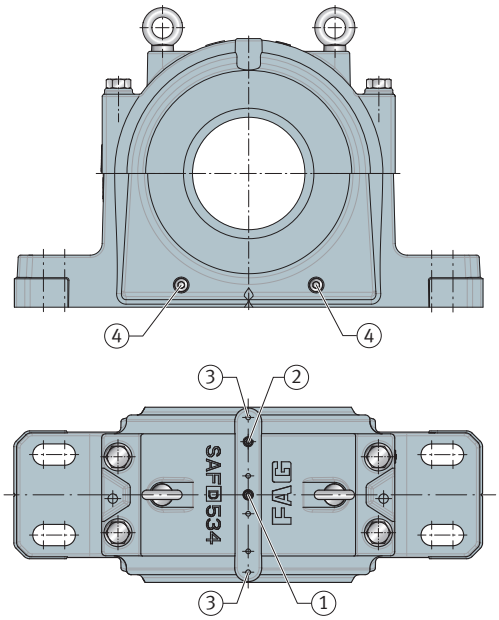
30 Recommended grease quantity for bearings at fill level 60 %

Housing	SAFD200	SAF500, SAFD500-WS	SAFD0
Spherical roller bearing	222	222..-K	230..-K
Bore code	Recommended grease quantity		
	≈ g		
09	–	85	–
10	–	110	–
11	–	140	–
13	–	220	–
15	–	250	–
16	370	370	–
17	375	375	–
18	400	400	–
20	600	600	–
22	800	800	–
24	1130	1130	600
26	1475	1475	800
28	1500	1500	1130
30	1700	1700	1150
32	1930	1930	1150
34	2400	2400	1500
36	2700	2700	1930
38	3390	3390	1930
40	3900	3900	2400
44	5200	5200	3900
48	–	–	3900
52	–	–	5200
56	–	–	6325

4.5.3 Relubrication

Bearings with circumferential lubrication groove

77 Positions for relubrication



1	Central lubrication hole, for bearings with lubrication groove	2	Lateral lubrication hole, for bearings without lubrication groove
3	Points for relubricating labyrinth seals	4	Grease outlet holes, 2 per side

For relubricating bearings with a circumferential lubrication groove, the grease is inserted into the housing through the central lubrication hole. For this type of relubrication, the grease acts on the actual bearing raceway.

There are two grease outlets with screw plugs on each side of the housing in the lower part of the housing.

Bearings without circumferential lubrication groove

For relubricating bearings that do not have a circumferential lubrication groove, the grease is inserted through a lubrication hole located on the side of the upper housing section. Depending on the housing size, this lubrication hole has either a thread of $^1/_8$" – 27 NPT or $^1/_4$" – 18 NPT. A screw plug seals the lubrication hole. Remove the screw plug and permanently replace it with one of the supplied grease nipples.

Additional lubrication holes for labyrinth seals

Lubrication holes for relubricating labyrinth seals can be installed in two other positions.

Supplied lubrication nipples with dust cap:

- Flat lubrication nipples $^1/_8$" – 27 NPT or $^1/_4$" – 18 NPT
- Tapered lubrication nipples $^1/_8$" – 27 NPT or $^1/_4$" – 18 NPT

At least one of the screw plugs in the grease outlet holes in the lower housing sections must be removed for the duration of relubrication process to prevent excessive greasing. Remove the screw plug on the side opposite the grease inlet for perfect relubrication. Removing this screw makes it easier for the fresh grease to pass through the bearing.

Dirt can penetrate into the inside of the housing if the grease outlet holes are opened.

4.6 Sealing

Labyrinth seals and Taconite seals are used to seal the shaft to the bearing housing. EC and ECR covers are used for housings closed on one side. These seals and covers are tailored to the radial labyrinth grooves in each side of the housing.

4.6.1 LER labyrinth seals

LER labyrinth seals enable a non-contact seal, making them suitable for high speeds. The seal is supplied with an O-ring made of NBR. This O-ring seals against the shaft and is suitable for temperatures of up to +100 °C.

Labyrinth seals allow shaft misalignment of up to 0,3 ° in all directions and are suitable for grease lubrication. If necessary, the labyrinth can be relubricated. A lubrication hole must be made in the upper housing section for each labyrinth seal for this purpose. Cast pilot holes in the housing indicate the perfect positions for the lubrication holes.

4.6.2 Taconite SuperTAC II TA seals

Taconite SuperTAC II TA seals consist of 2 rings, of which one is located in the housing and the other on the shaft. An axial and a radial labyrinth are formed between these rings. Both labyrinths are filled with grease. 2 spiral rings reinforce the radial labyrinth. The non-contact sealing makes Taconite seals suitable for high speeds. The two NBR O-rings seal the rings to the body or shaft and are suitable for temperatures of up to +100 °C.

Taconite SuperTAC II TA seals are specially designed for extreme operating conditions which may involve heavy soiling , including with abrasive particles.

Taconite SuperTAC II TA seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication. The seals have a lubricating nipple for relubrication.

Order seals separately.

4.6.3 Taconite MaxTAC NTC seals

Taconite MaxTAC NTC seals are suitable for extreme operating conditions in terms of dirt, dust and very small particles. The seals are combined seals consisting of a labyrinth seal and a V-ring. The V-ring prevents dirt from entering and facilitates grease leakage during relubrication. The V ring is made of NBR and is suitable for temperatures of up to +100 °C.

The MaxTAC NTC allows shaft misalignment of up to 0,3 ° in all directions and is suitable for grease lubrication. The seal has a lubricating nipple for relubrication.

Order seals separately.

4.6.4 EC covers

EC covers are used on housings which are closed on one side. The covers fit into the rectangular annular grooves in the housings. They are suitable for temperatures of up to +100 °C.

1 cover is supplied with each universal housing.

4.6.5 ECR covers

ECR covers are used on housings which are closed on one side. The covers fit into the rectangular annular grooves in the housings. They are suitable for temperatures of up to +200 °C.

1 cover is supplied with each universal housing.

4.6.6 Overview of properties and applications

The following overview compares the properties and applications of the seals and covers.

31 Standard seals and covers for SAF plummer block housings

Design features and suitability		Seals			Cover	
		Labyrinth seal	MaxTAC NTC	SuperTAC II TA		
Designation		LER	NTC	TA	EC	ECR
Material		Aluminium, NBR	Steel with Durotect Z (zinc phosphate coated), NBR	Steel with Durotect BS (burnished), NBR	NBR	Gray cast iron, FKM
Suitability for sealing against						
Dust		o	++	++	++	++
Fine solid particles		+	++	++	++	++
Coarse solid particles		+	++	++	++	++
Splinters		++	++	++	++	++
Spraying liquids		–	++	++	++	++
Application						
Long term temperature	°C	–40 - +100	–40 - +100	–40 - +100	–40 - +100	–40 - +200
	°F	–40 - +210	–40 - +210	–40 - +210	–40 - +210	–40 - +390
Circumferential velocity	m/s	No restriction	≤ 12	No restriction	Not applicable	Not applicable
Misalignment	°	≤ 0,3	≤ 0,3	≤ 0,5	Not applicable	Not applicable
Low friction		++	+	+	Not applicable	Not applicable
Axial shaft displacement (suitability as non-locating bearing)		+	+	+	Not applicable	Not applicable
Vertical arrangement		o	o	o	++	++
Grease relubrication		+	++	++	++	++
Oil lubrication		o	o	o	+	+
Sunlight		++	++	++	++	++
Application						
Tolerance class [1)]		h8 (h9)	h8 (h9)	h8 (h9)	Not applicable	Not applicable
Max. roughness	µm	Ra 3,2	Ra 3,2	Ra 3,2	Not applicable	Not applicable

++ highly suitable
+ suitable
o suitable with restrictions
– not suitable

1) The envelope condition Ⓔ applies here.

4.7 Dimensions, tolerances

4.7.1 Tolerances for the bearing seat

The bearing seat in the housings is machined to the tolerance class G7 to DIN EN ISO 286-1. The tolerance indicated applies to the condition on delivery, in other words before the screws connecting the upper and lower sections are loosened.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

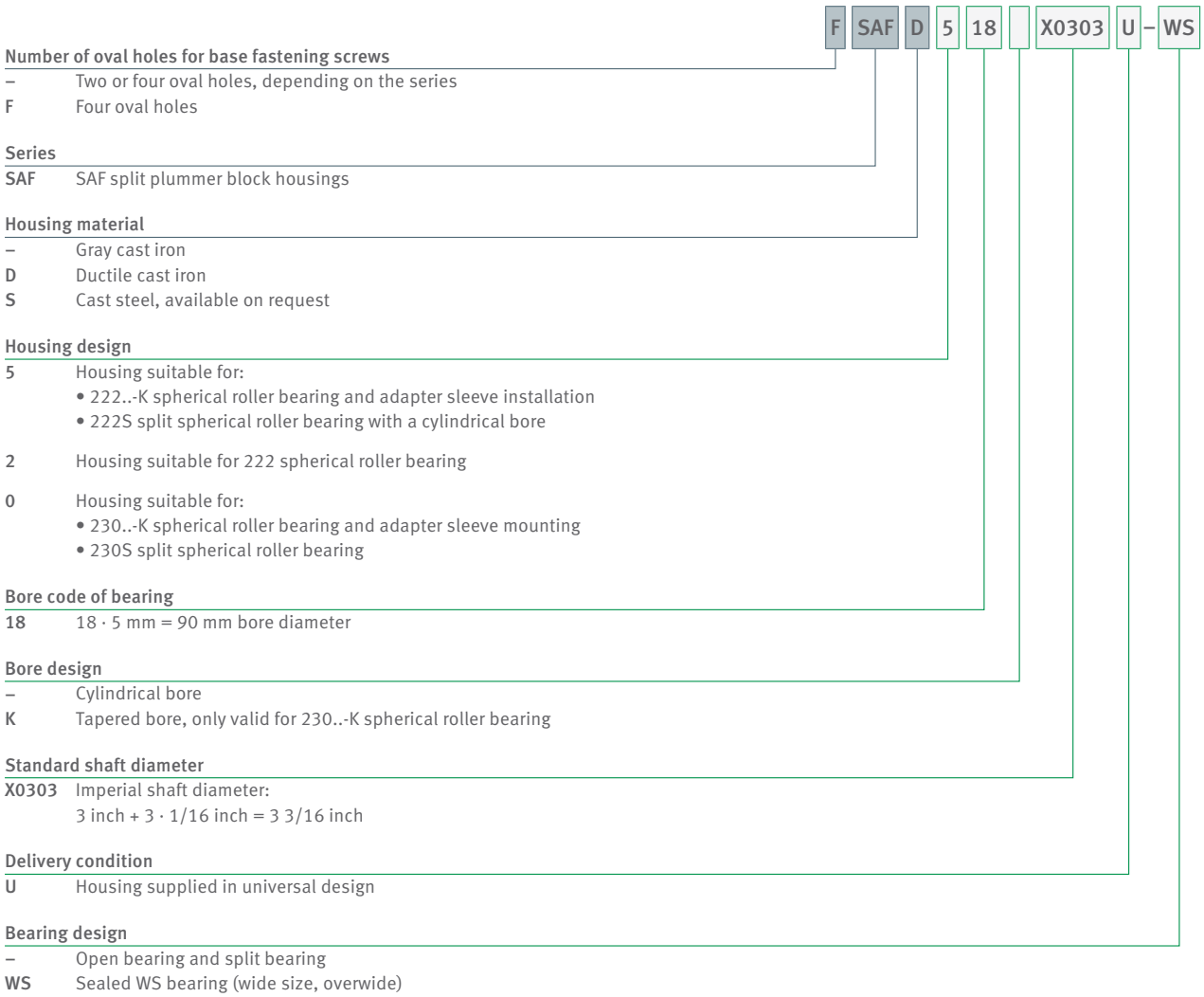
4.8 Structure of the order designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

Bearing housing

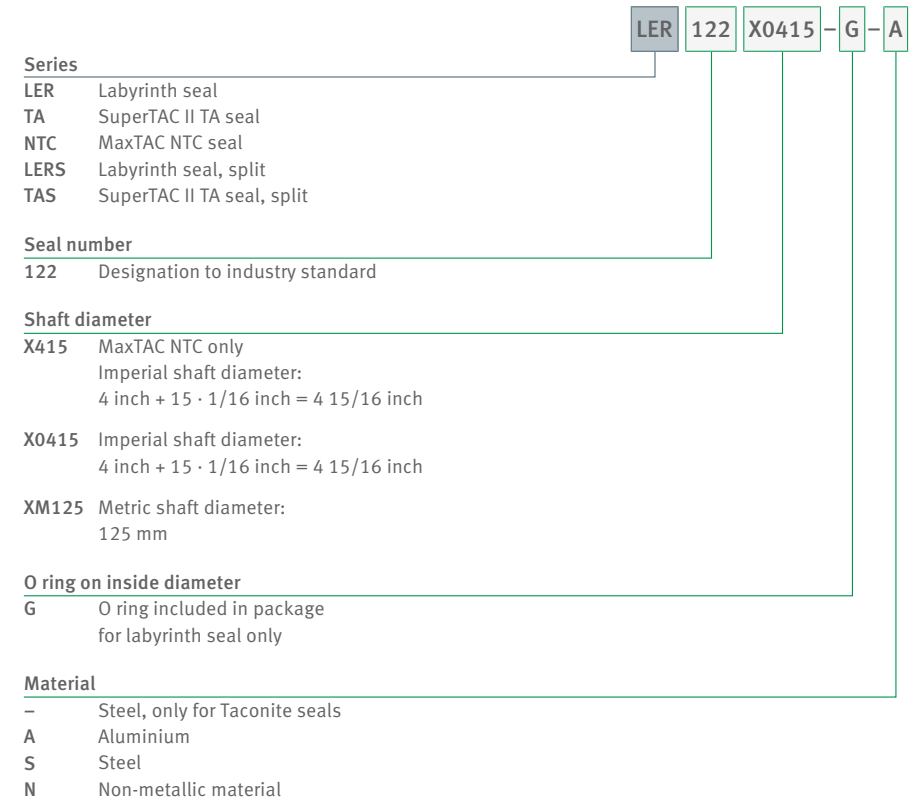
4

78 Structure of the designations of SAF plummer block housings



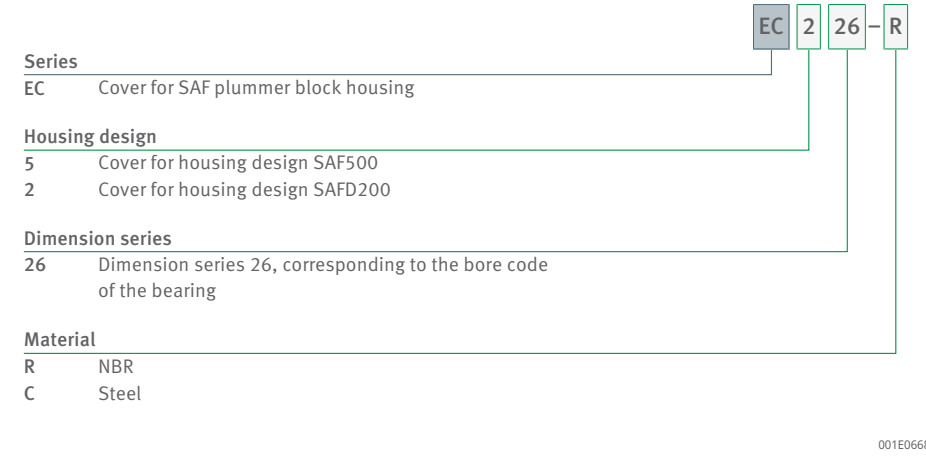
Seals

79 Structure of the designations of standard seals

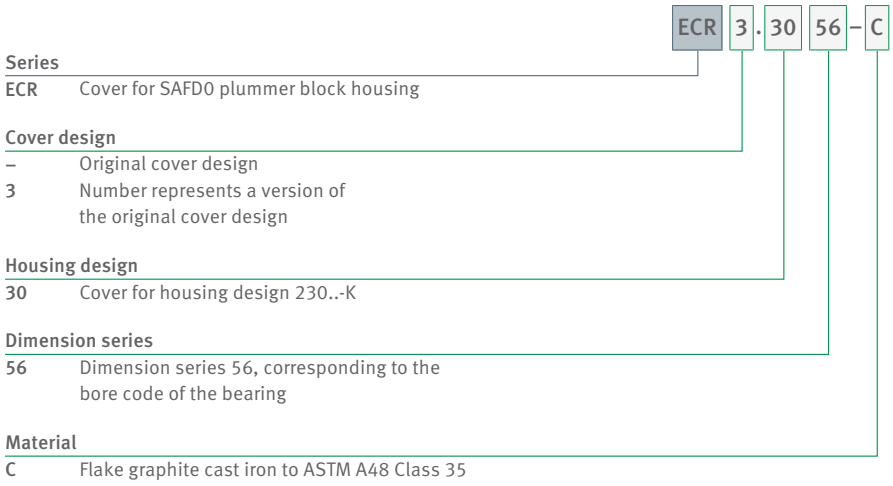


Cover

80 Structure of the designations of EC covers

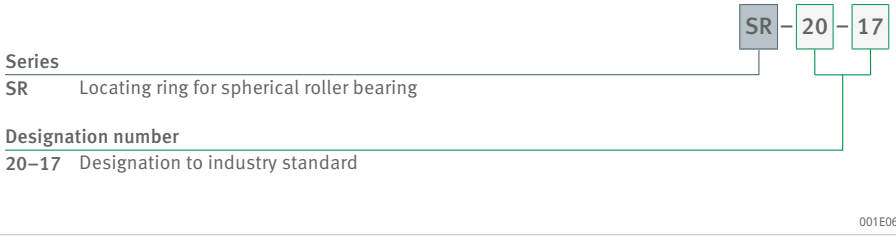


81 Structure of the designations of ECR covers



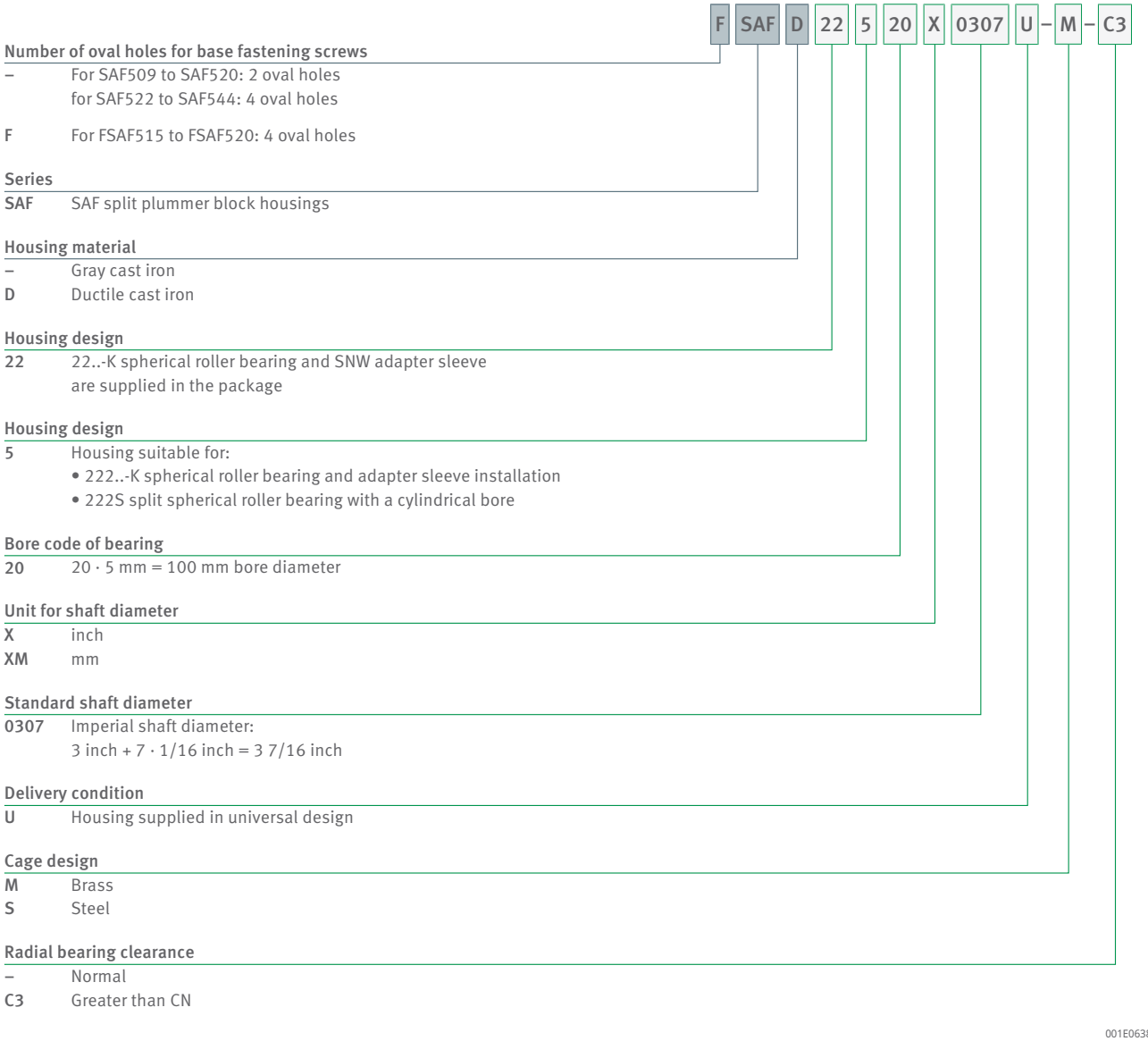
Locating rings

82 Structure of the designations of locating rings, suitable for the relevant spherical roller bearing



Kits

83 Structure of the designations of SAF500 kits



4.8.1 Specimen orders

Check the housing design before ordering an SAF split plummer block housing. The required accessories may be included or must be ordered separately depending on the required housing configuration.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- Standard series made of gray cast iron
- 4 oval holes for base fastening screws
- Housing closed on one side
- Shaft with a diameter of 4 inch

- Spherical roller bearing in the form of a locating or non-locating bearing
- Fastening with adapter sleeve with imperial thread
- Taconite MaxTAC NTC seals

Parts ordered:

Quantity	Designation	Order number
1	Plummer block housing	SAF522X0315U
1	Spherical roller bearing	22222-E1A-XL-M
2	Taconite MaxTAC NTC seal	NTC110X400
1	Adapter sleeve [1)]	SNW22X0400

[1)] For more information on this component, see medias, medias.schaeffler.com.

84 Configuration SAF522X0400U with Taconite MaxTAC NTC seal

001B1031

Example 2

SAF500 kit with the following properties:

- Split plummer block housing in standard series made of ductile cast iron
- 4 oval holes for base fastening screws
- Housing open on both sides
- Shaft with a diameter of 3 $^{15}/_{16}$ inch
- Spherical roller bearing 22222-E1-XL-K-C3 in the form of a locating or non-locating bearing
- Fastening with adapter sleeve with imperial thread

Parts ordered:

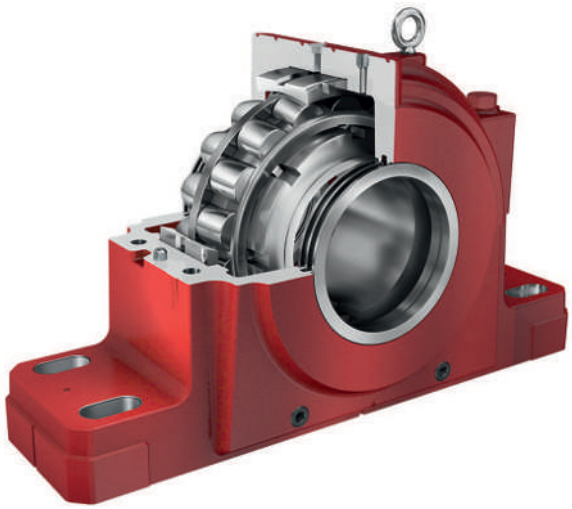
Quantity	Designation	Order number
1	SAF500 kit	SAFD22522X0315U-S-C3

The kit includes:

Quantity	Designation	Order number
1	Plummer block housing	SAFD522X0315U
1	Spherical roller bearing	22222-E1-XL-K-C3
1	Adapter sleeve [2)]	SNW22X0315

[2)] For more information on this component, see medias, medias.schaeffler.com.

85 Configuration SAFD22522X0315U-S-C3



001AE1D8

Example 3

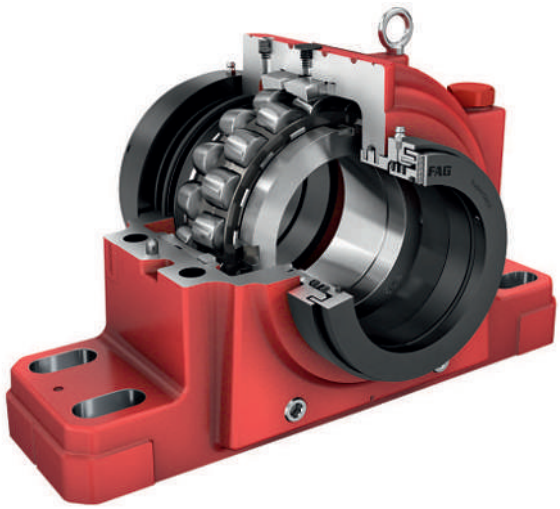
Example of a housing configuration, characteristics:

- Standard series made of ductile cast iron
- 4 oval holes for base fastening screws
- Housing open on both sides and with stepped shaft
- Shaft with a diameter of 4 $^{7}/_{8}$ inch and 4 $^{3}/_{16}$ inch
- Spherical roller bearing 22222-E1-XL in the form of a locating or non-locating bearing
- Fastening with locknut with imperial thread and lock washer
- SuperTAC II TA seals

Parts ordered:

Quantity	Designation	Order number
1	Plummer block housing	SAFD222U
1	Spherical roller bearing	22222-E1-XL
1	Taconite SuperTAC II TA seal	TA121X0414
1	Taconite SuperTAC II TA seal	TA113X0403
1	Locknut	AN24
1	Lock washer	W-24

86 Configuration SAFD222U with SuperTAC II TA seal



0019A215

Example 4

Example of a housing configuration, characteristics:

- Standard series made of ductile cast iron
- 4 oval holes for base fastening screws
- Housing open on both sides
- Shaft with a diameter of 3 $^{15}/_{16}$ inch
- Spherical ball bearing 1222-K-M in the form of a locating or non-locating bearing
- Locating rings suitable for spherical ball bearings
- Fastening with adapter sleeve with imperial thread

Parts ordered:

Quantity	Designation	Order number
1	Plummer block housing	SAFD522X0315U
1	Spherical ball bearing	1222-K-M
1	Adapter sleeve [3)]	SNW22X0315

[3)] For more information on this component, see medias, medias.schaeffler.com.

87 Configuration SAFD522X0315U

001B22AC

Example 5

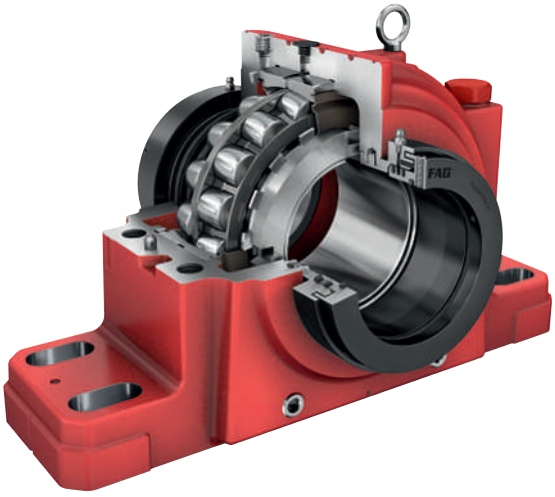
Example of a housing configuration, characteristics:

- Wide series made of ductile cast iron
- 4 oval holes for base fastening screws
- Housing open on both sides
- Shaft with a diameter of 3 $^{15}/_{16}$ inch
- Sealed spherical roller bearing WS22222-E1-XL-K-2RSR in the form of a locating or non-locating bearing
- Fastening with adapter sleeve with imperial thread
- Taconite SuperTAC II TA seals

Quantity	Designation	Order number
1	Plummer block housing	SAFD522X0315U-WS
1	Sealed spherical roller bearing	WS22222-E1-XL-K-2RSR
2	Taconite SuperTAC II TA seal	TA109X0315
1	Adapter sleeve [4)]	H322X315-T-WS

4) For more information on this component, see medias, medias.schaeffler.com.

88 Configuration SAFD522X0315U-WS with Taconite SuperTAC II TA seal

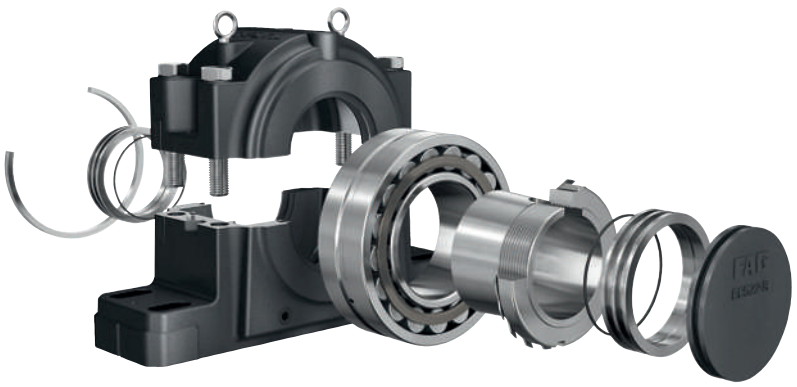


0019A247

4.8.2 SAF500 kits

Kits have been compiled to simplify the ordering process. All parts of the ready-to-install unit are combined in a single order number. The kits are suitable for shaft diameters from 1 $^{3}/_{8}$ inch to 8 inch.

89 Kit, package



001AD22F

A kit contains all the necessary parts and consists of the following:

- 1 housing made of gray cast iron or ductile cast iron SAF500
- 1 222..-K spherical roller bearing
- 1 SNW adapter sleeve with imperial thread, including locknut and locking washer
- 2 LER seals, suitable for the standard imperial shaft diameter
- 1 locating ring or 2 locating rings, suitable for the relevant spherical roller bearing, depending on the housing version
- 1 cover
- Selection of lubrication nipples

4.9 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

4.9.1 Eye bolts

Depending on the housing dimensions, there are threads for 2 eye bolts to ANSI B18.15 in every housing upper section. This eye bolt is intended as a locating point for the installation and removal of the housing. The load capacity of the eye bolts allows the housing including a bearing fitted in the housing to be lifted.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

4.9.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

The specified tightening torques apply to screws with standard threads to ASTM A449 Grade 5 with a friction coefficient of 0.15 ➤30|2.7.2.3.

4.10 Product tables

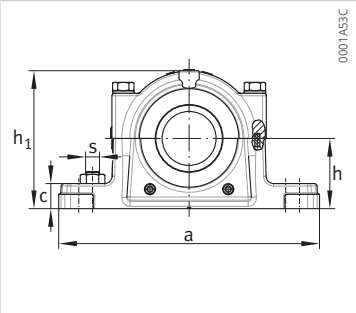
4.10.1 Explanations of the product tables

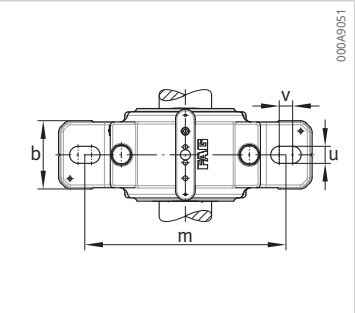
(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	inch	Length of the base
b	inch	Width of the base
c	inch	Height of the base
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
d_2	inch or mm	Larger diameter of the offset shaft
d_3	inch or mm	Smaller diameter of the stepped shaft
e	inch	Offset to the centre of the bearing
g	inch	Width of the housing body
G	–	Eye bolt thread
g_T	inch	Overall width, with 2 Taconite seals
g_{TD}	inch	Overall width, with Taconite seal and cover
h	inch	Shaft spacing
h_1	inch	Overall height
k	mm	Bearing seat width
m	inch	Distance between fastening holes
m	kg or lbs	Mass
n	inch	Short distance between fastening holes
s	inch	Screw size
u	inch	Width of the oval
v	inch	Length of the oval

4.10.2 SAF0, dimensions

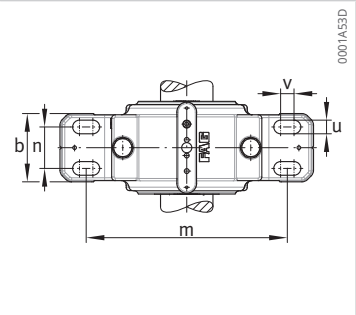
For spherical roller bearings with a tapered bore and adapter sleeve

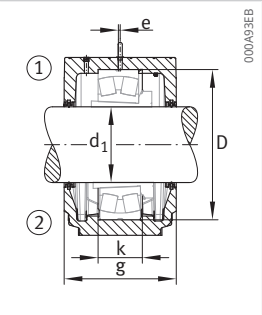
Inch size shaft

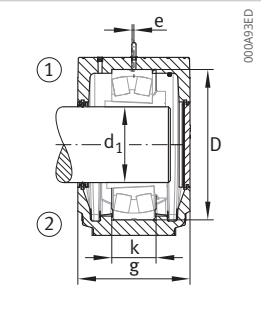




Housing	d_1	h	h_1	g	b	c
Ductile cast iron	**inch**	**inch**	**inch**	**inch**	**inch**	**inch**
SAFD024KX0403U	**4,1875**	4,5	8,4375	5,8125	4,375	1,75
FSAFD024KX0403U	**4,1875**	4,5	8,4375	5,8125	4,375	1,75
SAFD026KX0407U	**4,4375**	4,9375	9,625	6,4375	4,75	2
SAFD028KX0415U	**4,9375**	5,25	10,2	7,38	4,75	2,13
SAFD030KX0503U	**5,1875**	6	11,313	8,125	5,25	2,38
SAFD032KX0507U	**5,375**	6	11,313	8,125	5,25	2,38
SAFD034KX0515U	**5,9375**	6	11,75	7,375	5,875	2,375
SAFD036KX0607U	**6,4375**	6,6875	13,3125	8,5	6,25	2,625
SAFD038KX0615U	**6,9375**	6,6875	13,3125	8,5	6,25	2,625
SAFD040KX0703U	**7,1875**	7,0625	14,1875	9,25	6,75	2,75
SAFD044KX0715U	**7,9375**	7,875	15,5625	10,5	7,5	3,125
SAFD048KX0807U	**8,4375**	8,25	16,5	11	8	3,375
SAFD048KX0808U	**8,5**	8,25	16,5	11	8	3,375
SAFD048KX0815U	**8,9375**	8,25	16,5	11	8	3,375
SAFD048KX0900U	**9**	8,25	16,5	11	8	3,375
SAFD052KX0907U	**9,4375**	9,5	18,625	11,75	8,75	3,75
SAFD052KX0908U	**9,5**	9,5	18,625	11,75	8,75	3,75
SAFD056KX0915U	**9,9375**	9,875	19,5	13,4	9	4
SAFD056KX1000U	**10**	9,875	19,5	13,4	9	4
SAFD056KX1007U	**10,4375**	9,875	19,5	13,4	9	4
SAFD056KX1008U	**10,5**	9,875	19,5	13,4	9	4





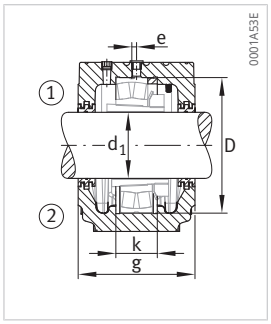


a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
15,25	12,375	0,75	1	–	0,875	180	55,8	0,2	20,8
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	20,8
16,5	13,5625	0,9375	0,875	2,75	0,75	200	61,8	0,2	31,4
16,5	13,875	0,625	0,875	2,75	0,75	210	62,8	0,2	37
18,38	15,313	0,6875	1	3,25	0,875	225	65,8	0,2	51,6
18,38	15,313	0,6875	1	3,25	0,875	240	89,8	0,2	47,7
20,125	16,5	0,875	1,125	3,375	1	260	86,6	0,2	52
22	18,3125	0,9375	1,125	3,75	1	280	102,8	0,2	76
22	18,3125	0,9375	1,125	3,75	1	290	84,8	0,2	78,2
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	88,1
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	131
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	157,1
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	156,9
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	156,7
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	156,2
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	204
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	203,7
34,25	27,875	1,625	1,654	5,5	1,5	420	147,8	0,2	248,4
34,25	27,875	1,625	1,654	5,5	1,5	420	147,8	0,2	247,9
34,25	27,875	1,625	1,654	5,5	1,5	420	147,8	0,2	246,2
34,25	27,875	1,625	1,654	5,5	1,5	420	147,8	0,2	299,1

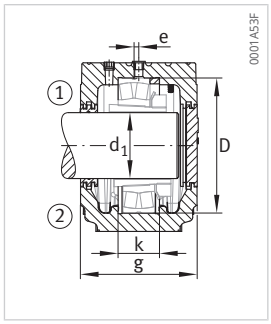
4.10.3 SAF0, accessories

For spherical roller bearings with a tapered bore and adapter sleeve

Inch size shaft

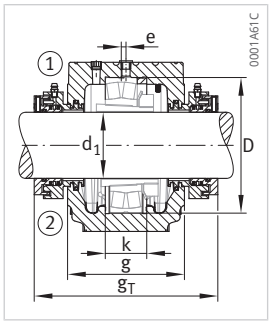


LER labyrinth seal

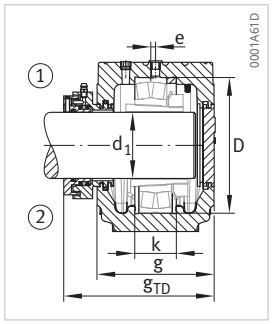


LER labyrinth seal, EC cover

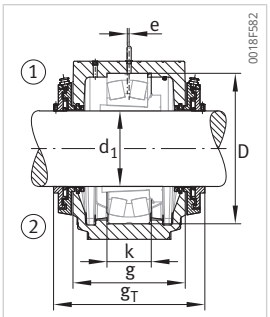
Housing	d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	inch	–	Metric	Imperial	–	Quantity
SAFD024KX0403U	**4,1875**	**23024-E1-K**	H3024×403	SNW3024×0403	SR-20-17	1
FSAFD024KX0403U	**4,1875**	**23024-E1-K**	H3024×403	SNW3024×0403	SR-20-17	1
SAFD026KX0407U	**4,4375**	**23026-E1-K**	H3026×407	SNW3026×0407	SR-22-19	1
SAFD028KX0415U	**4,9375**	**23028-E1-K**	H3028×415	SNW3028×0415	SR-0-20	2
SAFD030KX0503U	**5,1875**	**23030-E1-K**	H3030×503	SNW3030×0503	SR-0-21	1
SAFD032KX0507U	**5,375**	**23032-E1-K**	H3032×507	SNW3032×0507	SR-0-22	2
SAFD034KX0515U	**5,9375**	**23034-E1-K**	H3034×515	SNW3034×0515	SR-0-24	2
SAFD036KX0607U	**6,4375**	**23036-E1-K**	H3036×607	SNW3036×0607	SR-0-26-4	2
SAFD038KX0615U	**6,9375**	**23038-E1-K**	H3038×615	SNW3038×0615	SR-32-0-1	2
SAFD040KX0703U	**7,1875**	**23040-E1-K**	H3040×703	SNW3040×0703	SR-34-0-4	2
SAFD044KX0715U	**7,9375**	**23044-E1-K**	H3044XX715	SNW3044×0715	SR-38-32-2	2
SAFD048KX0807U	**8,4375**	**23048-E1-K**	H3048×807	SNP3048×0807	A8897	2
SAFD048KX0808U	**8,5**	**23048-E1-K**	H3048×808	SNP3048×0808	A8897	2
SAFD048KX0815U	**8,9375**	**23048-E1-K**	H3048×815	SNP3048×0815	A8897	2
SAFD048KX0900U	**9**	**23048-E1-K**	H3048×900	SNP3048×0900	A8897	2
SAFD052KX0907U	**9,4375**	**23052-E1-K**	H3052XX907	SNP3052×0907	A8898	2
SAFD052KX0908U	**9,5**	**23052-E1-K**	H3052XX908	SNP3052×0908	A8898	2
SAFD056KX0915U	**9,9375**	**23056-E1-K**	H3056×915	SNP3056×0915	A8819	2
SAFD056KX1000U	**10**	**23056-E1-K**	H3056×1000	SNP3056×1000	A8819	2
SAFD056KX1007U	**10,4375**	**23056-E1-K**	H3056×1007	SNP3056×1007	A8819	2
SAFD056KX1008U	**10,5**	**23056-E1-K**	H3056×1008	SNP3056×1008	A8819	2



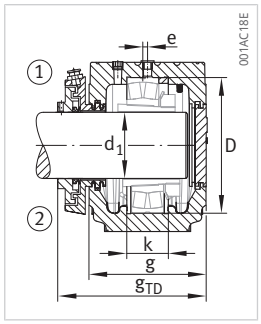
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal



Taconite MaxTAC NTC seal, EC cover

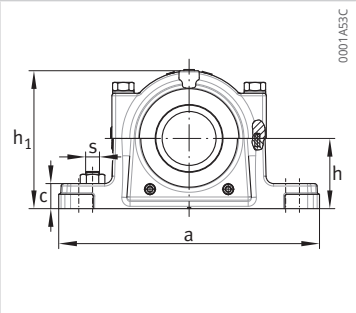
LER	EC	SuperTAC II TA [1]			MaxTAC NTC [1]		
		–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	–	inch	inch	–	inch	inch
LER113×0403-G-A	EC524-R	TA113×0403	10,2	8,65	NTC113×403	10,1	8,6
LER113×0403-G-A	EC524-R	TA113×0403	10,2	8,65	NTC113×403	10,1	8,6
LER117×0407-G-A	EC526-R	TA117×0407	11,25	9,5	NTC117×407	10,75	9,25
LER122×0415-G-A	EC528-R	TA122×0415	10,85	9,1	NTC122×415	10,35	8,85
LER125×0503-G-A	EC530-R	TA125×0503	11,85	10	NTC125×503	11,25	9,7
LER130×0507-G-A	EC532-R	TA130×0507	12,15	10,35	NTC130×507	11,8	10,15
LER140×0515-G-A	EC534-R	TA140×0515	13,15	11,2	NTC140×515	12,9	11,1
LER148×0607-G-A	EC536-R	TA148×0607	13,55	11,6	NTC148×607	12,7	11,15
LER155×0615-G-A	EC538-R	TA155×0615	14,55	12,5	NTC155×615	13,8	12,15
LER159×0703-G-A	EC540-R	TA159×0703	15,3	13,15	NTC159×703	15,55	13,3
LER167×0715-G-A	EC544-R	TA167×0715	16,1	13,9	NTC167×715	16,25	14
LER550×0807-G-S	ECR1.3048-C	TA550×0807	15,6	13,3	NTC550×807	15,55	13,3
LER551×0808-G-S	ECR1.3048-C	TA551×0808	15,6	13,3	NTC551×808	15,55	13,3
LER552×0815-G-S	ECR1.3048-C	TA552×0815	15,6	13,3	NTC552×815	15,55	13,3
LER513×0900-G-S	ECR1.3048-C	TA513×0900	15,6	13,3	NTC513×900	15,55	13,3
LER553×0907-G-S	ECR3052-C	TA553×0907	16,1	13,95	NTC553×907	15,65	13,7
LER178×0908-G-S	ECR3052-C	TA178×0908	16,1	13,95	NTC178×908	15,65	13,7
LER607×0915-G-S	ECR2.3056-C	TA607×0915	16,8	14,6	NTC607×915	16,5	14,4
LER568×1000-G-S	ECR2.3056-C	TA568×1000	16,8	14,6	NTC568×1000	16,5	14,4
LER606×1007-G-S	ECR3.3056-C	TA606×1007	16,8	14,6	NTC606×1007	16,5	14,4
LER519×1008-G-S	ECR3.3056-C	TA519×1008	16,8	14,6	NTC519×1008	16,5	14,4

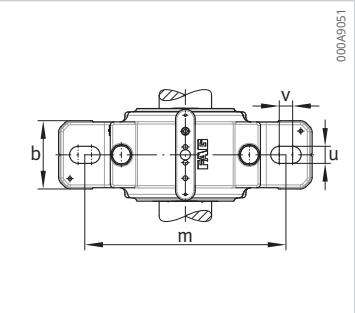
[1] Accessories to be ordered separately

4.10.4 SAFD200, dimensions

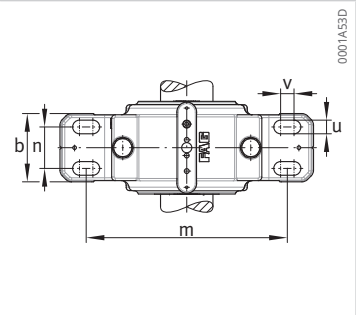
For spherical roller bearings with a cylindrical bore and stepped shaft

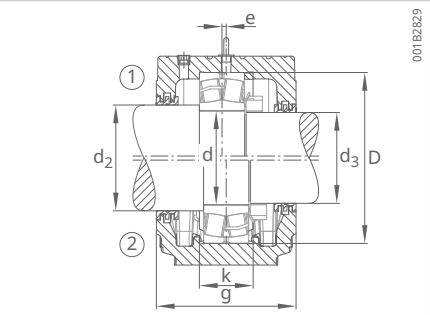
Inch size shaft

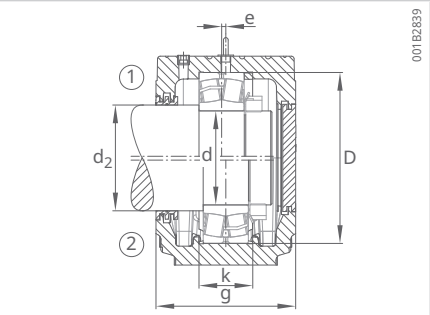




Housing	d_2	d_3	h	h_1	g	b	c
Ductile cast iron	**inch**	**inch**	**inch**	**inch**	**inch**	**inch**	**inch**
SAFD216U	**3,625**	3	3,5	6,875	4,75	3,5	1,25
FSAFD216U	**3,625**	3	3,5	6,875	4,75	3,5	1,25
SAFD217U	**3,9375**	3,1875	3,75	7,3125	4,8125	3,5	1,25
FSAFD217U	**3,9375**	3,1875	3,75	7,3125	4,8125	3,5	1,25
SAFD218U	**4,125**	3,375	4	7,75	5,5	3,875	1,625
FSAFD218U	**4,125**	3,375	4	7,75	5,5	3,875	1,625
SAFD220U	**4,5**	3,8125	4,5	8,4375	5,8125	4,375	1,75
FSAFD220U	**4,5**	3,8125	4,5	8,4375	5,8125	4,375	1,75
SAFD222U	**4,875**	4,1875	4,9375	9,625	6,4375	4,75	2
SAFD224U	**5,3125**	4,5625	5,25	10,25	7,125	4,75	2,125
SAFD226U	**5,875**	4,9375	6	11,5	7,75	5,125	3,375
SAFD228U	**6,25**	5,3125	6	11,75	7,375	5,875	2,375
SAFD230U	**6,625**	5,3125	6,3125	12,5	8,125	6,25	2,5
SAFD232U	**7**	6,0625	6,6875	13,3125	8,5	6,25	2,625
SAFD236U	**7**	6,0625	7,5	14,875	9,625	7,125	3
SAFD234U	**7,4375**	6,875	7,0625	14,1875	9,25	6,75	2,75
SAFD238U	**8,375**	7,25	7,875	15,5625	10,5	7,5	3,125
SAFD240U	**8,75**	7,625	8,25	16,5	11	8	3,375
SAFD244U	**9,5625**	8,3125	9,5	18,625	11,75	8,75	3,75







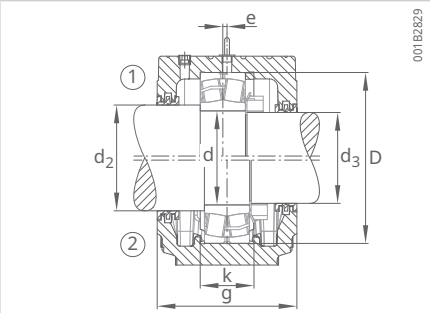
4

a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
13	10,3125	0,6875	0,875	–	0,625	140	42,8	0,2	13,2
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	0,2	13,4
13	10,4375	0,625	0,875	–	0,625	150	45,8	0,2	13,3
13	10,4375	0,625	0,75	2,125	0,625	150	45,8	0,2	13,6
13,75	11	0,625	0,875	–	0,625	160	49,8	0,2	16,3
13,75	11	0,625	0,75	2,125	0,625	160	49,8	0,2	16,5
15,25	12,375	0,75	1	–	0,75	180	55,8	0,2	20,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	20,8
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	0,2	37,6
16,5	13,875	0,625	0,875	2,75	0,75	215	67,8	0,2	35,5
18,375	15,3125	0,6875	1	3,25	0,875	230	73,8	0,2	47,5
20,125	16,5	0,875	1,125	3,375	1	250	77,8	0,2	51
21,25	17,625	0,625	1,125	3,75	1	270	82,8	0,2	61,6
22	18,3125	0,9375	1,125	3,75	1	290	89,8	0,2	73,1
26,75	22,25	1,375	1,125	4,625	1	320	95,8	0,2	105,1
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	88,2
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	121,8
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	152,1
32,75	26,3125	1,5675	1,625	5,25	1,5	400	117,8	0,2	200,7

4.10.5 SAFD200, accessories

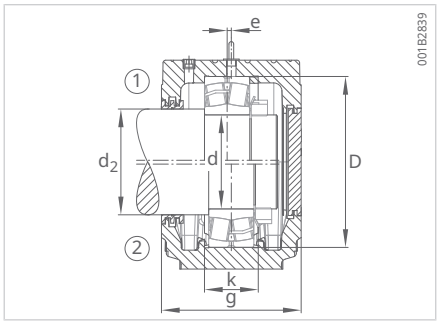
For spherical roller bearings with a cylindrical bore and stepped shaft

Inch size shaft

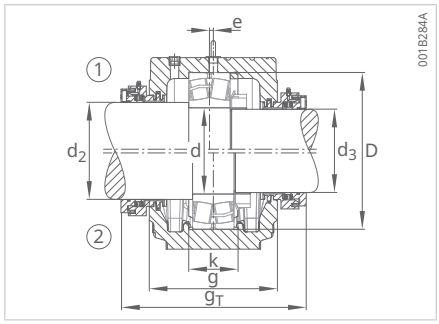


LER labyrinth seal

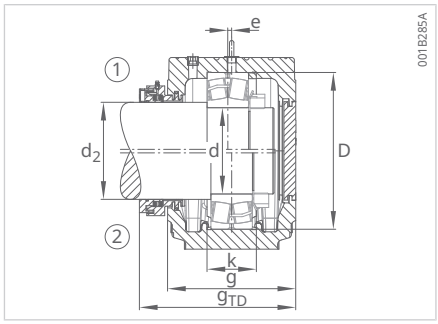
Housing	d_2	d_3	Spherical roller bearing [1]	Lock nut [1]	Locking washer [1]	Locating ring	
Ductile cast iron	inch	inch	–	Imperial	Imperial	–	Quantity
SAFD216U	**3,625**	3	**22216-E1**	W-16	AN16	SR-16-13	1
FSAFD216U	**3,625**	3	**22216-E1**	W-16	AN16	SR-16-13	1
SAFD217U	**3,9375**	3,1875	**22217-E1**	W-17	AN17	SR-17-14	1
FSAFD217U	**3,9375**	3,1875	**22217-E1**	W-17	AN17	SR-17-14	1
SAFD218U	**4,125**	3,375	**22218-E1**	W-18	AN18	SR-18-15	1
FSAFD218U	**4,125**	3,375	**22218-E1**	W-18	AN18	SR-18-15	1
SAFD220U	**4,5**	3,8125	**22220-E1**	W-20	AN20	SR-20-17	1
FSAFD220U	**4,5**	3,8125	**22220-E1**	W-20	AN20	SR-20-17	1
SAFD222U	**4,875**	4,1875	**22222-E1**	W-24	AN24	SR-22-19	1
SAFD224U	**5,3125**	4,5625	**22224-E1**	W-24	AN24	SR-24-20	1
SAFD226U	**5,875**	4,9375	**22226-E1**	W-26	AN26	SR-26-0	1
SAFD228U	**6,25**	5,3125	**22228-E1**	W-28	AN28	SR-28-0	1
SAFD230U	**6,625**	5,3125	**22230-E1**	W-30	AN30	SR-30-0	1
SAFD232U	**7**	6,0625	**22232-E1**	W-32	AN32	SR-32-0	1
SAFD236U	**7**	6,0625	**22236-E1**	W-36	AN36	SR-36-30	1
SAFD234U	**7,4375**	6,875	**22234-E1**	W-34	AN34	SR-34-0	1
SAFD238U	**8,375**	7,25	**22238-E1**	W-38	AN38	SR-38-32	1
SAFD240U	**8,75**	7,625	**22240-E1**	W-40	AN40	SR-40-34	1
SAFD244U	**9,5625**	8,3125	**22244-E1**	W-44	N44	SR-44-38	1



LER labyrinth seal, EC cover



Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover

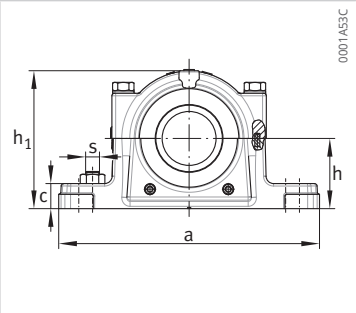
LER, d_2	LER, d_3	EC	SuperTAC II TA [1], d_2	SuperTAC II TA [1], d_3		
				–	g_T	g_{TD}
–	–	–	–	–	inch	inch
LER82×0310-G-A	LER54×0300-G-A	EC216-R	TA82×0310	TA54×0300	7,85	6,05
LER82×0310-G-A	LER54×0300-G-A	EC216-R	TA82×0310	TA54×0300	7,85	6,05
LER89×0315-G-A	LER63×0303-G-A	EC217-R	TA89×0315	TA63×303	8,05	6,15
LER89×0315-G-A	LER63×0303-G-A	EC217-R	TA89×0315	TA63×303	8,05	6,15
LER112×0402-G-A	LER191×0306-G-A	EC518-R	TA112×0402	TA191×0306	8,8	6,55
LER112×0402-G-A	LER191×0306-G-A	EC518-R	TA112×0402	TA191×306	8,8	6,55
LER118×0408-G-A	LER106×0313-G-A	EC220-R	TA118×0408	TA106×0313	9,45	7,2
LER118×0408-G-A	LER106×0313-G-A	EC220-R	TA118×0408	TA106×0313	9,45	7,2
LER121×0414-G-A	LER113×0403-G-A	EC524-R	TA121×0414	TA113×0403	9,7	7,7
LER127×0505-G-A	LER119×0409-G-A	EC524-R	TA127×0505	TA119×0409	10,7	8,5
LER136×0514-G-A	LER122×0415-G-A	EC528-R	TA136×0514	TA122×0415	11,5	9,25
LER144×0604-G-A	LER127×0505-G-A	EC530-R	TA144×0604	TA127×0505	11,3	9,95
LER151×0610-G-A	LER134×0512-G-A	EC230-R	TA151×0610	TA134×0512	12,2	9,7
LER156×0700-G-A	LER142×0601-G-A	EC534-R	TA156×0700	TA142×0601	12,5	10
LER165×0713-G-A	LER154×0614-G-A	EC534-R	TA156×0700	TA142×0601	12,5	10
LER161×0707-G-A	LER148×0607-G-A	EC536-R	TA161×0707	TA148×607	13,2	10,75
LER171×0806-G-A	LER160×0704-G-A	EC540-R	TA171×0806	TA160×0704	14,8	12,15
LER175×0812-G-A	LER164×0710-G-A	EC240-R	TA175×0812	TA164×0710	15,35	12,65
LER179×0909-G-A	LER170×0805-G-A	EC244-R	TA179×0909	TA170×0805	16,75	14,1

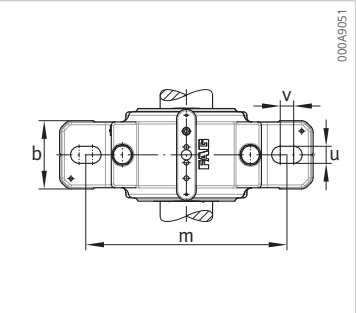
[1] Accessories to be ordered separately

4.10.6 SAF500, preferred sizes, dimensions

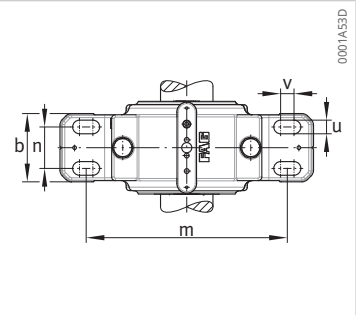
For spherical roller bearings with a tapered bore and adapter sleeve

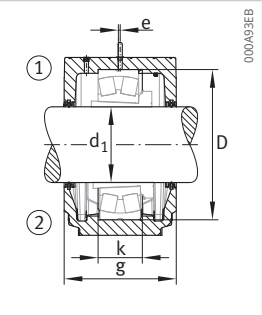
Inch size shaft

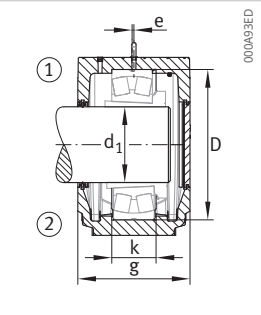




Housing		d_1	h	h_1	g	b	c
Ductile cast iron	Gray cast iron	inch	inch	inch	inch	inch	inch
SAFD509×0107U	**–**	**1,4375**	2,25	4,375	3,4375	2,375	0,8125
SAFD510×0111U	**–**	**1,6875**	2,5	4,75	3,4375	2,375	0,9375
SAFD511×0115U	**–**	**1,9375**	2,75	5,25	3,75	2,75	0,9375
SAFD513×0203U	**–**	**2,1875**	3	5,9375	4,25	3,125	1
FSAFD515×0207U	**FSAF515×0207U**	**2,4375**	3,25	6,375	4,625	3,125	1,125
SAFD515×0207U	**SAF515×0207U**	**2,4375**	3,25	6,375	4,625	3,125	1,125
SAFD516×0211U	**–**	**2,6875**	3,5	6,875	4,75	3,5	1,25
FSAFD517×0215U	**FSAF517×0215U**	**2,9375**	3,75	7,3125	4,8125	3,5	1,25
SAFD517×0215U	**SAF517×0215U**	**2,9375**	3,75	7,3125	4,8125	3,5	1,25
FSAFD518×0303U	**FSAF518×0303U**	**3,1875**	4	7,75	5,5	3,875	1,625
SAFD518×0303U	**SAF518×0303U**	**3,1875**	4	7,75	5,5	3,875	1,625
FSAFD520×0307U	**FSAF520×0307U**	**3,4375**	4,5	8,4375	5,8125	4,375	1,75
SAFD520×0307U	**SAF520×0307U**	**3,4375**	4,5	8,4375	5,8125	4,375	1,75
SAFD522×0315U	**SAF522×0315U**	**3,9375**	4,9375	9,625	6,4375	4,75	2
SAFD524×0403U	**SAF524×0403U**	**4,1875**	5,25	10,25	7,125	4,75	2,125
SAFD526×0407U	**SAF526×0407U**	**4,4375**	6	11,5	7,75	5,125	3,375
SAFD528×0415U	**SAF528×0415U**	**4,9375**	6	11,75	7,375	5,875	2,375
SAFD530×0503U	**–**	**5,1875**	6,3125	12,5	8,125	6,25	2,5
SAFD532×0507U	**SAF532×0507U**	**5,4375**	6,6875	13,3125	8,5	6,25	2,625
SAFD534×0515U	**SAF534×0515U**	**5,9375**	7,0625	14,1875	9,25	6,75	2,75
SAFD536×0607U	**SAF536×0607U**	**6,4375**	7,5	14,875	9,625	7,125	3
SAFD538×0615U	**SAF538×0615U**	**6,9375**	7,875	15,5625	10,5	7,5	3,125
SAFD540×0703U	**–**	**7,1875**	8,25	16,5	11	8	3,375
SAFD544×0715U	**SAF544×0715U**	**7,9375**	9,5	18,625	11,75	8,75	3,75





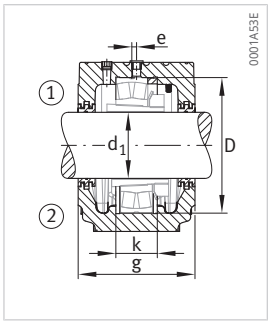


a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
8,25	6,625	0,375	0,625	–	0,5	85	28,8	0,1	3,6
8,25	6,75	0,25	0,625	–	0,5	90	29,8	0,1	4
9,625	7,625	0,25	0,75	–	0,625	100	30,8	0,1	5,3
11	8,8125	0,6875	0,75	–	0,625	120	39	0,15	8,6
11,25	9,125	0,5	0,625	1,875	0,5	130	37	0,1	10,6
11,25	9,125	0,5	0,75	–	0,625	130	37	0,1	10
13	10,3125	0,6875	0,875	–	0,75	140	42,8	0,2	13,7
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	0,2	14
13	10,4375	0,56	0,875	–	0,75	150	45,8	0,2	14
13,75	11	0,625	0,75	2,125	0,625	160	49,8	0,2	17
13,75	11	0,625	0,875	–	0,75	160	49,8	0,2	16,9
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	21,8
15,25	12,375	0,75	1	–	0,875	180	55,8	0,2	21,6
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	0,2	33,2
16,5	13,875	0,625	0,875	2,75	0,75	215	67,8	0,2	36,3
18,375	15,3125	0,6875	1	3,25	0,875	230	73,8	0,2	49,1
20,125	16,5	0,875	1,125	3,375	1	250	77,8	0,2	52,3
21,25	17,625	0,625	1,125	3,75	1	270	82,8	0,2	62,8
22	18,3125	0,9375	1,125	3,75	1	290	89,8	0,2	75
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	89,5
26,75	22,25	1,375	1,125	4,625	1	320	95,8	0,2	106,5
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	123,4
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	154
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	201,9

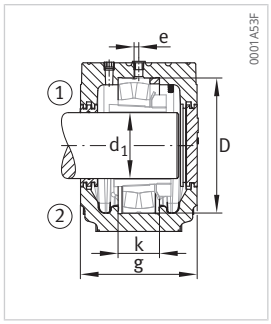
4.10.7 SAF500, preferred sizes, accessories

For spherical roller bearings with a tapered bore and adapter sleeve

Inch size shaft

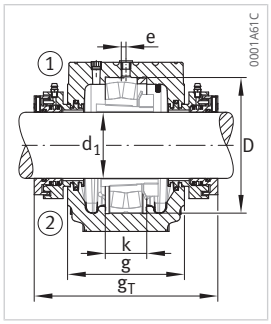


LER labyrinth seal

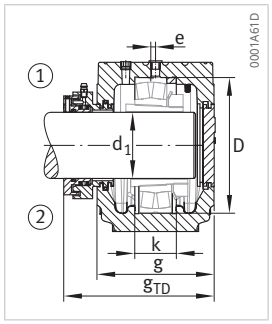


LER labyrinth seal, EC cover

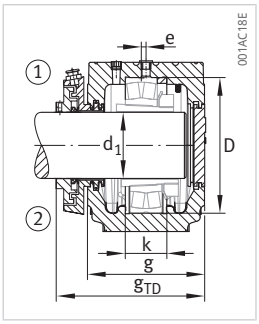
Housing		d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	Gray cast iron	inch	–	Metric	Imperial	–	Quantity
SAFD509×0107U	**–**	**1,4375**	**22209-E1-K**	H309×0107	SNW09×0107	SR-9-9	1
SAFD510×0111U	**–**	**1,6875**	**22210-E1-K**	H310×0111	SNW10×0111	SR-10-0	1
SAFD511×0115U	**–**	**1,9375**	**22211-E1-K**	H311×0115	SNW11×0115	SR-11-0	1
SAFD513×0203U	**–**	**2,1875**	**22213-E1-K**	H313×0203	SNW13×0203	SR-13-0	1
FSAFD515×0207U	**FSAF515×0207U**	**2,4375**	**22215-E1-K**	H315×0207	SNW15×0207	SR-15-0	1
SAFD515×0207U	**SAF515×0207U**	**2,4375**	**22215-E1-K**	H315×0207	SNW15×0207	SR-15-0	1
SAFD516×0211U	**–**	**2,6875**	**22216-E1-K**	H316×0211	SNW16×0211	SR-16-13	1
FSAFD517×0215U	**FSAF517×0215U**	**2,9375**	**22217-E1-K**	H317×0215	SNW17×0215	SR-17-14	1
SAFD517×0215U	**SAF517×0215U**	**2,9375**	**22217-E1-K**	H317×0215	SNW17×0215	SR-17-14	1
FSAFD518×0303U	**FSAF518×0303U**	**3,1875**	**22218-E1-K**	H318×0303	SNW18×0303	SR-18-15	1
SAFD518×0303U	**SAF518×0303U**	**3,1875**	**22218-E1-K**	H318×0303	SNW18×0303	SR-18-15	1
FSAFD520×0307U	**FSAF520×0307U**	**3,4375**	**22220-E1-K**	H320×0307	SNW20×0307	SR-20-17	1
SAFD520×0307U	**SAF520×0307U**	**3,4375**	**22220-E1-K**	H320×0307	SNW20×0307	SR-20-17	1
SAFD522×0315U	**SAF522×0315U**	**3,9375**	**22222-E1-K**	H322×0315	SNW22×0315	SR-22-19	1
SAFD524×0403U	**SAF524×0403U**	**4,1875**	**22224-E1-K**	H3124×0403	SNW24×0403	SR-24-20	1
SAFD526×0407U	**SAF526×0407U**	**4,4375**	**22226-E1-K**	H3126×0407	SNW26×0407	SR-26-0	1
SAFD528×0415U	**SAF528×0415U**	**4,9375**	**22228-E1-K**	H3128×0415	SNW28×0415	SR-28-0	1
SAFD530×0503U	**–**	**5,1875**	**22230-E1-K**	H3130×0503	SNW30×0503	SR-30-0	1
SAFD532×0507U	**SAF532×0507U**	**5,4375**	**22232-E1-K**	H3132×0507	SNW32×0507	SR-32-0	1
SAFD534×0515U	**SAF534×0515U**	**5,9375**	**22234-E1-K**	H3134×0515	SNW34×0515	SR-34-0	1
SAFD536×0607U	**SAF536×0607U**	**6,4375**	**22236-E1-K**	H3136×0607	SNW37×0607	SR-36-30	1
SAFD538×0615U	**SAF538×0615U**	**6,9375**	**22238-E1-K**	H3138×0615	SNW38×0615	SR-38-32	1
SAFD540×0703U	**–**	**7,1875**	**22240-E1-K**	H3140×0703	SNW40×0703	SR-40-34	1
SAFD544×0715U	**SAF544×0715U**	**7,9375**	**22244-E1-K**	H3144×0715	SNW44×0715	SR-44-38	1



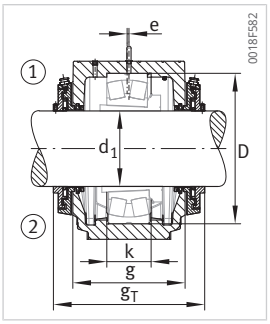
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



Taconite MaxTAC NTC seal

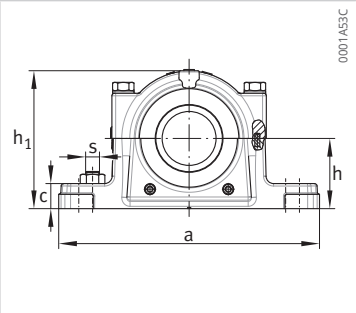
LER	EC	SuperTAC II TA [1]			MaxTAC NTC [1]		
		–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	–	inch	inch	–	inch	inch
LER17×0107-G-A	EC509-R	TA17×0107	5,9	4,7	–	–	–
LER20×0111-G-A	EC510-R	TA20×0111	5,85	4,65	–	–	–
LER24×0115-G-A	EC511-R	TA24×0115	6,35	5,05	–	–	–
LER29×0203-G-A	EC513-R	TA29×0203	6,9	5,6	NTC29×203	6,65	5,45
LER37×0207-G-A	EC515-R	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	TA37×0207	7,35	6	NTC37×207	7,35	6
LER44×0211-G-A	EC516-R	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER53×0215-G-A	EC517-R	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER188×0303-G-A	EC518-R	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER188×0303-G-A	EC518-R	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER102×0307-G-A	EC520-R	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER109×0315-G-A	EC522-R	TA109×0315	9,45	7,95	NTC109×315	9,3	7,85
LER113×0403-G-A	EC524-R	TA113×0403	10,2	8,65	NTC113×403	10,1	8,6
LER117×0407-G-A	EC526-R	TA117×0407	11,25	9,5	NTC117×407	10,75	9,25
LER122×0415-G-A	EC528-R	TA122×0415	10,85	9,1	NTC122×415	10,35	8,85
LER125×0503-G-A	EC530-R	TA125×0503	11,85	10	NTC125×503	11,25	9,7
LER130×0507-G-A	EC532-R	TA130×0507	12,15	10,35	NTC130×507	11,8	10,15
LER140×0515-G-A	EC534-R	TA140×0515	13,15	11,2	NTC140×515	12,9	11,1
LER148×0607-G-A	EC536-R	TA148×0607	13,55	11,6	NTC148×607	12,7	11,15
LER155×0615-G-A	EC538-R	TA155×0615	14,55	12,5	NTC155×615	13,8	12,15
LER159×0703-G-A	EC540-R	TA159×0703	15,3	13,15	NTC159×703	15,55	13,3
LER167×0715-G-A	EC544-R	TA167×0715	16,1	13,9	NTC167×715	16,25	14

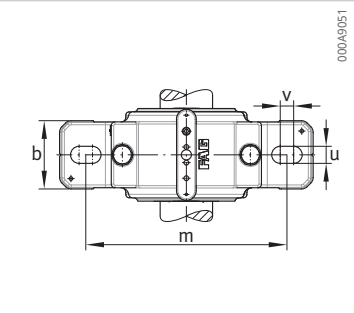
[1] Accessories to be ordered separately

4.10.8 SAF500, all sizes, dimensions

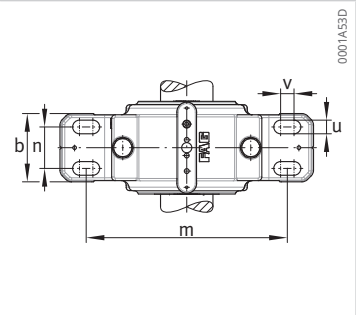
For spherical roller bearings with a tapered bore and adapter sleeve

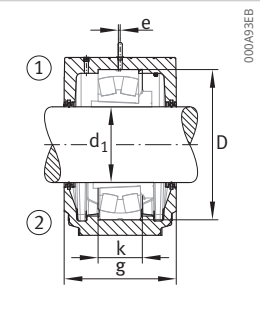
Inch size shaft

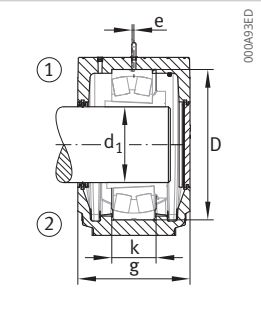




Housing		d_1	h	h_1	g	b	c
Ductile cast iron	Gray cast iron	inch	inch	inch	inch	inch	inch
SAFD509×0107U	–	1,375	2,25	4,375	3,4375	2,375	0,8125
SAFD509×0107U	–	1,4375	2,25	4,375	3,4375	2,375	0,8125
SAFD509×0107U	–	1,5	2,25	4,375	3,4375	2,375	0,8125
SAFD510×0111U	–	1,625	2,5	4,75	3,4375	2,375	0,9375
SAFD510×0111U	–	1,6875	2,5	4,75	3,4375	2,375	0,9375
SAFD510×0111U	–	1,75	2,5	4,75	3,4375	2,375	0,9375
SAFD511×0115U	–	1,8125	2,75	5,25	3,75	2,75	0,9375
SAFD511×0115U	–	1,875	2,75	5,25	3,75	2,75	0,9375
SAFD511×0115U	–	1,9375	2,75	5,25	3,75	2,75	0,9375
SAFD513×0203U	–	2	3	5,9375	4,25	3,125	1
SAFD511×0115U	–	2	2,75	5,25	3,75	2,75	0,9375
SAFD513×0203U	–	2,0625	3	5,9375	4,25	3,125	1
SAFD513×0203U	–	2,125	3	5,9375	4,25	3,125	1
SAFD513×0203U	–	2,1875	3	5,9375	4,25	3,125	1
SAFD513×0203U	–	2,25	3	5,9375	4,25	3,125	1
SAFD515×0207U	SAF515×0207U	2,375	3,25	6,375	4,625	3,125	1,125
FSAFD515×0207U	FSAF515×0207U	2,375	3,25	6,375	4,625	3,125	1,125
FSAFD515×0207U	FSAF515×0207U	2,4375	3,25	6,375	4,625	3,125	1,125
SAFD515×0207U	SAF515×0207U	2,4375	3,25	6,375	4,625	3,125	1,125
SAFD515×0207U	SAF515×0207U	2,5	3,25	6,375	4,625	3,125	1,125
FSAFD515×0207U	FSAF515×0207U	2,5	3,25	6,375	4,625	3,125	1,125
SAFD516×0211U	–	2,5625	3,5	6,875	4,75	3,5	1,25
FSAFD516×0211U	–	2,5625	3,5	6,875	4,75	3,5	1,25
SAFD516×0211U	–	2,625	3,5	6,875	4,75	3,5	1,25
FSAFD516×0211U	–	2,625	3,5	6,875	4,75	3,5	1,25
FSAFD516×0211U	–	2,6875	3,5	6,875	4,75	3,5	1,25
SAFD516×0211U	–	2,6875	3,5	6,875	4,75	3,5	1,25
SAFD516×0211U	–	2,75	3,5	6,875	4,75	3,5	1,25
FSAFD516×0211U	–	2,75	3,5	6,875	4,75	3,5	1,25
SAFD517×0215U	SAF517×0215U	2,9375	3,75	7,3125	4,8125	3,5	1,25
SAFD517×0215U	SAF517×0215U	2,9375	3,75	7,3125	4,8125	3,5	1,25
FSAFD517×0215U	FSAF517×0215U	2,9375	3,75	7,3125	4,8125	3,5	1,25
SAFD517×0215U	SAF517×0215U	2,9375	3,75	7,3125	4,8125	3,5	1,25
FSAFD517×0215U	FSAF517×0215U	3	3,75	7,3125	4,8125	3,5	1,25
FSAFD517×0215U	FSAF517×0215U	2,9375	3,75	7,3125	4,8125	3,5	1,25
FSAFD517×0215U	FSAF517×0215U	2,9375	3,75	7,3125	4,8125	3,5	1,25
SAFD517×0215U	SAF517×0215U	3	3,75	7,3125	4,8125	3,5	1,25
SAFD518×0303U	SAF518×0303U	3,215	4	7,75	5,5	3,875	1,625
FSAFD518×0303U	FSAF518×0303U	3,1875	4	7,75	5,5	3,875	1,625
SAFD518×0303U	SAF518×0303U	3,1875	4	7,75	5,5	3,875	1,625





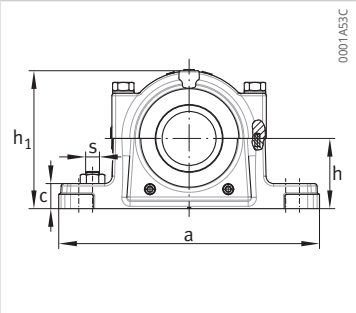


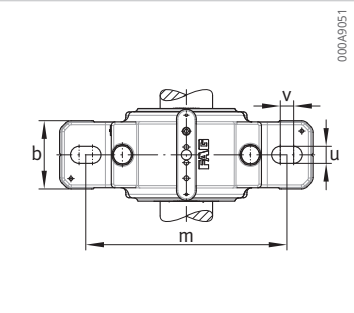
a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
8,25	6,625	0,375	0,625	–	0,5	85	28,8	0,1	3,6
8,25	6,625	0,375	0,625	–	0,5	85	28,8	0,1	3,6
8,25	6,625	0,375	0,625	–	0,5	85	28,8	0,1	3,6
8,25	6,75	0,25	0,625	–	0,5	90	29,8	0,1	4
8,25	6,75	0,25	0,625	–	0,5	90	29,8	0,1	4
8,25	6,75	0,25	0,625	–	0,5	90	29,8	0,1	4
9,625	7,625	0,25	0,75	–	0,625	100	30,8	0,1	5,3
9,625	7,625	0,25	0,75	–	0,625	100	30,8	0,1	5,3
9,625	7,625	0,25	0,75	–	0,625	100	30,8	0,1	5,3
11	8,8125	0,6875	0,75	–	0,625	120	39	0,15	8,6
9,625	7,625	0,25	0,75	–	0,625	100	30,8	0,1	5,3
11	8,8125	0,6875	0,75	–	0,625	120	39	0,15	8,6
11	8,8125	0,6875	0,75	–	0,625	120	39	0,15	8,6
11	8,8125	0,6875	0,75	–	0,625	120	39	0,15	8,6
11	8,8125	0,6875	0,75	–	0,625	120	39	0,15	8,6
11,25	9,125	0,5	0,75	–	0,625	130	37	0,1	10
11,25	9,125	0,5	0,625	1,875	0,5	130	37	0,1	10,6
11,25	9,125	0,5	0,625	1,875	0,5	130	37	0,1	10,6
11,25	9,125	0,5	0,75	–	0,625	130	37	0,1	10
11,25	9,125	0,5	0,75	–	0,625	130	37	0,1	10
11,25	9,125	0,5	0,625	1,875	0,5	130	37	0,1	10,6
13	10,3125	0,6875	0,875	–	0,75	140	42,8	0,2	13,7
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	0,2	13,9
13	10,3125	0,6875	0,875	–	0,75	140	42,8	0,2	13,7
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	0,2	13,9
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	0,2	13,9
13	10,3125	0,6875	0,875	–	0,75	140	42,8	0,2	13,7
13	10,3125	0,6875	0,875	–	0,75	140	42,8	0,2	13,7
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	0,2	13,9
13	10,4375	0,56	0,875	–	0,75	150	45,8	0,2	14
13	10,4375	0,56	0,875	–	0,75	150	45,8	0,2	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	0,2	14
13	10,4375	0,56	0,875	–	0,75	150	45,8	0,2	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	0,2	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	0,2	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	0,2	14
13	10,4375	0,56	0,875	–	0,75	150	45,8	0,2	14
13,75	11	0,625	0,875	–	0,75	160	49,8	0,2	16,9
13,75	11	0,625	0,75	2,125	0,625	160	49,8	0,2	17
13,75	11	0,625	0,875	–	0,75	160	49,8	0,2	16,9

4.10.8 SAF500, all sizes, dimensions

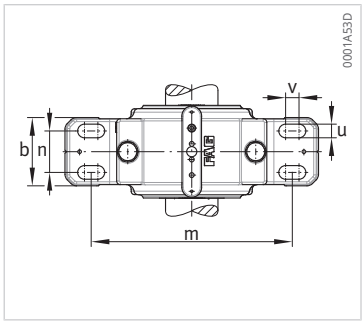
For spherical roller bearings with a tapered bore and adapter sleeve

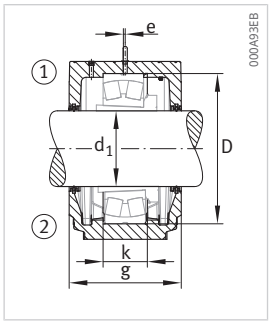
Inch size shaft

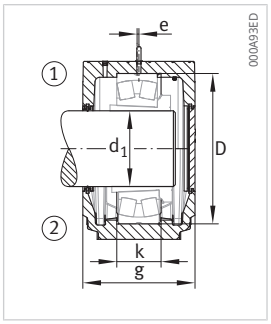




Housing		d_1	h	h_1	g	b	c
Ductile cast iron	Gray cast iron	inch	inch	inch	inch	inch	inch
FSAFD518×0303U	FSAF518×0303U	3,125	4	7,75	5,5	3,875	1,625
SAFD518×0303U	SAF518×0303U	3,25	4	7,75	5,5	3,875	1,625
FSAFD518×0303U	FSAF518×0303U	3,25	4	7,75	5,5	3,875	1,625
SAFD520×0307U	SAF520×0307U	3,3125	4,5	8,4375	5,8125	4,375	1,75
FSAFD520×0307U	FSAF520×0307U	3,3125	4,5	8,4375	5,8125	4,375	1,75
SAFD520×0307U	SAF520×0307U	3,375	4,5	8,4375	5,8125	4,375	1,75
FSAFD520×0307U	FSAF520×0307U	3,375	4,5	8,4375	5,8125	4,375	1,75
FSAFD520×0307U	FSAF520×0307U	3,4375	4,5	8,4375	5,8125	4,375	1,75
SAFD520×0307U	SAF520×0307U	3,4375	4,5	8,4375	5,8125	4,375	1,75
FSAFD520×0307U	FSAF520×0307U	3,4375	4,5	8,4375	5,8125	4,375	1,75
SAFD520×0307U	SAF520×0307U	3,5	4,5	8,4375	5,8125	4,375	1,75
SAFD522×0315U	SAF522×0315U	3,8125	4,9375	9,625	6,4375	4,75	2
SAFD522×0315U	SAF522×0315U	3,875	4,9375	9,625	6,4375	4,75	2
SAFD522×0315U	SAF522×0315U	3,9375	4,9375	9,625	6,4375	4,75	2
SAFD522×0315U	SAF522×0315U	4	4,9375	9,625	6,4375	4,75	2
SAFD524×0403U	SAF524×0403U	4,0625	5,25	10,25	7,125	4,75	2,125
SAFD524×0403U	SAF524×0403U	4,125	5,25	10,25	7,125	4,75	2,125
SAFD524×0403U	SAF524×0403U	4,1875	5,25	10,25	7,125	4,75	2,125
SAFD524×0403U	SAF524×0403U	4,25	5,25	10,25	7,125	4,75	2,125
SAFD526×0407U	SAF526×0407U	4,3125	6	11,5	7,75	5,125	3,375
SAFD526×0407U	SAF526×0407U	4,375	6	11,5	7,75	5,125	3,375
SAFD526×0407U	SAF526×0407U	4,4375	6	11,5	7,75	5,125	3,375
SAFD526×0407U	SAF526×0407U	4,5	6	11,5	7,75	5,125	3,375
SAFD528×0415U	SAF528×0415U	4,8125	6	11,75	7,375	5,875	2,375
SAFD528×0415U	SAF528×0415U	4,875	6	11,75	7,375	5,875	2,375
SAFD528×0415U	SAF528×0415U	4,9375	6	11,75	7,375	5,875	2,375
SAFD530×0503U	–	5,125	6,3125	12,5	8,125	6,25	2,5
SAFD530×0503U	–	5,1875	6,3125	12,5	8,125	6,25	2,5
SAFD530×0503U	–	5,25	6,3125	12,5	8,125	6,25	2,5
SAFD532×0507U	SAF532×0507U	5,375	6,6875	13,3125	8,5	6,25	2,625
SAFD532×0507U	SAF532×0507U	5,4375	6,6875	13,3125	8,5	6,25	2,625
SAFD532×0507U	SAF532×0507U	5,5	6,6875	13,3125	8,5	6,25	2,625
SAFD534×0515U	SAF534×0515U	5,8125	7,0625	14,1875	9,25	6,75	2,75
SAFD534×0515U	SAF534×0515U	5,875	7,0625	14,1875	9,25	6,75	2,75
SAFD534×0515U	SAF534×0515U	5,9375	7,0625	14,1875	9,25	6,75	2,75
SAFD534×0515U	SAF534×0515U	6	7,0625	14,1875	9,25	6,75	2,75
SAFD536×0607U	SAF536×0607U	6,3125	7,5	14,875	9,625	7,125	3
SAFD536×0607U	SAF536×0607U	6,375	7,5	14,875	9,625	7,125	3
SAFD536×0607U	SAF536×0607U	6,4375	7,5	14,875	9,625	7,125	3
SAFD536×0607U	SAF536×0607U	6,5	7,5	14,875	9,625	7,125	3





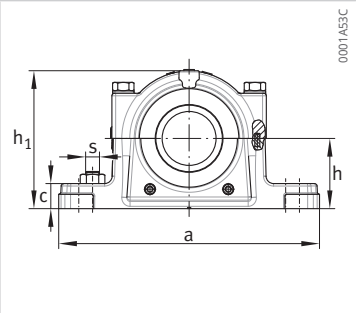


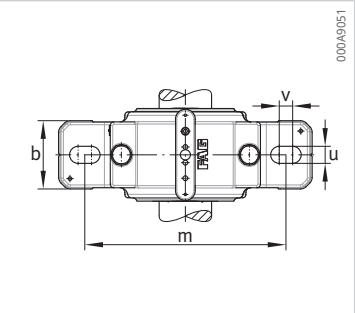
a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
13,75	11	0,625	0,75	2,125	0,625	160	49,8	0,2	17
13,75	11	0,625	0,875	–	0,75	160	49,8	0,2	16,9
13,75	11	0,625	0,75	2,125	0,625	160	49,8	0,2	17
15,25	12,375	0,75	1	–	0,875	180	55,8	0,2	21,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	21,8
15,25	12,375	0,75	1	–	0,875	180	55,8	0,2	21,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	21,8
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	21,8
15,25	12,375	0,75	1	–	0,875	180	55,8	0,2	21,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	0,2	21,8
15,25	12,375	0,75	1	–	0,875	180	55,8	0,2	21,6
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	0,2	33,2
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	0,2	33,2
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	0,2	33,2
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	0,2	33,2
16,5	13,875	0,625	0,875	2,75	0,75	215	67,8	0,2	36,3
16,5	13,875	0,625	0,875	2,75	0,75	215	67,8	0,2	36,3
16,5	13,875	0,625	0,875	2,75	0,75	215	67,8	0,2	36,3
16,5	13,875	0,625	0,875	2,75	0,75	215	67,8	0,2	36,3
18,375	15,3125	0,6875	1	3,25	0,875	230	73,8	0,2	49,1
18,375	15,3125	0,6875	1	3,25	0,875	230	73,8	0,2	49,1
18,375	15,3125	0,6875	1	3,25	0,875	230	73,8	0,2	49,1
18,375	15,3125	0,6875	1	3,25	0,875	230	73,8	0,2	49,1
20,125	16,5	0,875	1,125	3,375	1	250	77,8	0,2	52,3
20,125	16,5	0,875	1,125	3,375	1	250	77,8	0,2	52,3
20,125	16,5	0,875	1,125	3,375	1	250	77,8	0,2	52,3
21,25	17,625	0,625	1,125	3,75	1	270	82,8	0,2	62,8
21,25	17,625	0,625	1,125	3,75	1	270	82,8	0,2	62,8
21,25	17,625	0,625	1,125	3,75	1	270	82,8	0,2	62,8
22	18,3125	0,9375	1,125	3,75	1	290	89,8	0,2	75
22	18,3125	0,9375	1,125	3,75	1	290	89,8	0,2	75
22	18,3125	0,9375	1,125	3,75	1	290	89,8	0,2	75
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	89,5
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	89,5
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	89,5
24,75	20,5	1,125	1,125	4,25	1	310	95,8	0,2	89,5
26,75	22,25	1,375	1,125	4,625	1	320	95,8	0,2	106,5
26,75	22,25	1,375	1,125	4,625	1	320	95,8	0,2	106,5
26,75	22,25	1,375	1,125	4,625	1	320	95,8	0,2	106,5
26,75	22,25	1,375	1,125	4,625	1	320	95,8	0,2	106,5

4.10.8 SAF500, all sizes, dimensions

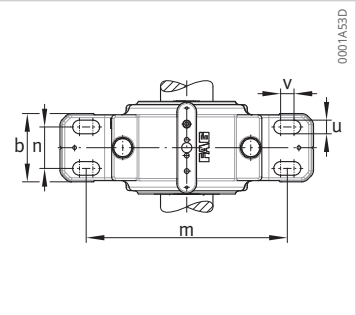
For spherical roller bearings with a tapered bore and adapter sleeve

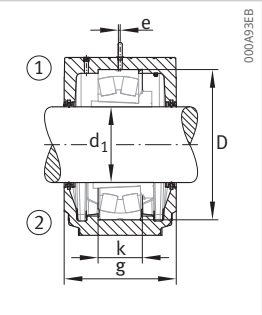
Inch size shaft

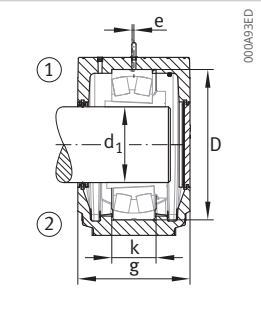




Housing		d_1	h	h_1	g	b	c
Ductile cast iron	**Gray cast iron**	**inch**	**inch**	**inch**	**inch**	**inch**	**inch**
SAFD538×0615U	**SAF538×0615U**	**6,8125**	7,875	15,5625	10,5	7,5	3,125
SAFD538×0615U	**SAF538×0615U**	**6,875**	7,875	15,5625	10,5	7,5	3,125
SAFD538×0615U	**SAF538×0615U**	**6,9375**	7,875	15,5625	10,5	7,5	3,125
SAFD538×0615U	**SAF538×0615U**	**7**	7,875	15,5625	10,5	7,5	3,125
SAFD540×0703U	**–**	**7,125**	8,25	16,5	11	8	3,375
SAFD540×0703U	**–**	**7,1875**	8,25	16,5	11	8	3,375
SAFD540×0703U	**–**	**7,25**	8,25	16,5	11	8	3,375
SAFD544×0715U	**SAF544×0715U**	**7,5**	9,5	18,625	11,75	8,75	3,75
SAFD544×0715U	**SAF544×0715U**	**7,8125**	9,5	18,625	11,75	8,75	3,75
SAFD544×0715U	**SAF544×0715U**	**7,875**	9,5	18,625	11,75	8,75	3,75
SAFD544×0715U	**SAF544×0715U**	**7,9375**	9,5	18,625	11,75	8,75	3,75
SAFD544×0715U	**SAF544×0715U**	**8**	9,5	18,625	11,75	8,75	3,75





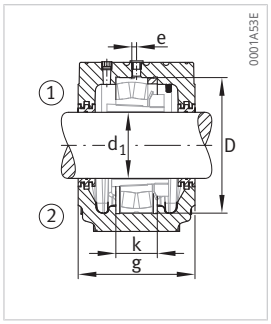


a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	123,4
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	123,4
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	123,4
28	23	1,375	1,375	4,5	1,25	340	101,8	0,2	123,4
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	154
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	154
29,5	23,75	1,25	1,375	5	1,25	360	107,8	0,2	154
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	201,9
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	201,9
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	201,9
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	201,9
32,75	26,3125	1,5625	1,625	5,25	1,5	400	117,8	0,2	201,9

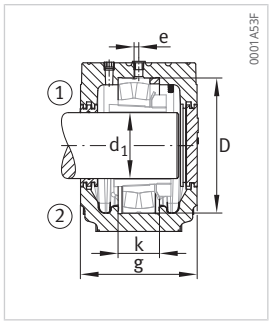
4.10.9 SAF500, all sizes, accessories

For spherical roller bearings with a tapered bore and adapter sleeve

Inch size shaft

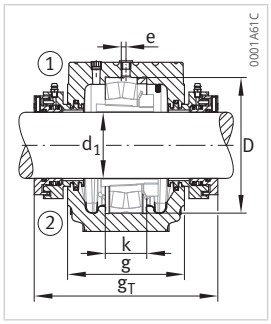


LER labyrinth seal

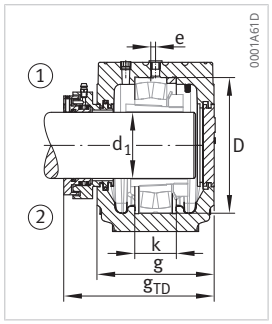


LER labyrinth seal, EC cover

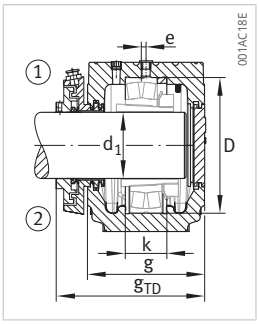
Housing		d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	Gray cast iron	inch	–	Metric	Imperial	–	Quantity
SAFD509×0107U	–	1,375	22209-E1-K	H309×0106	SNW09×0106	SR-9-9	1
SAFD509×0107U	–	1,4375	22209-E1-K	H309×0107	SNW09×0107	SR-9-9	1
SAFD509×0107U	–	1,5	22209-E1-K	H309×0108	SNW09×0108	SR-9-9	1
SAFD510×0111U	–	1,625	22210-E1-K	H310×0110	SNW10×0110	SR-10-0	1
SAFD510×0111U	–	1,6875	22210-E1-K	H310×0111	SNW10×0111	SR-10-0	1
SAFD510×0111U	–	1,75	22210-E1-K	H310×0112	SNW10×0112	SR-10-0	1
SAFD511×0115U	–	1,8125	22211-E1-K	H311×0113	SNW11×0113	SR-11-0	1
SAFD511×0115U	–	1,875	22211-E1-K	H311×0114	SNW11×0114	SR-11-0	1
SAFD511×0115U	–	1,9375	22211-E1-K	H311×0115	SNW11×0115	SR-11-0	1
SAFD513×0203U	–	2	22213-E1-K	H313×0200	SNW13×0200	SR-13-0	1
SAFD511×0115U	–	2	22211-E1-K	H311×0200	SNW11×0200	SR-11-0	1
SAFD513×0203U	–	2,0625	22213-E1-K	H313×0201	SNW13×0201	SR-13-0	1
SAFD513×0203U	–	2,125	22213-E1-K	H313×0202	SNW13×0202	SR-13-0	1
SAFD513×0203U	–	2,1875	22213-E1-K	H313×0203	SNW13×0203	SR-13-0	1
SAFD513×0203U	–	2,25	22213-E1-K	H313×0204	SNW13×0204	SR-13-0	1
SAFD515×0207U	SAF515×0207U	2,375	22215-E1-K	H315×0206	SNW15×0206	SR-15-0	1
FSAFD515×0207U	FSAF515×0207U	2,375	22215-E1-K	H315×0206	SNW15×0206	SR-15-0	1
FSAFD515×0207U	FSAF515×0207U	2,4375	22215-E1-K	H315×0207	SNW15×0207	SR-15-0	1
SAFD515×0207U	SAF515×0207U	2,4375	22215-E1-K	H315×0207	SNW15×0207	SR-15-0	1
SAFD515×0207U	SAF515×0207U	2,5	22215-E1-K	H315×0208	SNW15×0208	SR-15-0	1
FSAFD515×0207U	FSAF515×0207U	2,5	22215-E1-K	H315×0208	SNW15×0208	SR-15-0	1
SAFD516×0211U	–	2,5625	22216-E1-K	H316×0209	SNW16×0209	SR-16-13	1
FSAFD516×0211U	–	2,5625	22216-E1-K	H316×0209	SNW16×0209	SR-16-13	1
SAFD516×0211U	–	2,625	22216-E1-K	H316×0210	SNW16×0210	SR-16-13	1
FSAFD516×0211U	–	2,625	22216-E1-K	H316×0210	SNW16×0210	SR-16-13	1
FSAFD516×0211U	–	2,6875	22216-E1-K	H316×0211	SNW16×0211	SR-16-13	1
SAFD516×0211U	–	2,6875	22216-E1-K	H316×0211	SNW16×0211	SR-16-13	1
SAFD516×0211U	–	2,75	22216-E1-K	H316×0212	SNW16×0212	SR-16-13	1
FSAFD516×0211U	–	2,75	22216-E1-K	H316×0212	SNW16×0212	SR-16-13	1
SAFD517×0215U	SAF517×0215U	2,9375	22217-E1-K	H317×0213	SNW17×0213	SR-17-14	1
SAFD517×0215U	SAF517×0215U	2,9375	22217-E1-K	H317×0214	SNW17×0214	SR-17-14	1
FSAFD517×0215U	FSAF517×0215U	2,9375	22217-E1-K	H317×0215	SNW17×0215	SR-17-14	1
SAFD517×0215U	SAF517×0215U	2,9375	22217-E1-K	H317×0215	SNW17×0215	SR-17-14	1
FSAFD517×0215U	FSAF517×0215U	3	22217-E1-K	H317×0300	SNW17×0300	SR-17-14	1
FSAFD517×0215U	FSAF517×0215U	2,9375	22217-E1-K	H317×0214	SNW17×0214	SR-17-14	1
FSAFD517×0215U	FSAF517×0215U	2,9375	22217-E1-K	H317×0213	SNW17×0213	SR-17-14	1
SAFD517×0215U	SAF517×0215U	3	22217-E1-K	H317×0300	SNW17×0300	SR-17-14	1
SAFD518×0303U	SAF518×0303U	3,215	22218-E1-K	H318×0302	SNW18×0302	SR-18-15	1
FSAFD518×0303U	FSAF518×0303U	3,1875	22218-E1-K	H318×0303	SNW18×0303	SR-18-15	1
SAFD518×0303U	SAF518×0303U	3,1875	22218-E1-K	H318×0303	SNW18×0303	SR-18-15	1



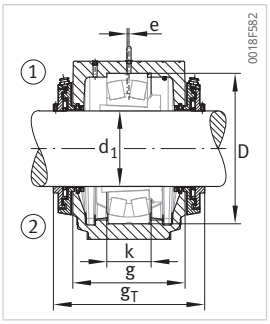
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



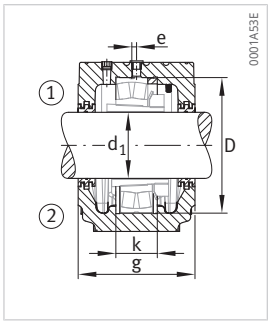
Taconite MaxTAC NTC seal

LER	EC	LER [1]	SuperTAC II TA [1]			MaxTAC NTC [1]		
			–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	2x	–	inch	inch	–	inch	inch
LER17×0107-G-A	EC509-R	LER16×0106-G-A	–	–	–	–	–	–
LER17×0107-G-A	EC509-R	–	TA17×0107	5,9	4,7	–	–	–
LER17×0107-G-A	EC509-R	LER18×0108-G-A	–	–	–	–	–	–
LER20×0111-G-A	EC510-R	LER19×0110-G-A	–	–	–	–	–	–
LER20×0111-G-A	EC510-R	–	TA20×0111	5,85	4,65	–	–	–
LER20×0111-G-A	EC510-R	LER21×0112-G-A	–	–	–	–	–	–
LER24×0115-G-A	EC511-R	LER22×0113-G-A	–	–	–	–	–	–
LER24×0115-G-A	EC511-R	LER23×0114-G-A	–	–	–	–	–	–
LER24×0115-G-A	EC511-R	–	TA24×0115	6,35	5,05	–	–	–
LER29×0203-G-A	EC513-R	LER26×0200-G-A	–	–	–	–	–	–
LER24×0115-G-A	EC511-R	LER25×0200-G-A	–	–	–	–	–	–
LER29×0203-G-A	EC513-R	LER27×0201-G-A	–	–	–	–	–	–
LER29×0203-G-A	EC513-R	LER28×0202-G-A	–	–	–	–	–	–
LER29×0203-G-A	EC513-R	–	TA29×0203	6,9	5,6	NTC29×203	6,65	5,45
LER29×0203-G-A	EC513-R	LER30×0204-G-A	–	–	–	–	–	–
LER37×0207-G-A	EC515-R	LER36×0206-G-A	–	–	–	–	–	–
LER37×0207-G-A	EC515-R	LER36×0206-G-A	–	–	–	–	–	–
LER37×0207-G-A	EC515-R	–	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	–	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	LER38×0208-G-A	–	–	–	–	–	–
LER37×0207-G-A	EC515-R	LER38×0208-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	LER42×0209-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	LER42×0209-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	LER43×0210-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	LER43×0210-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	–	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER44×0211-G-A	EC516-R	–	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER44×0211-G-A	EC516-R	LER45×0212-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	LER45×0212-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	LER51×0213-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	LER52×0214-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	–	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	–	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	LER54×0300-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	LER52×0214-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	LER51×0213-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	LER54×0300-G-A	–	–	–	–	–	–
LER188×0303-G-A	EC518-R	LER187×0302-G-A	–	–	–	–	–	–
LER188×0303-G-A	EC518-R	–	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER188×0303-G-A	EC518-R	–	TA188×0303	7,95	6,75	NTC188×303	8	6,75

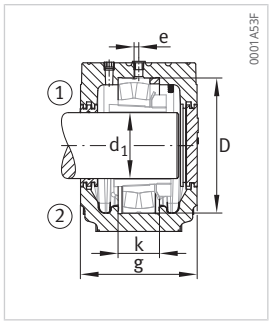
4.10.9 SAF500, all sizes, accessories

For spherical roller bearings with a tapered bore and adapter sleeve

Inch size shaft

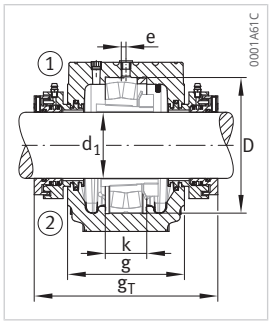


LER labyrinth seal

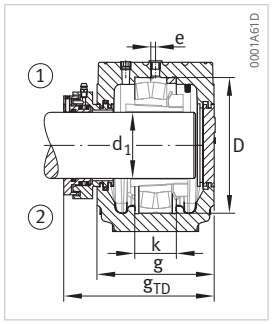


LER labyrinth seal, EC cover

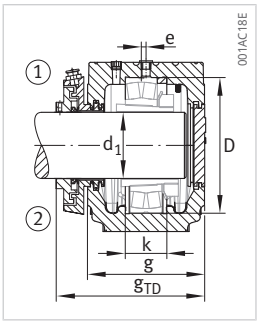
Housing		d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	Gray cast iron	inch	–	Metric	Imperial	–	Quantity
FSAFD518×0303U	FSAF518×0303U	3,125	22218-E1-K	H318×0302	SNW18×0302	SR-18-15	1
SAFD518×0303U	SAF518×0303U	3,25	22218-E1-K	H318×0304	SNW18×0304	SR-18-15	1
FSAFD518×0303U	FSAF518×0303U	3,25	22218-E1-K	H318×0304	SNW18×0304	SR-18-15	1
SAFD520×0307U	SAF520×0307U	3,3125	22220-E1-K	H320×0305	SNW20×0305	SR-20-17	1
FSAFD520×0307U	FSAF520×0307U	3,3125	22220-E1-K	H320×0305	SNW20×0305	SR-20-17	1
SAFD520×0307U	SAF520×0307U	3,375	22220-E1-K	H320×0306	SNW20×0306	SR-20-17	1
FSAFD520×0307U	FSAF520×0307U	3,375	22220-E1-K	H320×0306	SNW20×0306	SR-20-17	1
FSAFD520×0307U	FSAF520×0307U	3,4375	22220-E1-K	H320×0307	SNW20×0307	SR-20-17	1
SAFD520×0307U	SAF520×0307U	3,4375	22220-E1-K	H320×0307	SNW20×0307	SR-20-17	1
FSAFD520×0307U	FSAF520×0307U	3,4375	22220-E1-K	H320×0308	SNW20×0308	SR-20-17	1
SAFD520×0307U	SAF520×0307U	3,5	22220-E1-K	H320×0308	SNW20×0308	SR-20-17	1
SAFD522×0315U	SAF522×0315U	3,8125	22222-E1-K	H322×0313	SNW22×0313	SR-22-19	1
SAFD522×0315U	SAF522×0315U	3,875	22222-E1-K	H322×0314	SNW22×0314	SR-22-19	1
SAFD522×0315U	SAF522×0315U	3,9375	22222-E1-K	H322×0315	SNW22×0315	SR-22-19	1
SAFD522×0315U	SAF522×0315U	4	22222-E1-K	H322×0400	SNW22×0400	SR-22-19	1
SAFD524×0403U	SAF524×0403U	4,0625	22224-E1-K	H3124×0401	SNW24×0401	SR-24-20	1
SAFD524×0403U	SAF524×0403U	4,125	22224-E1-K	H3124×0402	SNW24×0402	SR-24-20	1
SAFD524×0403U	SAF524×0403U	4,1875	22224-E1-K	H3124×0403	SNW24×0403	SR-24-20	1
SAFD524×0403U	SAF524×0403U	4,25	22224-E1-K	H3124×0404	SNW24×0404	SR-24-20	1
SAFD526×0407U	SAF526×0407U	4,3125	22226-E1-K	H3126×0405	SNW26×0405	SR-26-0	1
SAFD526×0407U	SAF526×0407U	4,375	22226-E1-K	H3126×0406	SNW26×0406	SR-26-0	1
SAFD526×0407U	SAF526×0407U	4,4375	22226-E1-K	H3126×0407	SNW26×0407	SR-26-0	1
SAFD526×0407U	SAF526×0407U	4,5	22226-E1-K	H3126×0408	SNW26×0408	SR-26-0	1
SAFD528×0415U	SAF528×0415U	4,8125	22228-E1-K	H3128×0413	SNW28×0413	SR-28-0	1
SAFD528×0415U	SAF528×0415U	4,875	22228-E1-K	H3128×0414	SNW28×0414	SR-28-0	1
SAFD528×0415U	SAF528×0415U	4,9375	22228-E1-K	H3128×0415	SNW28×0415	SR-28-0	1
SAFD530×0503U	–	5,125	22230-E1-K	H3130×0502	SNW30×0502	SR-30-0	1
SAFD530×0503U	–	5,1875	22230-E1-K	H3130×0503	SNW30×0503	SR-30-0	1
SAFD530×0503U	–	5,25	22230-E1-K	H3130×0504	SNW30×0504	SR-30-0	1
SAFD532×0507U	SAF532×0507U	5,375	22232-E1-K	H3132×0506	SNW32×0506	SR-32-0	1
SAFD532×0507U	SAF532×0507U	5,4375	22232-E1-K	H3132×0507	SNW32×0507	SR-32-0	1
SAFD532×0507U	SAF532×0507U	5,5	22232-E1-K	H3132×0508	SNW32×0508	SR-32-0	1
SAFD534×0515U	SAF534×0515U	5,8125	22234-E1-K	H3134×0513	SNW34×0513	SR-34-0	1
SAFD534×0515U	SAF534×0515U	5,875	22234-E1-K	H3134×0514	SNW34×0514	SR-34-0	1
SAFD534×0515U	SAF534×0515U	5,9375	22234-E1-K	H3134×0515	SNW34×0515	SR-34-0	1
SAFD534×0515U	SAF534×0515U	6	22234-E1-K	H3134×0600	SNW34×0600	SR-34-0	1
SAFD536×0607U	SAF536×0607U	6,3125	22236-E1-K	H3136×0605	SNW36×0605	SR-36-30	1
SAFD536×0607U	SAF536×0607U	6,375	22236-E1-K	H3136×0606	SNW36×0606	SR-36-30	1
SAFD536×0607U	SAF536×0607U	6,4375	22236-E1-K	H3136×0607	SNW37×0607	SR-36-30	1
SAFD536×0607U	SAF536×0607U	6,5	22236-E1-K	H3136×0608	SNW36×0608	SR-36-30	1



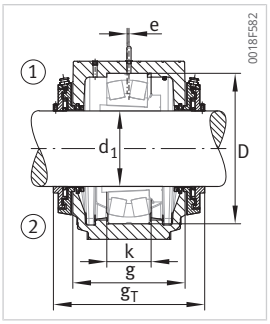
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



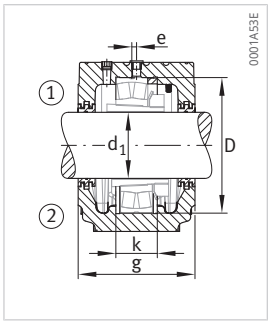
Taconite MaxTAC NTC seal

LER	EC	LER [1]	SuperTAC II TA [1]			MaxTAC NTC [1]		
			–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	2x	–	inch	inch	–	inch	inch
LER188×0303-G-A	EC518-R	LER187×0302-G-A	–	–	–	–	–	–
LER188×0303-G-A	EC518-R	LER189×0304-G-A	–	–	–	–	–	–
LER188×0303-G-A	EC518-R	LER189×0304-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	LER100×0305-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	LER100×0305-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	LER101×0306-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	LER101×0306-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	–	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	–	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	LER103×0308-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	LER103×0308-G-A	–	–	–	–	–	–
LER109×0315-G-A	EC522-R	LER107×0313-G-A	–	–	–	–	–	–
LER109×0315-G-A	EC522-R	LER108×0314-G-A	–	–	–	–	–	–
LER109×0315-G-A	EC522-R	–	TA109×0315	9,45	7,95	NTC109×315	9,3	7,85
LER109×0315-G-A	EC522-R	LER110×0400-G-A	–	–	–	–	–	–
LER113×0403-G-A	EC524-R	LER111×0401-G-A	–	–	–	–	–	–
LER113×0403-G-A	EC524-R	LER112×0402-G-A	–	–	–	–	–	–
LER113×0403-G-A	EC524-R	–	TA113×0403	10,2	8,65	NTC113×403	10,1	8,6
LER113×0403-G-A	EC524-R	LER114×0404-G-A	–	–	–	–	–	–
LER117×0407-G-A	EC526-R	LER115×0405-G-A	–	–	–	–	–	–
LER117×0407-G-A	EC526-R	LER116×0406-G-A	–	–	–	–	–	–
LER117×0407-G-A	EC526-R	–	TA117×0407	11,25	9,5	NTC117×407	10,75	9,25
LER117×0407-G-A	EC526-R	LER118×0408-G-A	–	–	–	–	–	–
LER122×0415-G-A	EC528-R	LER120×0413-G-A	–	–	–	–	–	–
LER122×0415-G-A	EC528-R	LER121×0414-G-A	–	–	–	–	–	–
LER122×0415-G-A	EC528-R	–	TA122×0415	10,85	9,1	NTC122×415	10,35	8,85
LER125×0503-G-A	EC530-R	LER124×0502-G-A	–	–	–	–	–	–
LER125×0503-G-A	EC530-R	–	TA125×0503	11,85	10	NTC125×503	11,25	9,7
LER125×0503-G-A	EC530-R	LER126×0504-G-A	–	–	–	–	–	–
LER130×0507-G-A	EC532-R	LER129×0506-G-A	–	–	–	–	–	–
LER130×0507-G-A	EC532-R	–	TA130×0507	12,15	10,35	NTC130×507	11,8	10,15
LER130×0507-G-A	EC532-R	LER131×0508-G-A	–	–	–	–	–	–
LER140×0515-G-A	EC534-R	LER138×0513-G-A	–	–	–	–	–	–
LER140×0515-G-A	EC534-R	LER139×0514-G-A	–	–	–	–	–	–
LER140×0515-G-A	EC534-R	–	TA140×0515	13,15	11,2	NTC140×515	12,9	11,1
LER140×0515-G-A	EC534-R	LER141×0600-G-A	–	–	–	–	–	–
LER148×0607-G-A	EC536-R	LER146×0605-G-A	–	–	–	–	–	–
LER148×0607-G-A	EC536-R	LER147×0606-G-A	–	–	–	–	–	–
LER148×0607-G-A	EC536-R	–	TA148×0607	13,55	11,6	NTC148×607	12,7	11,15
LER148×0607-G-A	EC536-R	LER149×0608-G-A	–	–	–	–	–	–

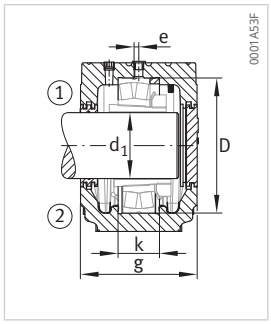
4.10.9 SAF500, all sizes, accessories

For spherical roller bearings with a tapered bore and adapter sleeve

Inch size shaft

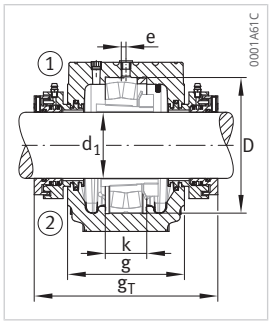


LER labyrinth seal

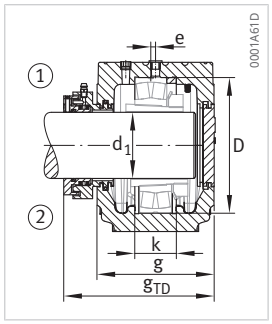


LER labyrinth seal, EC cover

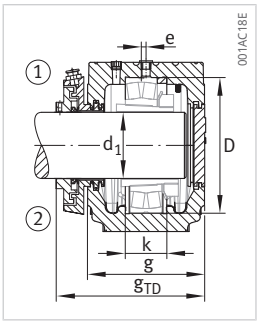
Housing		d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	Gray cast iron	inch	–	Metric	Imperial	–	Quantity
SAFD538×0615U	**SAF538×0615U**	**6,8125**	**22238-E1-K**	H3138×0613	SNW38×0613	SR-38-32	1
SAFD538×0615U	**SAF538×0615U**	**6,875**	**22238-E1-K**	H3138×0614	SNW38×0614	SR-38-32	1
SAFD538×0615U	**SAF538×0615U**	**6,9375**	**22238-E1-K**	H3138×0615	SNW38×0615	SR-38-32	1
SAFD538×0615U	**SAF538×0615U**	**7**	**22238-E1-K**	H3138×0700	SNW38×0700	SR-38-32	1
SAFD540×0703U	–	**7,125**	**22240-E1-K**	H3140×0702	SNW40×0702	SR-40-34	1
SAFD540×0703U	–	**7,1875**	**22240-E1-K**	H3140×0703	SNW40×0703	SR-40-34	1
SAFD540×0703U	–	**7,25**	**22240-E1-K**	H3140×0704	SNW40×0704	SR-40-34	1
SAFD544×0715U	**SAF544×0715U**	**7,5**	**22244-E1-K**	H3144×0708	SNW44×0708	SR-44-38	1
SAFD544×0715U	**SAF544×0715U**	**7,8125**	**22244-E1-K**	H3144×0713	SNW44×0713	SR-44-38	1
SAFD544×0715U	**SAF544×0715U**	**7,875**	**22244-E1-K**	H3144×0714	SNW44×0714	SR-44-38	1
SAFD544×0715U	**SAF544×0715U**	**7,9375**	**22244-E1-K**	H3144×0715	SNW44×0715	SR-44-38	1
SAFD544×0715U	**SAF544×0715U**	**8**	**22244-E1-K**	H3144×0800	SNW44×0800	SR-44-38	1



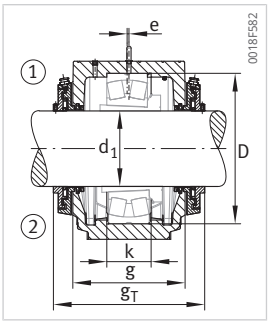
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



Taconite MaxTAC NTC seal

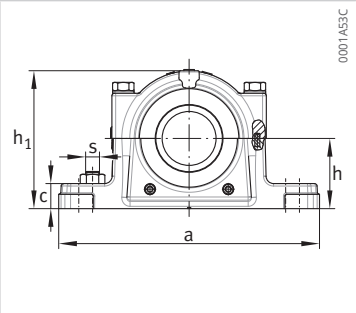
LER	EC	LER [1]	SuperTAC II TA [1]			MaxTAC NTC [1]		
			–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	2x	–	inch	inch	–	inch	inch
LER155×0615-G-A	EC538-R	LER153×0613-G-A	–	–	–	–	–	–
LER155×0615-G-A	EC538-R	LER154×0614-G-A	–	–	–	–	–	–
LER155×0615-G-A	EC538-R	–	TA155×0615	14,55	12,5	NTC155×615	13,8	12,15
LER155×0615-G-A	EC538-R	LER156×0700-G-A	–	–	–	–	–	–
LER159×0703-G-A	EC540-R	LER158×0702-G-A	–	–	–	–	–	–
LER159×0703-G-A	EC540-R	–	TA159×0703	15,3	13,15	NTC159×703	15,55	13,3
LER159×0703-G-A	EC540-R	LER160×0704-G-A	–	–	–	–	–	–
LER167×0715-G-A	EC544-R	LER584×0708-G-A	–	–	–	–	–	–
LER167×0715-G-A	EC544-R	LER165×0713-G-A	–	–	–	–	–	–
LER167×0715-G-A	EC544-R	LER166×0714-G-A	–	–	–	–	–	–
LER167×0715-G-A	EC544-R	–	TA167×0715	16,1	13,9	NTC167×715	16,25	14
LER167×0715-G-A	EC544-R	LER168×0800-G-A	–	–	–	–	–	–

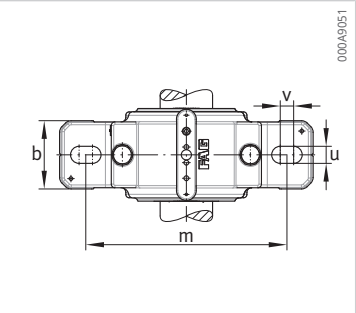
[1] Accessories to be ordered separately

4.10.10 SAF500-WS, dimensions

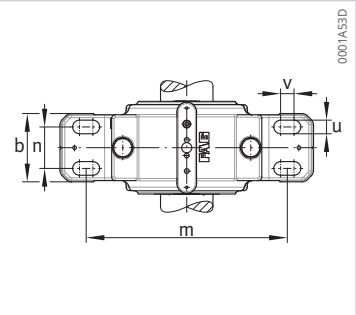
For sealed spherical roller bearings with a tapered bore and adapter sleeve

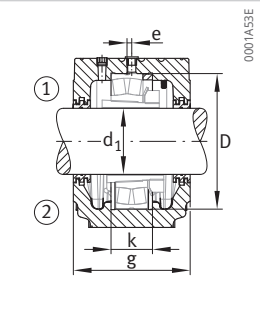
Inch size shaft

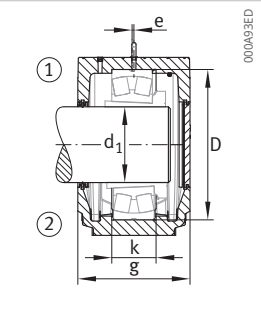




Housing	d_1	h	h_1	g	b	c
Ductile cast iron	**inch**	**inch**	**inch**	**inch**	**inch**	**inch**
SAFD513×0203U-WS	**2,1875**	3	5,9375	4,25	3,125	1
FSAFD515×0207U-WS	**2,4375**	3,25	6,375	4,625	3,125	1,125
SAFD515×0207U-WS	**2,4375**	3,25	6,375	4,625	3,125	1,125
SAFD516×0211U-WS	**2,6875**	3,5	6,875	4,75	3,5	1,25
FSAFD516×0211U-WS	**2,6875**	3,5	6,875	4,75	3,5	1,25
SAFD517×0215U-WS	**2,9375**	3,75	7,3125	5	3,5	1,25
FSAFD517×0215U-WS	**2,9375**	3,75	7,3125	5	3,5	1,25
SAFD518×0303U-WS	**3,1875**	4	7,75	5,875	3,875	1,625
FSAFD518×0303U-WS	**3,1875**	4	7,75	5,875	3,875	1,625
SAFD520×0307U-WS	**3,4375**	4,5	8,4375	6,125	4,375	1,75
FSAFD520×0307U-WS	**3,4375**	4,5	8,4375	6,125	4,375	1,75
SAFD522×0315U-WS	**3,9375**	4,9375	9,625	6,4375	4,75	2
SAFD524×0403U-WS	**4,1875**	5,25	10,25	7,375	4,75	2,125
SAFD526×0407U-WS	**4,4375**	6	11,5	7,75	5,125	3,375





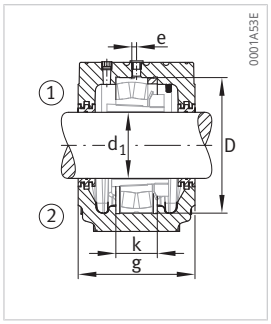


a	m	v	u	n	s	D	k	e	m
inch	inch	inch	inch	inch	inch	mm	mm	inch	kg
11	8,8125	0,6875	0,75	–	0,625	120	42	0,1	8,6
11,25	9,125	0,5	0,625	1,875	0,5	130	44	0,1	10,1
11,25	9,125	0,5	0,75	–	0,625	130	44	0,1	10
13	10,3125	0,6875	0,875	–	0,75	140	44,9	0,2	13,6
13	10,3125	0,6875	0,75	2,125	0,625	140	44,9	0,2	13,8
13	10,4375	0,56	0,875	–	0,75	150	48,9	0,2	14,7
13	10,4375	0,56	0,75	2,125	0,625	150	48,9	0,2	14,7
13,75	11	0,625	0,875	–	0,75	160	52,9	0,1	17,8
13,75	11	0,625	0,75	2,125	0,625	160	52,9	0,1	18,2
15,25	12,375	0,75	1	–	0,875	180	59,9	0,1	21,4
15,25	12,375	0,75	0,875	2,375	0,75	180	59,9	0,1	21,6
16,5	13,5625	0,9375	0,875	2,75	0,75	200	67,9	0,15	32,4
16,5	13,875	0,625	0,875	2,75	0,75	215	73,9	0,1	37,2
18,375	15,3125	0,6875	1	3,25	0,875	230	84,8	0,2	49,8

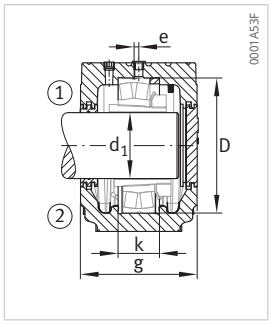
4.10.11 SAF500-WS, accessories

For sealed spherical roller bearings with a tapered bore and adapter sleeve

Inch size shaft

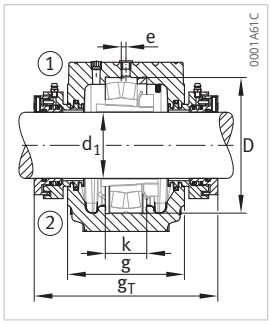


LER labyrinth seal

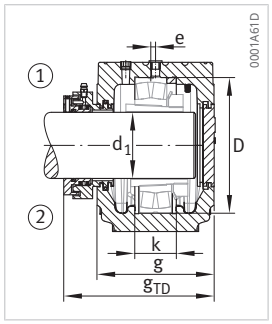


LER labyrinth seal, EC cover

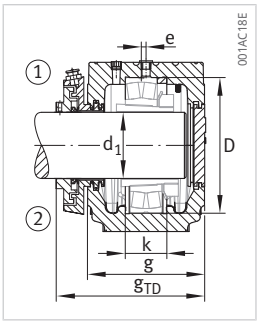
Housing	d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]	Locating ring	
Ductile cast iron	inch	–	Metric	–	Quantity
SAFD513×0203U-WS	**2,1875**	**WS22213-E1-K-2RSR**	H313×203-T-WS	FRM120/4	1
FSAFD515×0207U-WS	**2,4375**	**WS22215-E1-K-2RSR**	H315×207-T-WS	SR-15-0	1
SAFD515×0207U-WS	**2,4375**	**WS22215-E1-K-2RSR**	H315×207-T-WS	SR-15-0	1
SAFD516×0211U-WS	**2,6875**	**WS22216-E1-K-2RSR**	H316×211-T-WS	SR-16-13-1	1
FSAFD516×0211U-WS	**2,6875**	**WS22216-E1-K-2RSR**	H316×211-T-WS	SR-16-13-1	1
SAFD517×0215U-WS	**2,9375**	**WS22217-E1-K-2RSR**	H317×215-T-WS	SR-17-14-1	1
FSAFD517×0215U-WS	**2,9375**	**WS22217-E1-K-2RSR**	H317×215-T-WS	SR-17-14-1	1
SAFD518×0303U-WS	**3,1875**	**WS22218-E1-K-2RSR**	H318×303-T-WS	SR-18-15-1	1
FSAFD518×0303U-WS	**3,1875**	**WS22218-E1-K-2RSR**	H318×303-T-WS	SR-18-15-1	1
SAFD520×0307U-WS	**3,4375**	**WS22220-E1-K-2RSR**	H320×307-T-WS	SR-20-17-1	1
FSAFD520×0307U-WS	**3,4375**	**WS22220-E1-K-2RSR**	H320×307-T-WS	SR-20-17-1	1
SAFD522×0315U-WS	**3,9375**	**WS22222-E1-K-2RSR**	H322×315-T-WS	SR-22-19-1	1
SAFD524×0403U-WS	**4,1875**	**WS22224-E1-K-2RSR**	H3124×403-T-WS	SR-24-20-1	1
SAFD526×0407U-WS	**4,4375**	**WS22226-E1-K-2RSR**	H3126×407-T-WS	SR-26-0	1



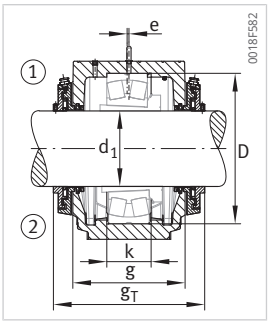
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



Taconite MaxTAC NTC seal

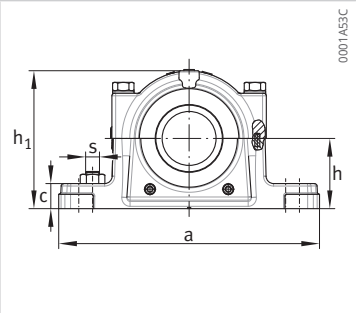
LER	EC	SuperTAC II TA [1]			MaxTAC NTC [1]		
		–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	–	inch	inch	–	inch	inch
LER29×0203-G-A	EC513-R	TA29×0203	6,9	5,6	NTC29×203	6,65	5,45
LER37×0207-G-A	EC515-R	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	TA37×0207	7,35	6	NTC37×207	7,35	6
LER44×0211-G-A	EC516-R	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER44×0211-G-A	EC516-R	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER53×0215-G-A	EC517-R	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER188×0303-G-A	EC518-R	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER188×0303-G-A	EC518-R	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER102×0307-G-A	EC520-R	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER109×0315-G-A	EC522-R	TA109×0315	9,45	7,95	NTC109×315	9,3	7,85
LER113×0403-G-A	EC524-R	TA113×0403	10,2	8,65	NTC113×403	10,1	8,6
LER117×0407-G-A	EC526-R	TA117×0407	11,25	9,5	NTC117×407	10,75	9,25

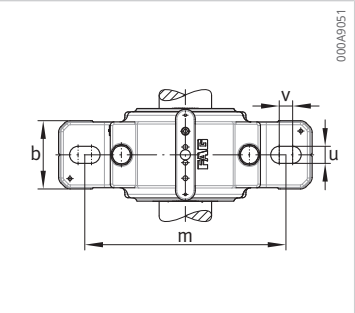
[1] Accessories to be ordered separately

4.10.12 SAF1500, preferred sizes, dimensions

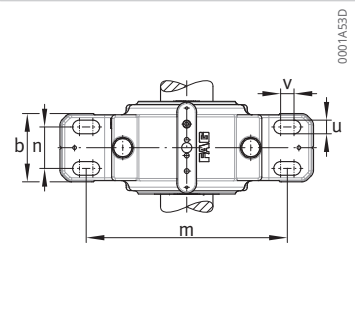
For spherical ball bearings with a tapered bore and adapter sleeve

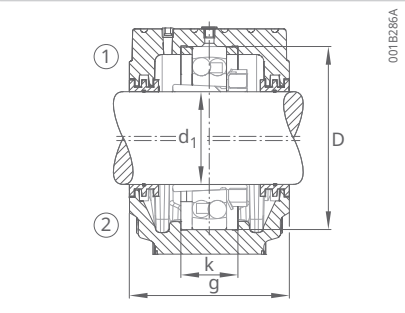
Inch size shaft

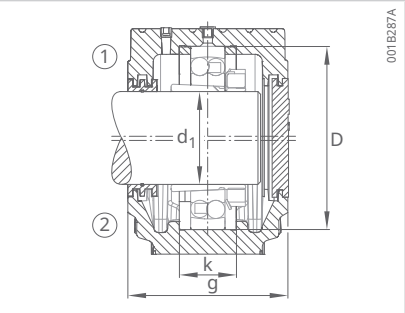




Housing		d_1	h	h_1	g	b	c
Ductile cast iron	Gray cast iron	inch	inch	inch	inch	inch	inch
SAFD1509×0107U	–	**1,4375**	2,25	4,375	3,4375	2,375	0,8125
SAFD1510×0111U	–	**1,6875**	2,5	4,75	3,4375	2,375	0,9375
SAFD1511×0115U	–	**1,9375**	2,75	5,25	3,75	2,75	0,9375
SAFD1513×0203U	–	**2,1875**	3	5,9375	4,25	3,125	1
SAFD1515×0207U	**SAF515×0207U**	**2,4375**	3,25	6,375	4,625	3,125	1,125
FSAFD1515×0207U	**FSAF515×0207U**	**2,4375**	3,25	6,375	4,625	3,125	1,125
SAFD1516×0211U	–	**2,6875**	3,5	6,875	4,75	3,5	1,25
FSAFD1516×0211U	–	**2,6875**	3,5	6,875	4,75	3,5	1,25
SAFD1517×0215U	**SAF517×0215U**	**2,9375**	3,75	7,3125	4,8125	3,5	1,25
FSAFD1517×0215U	**FSAF517×0215U**	**2,9375**	3,75	7,3125	4,8125	3,5	1,25
SAFD1518×0303U	**SAF518×0303U**	**3,1875**	4	7,75	5,5	3,875	1,625
FSAFD1518×0303U	**FSAF518×0303U**	**3,1875**	4	7,75	5,5	3,875	1,625
SAFD1520×0307U	**SAF520×0307U**	**3,4375**	4,5	8,4375	5,8125	4,375	1,75
FSAFD1520×0307U	**FSAF520×0307U**	**3,4375**	4,5	8,4375	5,8125	4,375	1,75
SAFD1522×0315U	**SAF522×0315U**	**3,9375**	4,9375	9,625	6,4375	4,75	2





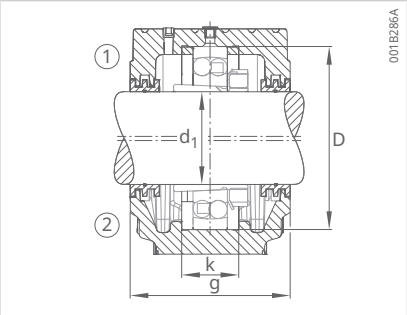


a	m	v	u	n	s	D	k	m
inch	inch	inch	inch	inch	inch	mm	mm	kg
8,25	6,625	0,375	0,625	–	0,5	85	28,8	3,6
8,25	6,75	0,25	0,625	–	0,5	90	29,8	4
9,625	7,625	0,25	0,75	–	0,625	100	30,8	5,3
11	8,8125	0,6875	0,75	–	0,625	120	39	8,6
11,25	9,125	0,5	0,75	–	0,625	130	37	10
11,25	9,125	0,5	0,625	1,875	0,5	130	37	10,6
13	10,3125	0,6875	0,875	–	0,75	140	42,8	13,7
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	13,9
13	10,4375	0,56	0,875	–	0,75	150	45,8	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	14
13,75	11	0,625	0,875	–	0,75	160	49,8	16,9
13,75	11	0,625	0,75	2,125	0,625	160	49,8	17
15,25	12,375	0,75	1	–	0,875	180	55,8	21,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	21,8
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	33,2

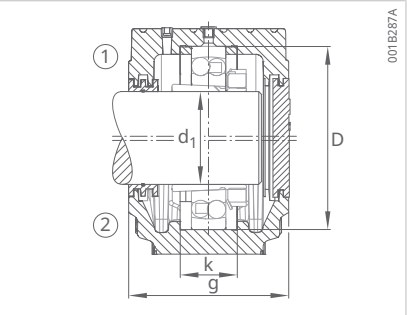
4.10.13 SAF1500, preferred sizes, accessories

For spherical ball bearings with a tapered bore and adapter sleeve

Inch size shaft

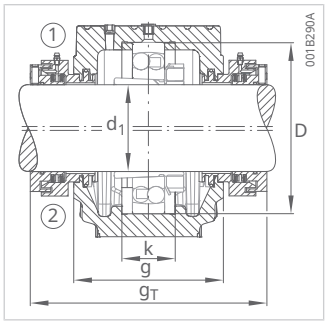


LER labyrinth seal

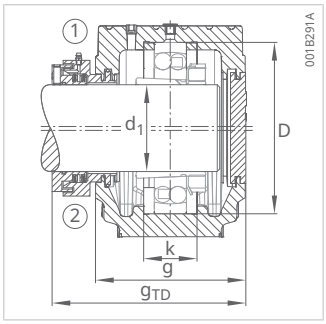


LER labyrinth seal, EC cover

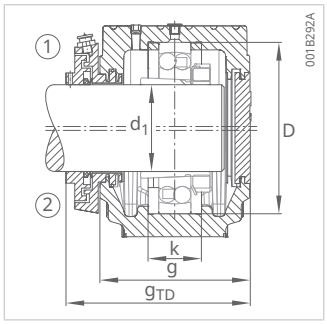
Housing		d_1	Spherical roller bearing [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	Gray cast iron	inch	–	Metric	Imperial	–	Quantity
SAFD1509×0107U	**–**	**1,4375**	**1209-K-TVH**	H209×107	SNW09×0107	SR-9-0	1
SAFD1510×0111U	**–**	**1,6875**	**1210-K-TVH**	H210×111	SNW10×0111	SR-10-8	1
SAFD1511×0115U	**–**	**1,9375**	**1211-K-TVH**	H211×115	SNW11×0115	SR-11-9	1
SAFD1513×0203U	**–**	**2,1875**	**1213-K-TVH**	H213×203	SNW13×0203	SR-13-0	2
SAFD1515×0207U	**SAF515×0207U**	**2,4375**	**1215-K-TVH**	H215×207	SNW15×0207	SR-15-0	2
FSAFD1515×0207U	**FSAF515×0207U**	**2,4375**	**1215-K-TVH**	H215×207	SNW15×0207	SR-15-0	2
SAFD1516×0211U	**–**	**2,6875**	**1216-K-TVH**	H216×211	SNW16×0211	SR-1516	2
FSAFD1516×0211U	**–**	**2,6875**	**1216-K-TVH**	H216×211	SNW16×0211	SR-1516	2
SAFD1517×0215U	**SAF517×0215U**	**2,9375**	**1217-K-TVH**	H217×215	SNW17×0215	SR-1517	2
FSAFD1517×0215U	**FSAF517×0215U**	**2,9375**	**1217-K-TVH**	H217×215	SNW17×0215	SR-1517	2
SAFD1518×0303U	**SAF518×0303U**	**3,1875**	**1218-K-TVH**	H218×303	SNW18×0303	SR-1518	2
FSAFD1518×0303U	**FSAF518×0303U**	**3,1875**	**1218-K-TVH**	H218×303	SNW18×0303	SR-1518	2
SAFD1520×0307U	**SAF520×0307U**	**3,4375**	**1220-K-M**	H220×307	SNW20×0307	SR-1520	2
FSAFD1520×0307U	**FSAF520×0307U**	**3,4375**	**1220-K-M**	H220×307	SNW20×0307	SR-1520	2
SAFD1522×0315U	**SAF522×0315U**	**3,9375**	**1222-K-M**	H222×315	SNW22×0315	SR-1522	2



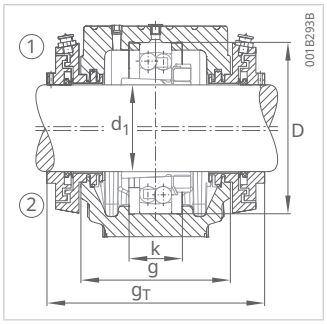
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



Taconite MaxTAC NTC seal

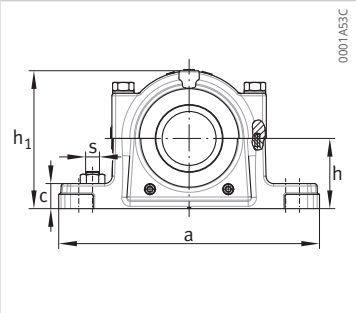
LER	EC	SuperTAC II TA [1]			MaxTAC NTC [1]		
		–	g_T	g_{TD}	–	g_T	g_{TD}
2x	–	–	inch	inch	–	inch	inch
LER17×0107-G-A	EC509-R	TA17×0107	5,9	4,7	–	–	–
LER20×0111-G-A	EC510-R	TA20×0111	5,85	4,65	–	–	–
LER24×0115-G-A	EC511-R	TA24×0115	6,35	5,05	–	–	–
LER29×0203-G-A	EC513-R	TA29×0203	6,9	5,6	NTC29×203	6,65	5,45
LER37×0207-G-A	EC515-R	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	TA37×0207	7,35	6	NTC37×207	7,35	6
LER44×0211-G-A	EC516-R	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER44×0211-G-A	EC516-R	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER53×0215-G-A	EC517-R	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER188×0303-G-A	EC518-R	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER188×0303-G-A	EC518-R	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER102×0307-G-A	EC520-R	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER109×0315-G-A	EC522-R	TA109×0315	9,45	7,95	NTC109×315	9,3	7,85

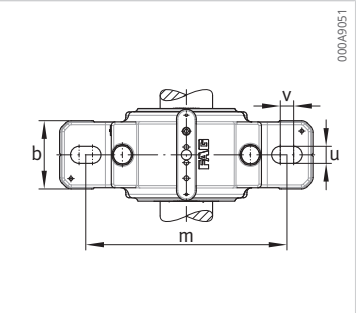
[1] Accessories to be ordered separately

4.10.14 SAF1500, all sizes, dimensions

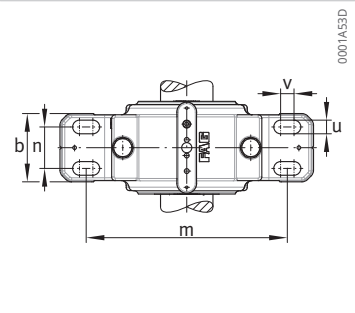
For spherical ball bearings with a tapered bore and adapter sleeve

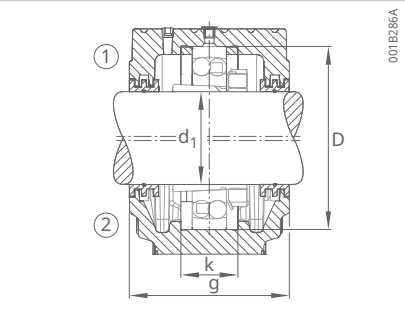
Inch size shaft

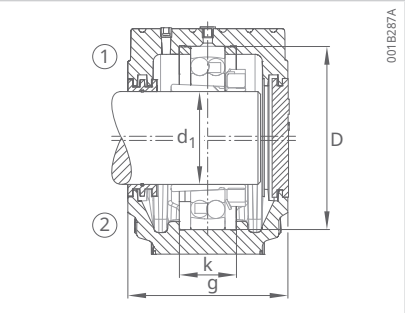




Housing		d_1	h	h_1	g	b	c
Ductile cast iron	Gray cast iron	inch	inch	inch	inch	inch	inch
SAFD1509×0107U	**–**	**1,4375**	2,25	4,375	3,4375	2,375	0,8125
SAFD1509×0107U	**–**	**1,5**	2,25	4,375	3,4375	2,375	0,8125
SAFD1510×0111U	**–**	**1,6875**	2,5	4,75	3,4375	2,375	0,9375
SAFD1510×0111U	**–**	**1,75**	2,5	4,75	3,4375	2,375	0,9375
SAFD1511×0115U	**–**	**1,9375**	2,75	5,25	3,75	2,75	0,9375
SAFD1511×0115U	**–**	**2**	2,75	5,25	3,75	2,75	0,9375
SAFD1513×0203U	**–**	**2,1875**	3	5,9375	4,25	3,125	1
SAFD1513×0203U	**–**	**2,25**	3	5,9375	4,25	3,125	1
SAFD1515×0207U	**SAF515×0207U**	**2,4375**	3,25	6,375	4,625	3,125	1,125
FSAFD1515×0207U	**FSAF515×0207U**	**2,4375**	3,25	6,375	4,625	3,125	1,125
SAFD1515×0207U	**SAF515×0207U**	**2,5**	3,25	6,375	4,625	3,125	1,125
FSAFD1515×0207U	**FSAF515×0207U**	**2,5**	3,25	6,375	4,625	3,125	1,125
SAFD1516×0211U	**–**	**2,6875**	3,5	6,875	4,75	3,5	1,25
FSAFD1516×0211U	**–**	**2,6875**	3,5	6,875	4,75	3,5	1,25
SAFD1516×0211U	**–**	**2,75**	3,5	6,875	4,75	3,5	1,25
FSAFD1516×0211U	**–**	**2,75**	3,5	6,875	4,75	3,5	1,25
SAFD1517×0215U	**SAF517×0215U**	**2,9375**	3,75	7,3125	4,8125	3,5	1,25
FSAFD1517×0215U	**FSAF517×0215U**	**3**	3,75	7,3125	4,8125	3,5	1,25
FSAFD1517×0215U	**FSAF517×0215U**	**2,9375**	3,75	7,3125	4,8125	3,5	1,25
SAFD1517×0215U	**SAF517×0215U**	**3**	3,75	7,3125	4,8125	3,5	1,25
SAFD1518×0303U	**SAF518×0303U**	**3,1875**	4	7,75	5,5	3,875	1,625
FSAFD1518×0303U	**FSAF518×0303U**	**3,1875**	4	7,75	5,5	3,875	1,625
SAFD1518×0303U	**SAF518×0303U**	**3,25**	4	7,75	5,5	3,875	1,625
FSAFD1518×0303U	**FSAF518×0303U**	**3,25**	4	7,75	5,5	3,875	1,625
SAFD1520×0307U	**SAF520×0307U**	**3,4375**	4,5	8,4375	5,8125	4,375	1,75
FSAFD1520×0307U	**FSAF520×0307U**	**3,4375**	4,5	8,4375	5,8125	4,375	1,75
SAFD1520×0307U	**SAF520×0307U**	**3,5**	4,5	8,4375	5,8125	4,375	1,75
FSAFD1520×0307U	**FSAF520×0307U**	**3,5**	4,5	8,4375	5,8125	4,375	1,75
SAFD1522×0315U	**SAF522×0315U**	**3,9375**	4,9375	9,625	6,4375	4,75	2
SAFD1522×0315U	**SAF522×0315U**	**4**	4,9375	9,625	6,4375	4,75	2





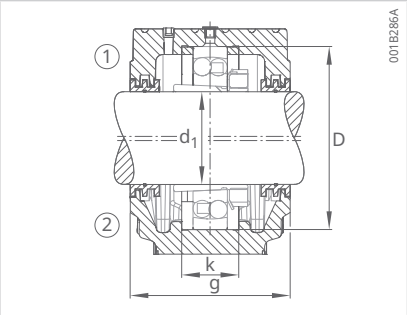


a	m	v	u	n	s	D	k	m
inch	inch	inch	inch	inch	inch	mm	mm	kg
8,25	6,625	0,375	0,625	–	0,5	85	28,8	3,6
8,25	6,625	0,375	0,625	–	0,5	85	28,8	3,6
8,25	6,75	0,25	0,625	–	0,5	90	29,8	4
8,25	6,75	0,25	0,625	–	0,5	90	29,8	4
9,625	7,625	0,25	0,75	–	0,625	100	30,8	5,3
9,625	7,625	0,25	0,75	–	0,625	100	30,8	5,3
11	8,8125	0,6875	0,75	–	0,625	120	39	8,6
11	8,8125	0,6875	0,75	–	0,625	120	39	8,6
11,25	9,125	0,5	0,75	–	0,625	130	37	10
11,25	9,125	0,5	0,625	1,875	0,5	130	37	10,6
11,25	9,125	0,5	0,75	–	0,625	130	37	10
11,25	9,125	0,5	0,625	1,875	0,5	130	37	10,6
13	10,3125	0,6875	0,875	–	0,75	140	42,8	13,7
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	13,9
13	10,3125	0,6875	0,875	–	0,75	140	42,8	13,7
13	10,3125	0,6875	0,75	2,125	0,625	140	42,8	13,9
13	10,4375	0,56	0,875	–	0,75	150	45,8	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	14
13	10,4375	0,56	0,75	2,125	0,625	150	45,8	14
13	10,4375	0,56	0,875	–	0,75	150	45,8	14
13,75	11	0,625	0,875	–	0,75	160	49,8	16,9
13,75	11	0,625	0,75	2,125	0,625	160	49,8	17
13,75	11	0,625	0,875	–	0,75	160	49,8	16,9
13,75	11	0,625	0,75	2,125	0,625	160	49,8	17
15,25	12,375	0,75	1	–	0,875	180	55,8	21,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	21,8
15,25	12,375	0,75	1	–	0,875	180	55,8	21,6
15,25	12,375	0,75	0,875	2,375	0,75	180	55,8	21,8
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	33,2
16,5	13,5625	0,9375	0,875	2,75	0,75	200	62,8	33,2

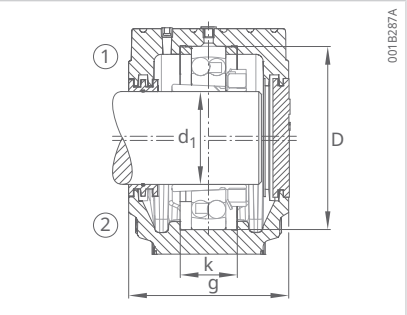
4.10.15 SAF1500, all sizes, accessories

For spherical ball bearings with a tapered bore and adapter sleeve

Inch size shaft

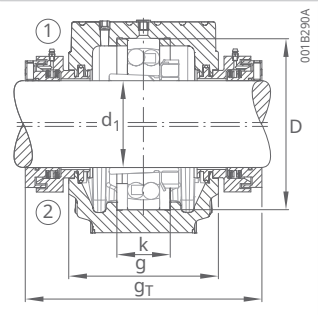


LER labyrinth seal

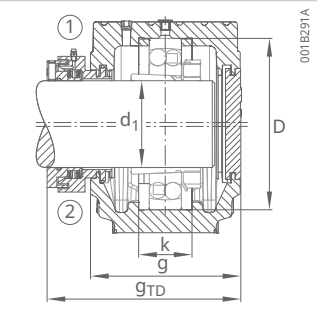


LER labyrinth seal, EC cover

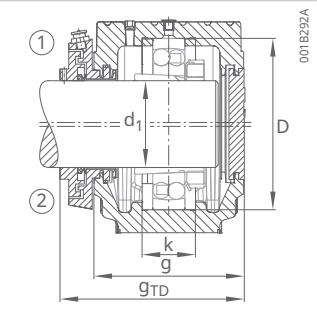
Housing		d_1	Spherical ball bearings [1]	Clamping sleeve, thread [1]		Locating ring	
Ductile cast iron	Gray cast iron	inch	–	Metric	Imperial	–	Quantity
SAFD1509×0107U	**–**	**1,4375**	**1209-K-TVH**	H209×107	SNW09×0107	SR-9-0	1
SAFD1509×0107U	**–**	**1,5**	**1209-K-TVH**	H209×108	SNW09×0108	SR-9-0	1
SAFD1510×0111U	**–**	**1,6875**	**1210-K-TVH**	H210×111	SNW10×0111	SR-10-8	1
SAFD1510×0111U	**–**	**1,75**	**1210-K-TVH**	H210×112	SNW10×0112	SR-10-8	1
SAFD1511×0115U	**–**	**1,9375**	**1211-K-TVH**	H211×115	SNW11×0115	SR-11-9	1
SAFD1511×0115U	**–**	**2**	**1211-K-TVH**	H211×200	SNW11×0200	SR-11-9	1
SAFD1513×0203U	**–**	**2,1875**	**1213-K-TVH**	H213×203	SNW13×0203	SR-13-0	2
SAFD1513×0203U	**–**	**2,25**	**1213-K-TVH**	H213×204	SNW13×0204	SR-13-0	2
SAFD1515×0207U	**SAF515×0207U**	**2,4375**	**1215-K-TVH**	H215×207	SNW15×0207	SR-15-0	2
FSAFD1515×0207U	**FSAF515×0207U**	**2,4375**	**1215-K-TVH**	H215×207	SNW15×0207	SR-15-0	2
SAFD1515×0207U	**SAF515×0207U**	**2,5**	**1215-K-TVH**	H215×208	SNW15×0208	SR-15-0	2
FSAFD1515×0207U	**FSAF515×0207U**	**2,5**	**1215-K-TVH**	H215×208	SNW15×0208	SR-15-0	2
SAFD1516×0211U	**–**	**2,6875**	**1216-K-TVH**	H216×211	SNW16×0211	SR-1516	2
FSAFD1516×0211U	**–**	**2,6875**	**1216-K-TVH**	H216×211	SNW16×0211	SR-1516	2
SAFD1516×0211U	**–**	**2,75**	**1216-K-TVH**	H216×212	SNW16×0212	SR-1516	2
FSAFD1516×0211U	**–**	**2,75**	**1216-K-TVH**	H216×212	SNW16×0212	SR-1516	2
SAFD1517×0215U	**SAF517×0215U**	**2,9375**	**1217-K-TVH**	H217×215	SNW17×0215	SR-1517	2
FSAFD1517×0215U	**FSAF517×0215U**	**3**	**1217-K-TVH**	H217×300	SNW17×0300	SR-1517	2
FSAFD1517×0215U	**FSAF517×0215U**	**2,9375**	**1217-K-TVH**	H217×215	SNW17×0215	SR-1517	2
SAFD1517×0215U	**SAF517×0215U**	**3**	**1217-K-TVH**	H217×300	SNW17×0300	SR-1517	2
SAFD1518×0303U	**SAF518×0303U**	**3,1875**	**1218-K-TVH**	H218×303	SNW18×0303	SR-1518	2
FSAFD1518×0303U	**FSAF518×0303U**	**3,1875**	**1218-K-TVH**	H218×303	SNW18×0303	SR-1518	2
SAFD1518×0303U	**SAF518×0303U**	**3,25**	**1218-K-TVH**	H218×304	SNW18×0304	SR-1518	2
FSAFD1518×0303U	**FSAF518×0303U**	**3,25**	**1218-K-TVH**	H218×304	SNW18×0304	SR-1518	2
SAFD1520×0307U	**SAF520×0307U**	**3,4375**	**1220-K-M**	H220×307	SNW20×0307	SR-1520	2
FSAFD1520×0307U	**FSAF520×0307U**	**3,4375**	**1220-K-M**	H220×307	SNW20×0307	SR-1520	2
SAFD1520×0307U	**SAF520×0307U**	**3,5**	**1220-K-M**	H220×308	SNW20×0308	SR-1520	2
FSAFD1520×0307U	**FSAF520×0307U**	**3,5**	**1220-K-M**	H220×308	SNW20×0308	SR-1520	2
SAFD1522×0315U	**SAF522×0315U**	**3,9375**	**1222-K-M**	H222×315	SNW22×0315	SR-1522	2
SAFD1522×0315U	**SAF522×0315U**	**4**	**1222-K-M**	H222×0400	SNW22×0400	SR-1522	2



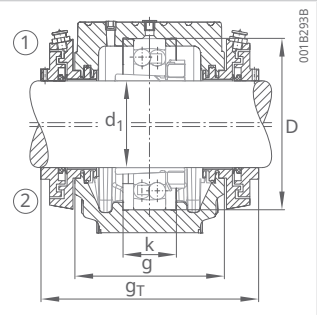
Taconite SuperTAC II TA seal



Taconite SuperTAC II TA seal, EC cover



Taconite MaxTAC NTC seal, EC cover



Taconite MaxTAC NTC seal

LER	EC	LER [1]	SuperTAC II TA [1]			MaxTAC NTC [1]		
			–	g_T	g_{TD}		g_T	g_{TD}
2x	–	2x	–	inch	inch		inch	inch
LER17×0107-G-A	EC509-R	–	TA17×0107	5,9	4,7	–	–	–
LER17×0107-G-A	EC509-R	LER18×0108-G-A	–	–	–	–	–	–
LER20×0111-G-A	EC510-R	–	TA20×0111	5,85	4,65	–	–	–
LER20×0111-G-A	EC510-R	LER21×0112-G-A	–	–	–	–	–	–
LER24×0115-G-A	EC511-R	–	TA24×0115	6,35	5,05	–	–	–
LER24×0115-G-A	EC511-R	LER25×0200-G-A	–	–	–	–	–	–
LER29×0203-G-A	EC513-R	–	TA29×0203	6,9	5,6	NTC29×203	6,65	5,45
LER29×0203-G-A	EC513-R	LER30×0204-G-A	–	–	–	–	–	–
LER37×0207-G-A	EC515-R	–	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	–	TA37×0207	7,35	6	NTC37×207	7,35	6
LER37×0207-G-A	EC515-R	LER38×0208-G-A	–	–	–	–	–	–
LER37×0207-G-A	EC515-R	LER38×0208-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	–	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER44×0211-G-A	EC516-R	–	TA44×0211	7,45	6,15	NTC44×211	7,45	6,15
LER44×0211-G-A	EC516-R	LER45×0212-G-A	–	–	–	–	–	–
LER44×0211-G-A	EC516-R	LER45×0212-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	–	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	LER54×0300-G-A	–	–	–	–	–	–
LER53×0215-G-A	EC517-R	–	TA53×0215	7,7	6,3	NTC53×215	7,7	6,3
LER53×0215-G-A	EC517-R	LER54×0300-G-A	–	–	–	–	–	–
LER188×0303-G-A	EC518-R	–	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER188×0303-G-A	EC518-R	–	TA188×0303	7,95	6,75	NTC188×303	8	6,75
LER188×0303-G-A	EC518-R	LER189×0304-G-A	–	–	–	–	–	–
LER188×0303-G-A	EC518-R	LER189×0304-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	–	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	–	TA102×0307	8,95	7,45	NTC102×307	8,7	7,3
LER102×0307-G-A	EC520-R	LER103×0308-G-A	–	–	–	–	–	–
LER102×0307-G-A	EC520-R	LER103×0308-G-A	–	–	–	–	–	–
LER109×0315-G-A	EC522-R	–	TA109×0315	9,45	7,95	NTC109×315	9,3	7,85
LER109×0315-G-A	EC522-R	LER110×0400-G-A	–	–	–	–	–	–

[1] Accessories to be ordered separately

5 S30 split plummer block housing

medias | Product catalogue |
S30 split plummer block housings |
https://www.schaeffler.de/std/21FB

S30 split plummer block housings can be used for various applications, such as sintering and pelletising equipment as well as paper processing machinery.

5.1 Housing design

Bearing housing

90 S30 split plummer block housings

00081B4D

5.1.1 Locating and non-locating bearings

The bearing seats in the housing are machined so that the bearings can move, in other words they act as non-locating bearings. Locating bearings can be achieved by the insertion of an FRM locating ring next to the bearing outer ring.

Locating rings must be ordered separately.

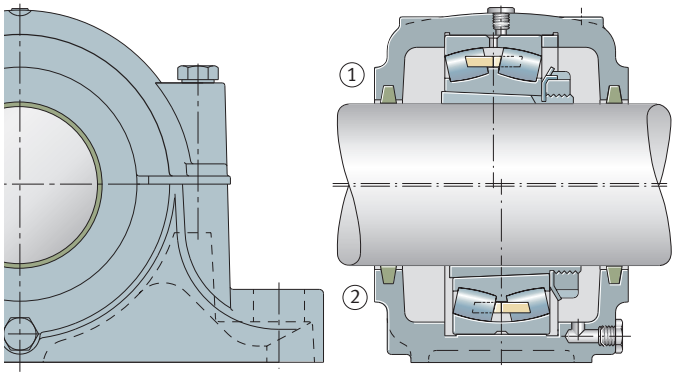
5.1.2 Housing configurations

The modular design of plummer block housings allows a wide range of possible combinations.

The following features can be considered for the housing configuration:

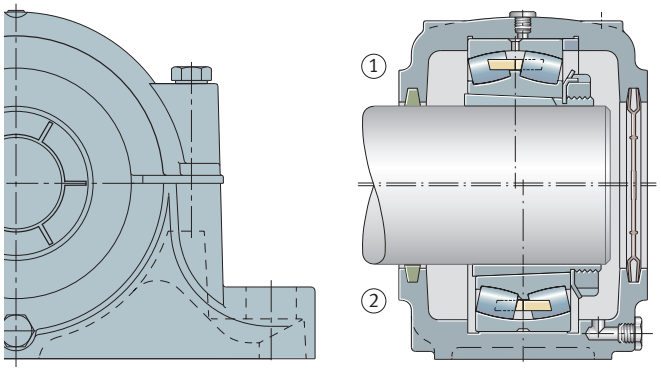
- Fastening: bearings with a tapered bore by an adapter sleeve on a shaft with a constant diameter or bearings with a cylindrical bore on a stepped shaft
- Continuous shaft or a housing closed on one side
- Design of the bearing as a locating bearing or a non-locating bearing
- Spherical roller bearing, split or unsplit.

91 S30 plummer block housing for spherical roller bearings with a tapered bore and adapter sleeve, felt seals on both sides



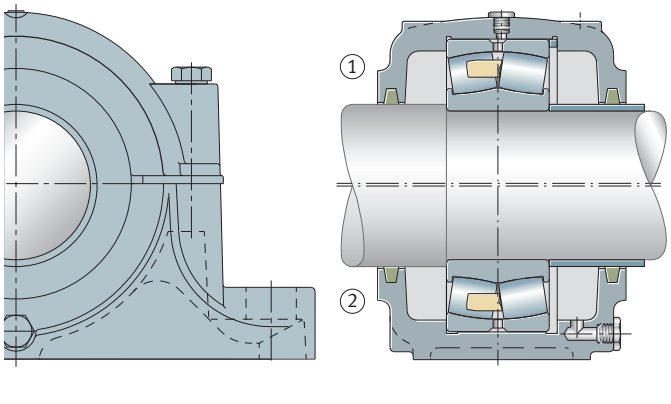
1	Locating bearing	2	Non-locating bearing

92 S30 plummer block housing for bearings with a tapered bore and adapter sleeve, felt seal and DK cover



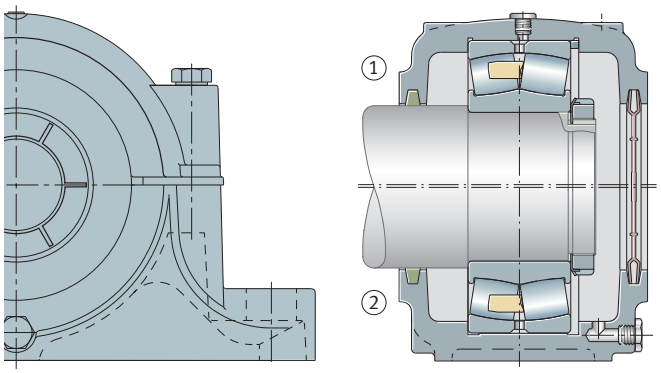
1	Locating bearing	2	Non-locating bearing

93 S30 plummer block housing for bearings with a cylindrical bore, felt seals on both sides



1	Locating bearing	2	Non-locating bearing

94 S30 plummer block housing for bearings with a tapered bore, felt seal and DK cover

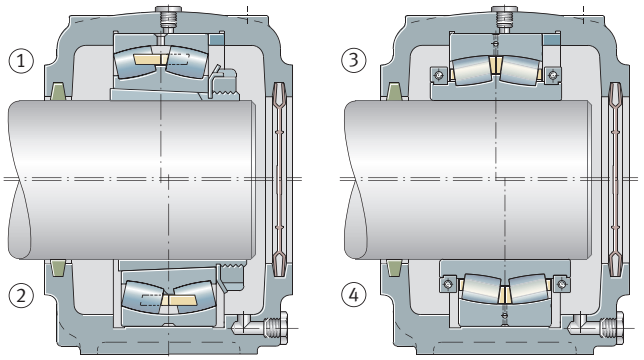


1	Locating bearing	2	Non-locating bearing

Installation of split spherical roller bearing

For S30 plummer block housings, an unsplit spherical roller bearing with a tapered bore and adapter sleeve can be replaced by a split spherical roller bearing.

95 S30 plummer block housings with split and unsplit spherical roller bearing



1	Locating bearing with unsplit bearing	2	Non-locating bearing with unsplit bearing
3	Locating bearing with split bearing	4	Non-locating bearing with split bearing

5.1.3 Accessories

Accessories must be ordered separately.

Seals

96 DK cover

000822F5

Locating rings

97 FRM locating ring

0001AD2A

5.2 Suitable bearings

The dimensions of S30 split plummer block housings are tailored to the spherical roller bearings of the series 230. The shaft diameters are 110 mm to 150 mm.

32 Suitable bearings for S30 split plummer block housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	23024..-K	23034..-K
Spherical roller bearings with a cylindrical bore	23024	23034

Split spherical roller bearings

In bearing replacement, unsplit spherical roller bearings with an adapter sleeve can be replaced by split spherical roller bearings. In many cases, the use of split spherical roller bearings significantly reduces the work associated with bearing replacement.

In order to ensure selection of the correct combination of housing and bearing, please contact us.

Further information

TPI 250 | Split spherical roller bearings |
https://www.schaeffler.de/std/1F81

5.3 Materials, corrosion protection

33 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron GJL	L	✓	RAL 7016, anthracite grey
Ductile cast iron GJS	D	o	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

5.4 Load capacity

The guide values for the housing rupture load apply to the standard housing material gray cast iron. 1,6 times this value applies for ductile cast iron.

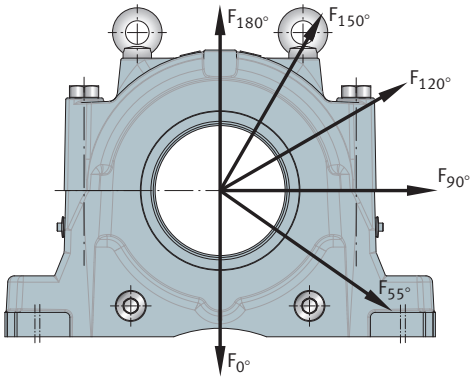
When determining the permissible static load, safety factors must be applied:

- safety factor of 6 relative to the guide values: housing rupture load for gray cast iron, ductile cast iron
- safety factor of 3 relative to the maximum load carrying capacity of the connecting screws and foot screws

Additional factors influencing load carrying capacity:

- If the housing is subjected to axial load, the permissible axial load of the bearing fitted must be taken into consideration.
- If an adapter sleeve is used to locate the bearing on the shaft, the axial retaining force of the bearing and adapter sleeve must also be taken into consideration.
- The housing must be secured by stoppers in the load direction for a load direction between 55 ° and 120 ° or for an axial load.

98 Load directions F for the guide values for the housing rupture load and the maximum load capacity of the connecting screws



34 Guide values for housing rupture load and maximum load capacity of connecting screws

Housing	Housing rupture load in load direction F [1]					Connecting screws			
						Thread	Max. load capacity of 2 screws in load direction F [2]		
	55 °	90 °	120 °	150 °	180 °		120 °	150 °	180 °
	kN	kN	kN	kN	kN	–	kN	kN	kN
S3024	540	320	245	215	270	M20	260	150	130
S3026	620	370	280	250	310	M20	260	150	130
S3028	700	420	315	280	350	M20	260	150	130
S3030	780	470	350	310	390	M20	260	150	130
S3032	860	520	390	345	430	M20	260	150	130
S3034	1000	600	450	400	500	M24	360	210	180

[1] For housings made from flake graphite cast iron. Housings made from spheroidal graphite cast iron have a housing rupture load that is 1,6 times higher than that of housings made from flake graphite cast iron.

[2] Maximum load carrying capacity in load direction F for maintaining contact at the parting surface between the upper and lower housing sections

5.5 Lubrication

S30 split plummer block housings are designed for grease lubrication.

5.5.1 Grease quantities

For initial greasing, the basic rule is that the bearing should be filled with grease to 100 % and the free volume of the housing to 60 %. The free volume is the space that remains in the housing once the bearing, adapter sleeve, shaft and seals have been fitted.

35 Recommended grease quantity for initial grease application

Housing	Recommended grease quantity
	≈ g
S3024	390
S3026	560
S3028	630
S3030	730
S3032	970
S3034	1100

5.5.2 Relubrication

There is a threaded hole M10×1 at the top of the housing. When delivered, this is sealed with a plastic stopper. A taper type or button type lubrication nipple can be screwed into this for relubrication.

The lubrication nipples are not included in the package for S30 plummer block housings.

5.6 Sealing

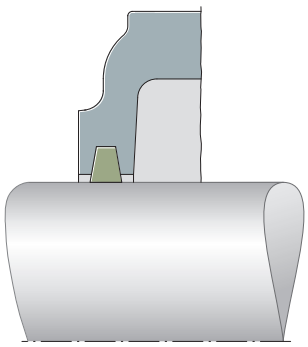
Seals

S30 plummer block housings are sealed with felt seals.

The seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication.

The oil-saturated felt strips must be inserted in the relevant slots in the housing. 2 felt strips are required for a housing closed on one side, whilst 4 felt strips are required for a continuous shaft.

99 Felt seal for S30 plummer block housing



0001AC55

Cover

A DK cover can be inserted instead of the felt strips is inserted in the slots on one side of the housing on a housing closed on one side. The covers are made of polyamide. They must be ordered separately.

5.7 Dimensions, tolerances

5.7.1 Tolerances for the bearing seat

The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1.

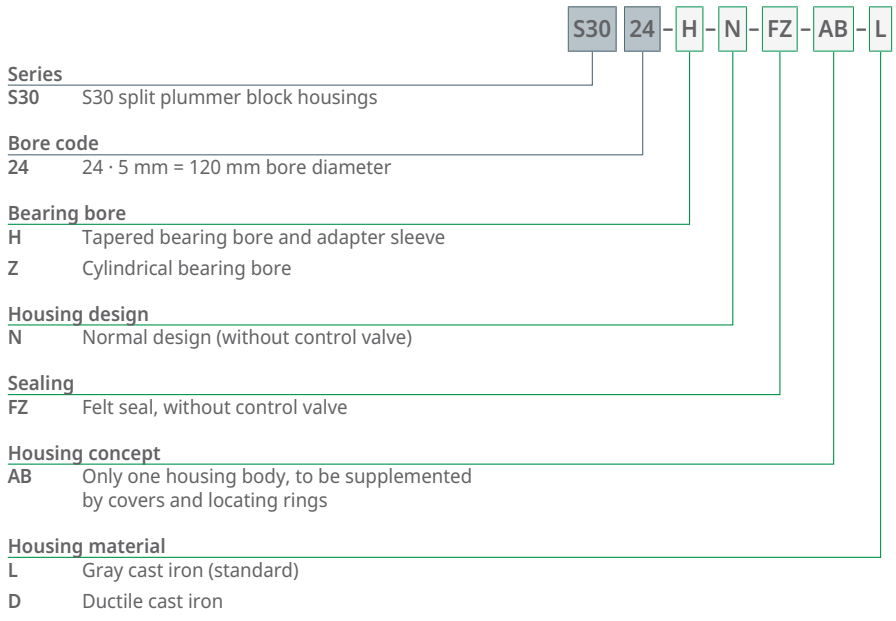
By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

5.8 Structure of the ordering designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

Bearing housing

100 Structure of the designations of S30 plummer block housings

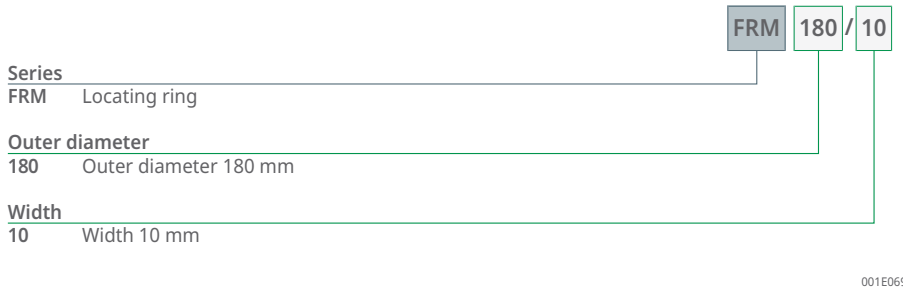


Cover

101 Structure of the designations of covers

Locating rings

102 Structure of the designations of locating rings



5.8.1 Specimen orders

For ordering a S30 split plummer block housing, the housing designation only describes the housing body. The other components such as covers or locating rings must be ordered separately in the required design. The roller bearing and, if necessary, the adapter sleeve must also be ordered separately.

Housings with an installed bearing produce non-locating bearings. These can be converted into locating bearings by adding locating rings.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

The felt strips for sealing the housing are supplied in the package and do not have to be ordered separately.

Example 1

Example of a housing configuration, characteristics:

- S30 plummer block housing made of gray cast iron, closed on one side
- 23024-E1-K-TVPB spherical roller bearing in the form of a locating bearing
- Secured by means of adapter sleeve
- Felt seal

36 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	S3024-H-N-FZ-AB-L
1	Spherical roller bearing	23024-E1-K-TVPB
1	Adapter sleeve	H3024
1	Locating ring	FRM180/10
1	Cover	DK127-135

Example 2

Example of a housing configuration, characteristics:

- S30 plummer block housing made of gray cast iron, for a continuous shaft
- 23034-E1-TVPB spherical roller bearing in the form of a non-locating bearing
- Felt seal

37 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	S3034-Z-N-FZ-AB-L
1	Spherical roller bearing	23034-E1-TVPB

5.9 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

5.9.1 Eye bolts

S3034 housings have 1 eye bolt in the upper part of the housing to DIN 580:2018. This is designed for use as an anchoring point for the installation and removal of the housing. The load capacity of the eye bolt allows lifting of the housing including a bearing fitted in the housing.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

5.9.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

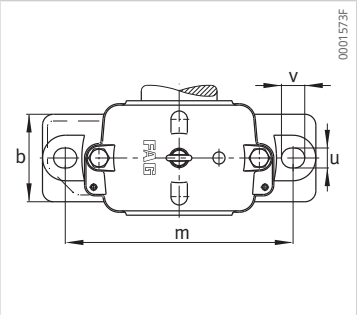
The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

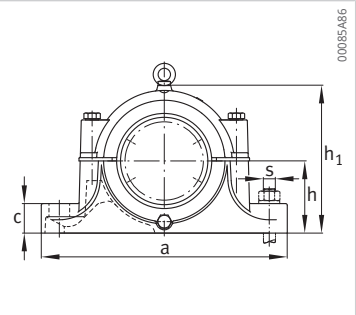
5.10 Product tables

5.10.1 Explanations of the product tables

(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
b	mm	Base width
c	mm	Base height
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
g	mm	Housing body width
g_3	mm	Wall thickness
h	mm	Distance to shaft axis
h_1	mm	Overall height
m	mm	Fixing hole spacing
m	kg	Mass
n	mm	Distance between fastening holes
n_F	–	Required number of felt strips: With a housing closed on one side/with a continuous shaft
s	inch or mm	Screw size
u	mm	Slot width
v	mm	Slot length

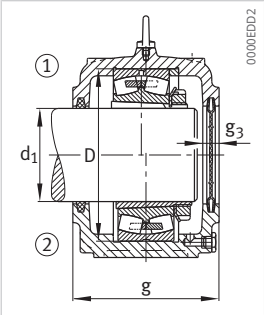
5.10.2 S30, for spherical roller bearings with a tapered bore and adapter sleeve



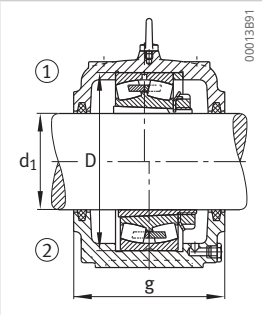


5

Housing	d_1	h	h_1	g	b	c	a	m	v	u	s		D	g_3	m
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	–	inch	mm	mm	≈ kg
S3024-H-N-FZ-AB-L	**110**	112	215	150	110	40	390	320	36	30	M24	1	180	18	16,5
S3026-H-N-FZ-AB-L	**115**	125	239	160	120	45	420	350	36	30	M24	1	200	18	19,3
S3028-H-N-FZ-AB-L	**125**	140	259	170	120	45	420	350	36	30	M24	1	210	21	24,6
S3030-H-N-FZ-AB-L	**135**	150	278	175	130	45	460	380	36	30	M24	1	225	21	29
S3032-H-N-FZ-AB-L	**140**	150	288	190	130	50	470	390	36	30	M24	1	240	21	37
S3034-H-N-FZ-AB-L	**150**	160	320	200	160	55	540	450	48	36	M30	$1\ ^1/_8$	260	25	45



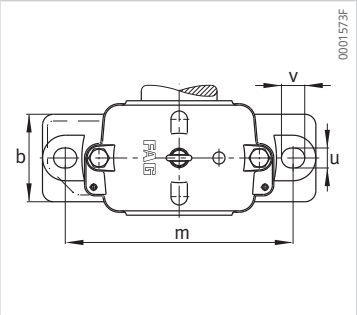
Felt strip, DK cover

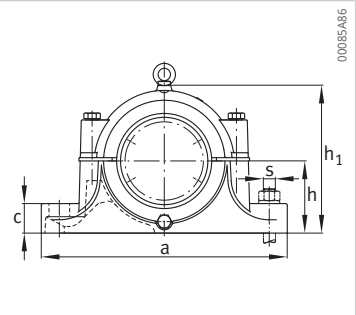


Felt strip

Bearing	Adapter sleeve	Locating ring	Cover	Felt strip	n_F
–	–	–	–	–	–
23024-E1-K-TVPB	H3024	FRM180/10	DK127-135	10×8,5×195	2 / 4
23026-E1-K-TVPB	H3026	FRM200/10	DK127-135	10×8,5×200	2 / 4
23028-E1-K-TVPB	H3028	FRM210/10	DK147-155	14×11×225	2 / 4
23030-E1-K-TVPB	H3030	FRM225/10	DK156-163	14×11×240	2 / 4
23032-E1-K-TVPB	H3032	FRM240/10	DK166-182	14×11×250	2 / 4
23034-E1-K-TVPB	H3034	FRM260/10	DK166-182	16×12×270	2 / 4

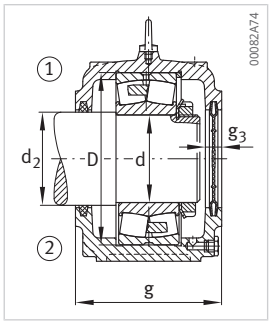
5.10.3 S30, for spherical roller bearings with a cylindrical bore



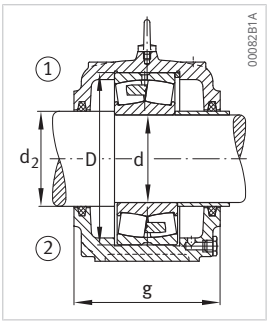


5

Housing	d	d_2	h	h_1	g	b	c	a	m	v	u	s		D	g_3	m
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm		inch	mm	mm	≈ kg
S3024-Z-N-FZ-AB-L	**120**	130	112	215	150	110	40	390	320	36	30	M24	1	180	18	16,5
S3026-Z-N-FZ-AB-L	**130**	145	125	239	160	120	45	420	350	36	30	M24	1	200	18	19,3
S3028-Z-N-FZ-AB-L	**140**	155	140	259	170	120	45	420	350	36	30	M24	1	210	21	24,6
S3030-Z-N-FZ-AB-L	**150**	160	150	278	175	130	45	460	380	36	30	M24	1	225	21	29
S3032-Z-N-FZ-AB-L	**160**	175	150	288	190	130	50	470	390	36	30	M24	1	240	21	37
S3034-Z-N-FZ-AB-L	**170**	185	160	320	200	160	55	540	450	48	36	M30	$1\,^1/_8$	260	25	45



Felt strip, DK cover



Felt strip

Bearing	Locating ring	Cover	Felt strip	n_F
–	–	–	–	–
23024	FRM180/10	DK147-155	10×8,5×410	1 / 1
23026	FRM200/10	DK156-163	10×8,5×445	1 / 1
23028	FRM210/10	DK166-182	14×11×470	1 / 1
23030	FRM225/10	DK185-197	14×11×505	1 / 2
23032	FRM240/10	DK200-212	14×11×535	1 / 2
23034	FRM260/10	DK200-212	16×12×565	1 / 2

6 SNS split plummer block housing

medias | Product catalogue |
SNS split plummer block housings |
https://www.schaeffler.de/std/21FC

SNS split plummer block housings and the relevant bearings form bearing units that can be tailored to a wide range of requirements, through the specific selection of seal design, lubrication type and bearing mounting. The housings are specifically designed for the location of spherical roller bearings.

6.1 Housing design

Bearing housing

103 SNS..-H and SNS..-Z plummer block housing, split, for labyrinth seal and Taconite seal

0001AD70

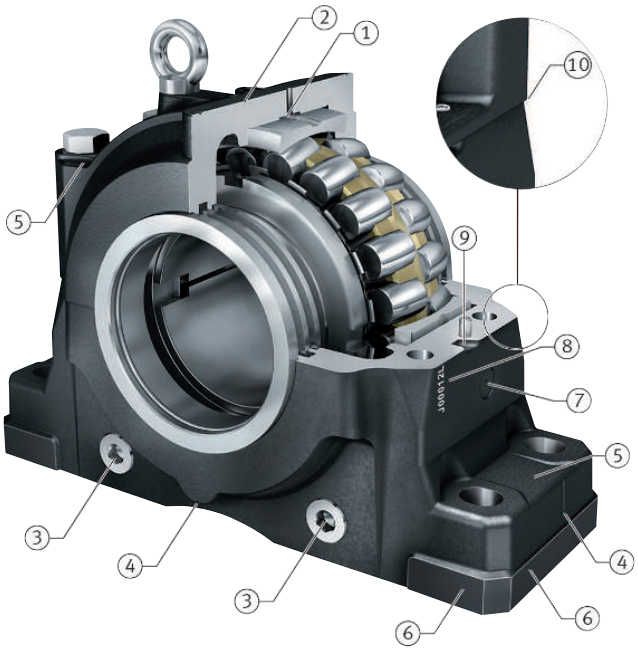
104 SNS..-B plummer block housing, split, for bolt-on seal

0008D681

6.1.1 Advantages of the detailed design

A large number of details in the housing design that are systematically tailored to practical applications ensures the particularly flexible and economical usage of SNS plummer block housings.

105 Advantages of the detailed design



0001ABF8

1	Lubrication hole with long axial lubrication groove	2	Pilot hole at predefined position for condition monitoring
3	Grease outlet holes	4	Indentations indicating shaft and bearing centres
5	Inclined run-off areas	6	Machined locating surfaces
7	Prepared mounting points for condition monitoring	8	Identification of the housing halves
9	Dismantling notches	10	Projection of upper housing section

6.1.1.1 Lubrication hole with long axial lubrication groove

A lubrication hole with a long axial lubrication groove in the upper housing section provides lubricant to all bearing series that can be installed regardless of the bearing position. Suitable lubrication nipples are included in the package.

6.1.1.2 Grease outlet holes

Grease outlet holes in the lower housing section allow the escape of used and excess grease during relubrication.

6.1.1.3 Indentations to mark the housing centre

Indentations on the end and side surfaces of the lower housing section mark the housing centre. The marks ensure rapid alignment and reduce assembly time.

6.1.1.4 Inclined run-off areas

Inclined run-off areas on the upper and lower housing sections help remove fluids and reduce dirt deposits. This means less cleaning work and a lower risk of corrosion.

6.1.1.5 Machined locating surfaces

Machined locating surfaces on the lower housing section allow the precise alignment of the housing. If the housing is coated, the locating surfaces also get a universal coating. Since the locating surfaces are raised, strips can be used as an alignment aid. One-off alignment is sufficient to allow rapid replacement of the housing when necessary.

6.1.1.6 Prepared mounting points for condition monitoring

To allow the simple use of condition monitoring, the lower housing section has a connection point on both sides for condition monitoring systems. The M6 thread and a flattened area with a diameter of 26 mm allow the connection of the Schaeffler SmartCheck, a modular online measuring system for machine monitoring. Alternatively, other vibration sensors can also be connected. If neither of the two connection points is accessible as a result of the surrounding structures, there is another suitable connection point on the upper section marked by cast pilot hole.

The Schaeffler SmartCheck must be ordered separately.

Further information

TPI 214 | Schaeffler SmartCheck |
https://www.schaeffler.de/std/1B6C

6.1.1.7 Identification of the housing halves

The upper and lower housing sections are tailored to each other and must not be mixed with other housing halves. The serial numbering on both halves of the housing ensures their correction allocation. Furthermore, the serial numbering allows traceability and simplified documentation.

6.1.1.8 Dismounting notches

Recessed dismounting notches on the interface of the housing halves make it easier to remove the upper housing section.

6.1.1.9 Projection of upper housing section

The projection of the upper housing ensures the following:

- It prevents the collection of fluids and contaminants at the interface of the two housing halves
- It reduces the risk of dirt entering the housing
- It reduces the risk of corrosion

6.1.2 Modular concept

The housings are designed in accordance with a modular concept. Each housing can be fitted with rolling bearings of various diameter series and width series if they have the correct outside diameter for the housing.

Depending on their design, the bearings can be located either directly on the shaft or by means of an adapter sleeve. This gives different shaft diameters for the same bearing size. The spacings between the shaft and housing body are compensated by means of appropriately matched seals.

6.1.3 Locating and non-locating bearings

The bearing seats in the housing are machined so that the bearings can move and can act as non-locating bearings. Locating bearings can be achieved by inserting NFR locating rings on both sides of the bearing outer ring. Depending on the combination of housing and bearing, 2, 4 or 6 locating rings may be required.

Locating rings must be ordered separately.

6.1.4 Housing configurations

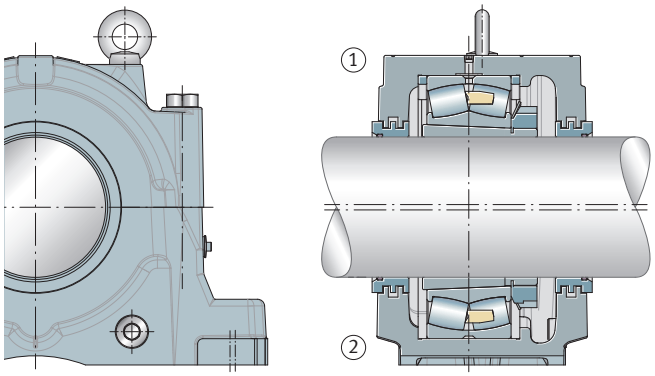
The modular design of plummer block housings allows a wide range of possible combinations.

The following features can be considered for the housing configuration:

- Fastening of bearings with a tapered bore by an adapter sleeve on a shaft with a constant diameter or bearings with a cylindrical bore on a stepped shaft
- Housing sealing by a labyrinth seal (unsplit or split), Taconite seal (unsplit or split) or bolt-on seal (suitable for housing versions SNS..-B)
- Continuous shaft or a housing closed on one side
- Design of the bearing as a locating bearing or a non-locating bearing
- Spherical roller bearing, split or unsplit.

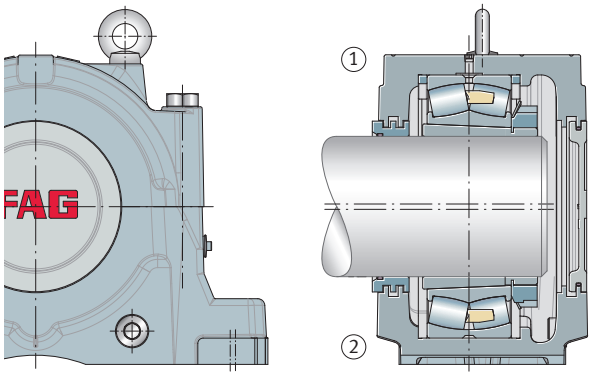
The resulting possible housing configurations, make it possible to meet various requirements for the bearings using standard components.

106 SNS plummer block housings for bearings with a tapered bore and adapter sleeve, 2 NTS labyrinth seals



1	Locating bearing	2	Non-locating bearing

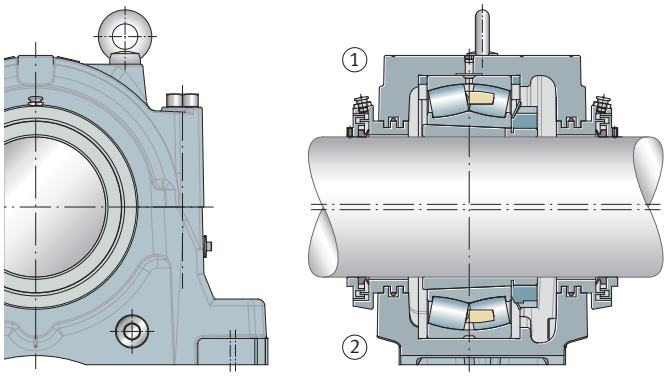
107 SNS plummer block housings for bearings with a tapered bore and adapter sleeve, NTS labyrinth seal and NDK cover



00019E4A

1	Locating bearing	2	Non-locating bearing

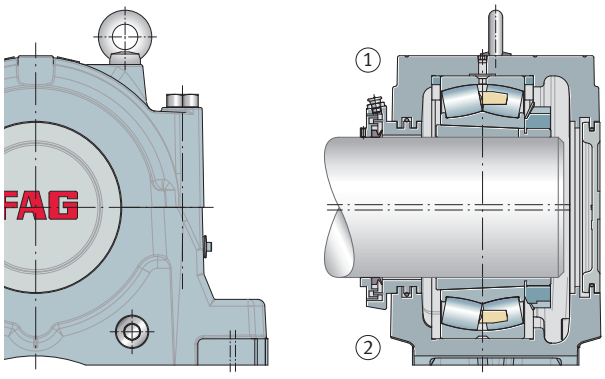
108 SNS plummer block housings for bearings with a tapered bore and adapter sleeve, 2 Taconite NTC seals



00019E47

1	Locating bearing	2	Non-locating bearing

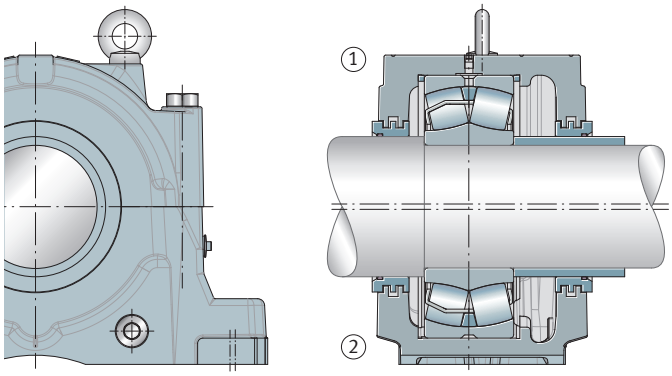
109 SNS plummer block housings for bearings with a tapered bore and adapter sleeve, Taconite NTC seal and NDK cover



00019E49

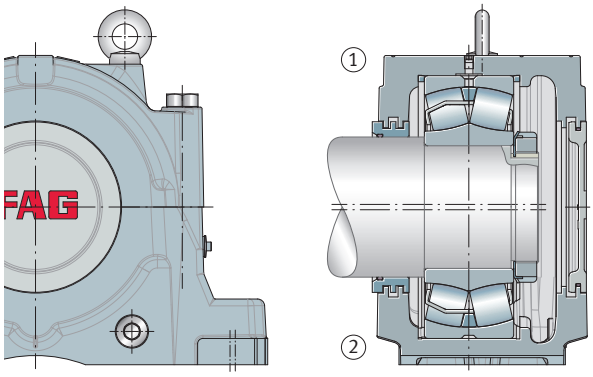
1	Locating bearing	2	Non-locating bearing

110 SNS plummer block housings for bearings with a cylindrical bore, 2 NTS labyrinth seals



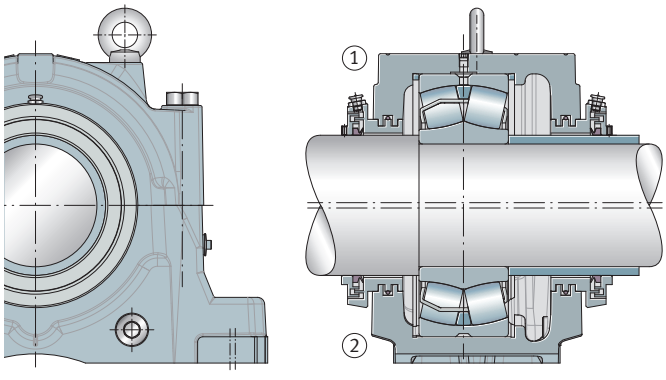
1	Locating bearing	2	Non-locating bearing

111 SNS plummer block housings for bearings with a cylindrical bore, NTS labyrinth seal and NDK cover



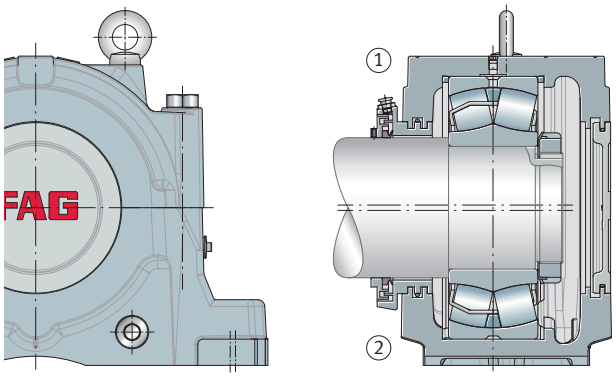
1	Locating bearing	2	Non-locating bearing

112 SNS plummer block housings for bearings with a cylindrical bore, 2 Taconite NTC seals



1	Locating bearing	2	Non-locating bearing

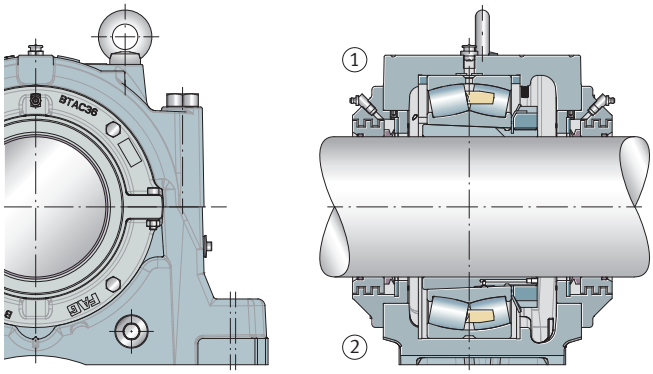
113 SNS plummer block housings for bearings with a cylindrical bore, Taconite NTC seal and NDK cover



00019E4D

1	Locating bearing	2	Non-locating bearing

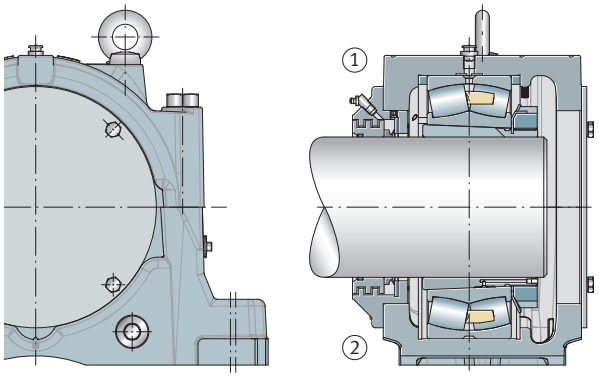
114 SNS plummer block housings for bearings with a tapered bore, 2 BTAC bolt-on seals



0008EB05

1	Locating bearing	2	Non-locating bearing

115 SNS plummer block housings for bearings with a tapered bore and adapter sleeve, BTAC bolt-on seal and BDK cover



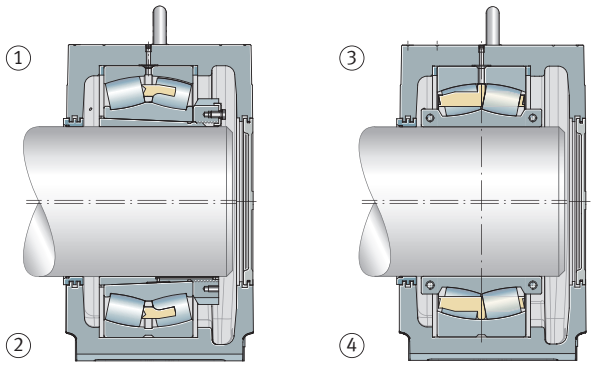
0008EB01

1	Locating bearing	2	Non-locating bearing

Installation of split spherical roller bearing

For SNS plummer block housings, an unsplit spherical roller bearing with a tapered bore and adapter sleeve can be replaced by a split spherical roller bearing.

116 SNS plummer block housings with split and unsplit spherical roller bearing



1	Locating bearing with unsplit bearing	2	Non-locating bearing with unsplit bearing
3	Locating bearing with split bearing	4	Non-locating bearing with split bearing

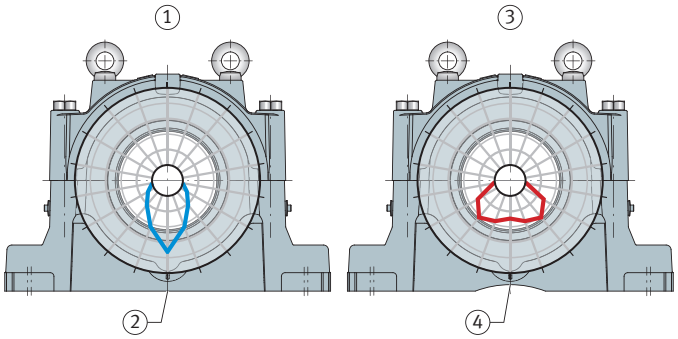
6.1.5 Longer bearing service life

The service life of the spherical roller bearings installed in SNS plummer block housing is up to 50 % longer than conventional plummer block housings. This increase in service life depends on the application, the load and the bearing design. Calculation of the bearing service life is based on the Bearinx calculation software. The reason for the longer bearing life is the improved load distribution in the bearing that is achieved as a result of the new housing design.

Conventional plummer block housings are designed such that the bearing load is introduced into the housing locating surface by a support directly below the bearing. If load is in the main load direction, namely with an orientation perpendicular to the housing locating face, an load distribution in the bearing is poor. Due to the narrow load zone, the load is distributed over only a small number of rolling elements, which results in a particularly high maximum load on the roller in the apex position.

A material recess below the bearing distributes the support over two symmetrically arranged support surfaces in SNS series plummer block housings. This leads to a significant improvement in the load distribution in the bearing. The load zone is wider and the load is distributed more uniformly over several rolling elements. The maximum then acts on the two neighbouring rollers of the roller in the apex position and is reduced. This results in a significant increase in the fatigue life of the roller bearings.

117 Influence of housing design on the load distribution in the bearing



00019EB8

1	Conventional plummer block housing with associated load distribution	2	Support directly below the bearing
3	SNS plummer block housing with associated load distribution	4	Material recess below the bearing

6.1.6 Accessories

Accessories must be ordered separately.

Seals

118 NTS labyrinth seal

0008D3A1

119 Taconite NTC seal

00019ED1

120 NTSG split labyrinth seal

0008CD51

121 BTAC bolt-on seal

0008CD3F

Cover

122 NDK cover

00019FA7

123 BDK cover

0008CD43

Locating rings

124 NFR locating ring

00019ED2

6.2 Suitable bearings

SNS split plummer block housings are suitable for the installation of spherical roller bearings with a tapered or cylindrical bore. The shaft diameters are 115 mm to 530 mm and 4 $^{7}/_{16}$ inch to 19 $^{1}/_{2}$ inch.

38 Suitable bearings for SNS split plummer block housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	22232..-K	22272..-K
	22326..-K	22356..-K
	23036-K	230/530..-K
	23134..-K	23196..-K
	23232..-K	23288..-K
Spherical roller bearings with a cylindrical bore	22232	22272
	22326	22356
	23036	230/530
	23134	23196
	23232	23288
	24036	240/530
Split spherical roller bearings	230SM160	230SM410
	230S.607	230S.1608
	231SM150	231SM410
	231S.515	231S.1500
	232SM260	232SM340
	232S.1200	232S.1308

Due to the modular concept, one or more series of plummer block housings are suitable for each series of spherical roller bearings.

Further information

39 Possible combinations of SNS plummer block housings with spherical roller bearings

Bearing type	Series	SNS2264		SNS30		SNS31			SNS32		SNS40	
		Suffix										
		H	Z	H	Z	H	Z	B	H	Z	H	Z
Spherical roller bearings with a tapered bore and adapter sleeve	222..-K	+		+		+						
	223..-K			+		+						
	230..-K			+								
	231..-K					+		+				
	232..-K			+					+			

Bearing type	Series	SNS2264		SNS30		SNS31			SNS32		SNS40	
		Suffix										
		H	Z	H	Z	H	Z	B	H	Z	H	Z
Spherical roller bearings with a cylindrical bore	222		+	+		+	+					
	223			+		+						
	230			+	+							
	231					+	+					
	232			+						+		
	240			+	+							+
Split spherical roller bearings	230S			+								
	231S					+		+				
	232S								+			

+ suitable

Split spherical roller bearings

In bearing replacement, unsplit spherical roller bearings with an adapter sleeve can be replaced by split spherical roller bearings. In many cases, the use of split spherical roller bearings significantly reduces the work associated with bearing replacement.

In order to ensure selection of the correct combination of housing and bearing, please contact us.

TPI 250 | Split spherical roller bearings |
https://www.schaeffler.de/std/1F81

6.3 Materials, corrosion protection

40 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Ductile cast iron GJS	D	✓	RAL 7016, anthracite grey
EN-GJS-400-18LT plummer block housings	LT	o	RAL 7016, anthracite grey

✓ available
o available as an option

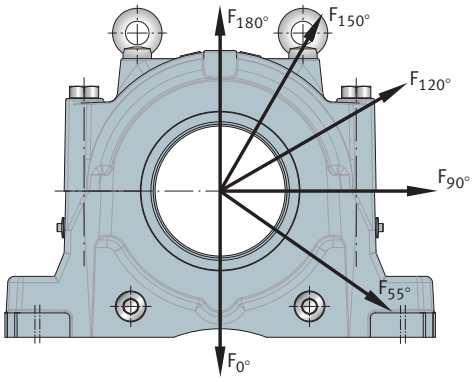
Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

6.4 Load capacity

When determining the permissible static load, safety factors must be applied:

- Safety factor of 6 compared to the guide values: housing rupture load for gray cast iron
- Safety factor of 5 compared to the guide values: housing rupture load for ductile cast iron
- No safety factor is required compared to the maximum load capacity of the connecting screws
- The housing must be secured by stoppers in the load direction for a load direction between 55 ° and 120 ° or for an axial load.

125 Load directions F for the guide values for the housing rupture load and the maximum load capacity of the connecting screws



41 Guide values for housing rupture load and maximum load capacity of connecting screws

Housing			Housing rupture load in load direction F							Connecting screws					
										G	M_A [1]	Max. load capacity of 4 screws in load direction F [2]			
			0 °	55 °	90 °	120 °	150 °	180 °	Axial			90 °	120 °	150 °	180 °
			kN	kN	kN	kN	kN	kN	kN	–	Nm	kN	kN	kN	kN
SNS3036	SNS3134	–	4050	4540	3040	1650	1190	1220	1170	M24	740	1380	760	618	725
SNS3038	SNS3136	–	4400	4940	3300	1800	1300	1330	1270	M24	740	1370	760	615	723
SNS3040	SNS3138	SNS3234	4880	5480	3660	1990	1440	1470	1410	M24	740	1500	830	670	790
–	–	SNS3236	4880	5480	3660	1990	1440	1470	1410	M24	740	1500	830	670	790
SNS3044	SNS3140	SNS3238	5800	6500	4350	2370	1700	1750	1670	M24	740	1510	836	677	796
SNS3048	SNS3144	SNS3240	6240	7560	5070	2880	2360	2380	1880	M24	740	1484	820	664	781
SNS3052	SNS3148	–	7330	8550	5720	3190	2470	2510	2080	M30	1450	2380	1315	1065	1252
–	–	SNS3244	7330	8550	5720	3190	2470	2510	1880	M30	1450	2380	1315	1065	1252
SNS3056	SNS3152	–	8740	9810	6550	3570	2580	2640	2520	M30	1450	2372	1310	1061	1250
–	–	SNS3248	8740	9810	6550	3570	2580	2640	1410	M30	1450	2372	1310	1061	1250
SNS3060	SNS3156	–	9200	10320	6900	3760	2710	2780	2652	M30	1450	2330	1290	1040	1230
SNS3064	SNS3160	–	10150	11390	7620	4150	3000	3070	2810	M30	1450	2220	1230	990	1170
–	–	SNS3252	10150	11390	7620	4150	3000	3070	2230	M30	1450	2220	1230	990	1170
–	–	SNS3256	10150	11390	7620	4150	3000	3070	2230	M30	1450	2220	1230	990	1170
SNS3068	SNS3164	–	11410	12810	8560	4660	3370	3450	3355	M30	1450	2220	1230	1000	1170
SNS3072	–	–	11410	12810	8560	4660	3370	3450	3355	M30	1450	2200	1230	1000	1170
–	–	SNS3260	11410	12810	8560	4660	3370	3450	2900	M30	1450	2200	1230	1000	1170
SNS2264, SNS3076	SNS3168	SNS4076	13300	14930	9980	5430	3930	4020	3480	M36	2600	3260	1800	1460	1720
–	–	SNS3264	13300	14930	9980	5430	3930	4020	2690	M36	2600	3260	1800	1460	1720
SNS3080	SNS3172	–	14740	16540	11060	6020	4350	4450	4250	M36	2600	3100	1710	1390	1630
–	–	SNS4080	14740	16540	11060	6020	4350	4450	4295	M36	2600	3100	1710	1390	1630
SNS3084	SNS3176	–	15940	17890	11960	6510	4710	4820	4600	M36	2600	3100	1710	1390	1630
–	–	SNS3268	15940	17890	11960	6510	4710	4820	3628	M36	2600	3100	1710	1390	1630
–	–	SNS4084	15940	17890	11960	6510	4710	4820	4200	M36	2600	3100	1710	1390	1630
SNS3088	SNS3180	–	18490	20750	13870	7560	5460	5590	5330	M36	2600	3200	1770	1430	1690
–	–	SNS3272	18490	20750	13870	7560	5460	5590	3200	M36	2600	3200	1770	1430	1690
–	–	SNS4088	18490	20750	13870	7560	5460	5590	4535	M36	2600	3200	1770	1430	1690
SNS3092	SNS3184	SNS4092	20180	22650	15140	8250	5960	6100	5650	M42	3700	4110	2270	1840	2160
SNS3096	–	SNS4096	20180	22650	15140	8250	5960	6100	5650	M42	3700	4110	2270	1840	2160
–	–	SNS3276	20180	22650	15140	8250	5960	6100	4530	M42	3700	4110	2270	1840	2160
SNS30/500	SNS3188	–	21940	24620	16460	8970	6480	6630	6320	M42	3700	4110	2270	1840	2160
–	–	SNS3280	21940	24620	16460	8970	6480	6630	4360	M42	3700	4110	2270	1840	2160
–	–	SNS40/500	21940	24620	16460	8970	6480	6630	6350	M42	3700	4110	2270	1840	2160
–	SNS3192	–	24150	27100	18120	9870	7130	7300	6962	M42	3700	4160	2300	1860	2190

Housing			Housing rupture load in load direction F							Connecting screws					
										G	M_A [1]	Max. load capacity of 4 screws in load direction F [2]			
			0 °	55 °	90 °	120 °	150 °	180 °	Axial			90 °	120 °	150 °	180 °
			kN	kN	kN	kN	kN	kN	kN	–	Nm	kN	kN	kN	kN
–	–	SNS3284	24150	27100	18120	9870	7130	7300	4747	M42	3700	4160	2300	1860	2190
SNS30/530	SNS3196	SNS40/530	26080	29270	19560	10660	7700	7880	7420	M42	3700	4160	2300	1860	2190
–	SNS3288	–	26080	29270	19560	10660	7700	7880	4500	M42	3700	4160	2300	1860	2190

1) The tightening torques are maximum values when utilising 90 % of the yield stress of the screw material and with a friction factor of 0,14. Schaeffler recommends that the screws should be tightened to 70 % of these values.

2) Maximum load carrying capacity in load direction F for maintaining contact at the parting surface between the upper and lower housing sections

G	–	Thread
M_A	Nm	Tightening torque

6.5 Lubrication

SNS split plummer block housings are mainly designed for grease lubrication. However, a design for oil lubrication can also be supplied to order.

6.5.1 Grease types

In order to achieve a long operating life and high operational security of the bearing arrangement, Schaeffler recommends the use of Arcanol rolling bearing greases. These rolling bearing greases have been designed and tested for bearing arrangement engineering.

6.5.2 Grease quantities

For initial greasing, the basic rule is that the bearing should be filled with grease to 100 % and the free volume of the housing to 60 %. The free volume is the space that remains in the housing once the bearing, adapter sleeve, shaft and seals have been fitted.

The recommended grease quantities specified in the following table take account of the following criteria:

- filling of the free volume to 60 %
- grease with a density of 0,9 g/cm³

Information on the recommended grease quantity for other housing sizes is available on request.

The recommended fill level for greasing the bearing may differ from100 %, depending on the application and speed.

42 Recommended grease quantity for SNS bearing housing

Housing	Bearing	Adapter sleeve	Free volume (100%)	Recommended grease quantity
			cm^3	≈g
SNS2264-H-D	22264..-K	H3064	27550	14900
SNS3036-H-D	23036..-K	H3036	2900	1600
SNS3038-H-D	23038..-K	H3038	3900	2250
SNS3040-H-D	23040..-K	H3040	5020	3050
SNS3044-H-D	23044..-K	H3044	5770	4260
SNS3048-H-D	23048..-K	H3048	7280	4550
SNS3052-H-D	23052..-K	H3052	9930	6750
SNS3056-H-D	23056..-K	H3056	12500	6750
SNS3060-H-D	23060..-K	H3060	14800	8000
SNS3064-H-D	23064..-K	H3064	19400	10500
SNS3068-H-D	23068..-K	H3068	24000	13000
SNS3072-H-D	23072..-K	H3072	30400	16450
SNS3076-H-D	23076..-K	H3076	32000	17300
SNS3080-H-D	23080..-K	H3080	33500	18100
SNS3084-H-D	23084..-K	H3084	35000	18900
SNS3088-H-D	23088..-K	H3088	43400	23450
SNS3092-H-D	23092..-K	H3092	56600	30600
SNS3096-H-D	23096..-K	H3096	55200	29850
SNS30/500-H-D	230/500..-K	H30/500	55200	29850
SNS30/530-H-D	230/530..-K	H30/530	67500	36450
SNS3134-H-D	23134..-K	H3134	2900	1600
SNS3136-H-D	23136..-K	H3136	3810	2060
SNS3138-H-D	23138..-K	H3138	5030	2720
SNS3140-H-D	23140..-K	H3140	6780	3660
SNS3144-H-D	23144..-K	H3144	7520	4060
SNS3148-H-D	23148..-K	H3148	10100	5990
SNS3152-H-D	23152..-K	H3152	11650	6300
SNS3156-H-D	23156..-K	H3156	12550	6800
SNS3160-H-D	23160..-K	H3160	17200	9300
SNS3164-H-D	23164..-K	H3164	21250	11500
SNS3168-H-D	23168..-K	H3168	27550	14900
SNS3172-H-D	23172..-K	H3172	28500	15400
SNS3176-H-D	23176..-K	H3176	28300	15300
SNS3180-H-D	23180..-K	H3180	36450	19700
SNS3184-H-D	23184..-K	H3184	45550	24600
SNS3188-H-D	23188..-K	H3188	48300	26100
SNS3192-H-D	23192..-K	H3192	52000	28100
SNS3196-H-D	23196..-K	H3196	55000	29700
SNS3234-H-D	23234..-K	H2334	5350	2900
SNS3236-H-D	23236..-K	H2336	5350	2900
SNS3238-H-D	23238..-K	H2338	6500	3510
SNS3240-H-D	23240..-K	H2340	6900	3760
SNS3244-H-D	23244..-K	H2344	10000	5400
SNS3248-H-D	23248..-K	H2348	10200	5550
SNS3252-H-D	23252..-K	H2352	15500	8400
SNS3256-H-D	23256..-K	H2356	15500	8400
SNS3260-H-D	23260..-K	H3260	19000	10300
SNS3264-H-D	23264..-K	H3264	25300	13700
SNS3268-H-D	23268..-K	H3268	25000	13500
SNS3272-H-D	23272..-K	H3272	26800	14500
SNS3276-H-D	23276..-K	H3276	38600	20850
SNS3280-H-D	23280..-K	H3280	44100	23850
SNS3284-H-D	23284..-K	H3284	45000	24350
SNS3288-H-D	23288..-K	H3288	49400	26700

6.5.3 Relubrication

For relubricating bearings with a circumferential lubrication groove, the grease is inserted into the housing through the central lubrication hole. With this type of relubrication, the grease acts on the bearing direct.

There are 2 relubrication points in the upper housing section for the relubrication of bearings without a lubrication groove. One of the two points has a threaded hole, while the other is a cast pilot hole.

The lubrication holes and lubrication nipples provided for relubrication differ depending on the design of the housing. On SNS..-H and SNS..-Z housings, the threaded holes are sealed with a screw plug. The screw plug must be removed and replaced permanently by one of the lubrication nipples. The 2 tapered lubrication nipples are already installed on SNS..-B housings.

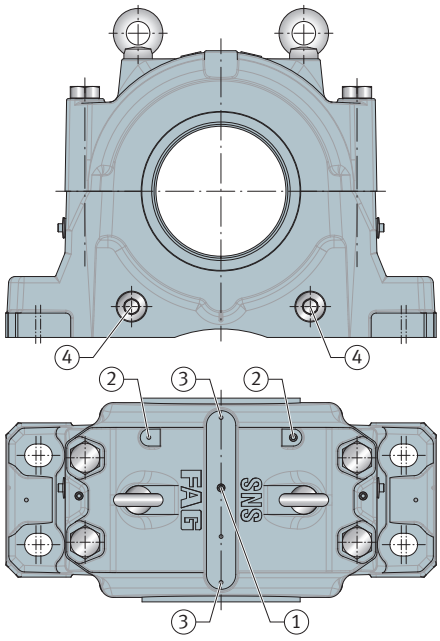
43 Lubrication holes and lubrication nipples

Housing	Thread	Lubrication nipples	Supplied quantity
SNS..-H	M10×1	Flat lubrication nipple to DIN 3404, with dust protection cap	1
SNS..-Z	M10×1	Tapered lubricating nipple to DIN 71412, with dust cap	1
SNS..-B	G $^{1}/_{4}$ ISO 228-1	Flat lubrication nipple to DIN 3404, type A	2

Lubrication holes can be drilled in 2 other marked positions for the relubrication of labyrinth seals.

The screw plugs must be removed from the two grease outlet holes in the lower housing section for the duration of relubrication to prevent excess greasing. This allows excess grease to escape. The grease outlet holes must then be sealed again using the screw plugs.

126 Positions for relubrication



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1	Central lubrication hole, for bearings with lubrication groove	2	Points for relubrication of bearings without lubrication groove
3	Points for relubricating labyrinth seals	4	Grease outlet holes

In poor environmental conditions, there is therefore a risk of dirt ingress into the housing when the grease outlet holes are opened.

6.5.4 Automatic lubricators

The housings can be directly connected to automatic lubricators, such as OPTIME C1 or CONCEPT1 from Schaeffler.

127 SNS plummer block housing with CONCEPT1 automatic lubricator

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We recommend that the grease outlet bore is left open when using an automatic lubricator to allow the escape of used grease.

Further information

TPI 252 | Lubricators |
https://www.schaeffler.de/std/1D4E

6.6 Sealing

Seals

The standard seals for sealing SNS..-H and SNS..-Z bearing housings are the labyrinth seal and the Taconite seal, each in split and unsplit versions. These seals tailor with the rectangular labyrinth grooves in each side of the housing.

The standard seals for sealing SNS..-B bearing housings are the BTAC bolt-on seals. These seals are bolted to the outside of housing.

Cover

For housings closed on one side, the NDK cover can be used for SNS..-H and SNS..-Z bearing housings.

For housings closed on one side, the BDK cover can be used for SNS..-B bearing housings.

Seals and covers must be ordered specially. Seals are supplied individually. 2 seals must be ordered for a continuous shaft.

6.6.1 NTS labyrinth seals

NTS labyrinth seals enable non-contact sealing. This makes them suitable for high circumferential velocities. The FKM O-ring made from fluoro rubber which is pressed between the labyrinth ring and shaft is suitable for temperatures of up to +200 °C.

The permissible shaft misalignment is 0,5° in both directions.

If necessary, the labyrinth can be relubricated. 1 lubrication hole must be drilled in the upper housing section for each labyrinth seal for this purpose. The optimal positions are indicated by cast pilot holes.

6.6.2 Taconite NTC seals

Taconite NTC seals are combination seals comprising a labyrinth seal and a V ring. These seals are suitable for extreme operating conditions in terms of contamination and dust. The V ring is made of NBR and is suitable for temperatures of up to +100 °C.

The seal is suitable for grease lubrication. The seal is fitted with 1 lubrication nipple for relubrication.

The permissible shaft misalignment is 0,5° in both directions.

6.6.3 NTSG split labyrinth seals

Labyrinth seals are also available in a split design for SNS plummer block housings. This can considerably reduce the work involved in fitting or replacing the seal, especially if a split bearing is also fitted.

The characteristics and operating ranges are the same as those of the NTS unsplit labyrinth seal.

NTSG split labyrinth seals are available in various sizes. Other sizes are available on request.

6.6.4 NTCG split Taconite seals

Taconite seals are also available in a split design for SNS plummer block housings. This can considerably reduce the work involved in fitting or replacing the seal, especially if a split bearing is also fitted.

Split Taconite seals are supplied on request.

6.6.5 BTAC bolt-on seal

BTAC bolt-on seals are combined seals, consisting of a labyrinth ring and a V ring. They are bolted to the outside of the housing.

The seals are suitable for extreme operating conditions in terms of dirt and dust. The V ring is made of NBR and is suitable for temperatures of up to +100 °C.

The maximum shaft misalignment is 0,4 ° to both sides.

The seal is suitable for grease lubrication. The seal is fitted with 1 lubrication nipple for relubrication.

The seal is available in sizes BTAC34 to BTAC68.

Special SNS..-B housings are required for using bolt-on seals.

6.6.6 NDK covers

NDK covers are used on housings which are closed on one side. The covers fit into the rectangular annular grooves in the housings. They are suitable for temperatures up to +200 °C.

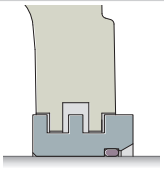
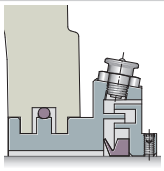
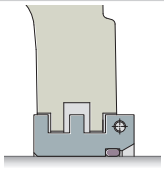
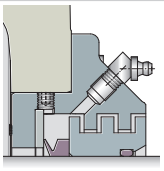
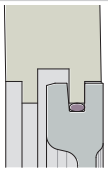
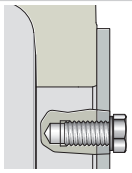
6.6.7 BDK covers

BDK covers are bolted to the outside of the housing. They fit the SNS..-B housings designed for the bolt-on seal.

6.6.8 Overview of properties and applications

The following overview compares the properties and applications of the seals and covers.

44 Standard seals and covers for SNS plummer block housings

Design features and suitability		Seals				Cover	
		Labyrinth seal	Taconite seal	Labyrinth seal	Bolt-on seal	Inserted	Bolted on
							
Designation		NTS	NTC	NTSG	BTAC	NDK	BDK
Material		Steel, FKM	Steel, NBR	Steel, FKM	Steel, NBR	Steel, FKM	Steel
Split seal		No	No	Yes	No	Not applicable	Not applicable
Quantity per pack [1]		1	1	1	1	1	1
Suitability for sealing against							
Dust		o	++	o	++	++	++
Fine solid particles		+	++	+	++	++	++
Coarse solid particles		+	++	+	++	++	++
Splinters		++	++	++	++	++	++
Spraying liquids		–	++	–	++	++	++
Application							
Long term temperature	°C	−20 - +200	−30 - +100	−20 - +200	−30 - +100	−20 - +200	Not applicable
	°F	−4 - +390	−22 ... +210)	−4 - +390	−22 - +210	−2 - +390	Not applicable
		Restricted due to FKM	Restricted due to NBR	Restricted due to FKM	Restricted due to NBR	Restricted due to FKM	Not applicable
Circumferential velocity	m/s	No restriction	≤ 12	No restriction	≤ 12	Not applicable	Not applicable
Misalignment	°	≤ 0,3	≤ 0,3	≤ 0,3	≤ 0,4	Not applicable	Not applicable
Low friction		++	+	++	+	Not applicable	Not applicable
Axial shaft displacement (suitability as non-locating bearing)		+	+	+	+	Not applicable	Not applicable
Vertical arrangement		–	o	–	–	++	++
Grease relubrication		+	+	+	++	++	++
Oil lubrication		–	o	–	–	+	-
Sunlight		++	++	++	++	++	++
Shaft requirements							
Tolerance class [2]		h8 (h9)	h8 (h9)	h8 (h9)	h8 (h9)	Not applicable	Not applicable
Max. roughness	µm	Ra 3.2	Ra 3.2	Ra 3.2	Ra 3.2	Not applicable	Not applicable

++ highly suitable
\+ suitable
o suitable with restrictions
– not suitable

[1] Only multiples of 2 are possible as an order quantity, as there are 2 items per pack

[2] The envelope condition Ⓔ applies here.

6.7 Dimensions, tolerances

6.7.1 Dimensions

The dimensions of the housings conform to ISO 113:2010.

SNS housings are interchangeable with SD31 housings.

The interchangeability with other series of housings is as follows:

- All bearing series that can be combined with SD5 and SD6 plummer block housings and can be installed in plummer block housings.
- Bearing series that can be combined with S30 plummer block housings can only be installed in some plummer block housings.

The dimensions of SD5, SD6 and S30 housings are not identical to the dimensions of SNS housings.

6.7.2 Tolerances for the bearing seat

The bearing seat in the housings is machined to the tolerance class G7 to DIN EN ISO 286-1. The tolerance indicated applies to the condition on delivery, in other words before the screws connecting the upper and lower sections are loosened.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

6.8 Structure of the order designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

Bearing housing

128 Structure of the designations of SNS plummer block housings

Series

SNS	SNS split plummer block housings

ISO dimension series of bearing

30	ISO dimension series 22, 23, 30 and 40
31	ISO dimension series 22, 23 and 31
32	ISO dimension series 22 and 32
40	ISO dimension series 40

Housing size

36	For bearings of ISO dimension series 30, 31 and 32
/530	From bore code ≥100, the bore diameter of the bearing is specified /530 = 530 mm bore diameter

Housing design

H	The housing is suitable for: • Spherical roller bearings with a tapered bore and clamping sleeve • Split spherical roller bearings • The following spherical roller bearings with a cylindrical bore: 22232, 22234, 22252, 22272, 22326, 22328, 22330, 22332, 22334, 22338, 22344, 22348, 22352, 22356, 23232, 24036
Z	The housing is suitable for: • Spherical roller bearings with a cylindrical bore, which are not assigned to housing design H
B	Design for bolt-on seal

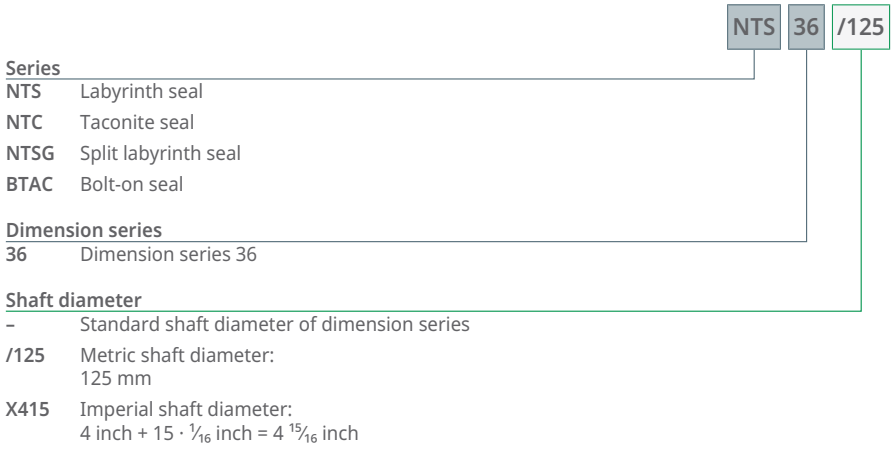
Housing material

D	Ductile cast iron (standard)
LT	Ductile cast iron (for low temperatures)

001E070B

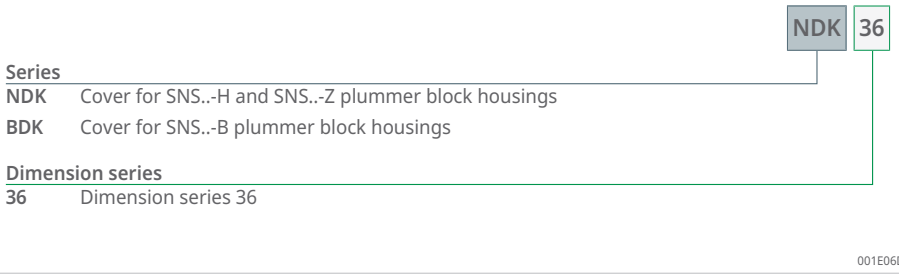
Seals

129 Structure of the designations of standard seals



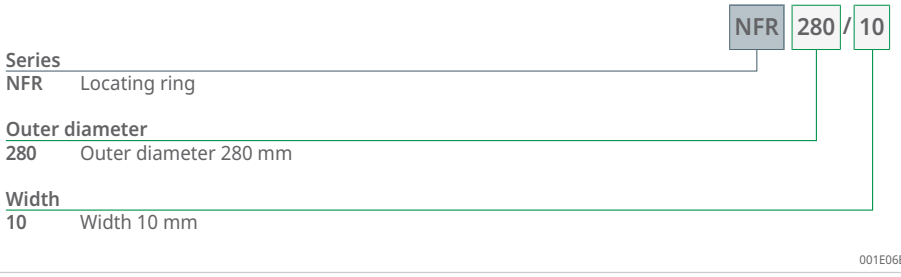
Cover

130 Structure of the designations of covers



Locating rings

131 Structure of the designations of locating rings



6.8.1 Specimen orders

For ordering a SNS split plummer block housing, the housing designation only describes the housing body. The other components such as seals, covers or locating rings must be ordered separately in the specific design required. The roller bearing and, if necessary, the adapter sleeve must also be ordered separately.

Housings with an installed bearing produce non-locating bearings. These can be converted into locating bearings by adding locating rings.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- SNS plummer block housing made of ductile cast iron, closed on one side
- 23136-E1A-XL-K-M spherical roller bearing in the form of a locating bearing
- Fastened with adapter sleeve to a shaft diameter of 160 mm
- Labyrinth seal

45 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	SNS3136-H-D
1	Spherical roller bearing	23136-E1A-XL-K-M
1	Adapter sleeve	H3136-HG
2	Locating rings	NFR300/10
1	Cover	NDK36
1	Labyrinth seal	NTS36

Example 2

Example of a housing configuration, characteristics:

- SNS plummer block housing made from ductile cast iron, continuous shaft
- 23136-E1A-XL-K-M spherical roller bearing in the form of a non-locating bearing
- Fastened with adapter sleeve to a shaft diameter of 160 mm
- Taconite seal

46 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	SNS3136-H-D
1	Spherical roller bearing	23136-E1A-XL-K-M
1	Adapter sleeve	H3136-HG
2	Taconite seals	NTC36

Example 3

Example of a housing configuration, characteristics:

- SNS plummer block housing made of ductile cast iron, closed on one side
- 23136-E1A-XL-K-M spherical roller bearing in the form of a non-locating bearing
- Fastened with adapter sleeve to a shaft diameter of 160 mm
- Bolt-on seal

47 Order

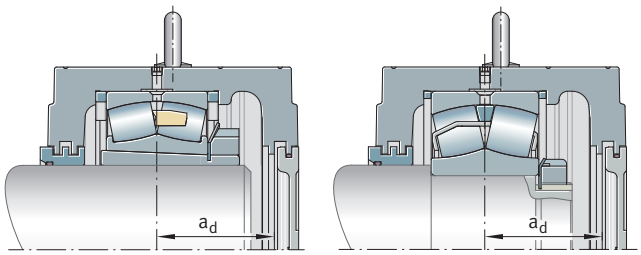
Quantity	Description	Ordering designation, short designation
1	Plummer block housing	SNS3136-H-D
1	Spherical roller bearing	23136-E1A-XL-K-M
1	Adapter sleeve	H3136-HG
1	Cover	BDK36
1	Bolt-on seal	BTAC36

6.9 Design of the adjacent construction

6.9.1 Permitted shaft ends

The length of the shaft in a housing which is closed on one side must be restricted so that there is no contact between the shaft and the cover. In this case, the movement of the bearing in a non-locating bearing must be taken into account. The basis for determining the maximum length of the shaft is the distance between the centre of the bearing seat and the cover.

132 Spacing between centre of the bearing seat and cover



a_d	Spacing between centre of the bearing seat and cover

48 Spacing between centre of the bearing seat and cover

Housing	Spacing a_d
	mm
SNS3036, SNS3134	107,5
SNS3038, SNS3136	113,5
SNS3040, SNS3138, SNS3234	117,5
SNS3236	117,5
SNS3044, SNS3138, SNS3238	127,5
SNS3048, SNS3144, SNS3240	133,5
SNS3052, SNS3148, SNS3244	141,5
SNS3056, SNS3152, SNS3248	151,5
SNS3060, SNS3156	154,5
SNS3064, SNS3160, SNS3252	175,5
SNS3256	175,5
SNS3068, SNS3160, SNS3260	186,5
SNS3072	186,5
SNS2264, SNS3076, SNS3168, SNS3264, SNS4076	202,5
SNS3080, SNS3168, SNS3268, SNS4080	208,5
SNS3084, SNS3176, SNS4084	208,5
SNS3088, SNS3176, SNS3272, SNS4088	223,5
SNS3092, SNS3184, SNS3276, SNS4092	243,5
SNS3096, SNS3188, SNS3276, SNS4096	243,5
SNS30/500, SNS4096	243,5
SNS30/530, SNS3192, SNS3284, SNS40/530	248,5
SNS3196, SNS3284	248,5

6.10 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

Further information

MON 84 | SNS split plummer block housings |
https://www.schaeffler.de/std/1D63

6.10.1 Eye bolts

There are 2 eye bolts to DIN 580:2018 in each housing upper section. These eye bolts are intended as a locating point for the installation and removal of the housing. The load capacity of the eye bolts allows the housing, including the bearing fitted in the housing, to be lifted.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

6.10.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

49 Nominal sizes of base fastening screws for SNS plummer block housings

Housing					Nominal screw diameter
SNS22	SNS30	SNS31	SNS32	SNS40	
–	SNS3036	SNS3134	SNS3234	–	M24
–	SNS3038	SNS3136	SNS3236	–	
–	SNS3040	SNS3138	–	–	
–	SNS3044	SNS3140	SNS3238	–	M30
–	SNS3048	SNS3144	SNS3240	–	
–	SNS3052	SNS3148	SNS3244	–	
SNS2264	SNS3056	SNS3152	SNS3248	SNS4076	M36
–	SNS3060	SNS3156	SNS3252	SNS4080	
–	SNS3064	SNS3160	SNS3256	SNS4084	
–	SNS3068	SNS3164	SNS3260	–	
–	SNS3072	SNS3168	SNS3264	–	
–	SNS3076	SNS3172	SNS3268	–	
–	SNS3080	SNS3176	–	–	
–	SNS3084	–	–	–	
–	SNS3088	SNS3180	SNS3272	SNS4088	M42
–	SNS3092	SNS3184	SNS3276	SNS4092	
–	SNS3096	SNS3188	SNS3280	SNS4096	
–	SNS30/500	SNS3192	SNS3284	SNS40/500	
–	SNS30/530	SNS3196	SNS3288	SNS40/530	M48

The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

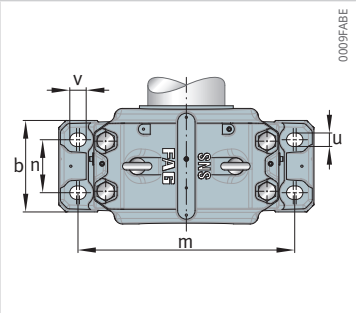
6.11 Product tables

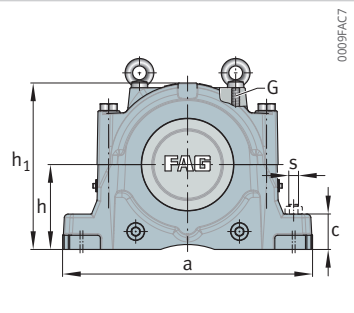
6.11.1 Explanations of the product tables

(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
b	mm	Base width
c	mm	Base height

d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
d_2	mm	Shaft shoulder diameter
e	mm	Centre offset
g	mm	Housing body width
G	–	Ring bolt thread
g_B	mm	Overall width, with bolt-on seal
g_{BD}	mm	Overall width, with bolt-on seal and cover
g_L	mm	Total width, with 2 labyrinth seals
g_{LD}	mm	Overall width, with 2 labyrinth seals and cover
g_T	mm	Total width, with 2 Taconite seals
h	mm	Distance to shaft axis
h_1	mm	Overall height
k	mm	Bearing seat width
m	mm	Fixing hole spacing
m	kg	Mass
n_S	–	Number of locating rings
s	inch or mm	Screw size
u	mm	Slot width
v	mm	Slot length

6.11.2 SNS, Dimensions





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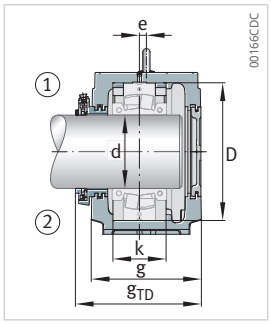
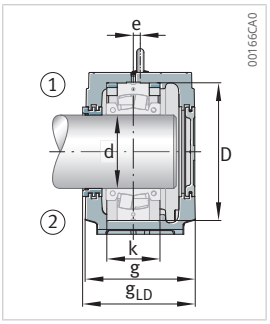
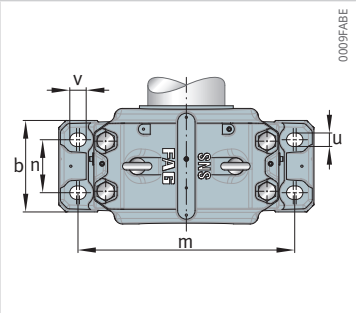
Housing	h	h_1	g	b	c	a
	mm	mm	mm	mm	mm	mm
SNS2264-H-D	340	675	400	360	120	950
SNS30/500-H-D	420	835	460	430	135	1220
SNS30/530-H-D	460	920	470	440	155	1330
SNS3036-H-D	170	333	230	180	70	510
SNS3038-H-D	180	353	240	190	75	530
SNS3040-H-D	190	375	260	210	80	560
SNS3044-H-D	210	411	280	230	85	610
SNS3048-H-D	220	434	290	240	90	640
SNS3052-H-D	240	474	310	260	95	700
SNS3056-H-D	260	516	320	280	100	770
SNS3060-H-D	280	550	320	280	105	790
SNS3064-H-D	300	591	350	310	110	830
SNS3068-H-D	320	631	370	330	115	880
SNS3072-H-D	320	631	370	330	115	880
SNS3076-H-D	340	675	400	360	120	950
SNS3080-H-D	350	695	400	360	120	1000
SNS3084-H-D	360	715	400	360	120	1040
SNS3088-H-D	380	755	430	390	125	1120
SNS3092-H-D	410	810	460	420	130	1170
SNS3096-H-D	410	810	460	420	130	1170
SNS3134-B-D	170	333	219	180	70	510
SNS3134-H-D	170	333	230	180	70	510
SNS3136-B-D	180	353	240	190	75	530
SNS3136-H-D	180	353	240	190	75	530
SNS3138-B-D	190	375	260	210	80	560
SNS3138-H-D	190	375	260	210	80	560
SNS3140-B-D	210	411	280	230	85	610
SNS3140-H-D	210	411	280	230	85	610
SNS3144-B-D	220	434	290	240	90	640
SNS3144-H-D	220	434	290	240	90	640
SNS3148-B-D	240	474	310	260	95	700
SNS3148-H-D	240	474	310	260	95	700
SNS3152-B-D	260	516	320	280	100	770
SNS3152-H-D	260	516	320	280	100	770
SNS3156-B-D	280	550	310	280	105	790
SNS3156-H-D	280	550	320	280	105	790
SNS3160-B-D	300	591	350	310	110	830
SNS3160-H-D	300	591	350	310	110	830
SNS3164-B-D	320	631	370	330	115	880
SNS3164-H-D	320	631	370	330	115	880

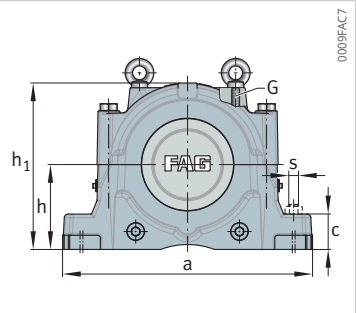




m	n	v	u	s	D	k	e	G	m
mm	mm	mm	mm	–	mm	mm	mm	–	≈ kg
810	220	50	42	M36	580	190	24	M30	466
1030	260	60	48	M42	720	226	35	M42	895
1110	260	70	56	M48	780	248	35	M48	1100
430	100	34	28	M24	280	108	14	M16	74
450	110	34	28	M24	290	115	15	M16	86
480	120	34	28	M24	310	122	10	M20	105
510	130	43	35	M30	340	130	10	M20	129
540	140	42	35	M30	360	140	12	M20	148
600	150	42	35	M30	400	148	12	M24	180
650	160	50	42	M36	420	166	13	M24	249
670	160	50	42	M36	460	168	16	M24	260
710	190	50	42	M36	480	181	22	M30	332
750	200	50	42	M36	520	197	23	M30	386
750	200	50	42	M36	540	198	23	M30	356
810	220	50	42	M36	560	180	24	M30	481
840	220	50	42	M36	600	192	30	M36	489
870	220	50	42	M36	620	194	30	M36	527
950	240	60	48	M42	650	200	30	M42	672
1000	260	60	48	M42	680	224	35	M42	849
1000	260	60	48	M42	700	224	35	M42	806
430	100	34	28	M24	280	108	14	M16	70
430	100	34	28	M24	280	108	14	M16	75
450	110	34	28	M24	300	116	15	M16	78
450	110	34	28	M24	300	116	15	M16	83
480	120	34	28	M24	320	124	10	M20	96
480	120	34	28	M24	320	124	10	M20	102
510	130	43	35	M30	340	132	10	M20	123
510	130	43	35	M30	340	132	10	M20	133
540	140	42	35	M30	370	140	12	M20	137
540	140	42	35	M30	370	140	12	M20	146
600	150	42	35	M30	400	148	12	M24	173
600	150	42	35	M30	400	148	12	M24	184
650	160	50	42	M36	440	164	13	M24	226
650	160	50	42	M36	440	164	13	M24	238
670	160	50	42	M36	460	166	16	M24	252
670	160	50	42	M36	460	166	16	M24	264
710	190	50	42	M36	500	180	22	M30	305
710	190	50	42	M36	500	180	22	M30	317
750	200	50	42	M36	540	196	23	M30	354
750	200	50	42	M36	540	196	23	M30	368

6.11.2 SNS, Dimensions





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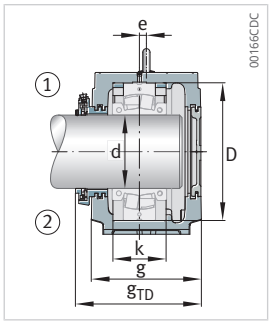
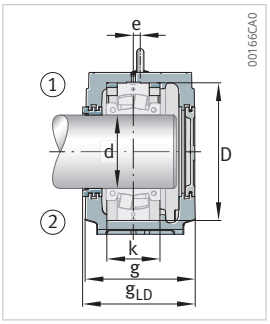
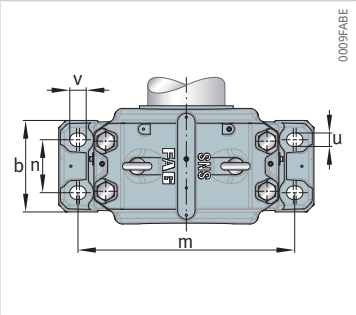
Housing	h	h_1	g	b	c	a
	mm	mm	mm	mm	mm	mm
SNS3168-B-D	340	675	400	360	120	950
SNS3168-H-D	340	675	400	360	120	950
SNS3172-H-D	350	695	400	360	120	1000
SNS3176-H-D	360	715	400	360	120	1040
SNS3180-H-D	380	755	430	390	125	1120
SNS3184-H-D	410	810	460	420	130	1170
SNS3188-H-D	420	835	460	430	135	1220
SNS3192-H-D	440	880	470	440	145	1280
SNS3196-H-D	460	920	470	440	155	1330
SNS3234-H-D	190	375	260	210	80	560
SNS3236-H-D	190	375	260	210	80	560
SNS3238-H-D	210	411	280	230	85	610
SNS3240-H-D	220	434	290	240	90	640
SNS3244-H-D	240	474	310	260	95	700
SNS3248-H-D	260	516	320	280	100	770
SNS3252-H-D	300	591	350	310	110	830
SNS3256-H-D	300	591	350	310	110	830
SNS3260-H-D	320	631	370	330	115	880
SNS3264-H-D	340	675	400	360	120	950
SNS3268-H-D	360	715	400	360	120	1040
SNS3272-H-D	380	755	430	390	125	1120
SNS3276-H-D	410	810	460	420	130	1170
SNS3280-H-D	420	835	460	430	135	1220
SNS3284-H-D	440	880	470	440	145	1280
SNS3288-H-D	460	920	470	440	155	1330

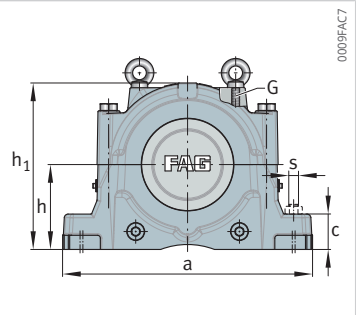




m	n	v	u	s	D	k	e	G	m
mm	mm	mm	mm	–	mm	mm	mm	–	≈ kg
810	220	50	42	M36	580	210	24	M30	446
810	220	50	42	M36	580	210	24	M30	461
840	220	50	42	M36	600	212	30	M36	498
870	220	50	42	M36	620	214	30	M36	534
950	240	60	48	M42	650	220	30	M42	681
1000	260	60	48	M42	700	244	35	M42	821
1030	260	60	48	M42	720	246	35	M42	895
1070	260	60	48	M42	760	260	35	M48	993
1110	260	70	56	M48	790	268	35	M48	1100
480	120	34	28	M24	310	122	10	M20	105
480	120	34	28	M24	320	124	10	M20	102
510	130	43	35	M30	340	132	10	M20	133
540	140	42	35	M30	360	140	12	M20	148
600	150	42	35	M30	400	164	12	M24	183
650	160	50	42	M36	440	180	13	M24	235
710	190	50	42	M36	480	194	22	M30	331
710	190	50	42	M36	500	196	22	M30	314
750	200	50	42	M36	540	212	23	M30	364
810	220	50	42	M36	580	228	24	M30	457
870	220	50	42	M36	620	244	30	M36	530
950	240	60	48	M42	650	252	30	M42	675
1000	260	60	48	M42	680	260	35	M42	841
1030	260	60	48	M42	720	276	35	M42	895
1070	260	60	48	M42	760	284	35	M48	984
1110	260	70	56	M48	790	300	35	M48	1100

6.11.3 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, metric shaft





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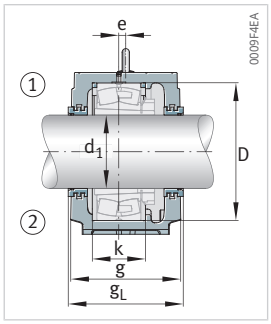
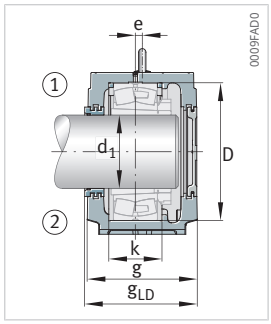
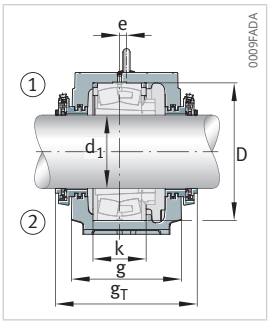
Housing	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	mm	–	–	–	–
SNS3134-H-D	**115**	22326..-K	H2326	NFR280/7,5	2
SNS3136-H-D	**125**	22328..-K	H2328	NFR300/7	2
SNS3138-H-D	**135**	22330..-K	H2330	NFR320/8	2
SNS3140-H-D	**140**	22332..-K	H2332(-HG)	NFR340/9	2
SNS3038-H-D	**140**	23232..-K	H2332(-HG)	NFR290/5,5	2
SNS3038-H-D	**140**	22232..-K	H3132(-HG)	NFR290/17,5	2
SNS3234-H-D	**150**	23234..-K	H2334(-HG)	NFR310/6	2
SNS3134-H-D	**150**	23134..-K	H3134(-HG)	NFR280/10	2
SNS3040-H-D	**150**	22234..-K	H3134(-HG)	NFR310/18	2
SNS3048-H-D	**150**	22334..-K	H2334(-HG)	NFR360/10	2
SNS3138-H-D	**160**	22236..-K	H3136(-HG)	NFR320/19	2
SNS3036-H-D	**160**	23036..-K	H3036(-HG)	NFR280/17	2
SNS3236-H-D	**160**	23236..-K	H2336(-HG)	NFR320/6	2
SNS3136-H-D	**160**	23136..-K	H3136(-HG)	NFR300/10	2
SNS3238-H-D	**170**	23238..-K	H2338(-HG)	NFR340/6	2
SNS3148-H-D	**170**	22338..-K	H2338(-HG)	NFR400/8	2
SNS3038-H-D	**170**	23038..-K	H3038(-HG)	NFR290/10	4
SNS3140-H-D	**170**	22238..-K	H3138(-HG)	NFR340/10	4
SNS3138-H-D	**170**	23138..-K	H3138(-HG)	NFR320/10	2
SNS3056-H-D	**180**	22340..-K	H2340(-HG)	NFR420/14	2
SNS3048-H-D	**180**	22240..-K	H3140(-HG)	NFR360/21	2
SNS3040-H-D	**180**	23040..-K	H3040(-HG)	NFR310/10	4
SNS3140-H-D	**180**	23140..-K	H3140(-HG)	NFR340/10	2
SNS3240-H-D	**180**	23240..-K	H2340(-HG)	NFR360/6	2
SNS3156-H-D	**200**	22344..-K	H2344X(-HG)	NFR460/10,5	2
SNS3148-H-D	**200**	22244..-K	H3144X(-HG)	NFR400/10	4
SNS3244-H-D	**200**	23244..-K	H2344X(-HG)	NFR400/10	2
SNS3144-H-D	**200**	23144..-K	H3144X(-HG)	NFR370/10	2
SNS3044-H-D	**200**	23044..-K	H3044X(-HG)	NFR340/10	4
SNS3148-H-D	**220**	23148..-K	H3148X(-HG)	NFR400/10	2
SNS3248-H-D	**220**	23248..-K	H2348X(-HG)	NFR440/10	2
SNS3152-H-D	**220**	22248..-K	H3148X(-HG)	NFR440/22	2
SNS3160-H-D	**220**	22348..-K	H2348X(-HG)	NFR500/12,5	2
SNS3048-H-D	**220**	23048..-K	H3048(-HG)	NFR360/12	4
SNS3252-H-D	**240**	23252..-K	H2352X(-HG)	NFR480/10	2
SNS3064-H-D	**240**	22252..-K	H3152X(-HG)	NFR480/25,5	2
SNS3164-H-D	**240**	22352..-K	H2352X(-HG)	NFR540/15,5	2
SNS3152-H-D	**240**	23152..-K	H3152X(-HG)	NFR440/10	2
SNS3052-H-D	**240**	23052..-K	H3052X(-HG)	NFR400/22	2
SNS3256-H-D	**260**	23256..-K	H2356X(-HG)	NFR500/10	2



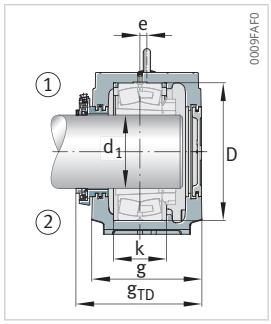
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

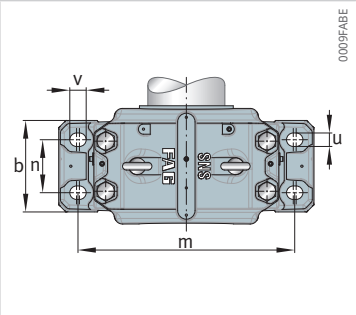


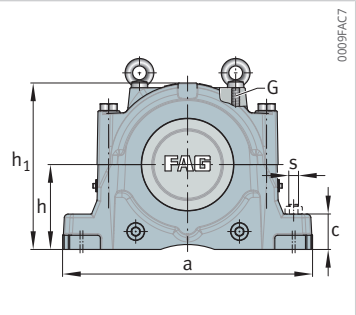
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS34/115	241	235,5	NTC34/115	299	264	NDK34
NTS36/125	251	245,5	NTC36/125	310	274,5	NDK36
NTS38/135	271	265,5	NTC38/135	330	294,5	NDK38
NTS40/140	291	285,5	NTC40/140	347	313	NDK40
NTS38/140	251	245,5	NTC38/140	312	275,5	NDK38
NTS38/140	251	245,5	NTC38/140	312	275,5	NDK38
NTS40/150	271	265,5	NTC40/150	327	293	NDK40
NTS34	241	235,5	NTC34	299	264,5	NDK34
NTS40/150	271	265,5	NTC40/150	326	293	NDK40
NTS48/150	301	295,5	NTC48/150	380	335	NDK48
NTS38/160	271	265,5	NTC38/160	329	294,5	NDK38
NTS36	241	235,5	NTC36	304	267	NDK36
NTS38/160	271	265,5	NTC38/160	330	294,5	NDK38
NTS36	251	245,5	NTC36	314	277	NDK36
NTS40/170	291	285,5	NTC40/170	347	313	NDK40
NTS48/170	321	315,5	NTC48/170	401	355	NDK48
NTS38	251	245,5	NTC38	312	275,5	NDK38
NTS40/170	291	285,5	NTC40/170	346	313	NDK40
NTS38	271	265,5	NTC38	334	297	NDK38
NTS56/180	331	325,5	NTC56/180	411	365	NDK56
NTS48/180	301	295,5	NTC48/180	380	335	NDK48
NTS40	271	265,5	NTC40	326	293	NDK40
NTS40	291	285,5	NTC40	354	317	NDK40
NTS48/180	301	295,5	NTC48/180	381	335	NDK48
NTS56/200	331	325,5	NTC56/200	411	365	NDK56
NTS48/200	321	315,5	NTC48/200	400	355	NDK48
NTS48/200	321	315,5	NTC48/200	401	355	NDK48
NTS44	301	295,5	NTC44	380	335	NDK44
NTS44	291	285,5	NTC44	370	325	NDK44
NTS48	321	315,5	NTC48	400	355	NDK48
NTS52/220	331	325,5	NTC52/220	410	365	NDK52
NTS52/220	331	325,5	NTC52/220	410	365	NDK52
NTS60/220	361	355,5	NTC60/220	441	395	NDK60
NTS48	301	295,5	NTC48	380	335	NDK48
NTS64/240	361	355,5	NTC64/240	440	395	NDK64
NTS64/240	361	355,5	NTC64/240	440	395	NDK64
NTS64/240	381	375,5	NTC64/240	460	415	NDK64
NTS52	331	325,5	NTC52	410	365	NDK52
NTS52	321	315,5	NTC52	401	355	NDK52
NTS60/260	361	355,5	NTC60/260	441	395	NDK60

6.11.3 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, metric shaft





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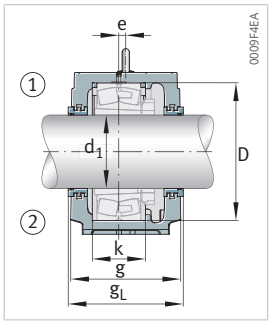
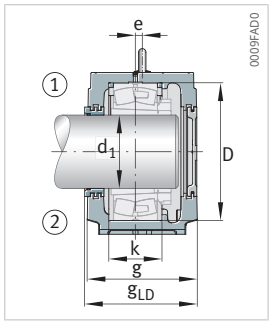
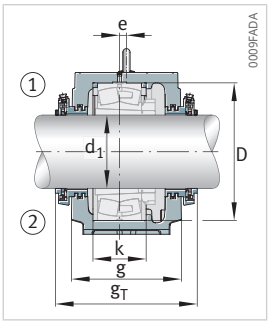
Housing	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	mm	–	–	–	–
SNS3160-H-D	**260**	22256..-K	H3156X(-HG)	NFR500/25	2
SNS3056-H-D	**260**	23056..-K	H3056(-HG)	NFR420/10	6
SNS3168-H-D	**260**	22356..-K	H2356X(-HG)	NFR580/17,5	2
SNS3156-H-D	**260**	23156..-K	H3156X(-HG)	NFR460/10	2
SNS3060-H-D	**280**	23060..-K	H3060(-HG)	NFR460/25	2
SNS3260-H-D	**280**	23260..-K	H3260(-HG)	NFR540/10	2
SNS3160-H-D	**280**	23160..-K	H3160(-HG)	NFR500/10	2
SNS3164-H-D	**280**	22260..-K	H3160(-HG)	NFR540/28	2
SNS3064-H-D	**300**	23064..-K	H3064-HG	NFR480/10	6
SNS2264-H-D	**300**	22264..-K	H3164-HG	NFR580/20	2
SNS3264-H-D	**300**	23264..-K	H3264-HG	NFR580/10	2
SNS3164-H-D	**300**	23164..-K	H3164-HG	NFR540/10	2
SNS3068-H-D	**320**	23068..-K	H3068-HG	NFR520/16	4
SNS3268-H-D	**320**	23268..-K	H3268-HG	NFR620/10	2
SNS3168-H-D	**320**	23168..-K	H3168-HG	NFR580/10	2
SNS3172-H-D	**340**	23172..-K	H3172-HG	NFR600/10	2
SNS3180-H-D	**340**	22272..-K	H3172-HG	NFR650/25	2
SNS3272-H-D	**340**	23272..-K	H3272-HG	NFR650/10	2
SNS3072-H-D	**340**	23072..-K	H3072-HG	NFR540/16	4
SNS3176-H-D	**360**	23176..-K	H3176-HG	NFR620/10	2
SNS3076-H-D	**360**	23076..-K	H3076-HG	NFR560/22,5	2
SNS3276-H-D	**360**	23276..-K	H3276-HG	NFR680/10	2
SNS3180-H-D	**380**	23180..-K	H3180-HG	NFR650/10	2
SNS3280-H-D	**380**	23280..-K	H3280-HG	NFR720/10	2
SNS3080-H-D	**380**	23080..-K	H3080-HG	NFR600/22	2
SNS3284-H-D	**400**	23284..-K	H3284-HG	NFR760/10	2
SNS3084-H-D	**400**	23084..-K	H3084X-HG	NFR620/22	2
SNS3184-H-D	**400**	23184..-K	H3184-HG	NFR700/10	2
SNS3288-H-D	**410**	23288..-K	H3288-HG	NFR790/10	2
SNS3188-H-D	**410**	23188..-K	H3188-HG	NFR720/10	2
SNS3088-H-D	**410**	23088..-K	H3088-HG	NFR650/21,5	2
SNS3192-H-D	**430**	23192..-K	H3192-HG	NFR760/10	2
SNS3092-H-D	**430**	23092..-K	H3092-HG	NFR680/30,5	2
SNS3096-H-D	**450**	23096..-K	H3096-HG	NFR700/29,5	2
SNS3196-H-D	**450**	23196..-K	H3196-HG	NFR790/10	2
SNS30/500-H-D	**470**	230/500..-K	H30/500-HG	NFR720/29,5	2
SNS30/530-H-D	**500**	230/530..-K	H30/530-HG	NFR780/31,5	2



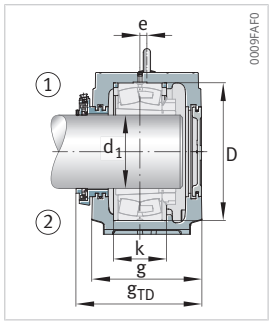
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

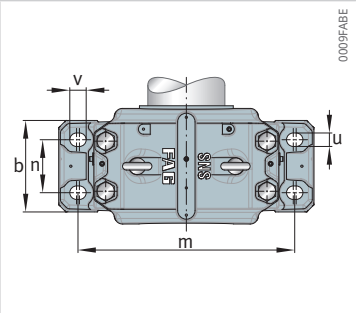


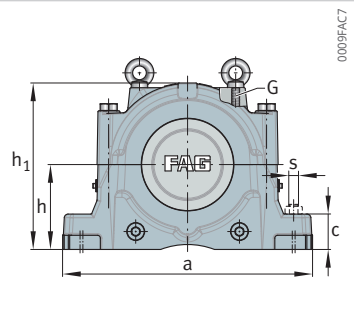
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS60/260	361	355,5	NTC60/260	440	395	NDK60
NTS56	331	325,5	NTC56	411	365	NDK56
NTS68/260	411	405,5	NTC68/260	490	445	NDK68
NTS56	331	325,5	NTC56	410	365	NDK56
NTS60	331	325,5	NTC60	411	365	NDK60
NTS64/280	381	375,5	NTC64/280	461	415	NDK64
NTS60	361	355,5	NTC60	440	395	NDK60
NTS64/280	381	375,5	NTC64/280	460	415	NDK64
NTS64	361	355,5	NTC64	441	395	NDK64
NTS68/300	411	405,5	NTC68/300	490	445	NDK68
NTS68/300	411	405,5	NTC68/300	491	445	NDK68
NTS64	381	375,5	NTC64	460	415	NDK64
NTS68	381	375,5	NTC68	461	415	NDK68
NTS76/320	411	405,5	NTC76/320	490	445	NDK76
NTS68	411	405,5	NTC68	490	445	NDK68
NTS72	411	405,5	NTC72	490	445	NDK72
NTS80/340	441	435,5	NTC80/340	520	475	NDK80
NTS80/340	441	435,5	NTC80/340	520	475	NDK80
NTS72	381	375,5	NTC72	461	415	NDK72
NTS76	411	405,5	NTC76	490	445	NDK76
NTS76	411	405,5	NTC76	491	445	NDK76
NTS92/360	471	465,5	NTC92/360	550	505	NDK92
NTS80	441	435,5	NTC80	520	475	NDK80
NTS88/380	471	465,5	NTC88/380	550	505	NDK88
NTS80	411	405,5	NTC80	491	445	NDK80
NTS92/400	481	475,5	NTC92/400	560	515	NDK92
NTS84	411	405,5	NTC84	491	445	NDK84
NTS84	471	465,5	NTC84	550	505	NDK84
NTS96/410	481	475,5	NTC96/410	560	515	NDK96
NTS88	471	465,5	NTC88	550	505	NDK88
NTS88	441	435,5	NTC88	521	475	NDK88
NTS92	481	475,5	NTC92	560	515	NDK92
NTS92	471	465,5	NTC92	551	505	NDK92
NTS96	471	465,5	NTC96	551	505	NDK96
NTS96	481	475,5	NTC96	560	515	NDK96
NTS500	471	465,5	NTC500	551	505	NDK500
NTS530	481	475,5	NTC530	561	515	NDK530

6.11.4 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, imperial shaft





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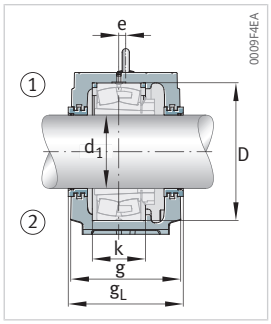
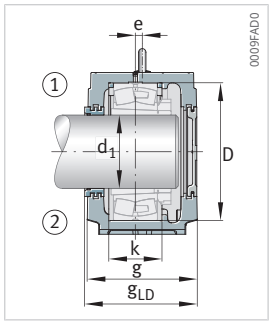
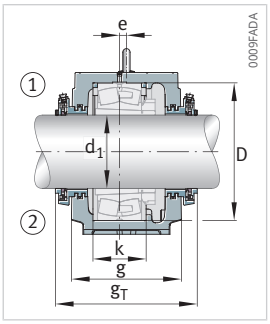
Housing	d_1	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	inch	mm	–	–	–	–
SNS3136-H-D	**5**	127	22328..-K	H2328×500	NFR300/7	2
SNS3234-H-D	**6**	152,4	23234..-K	H2334(-HG)X600	NFR310/6	2
SNS3048-H-D	**6**	152,4	22334..-K	H2334(-HG)X600	NFR360/10	2
SNS3134-H-D	**6**	152,4	23134..-K	H3134(-HG)X600	NFR280/10	2
SNS3040-H-D	**6**	152,4	22234..-K	H3134(-HG)X600	NFR310/18	2
SNS3056-H-D	**7**	177,8	22340..-K	H2340(-HG)X700	NFR420/14	2
SNS3238-H-D	**7**	177,8	23238..-K	H2338(-HG)X700	NFR340/6	2
SNS3040-H-D	**7**	177,8	23040..-K	H3040(-HG)X700	NFR310/10	4
SNS3148-H-D	**7**	177,8	22338..-K	H2338(-HG)X700	NFR400/8	2
SNS3048-H-D	**7**	177,8	22240..-K	H3140(-HG)X700	NFR360/21	2
SNS3038-H-D	**7**	177,8	23038..-K	H3038(-HG)X700	NFR290/10	4
SNS3140-H-D	**7**	177,8	23140..-K	H3140(-HG)X700	NFR340/10	2
SNS3240-H-D	**7**	177,8	23240..-K	H2340(-HG)X700	NFR360/6	2
SNS3140-H-D	**7**	177,8	22238..-K	H3138(-HG)X700	NFR340/10	4
SNS3044-H-D	**8**	203,2	23044..-K	H3044(-HG)X800	NFR340/10	4
SNS3148-H-D	**8**	203,2	22244..-K	H3144X(-HG)X800	NFR400/10	4
SNS3056-H-D	**10**	254	23056..-K	H3056(-HG)X1000	NFR420/10	6
SNS3160-H-D	**10**	254	22256..-K	H3156X(-HG)X1000	NFR500/25	2
SNS3156-H-D	**10**	254	23156..-K	H3156X(-HG)X1000	NFR460/10	2
SNS3160-H-D	**11**	279,4	23160..-K	H3160(-HG)X1100	NFR500/10	2
SNS3164-H-D	**11**	279,4	22260..-K	H3160(-HG)X1100	NFR540/28	2
SNS3060-H-D	**11**	279,4	23060..-K	H3060(-HG)X1100	NFR460/25	2
SNS3260-H-D	**11**	279,4	23260..-K	H3260(-HG)X1100	NFR540/10	2
SNS2264-H-D	**12**	304,8	22264..-K	H3164-HGX1200	NFR580/20	2
SNS3164-H-D	**12**	304,8	23164..-K	H3164-HGX1200	NFR540/10	2
SNS3264-H-D	**12**	304,8	23264..-K	H3264-HGX1200	NFR580/10	2
SNS3064-H-D	**12**	304,8	23064..-K	H3064-HGX1200	NFR480/10	6
SNS3180-H-D	**13**	330,2	22272..-K	H3172-HGX1300	NFR650/25	2
SNS3072-H-D	**13**	330,2	23072..-K	H3072-HGX1300	NFR540/16	4
SNS3276-H-D	**14**	355,6	23276..-K	H3276-HGX1400	NFR680/10	2
SNS3176-H-D	**14**	355,6	23176..-K	H3176-HGX1400	NFR620/10	2
SNS3076-H-D	**14**	355,6	23076..-K	H3076-HGX1400	NFR560/22,5	2
SNS3180-H-D	**15**	381	23180..-K	H3180-HGX1500	NFR650/10	2
SNS3280-H-D	**15**	381	23280..-K	H3280-HGX1500	NFR720/10	2
SNS3080-H-D	**15**	381	23080..-K	H3080-HGX1500	NFR600/22	2
SNS3092-H-D	**17**	431,8	23092..-K	H3092-HGX1700	NFR680/30,5	2
SNS3192-H-D	**17**	431,8	23192..-K	H3192-HGX1700	NFR760/10	2
SNS3096-H-D	**18**	457,2	23096..-K	H3096-HGX1800	NFR700/29,5	2
SNS3196-H-D	**18**	457,2	23196..-K	H3196-HGX1800	NFR790/10	2
SNS3048-H-D	**7 1/8**	180,975	22240..-K	H3140(-HG)X702	NFR360/21	2



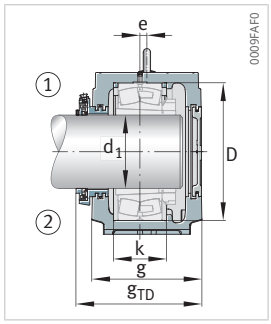
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

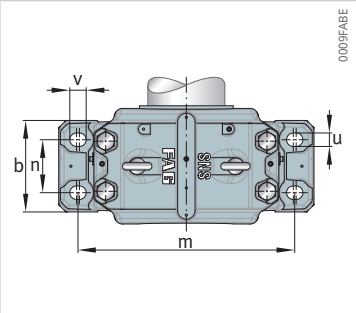


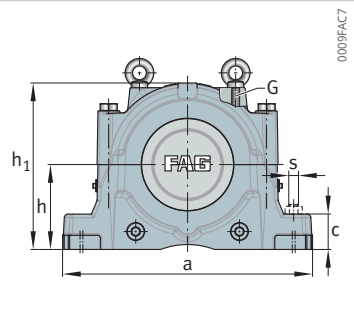
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS36×500	251	245,5	NTC36×500	310	274,5	NDK36
NTS40×600	271	265,5	NTC40×600	326	293	NDK40
NTS48×600	301	295,5	NTC48×600	380	335	NDK48
NTS34×600	241	235,5	NTC34×600	298	264	NDK34
NTS40×600	271	265,5	NTC40×600	326	293	NDK40
NTS56×700	331	325,5	NTC56×700	410	365	NDK56
NTS40×700	291	285,5	NTC40×700	346	313	NDK40
NTS40×700	271	265,5	NTC40×700	326	293	NDK40
NTS48×700	321	315,5	NTC48×700	400	355	NDK48
NTS48×700	301	295,5	NTC48×700	380	335	NDK48
NTS38×700	251	245,5	NTC38×700	311	275,5	NDK38
NTS40×700	291	285,5	NTC40×700	346	313	NDK40
NTS48×700	301	295,5	NTC48×700	380	335	NDK48
NTS40×700	291	285,5	NTC40×700	346	313	NDK40
NTS44×800	291	285,5	NTC44×800	370	325	NDK44
NTS48×800	321	315,5	NTC48×800	400	355	NDK48
NTS56×1000	331	325,5	NTC56×1000	410	365	NDK56
NTS60×1000	361	355,5	NTC60×1000	440	395	NDK60
NTS56×1000	331	325,5	NTC56×1000	410	365	NDK56
NTS60×1100	361	355,5	NTC60×1100	440	395	NDK60
NTS64×1100	381	375,5	NTC64×1100	460	415	NDK64
NTS60×1100	331	325,5	NTC60×1100	410	365	NDK60
NTS64×1100	381	375,5	NTC64×1100	460	415	NDK64
NTS68×1200	411	405,5	NTC68×1200	490	445	NDK68
NTS64×1200	381	375,5	NTC64×1200	460	415	NDK64
NTS68×1200	411	405,5	NTC68×1200	490	445	NDK68
NTS64×1200	361	355,5	NTC64×1200	440	395	NDK64
NTS80×1300	441	435,5	NTC80×1300	520	475	NDK80
NTS72×1300	381	375,5	NTC72×1300	460	415	NDK72
NTS92×1400	471	465,5	NTC92×1400	550	505	NDK92
NTS76×1400	411	405,5	NTC76×1400	490	445	NDK76
NTS76×1400	411	405,5	NTC76×1400	490	445	NDK76
NTS80×1500	441	435,5	NTC80×1500	520	475	NDK80
NTS88×1500	471	465,5	NTC88×1500	550	505	NDK88
NTS80×1500	411	405,5	NTC80×1500	490	445	NDK80
NTS92×1700	471	465,5	NTC92×1700	550	505	NDK92
NTS92×1700	481	475,5	NTC92×1700	560	515	NDK92
NTS96×1800	471	465,5	NTC96×1800	550	505	NDK96
NTS96×1800	481	475,5	NTC96×1800	560	515	NDK96
NTS48×702	301	295,5	NTC48×702	380	335	NDK48

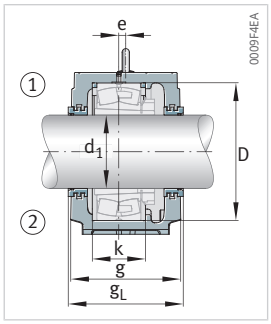
6.11.4 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, imperial shaft



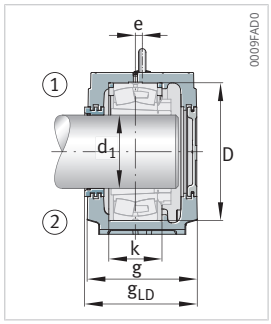


6

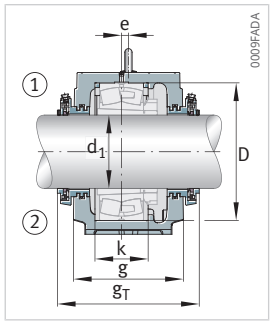
Housing	d_1	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	inch	mm	–	–	–	–
SNS3240-H-D	7 1/8	180,975	23240..-K	H2340(-HG)X702	NFR360/6	2
SNS3056-H-D	7 1/8	180,975	22340..-K	H2340(-HG)X702	NFR420/14	2
SNS3040-H-D	7 1/8	180,975	23040..-K	H3040(-HG)X702	NFR310/10	4
SNS3036-H-D	6 3/8	161,925	23036..-K	H3036(-HG)X606	NFR280/17	2
SNS3138-H-D	6 3/8	161,925	22236..-K	H3136(-HG)X606	NFR320/19	2
SNS3234-H-D	5 7/8	149,225	23234..-K	H2334(-HG)X514	NFR310/6	2
SNS3048-H-D	5 7/8	149,225	22334..-K	H2334(-HG)X514	NFR360/10	2
SNS3040-H-D	5 7/8	149,225	22234..-K	H3134(-HG)X514	NFR310/18	2
SNS3148-H-D	6 7/8	174,625	22338..-K	H2338(-HG)X614	NFR400/8	2
SNS3238-H-D	6 7/8	174,625	23238..-K	H2338(-HG)X614	NFR340/6	2
SNS3140-H-D	6 7/8	174,625	22238..-K	H3138(-HG)X614	NFR340/10	4
SNS3038-H-D	6 7/8	174,625	23038..-K	H3038(-HG)X614	NFR290/10	4
SNS3044-H-D	7 7/8	200,025	23044..-K	H3044(-HG)X714	NFR340/10	4
SNS3148-H-D	7 7/8	200,025	22244..-K	H3144X(-HG)X714	NFR400/10	4
SNS3138-H-D	5 3/16	131,7625	22330..-K	H2330×503	NFR320/8	2
SNS3040-H-D	7 3/16	182,5625	23040..-K	H3040(-HG)X703	NFR310/10	4
SNS3140-H-D	7 3/16	182,5625	23140..-K	H3140(-HG)X703	NFR340/10	2
SNS3048-H-D	7 3/16	182,5625	22240..-K	H3140(-HG)X703	NFR360/21	2
SNS3056-H-D	7 3/16	182,5625	22340..-K	H2340(-HG)X703	NFR420/14	2
SNS3240-H-D	7 3/16	182,5625	23240..-K	H2340(-HG)X703	NFR360/6	2
SNS3134-H-D	4 1/2	114,3	22326..-K	H2326×408	NFR280/7,5	2
SNS3140-H-D	5 1/2	139,7	22332..-K	H2332×508	NFR340/9	2
SNS3038-H-D	5 1/2	139,7	22232..-K	H3132×508	NFR290/17,5	2
SNS3036-H-D	6 1/2	165,1	23036..-K	H3036(-HG)X608	NFR280/17	2
SNS3136-H-D	6 1/2	165,1	23136..-K	H3136(-HG)X608	NFR300/10	2
SNS3236-H-D	6 1/2	165,1	23236..-K	H2336(-HG)X608	NFR320/6	2
SNS3138-H-D	6 1/2	165,1	22236..-K	H3136(-HG)X608	NFR320/19	2
SNS3048-H-D	8 1/2	215,9	23048..-K	H3048(-HG)X808	NFR360/12	4
SNS3152-H-D	8 1/2	215,9	22248..-K	H3148X(-HG)X808	NFR440/22	2
SNS3152-H-D	9 1/2	241,3	23152..-K	H3152X(-HG)X908	NFR440/10	2
SNS3252-H-D	9 1/2	241,3	23252..-K	H2352(-HG)X908	NFR480/10	2
SNS3164-H-D	9 1/2	241,3	22352..-K	H2352(-HG)X908	NFR540/15,5	2
SNS3052-H-D	9 1/2	241,3	23052..-K	H3052(-HG)X908	NFR400/22	2
SNS3064-H-D	9 1/2	241,3	22252..-K	H3152X(-HG)X908	NFR480/25,5	2
SNS3256-H-D	10 1/2	266,7	23256..-K	H2356(-HG)X1008	NFR500/10	2
SNS3056-H-D	10 1/2	266,7	23056..-K	H3056(-HG)X1008	NFR420/10	6
SNS3156-H-D	10 1/2	266,7	23156..-K	H3156X(-HG)X1008	NFR460/10	2
SNS3160-H-D	10 1/2	266,7	22256..-K	H3156X(-HG)X1008	NFR500/25	2
SNS3168-H-D	10 1/2	266,7	22356..-K	H2356(-HG)X1008	NFR580/17,5	2
SNS2264-H-D	11 1/2	292,1	22264..-K	H3164(-HG)X1108	NFR580/20	2



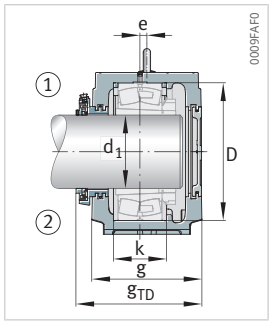
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

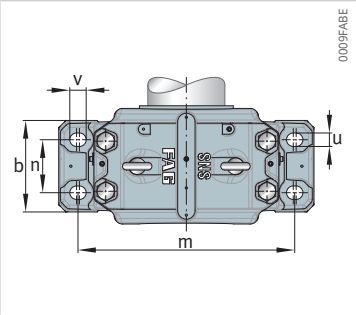


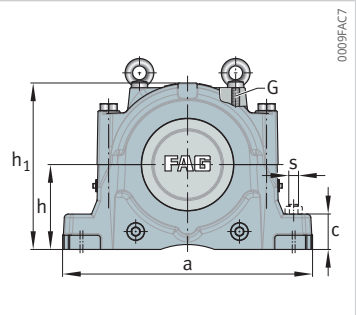
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS48×702	301	295,5	NTC48×702	380	335	NDK48
NTS56×702	331	325,5	NTC56×702	410	365	NDK56
NTS40×702	271	265,5	NTC40×702	326	293	NDK40
NTS36×606	241	235,5	NTC36×606	299	264,5	NDK36
NTS38×606	271	265,5	NTC38×606	329	294,5	NDK38
NTS40×514	271	265,5	NTC40×514	326	293	NDK40
NTS48×514	301	295,5	NTC48×514	380	335	NDK48
NTS40×514	271	265,5	NTC40×514	326	293	NDK40
NTS48×614	321	315,5	NTC48×614	400	355	NDK48
NTS40×614	291	285,5	NTC40×614	346	313	NDK40
NTS40×614	291	285,5	NTC40×614	346	313	NDK40
NTS38×614	251	245,5	NTC38×614	311	275,5	NDK38
NTS44×714	291	285,5	NTC44×714	370	325	NDK44
NTS48×714	321	315,5	NTC48×714	400	355	NDK48
NTS38×503	271	265,5	NTC38×503	330	294,5	NDK38
NTS40×703	271	265,5	NTC40×703	326	293	NDK40
NTS40×703	291	285,5	NTC40×703	346	313	NDK40
NTS48×703	301	295,5	NTC48×703	380	335	NDK48
NTS56×703	331	325,5	NTC56×703	410	365	NDK56
NTS48×703	301	295,5	NTC48×703	380	335	NDK48
NTS34×408	241	235,5	NTC34×408	299	264	NDK34
NTS40×508	291	285,5	NTC40×508	347	313	NDK40
NTS38×508	251	245,5	NTC38×508	312	275,5	NDK38
NTS36×608	241	235,5	NTC36×608	299	264,5	NDK36
NTS36×608	251	245,5	NTC36×608	309	274,5	NDK36
NTS38×608	271	265,5	NTC38×608	329	294,5	NDK38
NTS38×608	271	265,5	NTC38×608	329	294,5	NDK38
NTS48×808	301	295,5	NTC48×808	380	335	NDK48
NTS52×808	331	325,5	NTC52×808	410	365	NDK52
NTS52×908	331	325,5	NTC52×908	410	365	NDK52
NTS64×908	361	355,5	NTC64×908	440	395	NDK64
NTS64×908	381	375,5	NTC64×908	460	415	NDK64
NTS52×908	321	315,5	NTC52×908	400	355	NDK52
NTS64×908	361	355,5	NTC64×908	440	395	NDK64
NTS60×1008	361	355,5	NTC60×1008	440	395	NDK60
NTS56×1008	331	325,5	NTC56×1008	410	365	NDK56
NTS56×1008	331	325,5	NTC56×1008	410	365	NDK56
NTS60×1008	361	355,5	NTC60×1008	440	395	NDK60
NTS68×1008	411	405,5	NTC68×1008	490	445	NDK68
NTS68×1108	411	405,5	NTC68×1108	490	445	NDK68

6.11.4 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, imperial shaft





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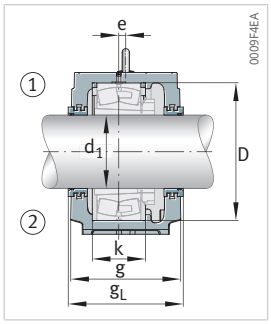
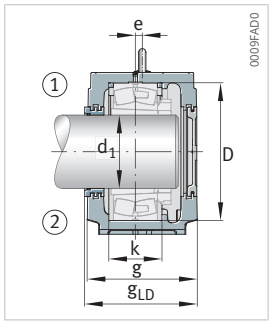
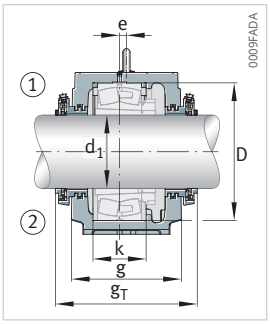
Housing	d_1	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	inch	mm	–	–	–	–
SNS3064-H-D	**11 1/2**	292,1	23064..-K	H3064(-HG)X1108	NFR480/10	6
SNS3068-H-D	**12 1/2**	317,5	23068..-K	H3068-HGX1208	NFR520/16	4
SNS3268-H-D	**12 1/2**	317,5	23268..-K	H3268-HGX1208	NFR620/10	2
SNS3168-H-D	**12 1/2**	317,5	23168..-K	H3168-HGX1208	NFR580/10	2
SNS3072-H-D	**13 1/2**	342,9	23072..-K	H3072-HGX1308	NFR540/16	4
SNS3172-H-D	**13 1/2**	342,9	23172..-K	H3172-HGX1308	NFR600/10	2
SNS3180-H-D	**13 1/2**	342,9	22272..-K	H3172-HGX1308	NFR650/25	2
SNS3272-H-D	**13 1/2**	342,9	23272..-K	H3272-HGX1308	NFR650/10	2
SNS3288-H-D	**16 1/2**	419,1	23288..-K	H3288-HGX1608	NFR790/10	2
SNS3188-H-D	**16 1/2**	419,1	23188..-K	H3188-HGX1608	NFR720/10	2
SNS3088-H-D	**16 1/2**	419,1	23088..-K	H3088-HGX1608	NFR650/21,5	2
SNS30/500-H-D	**18 1/2**	469,9	230/500..-K	H30/500-HGX1808	NFR720/29,5	2
SNS30/530-H-D	**19 1/2**	495,3	230/530..-K	H30/530-HGX1908	NFR780/31,5	2
SNS3138-H-D	**5 1/4**	133,35	22330..-K	H2330×504	NFR320/8	2
SNS3148-H-D	**6 3/4**	171,45	22338..-K	H2338(-HG)X612	NFR400/8	2
SNS3238-H-D	**6 3/4**	171,45	23238..-K	H2338(-HG)X612	NFR340/6	2
SNS3138-H-D	**6 3/4**	171,45	23138..-K	H3138(-HG)X612	NFR320/10	2
SNS3140-H-D	**6 3/4**	171,45	22238..-K	H3138(-HG)X612	NFR340/10	4
SNS3038-H-D	**6 3/4**	171,45	23038..-K	H3038(-HG)X612	NFR290/10	4
SNS3056-H-D	**7 1/4**	184,15	22340..-K	H2340(-HG)X704	NFR420/14	2
SNS3240-H-D	**7 1/4**	184,15	23240..-K	H2340(-HG)X704	NFR360/6	2
SNS3048-H-D	**7 1/4**	184,15	22240..-K	H3140(-HG)X704	NFR360/21	2
SNS3040-H-D	**7 1/4**	184,15	23040..-K	H3040(-HG)X704	NFR310/10	4
SNS3184-H-D	**15 3/4**	400,05	23184..-K	H3184-HGX1512	NFR700/10	2
SNS3284-H-D	**15 3/4**	400,05	23284..-K	H3284-HGX1512	NFR760/10	2
SNS3084-H-D	**15 3/4**	400,05	23084..-K	H3084X-HGX1512	NFR620/22	2
SNS3036-H-D	**6 5/16**	160,3375	23036..-K	H3036(-HG)X605	NFR280/17	2
SNS3138-H-D	**6 5/16**	160,3375	22236..-K	H3136(-HG)X605	NFR320/19	2
SNS3134-H-D	**4 7/16**	112,7125	22326..-K	H2326×407	NFR280/7,5	2
SNS3140-H-D	**5 7/16**	138,1125	22332..-K	H2332×507	NFR340/9	2
SNS3038-H-D	**5 7/16**	138,1125	22232..-K	H3132×507	NFR290/17,5	2
SNS3036-H-D	**6 7/16**	163,5125	23036..-K	H3036(-HG)X607	NFR280/17	2
SNS3138-H-D	**6 7/16**	163,5125	22236..-K	H3136(-HG)X607	NFR320/19	2
SNS3136-H-D	**6 7/16**	163,5125	23136..-K	H3136(-HG)X607	NFR300/10	2
SNS3236-H-D	**6 7/16**	163,5125	23236..-K	H2336(-HG)X607	NFR320/6	2
SNS3152-H-D	**8 7/16**	214,3125	22248..-K	H3148X(-HG)X807	NFR440/22	2
SNS3048-H-D	**8 7/16**	214,3125	23048..-K	H3048(-HG)X807	NFR360/12	4
SNS3252-H-D	**9 7/16**	239,7125	23252..-K	H2352(-HG)X907	NFR480/10	2
SNS3052-H-D	**9 7/16**	239,7125	23052..-K	H3052(-HG)X907	NFR400/22	2
SNS3152-H-D	**9 7/16**	239,7125	23152..-K	H3152X(-HG)X907	NFR440/10	2



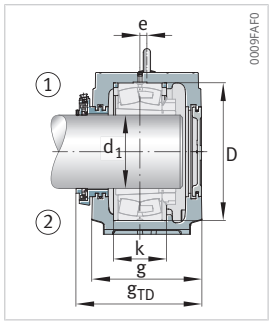
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

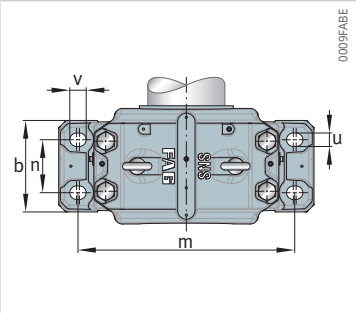


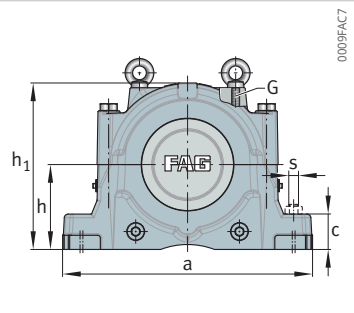
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS64×1108	361	355,5	NTC64×1108	440	395	NDK64
NTS68×1208	381	375,5	NTC68×1208	460	415	NDK68
NTS76×1208	411	405,5	NTC76×1208	490	445	NDK76
NTS68×1208	411	405,5	NTC68×1208	490	445	NDK68
NTS72×1308	381	375,5	NTC72×1308	460	415	NDK72
NTS72×1308	411	405,5	NTC72×1308	490	445	NDK72
NTS80×1308	441	435,5	NTC80×1308	520	475	NDK80
NTS80×1308	441	435,5	NTC80×1308	520	475	NDK80
NTS96×1608	481	475,5	NTC96×1608	560	515	NDK96
NTS88×1608	471	465,5	NTC88×1608	550	505	NDK88
NTS88×1608	441	435,5	NTC88×1608	520	475	NDK88
NTS500×1808	471	465,5	NTC500×1808	550	505	NDK500
NTS530×1908	481	475,5	NTC530×1908	560	515	NDK530
NTS38×504	271	265,5	NTC38×504	330	294,5	NDK38
NTS48×612	321	315,5	NTC48×612	400	355	NDK48
NTS40×612	291	285,5	NTC40×612	346	313	NDK40
NTS38×612	271	265,5	NTC38×612	329	294,5	NDK38
NTS40×612	291	285,5	NTC40×612	346	313	NDK40
NTS38×612	251	245,5	NTC38×612	311	275,5	NDK38
NTS56×704	331	325,5	NTC56×704	410	365	NDK56
NTS48×704	301	295,5	NTC48×704	380	335	NDK48
NTS48×704	301	295,5	NTC48×704	380	335	NDK48
NTS40×704	271	265,5	NTC40×704	326	293	NDK40
NTS84×1512	471	465,5	NTC84×1512	550	505	NDK84
NTS92×1512	481	475,5	NTC92×1512	560	515	NDK92
NTS84×1512	411	405,5	NTC84×1512	490	445	NDK84
NTS36×605	241	235,5	NTC36×605	299	264,5	NDK36
NTS38×605	271	265,5	NTC38×605	329	294,5	NDK38
NTS34×407	241	235,5	NTC34×407	299	264	NDK34
NTS40×507	291	285,5	NTC40×507	347	313	NDK40
NTS38×507	251	245,5	NTC38×507	312	275,5	NDK38
NTS36×607	241	235,5	NTC36×607	299	264,5	NDK36
NTS38×607	271	265,5	NTC38×607	329	294,5	NDK38
NTS36×607	251	245,5	NTC36×607	309	274,5	NDK36
NTS38×607	271	265,5	NTC38×607	329	294,5	NDK38
NTS52×807	331	325,5	NTC52×807	410	365	NDK52
NTS48×807	301	295,5	NTC48×807	380	335	NDK48
NTS64×907	361	355,5	NTC64×907	440	395	NDK64
NTS52×907	321	315,5	NTC52×907	400	355	NDK52
NTS52×907	331	325,5	NTC52×907	410	365	NDK52

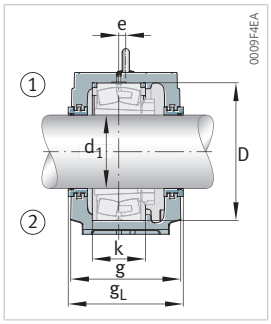
6.11.4 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, imperial shaft



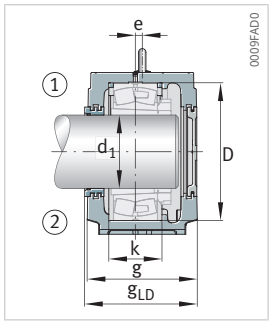


6

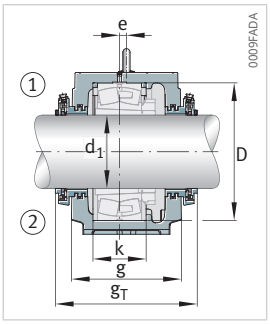
Housing	d_1	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	inch	mm	–	–	–	–
SNS3168-H-D	**10 7/16**	265,1125	22356..-K	H2356(-HG)X1007	NFR580/17,5	2
SNS3056-H-D	**10 7/16**	265,1125	23056..-K	H3056(-HG)X1007	NFR420/10	6
SNS3160-H-D	**10 7/16**	265,1125	22256..-K	H3156X(-HG)X1007	NFR500/25	2
SNS3156-H-D	**10 7/16**	265,1125	23156..-K	H3156X(-HG)X1007	NFR460/10	2
SNS3256-H-D	**10 7/16**	265,1125	23256..-K	H2356(-HG)X1007	NFR500/10	2
SNS3064-H-D	**11 7/16**	290,5125	23064..-K	H3064(-HG)X1107	NFR480/10	6
SNS3268-H-D	**12 7/16**	315,9125	23268..-K	H3268-HGX1207	NFR620/10	2
SNS3168-H-D	**12 7/16**	315,9125	23168..-K	H3168-HGX1207	NFR580/10	2
SNS3068-H-D	**12 7/16**	315,9125	23068..-K	H3068-HGX1207	NFR520/16	4
SNS3040-H-D	**5 13/16**	147,6375	22234..-K	H3134(-HG)X513	NFR310/18	2
SNS3234-H-D	**5 13/16**	147,6375	23234..-K	H2334(-HG)X513	NFR310/6	2
SNS3048-H-D	**5 13/16**	147,6375	22334..-K	H2334(-HG)X513	NFR360/10	2
SNS3038-H-D	**6 13/16**	173,0375	23038..-K	H3038(-HG)X613	NFR290/10	4
SNS3238-H-D	**6 13/16**	173,0375	23238..-K	H2338(-HG)X613	NFR340/6	2
SNS3148-H-D	**6 13/16**	173,0375	22338..-K	H2338(-HG)X613	NFR400/8	2
SNS3140-H-D	**6 13/16**	173,0375	22238..-K	H3138(-HG)X613	NFR340/10	4
SNS3148-H-D	**7 13/16**	198,4375	22244..-K	H3144X(-HG)X713	NFR400/10	4
SNS3044-H-D	**7 13/16**	198,4375	23044..-K	H3044(-HG)X713	NFR340/10	4
SNS3136-H-D	**4 15/16**	125,4125	22328..-K	H2328×415	NFR300/7	2
SNS3134-H-D	**5 15/16**	150,8125	23134..-K	H3134(-HG)X515	NFR280/10	2
SNS3040-H-D	**5 15/16**	150,8125	22234..-K	H3134(-HG)X515	NFR310/18	2
SNS3234-H-D	**5 15/16**	150,8125	23234..-K	H2334(-HG)X515	NFR310/6	2
SNS3048-H-D	**5 15/16**	150,8125	22334..-K	H2334(-HG)X515	NFR360/10	2
SNS3148-H-D	**6 15/16**	176,2125	22338..-K	H2338(-HG)X615	NFR400/8	2
SNS3238-H-D	**6 15/16**	176,2125	23238..-K	H2338(-HG)X615	NFR340/6	2
SNS3138-H-D	**6 15/16**	176,2125	23138..-K	H3138(-HG)X615	NFR320/10	2
SNS3140-H-D	**6 15/16**	176,2125	22238..-K	H3138(-HG)X615	NFR340/10	4
SNS3038-H-D	**6 15/16**	176,2125	23038..-K	H3038(-HG)X615	NFR290/10	4
SNS3156-H-D	**7 15/16**	201,6125	22344..-K	H2344(-HG)X715	NFR460/10,5	2
SNS3144-H-D	**7 15/16**	201,6125	23144..-K	H3144X(-HG)X715	NFR370/10	2
SNS3244-H-D	**7 15/16**	201,6125	23244..-K	H2344(-HG)X715	NFR400/10	2
SNS3044-H-D	**7 15/16**	201,6125	23044..-K	H3044(-HG)X715	NFR340/10	4
SNS3148-H-D	**7 15/16**	201,6125	22244..-K	H3144X(-HG)X715	NFR400/10	4
SNS3152-H-D	**8 15/16**	227,0125	22248..-K	H3148X(-HG)X815	NFR440/22	2
SNS3148-H-D	**8 15/16**	227,0125	23148..-K	H3148X(-HG)X815	NFR400/10	2
SNS3248-H-D	**8 15/16**	227,0125	23248..-K	H2348(-HG)X815	NFR440/10	2
SNS3048-H-D	**8 15/16**	227,0125	23048..-K	H3048(-HG)X815	NFR360/12	4
SNS3160-H-D	**8 15/16**	227,0125	22348..-K	H2348(-HG)X815	NFR500/12,5	2
SNS3056-H-D	**9 15/16**	252,4125	23056..-K	H3056(-HG)X915	NFR420/10	6
SNS3164-H-D	**10 15/16**	277,8125	22260..-K	H3160(-HG)X1015	NFR540/28	2



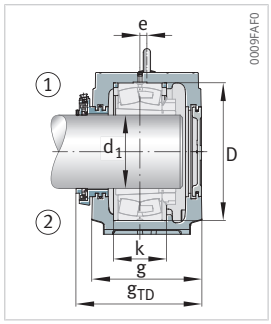
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

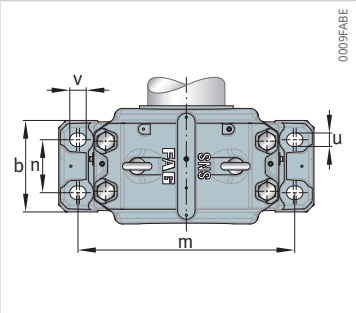


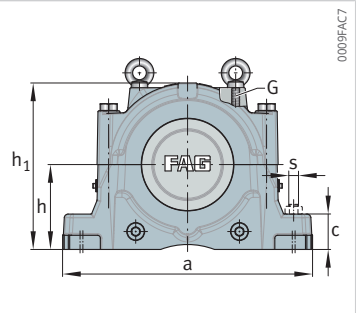
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS68×1007	411	405,5	NTC68×1007	490	445	NDK68
NTS56×1007	331	325,5	NTC56×1007	410	365	NDK56
NTS60×1007	361	355,5	NTC60×1007	440	395	NDK60
NTS56×1007	331	325,5	NTC56×1007	410	365	NDK56
NTS60×1007	361	355,5	NTC60×1007	440	395	NDK60
NTS64×1107	361	355,5	NTC64×1107	440	395	NDK64
NTS76×1207	411	405,5	NTC76×1207	490	445	NDK76
NTS68×1207	411	405,5	NTC68×1207	490	445	NDK68
NTS68×1207	381	375,5	NTC68×1207	460	415	NDK68
NTS40×513	271	265,5	NTC40×513	327	293	NDK40
NTS40×513	271	265,5	NTC40×513	326	293	NDK40
NTS48×513	301	295,5	NTC48×513	380	335	NDK48
NTS38×613	251	245,5	NTC38×613	311	275,5	NDK38
NTS40×613	291	285,5	NTC40×613	346	313	NDK40
NTS48×613	321	315,5	NTC48×613	400	355	NDK48
NTS40×613	291	285,5	NTC40×613	346	313	NDK40
NTS48×713	321	315,5	NTC48×713	400	355	NDK48
NTS44×713	291	285,5	NTC44×713	370	325	NDK44
NTS36×415	251	245,5	NTC36×415	310	274,5	NDK36
NTS34×515	241	235,5	NTC34×515	298	264	NDK34
NTS40×515	271	265,5	NTC40×515	326	293	NDK40
NTS40×515	271	265,5	NTC40×515	326	293	NDK40
NTS48×515	301	295,5	NTC48×515	380	335	NDK48
NTS48×615	321	315,5	NTC48×615	400	355	NDK48
NTS40×615	291	285,5	NTC40×615	346	313	NDK40
NTS38×615	271	265,5	NTC38×615	329	294,5	NDK38
NTS40×615	291	285,5	NTC40×615	346	313	NDK40
NTS38×615	251	245,5	NTC38×615	309	274,5	NDK38
NTS56×715	331	325,5	NTC56×715	410	365	NDK56
NTS44×715	301	295,5	NTC44×715	380	335	NDK44
NTS48×715	321	315,5	NTC48×715	400	355	NDK48
NTS44×715	291	285,5	NTC44×715	370	325	NDK44
NTS48×715	321	315,5	NTC48×715	400	355	NDK48
NTS52×815	331	325,5	NTC52×815	410	365	NDK52
NTS48×815	321	315,5	NTC48×815	400	355	NDK48
NTS52×815	331	325,5	NTC52×815	410	365	NDK52
NTS48×815	301	295,5	NTC48×815	380	335	NDK48
NTS60×815	361	355,5	NTC60×815	440	395	NDK60
NTS56×915	331	325,5	NTC56×915	410	365	NDK56
NTS64×1015	381	375,5	NTC64×1015	460	415	NDK64

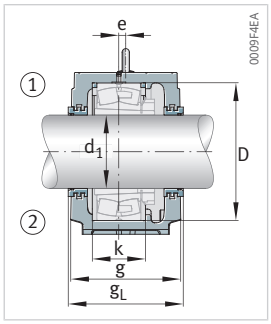
6.11.4 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, imperial shaft



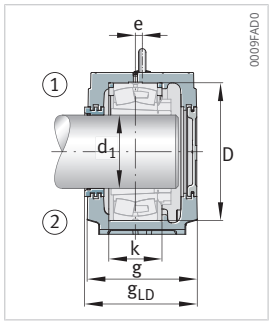


6

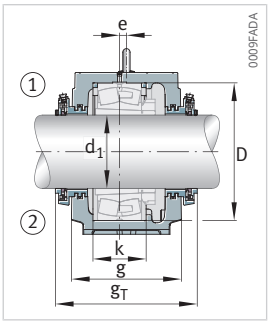
Housing	d_1	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	inch	mm	–	–	–	–
SNS3060-H-D	**10 15/16**	277,8125	23060..-K	H3060(-HG)X1015	NFR460/25	2
SNS3160-H-D	**10 15/16**	277,8125	23160..-K	H3160(-HG)X1015	NFR500/10	2
SNS3260-H-D	**10 15/16**	277,8125	23260..-K	H3260(-HG)X1015	NFR540/10	2
SNS3164-H-D	**11 15/16**	303,2125	23164..-K	H3164-HGX1115	NFR540/10	2
SNS3264-H-D	**11 15/16**	303,2125	23264..-K	H3264-HGX1115	NFR580/10	2
SNS3064-H-D	**11 15/16**	303,2125	23064..-K	H3064-HGX1115	NFR480/10	6
SNS3072-H-D	**12 15/16**	328,6125	23072..-K	H3072-HGX1215	NFR540/16	4
SNS3172-H-D	**13 7/16**	341,3125	23172..-K	H3172-HGX1307	NFR600/10	2
SNS3072-H-D	**13 7/16**	341,3125	23072..-K	H3072-HGX1307	NFR540/16	4
SNS3272-H-D	**13 7/16**	341,3125	23272..-K	H3272-HGX1307	NFR650/10	2
SNS3176-H-D	**13 15/16**	354,0125	23176..-K	H3176-HGX1315	NFR620/10	2
SNS3076-H-D	**13 15/16**	354,0125	23076..-K	H3076-HGX1315	NFR560/22,5	2
SNS3276-H-D	**13 15/16**	354,0125	23276..-K	H3276-HGX1315	NFR680/10	2



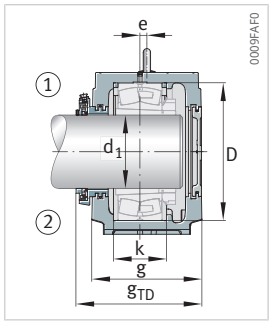
NTS labyrinth seal



NTS labyrinth seal, NDK cover



Taconite NTC seal

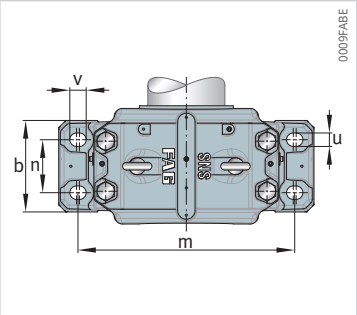


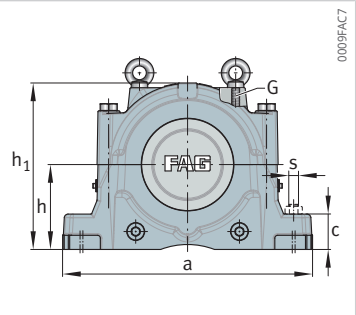
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS60×1015	331	325,5	NTC60×1015	410	365	NDK60
NTS60×1015	361	355,5	NTC60×1015	440	395	NDK60
NTS64×1015	381	375,5	NTC64×1015	460	415	NDK64
NTS64×1115	381	375,5	NTC64×1115	460	415	NDK64
NTS68×1115	411	405,5	NTC68×1115	490	445	NDK68
NTS64×1115	361	355,5	NTC64×1115	440	395	NDK64
NTS72×1215	381	375,5	NTC72×1215	460	415	NDK72
NTS72×1307	411	405,5	NTC72×1307	490	445	NDK72
NTS72×1307	381	375,5	NTC72×1307	460	415	NDK72
NTS80×1307	441	435,5	NTC80×1307	520	475	NDK80
NTS76×1315	411	405,5	NTC76×1315	490	445	NDK76
NTS76×1315	411	405,5	NTC76×1315	490	445	NDK76
NTS92×1315	471	465,5	NTC92×1315	550	505	NDK92

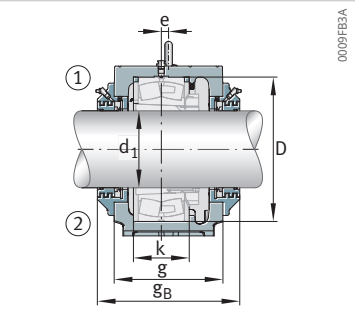
6.11.5 SNS, for spherical roller bearings with a tapered bore and adapter sleeve, metric shaft, bolt-on version



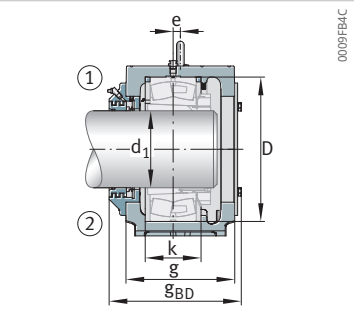


6

Housing	d_1	Bearing	Adapter sleeve	Locating ring	n_S
	mm	–	–	–	–
SNS3134-B-D	**150**	23134..-K	H3134(-HG)	NFR280/10	2
SNS3136-B-D	**160**	23136..-K	H3136(-HG)	NFR300/10	2
SNS3138-B-D	**170**	23138..-K	H3138(-HG)	NFR320/10	2
SNS3140-B-D	**180**	23140..-K	H3140(-HG)	NFR340/10	2
SNS3144-B-D	**200**	23144..-K	H3144X(-HG)	NFR370/10	2
SNS3148-B-D	**220**	23148..-K	H3148X(-HG)	NFR400/10	2
SNS3152-B-D	**240**	23152..-K	H3152X(-HG)	NFR440/10	2
SNS3156-B-D	**260**	23156..-K	H3156X(-HG)	NFR460/10	2
SNS3160-B-D	**280**	23160..-K	H3160(-HG)	NFR500/10	2
SNS3164-B-D	**300**	23164..-K	H3164-HG	NFR540/10	2
SNS3168-B-D	**320**	23168..-K	H3168-HG	NFR580/10	2



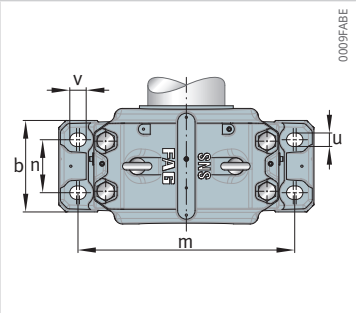
BTAC bolt-on seal

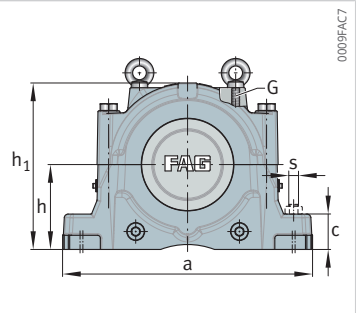


BTAC, bolt-on seal BDK cover

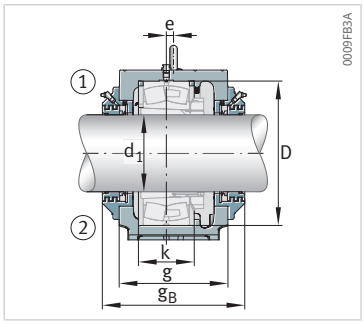
Bolt-on seal	g_B	g_{BD}	Cover
–	mm	mm	–
BTAC34	289	268	BDK34
BTAC36	296	275	BDK36
BTAC38	316	295	BDK38
BTAC40	336	315	BDK40
BTAC44	348	327	BDK44
BTAC48	378	353	BDK48
BTAC52	392	365	BDK52
BTAC56	392	365	BDK56
BTAC60	422	395	BDK60
BTAC64	442	415	BDK64
BTAC68	480	450	BDK68

6.11.6 SNS for spherical roller bearings with a tapered bore and adapter sleeve, metric shaft, bolt-on design

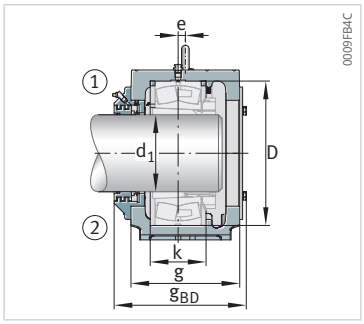




Housing	d_1	Bearing	Locating ring	n_S
	mm	–	–	–
SNS3134-B-D	**150**	231SM150-MA	NFR280/10	2
SNS3136-B-D	**160**	231SM160-MA	NFR300/10	2
SNS3138-B-D	**170**	231SM170-MA	NFR320/10	2
SNS3140-B-D	**180**	231SM180-MA	NFR340/10	2
SNS3144-B-D	**200**	231SM200-MA	NFR370/10	2
SNS3148-B-D	**220**	231SM220-MA	NFR400/10	2
SNS3152-B-D	**240**	231SM240-MA	NFR440/10	2
SNS3156-B-D	**260**	231SM260-MA	NFR460/10	2
SNS3160-B-D	**280**	231SM280-MA	NFR500/10	2
SNS3164-B-D	**300**	231SM300-MA	NFR540/10	2
SNS3168-B-D	**320**	231SM320-MA	NFR580/10	2



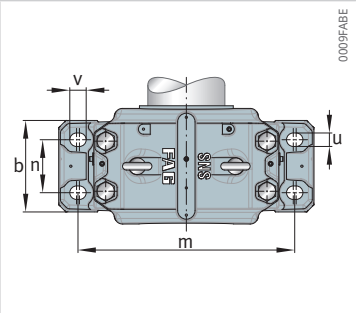
BTAC bolt-on seal

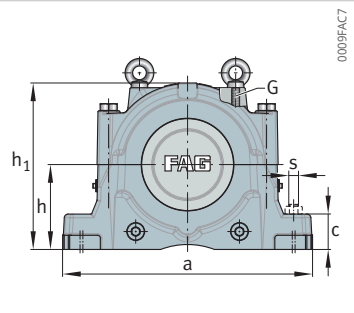


BTAC, bolt-on seal BDK cover

Bolt-on seal	g_B	g_{BD}	Cover
–	mm	mm	–
BTAC34	289	268	BDK34
BTAC36	296	275	BDK36
BTAC38	316	295	BDK38
BTAC40	336	315	BDK40
BTAC44	348	327	BDK44
BTAC48	378	353	BDK48
BTAC52	392	365	BDK52
BTAC56	392	365	BDK56
BTAC60	422	395	BDK60
BTAC64	442	415	BDK64
BTAC68	480	450	BDK68

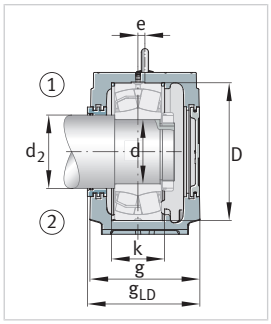
6.11.7 SNS for spherical roller bearings with a cylindrical bore, metric shaft



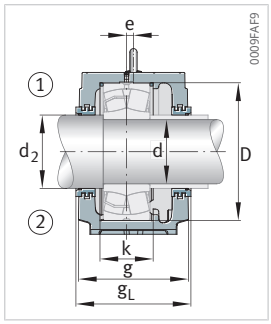


6

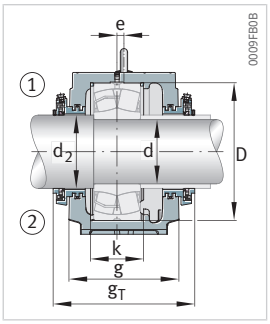
Housing	d	d_2	Bearing	Locating ring	n_S
	mm	mm	–	–	–
SNS3134-H-D	**130**	150	22326	NFR280/7,5	2
SNS3136-H-D	**140**	160	22328	NFR300/7	2
SNS3138-H-D	**150**	170	22330	NFR320/8	2
SNS3038-H-D	**160**	180	22232	NFR290/17,5	2
SNS3038-H-D	**160**	180	23232	NFR290/5,5	2
SNS3140-H-D	**160**	180	22332	NFR340/9	2
SNS3040-H-D	**170**	190	22234	NFR310/18	2
SNS3048-H-D	**170**	190	22334	NFR360/10	2
SNS3134-Z-D	**170**	180	23134	NFR280/10	2
SNS3234-Z-D	**170**	190	23234	NFR310/6	2
SNS3036-Z-D	**180**	200	24036	NFR280/4	2
SNS3036-Z-D	**180**	200	23036	NFR280/17	2
SNS3138-Z-D	**180**	200	22236	NFR320/19	2
SNS3136-Z-D	**180**	200	23136	NFR300/10	2
SNS3236-Z-D	**180**	200	23236	NFR320/6	2
SNS3148-H-D	**190**	210	22338	NFR400/8	2
SNS3238-Z-D	**190**	210	23238	NFR340/6	2
SNS3038-Z-D	**190**	200	23038	NFR290/10	4
SNS3038-Z-D	**190**	200	24038	NFR290/7,5	2
SNS3140-Z-D	**190**	210	22238	NFR340/10	4
SNS3138-Z-D	**190**	200	23138	NFR320/10	2
SNS3240-Z-D	**200**	220	23240	NFR360/6	2
SNS3048-H-D	**200**	220	22240	NFR360/21	2
SNS3040-Z-D	**200**	220	23040	NFR310/10	4
SNS3140-Z-D	**200**	220	23140	NFR340/10	2
SNS3040-Z-D	**200**	220	24040	NFR310/6,5	2
SNS3148-Z-D	**220**	240	22244	NFR400/10	4
SNS3144-Z-D	**220**	240	23144	NFR370/10	2
SNS3044-Z-D	**220**	240	24044	NFR340/6	2
SNS3044-Z-D	**220**	240	23044	NFR340/10	4
SNS3244-Z-D	**220**	240	23244	NFR400/10	2
SNS3156-H-D	**220**	240	22344	NFR460/10,5	2
SNS3048-Z-D	**240**	260	23048	NFR360/12	4
SNS3048-Z-D	**240**	260	24048	NFR360/11	2
SNS3248-Z-D	**240**	260	23248	NFR440/10	2
SNS3148-Z-D	**240**	260	23148	NFR400/10	2
SNS3160-H-D	**240**	260	22348	NFR500/12,5	2
SNS3152-Z-D	**240**	260	22248	NFR440/22	2
SNS3064-H-D	**260**	280	22252	NFR480/25,5	2
SNS3164-H-D	**260**	290	22352	NFR540/15,5	2



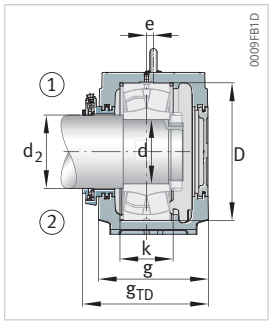
NTS labyrinth seal, NDK cover



NTS labyrinth seal



Taconite NTC seal

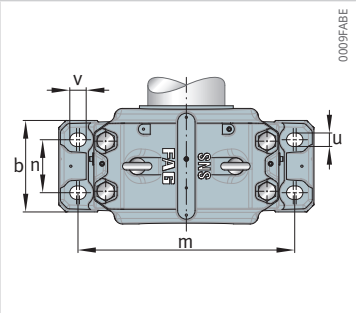


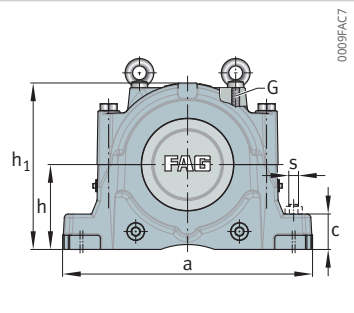
Taconite NTC seal, NDK cover

6

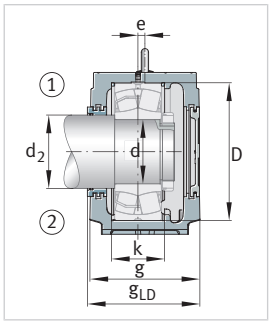
Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS34	241	235,5	NTC34	299	264,5	NDK34
NTS36	251	245,5	NTC36	314	277	NDK36
NTS38	271	265,5	NTC38	334	297	NDK38
NTS38/180	251	245,5	NTC38/180	314	277	NDK38
NTS38/180	251	245,5	NTC38/180	314	277	NDK38
NTS40	291	285,5	NTC40	354	317	NDK40
NTS40/190	271	265,5	NTC40/190	334	297	NDK40
NTS48/190	301	295,5	NTC48/190	380	335	NDK48
NTS40	241	235,5	NTC40	298	264	NDK40
NTS40/190	271	265,5	NTC40/190	326	293	NDK40
NTS44	241	235,5	NTC44	320	275	NDK44
NTS44	241	235,5	NTC44	320	275	NDK44
NTS44	271	265,5	NTC44	350	305	NDK44
NTS44	251	245,5	NTC44	330	285	NDK44
NTS44	271	265,5	NTC44	350	305	NDK44
NTS48/210	321	315,5	NTC48/210	402	329,5	NDK48
NTS48/210	291	285,5	NTC48/210	372	302,5	NDK48
NTS44	251	245,5	NTC44	330	285	NDK44
NTS44	251	245,5	NTC44	330	285	NDK44
NTS48/210	291	285,5	NTC48/210	372	302,5	NDK48
NTS44	271	265,5	NTC44	350	305	NDK44
NTS48	301	295,5	NTC48	380	335	NDK48
NTS48	301	295,5	NTC48	380	335	NDK48
NTS48	271	265,5	NTC48	350	305	NDK48
NTS48	291	285,5	NTC48	372	302,5	NDK48
NTS48	271	265,5	NTC48	350	305	NDK48
NTS56/240	321	315,5	NTC56/240	400	355	NDK56
NTS52	301	295,5	NTC52	380	335	NDK52
NTS52	291	285,5	NTC52	370	325	NDK52
NTS52	291	285,5	NTC52	370	325	NDK52
NTS56/240	321	315,5	NTC56/240	400	355	NDK56
NTS56/240	331	325,5	NTC56/240	410	365	NDK56
NTS56	301	295,5	NTC56	380	335	NDK56
NTS56	301	295,5	NTC56	380	335	NDK56
NTS60/260	331	325,5	NTC60/260	410	365	NDK60
NTS56	321	315,5	NTC56	400	355	NDK56
NTS60/260	361	355,5	NTC60/260	440	395	NDK60
NTS60/260	331	325,5	NTC60/260	410	365	NDK60
NTS64/280	361	355,5	NTC64/280	440	395	NDK64
NTS64/290	381	375,5	NTC64/290	460	415	NDK64

6.11.7 SNS for spherical roller bearings with a cylindrical bore, metric shaft

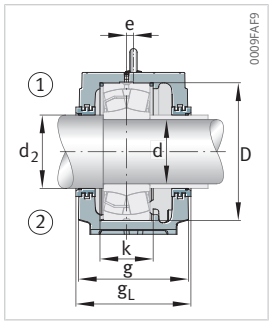




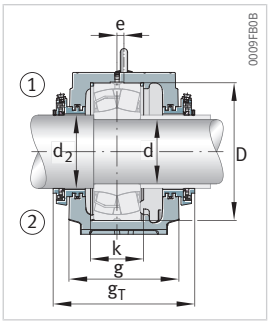
Housing	d	d_2	Bearing	Locating ring	n_S
	mm	mm	–	–	–
SNS3152-Z-D	**260**	280	23152	NFR440/10	2
SNS3252-Z-D	**260**	280	23252	NFR480/10	2
SNS3052-Z-D	**260**	280	23052	NFR400/22	2
SNS3052-Z-D	**260**	280	24052	NFR400/4	2
SNS3156-Z-D	**280**	300	23156	NFR460/10	2
SNS3056-Z-D	**280**	300	23056	NFR420/10	6
SNS3256-Z-D	**280**	300	23256	NFR500/10	2
SNS3160-Z-D	**280**	300	22256	NFR500/25	2
SNS3168-H-D	**280**	310	22356	NFR580/17,5	2
SNS3056-Z-D	**280**	300	24056	NFR420/13	2
SNS3160-Z-D	**300**	320	23160	NFR500/10	2
SNS3164-Z-D	**300**	320	22260	NFR540/28	2
SNS3260-Z-D	**300**	320	23260	NFR540/10	2
SNS3060-Z-D	**300**	320	24060	NFR460/4	2
SNS3060-Z-D	**300**	320	23060	NFR460/25	2
SNS3064-Z-D	**320**	340	23064	NFR480/10	6
SNS3064-Z-D	**320**	340	24064	NFR480/10,5	2
SNS3264-Z-D	**320**	340	23264	NFR580/10	2
SNS3164-Z-D	**320**	340	23164	NFR540/10	2
SNS2264-Z-D	**320**	340	22264	NFR580/20	2
SNS3168-Z-D	**340**	360	23168	NFR580/10	2
SNS3068-Z-D	**340**	360	24068	NFR520/8,5	2
SNS3068-Z-D	**340**	360	23068	NFR520/16	4
SNS3268-Z-D	**340**	370	23268	NFR620/10	2
SNS3272-Z-D	**360**	390	23272	NFR650/10	2
SNS3172-Z-D	**360**	380	23172	NFR600/10	2
SNS3072-Z-D	**360**	380	24072	NFR540/9	2
SNS3180-H-D	**360**	390	22272	NFR650/25	2
SNS3072-Z-D	**360**	380	23072	NFR540/16	4
SNS3276-Z-D	**380**	410	23276	NFR680/10	2
SNS3076-Z-D	**380**	400	23076	NFR560/22,5	2
SNS3176-Z-D	**380**	400	23176	NFR620/10	2
SNS4076-Z-D	**380**	400	24076	NFR560/10	2
SNS3080-Z-D	**400**	430	23080	NFR600/22	2
SNS3180-Z-D	**400**	430	23180	NFR650/10	2
SNS3280-Z-D	**400**	430	23280	NFR720/10	2
SNS4080-Z-D	**400**	430	24080	NFR600/10	2
SNS4084-Z-D	**420**	450	24084	NFR620/10	2
SNS3084-Z-D	**420**	450	23084	NFR620/22	2
SNS3184-Z-D	**420**	450	23184	NFR700/10	2



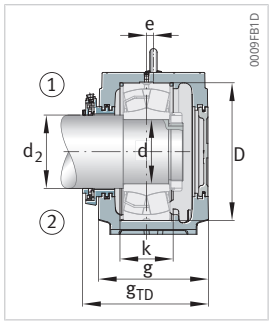
NTS labyrinth seal, NDK cover



NTS labyrinth seal



Taconite NTC seal

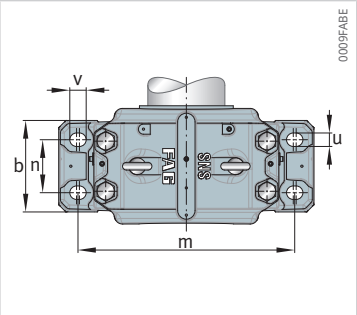


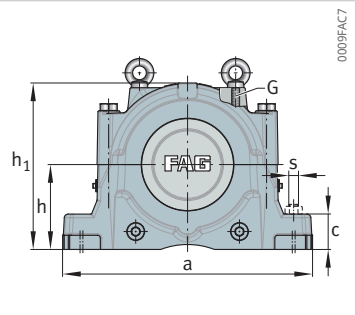
Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS60	331	325,5	NTC60	410	365	NDK60
NTS64/280	361	355,5	NTC64/280	440	395	NDK64
NTS60	321	315,5	NTC60	400	355	NDK60
NTS60	321	315,5	NTC60	400	355	NDK60
NTS64	331	325,5	NTC64	410	365	NDK64
NTS64	331	325,5	NTC64	410	365	NDK64
NTS68/300	361	355,5	NTC68/300	440	395	NDK68
NTS68/300	361	355,5	NTC68/300	440	395	NDK68
NTS68/310	411	405,5	NTC68/310	490	445	NDK68
NTS64	331	325,5	NTC64	410	365	NDK64
NTS68	361	355,5	NTC68	440	395	NDK68
NTS72/320	381	375,5	NTC72/320	460	415	NDK72
NTS72/320	381	375,5	NTC72/320	460	415	NDK72
NTS68	331	325,5	NTC68	410	365	NDK68
NTS68	331	325,5	NTC68	410	365	NDK68
NTS72	361	355,5	NTC72	440	395	NDK72
NTS72	361	355,5	NTC72	440	395	NDK72
NTS76/340	411	405,5	NTC76/340	490	445	NDK76
NTS72	381	375,5	NTC72	460	415	NDK72
NTS76/340	411	405,5	NTC76/340	490	445	NDK76
NTS76	411	405,5	NTC76	490	445	NDK76
NTS76	381	375,5	NTC76	460	415	NDK76
NTS76	381	375,5	NTC76	460	415	NDK76
NTS76/370	411	405,5	NTC76/370	490	445	NDK76
NTS80/390	441	435,5	NTC80/390	520	475	NDK80
NTS80	411	405,5	NTC80	490	445	NDK80
NTS80	381	375,5	NTC80	460	415	NDK80
NTS80/390	441	435,5	NTC80/390	520	475	NDK80
NTS80	381	375,5	NTC80	460	415	NDK80
NTS92/410	471	465,5	NTC92/410	550	505	NDK92
NTS84	411	405,5	NTC84	490	445	NDK84
NTS84	411	405,5	NTC84	490	445	NDK84
NTS84	411	405,5	NTC84	475	438	NDK84
NTS92	411	405,5	NTC92	490	445	NDK92
NTS92	441	435,5	NTC92	520	475	NDK92
NTS500/430	471	465,5	NTC500/430	550	505	NDK500
NTS92	411	405,5	NTC92	475	438	NDK92
NTS96	411	405,5	NTC96	475	438	NDK96
NTS96	411	405,5	NTC96	490	445	NDK96
NTS96	471	465,5	NTC96	550	505	NDK96

6.11.7 SNS for spherical roller bearings with a cylindrical bore, metric shaft





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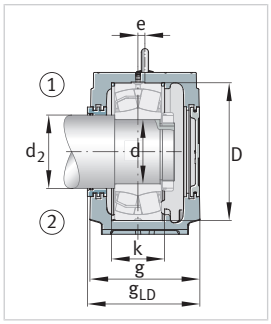
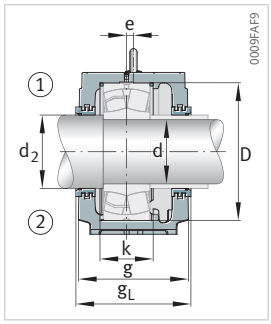
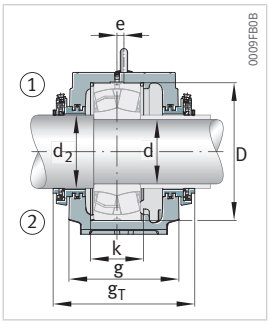
Housing	d	d_2	Bearing	Locating ring	n_S
	mm	mm	–	–	–
SNS3284-Z-D	**420**	460	23284	NFR760/10	2
SNS3188-Z-D	**440**	470	23188	NFR720/10	2
SNS4088-Z-D	**440**	470	24088	NFR650/10	2
SNS3088-Z-D	**440**	470	23088	NFR650/21,5	2
SNS3288-Z-D	**440**	480	23288	NFR790/10	2
SNS3192-Z-D	**460**	500	23192	NFR760/10	2
SNS3092-Z-D	**460**	500	23092	NFR680/30,5	2
SNS4092-Z-D	**460**	500	24092	NFR680/10	2
SNS3096-Z-D	**480**	510	23096	NFR700/29,5	2
SNS3196-Z-D	**480**	520	23196	NFR790/10	2
SNS4096-Z-D	**480**	510	24096	NFR700/10	2
SNS40/500-Z-D	**500**	530	240/500	NFR720/10	2
SNS30/500-Z-D	**500**	530	230/500	NFR720/29,5	2
SNS40/530-Z-D	**530**	560	240/530	NFR780/10	2
SNS30/530-Z-D	**530**	560	230/530	NFR780/31,5	2



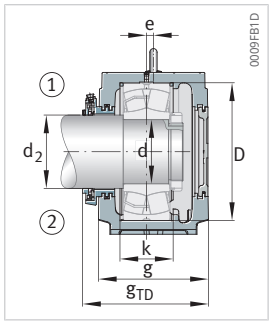
NTS labyrinth seal, NDK cover



NTS labyrinth seal



Taconite NTC seal



Taconite NTC seal, NDK cover

6

Labyrinth seal	g_L	g_{LD}	Taconite seal	g_T	g_{TD}	Cover
–	mm	mm	–	mm	mm	–
NTS530/460	481	475,5	NTC530/460	560	515	NDK530
NTS500	471	465,5	NTC500	550	505	NDK500
NTS500	441	435,5	NTC500	505	468	NDK500
NTS500	441	435,5	NTC500	520	475	NDK500
NTS560/480	481	475,5	NTC560/480	560	515	NDK560
NTS530	481	475,5	NTC530	560	515	NDK530
NTS530	471	465,5	NTC530	550	505	NDK530
NTS530	471	465,5	NTC530	535	498	NDK530
NTS530/510	471	465,5	NTC530/510	550	505	NDK530
NTS560/520	481	475,5	NTC560/520	560	515	NDK560
NTS530/510	471	465,5	NTC530/510	535	498	NDK530
NTS560	471	465,5	NTC560	535	498	NDK560
NTS560	471	465,5	NTC560	550	505	NDK560
NTS600	481	475,5	NTC600	545	508	NDK600
NTS600	481	475,5	NTC600	560	515	NDK600

7 RLE split plummer block housing

medias | Product catalogue |
RLE split plummer block housings |
https://www.schaeffler.de/std/21FD

RLE split plummer block housings were developed specifically for the bearings for rollers.

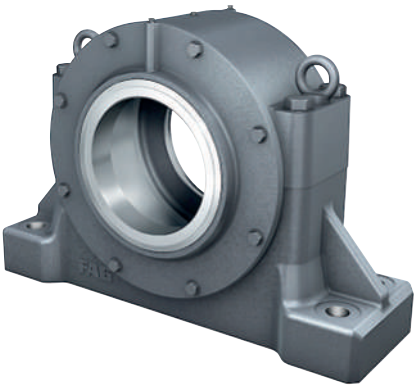
7.1 Housing design

Bearing housing

133 RLE..-AF, RLE..-AL. plummer block housing, closed on one side, with cover

0001ACB2

134 RLE..-BF, RLE..-BL. plummer block housing for a continuous shaft



00081EC9

7.1.1 Locating and non-locating bearings

The housings are supplied in a locating bearing or non-locating bearing design. For the locating bearing, the bearing is clamped between the housing covers. For the non-locating bearing, the bearing can align itself axially, since the covers have shorter centring collars.

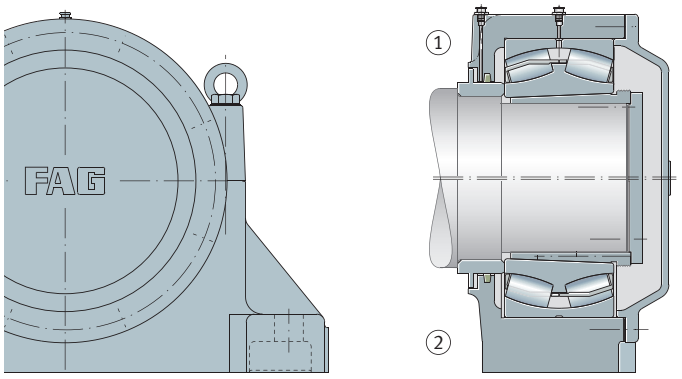
7.1.2 Housing configurations

Various housing configurations are possible for RLE plummer block housings, which are clearly described by the designation.

The following features can be considered for the housing configuration:

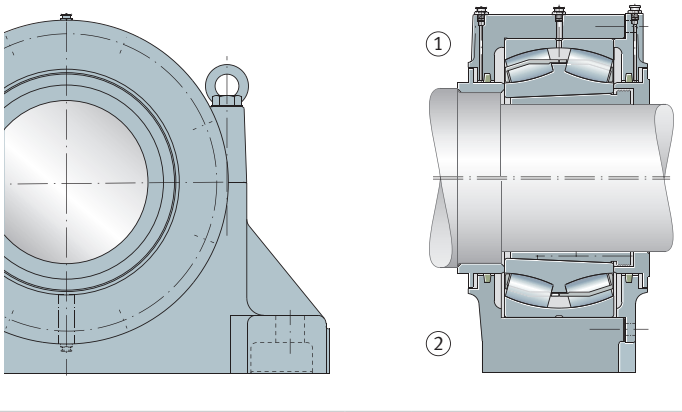
- Fastening of bearings with a tapered bore by a withdrawal sleeve on a shaft with a constant diameter or bearings with a cylindrical bore on a stepped shaft
- Continuous shaft or a housing closed on one side
- Design of the bearing as a locating or non-locating bearing

135 RLE plummer block housing for bearings with a tapered bore and withdrawal sleeve, design A



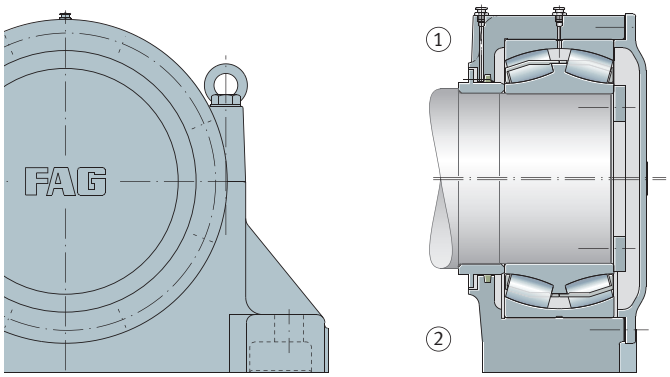
1	RLE..-AH-AF locating bearing	2	RLE..-AH-AL non-locating bearing

136 RLE plummer block housing for bearings with a tapered bore and withdrawal sleeve, design B



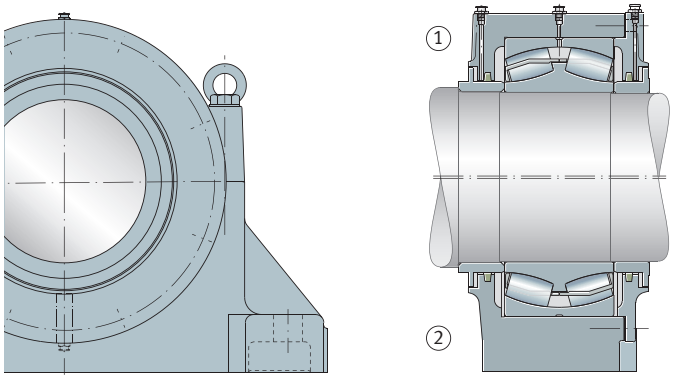
1	RLE..-AH-BF locating bearing	2	RLE..-AH-BL non-locating bearing

137 RLE plummer block housing for bearings with a cylindrical bore, version A



1	RLE..-Z-AF locating bearing	2	RLE..-Z-AL non-locating bearing

138 RLE plummer block housing for bearings with a cylindrical bore, version B



1	RLE..-Z-BF locating bearing	2	RLE..-Z-BL non-locating bearing

7.2 Suitable bearings

The dimensions of RLE split plummer block housings are tailored to the spherical roller bearings of the series 241. The shaft diameters are 180 mm to 500 mm.

50 Suitable bearings for RLE split plummer block housings

Bearing type	Size	
	from	to
Spherical roller bearings, tapered bore, withdrawal sleeve	24122..-K30	241/500..-K30
Spherical roller bearings with a cylindrical bore	24122	241/500

7.3 Materials, corrosion protection

51 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron GJL	L	✓	RAL 7016, anthracite grey
Ductile cast iron GJS	D	o	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

7.4 Lubrication

RLE split plummer block housings are designed for grease lubrication.

7.4.1 Relubrication

The housings have flat lubricating nipples to DIN 3404-A M10×1 St. The lubrication nipple in the centre of the housing is for relubricating the bearing. The grease is fed uniformly to both rows of rollers through the circumferential slot and three lubrication holes in the outer ring of the spherical roller bearings. The outer lubrication nipples are for relubricating the labyrinth seals.

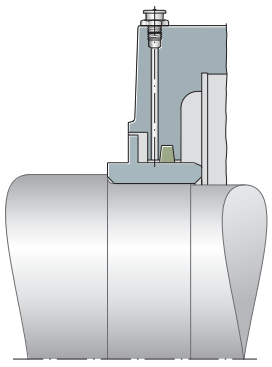
7.5 Sealing

RLE plummer block housings, design A are designed for the bearings on shaft ends. One side is closed by a cover and the other side is sealed against the shaft. Design B is designed for a continuous shafts where the housing is sealed against the shaft on both sides.

The seal is a combination of a labyrinth seal and a felt seal. The oil-soaked felt strip is inserted in the slot in the outer labyrinth ring. The labyrinth rings are unsplit.

The seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication.

139 Labyrinth with felt seal for RLE plummer block housings



0001AC59

7.6 Dimensions, tolerances

7.6.1 Tolerances for the bearing seat

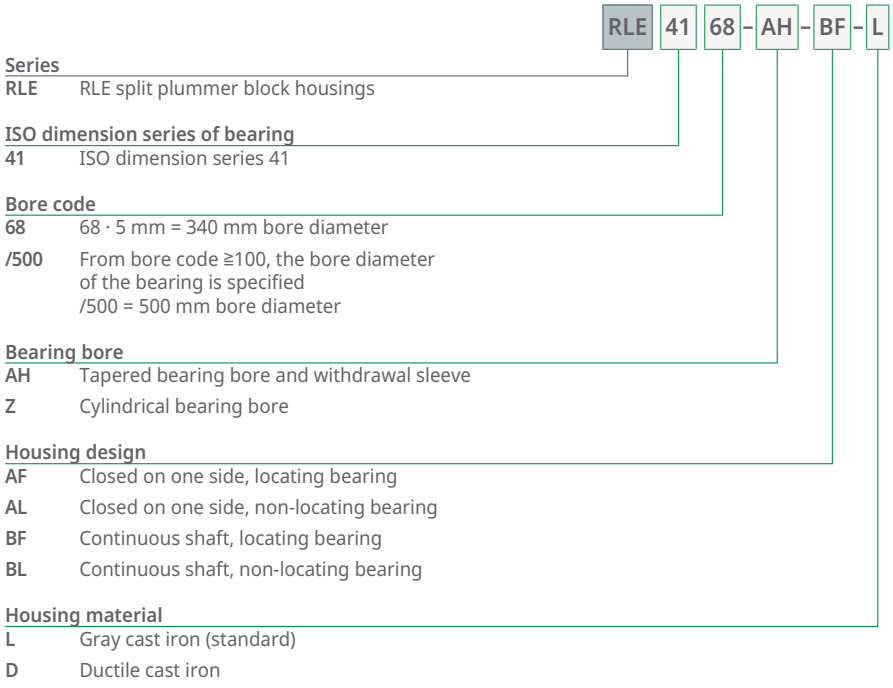
The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

7.7 Structure of the ordering designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

140 Structure of the designations of RLE plummer block housings



7.7.1 Specimen orders

For ordering an RLE split plummer block housing, the designation contains details of the size and location method of the bearing as well as the design and material used for the housing. The roller bearing and, if necessary, the withdrawal sleeve must be ordered separately.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- RLE plummer block housing made of gray cast iron, for a continuous shaft
- 24180-B-K30 spherical roller bearing in the form of a non-locating bearing
- Fastening with withdrawal sleeve

52 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	RLE4180-AH-BL-L
1	Spherical roller bearing	24180-B-K30
1	Withdrawal sleeve	AH24180-H

Example 2

Example of a housing configuration, characteristics:

- RLE plummer block housing made of gray cast iron, closed on one side
- 24180-B spherical roller bearing in the form of a locating bearing

53 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	RLE24180-Z-AF-L
1	Spherical roller bearing	24180-B

7.8 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

When using a spherical roller bearing with a tapered bore on a withdrawal sleeve and a continuous shaft, the supplied labyrinth ring must be adapted to the projection of the withdrawal sleeve during assembly. Failure to do so may result in bearing failure. Consult Schaeffler to ensure that the assembly is correct.

7.8.1 Eye bolts

There are 2 eye bolts to DIN 580:2018 in each housing upper section. These eye bolts are intended as a locating point for the installation and removal of the housing. The load capacity of the eye bolts allows the housing, including the bearing fitted in the housing, to be lifted.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

7.8.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

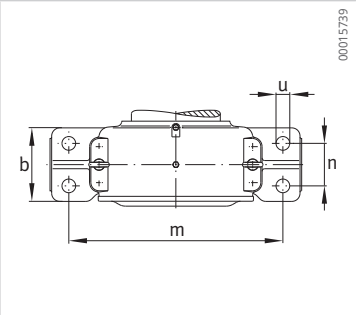
The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

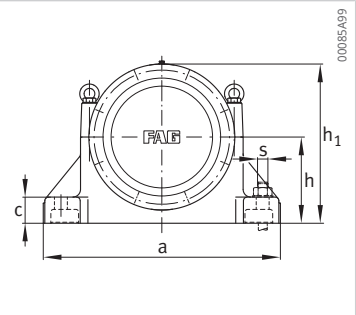
7.9 Product tables

7.9.1 Explanations of the product tables

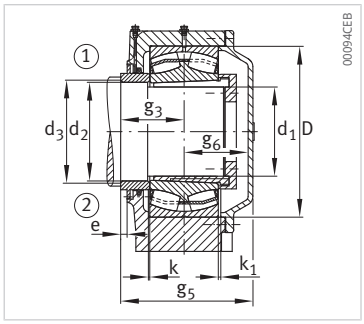
(3)	-	Locating bearing
(4)	-	Non-locating bearing
(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
b	mm	Base width
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
d_2	mm	Shaft shoulder diameter
d_3	mm	Diameter of shaft shoulder
d_4	mm	Diameter of shaft end
e	mm	Width of clamping elements
g_1	mm	Overall width
g_2	mm	Overall width
g_3	mm	Width over clamping element
g_5	mm	Overall width
g_6	mm	Spacing between the bearing centre and cover
h	mm	Distance to shaft axis
h_1	mm	Overall height
k	mm	Axial movement distance
k_1	mm	Axial movement distance
m	mm	Fixing hole spacing
m	kg	Mass
n	mm	Distance between fastening holes
s	inch or mm	Screw size
u	mm	Slot width
v	mm	Slot length

7.9.2 RLE, for spherical roller bearings with a tapered bore and withdrawal sleeve, stepped shaft

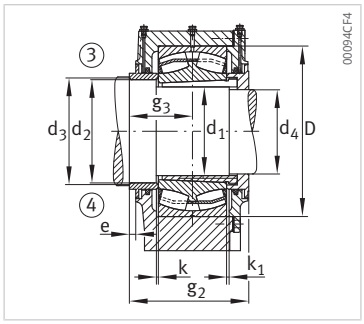




Housing	d_1	d_2	d_3	d_4	h	h_1	g_2	g_5	e
	mm	mm	mm	mm	mm	mm	mm	mm	mm
RLE4122-AH-BL-L	**110**	112	118	95	125	231	155	169,5	12
RLE4122-AH-BF-L	**110**	112	118	95	125	231	155	169,5	12
RLE4122-AH-AL-L	**110**	112	118	–	125	231	155	169,5	12
RLE4122-AH-AF-L	**110**	112	118	–	125	231	155	169,5	12
RLE4124-AH-BL-L	**120**	122	128	105	140	256	160	182	12
RLE4124-AH-BF-L	**120**	122	128	105	140	256	160	182	12
RLE4124-AH-AL-L	**120**	122	128	–	140	256	160	182	12
RLE4124-AH-AF-L	**120**	122	128	–	140	256	160	182	12
RLE4128-AH-AF-L	**140**	142	128	–	155	285	180	194	12
RLE4128-AH-BF-L	**140**	142	128	125	155	285	180	194	12
RLE4128-AH-BL-L	**140**	142	128	125	155	285	180	194	12
RLE4128-AH-AL-L	**140**	142	128	–	155	285	180	194	12
RLE4130-AH-BL-L	**150**	152	158	135	175	320	200	214	12
RLE4130-AH-BF-L	**150**	152	158	135	175	320	200	214	12
RLE4130-AH-AL-L	**150**	152	158	–	175	320	200	214	12
RLE4130-AH-AF-L	**150**	152	158	–	175	320	200	214	12
RLE4132-AH-BL-L	**160**	162	170	145	190	352	210	226	12
RLE4132-AH-BF-L	**160**	162	170	145	190	352	210	226	12
RLE4132-AH-AL-L	**160**	162	170	–	190	352	210	226	12
RLE4132-AH-AF-L	**160**	162	170	–	190	352	210	226	12
RLE4138-AH-BL-L	**180**	192	198	175	225	412	240	272	15
RLE4138-AH-BF-L	**180**	192	198	175	225	412	240	272	15
RLE4138-AH-AF-L	**180**	192	198	–	225	412	240	272	15
RLE4138-AH-AL-L	**180**	192	198	–	225	412	240	272	15
RLE4140-AH-BL-L	**190**	202	212	185	240	442	260	287	15
RLE4140-AH-AL-L	**190**	202	212	–	240	442	260	287	15
RLE4140-AH-BF-L	**190**	202	212	185	240	442	260	287	15
RLE4140-AH-AF-L	**190**	202	212	–	240	442	260	287	15
RLE4144-AH-AF-L	**200**	222	232	–	260	477	270	307	15
RLE4144-AH-BL-L	**200**	222	232	195	260	477	270	307	15
RLE4144-AH-BF-L	**200**	222	232	195	260	477	270	307	15
RLE4144-AH-AL-L	**200**	222	232	–	260	477	270	307	15
RLE4148-AH-AF-L	**220**	242	252	–	280	517	280	323	15
RLE4148-AH-BL-L	**220**	242	252	215	280	517	280	323	15
RLE4148-AH-BF-L	**220**	242	252	215	280	517	280	323	15
RLE4148-AH-AL-L	**220**	242	252	–	280	517	280	323	15
RLE4152-AH-BF-L	**240**	262	272	235	310	567	315	350	15
RLE4152-AH-AL-L	**240**	262	272	–	310	567	315	350	15
RLE4152-AH-AF-L	**240**	262	272	–	310	567	315	350	15
RLE4152-AH-BL-L	**240**	262	272	235	310	567	315	350	15



Design A

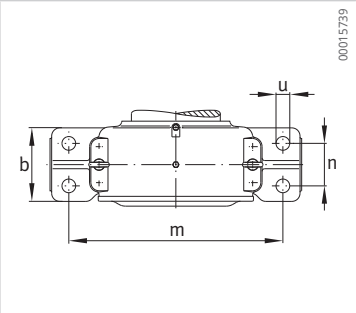


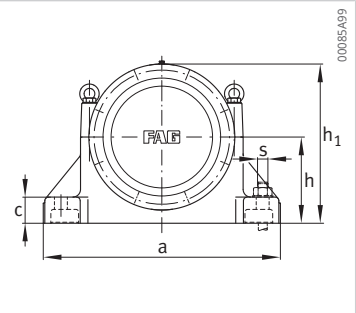
Design B

7

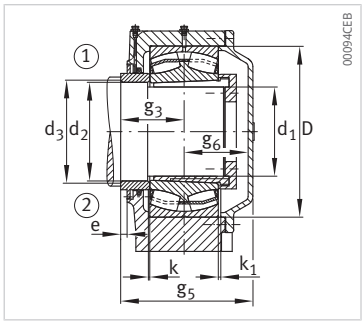
k	k_1	b	c	a	m	n	u	s	D	g_3	g_6	m
mm	mm	mm	mm	mm	mm	mm	mm	–	mm	mm	mm	≈ kg
1	3	100	40	365	305	50	25	M20	180	85	78,5	19
–	–	100	40	365	305	50	25	M20	180	85	78,5	19
1	3	100	40	365	305	50	25	M20	180	85	78,5	19
–	–	100	40	365	305	50	25	M20	180	85	78,5	19
1	3	100	42	400	335	85	25	M20	200	92	84	29
–	–	100	42	400	335	85	25	M20	200	92	84	29
1	3	100	42	400	335	85	25	M20	200	92	84	29
–	–	100	42	400	335	85	25	M20	200	92	84	29
–	–	120	48	460	390	60	25	M24	225	97	91	39
–	–	120	48	460	390	60	25	M24	225	97	91	39
1	3	120	48	460	390	60	25	M24	225	97	91	39
1	3	120	48	460	390	60	25	M24	225	97	91	39
2	4	140	55	500	420	75	30	M24	250	105	102	59
–	–	140	55	500	420	75	30	M24	250	105	102	59
2	4	140	55	500	420	75	30	M24	250	105	102	59
–	–	140	55	500	420	75	30	M24	250	105	102	59
1	4	150	58	525	450	75	36	M30	270	112	106	73
–	–	150	58	525	450	75	36	M30	270	112	106	73
1	4	150	58	525	450	75	36	M30	270	112	106	73
–	–	150	58	525	450	75	36	M30	270	112	106	73
1	4	180	70	600	510	100	36	M30	320	127	137	115
–	–	180	70	600	510	100	36	M30	320	127	137	115
–	–	180	70	600	510	100	36	M30	320	127	137	115
1	4	180	70	600	510	100	36	M30	320	127	137	115
1	4	195	75	690	580	105	42	M36	340	137	142	145
1	4	195	75	690	580	105	42	M36	340	137	142	145
–	–	195	75	690	580	105	42	M36	340	137	142	145
–	–	195	75	690	580	105	42	M36	340	137	142	145
–	–	210	80	720	610	120	42	M36	370	142	155	175
1	4	210	80	720	610	120	42	M36	370	142	155	175
–	–	210	80	720	610	120	42	M36	370	142	155	175
1	4	210	80	720	610	120	42	M36	370	142	155	175
–	–	225	85	770	655	130	42	M36	400	148	165	220
1	5	225	85	770	655	130	42	M36	400	148	165	220
–	–	225	85	770	655	130	42	M36	400	148	165	220
1	5	225	85	770	655	130	42	M36	400	148	165	220
–	–	250	95	860	730	140	49	M42	440	165	175	295
1	5	250	95	860	730	140	49	M42	440	165	175	295
–	–	250	95	860	730	140	49	M42	440	165	175	295
1	5	250	95	860	730	140	49	M42	440	165	175	295

7.9.2 RLE, for spherical roller bearings with a tapered bore and withdrawal sleeve, stepped shaft

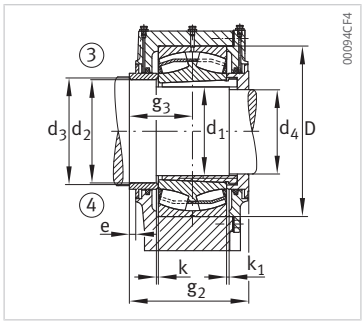




Housing	d_1	d_2	d_3	d_4	h	h_1	g_2	g_5	e
	mm	mm	mm	mm	mm	mm	mm	mm	mm
RLE4156-AH-AF-L	**260**	282	292	–	320	587	315	350	15
RLE4156-AH-BF-L	**260**	282	292	255	320	587	315	350	15
RLE4156-AH-BL-L	**260**	282	292	255	320	587	315	350	15
RLE4156-AH-AL-L	**260**	282	292	–	320	587	315	350	15
RLE4160-AH-BF-L	**280**	302	315	275	350	642	335	386	15
RLE4160-AH-AL-L	**280**	302	315	–	350	642	335	386	15
RLE4160-AH-BL-L	**280**	302	315	275	350	642	335	386	15
RLE4160-AH-AF-L	**280**	302	315	–	350	642	335	386	15
RLE4164-AH-BF-D	**300**	322	335	295	380	697	370	421	20
RLE4164-AH-AF-L	**300**	322	335	–	380	697	370	421	20
RLE4164-AH-BF-L	**300**	322	335	295	380	697	370	421	20
RLE4164-AH-BL-L	**300**	322	335	295	380	697	370	421	20
RLE4164-AH-AL-L	**300**	322	335	–	380	697	370	421	20
RLE4168-AH-AL-L	**320**	342	355	–	410	752	405	480	20
RLE4168-AH-BF-L	**320**	342	355	315	410	752	405	480	20
RLE4168-AH-BL-L	**320**	342	355	315	410	752	405	480	20
RLE4168-AH-AF-L	**320**	342	355	–	410	752	405	480	20
RLE4172-AH-BF-L	**340**	362	375	335	430	792	405	480	20
RLE4172-AH-BL-L	**340**	362	375	335	430	792	405	480	20
RLE4172-AH-AF-L	**340**	362	375	–	430	792	405	480	20
RLE4172-AH-AL-L	**340**	362	375	–	430	792	405	480	20
RLE4176-AH-AF-L	**360**	382	395	–	430	792	405	490	20
RLE4176-AH-BF-L	**360**	382	395	355	430	792	405	490	20
RLE4176-AH-BL-L	**360**	382	395	355	430	792	405	490	20
RLE4176-AH-AL-L	**360**	382	395	–	430	792	405	490	20
RLE4180-AH-BL-L	**380**	402	415	375	450	827	415	500	20
RLE4180-AH-AF-L	**380**	402	415	–	450	827	415	500	20
RLE4180-AH-BF-L	**380**	402	415	375	450	827	415	500	20
RLE4180-AH-AL-L	**380**	402	415	–	450	827	415	500	20
RLE4184-AH-AL-L	**400**	422	435	–	490	897	455	555	20
RLE4184-AH-AF-L	**400**	422	435	–	490	897	455	555	20
RLE4184-AH-BL-L	**400**	422	435	395	490	897	455	555	20
RLE4184-AH-BF-L	**400**	422	435	395	490	897	455	555	20
RLE4188-AH-AF-L	**420**	442	455	–	500	915	455	555	20
RLE4188-AH-AL-L	**420**	442	455	–	500	915	455	555	20
RLE4188-AH-BF-L	**420**	442	455	415	500	915	455	555	20
RLE4188-AH-BL-L	**420**	442	455	415	500	915	455	555	20
RLE4192-AH-AL-L	**440**	462	475	–	530	972	495	595	25
RLE4192-AH-BF-L	**440**	462	475	435	530	972	495	595	25
RLE4192-AH-AF-L	**440**	462	475	–	530	972	495	595	25



Design A

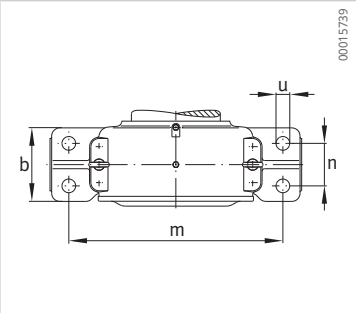


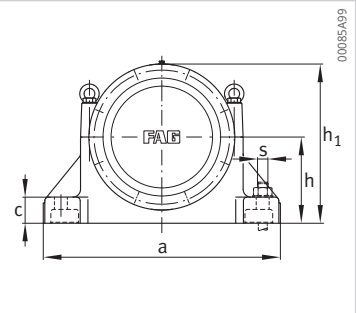
Design B

7

k	k_1	b	c	a	m	n	u	s	D	g_3	g_6	m
mm	mm	mm	mm	mm	mm	mm	mm	–	mm	mm	mm	≈ kg
–	–	250	100	880	750	145	49	M42	460	165	175	320
–	–	250	100	880	750	145	49	M42	460	165	175	320
2	5	250	100	880	750	145	49	M42	460	165	175	320
2	5	250	100	880	750	145	49	M42	460	165	175	320
–	–	280	110	940	800	165	49	M42	500	176	198	415
2	5	280	110	940	800	165	49	M42	500	176	198	415
2	5	280	110	940	800	165	49	M42	500	176	198	415
–	–	280	110	940	800	165	49	M42	500	176	198	415
2	5	305	120	1060	900	170	56	M48	540	196	213	550
–	–	305	120	1060	900	170	56	M48	540	196	213	550
2	5	305	120	1060	900	170	56	M48	540	196	213	550
2	5	305	120	1060	900	170	56	M48	540	196	213	550
2	5	305	120	1060	900	170	56	M48	540	196	213	550
2	5	340	130	1110	940	195	56	M48	580	210	258	685
–	–	340	130	1110	940	195	56	M48	580	210	258	685
2	5	340	130	1110	940	195	56	M48	580	210	258	685
–	–	340	130	1110	940	195	56	M48	580	210	258	685
–	–	340	135	1190	1010	195	56	M48	600	210	258	765
2	5	340	135	1190	1010	195	56	M48	600	210	258	765
–	–	340	135	1190	1010	195	56	M48	600	210	258	765
2	5	340	135	1190	1010	195	56	M48	600	210	258	765
–	–	340	140	1190	1010	195	68	M56	620	210	265	775
–	–	340	140	1190	1010	195	68	M56	620	210	265	775
2	5	340	140	1190	1010	195	68	M56	620	210	265	775
2	5	340	140	1190	1010	195	68	M56	620	210	265	775
2	5	350	145	1230	1040	200	68	M56	650	215	270	870
–	–	350	145	1230	1040	200	68	M56	650	215	270	870
–	–	350	145	1230	1040	200	68	M56	650	215	270	870
2	5	350	145	1230	1040	200	68	M56	650	215	270	870
2	5	390	155	1300	1110	225	65	M56	700	232	308	1100
–	–	390	155	1300	1110	225	65	M56	700	232	308	1100
2	5	390	155	1300	1110	225	65	M56	700	232	308	1100
–	–	390	155	1300	1110	225	65	M56	700	232	308	1100
–	–	390	155	1370	1165	225	68	M56	720	232	308	1150
2	5	390	155	1370	1165	225	68	M56	720	232	308	1150
–	–	390	155	1370	1165	225	68	M56	720	232	308	1150
2	5	390	155	1370	1165	225	68	M56	720	232	308	1150
2	6	420	160	1500	1270	240	76	M64	760	252	328	1400
–	–	420	160	1500	1270	240	76	M64	760	252	328	1400
–	–	420	160	1500	1270	240	76	M64	760	252	328	1400

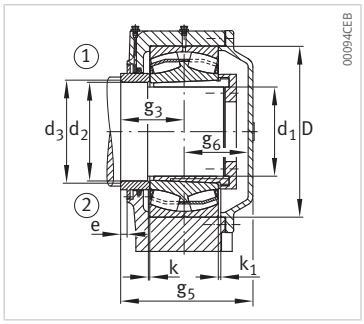
7.9.2 RLE, for spherical roller bearings with a tapered bore and withdrawal sleeve, stepped shaft



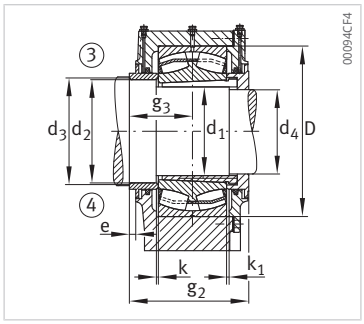


7

Housing	d_1	d_2	d_3	d_4	h	h_1	g_2	g_5	e
	mm	mm	mm	mm	mm	mm	mm	mm	mm
RLE4192-AH-BL-L	**440**	462	475	435	530	972	495	595	25
RLE4196-AH-BF-L	**460**	482	495	455	550	1012	500	610	25
RLE4196-AH-AL-L	**460**	482	495	–	550	1012	500	610	25
RLE4196-AH-AF-L	**460**	482	495	–	550	1012	500	610	25
RLE4196-AH-BL-L	**460**	482	495	455	550	1012	500	610	25
RLE41/500-AH-BL-L	**480**	502	515	475	550	1032	520	630	25
RLE41/500-AH-AL-L	**480**	502	515	–	550	1032	520	630	25
RLE41/500-AH-BF-L	**480**	502	515	475	550	1032	520	630	25
RLE41/500-AH-AF-L	**480**	502	515	–	550	1032	520	630	25



Design A

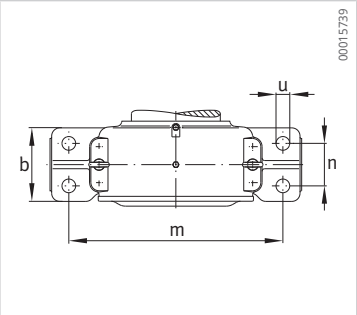


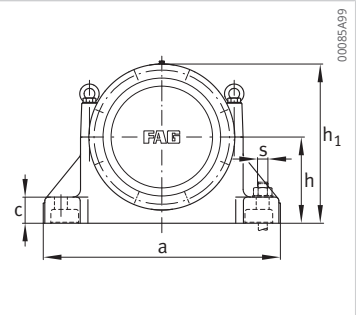
Design B

7

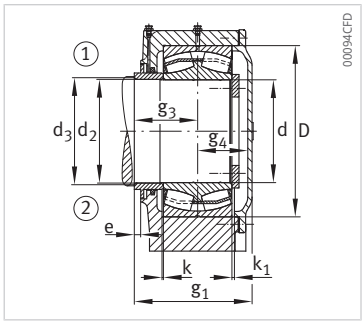
k	k_1	b	c	a	m	n	u	s	D	g_3	g_6	m
mm	mm	mm	mm	mm	mm	mm	mm	–	mm	mm	mm	≈ kg
2	6	420	160	1500	1270	240	76	M64	760	252	328	1400
–	–	430	165	1530	1300	245	76	M64	790	256	338	1550
2	6	430	165	1530	1300	245	76	M64	790	256	338	1550
–	–	430	165	1530	1300	245	76	M64	790	256	338	1550
2	6	430	165	1530	1300	245	76	M64	790	256	338	1550
2	6	450	180	1580	1300	260	76	M64	830	265	350	1600
2	6	450	180	1580	1300	260	76	M64	830	265	350	1600
–	–	450	180	1580	1300	260	76	M64	830	265	350	1600
–	–	450	180	1580	1300	260	76	M64	830	265	350	1600

7.9.3 RLE, for spherical roller bearings with a cylindrical bore, stepped shaft

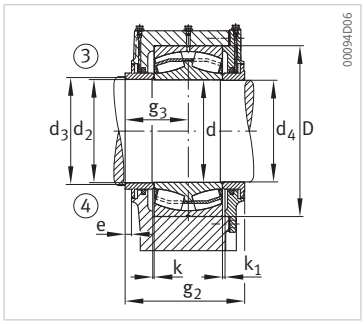




Housing	d_1	d_2	d_3	d_4	h	h_1	g_1	g_2	e
	mm	mm	mm	mm	mm	mm	mm	mm	mm
RLE4122-Z-AF-L	**110**	112	118	–	125	231	155	169,5	12
RLE4122-Z-AL-L	**110**	112	118	–	125	231	155	169,5	12
RLE4122-Z-BF-L	**110**	112	118	108	125	231	155	169,5	12
RLE4122-Z-BL-L	**110**	112	118	108	125	231	155	169,5	12
RLE4124-Z-AF-L	**120**	122	128	–	140	256	160	182	12
RLE4124-Z-AL-L	**120**	122	128	–	140	256	160	182	12
RLE4124-Z-BF-L	**120**	122	128	118	140	256	160	182	12
RLE4124-Z-BL-L	**120**	122	128	118	140	256	160	182	12
RLE4128-Z-AF-L	**140**	142	128	–	155	285	180	194	12
RLE4128-Z-AL-L	**140**	142	128	–	155	285	180	194	12
RLE4128-Z-BF-L	**140**	142	128	138	155	285	180	194	12
RLE4128-Z-BL-L	**140**	142	128	138	155	285	180	194	12
RLE4130-Z-AF-L	**150**	152	158	–	175	320	200	214	12
RLE4130-Z-AL-L	**150**	152	158	–	175	320	200	214	12
RLE4130-Z-BF-L	**150**	152	158	148	175	320	200	214	12
RLE4130-Z-BL-L	**150**	152	158	148	175	320	200	214	12
RLE4132-Z-AF-L	**160**	162	170	–	190	352	210	226	12
RLE4132-Z-AL-L	**160**	162	170	–	190	352	210	226	12
RLE4132-Z-BF-L	**160**	162	170	158	190	352	210	226	12
RLE4132-Z-BL-L	**160**	162	170	158	190	352	210	226	12
RLE4134-Z-AF-L	**170**	172	179	–	195	360	210	229	12
RLE4134-Z-AL-L	**170**	172	179	–	195	360	210	229	12
RLE4134-Z-BF-L	**170**	172	179	168	195	360	210	229	12
RLE4134-Z-BL-L	**170**	172	179	168	195	360	210	229	12
RLE4138-Z-AF-L	**190**	192	198	–	225	412	240	230	15
RLE4138-Z-AL-L	**190**	192	198	–	225	412	240	230	15
RLE4138-Z-BF-L	**190**	192	198	188	225	412	240	230	15
RLE4138-Z-BL-L	**190**	192	198	188	225	412	240	230	15
RLE4140-Z-AF-L	**200**	202	212	–	240	442	240	250	15
RLE4140-Z-AL-L	**200**	202	212	–	240	442	240	250	15
RLE4140-Z-BF-L	**200**	202	212	198	240	442	240	250	15
RLE4140-Z-BL-L	**200**	202	212	198	240	442	240	250	15
RLE4144-Z-AF-L	**220**	222	232	–	260	477	270	265	15
RLE4144-Z-AL-L	**220**	222	232	–	260	477	270	265	15
RLE4144-Z-BF-L	**220**	222	232	218	260	477	270	265	15
RLE4144-Z-BL-L	**220**	222	232	218	260	477	270	265	15
RLE4148-Z-AF-L	**240**	242	252	–	280	517	280	275	15
RLE4148-Z-AL-L	**240**	242	252	–	280	517	280	275	15
RLE4148-Z-BF-L	**240**	242	252	238	280	517	280	275	15
RLE4148-Z-BL-L	**240**	242	252	238	280	517	280	275	15



Design A

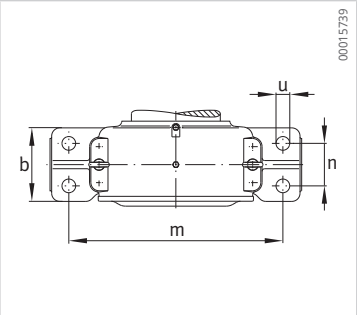


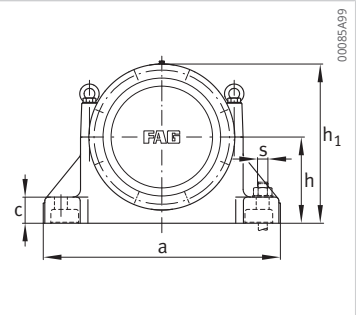
Design B

7

k	k_1	b	c	a	m	n	u	s	D	g_3	g_6	m
mm	mm	mm	mm	mm	mm	mm	mm	–	mm	mm	mm	≈ kg
–	–	100	40	365	305	50	25	M20	180	85	78,5	19
1	3	100	40	365	305	50	25	M20	180	85	78,5	19
–	–	100	40	365	305	50	25	M20	180	85	78,5	19
1	3	100	40	365	305	50	25	M20	180	85	78,5	19
–	–	100	42	400	335	85	25	M20	200	92	84	29
1	3	100	42	400	335	85	25	M20	200	92	84	29
–	–	100	42	400	335	85	25	M20	200	92	84	29
1	3	100	42	400	335	85	25	M20	200	92	84	29
–	–	120	48	460	390	60	25	M24	225	97	91	39
1	3	120	48	460	390	60	25	M24	225	97	91	39
–	–	120	48	460	390	60	25	M24	225	97	91	39
1	3	120	48	460	390	60	25	M24	225	97	91	39
–	–	140	55	500	420	75	30	M24	250	105	102	59
2	4	140	55	500	420	75	30	M24	250	105	102	59
–	–	140	55	500	420	75	30	M24	250	105	102	59
2	4	140	55	500	420	75	30	M24	250	105	102	59
–	–	150	58	525	450	75	36	M30	270	112	106	73
1	4	150	58	525	450	75	36	M30	270	112	106	73
–	–	150	58	525	450	75	36	M30	270	112	106	73
1	4	150	58	525	450	75	36	M30	270	112	106	73
–	–	150	60	560	480	75	36	M30	280	112	110	75
1	4	150	60	560	480	75	36	M30	280	112	110	75
–	–	150	60	560	480	75	36	M30	280	112	110	75
1	4	150	60	560	480	75	36	M30	280	112	110	75
–	–	180	70	600	510	100	36	M30	320	127	95	115
1	4	180	70	600	510	100	36	M30	320	127	95	115
–	–	180	70	600	510	100	36	M30	320	127	95	115
1	4	180	70	600	510	100	36	M30	320	127	95	115
1	4	195	75	690	580	105	42	M36	340	137	105	145
1	4	195	75	690	580	105	42	M36	340	137	105	145
1	4	195	75	690	580	105	42	M36	340	137	105	145
1	4	195	75	690	580	105	42	M36	340	137	105	145
1	4	210	80	720	610	120	42	M36	370	142	113	175
1	4	210	80	720	610	120	42	M36	370	142	113	175
1	4	210	80	720	610	120	42	M36	370	142	113	175
1	4	210	80	720	610	120	42	M36	370	142	113	175
1	5	225	85	770	655	130	42	M36	400	148	117	220
1	5	225	85	770	655	130	42	M36	400	148	117	220
–	–	225	85	770	655	130	42	M36	400	148	117	220
1	5	225	85	770	655	130	42	M36	400	148	117	220

7.9.3 RLE, for spherical roller bearings with a cylindrical bore, stepped shaft





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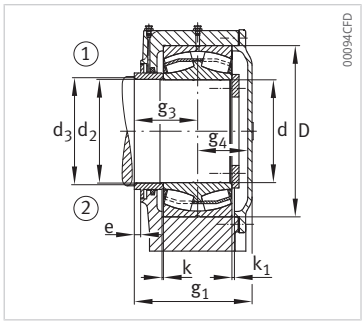
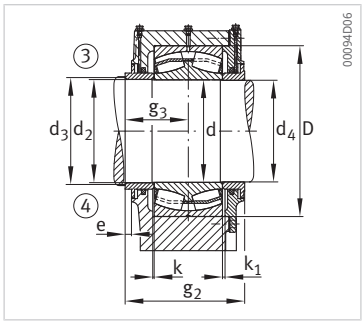
Housing	d_1	d_2	d_3	d_4	h	h_1	g_1	g_2	e
	mm	mm	mm	mm	mm	mm	mm	mm	mm
RLE4152-Z-AF-L	**260**	262	272	–	310	567	315	305	15
RLE4152-Z-AL-L	**260**	262	272	–	310	567	315	305	15
RLE4152-Z-BF-L	**260**	262	272	258	310	567	315	305	15
RLE4152-Z-BL-L	**260**	262	272	258	310	567	315	305	15
RLE4156-Z-AF-L	**280**	282	292	–	320	587	315	305	15
RLE4156-Z-AL-L	**280**	282	292	–	320	587	315	305	15
RLE4156-Z-BF-L	**280**	282	292	278	320	587	315	305	15
RLE4156-Z-BL-L	**280**	282	292	278	320	587	315	305	15
RLE4160-Z-AF-L	**300**	302	315	–	350	642	335	335	15
RLE4160-Z-AL-L	**300**	302	315	–	350	642	335	335	15
RLE4160-Z-BF-L	**300**	302	315	298	350	642	335	335	15
RLE4160-Z-BL-L	**300**	302	315	298	350	642	335	335	15
RLE4164-Z-AF-L	**320**	322	335	–	380	697	370	365	20
RLE4164-Z-AL-L	**320**	322	335	–	380	697	370	365	20
RLE4164-Z-BF-L	**320**	322	335	318	380	697	370	365	20
RLE4164-Z-BL-L	**320**	322	335	318	380	697	370	365	20
RLE4168-Z-AF-L	**340**	342	355	–	410	752	405	400	20
RLE4168-Z-AL-L	**340**	342	355	–	410	752	405	400	20
RLE4168-Z-BF-L	**340**	342	355	338	410	752	405	400	20
RLE4168-Z-BL-L	**340**	342	355	338	410	752	405	400	20
RLE4172-Z-AF-L	**360**	362	375	–	430	792	405	400	20
RLE4172-Z-AL-L	**360**	362	375	–	430	792	405	400	20
RLE4172-Z-BF-L	**360**	362	375	358	430	792	405	400	20
RLE4172-Z-BL-L	**360**	362	375	358	430	792	405	400	20
RLE4176-Z-AF-L	**380**	382	395	–	430	792	405	400	20
RLE4176-Z-AL-L	**380**	382	395	–	430	792	405	400	20
RLE4176-Z-BF-L	**380**	382	395	378	430	792	405	400	20
RLE4176-Z-BL-L	**380**	382	395	378	430	792	405	400	20
RLE4180-Z-AF-L	**400**	402	415	–	450	827	415	410	20
RLE4180-Z-AL-L	**400**	402	415	–	450	827	415	410	20
RLE4180-Z-BF-L	**400**	402	415	398	450	827	415	410	20
RLE4180-Z-BL-L	**400**	402	415	398	450	827	415	410	20
RLE4184-Z-AF-L	**420**	422	435	–	490	897	455	450	20
RLE4184-Z-AL-L	**420**	422	435	–	490	897	455	450	20
RLE4184-Z-BF-L	**420**	422	435	418	490	897	455	450	20
RLE4184-Z-BL-L	**420**	422	435	418	490	897	455	450	20
RLE4188-Z-AF-L	**440**	442	455	–	500	917	455	450	20
RLE4188-Z-AL-L	**440**	442	455	–	500	917	455	450	20
RLE4188-Z-BF-L	**440**	442	455	438	500	917	455	450	20
RLE4188-Z-BL-L	**440**	442	455	438	500	917	455	450	20



Design A

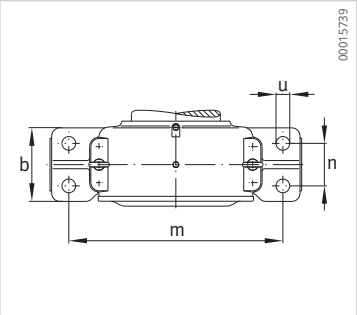


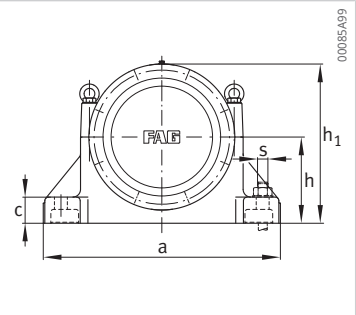
Design B

7

k	k_1	b	c	a	m	n	u	s	D	g_3	g_6	m
mm	mm	mm	mm	mm	mm	mm	mm	–	mm	mm	mm	≈ kg
1	5	250	95	860	730	140	49	M42	440	165	130	295
1	5	250	95	860	730	140	49	M42	440	165	130	295
1	5	250	95	860	730	140	49	M42	440	165	130	295
1	5	250	95	860	730	140	49	M42	440	165	130	295
2	5	250	100	880	750	145	49	M42	460	165	130	320
2	5	250	100	880	750	145	49	M42	460	165	130	320
–	–	250	100	880	750	145	49	M42	460	165	130	320
2	5	250	100	880	750	145	49	M42	460	165	130	320
2	5	280	110	940	800	165	49	M42	500	176	147	415
2	5	280	110	940	800	165	49	M42	500	176	147	415
–	–	280	110	940	800	165	49	M42	500	176	147	415
2	5	280	110	940	800	165	49	M42	500	176	147	415
2	5	305	120	1060	900	170	56	M48	540	196	157	550
2	5	305	120	1060	900	170	56	M48	540	196	157	550
–	–	305	120	1060	900	170	56	M48	540	196	157	550
2	5	305	120	1060	900	170	56	M48	540	196	157	550
2	5	340	130	1110	940	195	56	M48	580	210	178	685
2	5	340	130	1110	940	195	56	M48	580	210	178	685
–	–	340	130	1110	940	195	56	M48	580	210	178	685
2	5	340	130	1110	940	195	56	M48	580	210	178	685
–	–	340	135	1190	1010	195	56	M48	600	210	178	765
2	5	340	135	1190	1010	195	56	M48	600	210	178	765
–	–	340	135	1190	1010	195	56	M48	600	210	178	765
2	5	340	135	1190	1010	195	56	M48	600	210	178	765
–	–	340	140	1190	1010	195	68	M56	620	210	175	775
2	5	340	140	1190	1010	195	68	M56	620	210	175	775
–	–	340	140	1190	1010	195	68	M56	620	210	175	775
2	5	340	140	1190	1010	195	68	M56	620	210	175	775
–	–	350	145	1230	1040	200	68	M56	650	215	180	870
2	5	350	145	1230	1040	200	68	M56	650	215	180	870
–	–	350	145	1230	1040	200	68	M56	650	215	180	870
2	5	350	145	1230	1040	200	68	M56	650	215	180	870
–	–	390	155	1300	1110	225	68	M56	700	232	203	1100
2	5	390	155	1300	1110	225	68	M56	700	232	203	1100
–	–	390	155	1300	1110	225	68	M56	700	232	203	1100
2	5	390	155	1300	1110	225	68	M56	700	232	203	1100
–	–	390	155	1370	1165	225	68	M56	720	232	203	1150
2	5	390	155	1370	1165	225	68	M56	720	232	203	1150
–	–	390	155	1370	1165	225	68	M56	720	232	203	1150
2	5	390	155	1370	1165	225	68	M56	720	232	203	1150

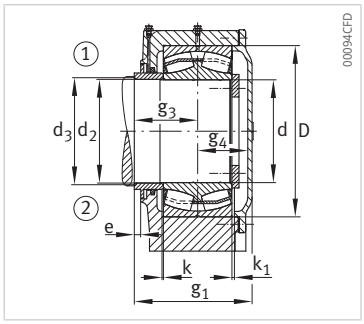
7.9.3 RLE, for spherical roller bearings with a cylindrical bore, stepped shaft



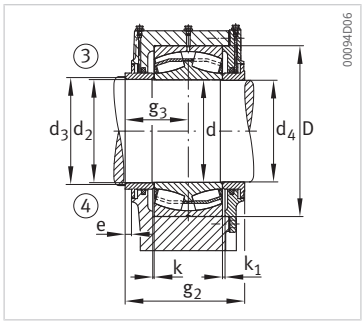


7

Housing	d_1	d_2	d_3	d_4	h	h_1	g_1	g_2	e
	mm	mm	mm	mm	mm	mm	mm	mm	mm
RLE4192-Z-AF-L	**460**	462	475	–	530	972	495	485	25
RLE4192-Z-AL-L	**460**	462	475	–	530	972	495	485	25
RLE4192-Z-BF-L	**460**	462	475	458	530	972	495	485	25
RLE4192-Z-BL-L	**460**	462	475	458	530	972	495	485	25
RLE4196-Z-AF-L	**480**	482	495	–	550	1012	500	500	25
RLE4196-Z-AL-L	**480**	482	495	–	550	1012	500	500	25
RLE4196-Z-BF-L	**480**	482	495	478	550	1012	500	500	25
RLE4196-Z-BL-L	**480**	482	495	478	550	1012	500	500	25
RLE41/500-Z-AF-L	**500**	502	515	–	550	1032	520	515	25
RLE41/500-Z-AL-L	**500**	502	515	–	550	1032	520	515	25
RLE41/500-Z-BF-L	**500**	502	515	475	550	1032	520	515	25
RLE41/500-Z-BL-L	**500**	502	515	475	550	1032	520	515	25



Design A



Design B

7

k	k_1	b	c	a	m	n	u	s	D	g_3	g_6	m
mm	mm	mm	mm	mm	mm	mm	mm	–	mm	mm	mm	≈ kg
–	–	420	160	1500	1270	240	76	M64	760	252	218	1400
2	6	420	160	1500	1270	240	76	M64	760	252	218	1400
–	–	420	160	1500	1270	240	76	M64	760	252	218	1400
2	6	420	160	1500	1270	240	76	M64	760	252	218	1400
–	–	430	165	1530	1300	245	76	M64	790	256	228	1550
2	6	430	165	1530	1300	245	76	M64	790	256	228	1550
–	–	430	165	1530	1300	245	76	M64	790	256	228	1550
2	6	430	165	1530	1300	245	76	M64	790	256	228	1550
–	–	450	180	1580	1300	260	76	M64	830	265	235	1600
2	6	450	180	1580	1300	260	76	M64	830	265	235	1600
–	–	450	180	1580	1300	260	76	M64	830	265	235	1600
2	6	450	180	1580	1300	260	76	M64	830	265	235	1600

8 KPG, KPGZ split plummer block housings

medias | Product catalogue |
KPG, KPGZ split plummer block housings |
https://www.schaeffler.de/std/21FE

KPG and KPGZ split plummer block housings have been specially developed to support the trunnion bearing on converters in steel production.

8.1 Housing design

Bearing housing

141 KPG, KPGZ split plummer block housing

0001AD19

8.1.1 Locating and non-locating bearings

The housings are supplied in the form of a locating or non-locating bearing design. The locating bearing on the drive side provides the axial guidance for the converter support ring.

The locating bearing design of KPG and KPGZ housings was originally designed for the installation of unsplit spherical roller bearings. For the locating bearing, the bearing is clamped between the housing covers.

A housing with a locating bearing design can also accommodate a split spherical roller bearing, replacing an unsplit bearing. As a result, the bearing can be replaced without dismantling the converter drive system.

The non-locating bearing design features unsplit bearings. The bearing outer ring can be move in the axial direction in a bearing bush.

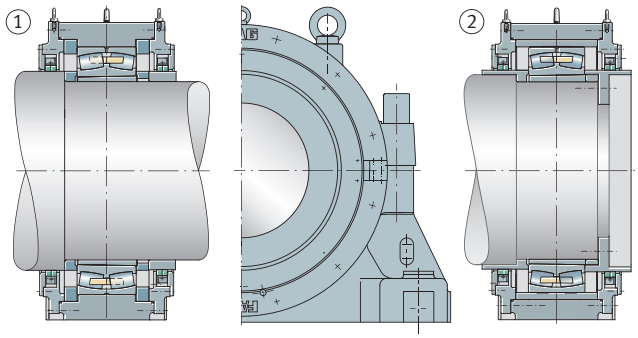
8.1.2 Housing configurations

KPG plummer block housings are used for bearings with a tapered bore and adjustment sleeve, KPGZ plummer block housings are used for bearings with a cylindrical bore.

The following features can be considered for the housing configuration:

- The housings are sealed with high-pressure packing
- Design of the bearing as a locating or non-locating bearing
- Spherical roller bearing, split or unsplit design

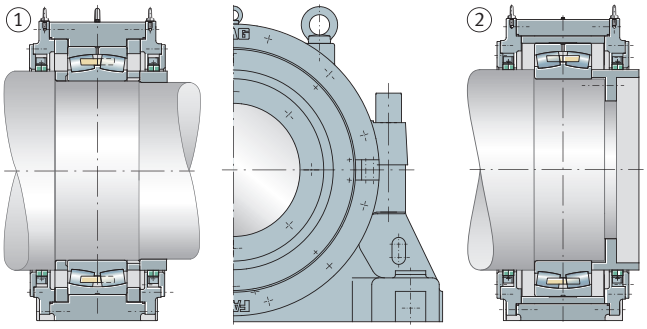
142 KPG plummer block housings for bearings with a tapered bore and adjustment sleeve



00082714

1	KPG..-F locating bearing	2	KPG..-L non-locating bearing

143 KPGZ plummer block housing for bearings with a cylindrical bore



00082861

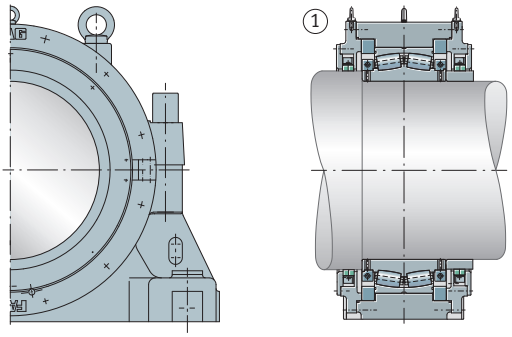
1	KPGZ..-F locating bearing	2	KPGZ..-L non-locating bearing

Installation of split spherical roller bearing

For KPG and KPGZ plummer block housings, an unsplit spherical roller bearing can be replaced by a split spherical roller bearing, but there is only possible as standard only for fixed bearing version F.

Housings in non-locating bearing design L, which allow the use of split spherical roller bearings, are available on request as a special solution.

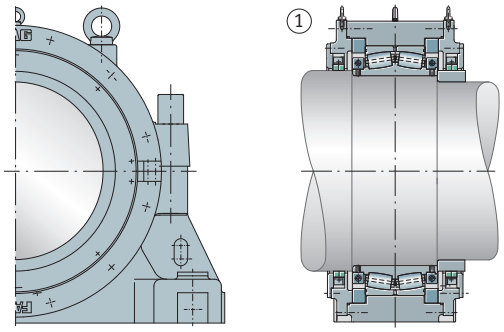
144 KPG plummer block housing for bearings with a tapered bore and adjustment sleeve, with split bearing



0008286B

1	KPG..-F locating bearing with split bearing		

145 KPGZ plummer block housing for bearings with a cylindrical bore, with split bearing



00082890

1	KPGZ..-F locating bearing with split bearing

8.1.3 Requirements for the trunnion bearing on converters

When filled, large converter vessels weigh several 100 t. The resulting loads must be supported by the trunnion bearing. As only slow swivel movements take place, the bearings must feature a high static load capacity. In addition, the shock loads that occur every day in converters must be managed.

The bearing must also be able to compensate for housing misalignments and the flexing of the construction. These are in addition to considerable changes in length that arise due to temperature changes when operating the converter as well as geometrical changes in the support ring.

The spherical roller bearing has established itself as the main roller bearing type for converters. In addition to its high radial and axial load capacity and its insensitivity to shocks, it also delivers a significant angular adjustment capacity. Spherical roller bearings specially developed for the application are used as converter bearings.

8.2 Suitable bearings

The dimensions of KPG and KPGZ split plummer block housings are tailored to series 249 spherical roller bearings. KPG plummer block housings are tailored to bearings with a tapered bore and adjustment sleeve and KPGZ plummer block housings to bearings with a cylindrical bore. The shaft diameters are 470 mm to 1320 mm.

54 Suitable bearings forKPG, KPGZ split plummer block housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	249/470..-K30	249/1250..-K30
Spherical roller bearings with a cylindrical bore	249/500	249/1320

Split spherical roller bearings

When replacing the bearing, an unsplit spherical roller bearing can be replaced with a split spherical roller bearing on the locating bearing side of the converter. As there is no need to dismantle the drive system, the assembly work is considerably reduced.

Further information

TPI 250 | Split spherical roller bearings |
https://www.schaeffler.de/std/1F81

8.3 Materials, corrosion protection

55 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Ductile cast iron GJS	D	✓	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

8.4 Lubrication

KPG and KPGZ split plummer block housings are designed for grease lubrication.

8.4.1 Grease types

Lithium soap greases with effective EP additives and anti-corrosion protection additives are suitable. We recommend Arcanol LOAD400.

8.4.2 Grease quantities

For initial greasing, the basic rule is that the bearing should be filled with grease to 100 % and the free volume of the housing to 60 %. The free volume is the space that remains in the housing once the bearing, adapter sleeve, shaft and seals have been fitted.

56 Recommended grease quantity for initial grease application

Housing		Recommended grease quantity		
		Locating bearing		Non-locating bearing
Designation		Unsplit	Split	Unsplit
		≈ kg	≈ kg	≈ kg
KPG49/470..-D	KPGZ49/500..-D	10	8	14
KPG49/500..-D	KPGZ49/530..-D	10	8	14
KPG49/530..-D	KPGZ49/560..-D	13	10	15
KPG49/570..-D	KPGZ49/600..-D	15	12	20
KPG49/600..-D	KPGZ49/630..-D	20	15	24
KPG49/630..-D	KPGZ49/670..-D	22	18	25
KPG49/670..-D	KPGZ49/710..-D	26	20	30
KPG49/710..-D	KPGZ49/750..-D	30	24	35
KPG49/750..-D	KPGZ49/800..-D	35	26	40
KPG49/800..-D	KPGZ49/850..-D	40	30	50
KPG49/850..-D	KPGZ49/900..-D	45	35	55
KPG49/900..-D	KPGZ49/950..-D	55	45	65
KPG49/950..-D	KPGZ49/1000..-D	65	50	80
KPG49/1000..-D	KPGZ49/1060..-D	75	60	95
KPG49/1060..-D	KPGZ49/1120..-D	80	65	100
KPG49/1120..-D	KPGZ49/1180..-D	95	75	110

Housing		Recommended grease quantity		
		Locating bearing		Non-locating bearing
Designation		Unsplit	Split	Unsplit
		≈ kg	≈ kg	≈ kg
KPG49/1180..-D	–	110	–	130
–	KPGZ49/1250..-D	110	85	130
KPG49/1250..-D	KPGZ49/1320..-D	125	100	170

8.4.3 Relubrication

Relubrication should be carried out using the same grease as for the initial lubrication. The bearings, displacement sleeve and seals are relubricated on the basis of a schedule.

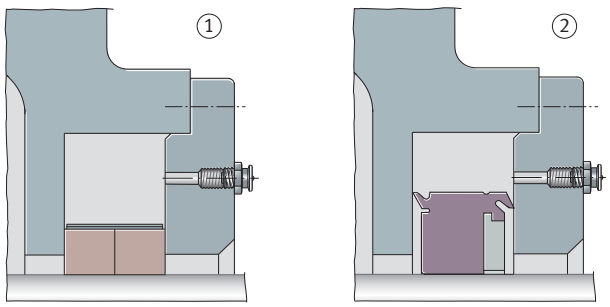
8.5 Sealing

KPG and KPGZ plummer block housings are sealed with high-pressure packings.

High-pressure packings allow shaft misalignment of up to 1,5 °.

Profiled rubber seals are available on request special seals.

146 Seals for KPG and KPGZ plummer block housings



1	High-pressure packing	2	Profiled rubber seal

8.6 Dimensions, tolerances

8.6.1 Tolerances for the bearing seat

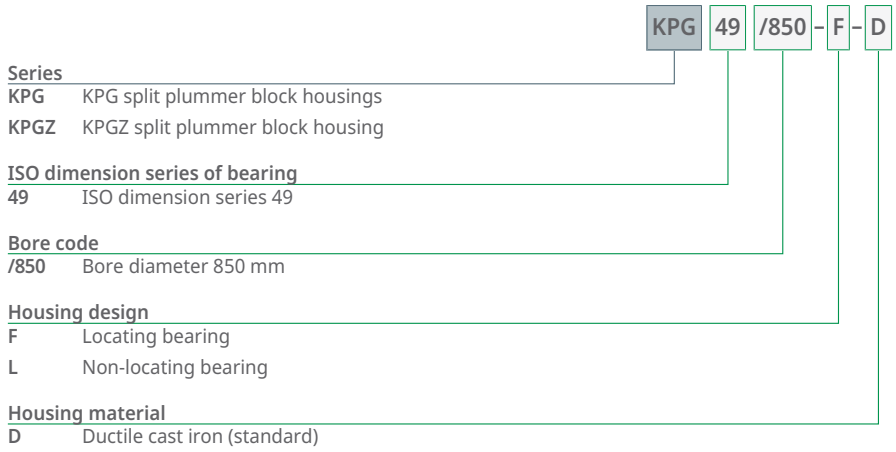
The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

8.7 Structure of the ordering designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

147 Designation structure of KPG and KPGZ plummer block housings



8.7.1 Specimen orders

For ordering KPG and KPGZ split plummer block housings, the designation contains details of the size of the bearing as well as the design and material of the housing. The roller bearing and, if necessary, the sleeve must be ordered separately.

The specimen order show the structure of orders for the selected housing combinations and the appropriate bearings. Allocation of housings, bearings and accessories for all housing sizes.

bespoke solutions are developed for each converter application on the basis of KPG and KPGZ split plummer block housings. The housings can be supplemented by additional components and modified where necessary. We therefore recommend contacting our engineering service before placing any order for these housings.

Example 1

Example of a housing configuration, characteristics:

- KPG plummer block housing made from ductile cast iron
- Z-541834.249/1060-B-K30 spherical roller bearing with pin cage in the form of a locating bearing
- Adapter sleeve

57 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	KPG49/1000-F-D
1	Spherical roller bearing	Z-541834.249/1060-B-K30
1	Adapter sleeve	Z-525000.KH

Example 2

Example of a housing configuration, characteristics:

- KPGZ plummer block housing made from ductile cast iron
- Z-533567.PRL split spherical roller bearing in the form of a locating bearing

58 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	KPGZ49/1000-F-D
1	Split spherical roller bearing	Z-533567.PRL

8.8 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

8.8.1 Eye bolts

There are 2 eye bolts to DIN 580:2018 in each housing upper section. These eye bolts are intended as a locating point for the installation and removal of the housing. The load capacity of the eye bolts allows the housing, including the bearing fitted in the housing, to be lifted.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

8.8.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

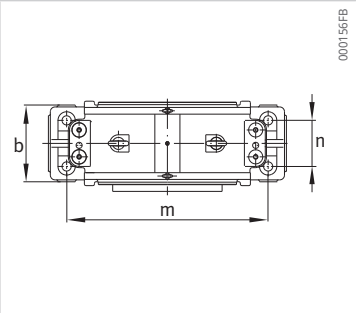
The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

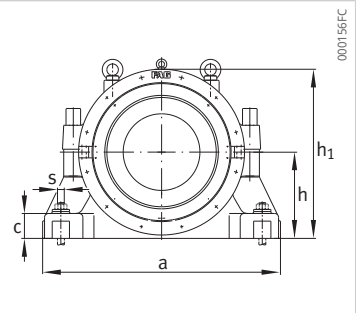
8.9 Product tables

8.9.1 Explanations of the product tables

a	mm	Base length
b	mm	Base width
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
d_2	mm	Shaft shoulder diameter
d_3	mm	Diameter of shaft end
d_4	mm	Diameter of shaft end
d_5	mm	Chamfer diameter
d_6	mm	Diameter of shaft shoulder
G	–	Ring bolt thread
g_1	mm	Overall width
g_2	mm	Distance from bearing centre
g_3	mm	Wall thickness
g_4	mm	Distance from bearing centre
h	mm	Distance to shaft axis
h_1	mm	Overall height
k	mm	Bearing seat width
m	mm	Fixing hole spacing
m	kg	Mass
n	mm	Distance between fastening holes
s	inch or mm	Screw size
s_2	-	Assembly hole to DIN EN ISO 4014
t	mm	Pitch circle diameter of the bores
u	mm	Slot width
v	mm	Slot length
w	mm	Distance from bearing centre to shaft shoulder
w_L	mm	Distance from bearing centre to shaft shoulder

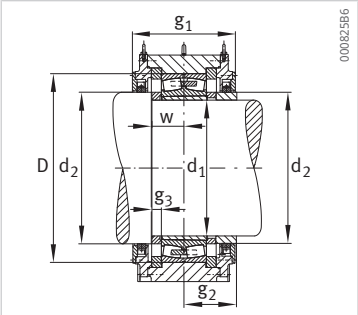
8.9.2 KPG, for spherical roller bearings with a tapered bore and sleeve, dimensions



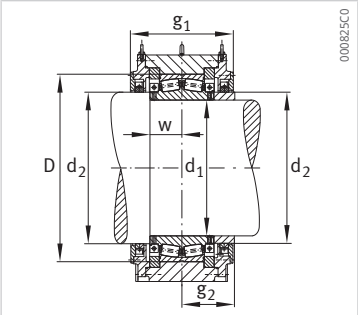


8

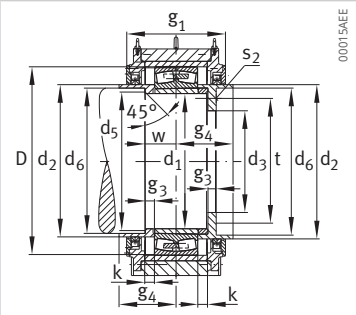
Housing	d_1	d_2	h	h_1	g_1	b	c	a	m	n	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	–
KPG49/470..-D	**470**	540	425	820	400	375	130	1170	975	230	M42
KPG49/500..-D	**500**	570	450	875	410	400	140	1240	1050	240	M42
KPG49/530..-D	**530**	600	475	930	420	420	145	1320	1100	255	M48
KPG49/570..-D	**570**	645	500	980	460	440	155	1400	1150	270	M52
KPG49/600..-D	**600**	675	535	1040	480	480	165	1500	1225	295	M56
KPG49/630..-D	**630**	720	570	1110	500	500	175	1570	1300	310	M56
KPG49/670..-D	**670**	760	600	1170	560	535	185	1660	1375	325	M64
KPG49/710..-D	**710**	800	630	1240	590	550	195	1750	1450	335	M64
KPG49/750..-D	**750**	860	670	1310	600	570	205	1850	1550	345	M72
KPG49/800..-D	**800**	910	710	1390	630	600	220	1960	1600	360	M72
KPG49/850..-D	**850**	960	740	1450	660	620	230	2060	1700	370	M80
KPG49/900..-D	**900**	1015	800	1550	680	660	250	2200	1820	390	M90
KPG49/950..-D	**950**	1065	830	1620	720	650	255	2330	1980	360	M90
KPG49/1000..-D	**1000**	1135	880	1710	780	740	275	2450	2000	460	M100
KPG49/1060..-D	**1060**	1195	920	1780	800	740	285	2560	2150	460	M100
KPG49/1120..-D	**1120**	1260	970	1880	820	780	300	2700	2300	480	M110
KPG49/1180..-D	**1180**	1330	1010	1985	850	820	320	2850	2400	510	M110
KPG49/1250..-D	**1250**	1400	1080	2100	900	850	340	3000	2500	520	M125



KPG49..-F, unsplit bearing, locating bearing



KPG49..-F, split bearing, locating bearing

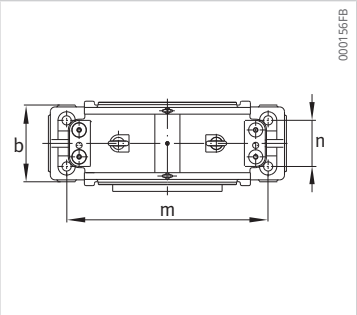


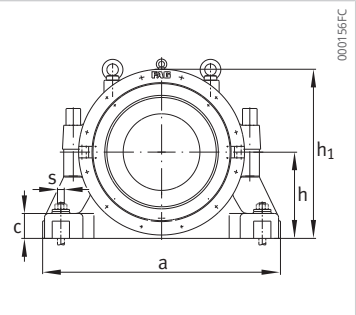
KPG49..-L, unsplit bearing, non-locating bearing

8

D	w	k	g_2	g_3	g_4	d_3	d_5	d_6	t	s_2	n_{S2}	m
mm	mm	mm	mm	mm	mm	mm	mm	mm	–	–	–	≈ kg
670	125	40	210	40	230	375	480	505	437,5	M20×70	8	945
710	130	40	215	40	235	400	510	535	465	M20×70	8	1050
750	135	40	220	40	240	420	540	565	490	M20×70	8	1365
800	145	40	240	45	260	450	580	610	525	M20×80	8	1575
850	155	40	250	46	270	475	612	640	552,5	M20×80	8	2205
900	165	40	260	50	280	505	642	675	587,5	M24×90	8	2625
950	175	50	290	53,5	317,5	535	682	715	622,5	M24×90	8	2835
1000	180	50	305	55	332,5	565	722	755	657,5	M30×100	8	2940
1060	185	50	310	56	337,5	600	762	805	700	M30×100	8	3465
1120	195	50	325	59	352,5	640	812	855	745	M30×110	8	3885
1180	200	60	340	60	375	675	862	905	787,5	M30×110	8	4515
1250	210	60	350	60	385	715	915	960	832,5	M36×110	8	5460
1320	230	70	370	72,5	412,5	750	965	1010	875	M36×130	8	5660
1400	245	60	400	77,5	435	795	1015	1070	927,5	M36×130	8	7140
1460	245	70	410	77,5	452,5	840	1075	1130	980	M42×140	8	8400
1540	260	70	420	82,5	462,5	885	1135	1190	1032,5	M42×140	8	9450
1630	275	70	435	87,5	477,5	940	1195	1255	1095	M42×150	8	11550
1720	290	70	460	90	502,5	990	1265	1325	1155	M42×180	8	13440

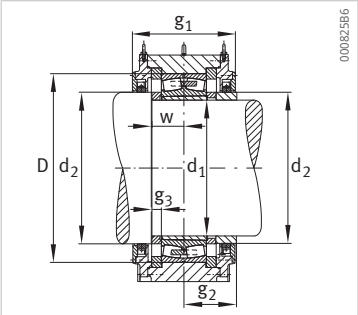
8.9.3 KPG, for spherical roller bearings with a tapered bore and sleeve, accessories



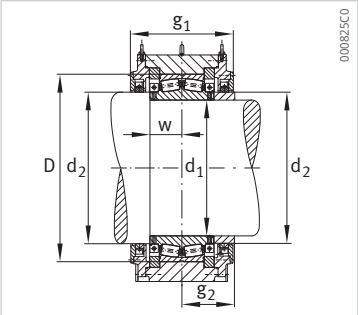


8

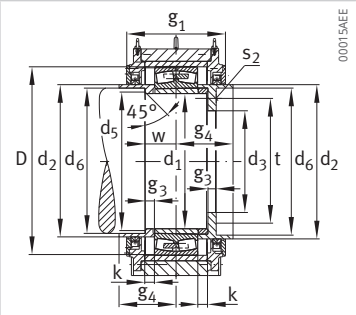
Housing	d_1	Bearing		
		MB cage	Pin cage	Adapter sleeve
	mm	–	–	–
KPG49/470..-D	**470**	Z-528741.PRL-K30	Z-541821.249/500-K30	Z-524974.KH
KPG49/500..-D	**500**	Z-528742.PRL-K30	Z-541822.249/530-K30	Z-524976.KH
KPG49/530..-D	**530**	Z-528743.PRL-K30	Z-541823.249/560-B-K30	Z-524978.KH
KPG49/570..-D	**570**	Z-528744.PRL-K30	Z-541824.249/600-B-K30	Z-524980.KH
KPG49/600..-D	**600**	–	Z-541825.249/630-K30	Z-524982.KH
KPG49/630..-D	**630**	Z-528746.PRL-K30	Z-541826.249/670-K30	Z-524984.KH
KPG49/670..-D	**670**	Z-528747.PRL-K30	Z-541827.249/710-B-K30	Z-524986.KH
KPG49/710..-D	**710**	Z-528748.PRL-K30	Z-541828.249/750-B-K30	Z-524988.KH
KPG49/750..-D	**750**	Z-528749.PRL-K30	Z-541829.249/800-B-K30	Z-524990.KH
KPG49/800..-D	**800**	Z-528750.PRL-K30	Z-541830.249/850-B-K30	Z-524992.KH
KPG49/850..-D	**850**	Z-528751.PRL-K30	Z-541831.249/900-B-K30	Z-524994.KH
KPG49/900..-D	**900**	Z-528752.PRL-K30	Z-541832.249/950-B-K30	Z-524996.KH
KPG49/950..-D	**950**	Z-528753.PRL-K30	Z-541833.249/1000-B-K30	Z-524998.KH
KPG49/1000..-D	**1000**	–	Z-541834.249/1060-B-K30	Z-525000.KH
KPG49/1060..-D	**1060**	–	Z-541835.249/1120-B-K30	Z-525001.KH
KPG49/1120..-D	**1120**	–	Z-541836.249/1180-B-K30	Z-525003.KH
KPG49/1180..-D	**1180**	–	Z-541837.249/1250-B-K30	Z-525005.KH
KPG49/1250..-D	**1250**	–	Z-541838.249/1320-B-K30	Z-525007.KH



KPG49..-F, unsplit bearing, locating bearing



KPG49..-F, split bearing, locating bearing

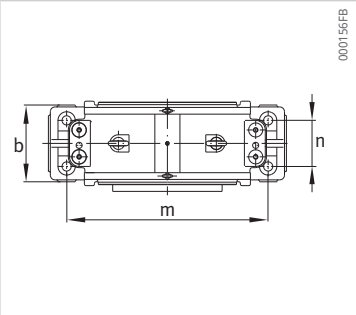


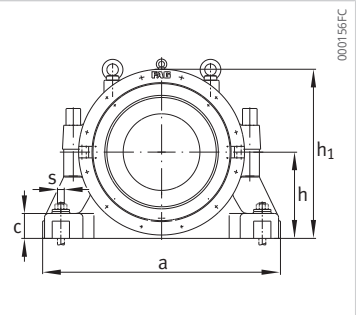
KPG49..-L, unsplit bearing, non-locating bearing

8

Split bearing (for locating bearing only)
–
Z-529173.PRL
Z-528441.PRL
Z-529223.PRL
Z-529224.PRL
Z-529225.PRL
Z-529226.PRL
Z-529227.PRL
Z-527943.PRL
Z-529228.PRL
Z-529229.PRL
Z-529230.PRL
Z-527254.PRL
Z-529231.PRL
Z-529232.PRL
Z-529233.01.PRL
Z-529234.PRL
–
Z-529215.PRL

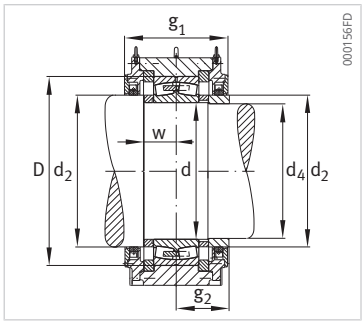
8.9.4 KPGZ, for spherical roller bearings with a cylindrical bore, dimensions



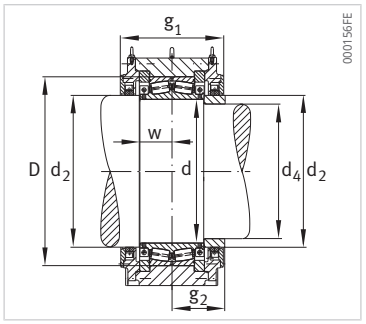


8

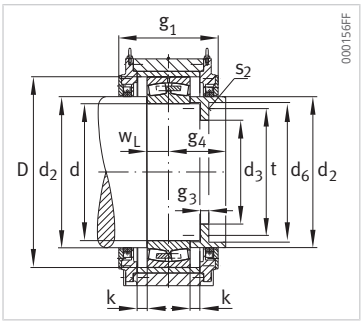
Housing	d	d_2	d_4	h	h_1	g_1	b	c	a	m	n	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	–
KPGZ49/500..-D	**500**	540	495	425	820	400	375	130	1170	975	230	M42
KPGZ49/530..-D	**530**	570	525	450	875	410	400	140	1240	1050	240	M42
KPGZ49/560..-D	**560**	600	555	475	930	420	420	145	1320	1100	255	M48
KPGZ49/600..-D	**600**	645	595	500	980	460	440	155	1400	1150	270	M52
KPGZ49/630..-D	**630**	675	625	535	1040	480	480	165	1500	1225	295	M56
KPGZ49/670..-D	**670**	720	665	570	1110	500	500	175	1570	1300	310	M56
KPGZ49/710..-D	**710**	760	695	600	1170	560	535	185	1660	1375	325	M64
KPGZ49/750..-D	**750**	800	745	630	1240	590	550	195	1750	1450	335	M64
KPGZ49/800..-D	**800**	860	795	670	1310	600	570	205	1850	1550	345	M72
KPGZ49/850..-D	**850**	910	845	710	1390	630	600	220	1960	1600	360	M72
KPGZ49/900..-D	**900**	960	895	740	1450	660	620	230	2060	1700	370	M80
KPGZ49/950..-D	**950**	1015	945	800	1550	680	660	250	2200	1820	390	M90
KPGZ49/1000..-D	**1000**	1065	985	830	1620	720	650	255	2330	1980	360	M90
KPGZ49/1060..-D	**1060**	1135	1055	880	1710	780	740	275	2450	2000	460	M100
KPGZ49/1120..-D	**1120**	1195	1115	920	1780	800	740	285	2560	2150	460	M100
KPGZ49/1180..-D	**1180**	1260	1175	970	1880	820	780	300	2700	2300	480	M110
KPGZ49/1250..-D	**1250**	1330	1245	1010	1985	850	820	320	2850	2400	510	M110
KPGZ49/1320..-D	**1320**	1400	1315	1080	2100	900	850	340	3000	2500	520	M125



KPGZ49..-F, unsplit bearing, locating bearing



KPGZ49..-F, split bearing, locating bearing

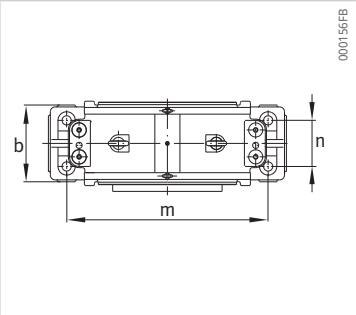


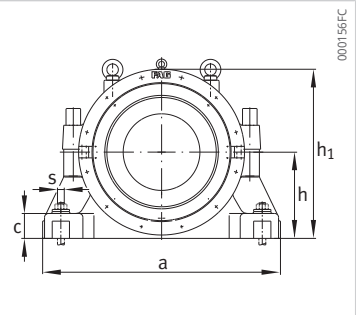
KPGZ49..-L, unsplit bearing, non-locating bearing

8

D	w	w_L	k	g_2	g_3	g_4	d_3	d_6	t	s_2	n_{S2}	m
mm	mm	mm	mm	mm	mm	mm	mm	mm	–	–	–	≈ kg
670	125	85	40	210	40	230	375	505	437,5	M20×70	8	900
710	130	90	40	215	40	235	400	535	465	M20×70	8	1000
750	135	95	40	220	40	240	420	565	490	M20×70	8	1300
800	145	100	40	240	45	260	450	610	525	M20×80	8	1500
850	155	109	40	250	46	270	475	640	552,5	M20×80	8	2100
900	163	115	40	260	48	280	505	675	587,5	M24×90	8	2500
950	175	122	50	290	54	318	535	715	622,5	M24×90	8	2700
1000	178	125	50	305	53	333	565	755	657,5	M30×100	8	2800
1060	185	129	50	310	56	338	600	805	700	M30×100	8	3300
1120	193	136	50	325	57	353	640	855	745	M30×110	8	3700
1180	195	140	60	340	55	375	675	905	787,5	M30×110	8	4300
1250	205	150	60	350	55	385	715	960	832,5	M36×110	8	5200
1320	225	158	70	370	68	413	750	1010	875	M36×130	8	5770
1400	238	168	60	400	70	435	795	1070	927,5	M36×130	8	6800
1460	238	168	70	410	70	453	840	1130	980	M42×140	8	8000
1540	250	178	70	420	73	463	885	1190	1032,5	M42×140	8	9000
1630	273	188	70	435	85	478	940	1255	1095	M42×150	8	11000
1720	290	200	70	460	90	503	990	1325	1155	M48×180	8	12800

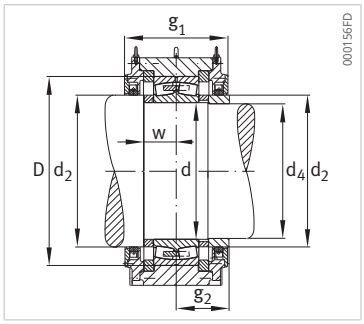
8.9.5 KPGZ, for spherical roller bearings with a cylindrical bore, accessories



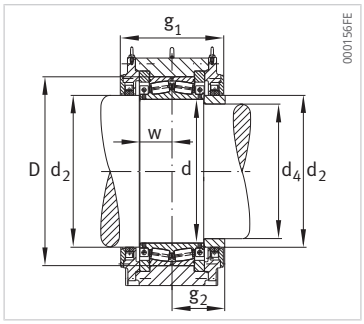


8

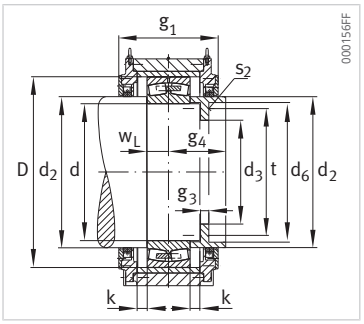
Housing	d	Bearing	
		MB cage	Pin cage
	mm	–	–
KPGZ49/500..-D	**500**	Z-528741.PRL	Z-541821.249/500
KPGZ49/530..-D	**530**	Z-528742.PRL	Z-541822.249/530
KPGZ49/560..-D	**560**	Z-528743.PRL	Z-541823.249/560-B
KPGZ49/600..-D	**600**	Z-528744.PRL	Z-541824.249/600-B
KPGZ49/630..-D	**630**	–	Z-541825.249/630
KPGZ49/670..-D	**670**	Z-528746.PRL	Z-541826.249/670
KPGZ49/710..-D	**710**	Z-528747.PRL	Z-541827.249/710-B
KPGZ49/750..-D	**750**	Z-528748.PRL	Z-541828.249/750-B
KPGZ49/800..-D	**800**	Z-528749.PRL	Z-541829.249/800-B
KPGZ49/850..-D	**850**	Z-528750.PRL	Z-541830.249/850-B
KPGZ49/900..-D	**900**	Z-528751.PRL	Z-541831.249/900-B
KPGZ49/950..-D	**950**	Z-528752.PRL	Z-541832.249/950-B
KPGZ49/1000..-D	**1000**	Z-528753.PRL	Z-541833.249/1000-B
KPGZ49/1060..-D	**1060**	–	Z-541834.249/1060-B
KPGZ49/1120..-D	**1120**	–	Z-541835.249/1120-B
KPGZ49/1180..-D	**1180**	–	Z-541836.249/1180-B
KPGZ49/1250..-D	**1250**	–	Z-541837.249/1250-B
KPGZ49/1320..-D	**1320**	–	Z-541838.249/1320-B



KPGZ49..-F, unsplit bearing, locating bearing



KPGZ49..-F, split bearing, locating bearing



KPGZ49..-L, unsplit bearing, non-locating bearing

8

Split bearing (for locating bearing only)

–
Z-537276.PRL
Z-537277.PRL
Z-537278.PRL
Z-533761.PRL
Z-537279.PRL
Z-537280.PRL
Z-526073.PRL
Z-533414.01.PRL
Z-532063.PRL
Z-537281.PRL
Z-537282.PRL
Z-534826.PRL
Z-533567.PRL
Z-537283.PRL
Z-537284.PRL
Z-536806.PRL
Z-537285.PRL
Z-545161.PRL

9 LOE split plummer block housing

medias | Product catalogue |
LOE split plummer block housings |
https://www.schaeffler.de/std/21FF

LOE split plummer block housings are designed for oil lubrication. They are suitable for high speed bearings, for example in fans.

9.1 Housing design

Bearing housing

148 LOE..-N-AF, LOE..-N-AL, split plummer block housing, closed on one side, with cover

0018FDD7

149 LOE..-N-BF, LOE..-N-BL split plummer block housing, for a continuous shaft

0018FDCF

9.1.1 Locating and non-locating bearings

The housings are supplied in a locating bearing or non-locating bearing design. For the locating bearing, the bearing is clamped between the housing covers. For the non-locating bearing, the bearing can align itself axially, since the covers have shorter centring collars.

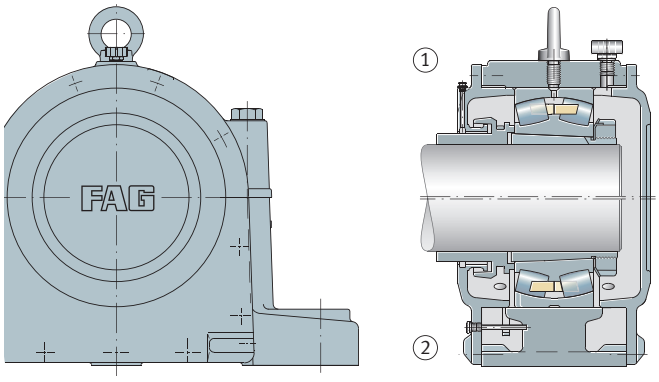
9.1.2 Housing configurations

Various housing configurations are possible for LOE plummer block housings, which are clearly described by the designation.

The following features can be considered for the housing configuration:

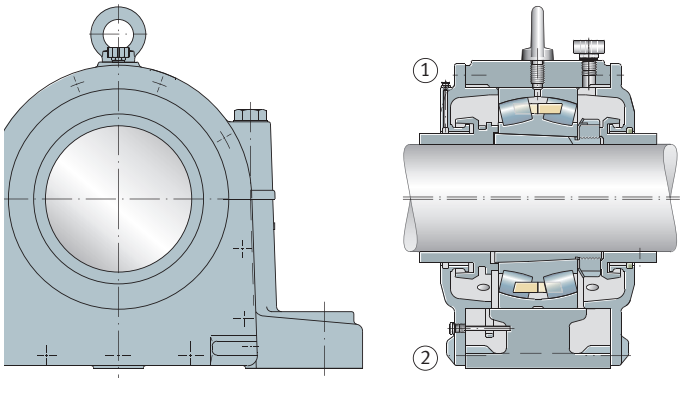
- Fastening of bearings with a tapered bore by an adapter sleeve on a shaft with a constant diameter or bearings with a cylindrical bore on a stepped shaft
- Continuous shaft or a housing closed on one side
- Design of the bearing as a locating or non-locating bearing
- Design with or without cooling coil

150 LOE5, LOE6 plummer block housing for bearings with a tapered bore and withdrawal sleeve, design A



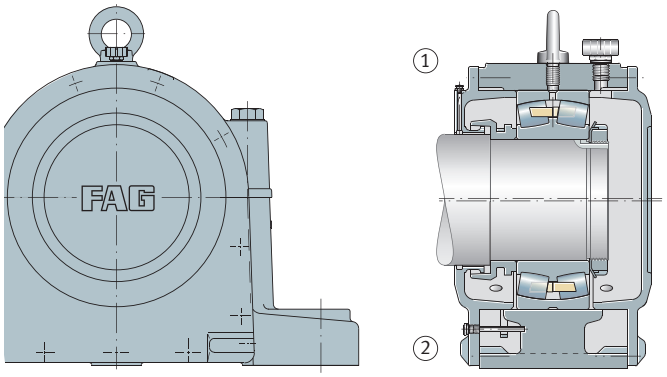
1	LOE..-N-AF locating bearing	2	LOE..-N-AL non-locating bearing

151 LOE5, LOE6 plummer block housing for bearings with a tapered bore and withdrawal sleeve, design B



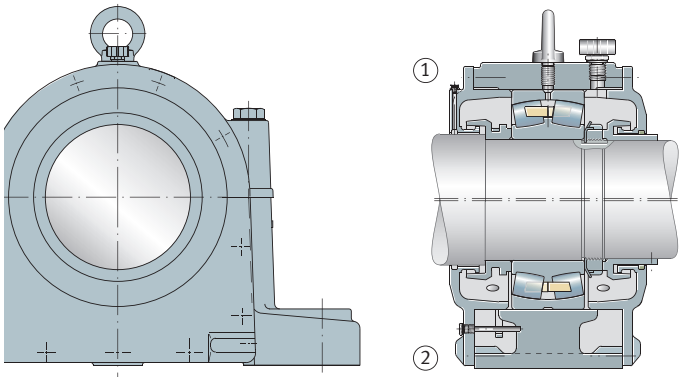
1	LOE..-N-BF locating bearing	2	LOE..-N-BL non-locating bearing

152 LOE2, LOE3 plummer block housing for bearings with a cylindrical bore, version A



1	LOE..-N-AF locating bearing	2	LOE..-N-AL non-locating bearing

153 LOE2, LOE3 plummer block housing for bearings with a cylindrical bore, version B



1	LOE..-N-BF locating bearing	2	LOE..-N-BL non-locating bearing

9.2 Suitable bearings

The dimensions of LOE split plummer block housing are tailored to the spherical roller bearings in series 222 and 223. The LOE2 and LOE3 housings are designed for bearings with a tapered bore, LOE5 and LOE6 housings for bearings with a tapered bore and adapter sleeve. The shaft diameters are 50 mm to 240 mm.

59 Suitable bearings for LOE split plummer block housings

Bearing type	Housing	Size	
		from	to
Spherical roller bearings with a tapered bore and adapter sleeve	LOE5	22217..-K	22248..-K
	LOE6	22314..-K	22336..-K
Spherical roller bearings with a cylindrical bore	LOE2	22214	22248
	LOE3	22310	22336

9.3 Materials, corrosion protection

60 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron GJL	L	✓	RAL 7016, anthracite grey
Ductile cast iron GJS	D	o	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

9.4 Lubrication

LOE split plummer block housings are designed for oil lubrication. They are suitable for high speed bearings. The oil delivers good heat dissipation from the bearing.

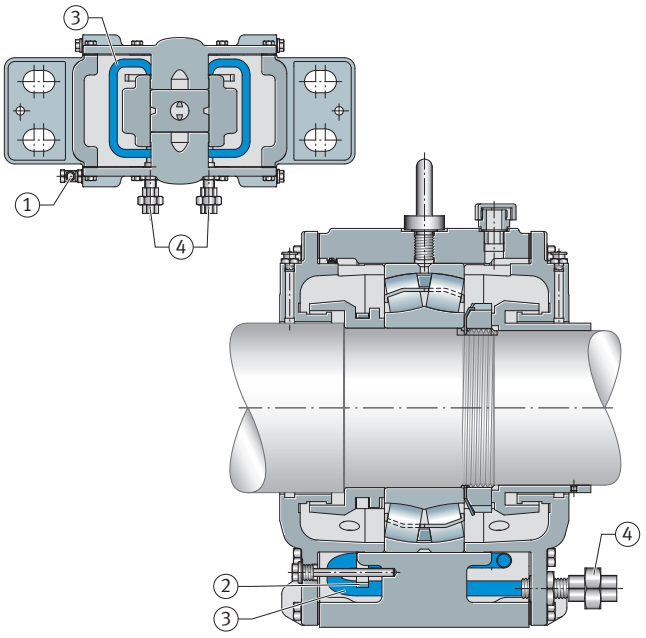
For split housings, the joint between the upper and lower sections of the housing must be sealed with a thin layer of a commercial sealant (with permanent elasticity). In addition, the housing covers must be sealed with the supplied sealing washer.

9.4.1 Oil bath lubrication

Oil bath lubrication is used as the lubrication method for the housings. The oil is supplied to the roller bearing from the oil sump in the lower section of the housing by a ring oiler. An angled oil level indicator is screwed to one cover.

If the requires cooling, the housings can be ordered in a design with a cooling coil (suffix E). The cooling coil is located in the area of the oil sump in the lower housing section and can be connected to an external cooling circuit.

154 LOE..-E plummer block housing with cooling coil



1	Angled oil level indicator	2	Ring oiler
3	Cooling coil	4	Connectors for external cooling circuit

9.4.2 Oil circulation lubrication

On request, a housing design for recirculating oil lubrication can be ordered using designation F-582158.LOU.

9.4.3 Oil quantity

The oil level can be read from the angled oil level indicator.

155 Reading the oil level on LOE devices

1	Angled oil level indicator		

61 Recommended oil quantity for initial filling

Housing	Recommended oil quantity	Oil level	
Designation	Initial filling	min.	Max.
	l	mm	mm
LOE217	1,4	50	65
LOE218	1,5	45	60
LOE219	1,6	55	70
LOE220	1,7	50	65
LOE222	2,1	50	70
LOE224	2,3	50	70
LOE226	2,3	55	75
LOE228	3,7	55	70
LOE230	4,2	65	90
LOE232	4,7	60	80
LOE234	6	90	105
LOE236	6	75	110
LOE238	7,2	70	100
LOE240	7,2	75	100
LOE244	8,2	80	110
LOE248	8,4	100	125
LOE310	0,9	50	65
LOE312	1	50	65
LOE314	1,4	50	65
LOE316	1,6	55	70
LOE318	2,3	65	85
LOE320	2,4	55	75
LOE322	2,4	45	65
LOE324	4,2	65	90
LOE326	3,7	75	105
LOE328	6,7	80	110
LOE330	6,2	75	110
LOE332	7	80	105
LOE334	7,2	80	105
LOE336	7,4	80	105
LOE517	1,4	50	65
LOE518	1,5	45	60
LOE519	1,6	55	70
LOE520	1,7	50	65
LOE522	2,1	50	70
LOE524	2,3	50	70
LOE526	2,3	55	75
LOE528	3,7	55	70
LOE530	4,2	65	90
LOE532	4,7	60	80
LOE534	6	90	105
LOE536	6	75	110
LOE538	7,2	70	100
LOE540	7,2	75	100
LOE544	8,2	80	110
LOE548	8,4	100	125
LOE614	1,4	50	65
LOE616	1,6	55	70
LOE618	2,3	65	85
LOE620	2,4	55	75
LOE622	2,4	45	65
LOE624	4,2	65	90
LOE626	3,7	75	105

Housing	Recommended oil quantity	Oil level	
Designation	Initial filling	min.	Max.
	l	mm	mm
LOE628	6,7	80	110
LOE630	6,2	75	110
LOE632	7	80	105
LOE634	7,2	80	105
LOE636	7,4	80	105

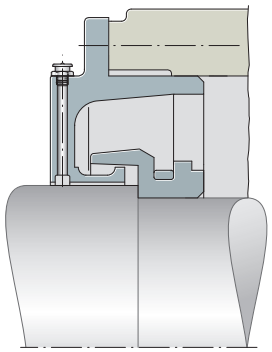
9.5 Sealing

LOE plummer block housings, design A are designed for the bearings on shaft ends. One side is closed by a cover and the other side is sealed against the shaft. Design B is designed for a continuous shafts where the housing is sealed against the shaft on both sides.

The seal used here is a labyrinth seal. The labyrinth rings are unsplit. The grease chamber in the labyrinth can be relubricated.

The labyrinth seal allows shaft misalignment of up to 0,25 ° to both sides and is suitable for oil lubrication.

156 Labyrinth seal for LOE plummer block housing



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9.6 Dimensions, tolerances

9.6.1 Tolerances for the bearing seat

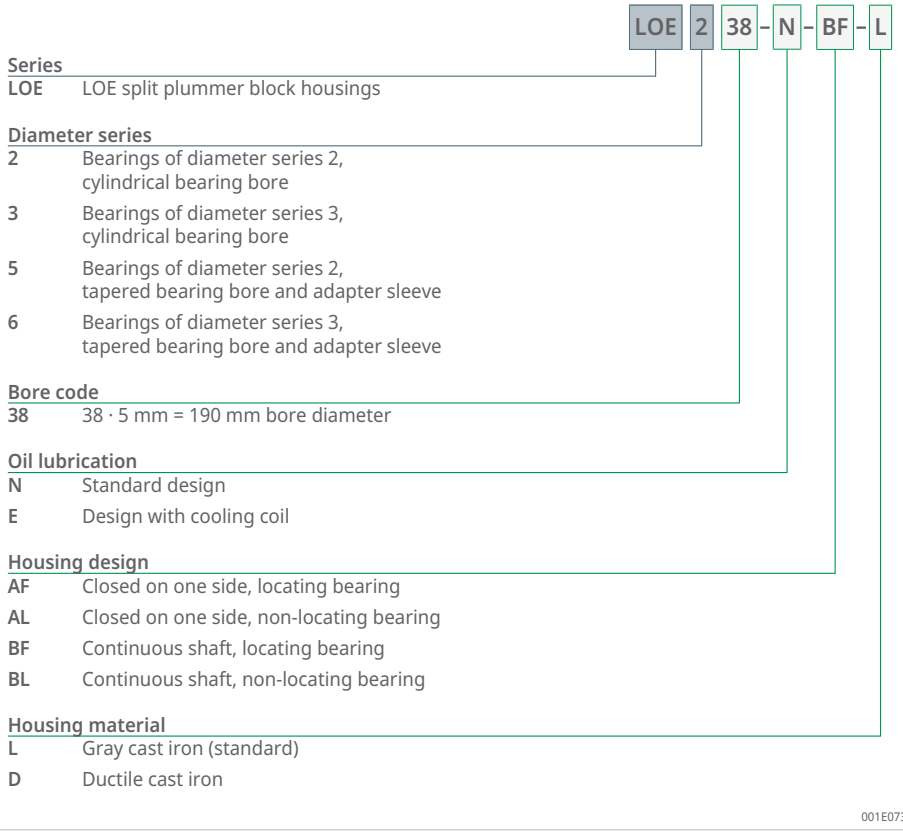
The bearing seat in the housings is machined to the tolerance class G6 to DIN EN ISO 286-1.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

9.7 Structure of the ordering designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

157 Structure of the designations of LOE plummer block housings



9.7.1 Specimen orders

For ordering a LOE split plummer block housing, the designation contains details of the size and location method of the bearing, the type of oil cooling and the design and material of the housing. The roller bearing and, if necessary, the adapter sleeve or locknut must be ordered separately.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- LOE plummer block housing made of gray cast iron, for a continuous shaft
- 22238-K-MB spherical roller bearing in the form of a locating bearing
- Secured by means of adapter sleeve
- Housing for oil cooling in standard design

62 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	LOE538-N-BF-L
1	Spherical roller bearing	22238-K-MB
1	Adapter sleeve	H3138-HG

Example 2

Example of a housing configuration, characteristics:

- LOE plummer block housing made of gray cast iron, for a continuous shaft
- 22238-MB spherical roller bearing in the form of a locating bearing
- Fastening with locknut
- Housing for oil cooling in standard design

63 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	LOE238-N-BF-L
1	Spherical roller bearing	22238-MB
1	Locknut	KM38
1	Tab washer	MB38

9.8 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

Further information

MON 111 | LOE, LOU plummer block housings |
https://www.schaeffler.de/std/2220

9.8.1 Eye bolts

There is 1 eye bolt to DIN 580:2018 in each housing upper section. This eye bolt is provided as a locating point for the installation and removal of the housing. The load capacity of the eye bolt allows the housing, including the bearing fitted in the housing, to be lifted.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

9.8.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

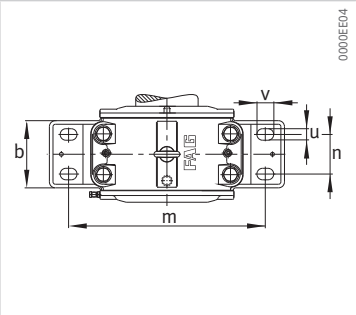
The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

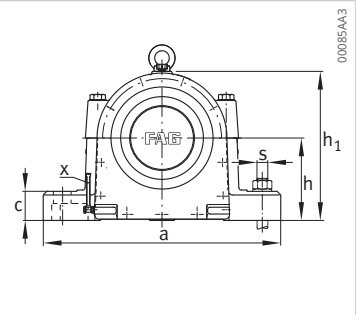
9.9 Product tables

9.9.1 Explanations of the product tables

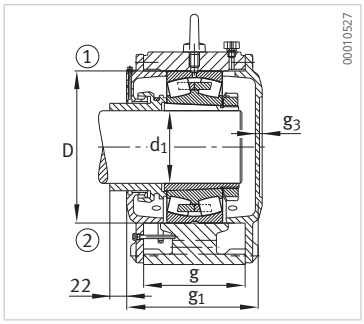
(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
b	mm	Base width
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
d_2	mm	Shaft shoulder diameter
d_4	mm	Diameter of shaft end
d_5	–	Thread of the shaft nut
g	mm	Housing body width
g_1	mm	Overall width
g_3	mm	Wall thickness
h	mm	Distance to shaft axis
h_1	mm	Overall height
m	mm	Fixing hole spacing
m	kg	Mass
n	mm	Distance between fastening holes
s	inch or mm	Screw size
u	mm	Slot width
v	mm	Slot length
w	mm	Distance from bearing centre to shaft shoulder
X	–	Angled oil level indicator
y	mm	Overhang of labyrinth ring

9.9.2 LOE, for spherical roller bearings with a tapered bore and adapter sleeve

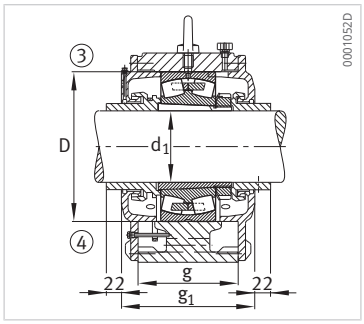




Housing	d_1	h	h_1	g	g_1	b	c	a	m	n
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
LOE614-N-AF-L	**60**	135	240	160	225	150	48	410	340	80
LOE614-N-AL-L	**60**	135	240	160	225	150	48	410	340	80
LOE614-N-BF-L	**60**	135	240	160	225	150	48	410	340	80
LOE614-N-BL-L	**60**	135	240	160	225	150	48	410	340	80
LOE616-N-AF-L	**70**	150	270	170	250	160	50	490	400	80
LOE616-N-AL-L	**70**	150	270	170	250	160	50	490	400	80
LOE616-N-BF-L	**70**	150	270	170	250	160	50	490	400	80
LOE616-N-BL-L	**70**	150	270	170	250	160	50	490	400	80
LOE517-N-AF-L	**75**	135	240	160	225	150	48	410	340	80
LOE517-N-AL-L	**75**	135	240	160	225	150	48	410	340	80
LOE517-N-BF-L	**75**	135	240	160	225	150	48	410	340	80
LOE517-N-BL-L	**75**	135	240	160	225	150	48	410	340	80
LOE518-N-AF-L	**80**	135	245	160	225	150	48	410	340	80
LOE518-N-AL-L	**80**	135	245	160	225	150	48	410	340	80
LOE518-N-BF-L	**80**	135	245	160	225	150	48	410	340	80
LOE518-N-BL-L	**80**	135	245	160	225	150	48	410	340	80
LOE618-N-AF-L	**80**	175	305	175	250	165	55	500	420	80
LOE618-N-AL-L	**80**	175	305	175	250	165	55	500	420	80
LOE618-N-BF-L	**80**	175	305	175	250	165	55	500	420	80
LOE618-N-BL-L	**80**	175	305	175	250	165	55	500	420	80
LOE519-N-AF-L	**85**	150	270	170	250	160	50	490	400	80
LOE519-N-AL-L	**85**	150	270	170	250	160	50	490	400	80
LOE519-N-BF-L	**85**	150	270	170	250	160	50	490	400	80
LOE519-N-BL-L	**85**	150	270	170	250	160	50	490	400	80
LOE520-N-AF-L	**90**	150	270	170	250	160	50	490	400	80
LOE520-N-AL-L	**90**	150	270	170	250	160	50	490	400	80
LOE520-N-BF-L	**90**	150	270	170	250	160	50	490	400	80
LOE520-N-BL-L	**90**	150	270	170	250	160	50	490	400	80
LOE620-N-AF-L	**90**	175	320	175	250	165	55	550	440	80
LOE620-N-AL-L	**90**	175	320	175	250	165	55	550	440	80
LOE620-N-BF-L	**90**	175	320	175	250	165	55	550	440	80
LOE620-N-BL-L	**90**	175	320	175	250	165	55	550	440	80
LOE522-N-AF-L	**100**	165	300	175	250	165	50	510	420	80
LOE522-N-AL-L	**100**	165	300	175	250	165	50	510	420	80
LOE522-N-BF-L	**100**	165	300	175	250	165	50	510	420	80
LOE522-N-BL-L	**100**	165	300	175	250	165	50	510	420	80
LOE622-N-AF-L	**100**	180	335	190	270	180	65	570	460	95
LOE622-N-AL-L	**100**	180	335	190	270	180	65	570	460	95
LOE622-N-BF-L	**100**	180	335	190	270	180	65	570	460	95
LOE622-N-BL-L	**100**	180	335	190	270	180	65	570	460	95



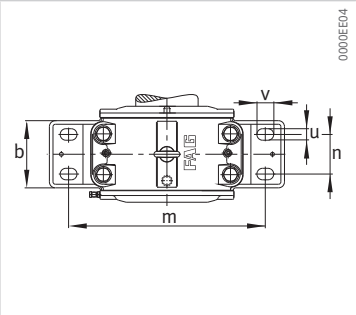
Design A

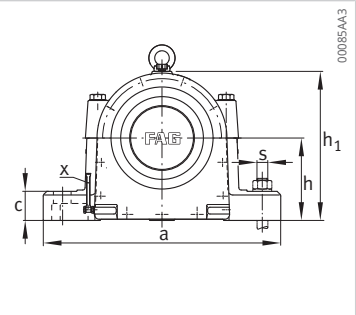


Design B

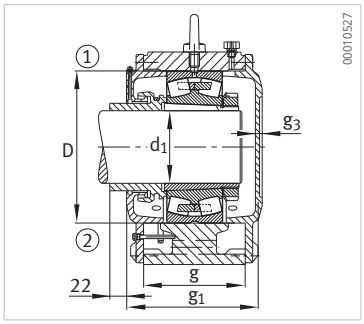
v	u	s	D	g_3	m	Bearing	Adapter sleeve
mm	mm	–	mm	mm	≈ kg	–	–
35	25	M20	150	18	45	22314-E1-K	H2314
35	25	M20	150	18	45	22314-E1-K	H2314
35	25	M20	150	–	45	22314-E1-K	H2314
35	25	M20	150	–	45	22314-E1-K	H2314
45	30	M24	170	18	60	22316-E1-K	H2316
45	30	M24	170	18	60	22316-E1-K	H2316
45	30	M24	170	–	60	22316-E1-K	H2316
45	30	M24	170	–	60	22316-E1-K	H2316
35	25	M20	150	18	45	22217-E1-K	H317
35	25	M20	150	18	45	22217-E1-K	H317
35	25	M20	150	–	45	22217-E1-K	H317
35	25	M20	150	–	45	22217-E1-K	H317
35	25	M20	160	17,5	47	22218-E1-K	H318
35	25	M20	160	17,5	47	22218-E1-K	H318
35	25	M20	160	–	47	22218-E1-K	H318
35	25	M20	160	–	47	22218-E1-K	H318
45	30	M24	190	18	73	22318-E1-K	H2318
45	30	M24	190	18	73	22318-E1-K	H2318
45	30	M24	190	–	73	22318-E1-K	H2318
45	30	M24	190	–	73	22318-E1-K	H2318
45	30	M24	170	18	60	22219-E1-K	H319
45	30	M24	170	18	60	22219-E1-K	H319
45	30	M24	170	–	60	22219-E1-K	H319
45	30	M24	170	–	60	22219-E1-K	H319
45	30	M24	180	20	67	22220-E1-K	H320
45	30	M24	180	20	67	22220-E1-K	H320
45	30	M24	180	–	67	22220-E1-K	H320
45	30	M24	180	–	67	22220-E1-K	H320
50	36	M30	215	18	81	22320-E1-K	H2320
50	36	M30	215	18	81	22320-E1-K	H2320
50	36	M30	215	–	81	22320-E1-K	H2320
50	36	M30	215	–	81	22320-E1-K	H2320
45	30	M24	200	18	74	22222-E1-K	H322
45	30	M24	200	18	74	22222-E1-K	H322
45	30	M24	200	–	74	22222-E1-K	H322
45	30	M24	200	–	74	22222-E1-K	H322
50	36	M30	240	20	100	22322-E1-K	H2322
50	36	M30	240	20	100	22322-E1-K	H2322
50	36	M30	240	–	100	22322-E1-K	H2322
50	36	M30	240	–	100	22322-E1-K	H2322

9.9.2 LOE, for spherical roller bearings with a tapered bore and adapter sleeve

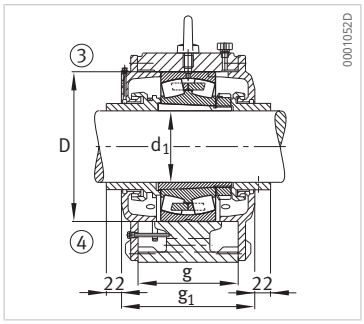




Housing	d_1	h	h_1	g	g_1	b	c	a	m	n
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
LOE524-N-AF-L	**110**	175	320	175	250	165	55	550	440	80
LOE524-N-AL-L	**110**	175	320	175	250	165	55	550	440	80
LOE524-N-BF-L	**110**	175	320	175	250	165	55	550	440	80
LOE524-N-BL-L	**110**	175	320	175	250	165	55	550	440	80
LOE624-N-AF-L	**110**	220	390	210	300	200	75	660	530	110
LOE624-N-AL-L	**110**	220	390	210	300	200	75	660	530	110
LOE624-N-BF-L	**110**	220	390	210	300	200	75	660	530	110
LOE624-N-BL-L	**110**	220	390	210	300	200	75	660	530	110
LOE526-N-AF-L	**115**	190	345	185	260	175	65	570	460	90
LOE526-N-AL-L	**115**	190	345	185	260	175	65	570	460	90
LOE526-N-BF-L	**115**	190	345	185	260	175	65	570	460	90
LOE526-N-BL-L	**115**	190	345	185	260	175	65	570	460	90
LOE626-N-AF-L	**115**	235	420	220	315	200	80	660	530	110
LOE626-N-AL-L	**115**	235	420	220	315	200	80	660	530	110
LOE626-N-BF-L	**115**	235	420	220	315	200	80	660	530	110
LOE626-N-BL-L	**115**	235	420	220	315	200	80	660	530	110
LOE528-N-AF-L	**125**	190	355	180	260	175	65	570	460	100
LOE528-N-AL-L	**125**	190	355	180	260	175	65	570	460	100
LOE528-N-BF-L	**125**	190	355	180	260	175	65	570	460	100
LOE528-N-BL-L	**125**	190	355	180	260	175	65	570	460	100
LOE628-N-AF-L	**125**	260	450	230	325	220	85	710	580	125
LOE628-N-AL-L	**125**	260	450	230	325	220	85	710	580	125
LOE628-N-BF-L	**125**	260	450	230	325	220	85	710	580	125
LOE628-N-BL-L	**125**	260	450	230	325	220	85	710	580	125
LOE530-N-AF-L	**135**	220	395	200	275	190	70	660	530	100
LOE530-N-AL-L	**135**	220	395	200	275	190	70	660	530	100
LOE530-N-BF-L	**135**	220	395	200	275	190	70	660	530	100
LOE530-N-BL-L	**135**	220	395	200	275	190	70	660	530	100
LOE630-N-AF-L	**135**	265	465	240	335	200	85	760	630	125
LOE630-N-AL-L	**135**	265	465	240	335	200	85	760	630	125
LOE630-N-BF-L	**135**	265	465	240	335	200	85	760	630	125
LOE630-N-BL-L	**135**	265	465	240	335	200	85	760	630	125
LOE532-N-AF-L	**140**	220	400	210	290	200	70	660	530	110
LOE532-N-AL-L	**140**	220	400	210	290	200	70	660	530	110
LOE532-N-BF-L	**140**	220	400	210	290	200	70	660	530	110
LOE532-N-BL-L	**140**	220	400	210	290	200	70	660	530	110
LOE632-N-AF-L	**140**	270	485	250	350	240	90	820	670	130
LOE632-N-AL-L	**140**	270	485	250	350	240	90	820	670	130
LOE632-N-BF-L	**140**	270	485	250	350	240	90	820	670	130
LOE632-N-BL-L	**140**	270	485	250	350	240	90	820	670	130



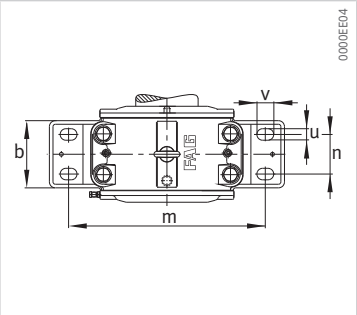
Design A

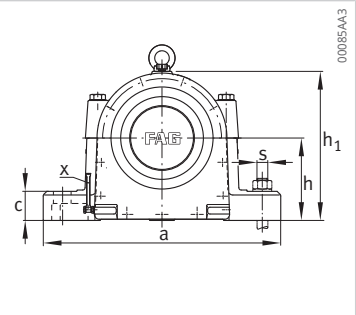


Design B

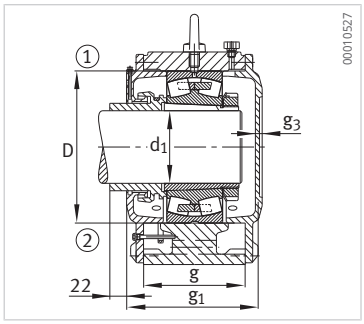
v	u	s	D	g_3	m	Bearing	Adapter sleeve
mm	mm	–	mm	mm	≈ kg	–	–
50	36	M30	215	18	80	22224-E1-K	H3124
50	36	M30	215	18	80	22224-E1-K	H3124
50	36	M30	215	–	80	22224-E1-K	H3124
50	36	M30	215	–	80	22224-E1-K	H3124
60	42	M36	260	18	130	22324-E1-K	H2324
60	42	M36	260	18	130	22324-E1-K	H2324
60	42	M36	260	–	130	22324-E1-K	H2324
60	42	M36	260	–	130	22324-E1-K	H2324
50	36	M30	230	18	93	22226-E1-K	H3126
50	36	M30	230	18	93	22226-E1-K	H3126
50	36	M30	230	–	93	22226-E1-K	H3126
50	36	M30	230	–	93	22226-E1-K	H3126
60	42	M36	280	18	142	22326-E1-K	H2326
60	42	M36	280	18	142	22326-E1-K	H2326
60	42	M36	280	–	142	22326-E1-K	H2326
60	42	M36	280	–	142	22326-E1-K	H2326
50	36	M30	250	18	100	22228-E1-K	H3128
50	36	M30	250	18	100	22228-E1-K	H3128
50	36	M30	250	–	100	22228-E1-K	H3128
50	36	M30	250	–	100	22228-E1-K	H3128
60	42	M36	300	19,5	170	22328-E1-K	H2328
60	42	M36	300	19,5	170	22328-E1-K	H2328
60	42	M36	300	–	170	22328-E1-K	H2328
60	42	M36	300	–	170	22328-E1-K	H2328
60	42	M36	270	18	125	22230-E1-K	H3130
60	42	M36	270	18	125	22230-E1-K	H3130
60	42	M36	270	–	125	22230-E1-K	H3130
60	42	M36	270	–	125	22230-E1-K	H3130
60	42	M36	320	18	200	22330-E1-K	H2330
60	42	M36	320	18	200	22330-E1-K	H2330
60	42	M36	320	–	200	22330-E1-K	H2330
60	42	M36	320	–	200	22330-E1-K	H2330
60	42	M36	290	20	135	22232-E1-K	H3132
60	42	M36	290	20	135	22232-E1-K	H3132
60	42	M36	290	–	135	22232-E1-K	H3132
60	42	M36	290	–	135	22232-E1-K	H3132
70	48	M42	340	20	240	22332-K-MB	H2332
70	48	M42	340	20	240	22332-K-MB	H2332
70	48	M42	340	–	240	22332-K-MB	H2332
70	48	M42	340	–	240	22332-K-MB	H2332

9.9.2 LOE, for spherical roller bearings with a tapered bore and adapter sleeve

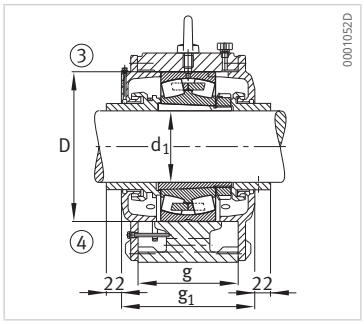




Housing	d_1	h	h_1	g	g_1	b	c	a	m	n
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
LOE534-N-AF-L	**150**	260	460	210	300	200	85	710	580	110
LOE534-N-AL-L	**150**	260	460	210	300	200	85	710	580	110
LOE534-N-BF-L	**150**	260	460	210	300	200	85	710	580	110
LOE534-N-BL-L	**150**	260	460	210	300	200	85	710	580	110
LOE634-N-AF-L	**150**	280	510	255	350	240	90	830	670	130
LOE634-N-AL-L	**150**	280	510	255	350	240	90	830	670	130
LOE634-N-BF-L	**150**	280	510	255	350	240	90	830	670	130
LOE634-N-BL-L	**150**	280	510	255	350	240	90	830	670	130
LOE536-N-AF-L	**160**	260	465	210	300	200	85	710	580	110
LOE536-N-AL-L	**160**	260	465	210	300	200	85	710	580	110
LOE536-N-BF-L	**160**	260	465	210	300	200	85	710	580	110
LOE536-N-BL-L	**160**	260	465	210	300	200	85	710	580	110
LOE636-N-AF-L	**160**	290	530	260	360	240	90	840	680	130
LOE636-N-AL-L	**160**	290	530	260	360	240	90	840	680	130
LOE636-N-BF-L	**160**	290	530	260	360	240	90	840	680	130
LOE636-N-BL-L	**160**	290	530	260	360	240	90	840	680	130
LOE538-N-AF-L	**170**	270	485	250	350	240	90	820	670	130
LOE538-N-AL-L	**170**	270	485	250	350	240	90	820	670	130
LOE538-N-BF-L	**170**	270	485	250	350	240	90	820	670	130
LOE538-N-BL-L	**170**	270	485	250	350	240	90	820	670	130
LOE540-N-AF-L	**180**	280	510	260	344	240	90	830	670	130
LOE540-N-AL-L	**180**	280	510	260	344	240	90	830	670	130
LOE540-N-BF-L	**180**	280	510	260	344	240	90	830	670	130
LOE540-N-BL-L	**180**	280	510	260	344	240	90	830	670	130
LOE544-N-AF-L	**200**	310	565	280	380	240	105	880	720	130
LOE544-N-AL-L	**200**	310	565	280	380	240	105	880	720	130
LOE544-N-BF-L	**200**	310	565	280	380	240	105	880	720	130
LOE544-N-BL-L	**200**	310	565	280	380	240	105	880	720	130
LOE548-N-AF-L	**220**	340	625	300	400	280	120	980	820	165
LOE548-N-AL-L	**220**	340	625	300	400	280	120	980	820	165
LOE548-N-BF-L	**220**	340	625	300	400	280	120	980	820	165
LOE548-N-BL-L	**220**	340	625	300	400	280	120	980	820	165



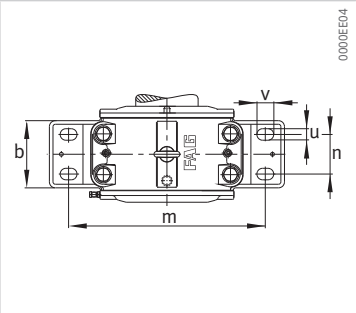
Design A

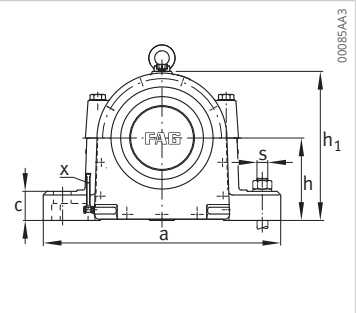


Design B

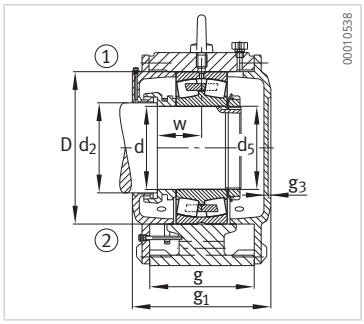
v	u	s	D	g_3	m	Bearing	Adapter sleeve
mm	mm	–	mm	mm	≈ kg	–	–
60	42	M36	310	18	160	22234-E1-K	H3134
60	42	M36	310	18	160	22234-E1-K	H3134
60	42	M36	310	–	160	22234-E1-K	H3134
60	42	M36	310	–	160	22234-E1-K	H3134
70	48	M42	360	18	270	22334-K-MB	H2334
70	48	M42	360	18	270	22334-K-MB	H2334
70	48	M42	360	–	270	22334-K-MB	H2334
70	48	M42	360	–	270	22334-K-MB	H2334
60	42	M36	320	20	200	22236-E1-K	H3136
60	42	M36	320	20	200	22236-E1-K	H3136
60	42	M36	320	–	200	22236-E1-K	H3136
60	42	M36	320	–	200	22236-E1-K	H3136
70	48	M42	380	20	330	22336-K-MB	H2336
70	48	M42	380	20	330	22336-K-MB	H2336
70	48	M42	380	–	330	22336-K-MB	H2336
70	48	M42	380	–	330	22336-K-MB	H2336
70	48	M42	340	20	230	22238-K-MB	H3138
70	48	M42	340	20	230	22238-K-MB	H3138
70	48	M42	340	–	230	22238-K-MB	H3138
70	48	M42	340	–	230	22238-K-MB	H3138
70	48	M42	360	20	250	22240-B-K-MB	H3140
70	48	M42	360	20	250	22240-B-K-MB	H3140
70	48	M42	360	–	250	22240-B-K-MB	H3140
70	48	M42	360	–	250	22240-B-K-MB	H3140
70	48	M42	400	20	310	22244-B-K-MB	H3144X
70	48	M42	400	20	310	22244-B-K-MB	H3144X
70	48	M42	400	–	310	22244-B-K-MB	H3144X
70	48	M42	400	–	310	22244-B-K-MB	H3144X
70	48	M42	440	20	385	22248-B-K-MB	H3148X
70	48	M42	440	20	385	22248-B-K-MB	H3148X
70	48	M42	440	–	385	22248-B-K-MB	H3148X
70	48	M42	440	–	385	22248-B-K-MB	H3148X

9.9.3 LOE, for spherical roller bearings with a cylindrical bore

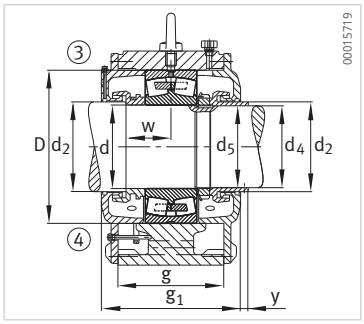




Housing	d	d_2	d_4	d_5	h	h_1	g	g_1	b	c	a	m	n
	mm	mm	mm	–	mm	mm	mm	mm	mm	mm	mm	mm	mm
LOE310-N-AF-L	**50**	55	–	M50×2	115	205	135	210	125	40	350	290	75
LOE310-N-AL-L	**50**	55	–	M50×2	115	205	135	210	125	40	350	290	75
LOE310-N-BF-L	**50**	55	47	M50×2	115	205	135	210	125	40	350	290	75
LOE310-N-BL-L	**50**	55	47	M50×2	115	205	135	210	125	40	350	290	75
LOE312-N-AF-L	**60**	65	–	M60×2	125	220	140	220	130	45	370	310	80
LOE312-N-AL-L	**60**	65	–	M60×2	125	220	140	220	130	45	370	310	80
LOE312-N-BF-L	**60**	65	57	M60×2	125	220	140	220	130	45	370	310	80
LOE312-N-BL-L	**60**	65	57	M60×2	125	220	140	220	130	45	370	310	80
LOE314-N-AF-L	**70**	75	–	M70×2	135	240	160	225	150	48	410	340	80
LOE314-N-AL-L	**70**	75	–	M70×2	135	240	160	225	150	48	410	340	80
LOE314-N-BF-L	**70**	75	67	M70×2	135	240	160	225	150	48	410	340	80
LOE314-N-BL-L	**70**	75	67	M70×2	135	240	160	225	150	48	410	340	80
LOE316-N-AF-L	**80**	85	–	M80×2	150	270	170	250	160	50	490	400	80
LOE316-N-AL-L	**80**	85	–	M80×2	150	270	170	250	160	50	490	400	80
LOE316-N-BF-L	**80**	85	77	M80×2	150	270	170	250	160	50	490	400	80
LOE316-N-BL-L	**80**	85	77	M80×2	150	270	170	250	160	50	490	400	80
LOE217-N-AF-L	**85**	90	–	M85×2	135	240	160	225	150	48	410	340	80
LOE217-N-AL-L	**85**	90	–	M85×2	135	240	160	225	150	48	410	340	80
LOE217-N-BF-L	**85**	90	82	M85×2	135	240	160	225	150	48	410	340	80
LOE217-N-BL-L	**85**	90	82	M85×2	135	240	160	225	150	48	410	340	80
LOE218-N-AF-L	**90**	95	–	M90×2	135	245	160	225	150	48	410	340	80
LOE218-N-AL-L	**90**	95	–	M90×2	135	245	160	225	150	48	410	340	80
LOE218-N-BF-L	**90**	95	87	M90×2	135	245	160	225	150	48	410	340	80
LOE218-N-BL-L	**90**	95	87	M90×2	135	245	160	225	150	48	410	340	80
LOE318-N-AF-L	**90**	95	–	M90×2	175	305	175	250	165	55	500	420	80
LOE318-N-AL-L	**90**	95	–	M90×2	175	305	175	250	165	55	500	420	80
LOE318-N-BF-L	**90**	95	87	M90×2	175	305	175	250	165	55	500	420	80
LOE318-N-BL-L	**90**	95	87	M90×2	175	305	175	250	165	55	500	420	80
LOE219-N-AF-L	**95**	100	–	M95×2	150	270	170	250	160	50	490	400	80
LOE219-N-AL-L	**95**	100	–	M95×2	150	270	170	250	160	50	490	400	80
LOE219-N-BF-L	**95**	100	92	M95×2	150	270	170	250	160	50	490	400	80
LOE219-N-BL-L	**95**	100	92	M95×2	150	270	170	250	160	50	490	400	80
LOE220-N-AF-L	**100**	110	–	M100×2	150	270	170	250	160	50	490	400	80
LOE220-N-AL-L	**100**	110	–	M100×2	150	270	170	250	160	50	490	400	80
LOE220-N-BF-L	**100**	110	97	M100×2	150	270	170	250	160	50	490	400	80
LOE220-N-BL-L	**100**	110	97	M100×2	150	270	170	250	160	50	490	400	80
LOE320-N-AF-L	**100**	106	–	M100×2	175	320	175	250	165	55	550	440	80
LOE320-N-AL-L	**100**	106	–	M100×2	175	320	175	250	165	55	550	440	80
LOE320-N-BF-L	**100**	106	97	M100×2	175	320	175	250	165	55	550	440	80
LOE320-N-BL-L	**100**	106	97	M100×2	175	320	175	250	165	55	550	440	80



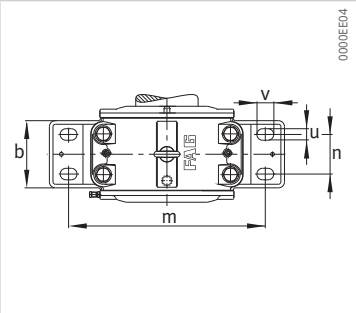
Design A

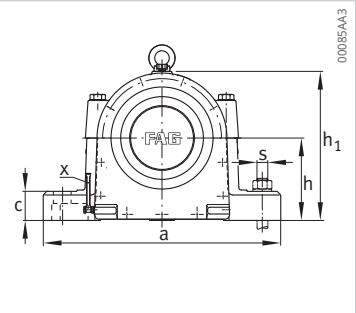


Design B

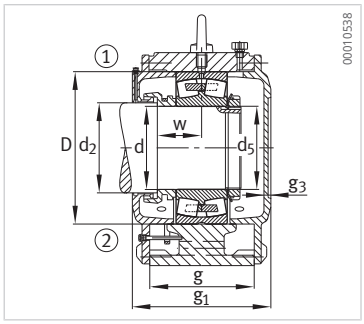
v	u	s	D	w	y	g_3	m	Bearing	Shaft nut	Adapter sleeve
mm	mm	–	mm	mm	mm	mm	≈ kg	–	–	–
30	20	M16	110	52	–	18	30	22310-E1	KM10	MB10
30	20	M16	110	52	–	18	30	22310-E1	KM10	MB10
30	20	M16	110	52	15	–	30	22310-E1	KM10	MB10
30	20	M16	110	52	15	–	30	22310-E1	KM10	MB10
30	20	M16	130	55	–	18	35	22312-E1	KM12	MB12
30	20	M16	130	55	–	18	35	22312-E1	KM12	MB12
30	20	M16	130	55	15	–	35	22312-E1	KM12	MB12
30	20	M16	130	55	15	–	35	22312-E1	KM12	MB12
35	25	M20	150	62	–	18	45	22314-E1	KM14	MB14
35	25	M20	150	62	–	18	45	22314-E1	KM14	MB14
35	25	M20	150	62	15	–	45	22314-E1	KM14	MB14
35	25	M20	150	62	15	–	45	22314-E1	KM14	MB14
45	30	M24	170	67	–	18	60	22316-E1	KM16	MB16
45	30	M24	170	67	–	18	60	22316-E1	KM16	MB16
45	30	M24	170	67	15	–	60	22316-E1	KM16	MB16
45	30	M24	170	67	15	–	60	22316-E1	KM16	MB16
35	25	M20	150	62	–	18	45	22217-E1	KM17	MB17
35	25	M20	150	62	–	18	45	22217-E1	KM17	MB17
35	25	M20	150	62	15	–	45	22217-E1	KM17	MB17
35	25	M20	150	62	15	–	45	22217-E1	KM17	MB17
35	25	M20	160	61	–	17,5	47	22218-E1	KM18	MB18
35	25	M20	160	61	–	17,5	47	22218-E1	KM18	MB18
35	25	M20	160	61	15	–	47	22218-E1	KM18	MB18
35	25	M20	160	61	15	–	47	22218-E1	KM18	MB18
45	30	M24	190	72	–	18	73	22318-E1	KM18	MB18
45	30	M24	190	72	–	18	73	22318-E1	KM18	MB18
45	30	M24	190	72	15	–	73	22318-E1	KM18	MB18
45	30	M24	190	72	15	–	73	22318-E1	KM18	MB18
45	30	M24	170	67	–	18	60	22219-E1	KM19	MB19
45	30	M24	170	67	–	18	60	22219-E1	KM19	MB19
45	30	M24	170	67	15	–	60	22219-E1	KM19	MB19
45	30	M24	170	67	15	–	60	22219-E1	KM19	MB19
45	30	M24	180	60	–	20	67	22220-E1	KM20	MB20
45	30	M24	180	60	–	20	67	22220-E1	KM20	MB20
45	30	M24	180	60	15	–	67	22220-E1	KM20	MB20
45	30	M24	180	60	15	–	67	22220-E1	KM20	MB20
50	36	M30	215	72	–	18	81	22320-E1	KM20	MB20
50	36	M30	215	72	–	18	81	22320-E1	KM20	MB20
50	36	M30	215	72	15	–	81	22320-E1	KM20	MB20
50	36	M30	215	72	15	–	81	22320-E1	KM20	MB20

9.9.3 LOE, for spherical roller bearings with a cylindrical bore

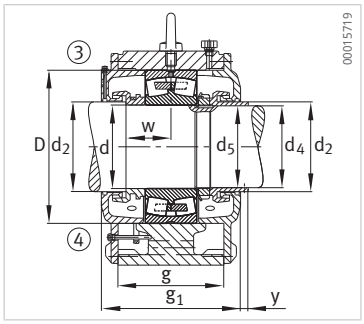




Housing	d	d_2	d_4	d_5	h	h_1	g	g_1	b	c	a	m	n
	mm	mm	mm	–	mm	mm	mm	mm	mm	mm	mm	mm	mm
LOE222-N-AF-L	**110**	116	–	M110×2	165	300	175	250	165	50	510	420	80
LOE222-N-AL-L	**110**	116	–	M110×2	165	300	175	250	165	50	510	420	80
LOE222-N-BF-L	**110**	116	107	M110×2	165	300	175	250	165	50	510	420	80
LOE222-N-BL-L	**110**	116	107	M110×2	165	300	175	250	165	50	510	420	80
LOE322-N-AF-L	**110**	120	–	M110×2	180	335	190	270	180	65	570	460	95
LOE322-N-AL-L	**110**	120	–	M110×2	180	335	190	270	180	65	570	460	95
LOE322-N-BF-L	**110**	120	107	M110×2	180	335	190	270	180	65	570	460	95
LOE322-N-BL-L	**110**	120	107	M110×2	180	335	190	270	180	65	570	460	95
LOE224-N-AF-L	**120**	126	–	M120×2	175	320	175	250	165	55	550	440	80
LOE224-N-AL-L	**120**	126	–	M120×2	175	320	175	250	165	55	550	440	80
LOE224-N-BF-L	**120**	126	117	M120×2	175	320	175	250	165	55	550	440	80
LOE224-N-BL-L	**120**	126	117	M120×2	175	320	175	250	165	55	550	440	80
LOE324-N-AF-L	**120**	126	–	M120×2	220	390	210	300	200	75	660	530	110
LOE324-N-AL-L	**120**	126	–	M120×2	220	390	210	300	200	75	660	530	110
LOE324-N-BF-L	**120**	126	117	M120×2	220	390	210	300	200	75	660	530	110
LOE324-N-BL-L	**120**	126	117	M120×2	220	390	210	300	200	75	660	530	110
LOE226-N-AF-L	**130**	136	–	M130×2	190	345	185	260	175	65	570	460	90
LOE226-N-AL-L	**130**	136	–	M130×2	190	345	185	260	175	65	570	460	90
LOE226-N-BF-L	**130**	136	127	M130×2	190	345	185	260	175	65	570	460	90
LOE226-N-BL-L	**130**	136	127	M130×2	190	345	185	260	175	65	570	460	90
LOE326-N-AF-L	**130**	140	–	M130×2	235	420	220	315	200	80	660	530	110
LOE326-N-AL-L	**130**	140	–	M130×2	235	420	220	315	200	80	660	530	110
LOE326-N-BF-L	**130**	140	127	M130×2	235	420	220	315	200	80	660	530	110
LOE326-N-BL-L	**130**	140	127	M130×2	235	420	220	315	200	80	660	530	110
LOE228-N-AF-L	**140**	150	–	M140×2	190	355	180	260	175	65	570	460	100
LOE228-N-AL-L	**140**	150	–	M140×2	190	355	180	260	175	65	570	460	100
LOE228-N-BF-L	**140**	150	137	M140×2	190	355	180	260	175	65	570	460	100
LOE228-N-BL-L	**140**	150	137	M140×2	190	355	180	260	175	65	570	460	100
LOE328-N-AF-L	**140**	146	–	M140×2	260	450	230	325	220	85	710	580	125
LOE328-N-AL-L	**140**	146	–	M140×2	260	450	230	325	220	85	710	580	125
LOE328-N-BF-L	**140**	146	137	M140×2	260	450	230	325	220	85	710	580	125
LOE328-N-BL-L	**140**	146	137	M140×2	260	450	230	325	220	85	710	580	125
LOE230-N-AF-L	**150**	156	–	M150×2	220	395	200	275	190	70	660	530	100
LOE230-N-AL-L	**150**	156	–	M150×2	220	395	200	275	190	70	660	530	100
LOE230-N-BF-L	**150**	156	147	M150×2	220	395	200	275	190	70	660	530	100
LOE230-N-BL-L	**150**	156	147	M150×2	220	395	200	275	190	70	660	530	100
LOE330-N-AF-L	**150**	160	–	M150×2	265	465	240	335	200	85	760	630	125
LOE330-N-AL-L	**150**	160	–	M150×2	265	465	240	335	200	85	760	630	125
LOE330-N-BF-L	**150**	160	147	M150×2	265	465	240	335	200	85	760	630	125
LOE330-N-BL-L	**150**	160	147	M150×2	265	465	240	335	200	85	760	630	125



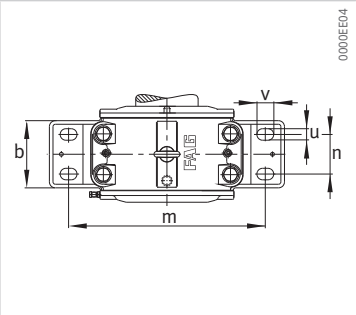
Design A

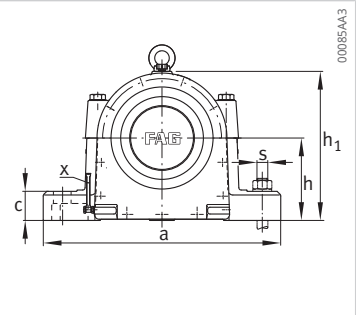


Design B

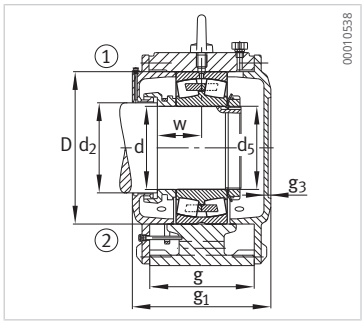
v	u	s	D	w	y	g_3	m	Bearing	Shaft nut	Adapter sleeve
mm	mm	–	mm	mm	mm	mm	≈ kg	–	–	–
45	30	M24	200	70	–	18	74	22222-E1	KM22	MB22
45	30	M24	200	70	–	18	74	22222-E1	KM22	MB22
45	30	M24	200	70	15	–	74	22222-E1	KM22	MB22
45	30	M24	200	70	15	–	74	22222-E1	KM22	MB22
50	36	M30	240	77	–	20	100	22322-E1	KM22	MB22
50	36	M30	240	77	–	20	100	22322-E1	KM22	MB22
50	36	M30	240	77	15	–	100	22322-E1	KM22	MB22
50	36	M30	240	77	15	–	100	22322-E1	KM22	MB22
50	36	M30	215	72	–	18	80	22224-E1	KM24	MB24
50	36	M30	215	72	–	18	80	22224-E1	KM24	MB24
50	36	M30	215	72	15	–	80	22224-E1	KM24	MB24
50	36	M30	215	72	15	–	80	22224-E1	KM24	MB24
60	42	M36	260	90	–	18	130	22324-E1	KM24	MB24
60	42	M36	260	90	–	18	130	22324-E1	KM24	MB24
60	42	M36	260	90	15	–	130	22324-E1	KM24	MB24
60	42	M36	260	90	15	–	130	22324-E1	KM24	MB24
50	36	M30	230	77	–	18	93	22226-E1	KM26	MB26
50	36	M30	230	77	–	18	93	22226-E1	KM26	MB26
50	36	M30	230	77	15	–	93	22226-E1	KM26	MB26
50	36	M30	230	77	15	–	93	22226-E1	KM26	MB26
60	42	M36	280	86	–	18	142	22326-E1	KM26	MB26
60	42	M36	280	86	–	18	142	22326-E1	KM26	MB26
60	42	M36	280	86	15	–	142	22326-E1	KM26	MB26
60	42	M36	280	86	15	–	142	22326-E1	KM26	MB26
50	36	M30	250	73	–	18	100	22228-E1	KM28	MB28
50	36	M30	250	73	–	18	100	22228-E1	KM28	MB28
50	36	M30	250	73	22	–	100	22228-E1	KM28	MB28
50	36	M30	250	73	22	–	100	22228-E1	KM28	MB28
60	42	M36	300	95	–	19,5	170	22328-E1	KM28	MB28
60	42	M36	300	95	–	19,5	170	22328-E1	KM28	MB28
60	42	M36	300	95	15	–	170	22328-E1	KM28	MB28
60	42	M36	300	95	15	–	170	22328-E1	KM28	MB28
60	42	M36	270	82	–	18	125	22230-E1	KM30	MB30
60	42	M36	270	82	–	18	125	22230-E1	KM30	MB30
60	42	M36	270	82	15	–	125	22230-E1	KM30	MB30
60	42	M36	270	82	15	–	125	22230-E1	KM30	MB30
60	42	M36	320	95	–	18	200	22330-E1	KM30	MB30
60	42	M36	320	95	–	18	200	22330-E1	KM30	MB30
60	42	M36	320	95	15	–	200	22330-E1	KM30	MB30
60	42	M36	320	95	15	–	200	22330-E1	KM30	MB30

9.9.3 LOE, for spherical roller bearings with a cylindrical bore

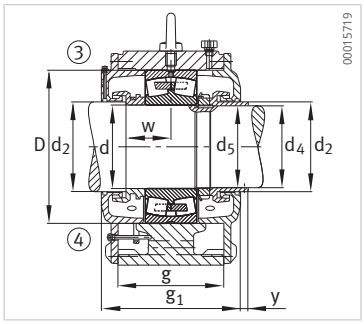




Housing	d	d_2	d_4	d_5	h	h_1	g	g_1	b	c	a	m	n
	mm	mm	mm	–	mm	mm	mm	mm	mm	mm	mm	mm	mm
LOE232-N-AF-L	**160**	170	–	M160×3	220	400	210	290	200	70	660	530	110
LOE232-N-AL-L	**160**	170	–	M160×3	220	400	210	290	200	70	660	530	110
LOE232-N-BF-L	**160**	170	155	M160×3	220	400	210	290	200	70	660	530	110
LOE232-N-BL-L	**160**	170	155	M160×3	220	400	210	290	200	70	660	530	110
LOE332-N-AF-L	**160**	166	–	M160×3	270	485	250	350	240	90	820	670	130
LOE332-N-AL-L	**160**	166	–	M160×3	270	485	250	350	240	90	820	670	130
LOE332-N-BF-L	**160**	166	155	M160×3	270	485	250	350	240	90	820	670	130
LOE332-N-BL-L	**160**	166	155	M160×3	270	485	250	350	240	90	820	670	130
LOE234-N-AF-L	**170**	176	–	M170×3	260	460	210	300	200	85	710	580	110
LOE234-N-AL-L	**170**	176	–	M170×3	260	460	210	300	200	85	710	580	110
LOE234-N-BF-L	**170**	176	165	M170×3	260	460	210	300	200	85	710	580	110
LOE234-N-BL-L	**170**	176	165	M170×3	260	460	210	300	200	85	710	580	110
LOE334-N-AF-L	**170**	180	–	M170×3	280	510	255	350	240	90	830	670	130
LOE334-N-AL-L	**170**	180	–	M170×3	280	510	255	350	240	90	830	670	130
LOE334-N-BF-L	**170**	180	165	M170×3	280	510	255	350	240	90	830	670	130
LOE334-N-BL-L	**170**	180	165	M170×3	280	510	255	350	240	90	830	670	130
LOE236-N-AF-L	**180**	190	–	M180×3	260	465	210	300	200	85	710	580	110
LOE236-N-AL-L	**180**	190	–	M180×3	260	465	210	300	200	85	710	580	110
LOE236-N-BF-L	**180**	190	175	M180×3	260	465	210	300	200	85	710	580	110
LOE236-N-BL-L	**180**	190	175	M180×3	260	465	210	300	200	85	710	580	110
LOE336-N-AF-L	**180**	190	–	M180×3	290	530	260	360	240	90	840	680	130
LOE336-N-AL-L	**180**	190	–	M180×3	290	530	260	360	240	90	840	680	130
LOE336-N-BF-L	**180**	190	175	M180×3	290	530	260	360	240	90	840	680	130
LOE336-N-BL-L	**180**	190	175	M180×3	290	530	260	360	240	90	840	680	130
LOE238-N-AF-L	**190**	196	–	M190×3	270	485	250	350	240	90	820	670	130
LOE238-N-AL-L	**190**	196	–	M190×3	270	485	250	350	240	90	820	670	130
LOE238-N-BF-L	**190**	196	185	M190×3	270	485	250	350	240	90	820	670	130
LOE238-N-BL-L	**190**	196	185	M190×3	270	485	250	350	240	90	820	670	130
LOE240-N-AF-L	**200**	210	–	M200×3	280	510	260	344	240	90	830	670	130
LOE240-N-AL-L	**200**	210	–	M200×3	280	510	260	344	240	90	830	670	130
LOE240-N-BF-L	**200**	210	195	M200×3	280	510	260	344	240	90	830	670	130
LOE240-N-BL-L	**200**	210	195	M200×3	280	510	260	344	240	90	830	670	130
LOE244-N-AF-L	**220**	230	–	TR220×4	310	565	280	380	240	105	880	720	130
LOE244-N-AL-L	**220**	230	–	TR220×4	310	565	280	380	240	105	880	720	130
LOE244-N-BF-L	**220**	230	212	TR220×4	310	565	280	380	240	105	880	720	130
LOE244-N-BL-L	**220**	230	212	TR220×4	310	565	280	380	240	105	880	720	130
LOE248-N-AF-L	**240**	260	–	TR240×4	340	615	300	400	280	120	980	820	165
LOE248-N-AL-L	**240**	260	–	TR240×4	340	615	300	400	280	120	980	820	165
LOE248-N-BF-L	**240**	260	235	TR240×4	340	615	300	400	280	120	980	820	165
LOE248-N-BL-L	**240**	260	235	TR240×4	340	615	300	400	280	120	980	820	165



Design A



Design B

v	u	s	D	w	y	g_3	m	Bearing	Shaft nut	Adapter sleeve
mm	mm	–	mm	mm	mm	mm	≈ kg	–	–	–
60	42	M36	290	80	–	20	136	22232-E1	KM32	MB32
60	42	M36	290	80	–	20	136	22232-E1	KM32	MB32
60	42	M36	290	80	22	–	136	22232-E1	KM32	MB32
60	42	M36	290	80	22	–	136	22232-E1	KM32	MB32
70	48	M42	340	100	–	20	240	22332-MB	KM32	MB32
70	48	M42	340	100	–	20	240	22332-MB	KM32	MB32
70	48	M42	340	100	15	–	240	22332-MB	KM32	MB32
70	48	M42	340	100	15	–	240	22332-MB	KM32	MB32
60	42	M36	310	90	–	18	160	22234-E1	KM34	MB34
60	42	M36	310	90	–	18	160	22234-E1	KM34	MB34
60	42	M36	310	90	15	–	160	22234-E1	KM34	MB34
60	42	M36	310	90	15	–	160	22234-E1	KM34	MB34
70	48	M42	360	105	–	18	270	22334-MB	KM34	MB34
70	48	M42	360	105	–	18	270	22334-MB	KM34	MB34
70	48	M42	360	105	15	–	270	22334-MB	KM34	MB34
70	48	M42	360	105	15	–	270	22334-MB	KM34	MB34
60	42	M36	320	90	–	20	200	22236-E1	KM36	MB36
60	42	M36	320	90	–	20	200	22236-E1	KM36	MB36
60	42	M36	320	90	22	–	200	22236-E1	KM36	MB36
60	42	M36	320	90	22	–	200	22236-E1	KM36	MB36
70	48	M42	380	108	–	20	330	22336-MB	KM36	MB36
70	48	M42	380	108	–	20	330	22336-MB	KM36	MB36
70	48	M42	380	108	15	–	330	22336-MB	KM36	MB36
70	48	M42	380	108	15	–	330	22336-MB	KM36	MB36
70	48	M42	340	95	–	20	230	22238-MB	KM38	MB38
70	48	M42	340	95	–	20	230	22238-MB	KM38	MB38
70	48	M42	340	95	15	–	230	22238-MB	KM38	MB38
70	48	M42	340	95	15	–	230	22238-MB	KM38	MB38
70	48	M42	360	100	–	20	250	22240-B-MB	KM40	MB40
70	48	M42	360	100	–	20	250	22240-B-MB	KM40	MB40
70	48	M42	360	100	15	–	250	22240-B-MB	KM40	MB40
70	48	M42	360	100	15	–	250	22240-B-MB	KM40	MB40
70	48	M42	400	108	–	20	310	22244-B-MB	HM44T	MB44
70	48	M42	400	108	–	20	310	22244-B-MB	HM44T	MB44
70	48	M42	400	108	15	–	310	22244-B-MB	HM44T	MB44
70	48	M42	400	108	15	–	310	22244-B-MB	HM44T	MB44
70	48	M42	440	120	–	20	385	22248-B-MB	HM48T	MB48
70	48	M42	440	120	–	20	385	22248-B-MB	HM48T	MB48
70	48	M42	440	120	22	–	385	22248-B-MB	HM48T	MB48
70	48	M42	440	120	22	–	385	22248-B-MB	HM48T	MB48

10 VRE3, VRE2 plummer block housing unit

medias | Product catalogue |
VRE3 plummer block housing |
https://www.schaeffler.de/std/2200

Plummer block housing units were originally developed for fan applications, for the flying bearing arrangement of high speed impellers.

The units are now used wherever a precise bearing that can be installed easily is required.

- Conveying equipment
- Test rigs
- Processing engineering machinery
- Belt drives
- Laboratory equipment
- Textile machinery
- Feed equipment

The VRE2 plummer bearing unit is only available on request.

10.1 Housing design

VRE3 plummer bearing units consist of a tubular VR3 plummer bearing housing with seals and a VRW3 shaft in roller bearings from series 3. The units are completely assembled and the bearings are greased. The units can therefore be installed straight into existing machinery without substantial preparation.

VRE2 plummer bearing units consist of a tubular VR2 plummer bearing housing with seals and a VRW2 shaft in roller bearings from series 2. The units are completely assembled and the bearings are greased.

Benefits

- Easy installation
- Low maintenance requirements
- Effective, low-friction seal against dust and moisture
- Bearing in a single housing, therefore no alignment work required
- Support of high tilting moments by two bearings with perfect spacing
- 6 designs mean they can be tailored to different stresses

Plummer block housing units

The complete VRE3 plummer block units are available with shaft diameters from 25 mm to 120 mm.

The complete VRE2 plummer block units are available with shaft diameters from 30 mm to 120 mm.

158 VRE3 and VRE2, plummer block units, unsplit, with bearings and shaft

0001AD1A

Plummer block housing

VR3 and VR2 plummer block housings contain covers, seals and fastening parts.

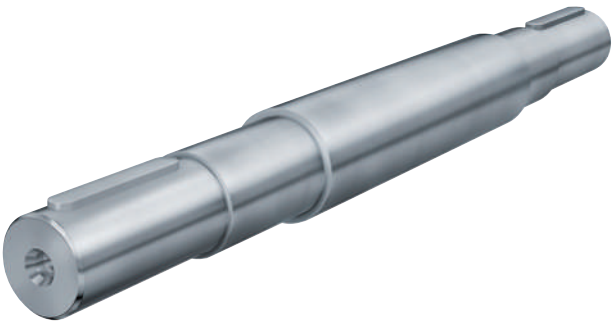
159 VR3 and VR2 plummer block housings, unsplit

0001ACB3

Shafts

VRW3 and VRW2 shafts contain all the required fastening parts.

160 VRW3 and VRW2 shafts



0001AD1B

10.1.1 Applications of the designs

Based on the specific bearing combination, the designs of VRE3 plummer block units have different applications.

64 Applications of the designs

Design	Application
A	• Predominantly radial load • High speed • Axial load from both directions, not alternating
B	• High radial load on one side • Axial load from one direction only
C	• High radial load on one side • High axial load from both directions
D	• Axial load from both directions
E	• High radial load on both sides • Axial load from both directions
F	• Predominantly radial load • High speed • Axial load from one direction, away from the spring

Maximum operating temperature

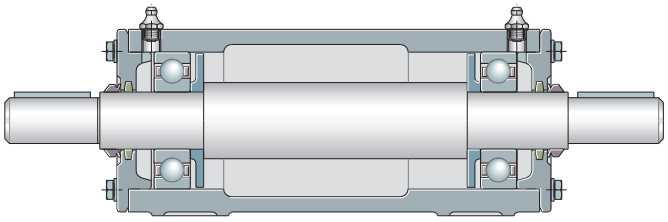
The maximum operating temperature for all designs is +100 °C.

10.1.2 Housing configurations

The 6 designs of VRE3 plummer block units represent different combinations of VR3housings, VRW3 shafts and one or more bearing types.

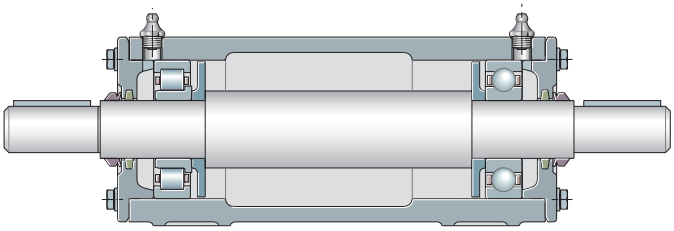
The 6 designs of VRE2 plummer block units represent different combinations of VR2housings, VRW2 shafts and one or more bearing types.

161 VRE3..-A plummer block unit, VR3..-A housing with 2 deep groove ball bearings, VRW3..-A shaft



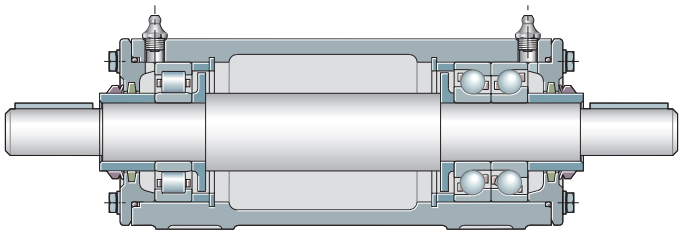
000102EA

162 VRE3..-B plummer block unit, VR3..-A housing with 1 deep groove ball bearing and 1 NJ cylindrical roller bearing in a floating design, VRW3..-A shaft



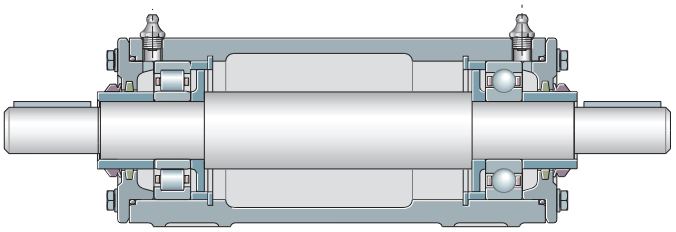
000102EF

163 VRE3..-C plummer block unit, VR3..-C housing in a locating/non-locating arrangement with 1 NU cylindrical roller bearing and 2 angular contact ball bearings in an O arrangement, VRW3..-C shaft



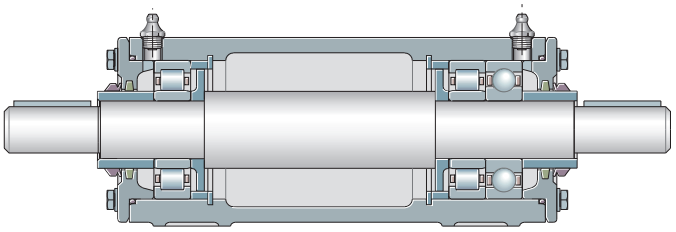
000102F4

164 VRE3..-D plummer block unit, VR3..-D housing in a locating/non-locating arrangement with 1 NU cylindrical roller bearing and 1 deep groove ball bearing, VRW3..-D shaft



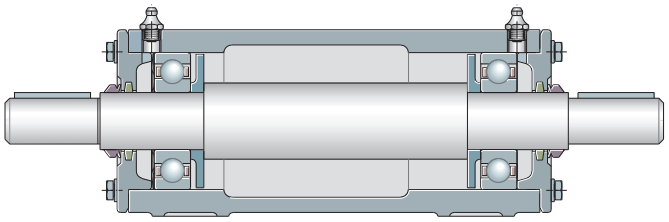
000102F9

165 VRE3..-E plummer block unit, VR3..-E housing in a locating/non-locating arrangement with 2 NU cylindrical roller bearing and 1 deep groove ball bearing, VRW3..-C shaft



000102FE

166 VRE3..-F plummer block unit, VR3..-F housing with 2 deep groove ball bearings in a floating arrangement, fixed axially using a spring between the outer ring and cover, VRW3..-F shaft5



000102E5

10.1.3 Designation

Special features are marked on the body body using a sprue.

- Position of cylindrical roller bearing on design B
- The position of the floating bearing on designs C, D and E
- The position of the ondular washer on design F

The position of the ondular washer is also marked on the VRW3..-F shaft.

10.2 Suitable bearings

Various combinations of deep groove ball bearings, cylindrical roller bearings and angular contact ball bearings are installed in plummer block housing units, depending on their design. Only bearings with a cylindrical bore are fitted.

65 Suitable bearings for VRE3 plummer block housing units

Plummer block housing unit		Bearing			
from	to	Quantity	Bearing type	Size	
				from	to
VRE305A	VRE324A	2	Deep groove ball bearing	6305	6324
VRE305B	VRE324B	1	Cylindrical roller bearing	NJ305-E	NJ324-E
		1	Deep groove ball bearing	6305	6324
VRE305C	VRE324C	1	Cylindrical roller bearing	NU305-E	NU324-E
		2	Angular contact ball bearing	7305-B..-UA	7324-B..-UA
VRE305D	VRE324D	1	Cylindrical roller bearing	NU305-E	NU324-E
		1	Deep groove ball bearing	6305	6324
VRE305E	VRE324E	2	Cylindrical roller bearing	NU305-E	NU324-E
		1	Deep groove ball bearing	6305	6324
VRE305F	VRE316F	2	Deep groove ball bearing	6305	6316

66 Suitable bearings for VRE2 plummer block housing units

Plummer block housing unit		Bearing			
from	to	Quantity	Bearing type	Size	
				from	to
VRE206A	VRE224A	2	Deep groove ball bearing	6206	6224
VRE206B	VRE224B	1	Cylindrical roller bearing	NJ206-E	NJ224-E
		1	Deep groove ball bearing	6206	6224
VRE206C	VRE224C	1	Cylindrical roller bearing	NU206-E	NU224-E
		2	Angular contact ball bearing	7206-B..-UA	7224-B..-UA
VRE206D	VRE224D	1	Cylindrical roller bearing	NU206-E	NU224-E
		1	Deep groove ball bearing	6206	6224
VRE206E	VRE224E	2	Cylindrical roller bearing	NU206-E	NU224-E
		1	Deep groove ball bearing	6206	6224
VRE205F	VRE216F	2	Deep groove ball bearing	6206	6224

10.3 Materials, corrosion protection

67 Standard material and universal coating

Assembly	Material	Availability	Universal coating, non-machined external surfaces
Housing	Gray cast iron GJL	✓	RAL 7016, anthracite grey
Cover	Steel	o	RAL 7016, anthracite grey
Control valves	Cast iron (gray cast iron or ductile cast iron)	✓	–
	Steel	✓	–
Shaft	Steel	✓	–

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

10.4 Lubrication

Plummer block units are fully assembled and greased on delivery.

10.4.1 Grease types

NLGI class 3lithium soap grease types whose mineral base oil has viscosity class ISO VG 100 are suitable.

10.4.2 Grease quantities

The grease quantities recommended for initial greasing depend on the size and design of the plummer block unit, the relubrication quantities and also the type of bearing.

68 Recommended grease quantities for initial greasing and relubrication

Plummer block housing unit		Grease quantity		
		Initial grease application	Relubrication	
			Non-locating bearing	Locating bearing
		g	g	g
VRE305	A, B, F	40	5	5
	C, E	45	5	10
	D	35	5	5
VRE306	A, B, F	60	5	5
	C, E	65	5	10
	D	55	5	5
VRE307	A, B, F	80	5	5
	C, E	90	5	10
	D	75	5	5
VRE308	A, B, F	110	10	10
	C, E	120	10	20
	D	100	10	10
VRE309	A, B, F	140	10	10
	C, E	155	10	20
	D	125	10	10
VRE310	A, B, F	190	10	10
	C, E	210	10	20
	D	170	10	10
VRE311	A, B, F	240	15	15
	C, E	265	15	25
	D	220	15	15
VRE312	A, B, F	300	15	15
	C, E	330	15	25
	D	270	15	15
VRE313	A, B, F	380	15	15
	C, E	420	15	25
	D	340	15	15
VRE314	A, B, F	480	20	20
	C, E	525	20	35
	D	425	20	20
VRE315	A, B, F	590	20	20
	C, E	650	20	35
	D	525	20	20

Plummer block housing unit		Grease quantity		
		Initial grease application	Relubrication	
			Non-locating bearing	Locating bearing
		g	g	g
VRE316	A, B, F	700	20	20
	C, E	770	20	40
	D	620	20	20
VRE317	A, B	830	25	25
	C, E	900	25	40
	D	750	25	25
VRE318	A, B	1000	25	25
	C, E	1050	25	50
	D	850	25	25
VRE319	A, B	1160	30	30
	C, E	1300	30	55
	D	1050	30	30
VRE320	A, B	1450	35	35
	C, E	1550	35	65
	D	1250	35	35
VRE322	A, B	1950	40	40
	C, E	1950	40	75
	D	1600	40	40
VRE324	A, B	2350	45	45
	C, E	2500	45	90
	D	2050	45	45

10.4.3 Relubrication

There is a M10×1 conical lubricating nipple at each bearing point to DIN 71412 for relubrication. Overgreasing the bearings is prevented by the washers mounted internally near the bearings. They move the grease exiting the bearings into the free housing volume between the bearing.

We recommend Arcanol MULTI3 for relubrication. If this grease is not available, a grease with the same soap base and NLGI class and the most similar base oil viscosity should be used.

The relubrication intervals depend on the speed and design of the plummer block unit. The values shown are for continuous operation.

69 Relubrication intervals for continuous operation

Plummer block housing unit		Speed value	Relubrication interval
		mm/min	h
VRE3	A, F	200000	3400
		300000	2350
		400000	1800
	B, C, D, E	200000	1050
		300000	600
		400000	350

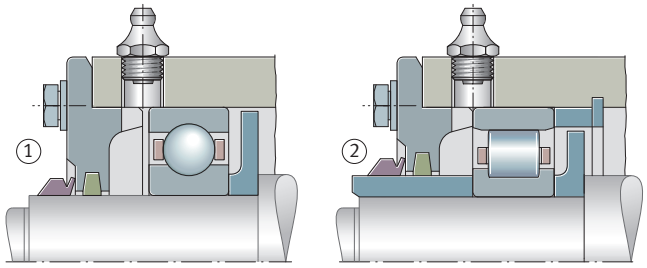
10.5 Sealing

Plummer block units are sealed on both sides by a combined seal comprising a felt seal and V ring. This achieves effective sealing against the ingress of contamination and the escape of lubricant.

The oil-soaked felt strips are inserted in the slots in the covers on the housing.

The external V rings are made from NBR and are suitable for temperatures up to +100 °C. The axial support of the V rings is required for a circumferential velocity of more than 8 m/s. As from a speed of 12 m/s, the seal lip lifts and the V ring then acts as a splash ring.

167 Felt seal with V ring for VRE3 plummer block units



1	Seals for designs A, B and F	2	Seals for designs C, D and E

10.6 Dimensions, tolerances

10.6.1 Tolerances for the bearing seat

The tolerance class of the bearing seat of plummer block units depends on the design of the associated housing

Design		Bearing seat tolerance class
VRE3 plummer block unit	VR3 housing	
A, B	A	G6
C	C	J6
D	D	J6
E	E	J6
F	F	G6

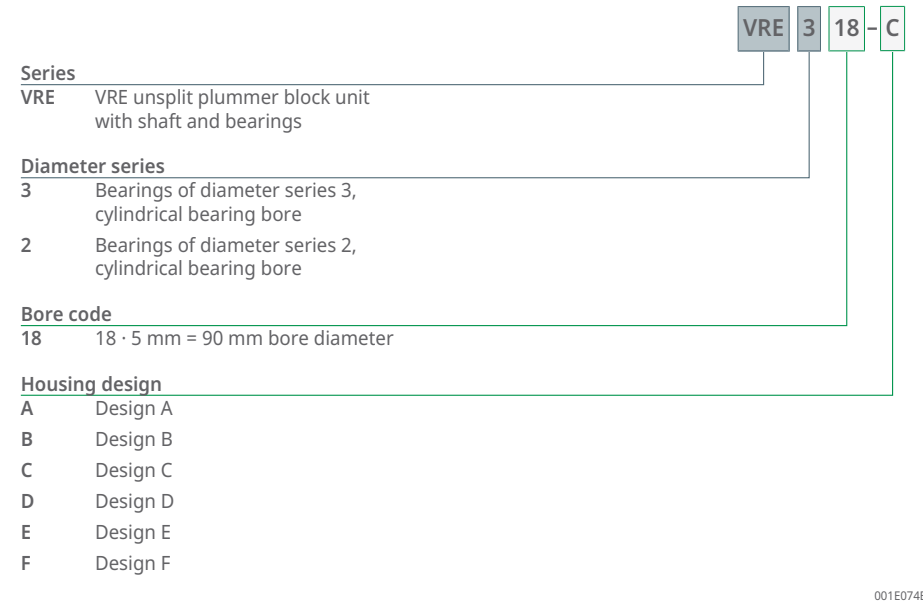
By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

10.7 Structure of the order designation

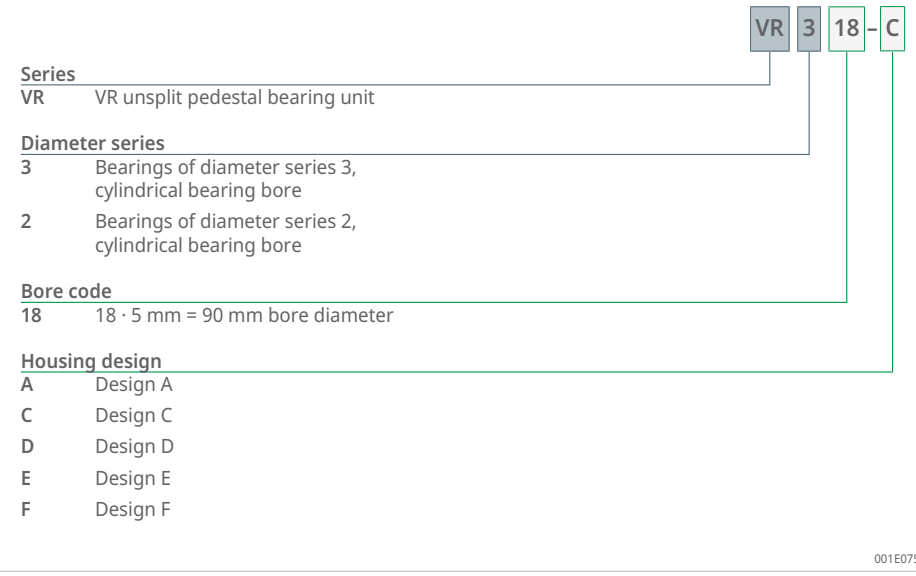
The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

Bearing housing

168 Designation structure of VRE3 and VRE2 plummer block housing units

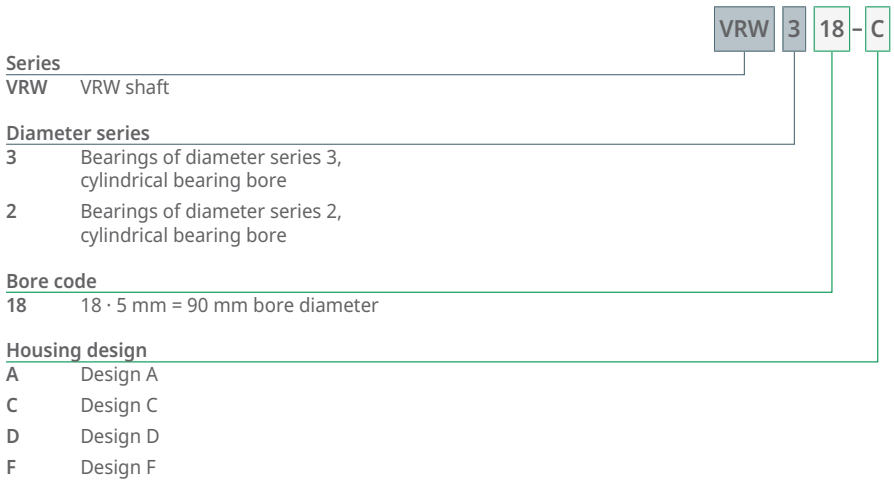


169 Designation structure of VR3 and VR2 plummer block housings



Shafts

170 Designation structure of VRW3 and VRW2 shafts



10.7.1 Specimen orders

VRE3 plummer block housing units should ideally be ordered as complete units. However, it is also possible to order only the VR3 housing or only the VRW3 shaft.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- VRE3 plummer bearing unit in design B
- VR314-A plummer block housing
- NJ314-E-TVP2 cylindrical roller bearing
- 6314-C3 deep groove ball bearing
- VRW314-A shaft

70 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing unit	VRE314-B

Example 2

Example of a housing configuration, characteristics:

- VR3 plummer block housing in design E
- Including covers, grease control valves, grease nipples, seals and fastening parts

The housing is suitable for a locating/non-locating bearing arrangement with 1 cylindrical roller bearing and 1 cylindrical roller bearing and 1 deep groove ball bearing on the opposite side.

71 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	VR322-E

Example 3

Example of a housing configuration, characteristics:

- VRW3 shaft in design A
- Including keys

The housing is suitable for a floating arrangement of 2 6314 deep groove ball bearings.

72 Order

Quantity	Description	Ordering designation, short designation
1	Shaft	VRW314-A

10.8 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28 | 2.7.

10.8.1 Eye bolts

There are 2 eye bolts to DIN 580:2018 on the housing. After mounting, the eye bolts can be replaced by the hexagon head bolts with seals supplied in the packaging.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

10.8.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30 | 2.7.2.3.

10.9 Product tables

10.9.1 Explanations of the product tables

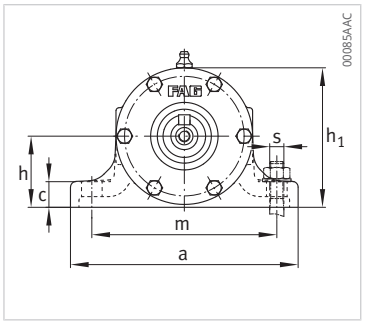
a	mm	Base length
b	mm	Base width
B	mm	Width
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_2	mm	Shaft shoulder diameter
d_4	mm	Diameter of shaft end
e	mm	Width of clamping elements
g	mm	Housing body width
g_2	mm	Overall width
h	mm	Distance to shaft axis
h_1	mm	Overall height
k	mm	Axial movement distance
m	mm	Fixing hole spacing
m	kg	Mass
n	mm	Distance between fastening holes
s	inch or mm	Screw size
w_1	mm	Total length of shaft
w_2	mm	Distance from shaft end
w_3	mm	Length of shaft journal
w_4	mm	Length of shaft end
w_5	mm	Length of shaft shoulder
w_7	mm	Length of shaft end

10.9.2 VRE3..-A

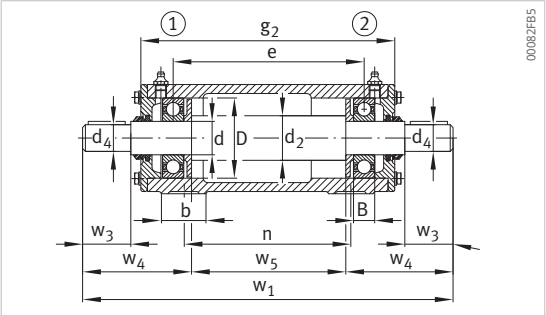
Design A

Unsplit

With bearings and shaft



Housing	d	d_2	d_4	w_1	w_2	w_3	w_4	w_5	w_7	h	h_1	g_2	b	c	a
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
VRE305-A	**25**	35	19	298	–	40	90,5	117	–	45	87	198	35	16	155
VRE306-A	**30**	40	24	345	–	50	102,5	140	–	50	98	225	40	18	160
VRE307-A	**35**	45	28	395	–	60	117,5	160	–	60	113	255	45	18	190
VRE308-A	**40**	50	32	501	–	80	143,5	214	–	60	118	317	52	20	190
VRE309-A	**45**	55	38	527	–	80	145,5	236	–	70	135	343	52	22	210
VRE310-A	**50**	60	42	625	–	110	179,5	266	–	70	138	381	60	25	210
VRE311-A	**55**	65	48	651	–	110	181,5	288	–	80	158	407	60	25	260
VRE312-A	**60**	70	48	701	–	110	183,5	334	–	80	162	457	70	25	260
VRE313-A	**65**	75	55	724	–	110	187,5	349	–	95	183	480	70	25	290
VRE314-A	**70**	80	60	811	–	140	223	365	–	95	188	500	70	25	290
VRE315-A	**75**	90	65	841	–	140	226	389	–	100	198	530	80	30	320
VRE316-A	**80**	95	70	861	–	140	228	405	–	112	217	550	80	30	320
VRE317-A	**85**	100	75	881	–	140	230	421	–	112	222	570	80	30	350
VRE318-A	**90**	105	80	971	–	170	263	445	–	112	227	600	85	30	350
VRE319-A	**95**	110	85	1004	–	170	267,5	469	–	125	248	633	90	35	400
VRE320-A	**100**	120	90	1036	–	170	268	500	–	130	260	665	95	40	400
VRE322-A	**110**	130	100	1133	–	210	313	507	–	150	295	678	95	40	450
VRE324-A	**120**	140	110	1160	–	210	318	524	–	160	320	705	100	40	500



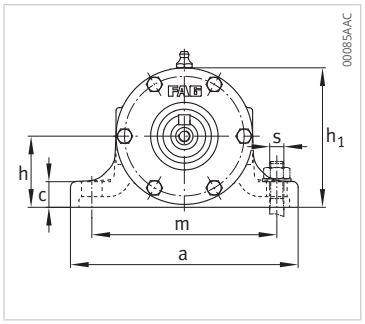
m	n	s	D	B	e	Shaft	Housing	Bearing position		m
mm	mm	-	mm	mm	mm	-	-	(1)	(2)	kg
120	135	M12	62	17	148	VRW305-A	VR305-A	6305-C3	6305-C3	7
130	150	M12	72	19	173	VRW306-A	VR306-A	6306-C3	6306-C3	9
150	175	M12	80	21	197	VRW307-A	VR307-A	6307-C3	6307-C3	13
150	225	M12	90	23	257	VRW308-A	VR308-A	6308-C3	6308-C3	18
170	250	M12	100	25	281	VRW309-A	VR309-A	6309-C3	6309-C3	24,3
170	275	M12	110	27	313	VRW310-A	VR310-A	6310-C3	6310-C3	30,3
210	300	M16	120	29	337	VRW311-A	VR311-A	6311-C3	6311-C3	38,6
210	340	M16	130	31	385	VRW312-A	VR312-A	6312-C3	6312-C3	51,2
230	360	M16	140	33	404	VRW313-A	VR313-A	6313-C3	6313-C3	58
230	380	M16	150	35	422	VRW314-A	VR314-A	6314-C3	6314-C3	66,9
260	400	M16	160	37	450	VRW315-A	VR315-A	6315-C3	6315-C3	84,8
260	420	M16	170	39	468	VRW316-A	VR316-A	6316-C3	6316-C3	90,8
290	440	M16	180	41	486	VRW317-A	VR317-A	6317-C3	6317-C3	114
290	460	M16	190	43	510	VRW318-A	VR318-A	6318-C3	6318-C3	128
320	480	M20	200	45	540	VRW319-A	VR319-A	6319-C3	6319-C3	156
320	500	M20	215	47	570	VRW320-A	VR320-A	6320-C3	6320-C3	177
380	520	M24	240	50	580	VRW322-A	VR322-A	6322-C3	6322-C3	226
410	540	M30	260	55	602	VRW324-A	VR324-A	6324-C3	6324-C3	276

10.9.3 VRE3..-B

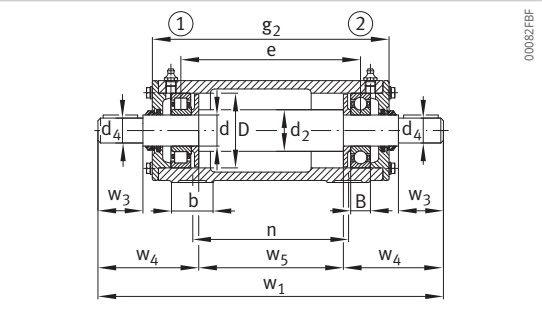
Design B

Unsplit

With bearings and shaft



Housing	d	d_2	d_4	w_1	w_2	w_3	w_4	w_5	w_7	h	h_1	g_2	b	c	a
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
VRE305-B	**25**	35	19	298	–	40	90,5	117	–	45	87	198	35	16	155
VRE306-B	**30**	40	24	345	–	50	102,5	140	–	50	98	225	40	18	160
VRE307-B	**35**	45	28	395	–	60	117,5	160	–	60	113	255	45	18	190
VRE308-B	**40**	50	32	501	–	80	143,5	214	–	60	118	317	52	20	190
VRE309-B	**45**	55	38	527	–	80	145,5	236	–	70	135	343	52	22	210
VRE310-B	**50**	60	42	625	–	110	179,5	266	–	70	138	381	60	25	210
VRE311-B	**55**	65	48	651	–	110	181,5	288	–	80	158	407	60	25	260
VRE312-B	**60**	70	48	701	–	110	183,5	334	–	80	162	457	70	25	260
VRE313-B	**65**	75	55	724	–	110	187,5	349	–	95	183	480	70	25	290
VRE314-B	**70**	80	60	811	–	140	223	365	–	95	188	500	70	25	290
VRE315-B	**75**	90	65	841	–	140	226	389	–	100	198	530	80	30	320
VRE316-B	**80**	95	70	861	–	140	228	405	–	112	217	550	80	30	320
VRE317-B	**85**	100	75	881	–	140	230	421	–	112	222	570	80	30	350
VRE318-B	**90**	105	80	971	–	170	263	445	–	112	227	600	85	30	350
VRE319-B	**95**	110	85	1004	–	170	267,5	469	–	125	248	633	90	35	400
VRE320-B	**100**	120	90	1036	–	170	268	500	–	130	260	665	95	40	400
VRE322-B	**110**	130	100	1133	–	210	313	507	–	150	295	678	95	40	450
VRE324-B	**120**	140	110	1160	–	210	318	524	–	160	320	705	100	40	500



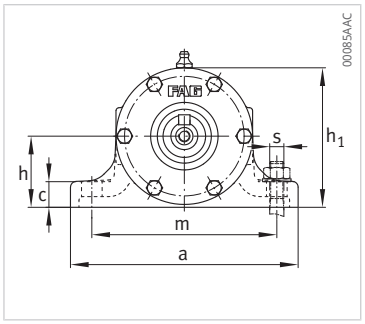
m	n	s	D	B	e	Shaft	Housing	Bearing position		m
mm	mm	-	mm	mm	mm	-	-	(1)	(2)	kg
120	135	M12	62	17	149	VRW305-A	VR305-A	NJ305-E-TVP2	6305-C3	7
130	150	M12	72	19	174	VRW306-A	VR306-A	NJ306-E-TVP2	6306-C3	9
150	175	M12	80	21	198	VRW307-A	VR307-A	NJ307-E-TVP2	6307-C3	13
150	225	M12	90	23	258	VRW308-A	VR308-A	NJ308-E-TVP2	6308-C3	18
170	250	M12	100	25	282	VRW309-A	VR309-A	NJ309-E-TVP2	6309-C3	24,4
170	275	M12	110	27	314	VRW310-A	VR310-A	NJ310-E-TVP2	6310-C3	30,3
210	300	M16	120	29	338	VRW311-A	VR311-A	NJ311-E-TVP2	6311-C3	38,7
210	340	M16	130	31	386	VRW312-A	VR312-A	NJ312-E-TVP2	6312-C3	51,4
230	360	M16	140	33	405	VRW313-A	VR313-A	NJ313-E-TVP2	6313-C3	58,2
230	380	M16	150	35	423	VRW314-A	VR314-A	NJ314-E-TVP2	6314-C3	67,1
260	400	M16	160	37	451	VRW315-A	VR315-A	NJ315-E-TVP2	6315-C3	85
260	420	M16	170	39	469	VRW316-A	VR316-A	NJ316-E-TVP2	6316-C3	91
290	440	M16	180	41	487	VRW317-A	VR317-A	NJ317-E-TVP2	6317-C3	115
290	460	M16	190	43	511	VRW318-A	VR318-A	NJ318-E-TVP2	6318-C3	128
320	480	M20	200	45	541	VRW319-A	VR319-A	NJ319-E-TVP2	6319-C3	157
320	500	M20	215	47	571	VRW320-A	VR320-A	NJ320-E-TVP2	6320-C3	177
380	520	M24	240	50	581	VRW322-A	VR322-A	NJ322-E-TVP2	6322-C3	226
410	540	M30	260	55	603	VRW324-A	VR324-A	NJ324-E-TVP2	6324-C3	277

10.9.4 VRE3..-C

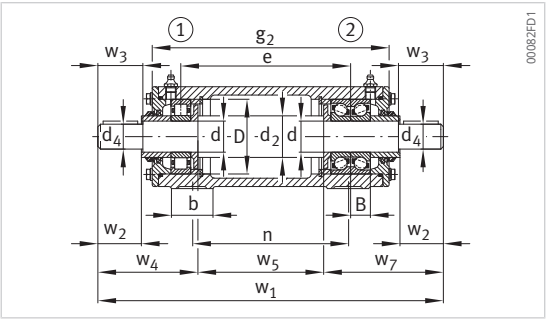
Design C

Unsplit

With bearings and shaft



Housing	d	d_2	d_4	w_1	w_2	w_3	w_4	w_5	w_7	h	h_1	g_2	b	c	a
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
VRE305-C	**25**	35	19	298	40	42	90	101	107	45	87	198	35	16	155
VRE306-C	**30**	40	24	349	50	52	104	122	123	50	98	225	40	18	160
VRE307-C	**35**	45	28	399	60	62	119	140	140	60	113	255	45	18	190
VRE308-C	**40**	50	32	501	80	82	143	192	166	60	118	317	52	20	190
VRE309-C	**45**	55	38	527	80	82	145	212	170	70	135	343	52	22	210
VRE310-C	**50**	60	42	625	110	112	179	240	206	70	138	381	60	25	210
VRE311-C	**55**	65	48	651	110	112	181	260	210	80	158	407	60	25	260
VRE312-C	**60**	70	48	706	110	112	185,5	304	216,5	80	162	457	70	25	260
VRE313-C	**65**	75	55	729	110	112	189,5	317	222,5	95	183	480	70	25	290
VRE314-C	**70**	80	60	811	140	143	222,5	331	257,5	95	188	500	70	25	290
VRE315-C	**75**	90	65	841	140	143	225,5	353	262,5	100	198	530	80	30	320
VRE316-C	**80**	95	70	861	140	143	227,5	367	266,5	112	217	550	80	30	320
VRE317-C	**85**	100	75	881	140	143	229,5	381	270,5	112	222	570	80	30	350
VRE318-C	**90**	105	80	975	170	173	264,5	403	307,5	112	227	600	85	30	350
VRE319-C	**95**	110	85	1008	170	173	269	425	314	125	248	633	90	35	400
VRE320-C	**100**	120	90	1040	170	173	269,5	454	316,5	130	260	665	95	40	400
VRE322-C	**110**	130	100	1133	210	213	312,5	458	362,5	150	295	678	95	40	450
VRE324-C	**120**	140	110	1160	210	213	317,5	470	372,5	160	320	705	100	40	500



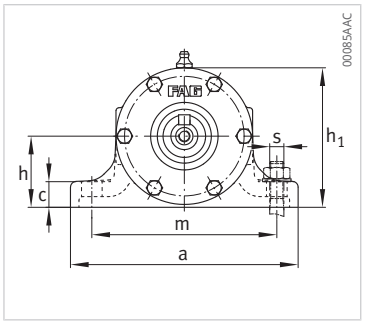
m	n	s	D	B	e	Shaft	Housing	Bearing position		m
mm	mm	-	mm	mm	mm	-	-	(1)	(2)	kg
120	135	M12	62	17	140,5	VRW305-C	VR305-C	NU305-E-TVP2-C3	2×7305-B-TVP-UA	7,4
130	150	M12	72	19	164,5	VRW306-C	VR306-C	NU306-E-TVP2-C3	2×7306-B-TVP-UA	9,4
150	175	M12	80	21	187,5	VRW307-C	VR307-C	NU307-E-TVP2-C3	2×7307-B-TVP-UA	13,6
150	225	M12	90	23	246,5	VRW308-C	VR308-C	NU308-E-TVP2-C3	2×7308-B-TVP-UA	19,1
170	250	M12	100	25	269,5	VRW309-C	VR309-C	NU309-E-TVP2-C3	2×7309-B-TVP-UA	25,3
170	275	M12	110	27	300,5	VRW310-C	VR310-C	NU310-E-TVP2-C3	2×7310-B-TVP-UA	31,7
210	300	M16	120	29	323,5	VRW311-C	VR311-C	NU311-E-TVP2-C3	2×7311-B-TVP-UA	40,2
210	340	M16	130	31	370,5	VRW312-C	VR312-C	NU312-E-TVP2-C3	2×7312-B-TVP-UA	53,8
230	360	M16	140	33	388,5	VRW313-C	VR313-C	NU313-E-TVP2-C3	2×7313-B-TVP-UA	60,8
230	380	M16	150	35	405,5	VRW314-C	VR314-C	NU314-E-TVP2-C3	2×7314-B-TVP-UA	70,4
260	400	M16	160	37	432,5	VRW315-C	VR315-C	NU315-E-TVP2-C3	2×7315-B-TVP-UA	89,3
260	420	M16	170	39	449,5	VRW316-C	VR316-C	NU316-E-TVP2-C3	2×7316-B-TVP-UA	96,2
290	440	M16	180	41	466,5	VRW317-C	VR317-C	NU317-E-TVP2-C3	2×7317-B-TVP-UA	120
290	460	M16	190	43	489,5	VRW318-C	VR318-C	NU318-E-TVP2-C3	2×7318-B-TVP-UA	134
320	480	M20	200	45	518,5	VRW319-C	VR319-C	NU319-E-TVP2-C3	2×7319-B-TVP-UA	164
320	500	M20	215	47	547,5	VRW320-C	VR320-C	NU320-E-TVP2-C3	2×7320-B-TVP-UA	186
380	520	M24	240	50	556	VRW322-C	VR322-C	NU322-E-TVP2-C3	2×7322-B-TVP-UA	238
410	540	M30	260	55	575,5	VRW324-C	VR324-C	NU324-E-TVP2-C3	2×7324-B-TVP-UA	294

10.9.5 VRE3..-D

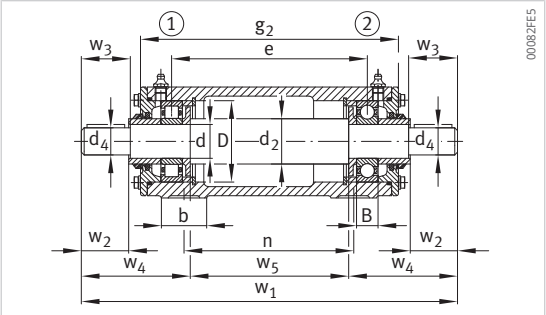
Design D

Unsplit

With bearings and shaft



Housing	d	d_2	d_4	w_1	w_2	w_3	w_4	w_5	w_7	h	h_1	g_2	b	c	a
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
VRE305-D	**25**	35	19	298	40	42	90	118	–	45	87	198	35	16	155
VRE306-D	**30**	40	24	349	50	52	104	141	–	50	98	225	40	18	160
VRE307-D	**35**	45	28	399	60	62	119	161	–	60	113	255	45	18	190
VRE308-D	**40**	50	32	501	80	82	143	215	–	60	118	317	52	20	190
VRE309-D	**45**	55	38	527	80	82	145	237	–	70	135	343	52	22	210
VRE310-D	**50**	60	42	625	110	112	179	267	–	70	138	381	60	25	210
VRE311-D	**55**	65	48	651	110	112	181	289	–	80	158	407	60	25	260
VRE312-D	**60**	70	48	706	110	112	185,5	335	–	80	162	457	70	25	260
VRE313-D	**65**	75	55	729	110	112	189,5	350	–	95	183	480	70	25	290
VRE314-D	**70**	80	60	811	140	143	222,5	366	–	95	188	500	70	25	290
VRE315-D	**75**	90	65	841	140	143	225,5	390	–	100	198	530	80	30	320
VRE316-D	**80**	95	70	861	140	143	227,5	406	–	112	217	550	80	30	320
VRE317-D	**85**	100	75	881	140	143	229,5	422	–	112	222	570	80	30	350
VRE318-D	**90**	105	80	975	170	173	264,5	446	–	112	227	600	85	30	350
VRE319-D	**95**	110	85	1008	170	173	269	470	–	125	248	633	90	35	400
VRE320-D	**100**	120	90	1040	170	173	269,5	501	–	130	260	665	95	40	400
VRE322-D	**110**	130	100	1133	210	213	312,5	508	–	150	295	678	95	40	450
VRE324-D	**120**	140	110	1160	210	213	317,5	525	–	160	320	705	100	40	500



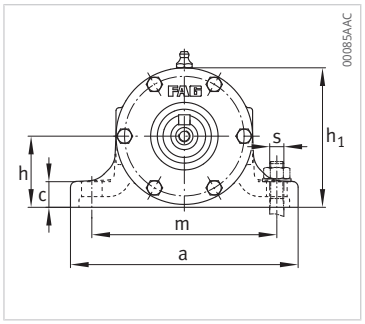
m	n	s	D	B	e	Shaft	Housing	Bearing position		m
mm	mm	-	mm	mm	mm	-	-	(1)	(2)	kg
120	135	M12	62	17	149	VRW305-D	VR305-D	NU305-E-TVP2-C3	6305-C3	7,2
130	150	M12	72	19	174	VRW306-D	VR306-D	NU306-E-TVP2-C3	6306-C3	9,2
150	175	M12	80	21	198	VRW307-D	VR307-D	NU307-E-TVP2-C3	6307-C3	13,3
150	225	M12	90	23	258	VRW308-D	VR308-D	NU308-E-TVP2-C3	6308-C3	18,8
170	250	M12	100	25	282	VRW309-D	VR309-D	NU309-E-TVP2-C3	6309-C3	24,8
170	275	M12	110	27	314	VRW310-D	VR310-D	NU310E-TVP2-C3	6310-C3	30,9
210	300	M16	120	29	338	VRW311-D	VR311-D	NU311-E-TVP2-C3	6311-C3	39,2
210	340	M16	130	31	386	VRW312-D	VR312-D	NU312-E-TVP2-C3	6312-C3	52,4
230	360	M16	140	33	405	VRW313-D	VR313-D	NU313-E-TVP2-C3	6313-C3	59,3
230	380	M16	150	35	423	VRW314-D	VR314-D	NU314-E-TVP2-C3	6314-C3	68
260	400	M16	160	37	451	VRW315-D	VR315-D	NU315-E-TVP2-C3	6315-C3	86,4
260	420	M16	170	39	469	VRW316-D	VR316-D	NU316-E-TVP2-C3	6316-C3	91,9
290	440	M16	180	41	487	VRW317-D	VR317-D	NU317-E-TVP2-C3	6317-C3	115
290	460	M16	190	43	511	VRW318-D	VR318-D	NU318-E-TVP2-C3	6318-C3	129
320	480	M20	200	45	541	VRW319-D	VR319-D	NU319-E-TVP2-C3	6319-C3	158
320	500	M20	215	47	571	VRW320-D	VR320-D	NU320-E-TVP2-C3	6320-C3	179
380	520	M24	240	50	581	VRW322-D	VR322-D	NU322-E-TVP2-C3	6322-C3	228
410	540	M30	260	55	603	VRW324-D	VR324-D	NU324-E-TVP2-C3	6324-C3	278

10.9.6 VRE3..-E

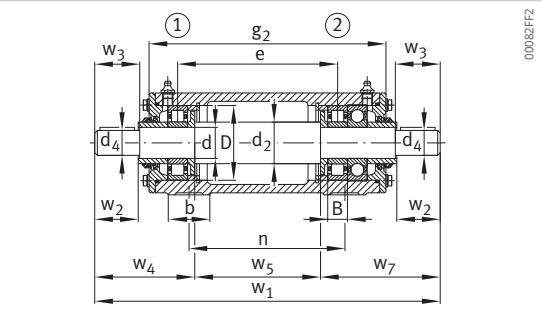
Design E

Unsplit

With bearings and shaft



Housing	d	d_2	d_4	w_1	w_2	w_3	w_4	w_5	w_7	h	h_1	g_2	b	c	a
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
VRE305-E	**25**	35	19	298	40	42	90	101	107	45	87	198	35	16	155
VRE306-E	**30**	40	24	349	50	52	104	122	123	50	98	225	40	18	160
VRE307-E	**35**	45	28	399	60	62	119	140	140	60	113	255	45	18	190
VRE308-E	**40**	50	32	501	80	82	143	192	166	60	118	317	52	20	190
VRE309-E	**45**	55	38	527	80	82	145	212	170	70	135	343	52	22	210
VRE310-E	**50**	60	42	625	110	112	179	240	206	70	138	381	60	25	210
VRE311-E	**55**	65	48	651	110	112	181	260	210	80	158	407	60	25	260
VRE312-E	**60**	70	48	706	110	112	185,5	304	216,5	80	162	457	70	25	260
VRE313-E	**65**	75	55	729	110	112	189,5	317	222,5	95	183	480	70	25	290
VRE314-E	**70**	80	60	811	140	143	222,5	331	257,5	95	188	500	70	25	290
VRE316-E	**80**	95	70	861	140	143	227,5	367	266,5	112	217	550	80	30	320
VRE317-E	**85**	100	75	881	140	143	229,5	381	270,5	112	222	570	80	30	350
VRE318-E	**90**	105	80	975	170	173	264,5	403	307,5	112	227	600	85	30	350
VRE319-E	**95**	110	85	1008	170	173	269	425	314	125	248	633	90	35	400
VRE320-E	**100**	120	90	1040	170	173	269,5	454	316,5	130	260	665	95	40	400
VRE322-E	**110**	130	100	1133	210	213	312,5	458	362,5	150	295	678	95	40	450
VRE324-E	**120**	140	110	1160	210	213	317,5	470	372,5	160	320	705	100	40	500



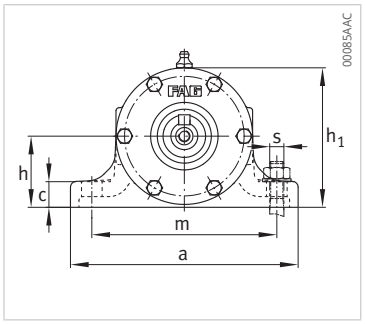
m	n	s	D	B	e	Shaft	Housing	Bearing position		m
mm	mm	-	mm	mm	mm	-	-	(1)	(2)	kg
120	135	M12	62	17	132	VRW305-C	VR305-E	NU305-E-TVP2-C3	NU305-E-TVP2-C3 + 6305-C3	7,5
130	150	M12	72	19	155	VRW306-C	VR306-E	NU306-E-TVP2-C3	NU306-E-TVP2-C3 + 6306-C3	9,4
150	175	M12	80	21	177	VRW307-C	VR307-E	NU307-E-TVP2-C3	NU307-E-TVP2-C3 + 6307-C3	13,6
150	225	M12	90	23	235	VRW308-C	VR308-E	NU308-E-TVP2-C3	NU308-E-TVP2-C3 + 6308-C3	19,2
170	250	M12	100	25	257	VRW309-C	VR309-E	NU309-E-TVP2-C3	NU309-E-TVP2-C3 + 6309-C3	25,3
170	275	M12	110	27	287	VRW310-C	VR310-E	NU310-E-TVP2-C3	NU310-E-TVP2-C3 + 6310-C3	31,8
210	300	M16	120	29	309	VRW311-C	VR311-E	NU311-E-TVP2-C3	NU311-E-TVP2-C3 + 6311-C3	40,2
210	340	M16	130	31	355	VRW312-C	VR312-E	NU312-E-TVP2-C3	NU312-E-TVP2-C3 + 6312-C3	53,7
230	360	M16	140	33	372	VRW313-C	VR313-E	NU313-E-TVP2-C3	NU313-E-TVP2-C3 + 6313-C3	60,8
230	380	M16	150	35	388	VRW314-C	VR314-E	NU314-E-TVP2-C3	NU314-E-TVP2-C3 + 6314-C3	70,6
260	420	M16	170	39	430	VRW316-C	VR316-E	NU316-E-TVP2-C3	NU316-E-TVP2-C3 + 6316-C3	95,3
290	440	M16	180	41	446	VRW317-C	VR317-E	NU317-E-TVP2-C3	NU317-E-TVP2-C3 + 6317-C3	120
290	460	M16	190	43	468	VRW318-C	VR318-E	NU318-E-TVP2-C3	NU318-E-TVP2-C3 + 6318-C3	134
320	480	M20	200	45	496	VRW319-C	VR319-E	NU319-E-TVP2-C3	NU319-E-TVP2-C3 + 6319-C3	164
320	500	M20	215	47	524	VRW320-C	VR320-E	NU320-E-TVP2-C3	NU320-E-TVP2-C3 + 6320-C3	186
380	520	M24	240	50	531	VRW322-C	VR322-E	NU322-E-TVP2-C3	NU322-E-TVP2-C3 + 6322-C3	238
410	540	M30	260	55	548	VRW324-C	VR324-E	NU324-E-TVP2-C3	NU324-E-TVP2-C3 + 6324-C3	291

10.9.7 VRE3..-F

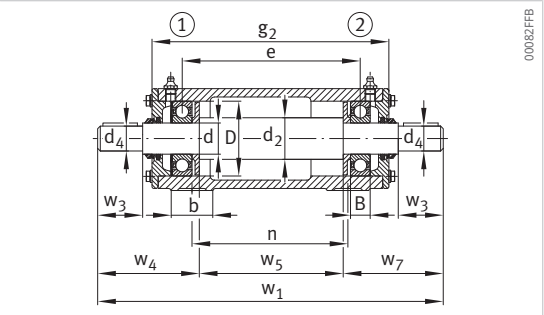
Design F

Unsplit

With bearings and shaft



Housing	d	d_2	d_4	w_1	w_2	w_3	w_4	w_5	w_7	h	h_1	g_2	b	c	a
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
VRE305-F	**25**	35	19	298	–	40	91,25	116,25	90,5	45	87	198	35	16	155
VRE306-F	**30**	40	24	345	–	50	103,25	139,25	102,5	50	98	225	40	18	160
VRE307-F	**35**	45	28	395	–	60	118,5	159	117,5	60	113	255	45	18	190
VRE308-F	**40**	50	32	501	–	80	144,5	213	143,5	60	118	317	52	20	190
VRE309-F	**45**	55	38	527	–	80	146,5	235	145,5	70	135	343	52	22	210
VRE310-F	**50**	60	42	625	–	110	180,5	265	179,5	70	138	381	60	25	210
VRE311-F	**55**	65	48	651	–	110	182,5	287	181,5	80	158	407	60	25	260
VRE312-F	**60**	70	48	701	–	110	184,5	333	183,5	80	162	457	70	25	260
VRE313-F	**65**	75	55	724	–	110	189	347,5	187,5	95	183	480	70	25	290
VRE314-F	**70**	80	60	811	–	140	224,5	363,5	223	95	188	500	70	25	290
VRE315-F	**75**	90	65	841	–	140	227,5	387,5	226	100	198	530	80	30	320
VRE316-F	**80**	95	70	861	–	140	229,5	403,5	228	112	217	550	80	30	320



m	n	s	D	B	e	Shaft	Housing	Bearing position		m
mm	mm	-	mm	mm	mm	-	-	(1)	(2)	kg
120	135	M12	62	17	147,25	VRW305-F	VR305-F	6305-C3	6305-C3	7
130	150	M12	72	19	172,25	VRW306-F	VR306-F	6306-C3	6306-C3	9
150	175	M12	80	21	196	VRW307-F	VR307-F	6307-C3	6307-C3	13
150	225	M12	90	23	256	VRW308-F	VR308-F	6308-C3	6308-C3	18
170	250	M12	100	25	280	VRW309-F	VR309-F	6309-C3	6309-C3	24,2
170	275	M12	110	27	312	VRW310-F	VR310-F	6310-C3	6310-C3	30,3
210	300	M16	120	29	336	VRW311-F	VR311-F	6311-C3	6311-C3	38,6
210	340	M16	130	31	384	VRW312-F	VR312-F	6312-C3	6312-C3	51,1
230	360	M16	140	33	402,5	VRW313-F	VR313-F	6313-C3	6313-C3	58
230	380	M16	150	35	420,5	VRW314-F	VR314-F	6314-C3	6314-C3	66,8
260	400	M16	160	37	448,5	VRW315-F	VR315-F	6315-C3	6315-C3	84,8
260	420	M16	170	39	466,5	VRW316-F	VR316-F	6316-C3	6316-C3	90,8

11 BND unsplit plummer block unit

medias | Product catalogue |
BND unsplit plummer block unit |
https://www.schaeffler.de/std/2202

11.1 Housing design

BND unsplit plummer block housings are suitable for very high loads. The housings, which were originally developed for conveyor belts, can also be used in materials processing, for example in hard crushers, sugar cane mill drives and rotor shafts in wind turbines.

Unsplit plummer bearings, closed on one side

171 BND..-AF and BND..-AL unsplit plummer block housing , closed on one side, with cover

0001ABA2

Unsplit pedestal bearings, for a continuous shaft

172 BND..-BF and BND..-BL unsplit plummer block housing for a continuous shafts

0001ABA4

11.1.1 Locating and non-locating bearings

The housings are supplied in the form of a locating or non-locating bearing design.

For the locating bearing, the bearing is clamped between the housing covers. For the non-locating bearing, the bearing can align itself axially, since the covers have shorter centring collars.

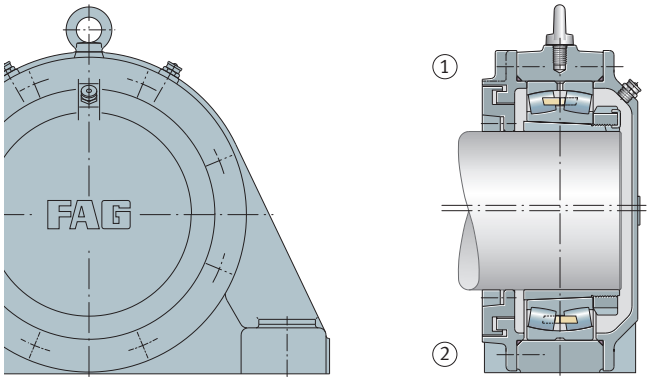
11.1.2 Housing configurations

Various housing configurations are possible for BND plummer block housings, which are clearly described by the designation.

The following features can be varied in the housing configuration:

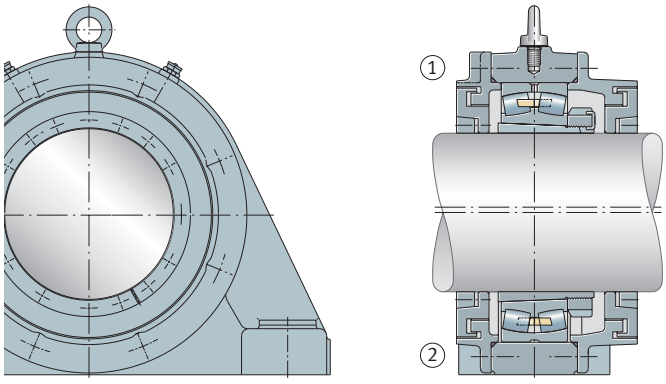
- Location method of the bearing:
  - Bearings with a tapered bore by means of an adapter sleeve on a shaft with a constant diameter
  - Bearings with a tapered bore by means of an adapter sleeve on a stepped shaft
  - Bearings with a cylindrical bore direct on a stepped shaft
- Housing seal using a labyrinth seal or Taconite seal
- Continuous shaft or a housing closed on one side
- Design of the bearing in the form of a locating or non-locating bearing.

173 BND plummer block housing for bearing with a tapered bore and adapter sleeve (labyrinth seal), design A



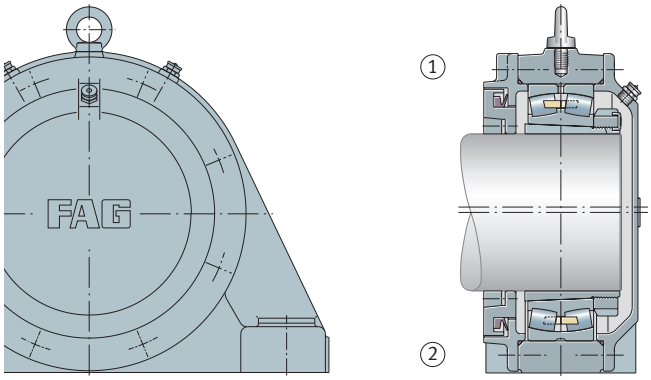
1	BND..-H-W-Y-AF locating bearing	2	BND..-H-W-Y-AL non-locating bearing

174 BND plummer block housing for bearing with a tapered bore and adapter sleeve (labyrinth seal), design B



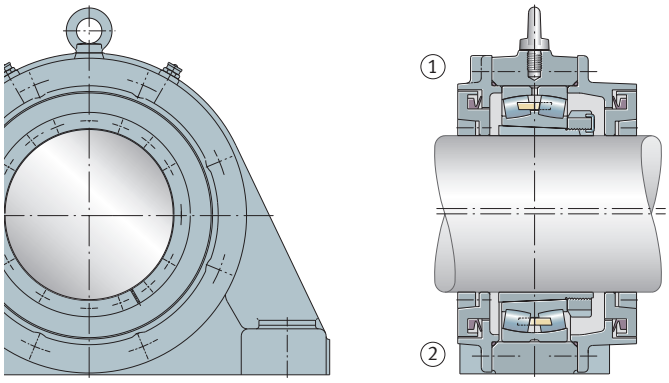
1	BND..-H-W-Y-BF locating bearing	2	BND..-H-W-Y-BL non-locating bearing

175 BND plummer block housing for bearing with a tapered bore and adapter sleeve (Taconite seal), design A



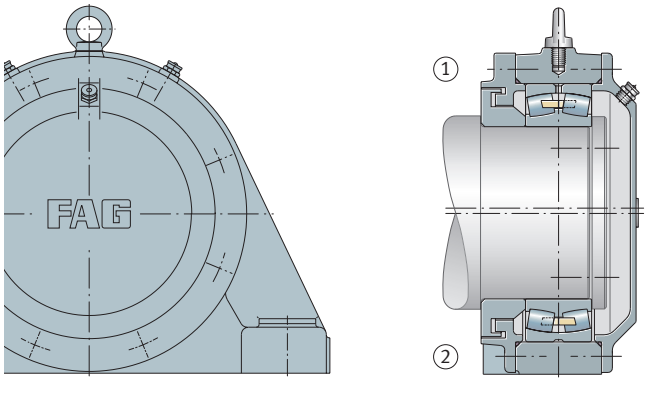
1	BND..-H-W-T-AF locating bearing	2	BND..-H-W-T-AL non-locating bearing

176 BND plummer block housing for bearings with a tapered bore and adapter sleeve (Taconite seal), design B



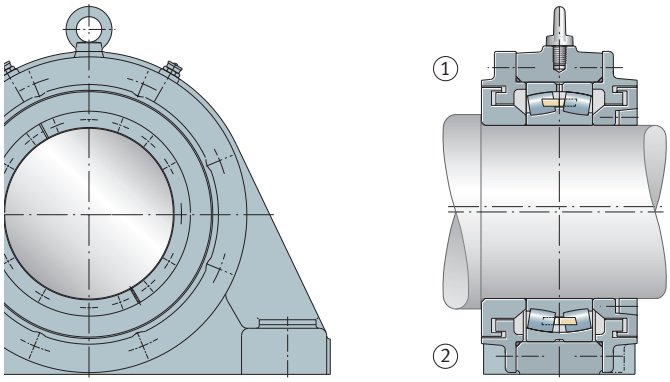
1	BND..-H-W-T-BF locating bearing	2	BND..-H-W-T-BL non-locating bearing

177 BND plummer block housing for bearings with a cylindrical bore (labyrinth seal), design A



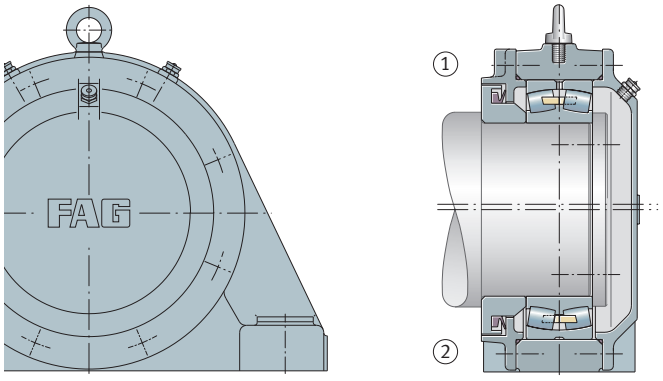
1	BND..-Z-Y-AF locating bearing	2	BND..-Z-Y-AL non-locating bearing

178 BND plummer block housing for bearings with a cylindrical bore (labyrinth seal), design B



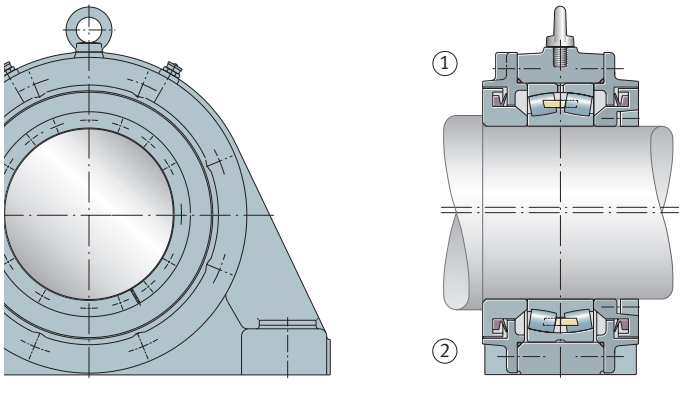
1	BND..-Z-Y-BF locating bearing	2	BND..-Z-Y-BL non-locating bearing

179 BND plummer block housing for bearings with a cylindrical bore (Taconite seal), design A



1	BND..-Z-T-AF locating bearing	2	BND..-Z-T-AL non-locating bearing

180 BND plummer block housing for bearings with a cylindrical bore, (Taconite seal), design B



1	BND..-Z-T-BF locating bearing	2	BND..-Z-T-BL non-locating bearing

11.2 Suitable bearings

The dimensions of BND split plummer block housings are tailored to 222, 230, 231 and 232 spherical roller bearings. The shaft diameters are 60 mm to 420 mm.

73 Suitable bearings for BND unsplit plummer block housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	22213..-K	22284..-K
	23024..-K	23084..-K
	23122..-K	23184..-K
	23222..-K	23284..-K
Spherical roller bearings with a cylindrical bore	22213	22284
	23024	23084
	23122	23184
	23222	23284

11.3 Materials, corrosion protection

74 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Ductile cast iron GJS	D	✓	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

11.4 Load capacity

Guide values are provided for the rupture load of BND plummer block housings. The guide values apply purely to static loads. They apply to the standard material ductile cast iron (suffix D) and to cast steel (suffix S).

The guide values only apply if the flatness of the mounting surface to DIN EN ISO 1101:2017 satisfies tolerance level IT7 to DIN EN ISO 286-1:2019 (measured across the diagonal). The requirement for supporting loads is that the housing base surface has full, rigid support.

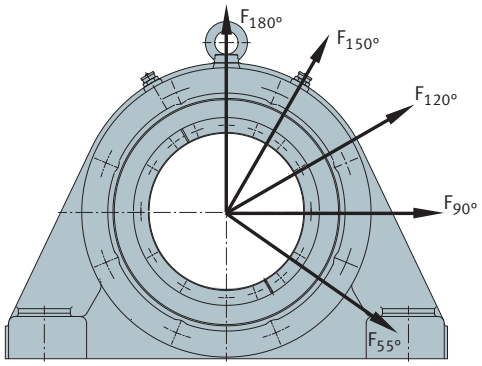
A safety factor of 6 must be used for defining the maximum load.

In the axial direction the housing may be subjected to a maximum of 20 % of the housing rupture load $F_{180°}$.

If the housing is subjected to axial load, the maximum axial load of the fitted bearing must be taken into consideration. If the bearing is located on the shaft using an adapter sleeve, the axial retaining force of the bearing and adapter sleeve must also be taken into consideration.

We recommend that housings should be secured in the load direction by means of stops or dowels if the load direction is between 55 ° and 120 ° or for an axial load.

181 Load directions F of the guide values for housing rupture load



75 Housing rupture load in load direction F

Housing		Housing rupture load in load direction F				
Designation		55°	90°	120°	150°	180°
		kN	kN	kN	kN	kN
BND2213	–	665	530	440	350	440
BND2215	–	880	705	580	465	580
BND2218	–	1125	900	745	595	745
BND2220	BND3024	1325	1070	1020	815	1020
BND2222	BND3026	1580	1275	1120	920	1120
BND2224	BND3028	1785	1430	1225	1020	1225
BND2226	BND3030	2040	1630	2140	1735	2140
–	BND3032	2295	1835	2245	1835	2245
BND2228	BND3034	2755	2195	2245	1835	2245
BND2230	BND3036	3060	2450	2245	1835	2245
BND2232	BND3038	3365	2705	2450	1940	2450
BND2234	BND3040	3925	3110	3060	2450	3060
BND2236	–	4435	3570	3470	2755	3470
BND2238	BND3044	4435	3570	3470	2755	3470
BND2240	BND3048	5050	4030	4895	3875	4895
BND2244	BND3052	5660	4540	5000	3980	5000

Housing		Housing rupture load in load direction F				
Designation		55°	90°	120°	150°	180°
		kN	kN	kN	kN	kN
–	BND3056	6580	5255	6120	4895	6120
BND2248	BND3060	7295	5815	6325	5100	6325
BND2252	BND3064	8000	6425	6835	5400	6835
BND2256	BND3068	8825	7040	6835	5400	6835
BND2260	BND3072	9640	7700	8160	6530	8160
–	BND3076	10810	8670	8365	8770	8365
BND2264	BND3080	12035	9690	9080	7240	9080
BND2268	BND3084	13360	10760	9280	7345	9280
BND2272	–	15700	12570	10370	8325	10370
BND2276	–	16600	13280	10960	8800	10960
BND2280	–	19750	15800	13030	10470	13030
BND2284	–	21540	17240	14220	11420	14220
BND3122	–	1900	1530	1685	1345	1685
BND3124	BND3222	1875	1500	1685	1345	1685
BND3126	BND3224	2020	1610	1735	1385	1735
BND3128	BND3226	2295	1835	1735	1385	1735
BND3130	BND3228	2755	2195	1785	1430	1785
BND3132	BND3230	3110	2500	1835	1470	1835
BND3134	BND3232	3570	2855	1940	1550	1940
BND3136	BND3234	4285	3415	2040	1630	2040
BND3138	BND3236	4590	3725	2140	1715	2140
BND3140	BND3238	5610	4540	2295	1835	2295
BND3144	BND3240	6120	4935	2550	2040	2550
BND3148	BND3244	6835	5510	3060	2450	3060
BND3152	BND3248	7650	6170	3570	2855	3570
BND3156	BND3252	9385	7550	4180	3365	4180
BND3160	BND3256	10200	8260	4490	3570	4490
BND3164	BND3260	11935	9535	5100	4080	5100
BND3168	BND3264	14280	11375	5815	4590	5815
BND3172	–	14485	11630	6630	5300	6630
BND3176	BND3268	16320	13055	6630	5300	6630
BND3180	BND3272	17850	14280	7345	5815	7345
–	BND3276	18870	15050	8160	6530	8160
BND3184	–	19380	15600	8160	6530	8160
–	BND3280	22440	17950	9280	7445	9280
–	BND3284	24480	19380	10710	8570	10710

11.5 Lubrication

BND unsplit plummer block housings are designed for grease lubrication.

11.5.1 Grease types

Lithium soap greases of NLGI classes 2 and 3 are suitable.

For low loads, for example Arcanol MULTI3, for high and very high loads Arcanol MULTITOP and LOAD400.

11.5.2 Grease quantities

The cavities in the bearing, the housing and the labyrinths are completely filled with grease during the initial lubrication process. The recommended grease quantities apply to this.

For relubrication, we recommend that the grease quantity is around 10 % of the grease quantity used in the initial process. If the ambient conditions result in heavy soiling of the housings, relubrication should be carried out on a daily basis using smaller quantities.

76 Recommended grease quantity

Bearing bore	Grease quantity for initial greasing	
	BND22, BND31, BND32	BND30
mm	g	g
65	700	–
75	800	–
90	900	–
100	950	–
110	1000	–
120	1100	500
130	1250	600
140	1400	700
150	1700	800
160	1900	900
170	2200	1000
180	2500	1200
190	3000	1300
200	3600	1600
220	4200	1900
240	5000	2100
260	6000	2500
280	7000	3000
300	8000	3500
320	9000	4100
340	10500	4800
360	12000	5500
380	13000	6200
400	14500	7000
420	16000	8000

11.5.3 Relubrication

The housings have button head lubrication nipples with a standard head diameter to DIN 3404:1988 of 22 mm. The grease is fed uniformly to both rows of rollers through the circumferential slot and 3 lubrication holes in the outer ring of the spherical roller bearings.

The relubrication intervals should be tailored to the ambient conditions. The bearings should be relubricated after an interval of no more than 4 weeks.

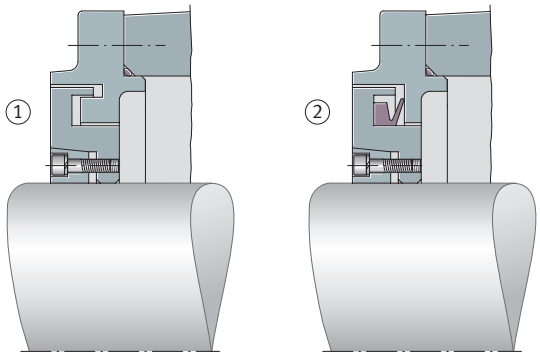
11.6 Sealing

Plummer block housings in design A are intended for the bearings at shaft ends. One side is closed by a cover and the other side is sealed against the shaft. Design B is intended for a continuous shafts, in which case the housing is sealed against the shaft on both sides.

Labyrinth seals (suffix Y) or Taconite seals (suffix T) can be used. Where shafts of constant diameter are used, the seals are located on the shaft by means of slotted conical clamping rings made from laminated fabric. Labyrinth rings,Taconite seals and covers are unsplit.

The seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication.

182 Seals for BND plummer block housing



1	Labyrinth seal	2	Taconite seal

11.7 Dimensions, tolerances

11.7.1 Tolerances for the bearing seat

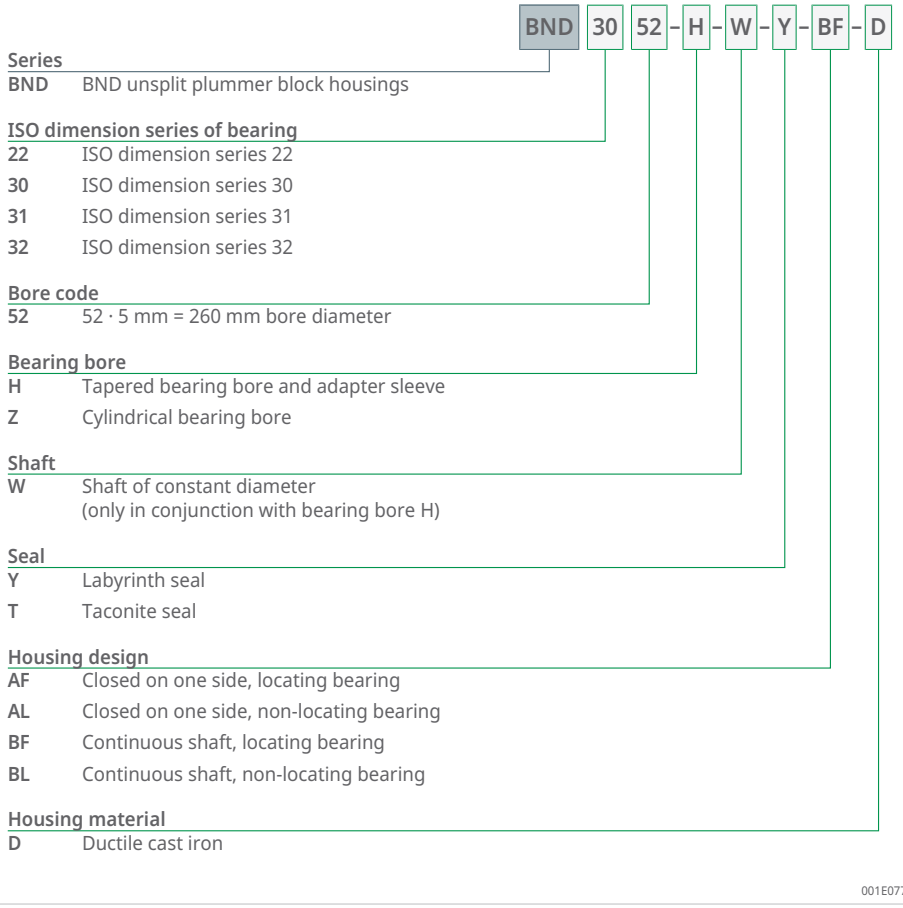
The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1. The tolerance indicated applies to the condition on delivery, in other words before the screws connecting the upper and lower sections are loosened.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

11.8 Structure of the order designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

183 Structure of the designations of BND plummer block housings



11.8.1 Specimen orders

For ordering a BND unsplit plummer block housing, the designation contains details of the size and location method of the bearing, the shaft design, the sealing design and the design and material of the housing. The roller bearing and, if necessary, the adapter sleeve must be ordered separately.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- BND plummer block housing made from ductile cast iron
- Closed on one side
- For a shaft with constant diameter
- Taconite seal
- 23260-BEA-XL-K-MB1 spherical roller bearing in the form of a non-locating bearing
- Secured by means of adapter sleeve

77 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	BND3260-H-W-T-AL-D
1	Spherical roller bearing	23260-BEA-XL-K-MB1
1	Adapter sleeve	H3260-HG

Example 2

Example of a housing configuration, characteristics:

- BND plummer block housing made from ductile cast iron
- Closed on one side
- Labyrinth seal
- 23148-BE-XL-K spherical roller bearing in the form of a locating bearing

78 Order

Quantity	Description	Ordering designation, short designation
1	Plummer block housing	BND3148-Z-Y-AF-D
1	Spherical roller bearing	23148-BE-XL-K

11.9 Design of the adjacent construction

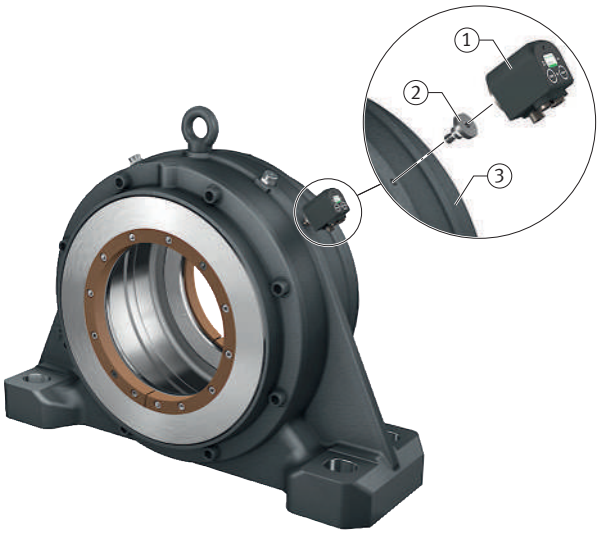
11.9.1 Prepared mounting points for condition monitoring

Housing as from size BND2220 have an M8 thread and a flat surface for the easy connection of conditioning monitoring systems. The M8 thread enables the direct connection of sensors, such as temperature sensors.

The Schaeffler SmartCheck modular online monitoring system has an M6 connection thread. An adapter can be supplied to enable it to be installed on an M8 internal thread.

Order designation for the adapter M8 external thread, M6 internal thread): SMART-CHECK.MOUNT-ADAPTER-M6TOM8

184 BND plummer block housing with Schaeffler SmartCheck



1	Schaeffler SmartCheck	2	Adapter
3	Plummer block housing		

11.10 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

11.10.1 Eye bolts

The housing has an eye bolt to DIN 580:2018. This is designed for use as an anchoring point for the installation and removal of the housing. The load capacity of the eye bolt allows lifting of the housing including a bearing fitted in the housing.

The load on eye bolts must not exceed the weight of the housing and the installed bearing.

Eye bolts must always be screwed fully into the housing.

11.10.2 Base fastening screws

Base fastening screws are used for securing the housings to the mounting surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

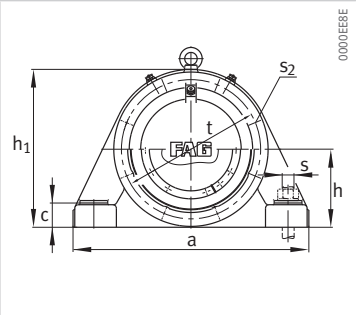
The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

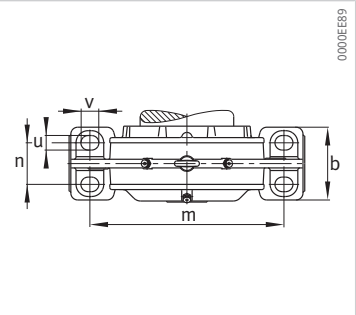
11.11 Product tables

11.11.1 Explanations of the product tables

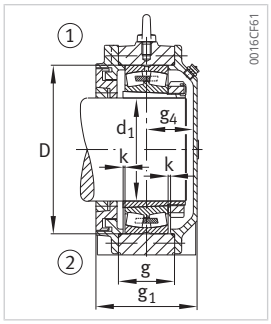
(3)	-	Locating bearing
(4)	-	Non-locating bearing
(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
b	mm	Base width
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
g	mm	Housing body width
g_1	mm	Overall width
g_2	mm	Overall width
g_3	mm	Width over clamping element
h	mm	Distance to shaft axis
h_1	mm	Overall height
k	mm	Axial movement distance
m	mm	Fixing hole spacing
m	kg	Mass
n	–	Number of screw mounting holes
n	mm	Distance between fastening holes
s	inch or mm	Screw size
s_2	-	Fastening hole
t	mm	Pitch circle diameter of the bores
u	mm	Slot width
v	mm	Slot length

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

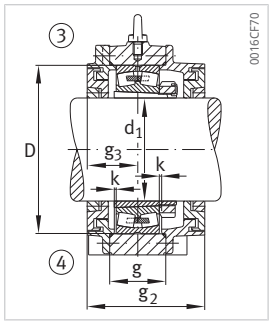




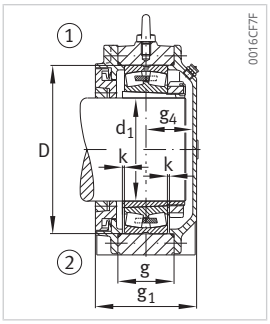
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2213-H-W-Y-AF-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-Y-AL-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-Y-BF-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-Y-BL-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-T-AF-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-T-AL-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-T-BF-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2213-H-W-T-BL-D	**60**	80	155	44	70	22	235	185	40	20	15	M10
BND2215-H-W-Y-AF-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-Y-AL-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-Y-BF-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-Y-BL-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-T-AF-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-T-AL-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-T-BF-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2215-H-W-T-BL-D	**65**	90	180	45	85	35	285	225	45	28	20	M16
BND2218-H-W-Y-AF-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-Y-AL-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-Y-BF-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-Y-BL-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-T-AF-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-T-AL-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-T-BF-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2218-H-W-T-BL-D	**80**	110	220	55	110	38	370	290	60	32	23	M20
BND2220-H-W-Y-AF-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-Y-AL-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-Y-BF-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-Y-BL-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-T-AF-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-T-AL-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-T-BF-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2220-H-W-T-BL-D	**90**	130	255	65	120	40	400	320	65	35	30	M24
BND2222-H-W-Y-AF-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-Y-AF-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-Y-AF-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-Y-AL-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-Y-AL-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-Y-AL-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-Y-BF-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-Y-BF-D	**100**	130	263	80	140	40	400	300	80	35	25	M20



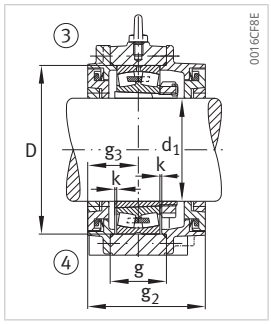
Design A,
Labyrinth seal



Design B,
Labyrinth seal



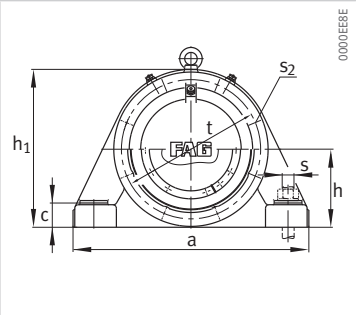
Design A,
Taconite seal

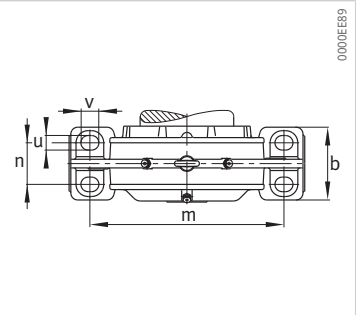


Design B,
Taconite seal

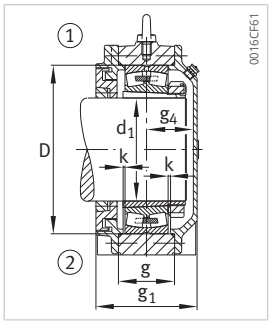
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
120	–	100	–	–	39	135	6	M6	22213..-K	H313	9,4
120	2	100	–	–	39	135	6	M6	22213..-K	H313	9,4
120	–	–	120	55	–	135	6	M6	22213..-K	H313	9,4
120	2	–	120	55	–	135	6	M6	22213..-K	H313	9,4
120	–	100	–	–	39	135	6	M6	22213..-K	H313	9,4
120	2	100	–	–	39	135	6	M6	22213..-K	H313	9,4
120	–	–	120	55	–	135	6	M6	22213..-K	H313	9,4
120	2	–	120	55	–	135	6	M6	22213..-K	H313	9,4
130	–	105	–	–	44	155	6	M6	22215..-K	H315	14,5
130	2	105	–	–	44	155	6	M6	22215..-K	H315	14,5
130	–	–	125	55	–	155	6	M6	22215..-K	H315	14,5
130	2	–	125	55	–	155	6	M6	22215..-K	H315	14,5
130	–	105	–	–	44	155	6	M6	22215..-K	H315	14,5
130	2	105	–	–	44	155	6	M6	22215..-K	H315	14,5
130	–	–	125	55	–	155	6	M6	22215..-K	H315	14,5
130	2	–	125	55	–	155	6	M6	22215..-K	H315	14,5
160	–	104	–	–	45	185	6	M8	22218..-K	H318	26,7
160	2	104	–	–	45	185	6	M8	22218..-K	H318	26,7
160	–	–	123	53	–	185	6	M8	22218..-K	H318	26,7
160	2	–	123	53	–	185	6	M8	22218..-K	H318	26,7
160	–	104	–	–	45	185	6	M8	22218..-K	H318	26,7
160	2	104	–	–	45	185	6	M8	22218..-K	H318	26,7
160	–	–	123	53	–	185	6	M8	22218..-K	H318	26,7
160	2	–	123	53	–	185	6	M8	22218..-K	H318	26,7
180	–	141	–	–	57	205	6	M8	22220..-K	H320	28
180	2,5	141	–	–	57	205	6	M8	22220..-K	H320	28
180	–	–	168	78	–	205	6	M8	22220..-K	H320	28
180	2,5	–	168	78	–	205	6	M8	22220..-K	H320	28
180	–	141	–	–	57	205	6	M8	22220..-K	H320	28
180	2,5	141	–	–	57	205	6	M8	22220..-K	H320	28
180	–	–	168	78	–	205	6	M8	22220..-K	H320	28
180	2,5	–	168	78	–	205	6	M8	22220..-K	H320	28
200	–	149,2	–	–	60	230	6	M12	22222..-K	H322	48
180	–	146	–	–	62	215	8	M12	23122..-K	H3122	38
200	–	166	–	–	68	230	6	M12	23222..-K	H2322	33,4
200	2,5	149,2	–	–	60	230	6	M12	22222..-K	H322	48
180	2,5	146	–	–	62	215	8	M12	23122..-K	H3122	38
200	2,5	166	–	–	68	230	6	M12	23222..-K	H2322	33,4
200	–	–	172	82,6	–	230	6	M12	22222..-K	H322	48
180	–	–	171	78	–	215	8	M12	23122..-K	H3122	38

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

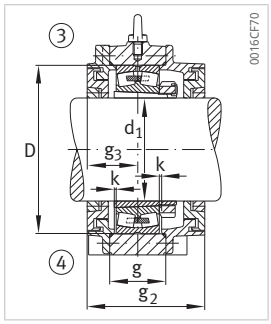




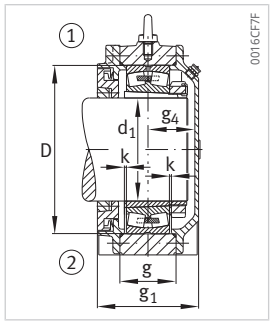
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3222-H-W-Y-BF-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-Y-BL-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-Y-BL-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-Y-BL-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-T-AF-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-T-AF-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-T-AF-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-T-AL-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-T-AL-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-T-AL-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-T-BF-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-T-BF-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-T-BF-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2222-H-W-T-BL-D	**100**	140	280	73	130	42	440	350	70	35	30	M24
BND3122-H-W-T-BL-D	**100**	130	263	80	140	40	400	300	80	35	25	M20
BND3222-H-W-T-BL-D	**100**	140	280	89,8	130	42	440	350	70	35	30	M24
BND2224-H-W-Y-AF-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-Y-AF-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-Y-AF-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-Y-AF-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2224-H-W-Y-AL-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-Y-AL-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-Y-AL-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-Y-AL-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2224-H-W-Y-BF-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-Y-BF-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-Y-BF-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-Y-BF-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2224-H-W-Y-BL-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-Y-BL-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-Y-BL-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-Y-BL-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2224-H-W-T-AF-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-T-AF-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-T-AF-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-T-AF-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2224-H-W-T-AL-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-T-AL-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-T-AL-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-T-AL-D	**110**	150	300	95	140	42	470	370	75	35	30	M24



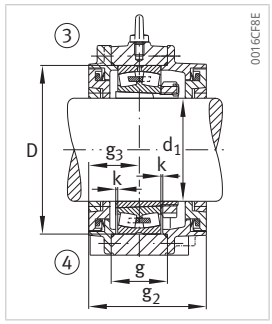
Design A,
Labyrinth seal



Design B,
Labyrinth seal



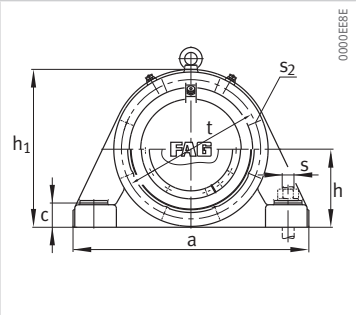
Design A,
Taconite seal

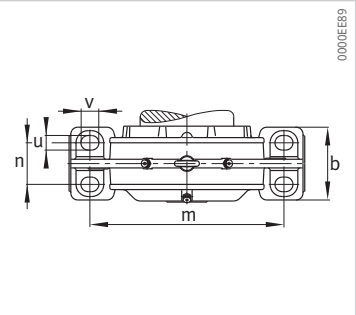


Design B,
Taconite seal

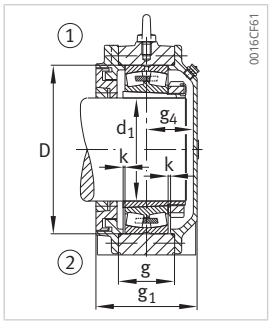
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
200	–	–	194	91	–	230	6	M12	23222..-K	H2322	33,4
200	2,5	–	172	82,6	–	230	6	M12	22222..-K	H322	48
180	2,5	–	171	78	–	215	8	M12	23122..-K	H3122	38
200	2,5	–	194	91	–	230	6	M12	23222..-K	H2322	33,4
200	–	149,2	–	–	60	230	6	M12	22222..-K	H322	48
180	–	146	–	–	62	215	8	M12	23122..-K	H3122	38
200	–	166	–	–	68	230	6	M12	23222..-K	H2322	33,4
200	2,5	149,2	–	–	60	230	6	M12	22222..-K	H322	48
180	2,5	146	–	–	62	215	8	M12	23122..-K	H3122	38
200	2,5	166	–	–	68	230	6	M12	23222..-K	H2322	33,4
200	–	–	172	82,6	–	230	6	M12	22222..-K	H322	48
180	–	–	171	78	–	215	8	M12	23122..-K	H3122	38
200	–	–	194	91	–	230	6	M12	23222..-K	H2322	33,4
200	2,5	–	172	82,6	–	230	6	M12	22222..-K	H322	48
180	2,5	–	171	78	–	215	8	M12	23122..-K	H3122	38
200	2,5	–	194	91	–	230	6	M12	23222..-K	H2322	33,4
215	–	143	–	–	61	245	6	M12	22224..-K	H3124	55,8
180	–	126	–	–	57	205	8	M8	23024..-K	H3024	19
200	–	160	–	–	74	235	8	M12	23124..-K	H3124	47,5
215	–	161	–	–	70	245	6	M12	23224..-K	H2324	38
215	2,5	143	–	–	61	245	6	M12	22224..-K	H3124	55,8
180	2,5	126	–	–	57	205	8	M8	23024..-K	H3024	19
200	2,5	160	–	–	74	235	8	M12	23124..-K	H3124	47,5
215	2,5	161	–	–	70	245	6	M12	23224..-K	H2324	38
215	–	–	168	74	–	245	6	M12	22224..-K	H3124	55,8
180	–	–	146	63	–	205	8	M8	23024..-K	H3024	19
200	–	–	180	80	–	235	8	M12	23124..-K	H3124	47,5
215	–	–	186	83	–	245	6	M12	23224..-K	H2324	38
215	2,5	–	168	74	–	245	6	M12	22224..-K	H3124	55,8
180	2,5	–	146	63	–	205	8	M8	23024..-K	H3024	19
200	2,5	–	180	80	–	235	8	M12	23124..-K	H3124	47,5
215	2,5	–	186	83	–	245	6	M12	23224..-K	H2324	38
215	–	143	–	–	61	245	6	M12	22224..-K	H3124	55,8
180	–	126	–	–	57	205	8	M8	23024..-K	H3024	19
200	–	160	–	–	74	235	8	M12	23124..-K	H3124	47,5
215	–	161	–	–	70	245	6	M12	23224..-K	H2324	38
215	2,5	143	–	–	61	245	6	M12	22224..-K	H3124	55,8
180	2,5	126	–	–	57	205	8	M8	23024..-K	H3024	19
200	2,5	160	–	–	74	235	8	M12	23124..-K	H3124	47,5
215	2,5	161	–	–	70	245	6	M12	23224..-K	H2324	38

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

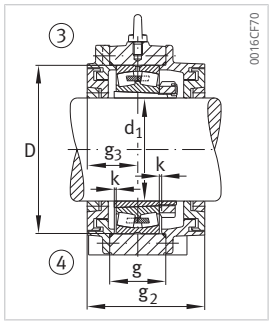




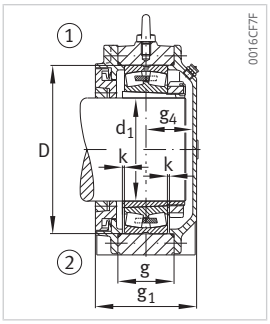
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2224-H-W-T-BF-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-T-BF-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-T-BF-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-T-BF-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2224-H-W-T-BL-D	**110**	150	300	77	140	42	470	370	75	35	30	M24
BND3024-H-W-T-BL-D	**110**	115	230	60	110	35	370	300	60	35	25	M20
BND3124-H-W-T-BL-D	**110**	140	280	85	150	40	410	330	80	35	25	M20
BND3224-H-W-T-BL-D	**110**	150	300	95	140	42	470	370	75	35	30	M24
BND2226-H-W-Y-AF-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-Y-AF-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-Y-AF-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-Y-AF-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-Y-AL-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-Y-AL-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-Y-AL-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-Y-AL-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-Y-BF-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-Y-BF-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-Y-BF-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-Y-BF-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-Y-BL-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-Y-BL-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-Y-BL-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-Y-BL-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-T-AF-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-T-AF-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-T-AF-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-T-AF-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-T-AL-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-T-AL-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-T-AL-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-T-AL-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-T-BF-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-T-BF-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-T-BF-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-T-BF-D	**115**	160	315	100	150	45	500	400	85	35	30	M24
BND2226-H-W-T-BL-D	**115**	160	315	84	150	45	500	400	85	35	30	M24
BND3026-H-W-T-BL-D	**115**	130	260	70	120	38	410	340	60	35	25	M20
BND3126-H-W-T-BL-D	**115**	145	295	90	150	40	430	350	80	35	25	M20
BND3226-H-W-T-BL-D	**115**	160	315	100	150	45	500	400	85	35	30	M24



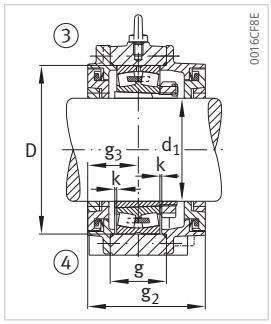
Design A,
Labyrinth seal



Design B,
Labyrinth seal



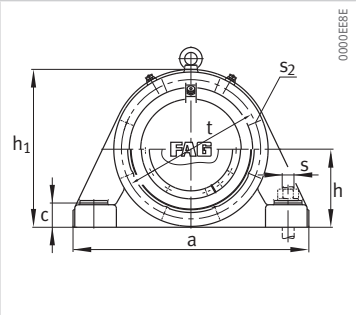
Design A,
Taconite seal

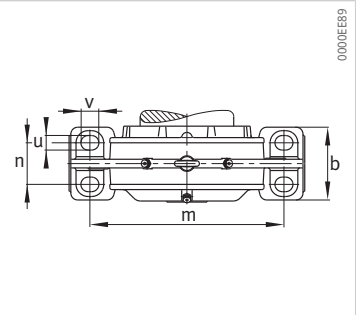


Design B,
Taconite seal

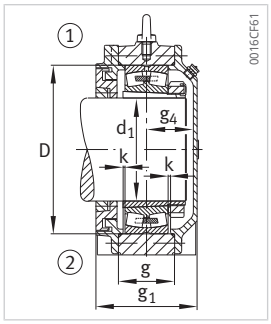
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
215	–	–	168	74	–	245	6	M12	22224..-K	H3124	55,8
180	–	–	146	63	–	205	8	M8	23024..-K	H3024	19
200	–	–	180	80	–	235	8	M12	23124..-K	H3124	47,5
215	–	–	186	83	–	245	6	M12	23224..-K	H2324	38
215	2,5	–	168	74	–	245	6	M12	22224..-K	H3124	55,8
180	2,5	–	146	63	–	205	8	M8	23024..-K	H3024	19
200	2,5	–	180	80	–	235	8	M12	23124..-K	H3124	47,5
215	2,5	–	186	83	–	245	6	M12	23224..-K	H2324	38
230	–	172	–	–	69	260	6	M12	22226..-K	H3126	61,4
200	–	133	–	–	59	225	8	M10	23026..-K	H3026	28,5
210	–	165	–	–	72	245	8	M12	23126..-K	H3126	52,5
230	–	188	–	–	77	260	6	M12	23226..-K	H2326	47,8
230	3	172	–	–	69	260	6	M12	22226..-K	H3126	61,4
200	2	133	–	–	59	225	8	M10	23026..-K	H3026	28,5
210	3	165	–	–	72	245	8	M12	23126..-K	H3126	52,5
230	3	188	–	–	77	260	6	M12	23226..-K	H2326	47,8
230	–	–	202	95	–	260	6	M12	22226..-K	H3126	61,4
200	–	–	156	68	–	225	8	M10	23026..-K	H3026	28,5
210	–	–	190	85	–	245	8	M12	23126..-K	H3126	52,5
230	–	–	218	103	–	260	6	M12	23226..-K	H2326	47,8
230	3	–	202	95	–	260	6	M12	22226..-K	H3126	61,4
200	2	–	156	68	–	225	8	M10	23026..-K	H3026	28,5
210	3	–	190	85	–	245	8	M12	23126..-K	H3126	52,5
230	3	–	218	103	–	260	6	M12	23226..-K	H2326	47,8
230	–	172	–	–	69	260	6	M12	22226..-K	H3126	61,4
200	–	133	–	–	59	225	8	M10	23026..-K	H3026	28,5
210	–	165	–	–	72	245	8	M12	23126..-K	H3126	52,5
230	–	188	–	–	77	260	6	M12	23226..-K	H2326	47,8
230	3	172	–	–	69	260	6	M12	22226..-K	H3126	61,4
200	2	133	–	–	59	225	8	M10	23026..-K	H3026	28,5
210	3	165	–	–	72	245	8	M12	23126..-K	H3126	52,5
230	3	188	–	–	77	260	6	M12	23226..-K	H2326	47,8
230	–	–	202	95	–	260	6	M12	22226..-K	H3126	61,4
200	–	–	156	68	–	225	8	M10	23026..-K	H3026	28,5
210	–	–	190	85	–	245	8	M12	23126..-K	H3126	52,5
230	–	–	218	103	–	260	6	M12	23226..-K	H2326	47,8
230	3	–	202	95	–	260	6	M12	22226..-K	H3126	61,4
200	2	–	156	68	–	225	8	M10	23026..-K	H3026	28,5
210	3	–	190	85	–	245	8	M12	23126..-K	H3126	52,5
230	3	–	218	103	–	260	6	M12	23226..-K	H2326	47,8

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

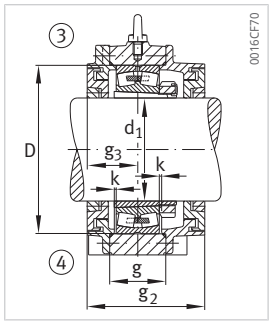




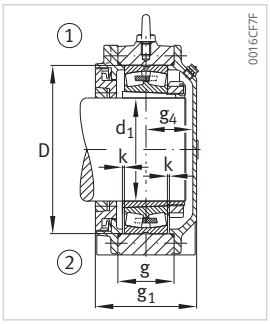
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2228-H-W-Y-AF-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-Y-AF-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-Y-AF-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-Y-AF-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-Y-AL-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-Y-AL-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-Y-AL-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-Y-AL-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-Y-BF-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-Y-BF-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-Y-BF-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-Y-BF-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-Y-BL-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-Y-BL-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-Y-BL-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-Y-BL-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-T-AF-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-T-AF-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-T-AF-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-T-AF-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-T-AL-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-T-AL-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-T-AL-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-T-AL-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-T-BF-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-T-BF-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-T-BF-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-T-BF-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2228-H-W-T-BL-D	**125**	170	345	88	160	50	530	430	85	35	30	M24
BND3028-H-W-T-BL-D	**125**	140	275	70	130	40	430	360	70	35	30	M24
BND3128-H-W-T-BL-D	**125**	155	315	95	160	45	470	380	85	35	25	M20
BND3228-H-W-T-BL-D	**125**	170	345	108	160	50	530	430	85	35	30	M24
BND2230-H-W-Y-AF-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-Y-AF-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-Y-AF-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-Y-AF-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2230-H-W-Y-AL-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-Y-AL-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-Y-AL-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-Y-AL-D	**135**	180	365	120	170	54	550	450	90	45	36	M30



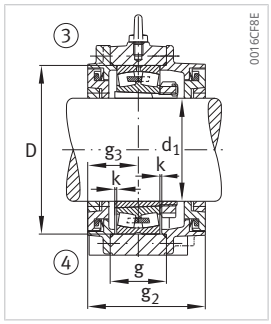
Design A,
Labyrinth seal



Design B,
Labyrinth seal



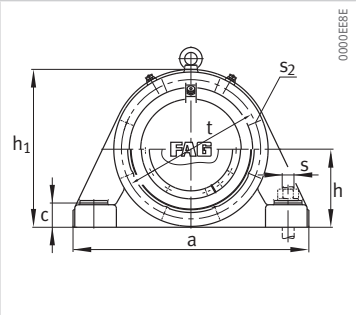
Design A,
Taconite seal

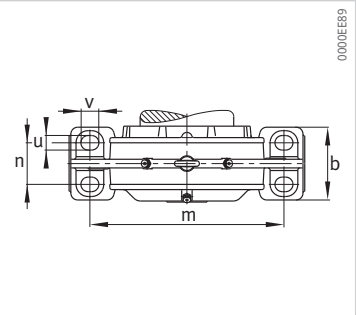


Design B,
Taconite seal

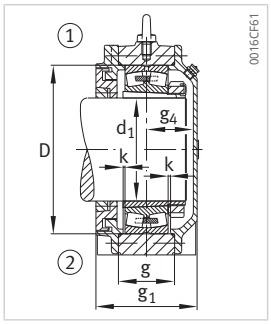
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
250	–	166	–	–	70	285	6	M16	22228..-K	H3128	66,3
210	–	136	–	–	61	240	8	M10	23028..-K	H3028	33
225	–	170	–	–	77	270	8	M12	23128..-K	H3128	57
250	–	186	–	–	80	285	6	M16	23228..-K	H2328	61,5
250	3	166	–	–	70	285	6	M16	22228..-K	H3128	66,3
210	3	136	–	–	61	240	8	M10	23028..-K	H3028	33
225	3	170	–	–	77	270	8	M12	23128..-K	H3128	57
250	3	186	–	–	80	285	6	M16	23228..-K	H2328	61,5
250	–	–	196	88	–	285	6	M16	22228..-K	H3128	66,3
210	–	–	156	68	–	240	8	M10	23028..-K	H3028	33
225	–	–	190	85	–	270	8	M12	23128..-K	H3128	57
250	–	–	216	98	–	285	6	M16	23228..-K	H2328	61,5
250	3	–	196	88	–	285	6	M16	22228..-K	H3128	66,3
210	3	–	156	68	–	240	8	M10	23028..-K	H3028	33
225	3	–	190	85	–	270	8	M12	23128..-K	H3128	57
250	3	–	216	98	–	285	6	M16	23228..-K	H2328	61,5
250	–	166	–	–	70	285	6	M16	22228..-K	H3128	66,3
210	–	136	–	–	61	240	8	M10	23028..-K	H3028	33
225	–	170	–	–	77	270	8	M12	23128..-K	H3128	57
250	–	186	–	–	80	285	6	M16	23228..-K	H2328	61,5
250	3	166	–	–	70	285	6	M16	22228..-K	H3128	66,3
210	3	136	–	–	61	240	8	M10	23028..-K	H3028	33
225	3	170	–	–	77	270	8	M12	23128..-K	H3128	57
250	3	186	–	–	80	285	6	M16	23228..-K	H2328	61,5
250	–	–	196	88	–	285	6	M16	22228..-K	H3128	66,3
210	–	–	156	68	–	240	8	M10	23028..-K	H3028	33
225	–	–	190	85	–	270	8	M12	23128..-K	H3128	57
250	–	–	216	98	–	285	6	M16	23228..-K	H2328	61,5
250	3	–	196	88	–	285	6	M16	22228..-K	H3128	66,3
210	3	–	156	68	–	240	8	M10	23028..-K	H3028	33
225	3	–	190	85	–	270	8	M12	23128..-K	H3128	57
250	3	–	216	98	–	285	6	M16	23228..-K	H2328	61,5
270	–	177	–	–	76	305	6	M16	22230..-K	H3130	81,5
225	–	149	–	–	64	250	8	M12	23030..-K	H3030	38
250	–	186	–	–	85	285	8	M12	23130..-K	H3130	66,4
270	–	200	–	–	87	305	6	M16	23230..-K	H2330	71
270	3	177	–	–	76	305	6	M16	22230..-K	H3130	81,5
225	2	149	–	–	64	250	8	M12	23030..-K	H3030	38
250	2	186	–	–	85	285	8	M12	23130..-K	H3130	66,4
270	3	200	–	–	87	305	6	M16	23230..-K	H2330	71

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

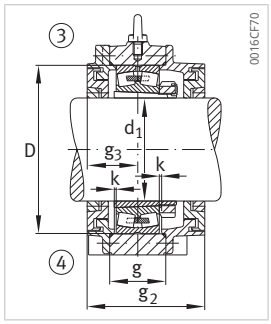




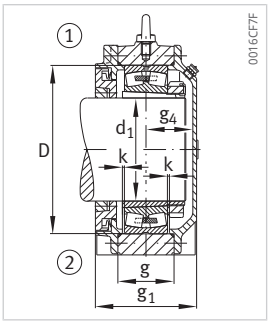
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2230-H-W-Y-BF-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-Y-BF-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-Y-BF-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-Y-BF-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2230-H-W-Y-BL-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-Y-BL-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-Y-BL-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-Y-BL-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2230-H-W-T-AF-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-T-AF-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-T-AF-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-T-AF-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2230-H-W-T-AL-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-T-AL-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-T-AL-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-T-AL-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2230-H-W-T-BF-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-T-BF-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-T-BF-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-T-BF-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2230-H-W-T-BL-D	**135**	180	365	97	170	54	550	450	90	45	36	M30
BND3030-H-W-T-BL-D	**135**	145	290	80	140	42	455	370	80	35	30	M24
BND3130-H-W-T-BL-D	**135**	170	345	102	200	60	580	450	110	35	30	M24
BND3230-H-W-T-BL-D	**135**	180	365	120	170	54	550	450	90	45	36	M30
BND2232-H-W-Y-AF-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-Y-AF-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-Y-AF-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-Y-AF-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2232-H-W-Y-AL-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-Y-AL-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-Y-AL-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-Y-AL-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2232-H-W-Y-BF-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-Y-BF-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-Y-BF-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-Y-BF-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2232-H-W-Y-BL-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-Y-BL-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-Y-BL-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-Y-BL-D	**140**	190	385	130	180	58	600	490	105	45	36	M30



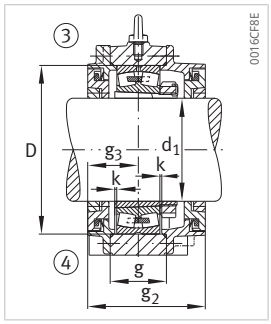
Design A,
Labyrinth seal



Design B,
Labyrinth seal



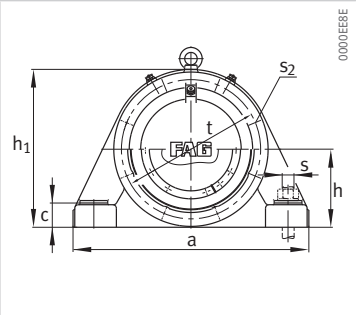
Design A,
Taconite seal

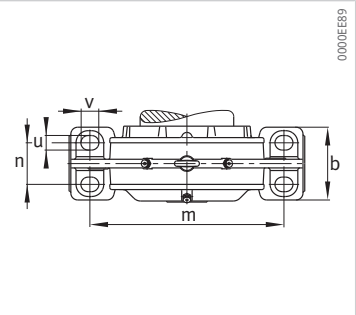


Design B,
Taconite seal

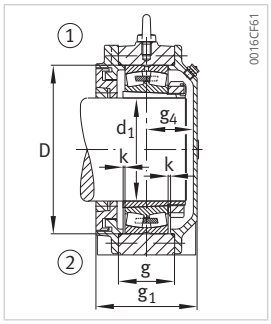
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
270	–	–	202	93,5	–	305	6	M16	22230..-K	H3130	81,5
225	–	–	176	78	–	250	8	M12	23030..-K	H3030	38
250	–	–	206	93	–	285	8	M12	23130..-K	H3130	66,4
270	–	–	225	105	–	305	6	M16	23230..-K	H2330	71
270	3	–	202	93,5	–	305	6	M16	22230..-K	H3130	81,5
225	2	–	176	78	–	250	8	M12	23030..-K	H3030	38
250	2	–	206	93	–	285	8	M12	23130..-K	H3130	66,4
270	3	–	225	105	–	305	6	M16	23230..-K	H2330	71
270	–	177	–	–	76	305	6	M16	22230..-K	H3130	81,5
225	–	149	–	–	64	250	8	M12	23030..-K	H3030	38
250	–	186	–	–	85	285	8	M12	23130..-K	H3130	66,4
270	–	200	–	–	87	305	6	M16	23230..-K	H2330	71
270	3	177	–	–	76	305	6	M16	22230..-K	H3130	81,5
225	2	149	–	–	64	250	8	M12	23030..-K	H3030	38
250	2	186	–	–	85	285	8	M12	23130..-K	H3130	66,4
270	3	200	–	–	87	305	6	M16	23230..-K	H2330	71
270	–	–	202	93,5	–	305	6	M16	22230..-K	H3130	81,5
225	–	–	176	78	–	250	8	M12	23030..-K	H3030	38
250	–	–	206	93	–	285	8	M12	23130..-K	H3130	66,4
270	–	–	225	105	–	305	6	M16	23230..-K	H2330	71
270	3	–	202	93,5	–	305	6	M16	22230..-K	H3130	81,5
225	2	–	176	78	–	250	8	M12	23030..-K	H3030	38
250	2	–	206	93	–	285	8	M12	23130..-K	H3130	66,4
270	3	–	225	105	–	305	6	M16	23230..-K	H2330	71
290	–	189	–	–	78	330	6	M16	22232..-K	H3132(-HG)	96
240	–	156	–	–	71	265	8	M12	23032..-K	H3032(-HG)	42,8
270	–	200	–	–	87	310	6	M16	23132..-K	H3132(-HG)	76
290	–	213	–	–	90	330	6	M16	23232..-K	H2332(-HG)	85,5
290	3	189	–	–	78	330	6	M16	22232..-K	H3132(-HG)	96
240	3	156	–	–	71	265	8	M12	23032..-K	H3032(-HG)	42,8
270	3	200	–	–	87	310	6	M16	23132..-K	H3132(-HG)	76
290	3	213	–	–	90	330	6	M16	23232..-K	H2332(-HG)	85,5
290	–	–	226	103	–	330	6	M16	22232..-K	H3132(-HG)	96
240	–	–	181	78	–	265	8	M12	23032..-K	H3032(-HG)	42,8
270	–	–	230	105	–	310	6	M16	23132..-K	H3132(-HG)	76
290	–	–	250	115	–	330	6	M16	23232..-K	H2332(-HG)	85,5
290	3	–	226	103	–	330	6	M16	22232..-K	H3132(-HG)	96
240	3	–	181	78	–	265	8	M12	23032..-K	H3032(-HG)	42,8
270	3	–	230	105	–	310	6	M16	23132..-K	H3132(-HG)	76
290	3	–	250	115	–	330	6	M16	23232..-K	H2332(-HG)	85,5

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

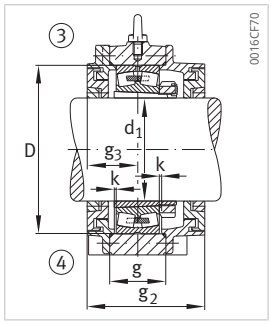




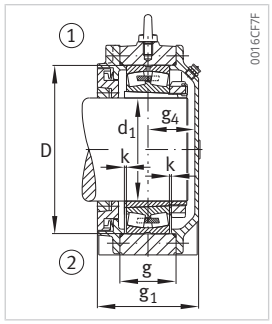
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2232-H-W-T-AF-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-T-AF-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-T-AF-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-T-AF-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2232-H-W-T-AL-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-T-AL-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-T-AL-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-T-AL-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2232-H-W-T-BF-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-T-BF-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-T-BF-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-T-BF-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2232-H-W-T-BL-D	**140**	190	385	106	180	58	600	490	105	45	36	M30
BND3032-H-W-T-BL-D	**140**	155	310	85	150	45	480	390	90	35	30	M24
BND3132-H-W-T-BL-D	**140**	180	360	110	200	55	540	430	110	35	30	M24
BND3232-H-W-T-BL-D	**140**	190	385	130	180	58	600	490	105	45	36	M30
BND2234-H-W-Y-AF-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-Y-AF-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-Y-AF-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-Y-AF-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2234-H-W-Y-AL-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-Y-AL-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-Y-AL-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-Y-AL-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2234-H-W-Y-BF-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-Y-BF-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-Y-BF-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-Y-BF-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2234-H-W-Y-BL-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-Y-BL-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-Y-BL-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-Y-BL-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2234-H-W-T-AF-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-T-AF-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-T-AF-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-T-AF-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2234-H-W-T-AL-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-T-AL-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-T-AL-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-T-AL-D	**150**	200	405	135	200	62	640	525	110	45	36	M30



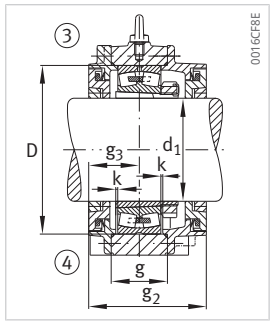
Design A,
Labyrinth seal



Design B,
Labyrinth seal



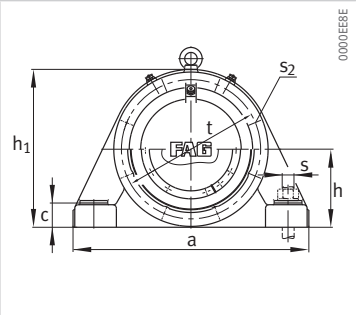
Design A,
Taconite seal

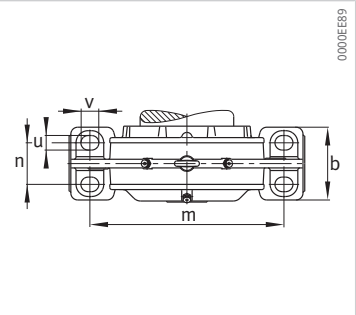


Design B,
Taconite seal

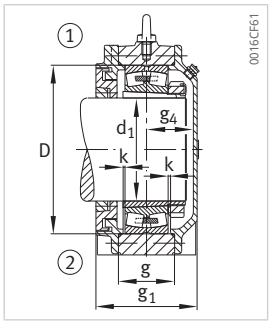
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
290	–	189	–	–	78	330	6	M16	22232..-K	H3132(-HG)	96
240	–	156	–	–	71	265	8	M12	23032..-K	H3032(-HG)	42,8
270	–	200	–	–	87	310	6	M16	23132..-K	H3132(-HG)	76
290	–	213	–	–	90	330	6	M16	23232..-K	H2332(-HG)	85,5
290	3	189	–	–	78	330	6	M16	22232..-K	H3132(-HG)	96
240	3	156	–	–	71	265	8	M12	23032..-K	H3032(-HG)	42,8
270	3	200	–	–	87	310	6	M16	23132..-K	H3132(-HG)	76
290	3	213	–	–	90	330	6	M16	23232..-K	H2332(-HG)	85,5
290	–	–	226	103	–	330	6	M16	22232..-K	H3132(-HG)	96
240	–	–	181	78	–	265	8	M12	23032..-K	H3032(-HG)	42,8
270	–	–	230	105	–	310	6	M16	23132..-K	H3132(-HG)	76
290	–	–	250	115	–	330	6	M16	23232..-K	H2332(-HG)	85,5
290	3	–	226	103	–	330	6	M16	22232..-K	H3132(-HG)	96
240	3	–	181	78	–	265	8	M12	23032..-K	H3032(-HG)	42,8
270	3	–	230	105	–	310	6	M16	23132..-K	H3132(-HG)	76
290	3	–	250	115	–	330	6	M16	23232..-K	H2332(-HG)	85,5
310	–	216	–	–	88	350	8	M16	22234..-K	H3134(-HG)	101
260	–	166	–	–	75	285	8	M12	23034..-K	H3034(-HG)	66,5
280	–	215	–	–	90	330	8	M12	23134..-K	H3134(-HG)	94,5
310	–	240	–	–	100	350	8	M16	23234..-K	H2334(-HG)	113,5
310	3	216	–	–	88	350	8	M16	22234..-K	H3134(-HG)	101
260	2	166	–	–	75	285	8	M12	23034..-K	H3034(-HG)	66,5
280	3	215	–	–	90	330	8	M12	23134..-K	H3134(-HG)	94,5
310	3	240	–	–	100	350	8	M16	23234..-K	H2334(-HG)	113,5
310	–	–	256	118	–	350	8	M16	22234..-K	H3134(-HG)	101
260	–	–	191	83	–	285	8	M12	23034..-K	H3034(-HG)	66,5
280	–	–	250	115	–	330	8	M12	23134..-K	H3134(-HG)	94,5
310	–	–	280	130	–	350	8	M16	23234..-K	H2334(-HG)	113,5
310	3	–	256	118	–	350	8	M16	22234..-K	H3134(-HG)	101
260	2	–	191	83	–	285	8	M12	23034..-K	H3034(-HG)	66,5
280	3	–	250	115	–	330	8	M12	23134..-K	H3134(-HG)	94,5
310	3	–	280	130	–	350	8	M16	23234..-K	H2334(-HG)	113,5
310	–	216	–	–	88	350	8	M16	22234..-K	H3134(-HG)	101
260	–	166	–	–	75	285	8	M12	23034..-K	H3034(-HG)	66,5
280	–	215	–	–	90	330	8	M12	23134..-K	H3134(-HG)	94,5
310	–	240	–	–	100	350	8	M16	23234..-K	H2334(-HG)	113,5
310	3	216	–	–	88	350	8	M16	22234..-K	H3134(-HG)	101
260	2	166	–	–	75	285	8	M12	23034..-K	H3034(-HG)	66,5
280	3	215	–	–	90	330	8	M12	23134..-K	H3134(-HG)	94,5
310	3	240	–	–	100	350	8	M16	23234..-K	H2334(-HG)	113,5

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

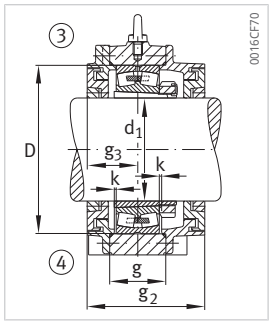




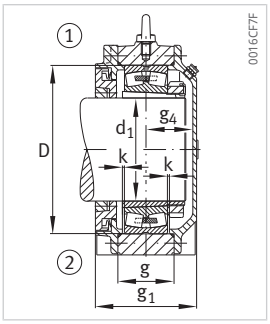
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2234-H-W-T-BF-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-T-BF-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-T-BF-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-T-BF-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2234-H-W-T-BL-D	**150**	200	405	111	200	62	640	525	110	45	36	M30
BND3034-H-W-T-BL-D	**150**	165	330	90	160	50	510	420	95	45	36	M30
BND3134-H-W-T-BL-D	**150**	190	380	120	200	55	570	470	110	45	36	M30
BND3234-H-W-T-BL-D	**150**	200	405	135	200	62	640	525	110	45	36	M30
BND2236-H-W-Y-AF-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-Y-AF-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-Y-AF-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-Y-AF-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-Y-AL-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-Y-AL-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-Y-AL-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-Y-AL-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-Y-BF-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-Y-BF-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-Y-BF-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-Y-BF-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-Y-BL-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-Y-BL-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-Y-BL-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-Y-BL-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-T-AF-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-T-AF-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-T-AF-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-T-AF-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-T-AL-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-T-AL-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-T-AL-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-T-AL-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-T-BF-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-T-BF-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-T-BF-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-T-BF-D	**160**	210	425	138	210	65	680	550	120	45	36	M30
BND2236-H-W-T-BL-D	**160**	210	425	112	210	65	680	550	120	45	36	M30
BND3036-H-W-T-BL-D	**160**	180	360	100	170	52	540	450	100	45	36	M30
BND3136-H-W-T-BL-D	**160**	200	395	125	200	58	600	490	105	45	36	M30
BND3236-H-W-T-BL-D	**160**	210	425	138	210	65	680	550	120	45	36	M30



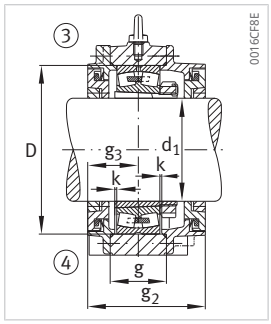
Design A,
Labyrinth seal



Design B,
Labyrinth seal



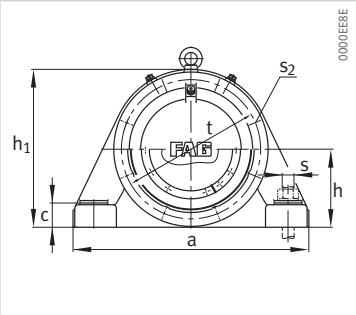
Design A,
Taconite seal

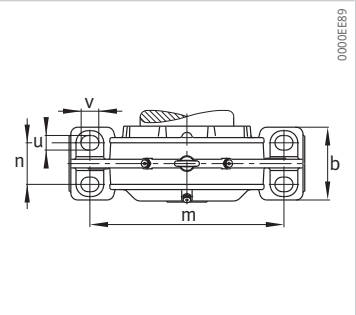


Design B,
Taconite seal

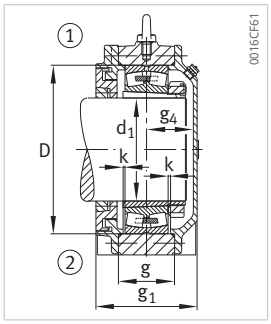
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
310	–	–	256	118	–	350	8	M16	22234..-K	H3134(-HG)	101
260	–	–	191	83	–	285	8	M12	23034..-K	H3034(-HG)	66,5
280	–	–	250	115	–	330	8	M12	23134..-K	H3134(-HG)	94,5
310	–	–	280	130	–	350	8	M16	23234..-K	H2334(-HG)	113,5
310	3	–	256	118	–	350	8	M16	22234..-K	H3134(-HG)	101
260	2	–	191	83	–	285	8	M12	23034..-K	H3034(-HG)	66,5
280	3	–	250	115	–	330	8	M12	23134..-K	H3134(-HG)	94,5
310	3	–	280	130	–	350	8	M16	23234..-K	H2334(-HG)	113,5
320	–	214	–	–	92	370	8	M16	22236..-K	H3136(-HG)	126
280	–	176	–	–	80	310	8	M12	23036..-K	H3036(-HG)	65,5
300	–	220	–	–	95	350	8	M16	23136..-K	H3136(-HG)	104
320	–	240	–	–	105	370	8	M16	23236..-K	H2336(-HG)	133
320	3	214	–	–	92	370	8	M16	22236..-K	H3136(-HG)	126
280	3	176	–	–	80	310	8	M12	23036..-K	H3036(-HG)	65,5
300	3	220	–	–	95	350	8	M16	23136..-K	H3136(-HG)	104
320	3	240	–	–	105	370	8	M16	23236..-K	H2336(-HG)	133
320	–	–	248	114	–	370	8	M16	22236..-K	H3136(-HG)	126
280	–	–	201	88	–	310	8	M12	23036..-K	H3036(-HG)	65,5
300	–	–	250	115	–	350	8	M16	23136..-K	H3136(-HG)	104
320	–	–	274	127	–	370	8	M16	23236..-K	H2336(-HG)	133
320	3	–	248	114	–	370	8	M16	22236..-K	H3136(-HG)	126
280	3	–	201	88	–	310	8	M12	23036..-K	H3036(-HG)	65,5
300	3	–	250	115	–	350	8	M16	23136..-K	H3136(-HG)	104
320	3	–	274	127	–	370	8	M16	23236..-K	H2336(-HG)	133
320	–	214	–	–	92	370	8	M16	22236..-K	H3136(-HG)	126
280	–	176	–	–	80	310	8	M12	23036..-K	H3036(-HG)	65,5
300	–	220	–	–	95	350	8	M16	23136..-K	H3136(-HG)	104
320	–	240	–	–	105	370	8	M16	23236..-K	H2336(-HG)	133
320	3	214	–	–	92	370	8	M16	22236..-K	H3136(-HG)	126
280	3	176	–	–	80	310	8	M12	23036..-K	H3036(-HG)	65,5
300	3	220	–	–	95	350	8	M16	23136..-K	H3136(-HG)	104
320	3	240	–	–	105	370	8	M16	23236..-K	H2336(-HG)	133
320	–	–	248	114	–	370	8	M16	22236..-K	H3136(-HG)	126
280	–	–	201	88	–	310	8	M12	23036..-K	H3036(-HG)	65,5
300	–	–	250	115	–	350	8	M16	23136..-K	H3136(-HG)	104
320	–	–	274	127	–	370	8	M16	23236..-K	H2336(-HG)	133
320	3	–	248	114	–	370	8	M16	22236..-K	H3136(-HG)	126
280	3	–	201	88	–	310	8	M12	23036..-K	H3036(-HG)	65,5
300	3	–	250	115	–	350	8	M16	23136..-K	H3136(-HG)	104
320	3	–	274	127	–	370	8	M16	23236..-K	H2336(-HG)	133

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

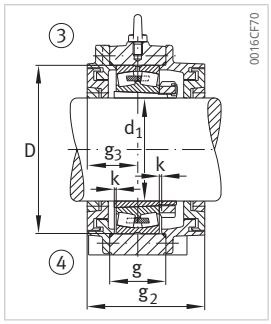




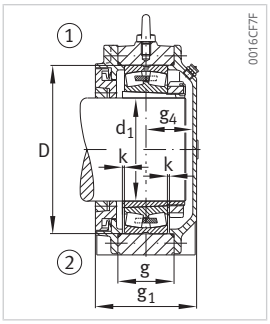
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2238-H-W-Y-AF-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-Y-AF-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-Y-AF-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-Y-AF-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-Y-AL-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-Y-AL-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-Y-AL-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-Y-AL-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-Y-BF-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-Y-BF-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-Y-BF-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-Y-BF-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-Y-BL-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-Y-BL-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-Y-BL-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-Y-BL-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-T-AF-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-T-AF-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-T-AF-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-T-AF-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-T-AL-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-T-AL-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-T-AL-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-T-AL-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-T-BF-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-T-BF-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-T-BF-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-T-BF-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2238-H-W-T-BL-D	**170**	220	455	115	220	85	710	560	120	52	42	M36
BND3038-H-W-T-BL-D	**170**	185	370	105	180	55	570	480	105	45	36	M30
BND3138-H-W-T-BL-D	**170**	210	425	130	210	65	680	550	120	45	36	M30
BND3238-H-W-T-BL-D	**170**	220	455	143	220	85	710	560	120	52	42	M36
BND2240-H-W-Y-AF-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-Y-AF-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-Y-AF-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-Y-AF-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2240-H-W-Y-AL-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-Y-AL-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-Y-AL-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-Y-AL-D	**180**	235	475	158	240	75	780	640	140	52	42	M36



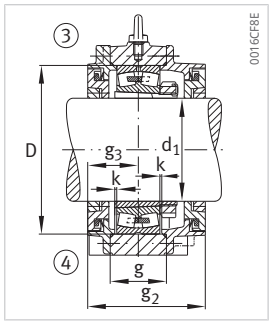
Design A,
Labyrinth seal



Design B,
Labyrinth seal



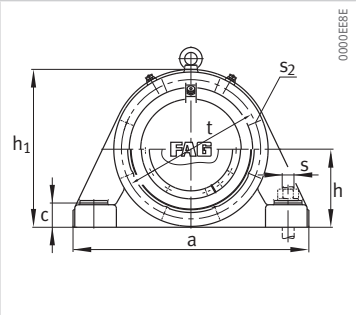
Design A,
Taconite seal

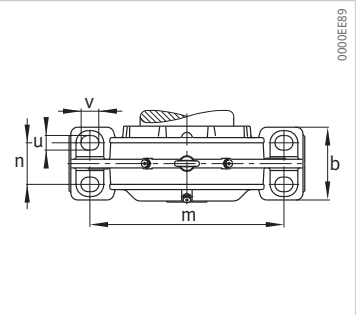


Design B,
Taconite seal

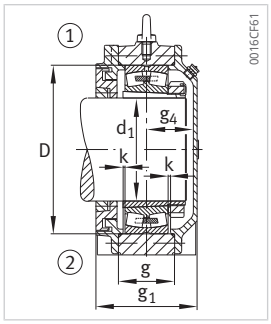
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
340	–	222	–	–	98	380	8	M16	22238..-K	H3138(-HG)	168
290	–	181	–	–	80	325	8	M12	23038..-K	H3038(-HG)	82
320	–	232	–	–	98	370	8	M16	23138..-K	H3138(-HG)	119
340	–	250	–	–	112	380	8	M16	23238..-K	H2338(-HG)	161,5
340	3	222	–	–	98	380	8	M16	22238..-K	H3138(-HG)	168
290	3	181	–	–	80	325	8	M12	23038..-K	H3038(-HG)	82
320	3	232	–	–	98	370	8	M16	23138..-K	H3138(-HG)	119
340	3	250	–	–	112	380	8	M16	23238..-K	H2338(-HG)	161,5
340	–	–	258	114	–	380	8	M16	22238..-K	H3138(-HG)	168
290	–	–	211	93	–	325	8	M12	23038..-K	H3038(-HG)	82
320	–	–	266	123	–	370	8	M16	23138..-K	H3138(-HG)	119
340	–	–	286	128	–	380	8	M16	23238..-K	H2338(-HG)	161,5
340	3	–	258	114	–	380	8	M16	22238..-K	H3138(-HG)	168
290	3	–	211	93	–	325	8	M12	23038..-K	H3038(-HG)	82
320	3	–	266	123	–	370	8	M16	23138..-K	H3138(-HG)	119
340	3	–	286	128	–	380	8	M16	23238..-K	H2338(-HG)	161,5
340	–	222	–	–	98	380	8	M16	22238..-K	H3138(-HG)	168
290	–	181	–	–	80	325	8	M12	23038..-K	H3038(-HG)	82
320	–	232	–	–	98	370	8	M16	23138..-K	H3138(-HG)	119
340	–	250	–	–	112	380	8	M16	23238..-K	H2338(-HG)	161,5
340	3	222	–	–	98	380	8	M16	22238..-K	H3138(-HG)	168
290	3	181	–	–	80	325	8	M12	23038..-K	H3038(-HG)	82
320	3	232	–	–	98	370	8	M16	23138..-K	H3138(-HG)	119
340	3	250	–	–	112	380	8	M16	23238..-K	H2338(-HG)	161,5
340	–	–	258	114	–	380	8	M16	22238..-K	H3138(-HG)	168
290	–	–	211	93	–	325	8	M12	23038..-K	H3038(-HG)	82
320	–	–	266	123	–	370	8	M16	23138..-K	H3138(-HG)	119
340	–	–	286	128	–	380	8	M16	23238..-K	H2338(-HG)	161,5
340	3	–	258	114	–	380	8	M16	22238..-K	H3138(-HG)	168
290	3	–	211	93	–	325	8	M12	23038..-K	H3038(-HG)	82
320	3	–	266	123	–	370	8	M16	23138..-K	H3138(-HG)	119
340	3	–	286	128	–	380	8	M16	23238..-K	H2338(-HG)	161,5
360	–	230	–	–	99	420	8	M16	22240..-K	H3140(-HG)	179
310	–	196	–	–	90	340	8	M16	23040..-K	H3040(-HG)	91
340	–	242	–	–	108	380	8	M16	23140..-K	H3140(-HG)	162
360	–	260	–	–	114	420	8	M16	23240..-K	H2340(-HG)	195,5
360	4	230	–	–	99	420	8	M16	22240..-K	H3140(-HG)	179
310	3	196	–	–	90	340	8	M16	23040..-K	H3040(-HG)	91
340	3	242	–	–	108	380	8	M16	23140..-K	H3140(-HG)	162
360	4	260	–	–	114	420	8	M16	23240..-K	H2340(-HG)	195,5

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

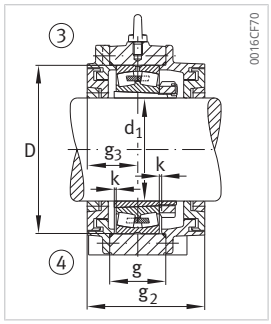




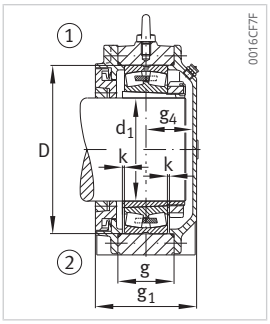
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2240-H-W-Y-BF-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-Y-BF-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-Y-BF-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-Y-BF-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2240-H-W-Y-BL-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-Y-BL-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-Y-BL-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-Y-BL-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2240-H-W-T-AF-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-T-AF-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-T-AF-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-T-AF-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2240-H-W-T-AL-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-T-AL-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-T-AL-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-T-AL-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2240-H-W-T-BF-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-T-BF-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-T-BF-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-T-BF-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2240-H-W-T-BL-D	**180**	235	475	128	240	75	780	640	140	52	42	M36
BND3040-H-W-T-BL-D	**180**	200	400	110	190	60	600	510	110	45	36	M30
BND3140-H-W-T-BL-D	**180**	220	455	135	220	85	710	560	120	52	42	M36
BND3240-H-W-T-BL-D	**180**	235	475	158	240	75	780	640	140	52	42	M36
BND2244-H-W-Y-AF-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-Y-AF-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-Y-AF-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-Y-AF-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2244-H-W-Y-AL-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-Y-AL-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-Y-AL-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-Y-AL-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2244-H-W-Y-BF-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-Y-BF-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-Y-BF-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-Y-BF-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2244-H-W-Y-BL-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-Y-BL-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-Y-BL-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-Y-BL-D	**200**	260	525	175	250	80	850	700	140	52	42	M36



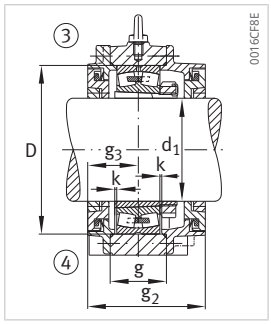
Design A,
Labyrinth seal



Design B,
Labyrinth seal



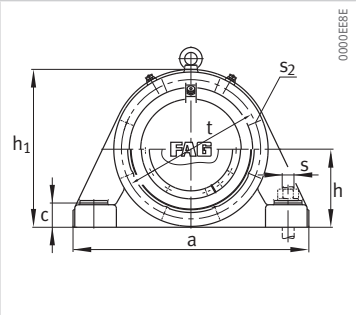
Design A,
Taconite seal

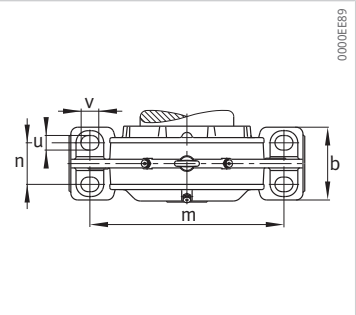


Design B,
Taconite seal

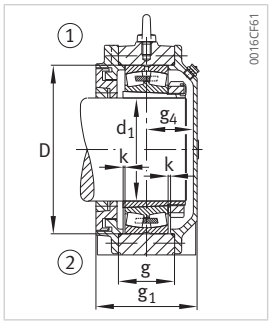
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
360	–	–	269	123	–	420	8	M16	22240..-K	H3140(-HG)	179
310	–	–	226	98	–	340	8	M16	23040..-K	H3040(-HG)	91
340	–	–	278	124	–	380	8	M16	23140..-K	H3140(-HG)	162
360	–	–	299	138	–	420	8	M16	23240..-K	H2340(-HG)	195,5
360	4	–	269	123	–	420	8	M16	22240..-K	H3140(-HG)	179
310	3	–	226	98	–	340	8	M16	23040..-K	H3040(-HG)	91
340	3	–	278	124	–	380	8	M16	23140..-K	H3140(-HG)	162
360	4	–	299	138	–	420	8	M16	23240..-K	H2340(-HG)	195,5
360	–	230	–	–	99	420	8	M16	22240..-K	H3140(-HG)	179
310	–	196	–	–	90	340	8	M16	23040..-K	H3040(-HG)	91
340	–	242	–	–	108	380	8	M16	23140..-K	H3140(-HG)	162
360	–	260	–	–	114	420	8	M16	23240..-K	H2340(-HG)	195,5
360	4	230	–	–	99	420	8	M16	22240..-K	H3140(-HG)	179
310	3	196	–	–	90	340	8	M16	23040..-K	H3040(-HG)	91
340	3	242	–	–	108	380	8	M16	23140..-K	H3140(-HG)	162
360	4	260	–	–	114	420	8	M16	23240..-K	H2340(-HG)	195,5
360	–	–	269	123	–	420	8	M16	22240..-K	H3140(-HG)	179
310	–	–	226	98	–	340	8	M16	23040..-K	H3040(-HG)	91
340	–	–	278	124	–	380	8	M16	23140..-K	H3140(-HG)	162
360	–	–	299	138	–	420	8	M16	23240..-K	H2340(-HG)	195,5
360	4	–	269	123	–	420	8	M16	22240..-K	H3140(-HG)	179
310	3	–	226	98	–	340	8	M16	23040..-K	H3040(-HG)	91
340	3	–	278	124	–	380	8	M16	23140..-K	H3140(-HG)	162
360	4	–	299	138	–	420	8	M16	23240..-K	H2340(-HG)	195,5
400	–	264	–	–	112	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	–	206	–	–	95	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	–	252	–	–	110	420	8	M16	23144..-K	H3144X(-HG)	181
400	–	279	–	–	122	445	8	M20	23244..-K	H2344X(-HG)	229
400	4	264	–	–	112	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	3	206	–	–	95	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	4	252	–	–	110	420	8	M16	23144..-K	H3144X(-HG)	181
400	4	279	–	–	122	445	8	M20	23244..-K	H2344X(-HG)	229
400	–	–	314	142	–	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	–	–	241	103	–	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	–	–	291	134	–	420	8	M16	23144..-K	H3144X(-HG)	181
400	–	–	329	147	–	445	8	M20	23244..-K	H2344X(-HG)	229
400	4	–	314	142	–	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	3	–	241	103	–	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	4	–	291	134	–	420	8	M16	23144..-K	H3144X(-HG)	181
400	4	–	329	147	–	445	8	M20	23244..-K	H2344X(-HG)	229

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

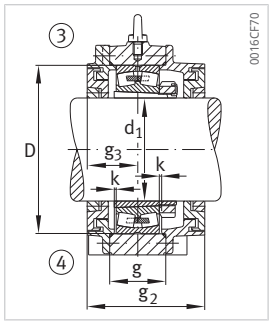




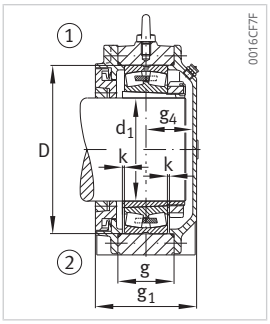
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2244-H-W-T-AF-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-T-AF-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-T-AF-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-T-AF-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2244-H-W-T-AL-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-T-AL-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-T-AL-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-T-AL-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2244-H-W-T-BF-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-T-BF-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-T-BF-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-T-BF-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2244-H-W-T-BL-D	**200**	270	550	140	250	80	890	720	140	52	42	M36
BND3044-H-W-T-BL-D	**200**	215	430	115	200	65	640	540	115	52	42	M36
BND3144-H-W-T-BL-D	**200**	235	475	150	240	75	780	640	140	52	42	M36
BND3244-H-W-T-BL-D	**200**	260	525	175	250	80	850	700	140	52	42	M36
BND2248-H-W-Y-AF-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-Y-AF-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-Y-AF-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-Y-AF-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2248-H-W-Y-AL-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-Y-AL-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-Y-AL-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-Y-AL-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2248-H-W-Y-BF-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-Y-BF-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-Y-BF-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-Y-BF-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2248-H-W-Y-BL-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-Y-BL-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-Y-BL-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-Y-BL-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2248-H-W-T-AF-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-T-AF-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-T-AF-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-T-AF-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2248-H-W-T-AL-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-T-AL-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-T-AL-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-T-AL-D	**220**	290	585	190	250	90	900	750	140	52	42	M36



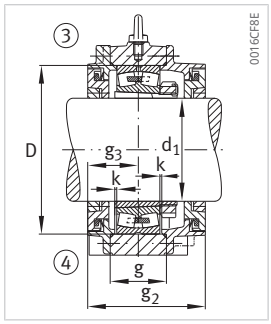
Design A,
Labyrinth seal



Design B,
Labyrinth seal



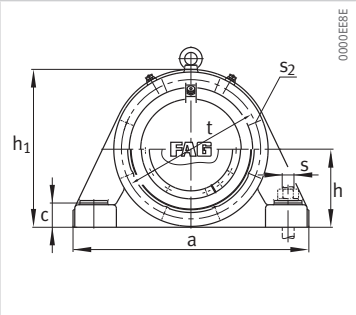
Design A,
Taconite seal

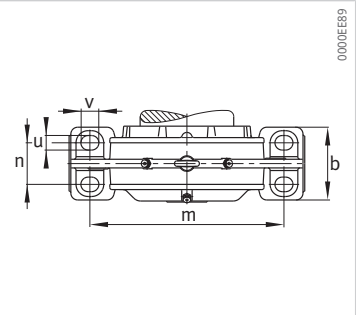


Design B,
Taconite seal

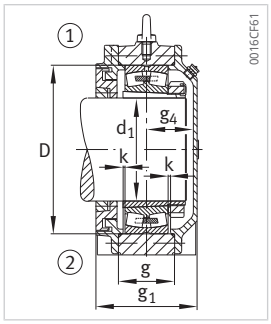
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
400	–	264	–	–	112	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	–	206	–	–	95	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	–	252	–	–	110	420	8	M16	23144..-K	H3144X(-HG)	181
400	–	279	–	–	122	445	8	M20	23244..-K	H2344X(-HG)	229
400	4	264	–	–	112	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	3	206	–	–	95	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	4	252	–	–	110	420	8	M16	23144..-K	H3144X(-HG)	181
400	4	279	–	–	122	445	8	M20	23244..-K	H2344X(-HG)	229
400	–	–	314	142	–	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	–	–	241	103	–	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	–	–	291	134	–	420	8	M16	23144..-K	H3144X(-HG)	181
400	–	–	329	147	–	445	8	M20	23244..-K	H2344X(-HG)	229
400	4	–	314	142	–	455	8	M20	22244..-K	H3144X(-HG)	275,5
340	3	–	241	103	–	375	8	M16	23044..-K	H3044X(-HG)	95,5
370	4	–	291	134	–	420	8	M16	23144..-K	H3144X(-HG)	181
400	4	–	329	147	–	445	8	M20	23244..-K	H2344X(-HG)	229
440	–	268	–	–	120	510	8	M20	22248..-K	H3148X(-HG)	300
360	–	216	–	–	100	400	8	M16	23048..-K	H3048(-HG)	122
400	–	284	–	–	122	455	8	M20	23148..-K	H3148X(-HG)	265
440	–	308	–	–	140	510	8	M20	23248..-K	H2348X(-HG)	313
440	4	268	–	–	120	510	8	M20	22248..-K	H3148X(-HG)	300
360	4	216	–	–	100	400	8	M16	23048..-K	H3048(-HG)	122
400	4	284	–	–	122	455	8	M20	23148..-K	H3148X(-HG)	265
440	4	308	–	–	140	510	8	M20	23248..-K	H2348X(-HG)	313
440	–	–	311	138	–	510	8	M20	22248..-K	H3148X(-HG)	300
360	–	–	251	108	–	400	8	M16	23048..-K	H3048(-HG)	122
400	–	–	334	152	–	455	8	M20	23148..-K	H3148X(-HG)	265
440	–	–	351	158	–	510	8	M20	23248..-K	H2348X(-HG)	313
440	4	–	311	138	–	510	8	M20	22248..-K	H3148X(-HG)	300
360	4	–	251	108	–	400	8	M16	23048..-K	H3048(-HG)	122
400	4	–	334	152	–	455	8	M20	23148..-K	H3148X(-HG)	265
440	4	–	351	158	–	510	8	M20	23248..-K	H2348X(-HG)	313
440	–	268	–	–	120	510	8	M20	22248..-K	H3148X(-HG)	300
360	–	216	–	–	100	400	8	M16	23048..-K	H3048(-HG)	122
400	–	284	–	–	122	455	8	M20	23148..-K	H3148X(-HG)	265
440	–	308	–	–	140	510	8	M20	23248..-K	H2348X(-HG)	313
440	4	268	–	–	120	510	8	M20	22248..-K	H3148X(-HG)	300
360	4	216	–	–	100	400	8	M16	23048..-K	H3048(-HG)	122
400	4	284	–	–	122	455	8	M20	23148..-K	H3148X(-HG)	265
440	4	308	–	–	140	510	8	M20	23248..-K	H2348X(-HG)	313

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

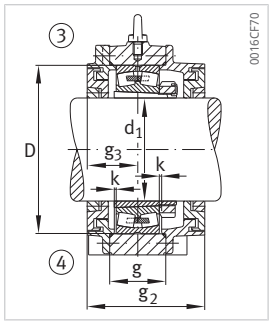




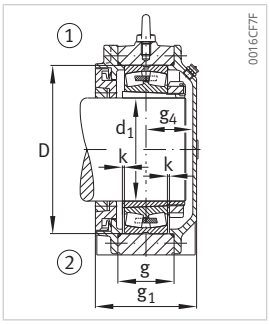
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2248-H-W-T-BF-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-T-BF-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-T-BF-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-T-BF-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2248-H-W-T-BL-D	**220**	290	585	150	250	90	900	750	140	52	42	M36
BND3048-H-W-T-BL-D	**220**	225	455	120	210	70	680	560	120	52	42	M36
BND3148-H-W-T-BL-D	**220**	270	550	160	250	80	890	720	140	52	42	M36
BND3248-H-W-T-BL-D	**220**	290	585	190	250	90	900	750	140	52	42	M36
BND2252-H-W-Y-AF-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-Y-AF-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-Y-AF-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-Y-AF-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-Y-AL-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-Y-AL-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-Y-AL-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-Y-AL-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-Y-BF-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-Y-BF-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-Y-BF-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-Y-BF-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-Y-BL-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-Y-BL-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-Y-BL-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-Y-BL-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-T-AF-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-T-AF-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-T-AF-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-T-AF-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-T-AL-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-T-AL-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-T-AL-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-T-AL-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-T-BF-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-T-BF-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-T-BF-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-T-BF-D	**240**	310	625	205	290	95	960	800	160	52	42	M36
BND2252-H-W-T-BL-D	**240**	310	625	161	290	95	960	800	160	52	42	M36
BND3052-H-W-T-BL-D	**240**	250	500	130	220	75	720	600	130	52	42	M36
BND3152-H-W-T-BL-D	**240**	290	585	174	250	90	900	750	140	52	42	M36
BND3252-H-W-T-BL-D	**240**	310	625	205	290	95	960	800	160	52	42	M36



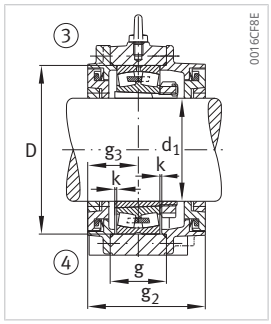
Design A,
Labyrinth seal



Design B,
Labyrinth seal



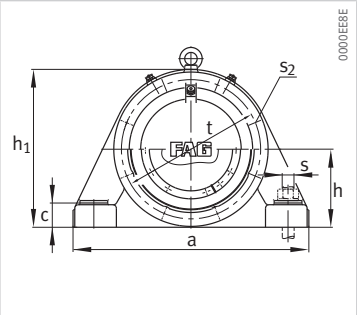
Design A,
Taconite seal

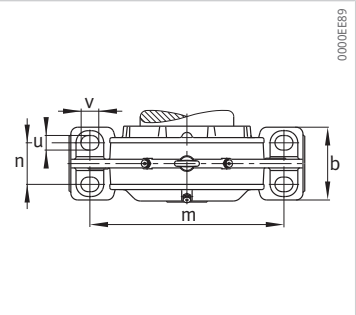


Design B,
Taconite seal

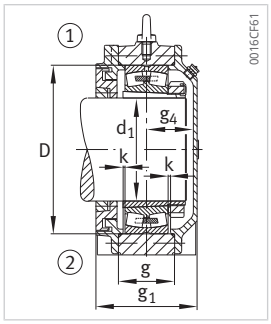
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
440	–	–	311	138	–	510	8	M20	22248..-K	H3148X(-HG)	300
360	–	–	251	108	–	400	8	M16	23048..-K	H3048(-HG)	122
400	–	–	334	152	–	455	8	M20	23148..-K	H3148X(-HG)	265
440	–	–	351	158	–	510	8	M20	23248..-K	H2348X(-HG)	313
440	4	–	311	138	–	510	8	M20	22248..-K	H3148X(-HG)	300
360	4	–	251	108	–	400	8	M16	23048..-K	H3048(-HG)	122
400	4	–	334	152	–	455	8	M20	23148..-K	H3148X(-HG)	265
440	4	–	351	158	–	510	8	M20	23248..-K	H2348X(-HG)	313
480	–	286	–	–	126	535	8	M20	22252..-K	H3152X(-HG)	352
400	–	226	–	–	103	440	8	M16	23052..-K	H3052X(-HG)	151
440	–	292	–	–	132	510	8	M20	23152..-K	H3152X(-HG)	294
480	–	330	–	–	148	535	8	M20	23252..-K	H2352X(-HG)	360
480	3	286	–	–	126	535	8	M20	22252..-K	H3152X(-HG)	352
400	4	226	–	–	103	440	8	M16	23052..-K	H3052X(-HG)	151
440	4	292	–	–	132	510	8	M20	23152..-K	H3152X(-HG)	294
480	3	330	–	–	148	535	8	M20	23252..-K	H2352X(-HG)	360
480	–	–	326	148	–	535	8	M20	22252..-K	H3152X(-HG)	352
400	–	–	261	113	–	440	8	M16	23052..-K	H3052X(-HG)	151
440	–	–	335	150	–	510	8	M20	23152..-K	H3152X(-HG)	294
480	–	–	370	170	–	535	8	M20	23252..-K	H2352X(-HG)	360
480	3	–	326	148	–	535	8	M20	22252..-K	H3152X(-HG)	352
400	4	–	261	113	–	440	8	M16	23052..-K	H3052X(-HG)	151
440	4	–	335	150	–	510	8	M20	23152..-K	H3152X(-HG)	294
480	3	–	370	170	–	535	8	M20	23252..-K	H2352X(-HG)	360
480	–	286	–	–	126	535	8	M20	22252..-K	H3152X(-HG)	352
400	–	226	–	–	103	440	8	M16	23052..-K	H3052X(-HG)	151
440	–	292	–	–	132	510	8	M20	23152..-K	H3152X(-HG)	294
480	–	330	–	–	148	535	8	M20	23252..-K	H2352X(-HG)	360
480	3	286	–	–	126	535	8	M20	22252..-K	H3152X(-HG)	352
400	4	226	–	–	103	440	8	M16	23052..-K	H3052X(-HG)	151
440	4	292	–	–	132	510	8	M20	23152..-K	H3152X(-HG)	294
480	3	330	–	–	148	535	8	M20	23252..-K	H2352X(-HG)	360
480	–	–	326	148	–	535	8	M20	22252..-K	H3152X(-HG)	352
400	–	–	261	113	–	440	8	M16	23052..-K	H3052X(-HG)	151
440	–	–	335	150	–	510	8	M20	23152..-K	H3152X(-HG)	294
480	–	–	370	170	–	535	8	M20	23252..-K	H2352X(-HG)	360
480	3	–	326	148	–	535	8	M20	22252..-K	H3152X(-HG)	352
400	4	–	261	113	–	440	8	M16	23052..-K	H3052X(-HG)	151
440	4	–	335	150	–	510	8	M20	23152..-K	H3152X(-HG)	294
480	3	–	370	170	–	535	8	M20	23252..-K	H2352X(-HG)	360

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

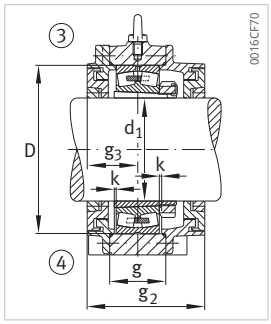




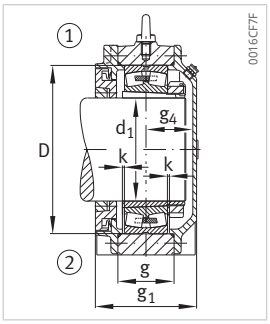
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2256-H-W-Y-AF-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-Y-AF-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-Y-AF-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-Y-AF-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-Y-AL-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-Y-AL-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-Y-AL-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-Y-AL-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-Y-BF-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-Y-BF-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-Y-BF-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-Y-BF-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-Y-BL-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-Y-BL-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-Y-BL-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-Y-BL-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-T-AF-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-T-AF-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-T-AF-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-T-AF-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-T-AL-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-T-AL-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-T-AL-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-T-AL-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-T-BF-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-T-BF-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-T-BF-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-T-BF-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2256-H-W-T-BL-D	**260**	320	645	160	300	100	1000	840	170	52	42	M36
BND3056-H-W-T-BL-D	**260**	260	520	135	240	80	760	630	140	52	42	M36
BND3156-H-W-T-BL-D	**260**	290	585	176	250	90	900	750	140	52	42	M36
BND3256-H-W-T-BL-D	**260**	320	645	206	300	100	1000	840	170	52	42	M36
BND2260-H-W-Y-AF-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-Y-AF-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-Y-AF-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-Y-AF-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2260-H-W-Y-AL-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-Y-AL-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-Y-AL-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-Y-AL-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48



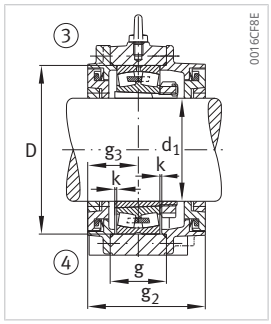
Design A,
Labyrinth seal



Design B,
Labyrinth seal



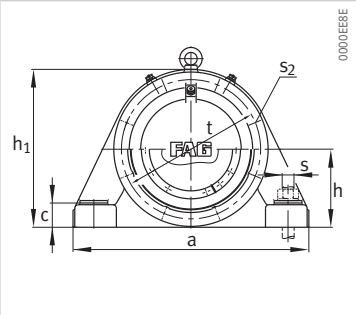
Design A,
Taconite seal

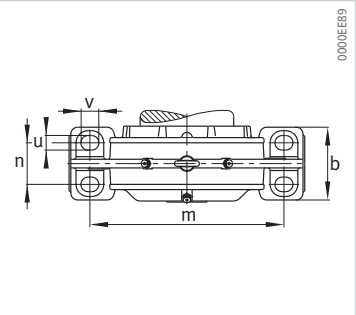


Design B,
Taconite seal

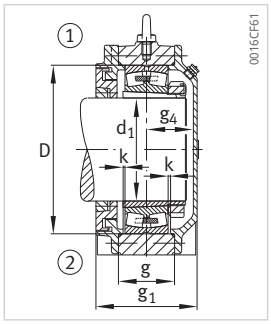
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
500	–	297	–	–	128	555	8	M24	22256..-K	H3156X(-HG)	396
420	–	236	–	–	108	460	8	M16	23056..-K	H3056(-HG)	170
460	–	294	–	–	133	510	8	M20	23156..-K	H3156X(-HG)	317
500	–	343	–	–	151	555	8	M24	23256..-K	H2356X(-HG)	462
500	4	297	–	–	128	555	8	M24	22256..-K	H3156X(-HG)	396
420	4	236	–	–	108	460	8	M16	23056..-K	H3056(-HG)	170
460	4	294	–	–	133	510	8	M20	23156..-K	H3156X(-HG)	317
500	4	343	–	–	151	555	8	M24	23256..-K	H2356X(-HG)	462
500	–	–	354	157	–	555	8	M24	22256..-K	H3156X(-HG)	396
420	–	–	281	118	–	460	8	M16	23056..-K	H3056(-HG)	170
460	–	–	337	151	–	510	8	M20	23156..-K	H3156X(-HG)	317
500	–	–	400	180	–	555	8	M24	23256..-K	H2356X(-HG)	462
500	4	–	354	157	–	555	8	M24	22256..-K	H3156X(-HG)	396
420	4	–	281	118	–	460	8	M16	23056..-K	H3056(-HG)	170
460	4	–	337	151	–	510	8	M20	23156..-K	H3156X(-HG)	317
500	4	–	400	180	–	555	8	M24	23256..-K	H2356X(-HG)	462
500	–	297	–	–	128	555	8	M24	22256..-K	H3156X(-HG)	396
420	–	236	–	–	108	460	8	M16	23056..-K	H3056(-HG)	170
460	–	294	–	–	133	510	8	M20	23156..-K	H3156X(-HG)	317
500	–	343	–	–	151	555	8	M24	23256..-K	H2356X(-HG)	462
500	4	297	–	–	128	555	8	M24	22256..-K	H3156X(-HG)	396
420	4	236	–	–	108	460	8	M16	23056..-K	H3056(-HG)	170
460	4	294	–	–	133	510	8	M20	23156..-K	H3156X(-HG)	317
500	4	343	–	–	151	555	8	M24	23256..-K	H2356X(-HG)	462
500	–	–	354	157	–	555	8	M24	22256..-K	H3156X(-HG)	396
420	–	–	281	118	–	460	8	M16	23056..-K	H3056(-HG)	170
460	–	–	337	151	–	510	8	M20	23156..-K	H3156X(-HG)	317
500	–	–	400	180	–	555	8	M24	23256..-K	H2356X(-HG)	462
500	4	–	354	157	–	555	8	M24	22256..-K	H3156X(-HG)	396
420	4	–	281	118	–	460	8	M16	23056..-K	H3056(-HG)	170
460	4	–	337	151	–	510	8	M20	23156..-K	H3156X(-HG)	317
500	4	–	400	180	–	555	8	M24	23256..-K	H2356X(-HG)	462
540	–	317	–	–	149	600	8	M24	22260..-K	H3160(-HG)	460
460	–	261	–	–	121	510	8	M16	23060..-K	H3060(-HG)	208
500	–	327	–	–	143	555	8	M24	23160..-K	H3160(-HG)	377
540	–	369	–	–	175	600	8	M24	23260..-K	H3260(-HG)	538
540	4	317	–	–	149	600	8	M24	22260..-K	H3160(-HG)	460
460	4	261	–	–	121	510	8	M16	23060..-K	H3060(-HG)	208
500	4	327	–	–	143	555	8	M24	23160..-K	H3160(-HG)	377
540	4	369	–	–	175	600	8	M24	23260..-K	H3260(-HG)	538

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

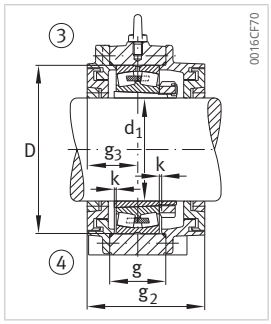




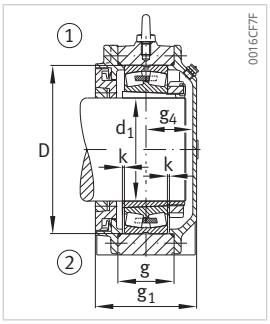
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2260-H-W-Y-BF-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-Y-BF-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-Y-BF-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-Y-BF-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2260-H-W-Y-BL-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-Y-BL-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-Y-BL-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-Y-BL-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2260-H-W-T-AF-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-T-AF-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-T-AF-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-T-AF-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2260-H-W-T-AL-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-T-AL-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-T-AL-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-T-AL-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2260-H-W-T-BF-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-T-BF-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-T-BF-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-T-BF-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2260-H-W-T-BL-D	**280**	350	695	178	330	105	1100	920	180	75	56	M48
BND3060-H-W-T-BL-D	**280**	285	570	140	250	85	820	690	150	52	42	M36
BND3160-H-W-T-BL-D	**280**	320	645	190	300	100	1000	840	170	52	42	M36
BND3260-H-W-T-BL-D	**280**	350	705	230	330	105	1100	920	180	75	56	M48
BND2264-H-W-Y-AF-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-Y-AF-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-Y-AF-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-Y-AF-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2264-H-W-Y-AL-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-Y-AL-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-Y-AL-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-Y-AL-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2264-H-W-Y-BF-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-Y-BF-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-Y-BF-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-Y-BF-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2264-H-W-Y-BL-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-Y-BL-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-Y-BL-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-Y-BL-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48



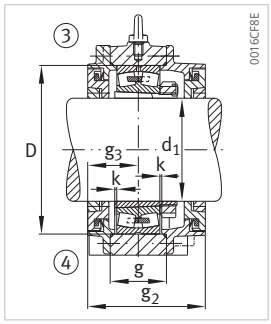
Design A,
Labyrinth seal



Design B,
Labyrinth seal



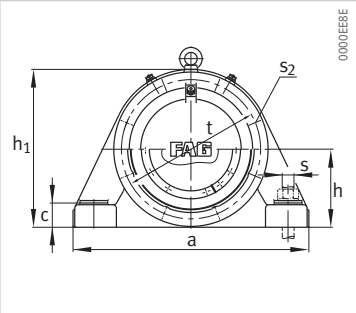
Design A,
Taconite seal

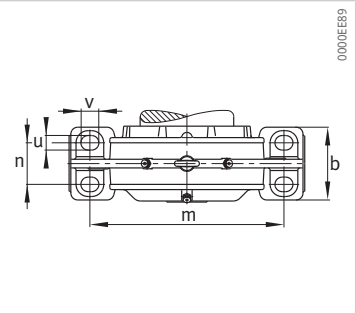


Design B,
Taconite seal

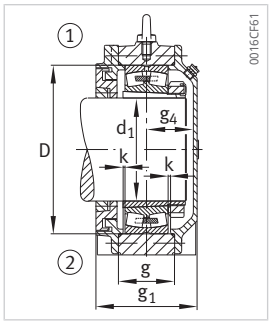
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
540	–	–	352	156	–	600	8	M24	22260..-K	H3160(-HG)	460
460	–	–	296	128	–	510	8	M16	23060..-K	H3060(-HG)	208
500	–	–	384	172	–	555	8	M24	23160..-K	H3160(-HG)	377
540	–	–	404	182	–	600	8	M24	23260..-K	H3260(-HG)	538
540	4	–	352	156	–	600	8	M24	22260..-K	H3160(-HG)	460
460	4	–	296	128	–	510	8	M16	23060..-K	H3060(-HG)	208
500	4	–	384	172	–	555	8	M24	23160..-K	H3160(-HG)	377
540	4	–	404	182	–	600	8	M24	23260..-K	H3260(-HG)	538
540	–	317	–	–	149	600	8	M24	22260..-K	H3160(-HG)	460
460	–	261	–	–	121	510	8	M16	23060..-K	H3060(-HG)	208
500	–	327	–	–	143	555	8	M24	23160..-K	H3160(-HG)	377
540	–	369	–	–	175	600	8	M24	23260..-K	H3260(-HG)	538
540	4	317	–	–	149	600	8	M24	22260..-K	H3160(-HG)	460
460	4	261	–	–	121	510	8	M16	23060..-K	H3060(-HG)	208
500	4	327	–	–	143	555	8	M24	23160..-K	H3160(-HG)	377
540	4	369	–	–	175	600	8	M24	23260..-K	H3260(-HG)	538
540	–	–	352	156	–	600	8	M24	22260..-K	H3160(-HG)	460
460	–	–	296	128	–	510	8	M16	23060..-K	H3060(-HG)	208
500	–	–	384	172	–	555	8	M24	23160..-K	H3160(-HG)	377
540	–	–	404	182	–	600	8	M24	23260..-K	H3260(-HG)	538
540	4	–	352	156	–	600	8	M24	22260..-K	H3160(-HG)	460
460	4	–	296	128	–	510	8	M16	23060..-K	H3060(-HG)	208
500	4	–	384	172	–	555	8	M24	23160..-K	H3160(-HG)	377
540	4	–	404	182	–	600	8	M24	23260..-K	H3260(-HG)	538
580	–	333	–	–	158	640	8	M24	22264..-K	H3164-HG	570
480	–	266	–	–	123	530	8	M16	23064..-K	H3064-HG	236
540	–	359	–	–	161	590	8	M24	23164..-K	H3164-HG	476,5
580	–	391	–	–	187	640	8	M24	23264..-K	H3264-HG	630
580	5	333	–	–	158	640	8	M24	22264..-K	H3164-HG	570
480	4	266	–	–	123	530	8	M16	23064..-K	H3064-HG	236
540	4	359	–	–	161	590	8	M24	23164..-K	H3164-HG	476,5
580	5	391	–	–	187	640	8	M24	23264..-K	H3264-HG	630
580	–	–	381	163	–	640	8	M24	22264..-K	H3164-HG	570
480	–	–	311	133	–	530	8	M16	23064..-K	H3064-HG	236
540	–	–	412	186	–	590	8	M24	23164..-K	H3164-HG	476,5
580	–	–	439	192	–	640	8	M24	23264..-K	H3264-HG	630
580	5	–	381	163	–	640	8	M24	22264..-K	H3164-HG	570
480	4	–	311	133	–	530	8	M16	23064..-K	H3064-HG	236
540	4	–	412	186	–	590	8	M24	23164..-K	H3164-HG	476,5
580	5	–	439	192	–	640	8	M24	23264..-K	H3264-HG	630

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

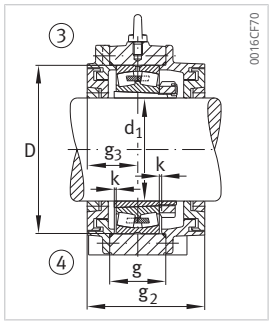




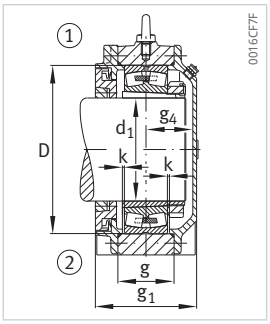
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2264-H-W-T-AF-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-T-AF-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-T-AF-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-T-AF-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2264-H-W-T-AL-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-T-AL-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-T-AL-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-T-AL-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2264-H-W-T-BF-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-T-BF-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-T-BF-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-T-BF-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2264-H-W-T-BL-D	**300**	370	745	180	360	115	1150	960	200	75	56	M48
BND3064-H-W-T-BL-D	**300**	295	590	150	260	90	860	730	160	52	42	M36
BND3164-H-W-T-BL-D	**300**	350	700	210	300	100	1150	940	160	52	42	M36
BND3264-H-W-T-BL-D	**300**	370	745	238	360	115	1150	960	200	75	56	M48
BND2268-H-W-Y-AF-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-Y-AF-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-Y-AF-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-Y-AF-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2268-H-W-Y-AL-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-Y-AL-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-Y-AL-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-Y-AL-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2268-H-W-Y-BF-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-Y-BF-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-Y-BF-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-Y-BF-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2268-H-W-Y-BL-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-Y-BL-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-Y-BL-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-Y-BL-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2268-H-W-T-AF-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-T-AF-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-T-AF-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-T-AF-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2268-H-W-T-AL-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-T-AL-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-T-AL-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-T-AL-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56



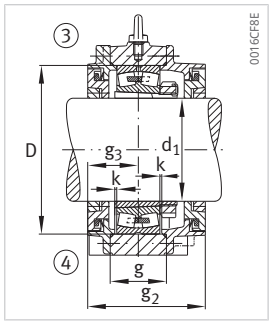
Design A,
Labyrinth seal



Design B,
Labyrinth seal



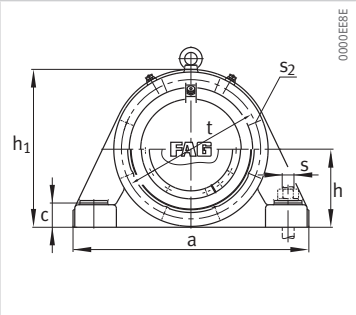
Design A,
Taconite seal

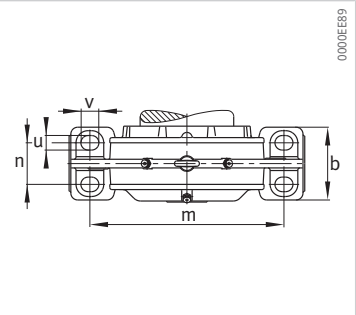


Design B,
Taconite seal

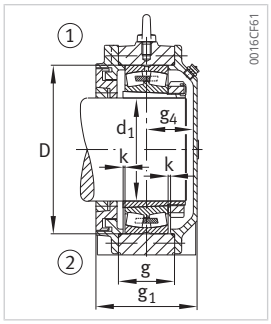
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
580	–	333	–	–	158	640	8	M24	22264..-K	H3164-HG	570
480	–	266	–	–	123	530	8	M16	23064..-K	H3064-HG	236
540	–	359	–	–	161	590	8	M24	23164..-K	H3164-HG	476,5
580	–	391	–	–	187	640	8	M24	23264..-K	H3264-HG	630
580	5	333	–	–	158	640	8	M24	22264..-K	H3164-HG	570
480	4	266	–	–	123	530	8	M16	23064..-K	H3064-HG	236
540	4	359	–	–	161	590	8	M24	23164..-K	H3164-HG	476,5
580	5	391	–	–	187	640	8	M24	23264..-K	H3264-HG	630
580	–	–	381	163	–	640	8	M24	22264..-K	H3164-HG	570
480	–	–	311	133	–	530	8	M16	23064..-K	H3064-HG	236
540	–	–	412	186	–	590	8	M24	23164..-K	H3164-HG	476,5
580	–	–	439	192	–	640	8	M24	23264..-K	H3264-HG	630
580	5	–	381	163	–	640	8	M24	22264..-K	H3164-HG	570
480	4	–	311	133	–	530	8	M16	23064..-K	H3064-HG	236
540	4	–	412	186	–	590	8	M24	23164..-K	H3164-HG	476,5
580	5	–	439	192	–	640	8	M24	23264..-K	H3264-HG	630
620	–	375	–	–	176	680	8	M30	22268..-K	H3168-HG	600
520	–	276	–	–	132	565	8	M20	23068..-K	H3068-HG	284
580	–	373	–	–	178	640	8	M24	23168..-K	H3168-HG	492
620	–	434	–	–	205	680	8	M30	23268..-K	H3268-HG	715
620	5	375	–	–	176	680	8	M30	22268..-K	H3168-HG	600
520	5	276	–	–	132	565	8	M20	23068..-K	H3068-HG	284
580	5	373	–	–	178	640	8	M24	23168..-K	H3168-HG	492
620	5	434	–	–	205	680	8	M30	23268..-K	H3268-HG	715
620	–	–	430	187,5	–	680	8	M30	22268..-K	H3168-HG	600
520	–	–	311	133	–	565	8	M20	23068..-K	H3068-HG	284
580	–	–	421	183	–	640	8	M24	23168..-K	H3168-HG	492
620	–	–	489	217	–	680	8	M30	23268..-K	H3268-HG	715
620	5	–	430	187,5	–	680	8	M30	22268..-K	H3168-HG	600
520	5	–	311	133	–	565	8	M20	23068..-K	H3068-HG	284
580	5	–	421	183	–	640	8	M24	23168..-K	H3168-HG	492
620	5	–	489	217	–	680	8	M30	23268..-K	H3268-HG	715
620	–	375	–	–	176	680	8	M30	22268..-K	H3168-HG	600
520	–	276	–	–	132	565	8	M20	23068..-K	H3068-HG	284
580	–	373	–	–	178	640	8	M24	23168..-K	H3168-HG	492
620	–	434	–	–	205	680	8	M30	23268..-K	H3268-HG	715
620	5	375	–	–	176	680	8	M30	22268..-K	H3168-HG	600
520	5	276	–	–	132	565	8	M20	23068..-K	H3068-HG	284
580	5	373	–	–	178	640	8	M24	23168..-K	H3168-HG	492
620	5	434	–	–	205	680	8	M30	23268..-K	H3268-HG	715

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

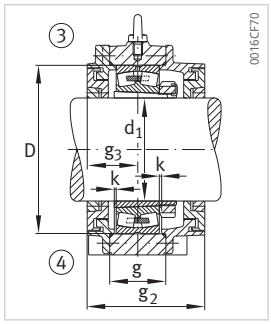




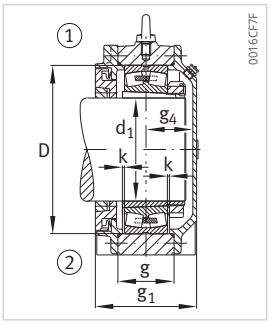
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2268-H-W-T-BF-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-T-BF-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-T-BF-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-T-BF-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2268-H-W-T-BL-D	**320**	390	790	201	380	125	1200	990	200	85	64	M56
BND3068-H-W-T-BL-D	**320**	315	630	160	270	95	900	770	170	52	42	M36
BND3168-H-W-T-BL-D	**320**	370	745	220	360	115	1150	960	200	75	56	M48
BND3268-H-W-T-BL-D	**320**	390	790	260	380	125	1200	990	200	85	64	M56
BND2272-H-W-Y-AF-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-Y-AF-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-Y-AF-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-Y-AF-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-Y-AL-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-Y-AL-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-Y-AL-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-Y-AL-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-Y-BF-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-Y-BF-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-Y-BF-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-Y-BF-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-Y-BL-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-Y-BL-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-Y-BL-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-Y-BL-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-T-AF-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-T-AF-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-T-AF-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-T-AF-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-T-AL-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-T-AL-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-T-AL-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-T-AL-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-T-BF-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-T-BF-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-T-BF-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-T-BF-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64
BND2272-H-W-T-BL-D	**340**	410	820	205	400	130	1280	1040	210	90	72	M64
BND3072-H-W-T-BL-D	**340**	330	660	170	280	100	960	820	180	52	42	M36
BND3172-H-W-T-BL-D	**340**	380	760	225	370	115	1200	1000	200	75	56	M48
BND3272-H-W-T-BL-D	**340**	410	820	267	400	130	1280	1040	210	90	72	M64



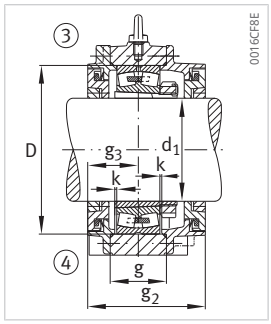
Design A,
Labyrinth seal



Design B,
Labyrinth seal



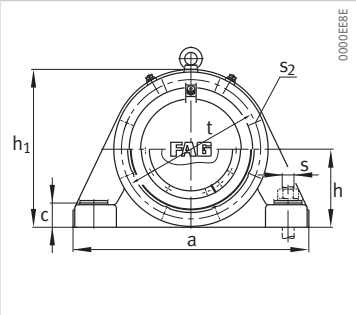
Design A,
Taconite seal

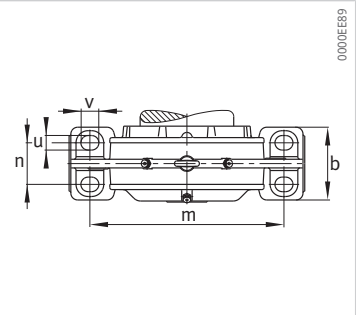


Design B,
Taconite seal

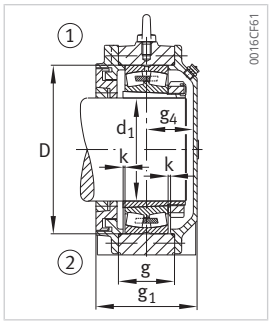
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
620	–	–	430	187,5	–	680	8	M30	22268..-K	H3168-HG	600
520	–	–	311	133	–	565	8	M20	23068..-K	H3068-HG	284
580	–	–	421	183	–	640	8	M24	23168..-K	H3168-HG	492
620	–	–	489	217	–	680	8	M30	23268..-K	H3268-HG	715
620	5	–	430	187,5	–	680	8	M30	22268..-K	H3168-HG	600
520	5	–	311	133	–	565	8	M20	23068..-K	H3068-HG	284
580	5	–	421	183	–	640	8	M24	23168..-K	H3168-HG	492
620	5	–	489	217	–	680	8	M30	23268..-K	H3268-HG	715
650	–	375	–	–	178	710	8	M30	22272..-K	H3172-HG	654
540	–	290	–	–	138	590	8	M20	23072..-K	H3072-HG	312
600	–	400	–	–	188	650	8	M24	23172..-K	H3172-HG	568
650	–	437	–	–	209	710	8	M30	23272..-K	H3272-HG	900
650	5	375	–	–	178	710	8	M30	22272..-K	H3172-HG	654
540	5	290	–	–	138	590	8	M20	23072..-K	H3072-HG	312
600	4	400	–	–	188	650	8	M24	23172..-K	H3172-HG	568
650	5	437	–	–	209	710	8	M30	23272..-K	H3272-HG	900
650	–	–	435	185	–	710	8	M30	22272..-K	H3172-HG	654
540	–	–	325	140	–	590	8	M20	23072..-K	H3072-HG	312
600	–	–	450	200	–	650	8	M24	23172..-K	H3172-HG	568
650	–	–	497	216	–	710	8	M30	23272..-K	H3272-HG	900
650	5	–	435	185	–	710	8	M30	22272..-K	H3172-HG	654
540	5	–	325	140	–	590	8	M20	23072..-K	H3072-HG	312
600	4	–	450	200	–	650	8	M24	23172..-K	H3172-HG	568
650	5	–	497	216	–	710	8	M30	23272..-K	H3272-HG	900
650	–	375	–	–	178	710	8	M30	22272..-K	H3172-HG	654
540	–	290	–	–	138	590	8	M20	23072..-K	H3072-HG	312
600	–	400	–	–	188	650	8	M24	23172..-K	H3172-HG	568
650	–	437	–	–	209	710	8	M30	23272..-K	H3272-HG	900
650	5	375	–	–	178	710	8	M30	22272..-K	H3172-HG	654
540	5	290	–	–	138	590	8	M20	23072..-K	H3072-HG	312
600	4	400	–	–	188	650	8	M24	23172..-K	H3172-HG	568
650	5	437	–	–	209	710	8	M30	23272..-K	H3272-HG	900
650	–	–	435	185	–	710	8	M30	22272..-K	H3172-HG	654
540	–	–	325	140	–	590	8	M20	23072..-K	H3072-HG	312
600	–	–	450	200	–	650	8	M24	23172..-K	H3172-HG	568
650	–	–	497	216	–	710	8	M30	23272..-K	H3272-HG	900
650	5	–	435	185	–	710	8	M30	22272..-K	H3172-HG	654
540	5	–	325	140	–	590	8	M20	23072..-K	H3072-HG	312
600	4	–	450	200	–	650	8	M24	23172..-K	H3172-HG	568
650	5	–	497	216	–	710	8	M30	23272..-K	H3272-HG	900

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

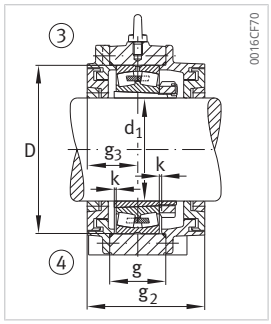




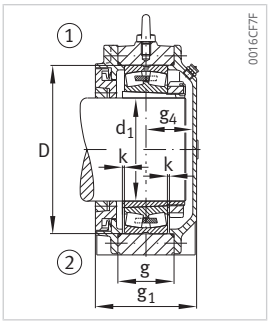
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2276-H-W-Y-AF-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-Y-AF-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-Y-AF-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-Y-AF-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-Y-AL-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-Y-AL-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-Y-AL-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-Y-AL-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-Y-BF-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-Y-BF-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-Y-BF-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-Y-BF-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-Y-BL-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-Y-BL-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-Y-BL-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-Y-BL-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-T-AF-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-T-AF-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-T-AF-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-T-AF-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-T-AL-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-T-AL-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-T-AL-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-T-AL-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-T-BF-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-T-BF-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-T-BF-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-T-BF-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND2276-H-W-T-BL-D	**360**	425	865	230	405	135	1350	1100	225	90	72	M64
BND3076-H-W-T-BL-D	**360**	340	680	160	300	105	1000	840	190	75	56	M48
BND3176-H-W-T-BL-D	**360**	390	790	230	380	125	1200	1000	200	85	64	M56
BND3276-H-W-T-BL-D	**360**	425	860	295	405	135	1350	1100	225	90	72	M64
BND3080-H-W-Y-AF-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-Y-AF-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-Y-AF-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-Y-AL-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-Y-AL-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-Y-AL-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-Y-BF-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-Y-BF-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64



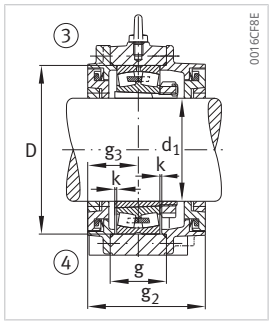
Design A,
Labyrinth seal



Design B,
Labyrinth seal



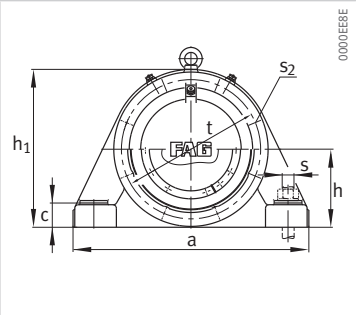
Design A,
Taconite seal

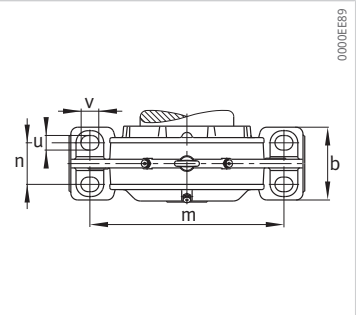


Design B,
Taconite seal

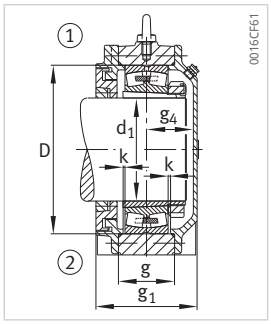
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
680	–	433	–	–	218	745	8	M30	22276..-K	H3176-HG	850
560	–	294	–	–	141	610	8	M20	23076..-K	H3076-HG	340
620	–	404	–	–	190	680	8	M30	23176..-K	H3176-HG	682
680	–	489	–	–	244	745	8	M30	23276..-K	H3276-HG	1045
680	5	433	–	–	218	745	8	M30	22276..-K	H3176-HG	850
560	7	294	–	–	141	610	8	M20	23076..-K	H3076-HG	340
620	5	404	–	–	190	680	8	M30	23176..-K	H3176-HG	682
680	5	489	–	–	244	745	8	M30	23276..-K	H3276-HG	1045
680	–	–	470	203	–	745	8	M30	22276..-K	H3176-HG	850
560	–	–	329	142	–	610	8	M20	23076..-K	H3076-HG	340
620	–	–	459	202	–	680	8	M30	23176..-K	H3176-HG	682
680	–	–	529	232	–	745	8	M30	23276..-K	H3276-HG	1045
680	5	–	470	203	–	745	8	M30	22276..-K	H3176-HG	850
560	7	–	329	142	–	610	8	M20	23076..-K	H3076-HG	340
620	5	–	459	202	–	680	8	M30	23176..-K	H3176-HG	682
680	5	–	529	232	–	745	8	M30	23276..-K	H3276-HG	1045
680	–	433	–	–	218	745	8	M30	22276..-K	H3176-HG	850
560	–	294	–	–	141	610	8	M20	23076..-K	H3076-HG	340
620	–	404	–	–	190	680	8	M30	23176..-K	H3176-HG	682
680	–	489	–	–	244	745	8	M30	23276..-K	H3276-HG	1045
680	5	433	–	–	218	745	8	M30	22276..-K	H3176-HG	850
560	7	294	–	–	141	610	8	M20	23076..-K	H3076-HG	340
620	5	404	–	–	190	680	8	M30	23176..-K	H3176-HG	682
680	5	489	–	–	244	745	8	M30	23276..-K	H3276-HG	1045
680	–	–	470	203	–	745	8	M30	22276..-K	H3176-HG	850
560	–	–	329	142	–	610	8	M20	23076..-K	H3076-HG	340
620	–	–	459	202	–	680	8	M30	23176..-K	H3176-HG	682
680	–	–	529	232	–	745	8	M30	23276..-K	H3276-HG	1045
680	5	–	470	203	–	745	8	M30	22276..-K	H3176-HG	850
560	7	–	329	142	–	610	8	M20	23076..-K	H3076-HG	340
620	5	–	459	202	–	680	8	M30	23176..-K	H3176-HG	682
680	5	–	529	232	–	745	8	M30	23276..-K	H3276-HG	1045
600	–	310	–	–	145	650	8	M20	23080..-K	H3080-HG	379
650	–	405	–	–	193	710	8	M30	23180..-K	H3180-HG	710
720	–	504	–	–	237	790	8	M30	23280..-K	H3280-HG	1140
600	7	310	–	–	145	650	8	M20	23080..-K	H3080-HG	379
650	5	405	–	–	193	710	8	M30	23180..-K	H3180-HG	710
720	5	504	–	–	237	790	8	M30	23280..-K	H3280-HG	1140
600	–	–	355	150	–	650	8	M20	23080..-K	H3080-HG	379
650	–	–	465	200	–	710	8	M30	23180..-K	H3180-HG	710

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

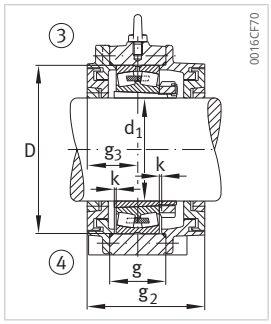




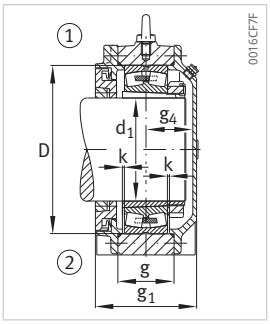
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3280-H-W-Y-BF-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-Y-BL-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-Y-BL-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-Y-BL-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-T-AF-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-T-AF-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-T-AF-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-T-AL-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-T-AL-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-T-AL-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-T-BF-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-T-BF-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-T-BF-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND3080-H-W-T-BL-D	**380**	360	720	175	320	110	1060	900	200	75	56	M48
BND3180-H-W-T-BL-D	**380**	410	820	235	400	130	1280	1040	210	90	72	M64
BND3280-H-W-T-BL-D	**380**	450	900	300	450	145	1430	1160	240	90	72	M64
BND2284-H-W-Y-AF-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-Y-AF-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-Y-AF-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-Y-AF-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64
BND2284-H-W-Y-AL-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-Y-AL-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-Y-AL-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-Y-AL-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64
BND2284-H-W-Y-BF-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-Y-BF-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-Y-BF-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-Y-BF-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64
BND2284-H-W-Y-BL-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-Y-BL-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-Y-BL-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-Y-BL-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64
BND2284-H-W-T-AF-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-T-AF-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-T-AF-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-T-AF-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64
BND2284-H-W-T-AL-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-T-AL-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-T-AL-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-T-AL-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64



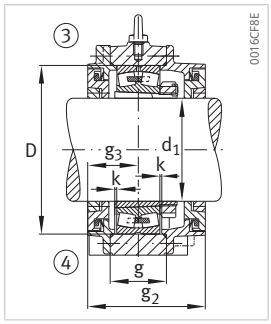
Design A,
Labyrinth seal



Design B,
Labyrinth seal



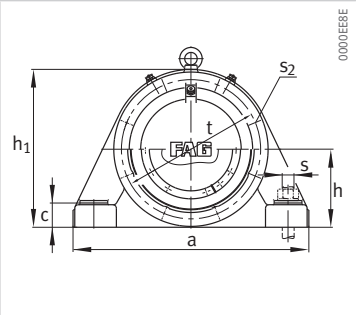
Design A,
Taconite seal

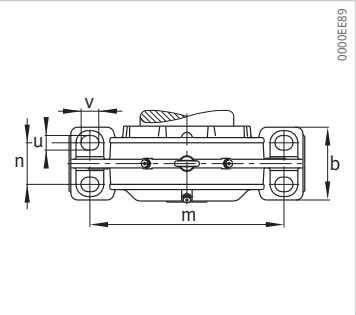


Design B,
Taconite seal

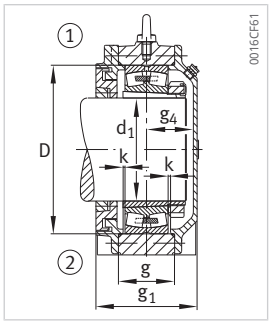
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
720	–	–	569	252	–	790	8	M30	23280..-K	H3280-HG	1140
600	7	–	355	150	–	650	8	M20	23080..-K	H3080-HG	379
650	5	–	465	200	–	710	8	M30	23180..-K	H3180-HG	710
720	5	–	569	252	–	790	8	M30	23280..-K	H3280-HG	1140
600	–	310	–	–	145	650	8	M20	23080..-K	H3080-HG	379
650	–	405	–	–	193	710	8	M30	23180..-K	H3180-HG	710
720	–	504	–	–	237	790	8	M30	23280..-K	H3280-HG	1140
600	7	310	–	–	145	650	8	M20	23080..-K	H3080-HG	379
650	5	405	–	–	193	710	8	M30	23180..-K	H3180-HG	710
720	5	504	–	–	237	790	8	M30	23280..-K	H3280-HG	1140
600	–	–	355	150	–	650	8	M20	23080..-K	H3080-HG	379
650	–	–	465	200	–	710	8	M30	23180..-K	H3180-HG	710
720	–	–	569	252	–	790	8	M30	23280..-K	H3280-HG	1140
600	7	–	355	150	–	650	8	M20	23080..-K	H3080-HG	379
650	5	–	465	200	–	710	8	M30	23180..-K	H3180-HG	710
720	5	–	569	252	–	790	8	M30	23280..-K	H3280-HG	1140
760	–	433	–	–	202	835	8	M30	22284..-K	H3184-HG	995
620	–	310	–	–	149	670	8	M20	23084..-K	H3084X-HG	413
700	–	440	–	–	215	760	8	M30	23184..-K	H3184-HG	900
760	–	510	–	–	240	835	8	M30	23284..-K	H3284-HG	1240
760	5	433	–	–	202	835	8	M30	22284..-K	H3184-HG	995
620	7	310	–	–	149	670	8	M20	23084..-K	H3084X-HG	413
700	7	440	–	–	215	760	8	M30	23184..-K	H3184-HG	900
760	5	510	–	–	240	835	8	M30	23284..-K	H3284-HG	1240
760	–	–	498	216,5	–	835	8	M30	22284..-K	H3184-HG	995
620	–	–	350	150	–	670	8	M20	23084..-K	H3084X-HG	413
700	–	–	510	210	–	760	8	M30	23184..-K	H3184-HG	900
760	–	–	575	255	–	835	8	M30	23284..-K	H3284-HG	1240
760	5	–	498	216,5	–	835	8	M30	22284..-K	H3184-HG	995
620	7	–	350	150	–	670	8	M20	23084..-K	H3084X-HG	413
700	7	–	510	210	–	760	8	M30	23184..-K	H3184-HG	900
760	5	–	575	255	–	835	8	M30	23284..-K	H3284-HG	1240
760	–	433	–	–	202	835	8	M30	22284..-K	H3184-HG	995
620	–	310	–	–	149	670	8	M20	23084..-K	H3084X-HG	413
700	–	440	–	–	215	760	8	M30	23184..-K	H3184-HG	900
760	–	510	–	–	240	835	8	M30	23284..-K	H3284-HG	1240
760	5	433	–	–	202	835	8	M30	22284..-K	H3184-HG	995
620	7	310	–	–	149	670	8	M20	23084..-K	H3084X-HG	413
700	7	440	–	–	215	760	8	M30	23184..-K	H3184-HG	900
760	5	510	–	–	240	835	8	M30	23284..-K	H3284-HG	1240

11.11.2 BND, for bearings with a tapered bore and adapter sleeve

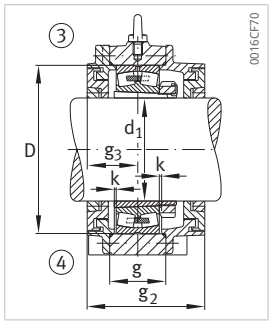




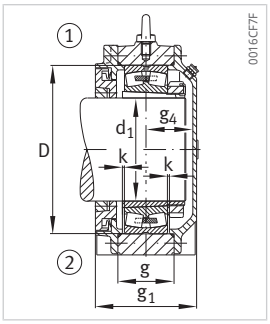
Housing	d_1	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2284-H-W-T-BF-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-T-BF-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-T-BF-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-T-BF-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64
BND2284-H-W-T-BL-D	**400**	470	950	238	470	150	1500	1220	255	90	72	M64
BND3084-H-W-T-BL-D	**400**	375	755	180	340	115	1100	940	210	75	56	M48
BND3184-H-W-T-BL-D	**400**	450	900	260	420	135	1350	1100	210	85	64	M56
BND3284-H-W-T-BL-D	**400**	470	950	315	470	150	1500	1220	255	90	72	M64



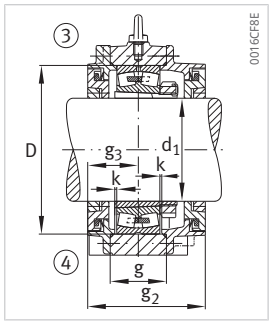
Design A,
Labyrinth seal



Design B,
Labyrinth seal



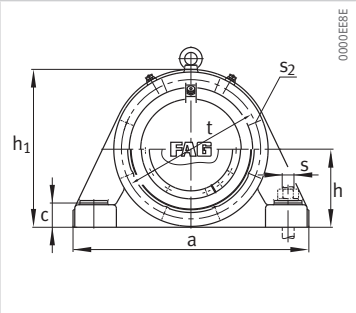
Design A,
Taconite seal

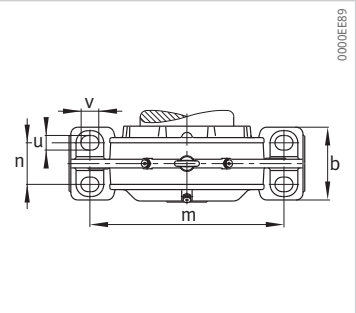


Design B,
Taconite seal

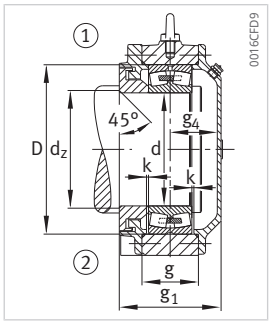
D	k	g_1	g_2	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
760	–	–	498	216,5	–	835	8	M30	22284..-K	H3184-HG	995
620	–	–	350	150	–	670	8	M20	23084..-K	H3084X-HG	413
700	–	–	510	210	–	760	8	M30	23184..-K	H3184-HG	900
760	–	–	575	255	–	835	8	M30	23284..-K	H3284-HG	1240
760	5	–	498	216,5	–	835	8	M30	22284..-K	H3184-HG	995
620	7	–	350	150	–	670	8	M20	23084..-K	H3084X-HG	413
700	7	–	510	210	–	760	8	M30	23184..-K	H3184-HG	900
760	5	–	575	255	–	835	8	M30	23284..-K	H3284-HG	1240

11.11.3 BND, for bearings with a cylindrical bore

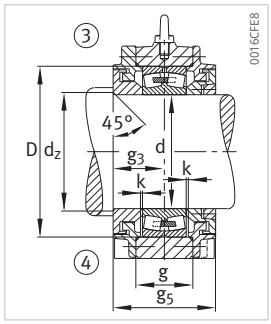




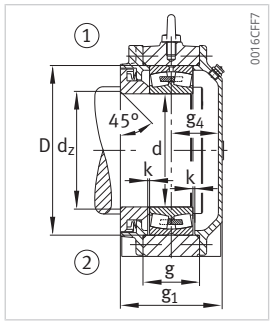
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2213-Z-T-AF-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-T-AL-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-T-BF-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-T-BL-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-Y-AF-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-Y-AL-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-Y-BF-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2213-Z-Y-BL-D	**65**	71	80	155	44	70	22	235	185	40	20	15	M10
BND2215-Z-T-AF-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-T-AL-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-T-BF-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-T-BL-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-Y-AF-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-Y-AL-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-Y-BF-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2215-Z-Y-BL-D	**75**	81	90	180	45	85	35	285	225	45	28	20	M16
BND2218-Z-T-AF-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-T-AL-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-T-BF-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-T-BL-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-Y-AF-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-Y-AL-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-Y-BF-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2218-Z-Y-BL-D	**90**	98	110	220	55	110	38	370	290	60	32	23	M20
BND2220-Z-T-AF-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-T-AL-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-T-BF-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-T-BL-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-Y-AF-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-Y-AL-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-Y-BF-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2220-Z-Y-BL-D	**100**	108	130	255	65	120	40	400	320	65	35	30	M24
BND2222-Z-T-AF-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-T-AL-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-T-BF-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-T-BL-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-Y-AF-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-Y-AL-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-Y-BF-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24
BND2222-Z-Y-BL-D	**110**	118	140	280	73	130	42	440	350	70	35	30	M24



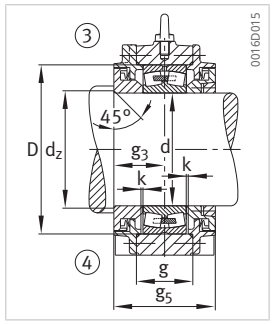
Design A
Labyrinth seal



Design B
Labyrinth seal



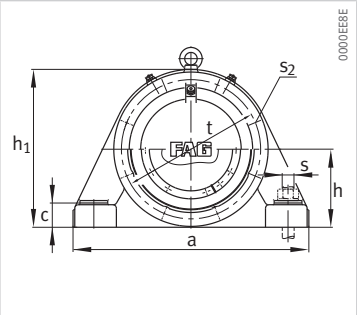
Design A
Taconite seal

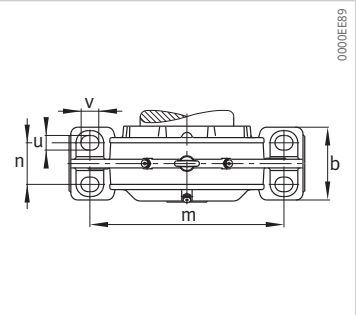


Design B
Taconite seal

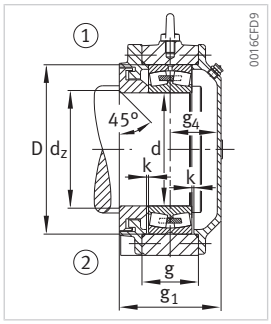
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
120	–	100	–	39	–	135	6	M6	22213	9,4
120	2	100	–	39	–	135	6	M6	22213	9,4
120	–	–	110	55	–	135	6	M6	22213	9,4
120	2	–	110	55	–	135	6	M6	22213	9,4
120	–	100	–	39	–	135	6	M6	22213	9,4
120	2	100	–	39	–	135	6	M6	22213	9,4
120	–	–	110	55	–	135	6	M6	22213	9,4
120	2	–	110	55	–	135	6	M6	22213	9,4
130	–	105	–	44	–	155	6	M6	22215	14,5
130	2	105	–	44	–	155	6	M6	22215	14,5
130	–	–	110	55	–	155	6	M6	22215	14,5
130	2	–	110	55	–	155	6	M6	22215	14,5
130	–	105	–	44	–	155	6	M6	22215	14,5
130	2	105	–	44	–	155	6	M6	22215	14,5
130	–	–	110	55	–	155	6	M6	22215	14,5
130	2	–	110	55	–	155	6	M6	22215	14,5
160	–	104	–	45	–	185	6	M8	22218	26,7
160	2	104	–	45	–	185	6	M8	22218	26,7
160	–	–	106	53	–	185	6	M8	22218	26,7
160	2	–	106	53	–	185	6	M8	22218	26,7
160	–	104	–	45	–	185	6	M8	22218	26,7
160	2	104	–	45	–	185	6	M8	22218	26,7
160	–	–	106	53	–	185	6	M8	22218	26,7
160	2	–	106	53	–	185	6	M8	22218	26,7
180	–	141	–	57	–	205	6	M8	22220	28
180	2,5	141	–	57	–	205	6	M8	22220	28
180	–	–	156	78	–	205	6	M8	22220	28
180	2,5	–	156	78	–	205	6	M8	22220	28
180	–	141	–	57	–	205	6	M8	22220	28
180	2,5	141	–	57	–	205	6	M8	22220	28
180	–	–	156	78	–	205	6	M8	22220	28
180	2,5	–	156	78	–	205	6	M8	22220	28
200	–	149,2	–	60	–	230	6	M12	22222	48
200	2,5	149,2	–	60	–	230	6	M12	22222	48
200	–	–	165,2	82,6	–	230	6	M12	22222	48
200	2,5	–	165,2	82,6	–	230	6	M12	22222	48
200	–	149,2	–	60	–	230	6	M12	22222	48
200	2,5	149,2	–	60	–	230	6	M12	22222	48
200	–	–	165,2	82,6	–	230	6	M12	22222	48
200	2,5	–	165,2	82,6	–	230	6	M12	22222	48

11.11.3 BND, for bearings with a cylindrical bore

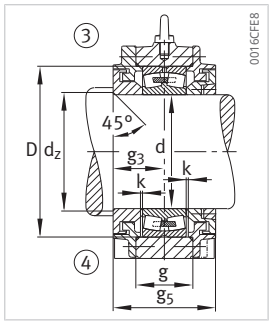




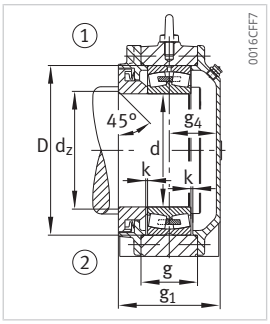
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2224-Z-T-AF-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-T-AL-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-T-BF-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-T-BL-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-Y-AF-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-Y-AL-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-Y-BF-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2224-Z-Y-BL-D	**120**	128	150	300	77	140	42	470	370	75	35	30	M24
BND2226-Z-T-AF-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-T-AL-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-T-BF-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-T-BL-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-Y-AF-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-Y-AL-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-Y-BF-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2226-Z-Y-BL-D	**130**	142	160	315	84	150	45	500	400	85	35	30	M24
BND2228-Z-T-AF-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-T-AL-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-T-BF-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-T-BL-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-Y-AF-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-Y-AL-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-Y-BF-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2228-Z-Y-BL-D	**140**	152	170	345	88	160	50	530	430	85	35	30	M24
BND2230-Z-T-AF-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-T-AL-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-T-BF-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-T-BL-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-Y-AF-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-Y-AL-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-Y-BF-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2230-Z-Y-BL-D	**150**	162	180	365	97	170	54	550	450	90	45	36	M30
BND2232-Z-T-AF-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-T-AL-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-T-BF-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-T-BL-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-Y-AF-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-Y-AL-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-Y-BF-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30
BND2232-Z-Y-BL-D	**160**	172	190	385	106	180	58	600	490	105	45	36	M30



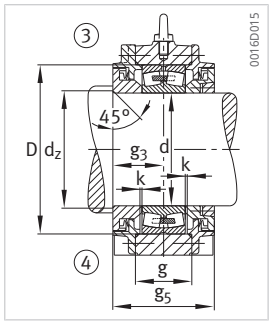
Design A
Labyrinth seal



Design B
Labyrinth seal



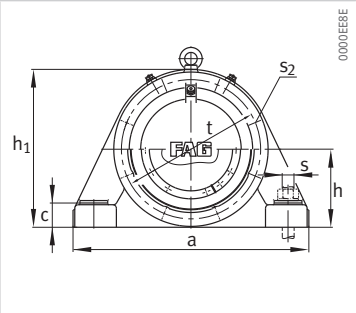
Design A
Taconite seal

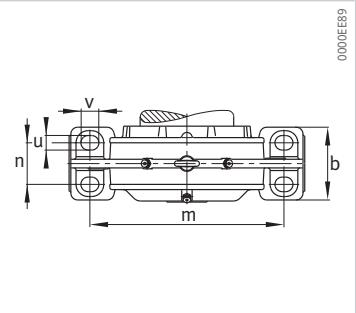


Design B
Taconite seal

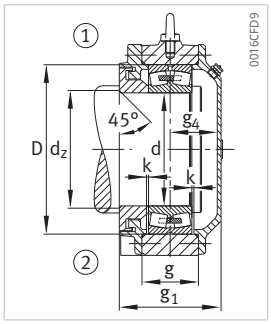
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
215	–	143	–	61	–	245	6	M12	22224	55,8
215	2,5	143	–	61	–	245	6	M12	22224	55,8
215	–	–	148	74	–	245	6	M12	22224	55,8
215	2,5	–	148	74	–	245	6	M12	22224	55,8
215	–	143	–	61	–	245	6	M12	22224	55,8
215	2,5	143	–	61	–	245	6	M12	22224	55,8
215	–	–	148	74	–	245	6	M12	22224	55,8
215	2,5	–	148	74	–	245	6	M12	22224	55,8
230	–	172	–	69	–	260	6	M12	22226	61,4
230	3	172	–	69	–	260	6	M12	22226	61,4
230	–	–	190	95	–	260	6	M12	22226	61,4
230	3	–	190	95	–	260	6	M12	22226	61,4
230	–	172	–	69	–	260	6	M12	22226	61,4
230	3	172	–	69	–	260	6	M12	22226	61,4
230	–	–	190	95	–	260	6	M12	22226	61,4
230	3	–	190	95	–	260	6	M12	22226	61,4
250	–	166	–	70	–	285	6	M16	22228	66,3
250	3	166	–	70	–	285	6	M16	22228	66,3
250	–	–	176	88	–	285	6	M16	22228	66,3
250	3	–	176	88	–	285	6	M16	22228	66,3
250	–	166	–	70	–	285	6	M16	22228	66,3
250	3	166	–	70	–	285	6	M16	22228	66,3
250	–	–	176	88	–	285	6	M16	22228	66,3
250	3	–	176	88	–	285	6	M16	22228	66,3
270	–	177	–	76	–	305	6	M16	22230	81,5
270	3	177	–	76	–	305	6	M16	22230	81,5
270	–	–	187	93,5	–	305	6	M16	22230	81,5
270	3	–	187	93,5	–	305	6	M16	22230	81,5
270	–	177	–	76	–	305	6	M16	22230	81,5
270	3	177	–	76	–	305	6	M16	22230	81,5
270	–	–	187	93,5	–	305	6	M16	22230	81,5
270	3	–	187	93,5	–	305	6	M16	22230	81,5
290	–	189	–	78	–	330	6	M16	22232	96
290	3	189	–	78	–	330	6	M16	22232	96
290	–	–	206	103	–	330	6	M16	22232	96
290	3	–	206	103	–	330	6	M16	22232	96
290	–	189	–	78	–	330	6	M16	22232	96
290	3	189	–	78	–	330	6	M16	22232	96
290	–	–	206	103	–	330	6	M16	22232	96
290	3	–	206	103	–	330	6	M16	22232	96

11.11.3 BND, for bearings with a cylindrical bore

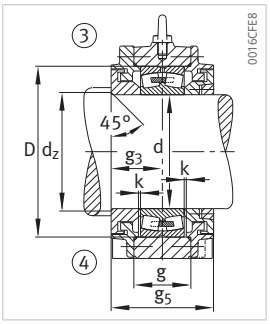




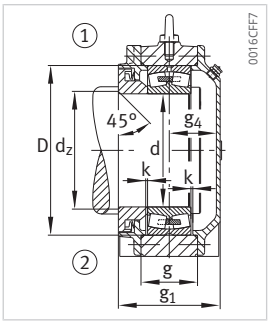
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2234-Z-T-AF-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-T-AL-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-T-BF-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-T-BL-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-Y-AF-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-Y-AL-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-Y-BF-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2234-Z-Y-BL-D	**170**	186	200	405	111	200	62	640	525	110	45	36	M30
BND2236-Z-T-AF-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-T-AL-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-T-BF-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-T-BL-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-Y-AF-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-Y-AL-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-Y-BF-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2236-Z-Y-BL-D	**180**	196	210	425	112	210	65	680	550	120	45	36	M30
BND2238-Z-T-AF-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-T-AL-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-T-BF-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-T-BL-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-Y-AF-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-Y-AL-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-Y-BF-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2238-Z-Y-BL-D	**190**	206	220	455	115	220	85	710	560	120	52	42	M36
BND2240-Z-T-AF-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-T-AL-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-T-BF-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-T-BL-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-Y-AF-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-Y-AL-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-Y-BF-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2240-Z-Y-BL-D	**200**	216	235	475	128	240	75	780	640	140	52	42	M36
BND2244-Z-T-AF-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-T-AL-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-T-BF-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-T-BL-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-Y-AF-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-Y-AL-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-Y-BF-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36
BND2244-Z-Y-BL-D	**220**	236	270	550	140	250	80	890	720	140	52	42	M36



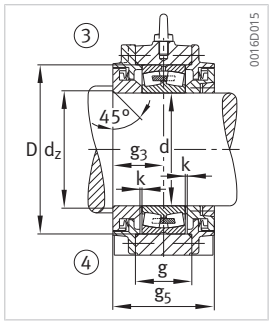
Design A
Labyrinth seal



Design B
Labyrinth seal



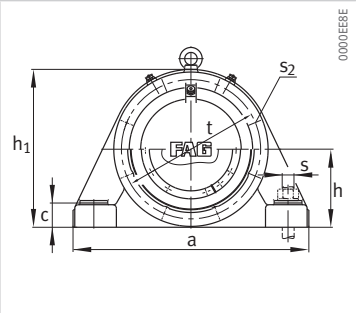
Design A
Taconite seal

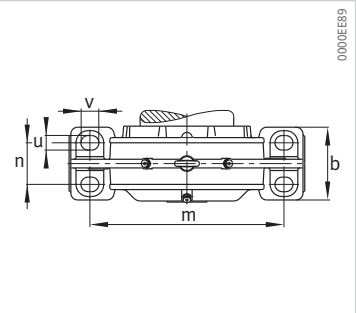


Design B
Taconite seal

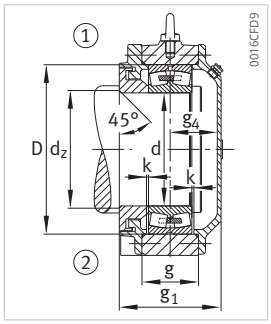
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
310	–	216	–	88	–	350	8	M16	22234	101
310	3	216	–	88	–	350	8	M16	22234	101
310	–	–	236	118	–	350	8	M16	22234	101
310	3	–	236	118	–	350	8	M16	22234	101
310	–	216	–	88	–	350	8	M16	22234	101
310	3	216	–	88	–	350	8	M16	22234	101
310	–	–	236	118	–	350	8	M16	22234	101
310	3	–	236	118	–	350	8	M16	22234	101
320	–	214	–	92	–	370	8	M16	22236	126
320	3	214	–	92	–	370	8	M16	22236	126
320	–	–	228	114	–	370	8	M16	22236	126
320	3	–	228	114	–	370	8	M16	22236	126
320	–	214	–	92	–	370	8	M16	22236	126
320	3	214	–	92	–	370	8	M16	22236	126
320	–	–	228	114	–	370	8	M16	22236	126
320	3	–	228	114	–	370	8	M16	22236	126
340	–	222	–	98	–	380	8	M16	22238	168
340	3	222	–	98	–	380	8	M16	22238	168
340	–	–	228	114	–	380	8	M16	22238	168
340	3	–	228	114	–	380	8	M16	22238	168
340	–	222	–	98	–	380	8	M16	22238	168
340	3	222	–	98	–	380	8	M16	22238	168
340	–	–	228	114	–	380	8	M16	22238	168
340	3	–	228	114	–	380	8	M16	22238	168
360	–	230	–	99	–	420	8	M16	22240	179
360	4	230	–	99	–	420	8	M16	22240	179
360	–	–	246	123	–	420	8	M16	22240	179
360	4	–	246	123	–	420	8	M16	22240	179
360	–	230	–	99	–	420	8	M16	22240	179
360	4	230	–	99	–	420	8	M16	22240	179
360	–	–	246	123	–	420	8	M16	22240	179
360	4	–	246	123	–	420	8	M16	22240	179
400	–	264	–	112	–	455	8	M20	22244	275,5
400	4	264	–	112	–	455	8	M20	22244	275,5
400	–	–	284	142	–	455	8	M20	22244	275,5
400	4	–	284	142	–	455	8	M20	22244	275,5
400	–	264	–	112	–	455	8	M20	22244	275,5
400	4	264	–	112	–	455	8	M20	22244	275,5
400	–	–	284	142	–	455	8	M20	22244	275,5
400	4	–	284	142	–	455	8	M20	22244	275,5

11.11.3 BND, for bearings with a cylindrical bore

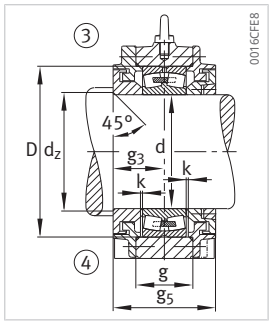




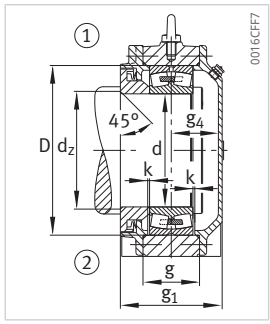
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2248-Z-T-AF-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-T-AL-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-T-BF-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-T-BL-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-Y-AF-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-Y-AL-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-Y-BF-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2248-Z-Y-BL-D	**240**	256	290	585	150	250	90	900	750	140	52	42	M36
BND2252-Z-T-AF-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-T-AL-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-T-BF-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-T-BL-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-Y-AF-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-Y-AL-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-Y-BF-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2252-Z-Y-BL-D	**260**	280	310	625	161	290	95	960	800	160	52	42	M36
BND2256-Z-T-AF-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-T-AL-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-T-BF-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-T-BL-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-Y-AF-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-Y-AL-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-Y-BF-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2256-Z-Y-BL-D	**280**	300	320	645	160	300	100	1000	840	170	52	42	M36
BND2260-Z-T-AF-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-T-AL-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-T-BF-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-T-BL-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-Y-AF-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-Y-AL-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-Y-BF-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2260-Z-Y-BL-D	**300**	320	350	695	178	330	105	1100	920	180	75	56	M48
BND2264-Z-T-AF-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-T-AL-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-T-BF-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-T-BL-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-Y-AF-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-Y-AL-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-Y-BF-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48
BND2264-Z-Y-BL-D	**320**	340	370	745	180	360	115	1150	960	200	75	56	M48



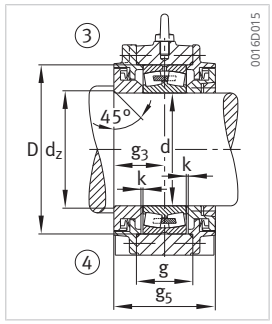
Design A
Labyrinth seal



Design B
Labyrinth seal



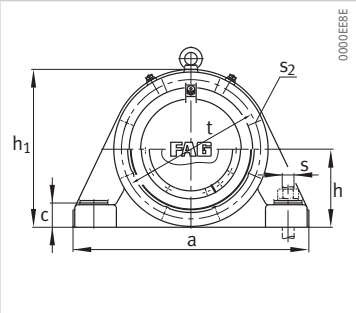
Design A
Taconite seal

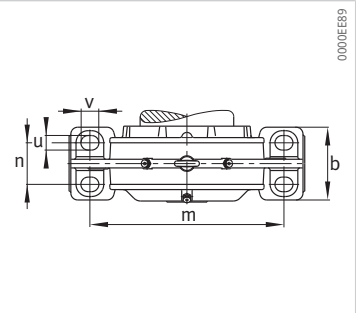


Design B
Taconite seal

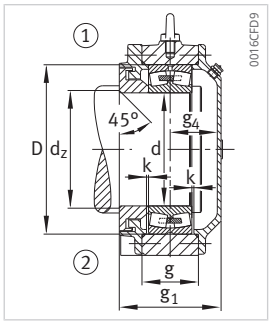
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
440	–	268	–	120	–	510	8	M20	22248	300
440	4	268	–	120	–	510	8	M20	22248	300
440	–	–	276	138	–	510	8	M20	22248	300
440	4	–	276	138	–	510	8	M20	22248	300
440	–	268	–	120	–	510	8	M20	22248	300
440	4	268	–	120	–	510	8	M20	22248	300
440	–	–	276	138	–	510	8	M20	22248	300
440	4	–	276	138	–	510	8	M20	22248	300
480	–	286	–	126	–	535	8	M20	22252	352
480	3	286	–	126	–	535	8	M20	22252	352
480	–	–	296	148	–	535	8	M20	22252	352
480	3	–	296	148	–	535	8	M20	22252	352
480	–	286	–	126	–	535	8	M20	22252	352
480	3	286	–	126	–	535	8	M20	22252	352
480	–	–	296	148	–	535	8	M20	22252	352
480	3	–	296	148	–	535	8	M20	22252	352
500	–	297	–	128	–	555	8	M24	22256	396
500	4	297	–	128	–	555	8	M24	22256	396
500	–	–	314	157	–	555	8	M24	22256	396
500	4	–	314	157	–	555	8	M24	22256	396
500	–	297	–	128	–	555	8	M24	22256	396
500	4	297	–	128	–	555	8	M24	22256	396
500	–	–	314	157	–	555	8	M24	22256	396
500	4	–	314	157	–	555	8	M24	22256	396
540	–	317	–	149	–	600	8	M24	22260	460
540	4	317	–	149	–	600	8	M24	22260	460
540	–	–	312	156	–	600	8	M24	22260	460
540	4	–	312	156	–	600	8	M24	22260	460
540	–	317	–	149	–	600	8	M24	22260	460
540	4	317	–	149	–	600	8	M24	22260	460
540	–	–	312	156	–	600	8	M24	22260	460
540	4	–	312	156	–	600	8	M24	22260	460
580	–	333	–	158	–	640	8	M24	22264	570
580	5	333	–	158	–	640	8	M24	22264	570
580	–	–	326	163	–	640	8	M24	22264	570
580	5	–	326	163	–	640	8	M24	22264	570
580	–	333	–	158	–	640	8	M24	22264	570
580	5	333	–	158	–	640	8	M24	22264	570
580	–	–	326	163	–	640	8	M24	22264	570
580	5	–	326	163	–	640	8	M24	22264	570

11.11.3 BND, for bearings with a cylindrical bore

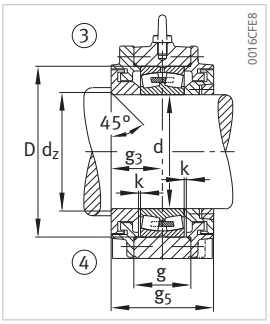




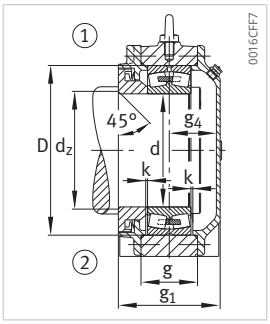
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND2268-Z-T-AF-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-T-AL-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-T-BF-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-T-BL-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-Y-AF-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-Y-AL-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-Y-BF-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2268-Z-Y-BL-D	**340**	364	390	790	201	380	125	1200	990	200	85	64	M56
BND2272-Z-T-AF-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-T-AL-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-T-BF-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-T-BL-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-Y-AF-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-Y-AL-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-Y-BF-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2272-Z-Y-BL-D	**360**	384	410	820	205	400	130	1280	1040	210	90	72	M64
BND2276-Z-T-AF-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-T-AL-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-T-BF-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-T-BL-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-Y-AF-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-Y-AL-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-Y-BF-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2276-Z-Y-BL-D	**380**	404	425	865	230	405	135	1350	1100	225	90	72	M64
BND2280-Z-T-AF-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-T-AL-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-T-BF-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-T-BL-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-Y-AF-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-Y-AL-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-Y-BF-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2280-Z-Y-BL-D	**400**	424	450	900	229	450	145	1430	1160	240	90	72	M64
BND2284-Z-T-AF-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-T-AL-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-T-BF-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-T-BL-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-Y-AF-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-Y-AL-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-Y-BF-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64
BND2284-Z-Y-BL-D	**420**	450	470	950	238	470	150	1500	1220	255	90	72	M64



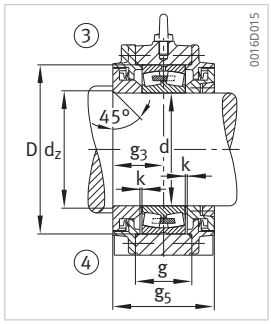
Design A
Labyrinth seal



Design B
Labyrinth seal



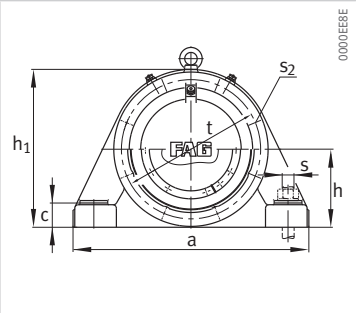
Design A
Taconite seal

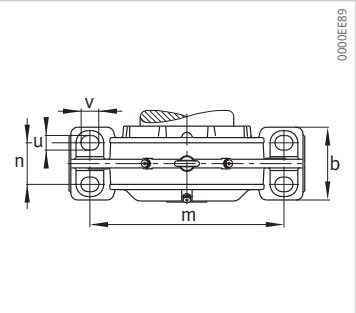


Design B
Taconite seal

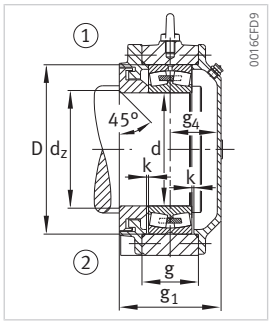
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
620	–	375	–	176	–	680	8	M30	22268	600
620	5	375	–	176	–	680	8	M30	22268	600
620	–	–	375	187,5	–	680	8	M30	22268	600
620	5	–	375	187,5	–	680	8	M30	22268	600
620	–	375	–	176	–	680	8	M30	22268	600
620	5	375	–	176	–	680	8	M30	22268	600
620	–	–	375	187,5	–	680	8	M30	22268	600
620	5	–	375	187,5	–	680	8	M30	22268	600
650	–	375	–	178	–	710	8	M30	22272	654
650	5	375	–	178	–	710	8	M30	22272	654
650	–	–	370	185	–	710	8	M30	22272	654
650	5	–	370	185	–	710	8	M30	22272	654
650	–	375	–	178	–	710	8	M30	22272	654
650	5	375	–	178	–	710	8	M30	22272	654
650	–	–	370	185	–	710	8	M30	22272	654
650	5	–	370	185	–	710	8	M30	22272	654
680	–	433	–	218	–	745	8	M30	22276	850
680	5	433	–	218	–	745	8	M30	22276	850
680	–	–	406	203	–	745	8	M30	22276	850
680	5	–	406	203	–	745	8	M30	22276	850
680	–	433	–	218	–	745	8	M30	22276	850
680	5	433	–	218	–	745	8	M30	22276	850
680	–	–	406	203	–	745	8	M30	22276	850
680	5	–	406	203	–	745	8	M30	22276	850
720	–	433	–	202	–	790	8	M30	22280	890
720	5	433	–	202	–	790	8	M30	22280	890
720	–	–	433	216,5	–	790	8	M30	22280	890
720	5	–	433	216,5	–	790	8	M30	22280	890
720	–	433	–	202	–	790	8	M30	22280	890
720	5	433	–	202	–	790	8	M30	22280	890
720	–	–	433	216,5	–	790	8	M30	22280	890
720	5	–	433	216,5	–	790	8	M30	22280	890
760	–	433	–	202	–	835	8	M30	22284	995
760	5	433	–	202	–	835	8	M30	22284	995
760	–	–	433	216,5	–	835	8	M30	22284	995
760	5	–	433	216,5	–	835	8	M30	22284	995
760	–	433	–	202	–	835	8	M30	22284	995
760	5	433	–	202	–	835	8	M30	22284	995
760	–	–	433	216,5	–	835	8	M30	22284	995
760	5	–	433	216,5	–	835	8	M30	22284	995

11.11.3 BND, for bearings with a cylindrical bore

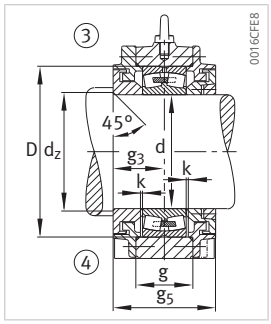




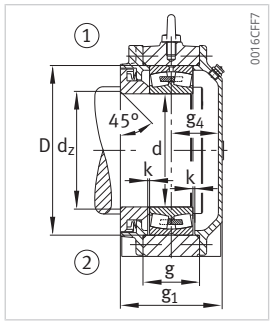
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3024-Z-T-AF-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-T-AL-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-T-BF-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-T-BL-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-Y-AF-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-Y-AL-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-Y-BF-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3024-Z-Y-BL-D	**120**	128	115	230	60	110	35	370	300	60	35	25	M20
BND3026-Z-T-AF-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-T-AL-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-T-BF-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-T-BL-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-Y-AF-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-Y-AL-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-Y-BF-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3026-Z-Y-BL-D	**130**	138	130	260	70	120	38	410	340	60	35	25	M20
BND3028-Z-T-AF-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-T-AL-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-T-BF-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-T-BL-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-Y-AF-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-Y-AL-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-Y-BF-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3028-Z-Y-BL-D	**140**	148	140	275	70	130	40	430	360	70	35	30	M24
BND3030-Z-T-AF-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-T-AL-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-T-BF-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-T-BL-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-Y-AF-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-Y-AL-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-Y-BF-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3030-Z-Y-BL-D	**150**	158	145	290	80	140	42	455	370	80	35	30	M24
BND3032-Z-T-AF-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-T-AL-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-T-BF-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-T-BL-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-Y-AF-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-Y-AL-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-Y-BF-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24
BND3032-Z-Y-BL-D	**160**	168	155	310	85	150	45	480	390	90	35	30	M24



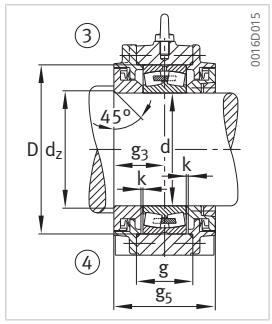
Design A
Labyrinth seal



Design B
Labyrinth seal



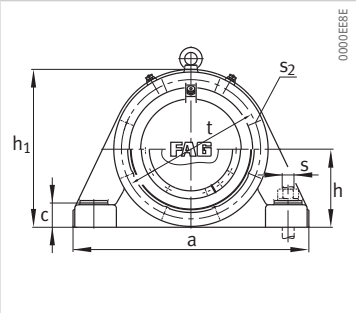
Design A
Taconite seal

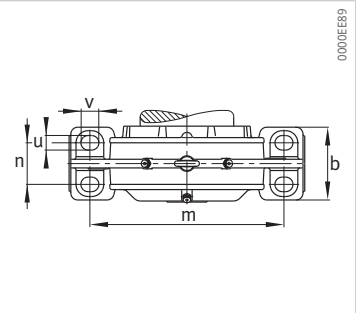


Design B
Taconite seal

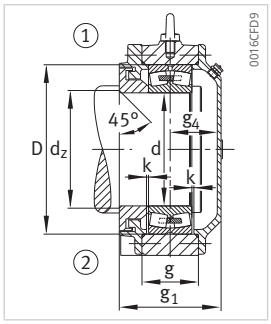
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
180	–	126	–	57	–	205	8	M8	23024	19
180	2,5	126	–	57	–	205	8	M8	23024	19
180	–	–	126	63	–	205	8	M8	23024	19
180	2,5	–	126	63	–	205	8	M8	23024	19
180	–	126	–	57	–	205	8	M8	23024	19
180	2,5	126	–	57	–	205	8	M8	23024	19
180	–	–	126	63	–	205	8	M8	23024	19
180	2,5	–	126	63	–	205	8	M8	23024	19
200	–	133	–	59	–	225	8	M10	23026	28,5
200	2	133	–	59	–	225	8	M10	23026	28,5
200	–	–	136	68	–	225	8	M10	23026	28,5
200	2	–	136	68	–	225	8	M10	23026	28,5
200	–	133	–	59	–	225	8	M10	23026	28,5
200	2	133	–	59	–	225	8	M10	23026	28,5
200	–	–	136	68	–	225	8	M10	23026	28,5
200	2	–	136	68	–	225	8	M10	23026	28,5
210	–	136	–	61	–	240	8	M10	23028	33
210	3	136	–	61	–	240	8	M10	23028	33
210	–	–	136	68	–	240	8	M10	23028	33
210	3	–	136	68	–	240	8	M10	23028	33
210	–	136	–	61	–	240	8	M10	23028	33
210	3	136	–	61	–	240	8	M10	23028	33
210	–	–	136	68	–	240	8	M10	23028	33
210	3	–	136	68	–	240	8	M10	23028	33
225	–	149	–	64	–	250	8	M12	23030	38
225	2	149	–	64	–	250	8	M12	23030	38
225	–	–	156	78	–	250	8	M12	23030	38
225	2	–	156	78	–	250	8	M12	23030	38
225	–	149	–	64	–	250	8	M12	23030	38
225	2	149	–	64	–	250	8	M12	23030	38
225	–	–	156	78	–	250	8	M12	23030	38
225	2	–	156	78	–	250	8	M12	23030	38
240	–	156	–	71	–	265	8	M12	23032	42,8
240	3	156	–	71	–	265	8	M12	23032	42,8
240	–	–	156	78	–	265	8	M12	23032	42,8
240	3	–	156	78	–	265	8	M12	23032	42,8
240	–	156	–	71	–	265	8	M12	23032	42,8
240	3	156	–	71	–	265	8	M12	23032	42,8
240	–	–	156	78	–	265	8	M12	23032	42,8
240	3	–	156	78	–	265	8	M12	23032	42,8

11.11.3 BND, for bearings with a cylindrical bore

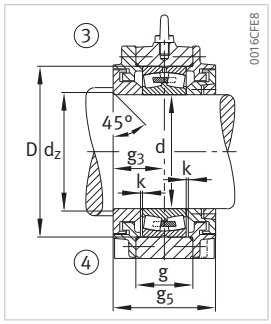




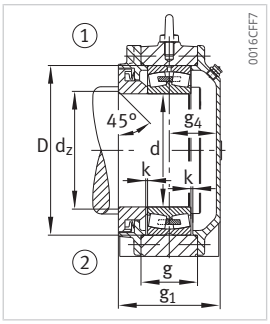
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3034-Z-T-AF-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-T-AL-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-T-BF-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-T-BL-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-Y-AF-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-Y-AL-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-Y-BF-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3034-Z-Y-BL-D	**170**	178	165	330	90	160	50	510	420	95	45	36	M30
BND3036-Z-T-AF-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-T-AL-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-T-BF-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-T-BL-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-Y-AF-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-Y-AL-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-Y-BF-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3036-Z-Y-BL-D	**180**	188	180	360	100	170	52	540	450	100	45	36	M30
BND3038-Z-T-AF-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-T-AL-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-T-BF-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-T-BL-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-Y-AF-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-Y-AL-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-Y-BF-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3038-Z-Y-BL-D	**190**	198	185	370	105	180	55	570	480	105	45	36	M30
BND3040-Z-T-AF-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-T-AL-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-T-BF-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-T-BL-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-Y-AF-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-Y-AL-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-Y-BF-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3040-Z-Y-BL-D	**200**	208	200	400	110	190	60	600	510	110	45	36	M30
BND3044-Z-T-AF-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-T-AL-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-T-BF-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-T-BL-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-Y-AF-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-Y-AL-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-Y-BF-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36
BND3044-Z-Y-BL-D	**220**	232	215	430	115	200	65	640	540	115	52	42	M36



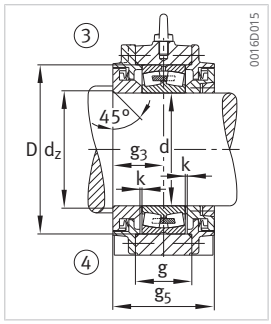
Design A
Labyrinth seal



Design B
Labyrinth seal



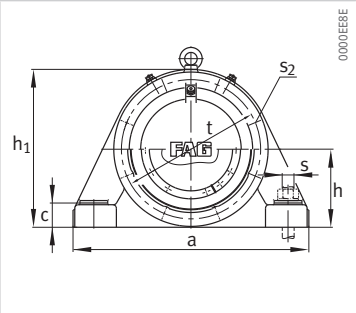
Design A
Taconite seal

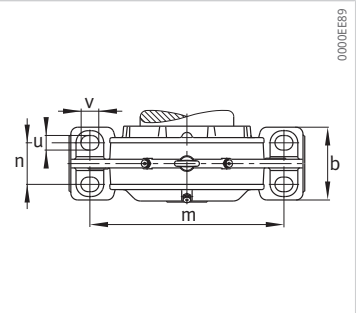


Design B
Taconite seal

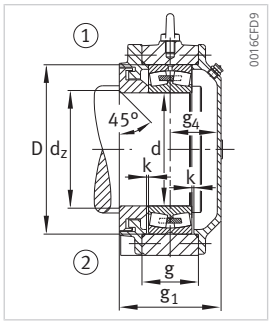
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
260	–	166	–	75	–	285	8	M12	23034	66,5
260	2	166	–	75	–	285	8	M12	23034	66,5
260	–	–	166	83	–	285	8	M12	23034	66,5
260	2	–	166	83	–	285	8	M12	23034	66,5
260	–	166	–	75	–	285	8	M12	23034	66,5
260	2	166	–	75	–	285	8	M12	23034	66,5
260	–	–	166	83	–	285	8	M12	23034	66,5
260	2	–	166	83	–	285	8	M12	23034	66,5
280	–	176	–	80	–	310	8	M12	23036	65,5
280	3	176	–	80	–	310	8	M12	23036	65,5
280	–	–	176	88	–	310	8	M12	23036	65,5
280	3	–	176	88	–	310	8	M12	23036	65,5
280	–	176	–	80	–	310	8	M12	23036	65,5
280	3	176	–	80	–	310	8	M12	23036	65,5
280	–	–	176	88	–	310	8	M12	23036	65,5
280	3	–	176	88	–	310	8	M12	23036	65,5
290	–	181	–	80	–	325	8	M12	23038	82
290	3	181	–	80	–	325	8	M12	23038	82
290	–	–	186	93	–	325	8	M12	23038	82
290	3	–	186	93	–	325	8	M12	23038	82
290	–	181	–	80	–	325	8	M12	23038	82
290	3	181	–	80	–	325	8	M12	23038	82
290	–	–	186	93	–	325	8	M12	23038	82
290	3	–	186	93	–	325	8	M12	23038	82
310	–	196	–	90	–	340	8	M16	23040	91
310	3	196	–	90	–	340	8	M16	23040	91
310	–	–	196	98	–	340	8	M16	23040	91
310	3	–	196	98	–	340	8	M16	23040	91
310	–	196	–	90	–	340	8	M16	23040	91
310	3	196	–	90	–	340	8	M16	23040	91
310	–	–	196	98	–	340	8	M16	23040	91
310	3	–	196	98	–	340	8	M16	23040	91
340	–	206	–	95	–	375	8	M16	23044	95,5
340	3	206	–	95	–	375	8	M16	23044	95,5
340	–	–	206	103	–	375	8	M16	23044	95,5
340	3	–	206	103	–	375	8	M16	23044	95,5
340	–	206	–	95	–	375	8	M16	23044	95,5
340	3	206	–	95	–	375	8	M16	23044	95,5
340	–	–	206	103	–	375	8	M16	23044	95,5
340	3	–	206	103	–	375	8	M16	23044	95,5

11.11.3 BND, for bearings with a cylindrical bore

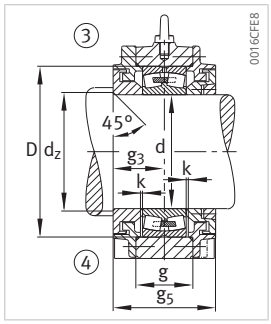




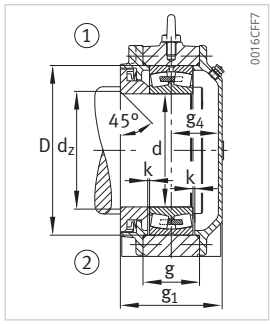
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3048-Z-T-AF-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-T-AL-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-T-BF-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-T-BL-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-Y-AF-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-Y-AL-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-Y-BF-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3048-Z-Y-BL-D	**240**	252	225	455	120	210	70	680	560	120	52	42	M36
BND3052-Z-T-AF-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-T-AL-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-T-BF-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-T-BL-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-Y-AF-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-Y-AL-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-Y-BF-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3052-Z-Y-BL-D	**260**	276	250	500	130	220	75	720	600	130	52	42	M36
BND3056-Z-T-AF-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-T-AL-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-T-BF-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-T-BL-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-Y-AF-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-Y-AL-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-Y-BF-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3056-Z-Y-BL-D	**280**	296	260	520	135	240	80	760	630	140	52	42	M36
BND3060-Z-T-AF-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-T-AL-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-T-BF-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-T-BL-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-Y-AF-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-Y-AL-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-Y-BF-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3060-Z-Y-BL-D	**300**	316	285	570	140	250	85	820	690	150	52	42	M36
BND3064-Z-T-AF-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-T-AL-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-T-BF-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-T-BL-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-Y-AF-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-Y-AL-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-Y-BF-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36
BND3064-Z-Y-BL-D	**320**	336	295	590	150	260	90	860	730	160	52	42	M36



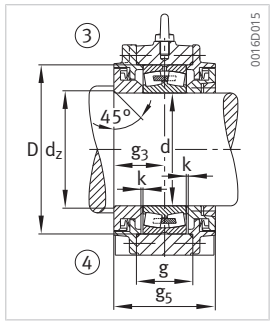
Design A
Labyrinth seal



Design B
Labyrinth seal



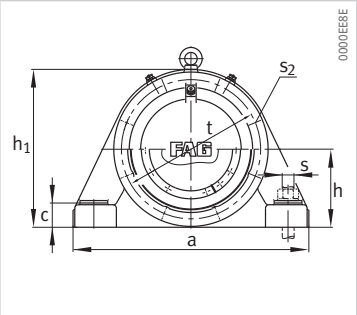
Design A
Taconite seal

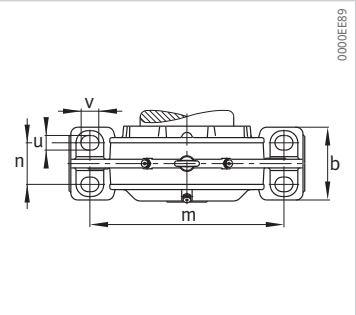


Design B
Taconite seal

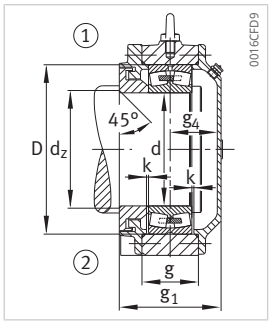
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
360	–	216	–	100	–	400	8	M16	23048	122
360	4	216	–	100	–	400	8	M16	23048	122
360	–	–	216	108	–	400	8	M16	23048	122
360	4	–	216	108	–	400	8	M16	23048	122
360	–	216	–	100	–	400	8	M16	23048	122
360	4	216	–	100	–	400	8	M16	23048	122
360	–	–	216	108	–	400	8	M16	23048	122
360	4	–	216	108	–	400	8	M16	23048	122
400	–	226	–	103	–	440	8	M16	23052	151
400	4	226	–	103	–	440	8	M16	23052	151
400	–	–	226	113	–	440	8	M16	23052	151
400	4	–	226	113	–	440	8	M16	23052	151
400	–	226	–	103	–	440	8	M16	23052	151
400	4	226	–	103	–	440	8	M16	23052	151
400	–	–	226	113	–	440	8	M16	23052	151
400	4	–	226	113	–	440	8	M16	23052	151
420	–	236	–	108	–	460	8	M16	23056	170
420	4	236	–	108	–	460	8	M16	23056	170
420	–	–	236	118	–	460	8	M16	23056	170
420	4	–	236	118	–	460	8	M16	23056	170
420	–	236	–	108	–	460	8	M16	23056	170
420	4	236	–	108	–	460	8	M16	23056	170
420	–	–	236	118	–	460	8	M16	23056	170
420	4	–	236	118	–	460	8	M16	23056	170
460	–	261	–	121	–	510	8	M16	23060	208
460	4	261	–	121	–	510	8	M16	23060	208
460	–	–	256	128	–	510	8	M16	23060	208
460	4	–	256	128	–	510	8	M16	23060	208
460	–	261	–	121	–	510	8	M16	23060	208
460	4	261	–	121	–	510	8	M16	23060	208
460	–	–	256	128	–	510	8	M16	23060	208
460	4	–	256	128	–	510	8	M16	23060	208
480	–	266	–	123	–	530	8	M16	23064	236
480	4	266	–	123	–	530	8	M16	23064	236
480	–	–	266	133	–	530	8	M16	23064	236
480	4	–	266	133	–	530	8	M16	23064	236
480	–	266	–	123	–	530	8	M16	23064	236
480	4	266	–	123	–	530	8	M16	23064	236
480	–	–	266	133	–	530	8	M16	23064	236
480	4	–	266	133	–	530	8	M16	23064	236

11.11.3 BND, for bearings with a cylindrical bore

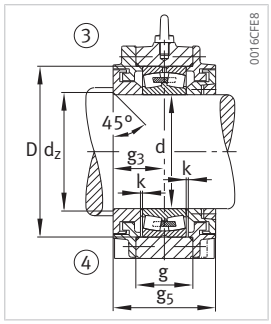




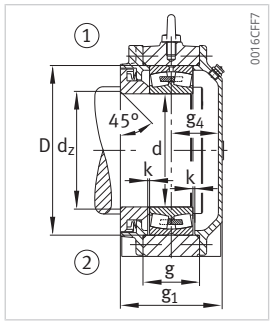
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3068-Z-T-AF-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-T-AL-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-T-BF-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-T-BL-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-Y-AF-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-Y-AL-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-Y-BF-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3068-Z-Y-BL-D	**340**	360	315	630	160	270	95	900	770	170	52	42	M36
BND3072-Z-T-AF-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-T-AL-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-T-BF-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-T-BL-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-Y-AF-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-Y-AL-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-Y-BF-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3072-Z-Y-BL-D	**360**	380	330	660	170	280	100	960	820	180	52	42	M36
BND3076-Z-T-AF-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-T-AL-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-T-BF-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-T-BL-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-Y-AF-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-Y-AL-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-Y-BF-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3076-Z-Y-BL-D	**380**	400	340	680	160	300	105	1000	840	190	75	56	M48
BND3080-Z-T-AF-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-T-AL-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-T-BF-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-T-BL-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-Y-AF-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-Y-AL-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-Y-BF-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3080-Z-Y-BL-D	**400**	420	360	720	175	320	110	1060	900	200	75	56	M48
BND3084-Z-T-AF-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-T-AL-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-T-BF-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-T-BL-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-Y-AF-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-Y-AL-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-Y-BF-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48
BND3084-Z-Y-BL-D	**420**	440	375	755	180	340	115	1100	940	210	75	56	M48



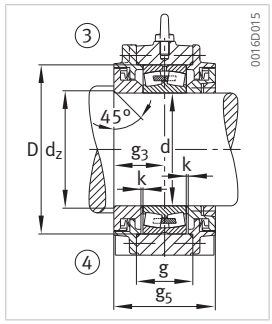
Design A
Labyrinth seal



Design B
Labyrinth seal



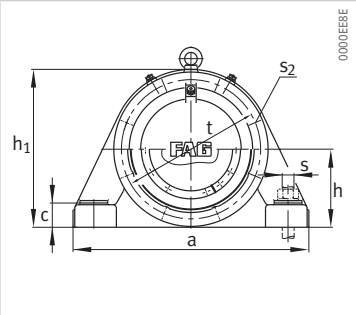
Design A
Taconite seal

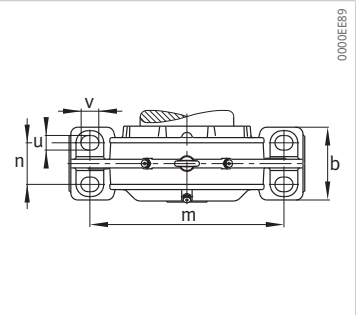


Design B
Taconite seal

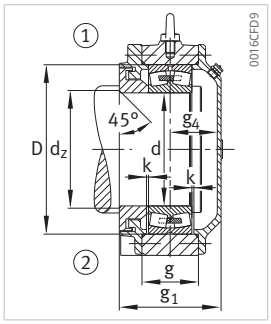
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
520	–	276	–	132	–	565	8	M20	23068	284
520	5	276	–	132	–	565	8	M20	23068	284
520	–	–	266	133	–	565	8	M20	23068	284
520	5	–	266	133	–	565	8	M20	23068	284
520	–	276	–	132	–	565	8	M20	23068	284
520	5	276	–	132	–	565	8	M20	23068	284
520	–	–	266	133	–	565	8	M20	23068	284
520	5	–	266	133	–	565	8	M20	23068	284
540	–	290	–	138	–	590	8	M20	23072	312
540	5	290	–	138	–	590	8	M20	23072	312
540	–	–	280	140	–	590	8	M20	23072	312
540	5	–	280	140	–	590	8	M20	23072	312
540	–	290	–	138	–	590	8	M20	23072	312
540	5	290	–	138	–	590	8	M20	23072	312
540	–	–	280	140	–	590	8	M20	23072	312
540	5	–	280	140	–	590	8	M20	23072	312
560	–	294	–	141	–	610	8	M20	23076	340
560	7	294	–	141	–	610	8	M20	23076	340
560	–	–	284	142	–	610	8	M20	23076	340
560	7	–	284	142	–	610	8	M20	23076	340
560	–	294	–	141	–	610	8	M20	23076	340
560	7	294	–	141	–	610	8	M20	23076	340
560	–	–	284	142	–	610	8	M20	23076	340
560	7	–	284	142	–	610	8	M20	23076	340
600	–	310	–	145	–	650	8	M20	23080	379
600	7	310	–	145	–	650	8	M20	23080	379
600	–	–	300	150	–	650	8	M20	23080	379
600	7	–	300	150	–	650	8	M20	23080	379
600	–	310	–	145	–	650	8	M20	23080	379
600	7	310	–	145	–	650	8	M20	23080	379
600	–	–	300	150	–	650	8	M20	23080	379
600	7	–	300	150	–	650	8	M20	23080	379
620	–	310	–	149	–	670	8	M20	23084	413
620	7	310	–	149	–	670	8	M20	23084	413
620	–	–	300	150	–	670	8	M20	23084	413
620	7	–	300	150	–	670	8	M20	23084	413
620	–	310	–	149	–	670	8	M20	23084	413
620	7	310	–	149	–	670	8	M20	23084	413
620	–	–	300	150	–	670	8	M20	23084	413
620	7	–	300	150	–	670	8	M20	23084	413

11.11.3 BND, for bearings with a cylindrical bore

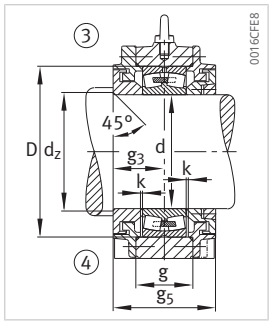




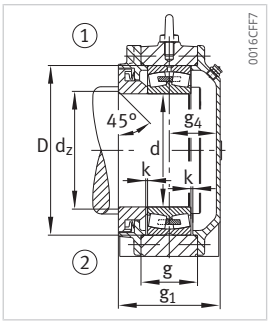
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3122-Z-T-AF-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-T-AL-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-T-BF-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-T-BL-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-Y-AF-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-Y-AL-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-Y-BF-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3122-Z-Y-BL-D	**110**	118	130	263	80	140	40	400	300	80	35	25	M20
BND3124-Z-T-AF-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-T-AL-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-T-BF-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-T-BL-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-Y-AF-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-Y-AL-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-Y-BF-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3124-Z-Y-BL-D	**120**	128	140	280	85	150	40	410	330	80	35	25	M20
BND3126-Z-T-AF-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-T-AL-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-T-BF-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-T-BL-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-Y-AF-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-Y-AL-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-Y-BF-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3126-Z-Y-BL-D	**130**	138	145	295	90	150	40	430	350	80	35	25	M20
BND3128-Z-T-AF-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-T-AL-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-T-BF-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-T-BL-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-Y-AF-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-Y-AL-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-Y-BF-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3128-Z-Y-BL-D	**140**	148	155	315	95	160	45	470	380	85	35	25	M20
BND3130-Z-T-AF-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-T-AL-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-T-BF-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-T-BL-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-Y-AF-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-Y-AL-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-Y-BF-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24
BND3130-Z-Y-BL-D	**150**	158	170	345	102	200	60	580	450	110	35	30	M24



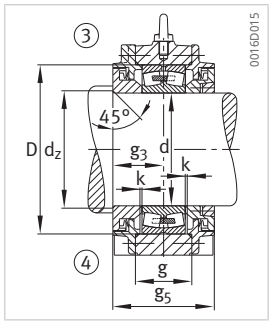
Design A
Labyrinth seal



Design B
Labyrinth seal



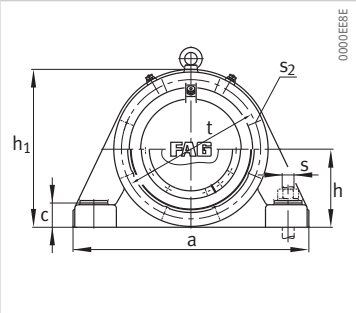
Design A
Taconite seal

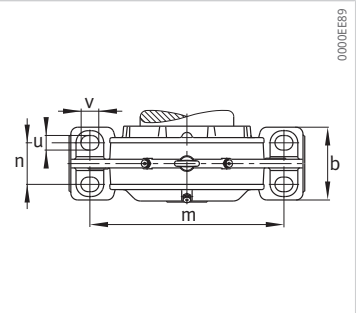


Design B
Taconite seal

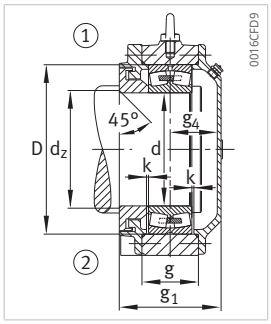
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
180	–	146	–	62	–	215	8	M12	23122	38
180	2,5	146	–	62	–	215	8	M12	23122	38
180	–	–	156	78	–	215	8	M12	23122	38
180	2,5	–	156	78	–	215	8	M12	23122	38
180	–	146	–	62	–	215	8	M12	23122	38
180	2,5	146	–	62	–	215	8	M12	23122	38
180	–	–	156	78	–	215	8	M12	23122	38
180	2,5	–	156	78	–	215	8	M12	23122	38
200	–	160	–	74	–	235	8	M12	23124	47,5
200	2,5	160	–	74	–	235	8	M12	23124	47,5
200	–	–	160	80	–	235	8	M12	23124	47,5
200	2,5	–	160	80	–	235	8	M12	23124	47,5
200	–	160	–	74	–	235	8	M12	23124	47,5
200	2,5	160	–	74	–	235	8	M12	23124	47,5
200	–	–	160	80	–	235	8	M12	23124	47,5
200	2,5	–	160	80	–	235	8	M12	23124	47,5
210	–	165	–	72	–	245	8	M12	23126	52,5
210	3	165	–	72	–	245	8	M12	23126	52,5
210	–	–	170	85	–	245	8	M12	23126	52,5
210	3	–	170	85	–	245	8	M12	23126	52,5
210	–	165	–	72	–	245	8	M12	23126	52,5
210	3	165	–	72	–	245	8	M12	23126	52,5
210	–	–	170	85	–	245	8	M12	23126	52,5
210	3	–	170	85	–	245	8	M12	23126	52,5
225	–	170	–	77	–	270	8	M12	23128	57
225	3	170	–	77	–	270	8	M12	23128	57
225	–	–	170	85	–	270	8	M12	23128	57
225	3	–	170	85	–	270	8	M12	23128	57
225	–	170	–	77	–	270	8	M12	23128	57
225	3	170	–	77	–	270	8	M12	23128	57
225	–	–	170	85	–	270	8	M12	23128	57
225	3	–	170	85	–	270	8	M12	23128	57
250	–	186	–	85	–	285	8	M12	23130	66,4
250	2	186	–	85	–	285	8	M12	23130	66,4
250	–	–	186	93	–	285	8	M12	23130	66,4
250	2	–	186	93	–	285	8	M12	23130	66,4
250	–	186	–	85	–	285	8	M12	23130	66,4
250	2	186	–	85	–	285	8	M12	23130	66,4
250	–	–	186	93	–	285	8	M12	23130	66,4
250	2	–	186	93	–	285	8	M12	23130	66,4

11.11.3 BND, for bearings with a cylindrical bore

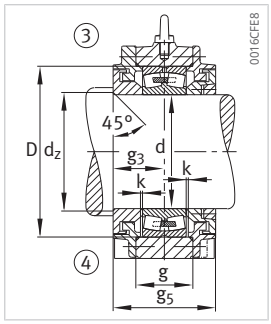




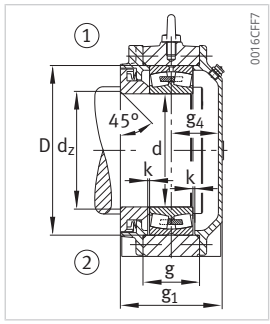
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3132-Z-T-AF-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-T-AL-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-T-BF-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-T-BL-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-Y-AF-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-Y-AL-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-Y-BF-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3132-Z-Y-BL-D	**160**	168	180	360	110	200	55	540	430	110	35	30	M24
BND3134-Z-T-AF-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-T-AL-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-T-BF-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-T-BL-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-Y-AF-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-Y-AL-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-Y-BF-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3134-Z-Y-BL-D	**170**	178	190	380	120	200	55	570	470	110	45	36	M30
BND3136-Z-T-AF-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-T-AL-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-T-BF-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-T-BL-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-Y-AF-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-Y-AL-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-Y-BF-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3136-Z-Y-BL-D	**180**	192	200	395	125	200	58	600	490	105	45	36	M30
BND3138-Z-T-AF-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-T-AL-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-T-BF-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-T-BL-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-Y-AF-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-Y-AL-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-Y-BF-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3138-Z-Y-BL-D	**190**	202	210	425	130	210	65	680	550	120	45	36	M30
BND3140-Z-T-AF-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-T-AL-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-T-BF-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-T-BL-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-Y-AF-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-Y-AL-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-Y-BF-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36
BND3140-Z-Y-BL-D	**200**	212	220	455	135	220	85	710	560	120	52	42	M36



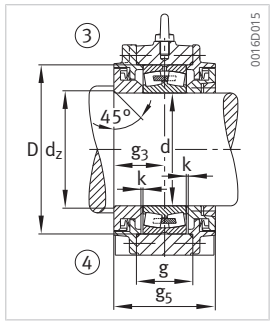
Design A
Labyrinth seal



Design B
Labyrinth seal



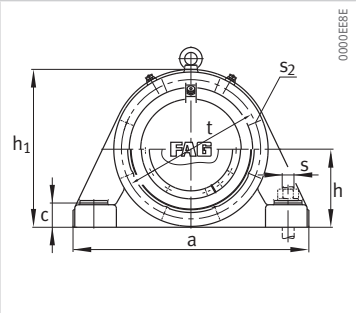
Design A
Taconite seal

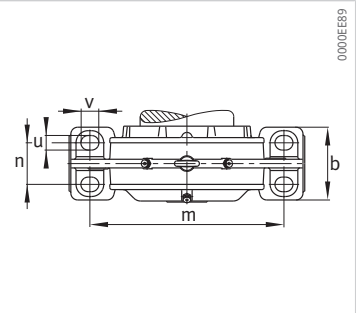


Design B
Taconite seal

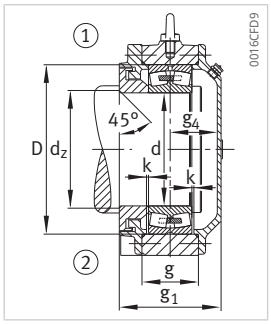
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
270	–	200	–	87	–	310	6	M16	23132	76
270	3	200	–	87	–	310	6	M16	23132	76
270	–	–	210	105	–	310	6	M16	23132	76
270	3	–	210	105	–	310	6	M16	23132	76
270	–	200	–	87	–	310	6	M16	23132	76
270	3	200	–	87	–	310	6	M16	23132	76
270	–	–	210	105	–	310	6	M16	23132	76
270	3	–	210	105	–	310	6	M16	23132	76
280	–	215	–	90	–	330	8	M12	23134	94,5
280	3	215	–	90	–	330	8	M12	23134	94,5
280	–	–	230	115	–	330	8	M12	23134	94,5
280	3	–	230	115	–	330	8	M12	23134	94,5
280	–	215	–	90	–	330	8	M12	23134	94,5
280	3	215	–	90	–	330	8	M12	23134	94,5
280	–	–	230	115	–	330	8	M12	23134	94,5
280	3	–	230	115	–	330	8	M12	23134	94,5
300	–	220	–	95	–	350	8	M16	23136	104
300	3	220	–	95	–	350	8	M16	23136	104
300	–	–	230	115	–	350	8	M16	23136	104
300	3	–	230	115	–	350	8	M16	23136	104
300	–	220	–	95	–	350	8	M16	23136	104
300	3	220	–	95	–	350	8	M16	23136	104
300	–	–	230	115	–	350	8	M16	23136	104
300	3	–	230	115	–	350	8	M16	23136	104
320	–	232	–	98	–	370	8	M16	23138	119
320	3	232	–	98	–	370	8	M16	23138	119
320	–	–	246	123	–	370	8	M16	23138	119
320	3	–	246	123	–	370	8	M16	23138	119
320	–	232	–	98	–	370	8	M16	23138	119
320	3	232	–	98	–	370	8	M16	23138	119
320	–	–	246	123	–	370	8	M16	23138	119
320	3	–	246	123	–	370	8	M16	23138	119
340	–	242	–	108	–	380	8	M16	23140	162
340	3	242	–	108	–	380	8	M16	23140	162
340	–	–	248	124	–	380	8	M16	23140	162
340	3	–	248	124	–	380	8	M16	23140	162
340	–	242	–	108	–	380	8	M16	23140	162
340	3	242	–	108	–	380	8	M16	23140	162
340	–	–	248	124	–	380	8	M16	23140	162
340	3	–	248	124	–	380	8	M16	23140	162

11.11.3 BND, for bearings with a cylindrical bore

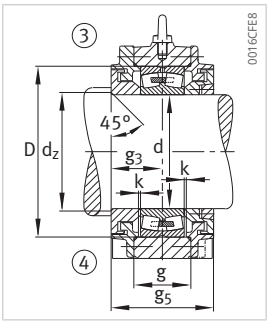




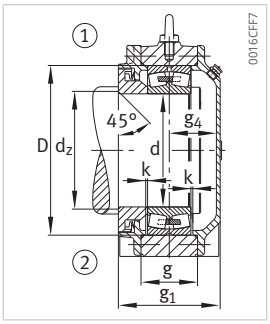
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3144-Z-T-AF-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-T-AL-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-T-BF-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-T-BL-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-Y-AF-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-Y-AL-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-Y-BF-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3144-Z-Y-BL-D	**220**	236	235	475	150	240	75	780	640	140	52	42	M36
BND3148-Z-T-AF-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-T-AL-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-T-BF-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-T-BL-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-Y-AF-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-Y-AL-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-Y-BF-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3148-Z-Y-BL-D	**240**	256	270	550	160	250	80	890	720	140	52	42	M36
BND3152-Z-T-AF-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-T-AL-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-T-BF-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-T-BL-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-Y-AF-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-Y-AL-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-Y-BF-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3152-Z-Y-BL-D	**260**	276	290	585	174	250	90	900	750	140	52	42	M36
BND3156-Z-T-AF-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-T-AL-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-T-BF-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-T-BL-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-Y-AF-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-Y-AL-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-Y-BF-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3156-Z-Y-BL-D	**280**	300	290	585	176	250	90	900	750	140	52	42	M36
BND3160-Z-T-AF-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-T-AL-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-T-BF-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-T-BL-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-Y-AF-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-Y-AL-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-Y-BF-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36
BND3160-Z-Y-BL-D	**300**	320	320	645	190	300	100	1000	840	170	52	42	M36



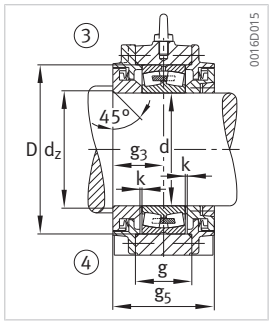
Design A
Labyrinth seal



Design B
Labyrinth seal



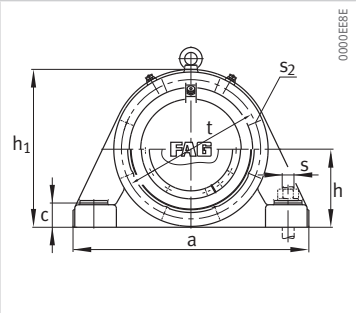
Design A
Taconite seal

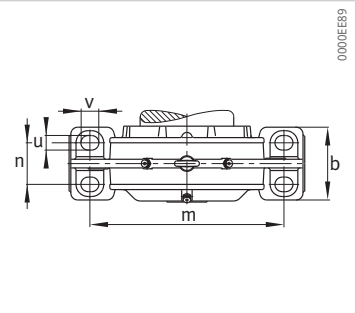


Design B
Taconite seal

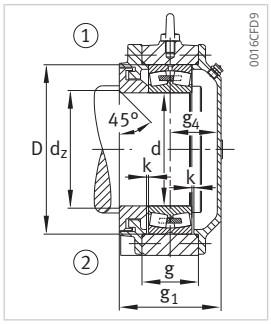
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
370	–	252	–	110	–	420	8	M16	23144	181
370	4	252	–	110	–	420	8	M16	23144	181
370	–	–	268	134	–	420	8	M16	23144	181
370	4	–	268	134	–	420	8	M16	23144	181
370	–	252	–	110	–	420	8	M16	23144	181
370	4	252	–	110	–	420	8	M16	23144	181
370	–	–	268	134	–	420	8	M16	23144	181
370	4	–	268	134	–	420	8	M16	23144	181
400	–	284	–	122	–	455	8	M20	23148	265
400	4	284	–	122	–	455	8	M20	23148	265
400	–	–	304	152	–	455	8	M20	23148	265
400	4	–	304	152	–	455	8	M20	23148	265
400	–	284	–	122	–	455	8	M20	23148	265
400	4	284	–	122	–	455	8	M20	23148	265
400	–	–	304	152	–	455	8	M20	23148	265
400	4	–	304	152	–	455	8	M20	23148	265
440	–	292	–	132	–	510	8	M20	23152	294
440	4	292	–	132	–	510	8	M20	23152	294
440	–	–	300	150	–	510	8	M20	23152	294
440	4	–	300	150	–	510	8	M20	23152	294
440	–	292	–	132	–	510	8	M20	23152	294
440	4	292	–	132	–	510	8	M20	23152	294
440	–	–	300	150	–	510	8	M20	23152	294
440	4	–	300	150	–	510	8	M20	23152	294
460	–	294	–	133	–	510	8	M20	23156	317
460	4	294	–	133	–	510	8	M20	23156	317
460	–	–	302	151	–	510	8	M20	23156	317
460	4	–	302	151	–	510	8	M20	23156	317
460	–	294	–	133	–	510	8	M20	23156	317
460	4	294	–	133	–	510	8	M20	23156	317
460	–	–	302	151	–	510	8	M20	23156	317
460	4	–	302	151	–	510	8	M20	23156	317
500	–	327	–	143	–	555	8	M24	23160	377
500	4	327	–	143	–	555	8	M24	23160	377
500	–	–	344	172	–	555	8	M24	23160	377
500	4	–	344	172	–	555	8	M24	23160	377
500	–	327	–	143	–	555	8	M24	23160	377
500	4	327	–	143	–	555	8	M24	23160	377
500	–	–	344	172	–	555	8	M24	23160	377
500	4	–	344	172	–	555	8	M24	23160	377

11.11.3 BND, for bearings with a cylindrical bore

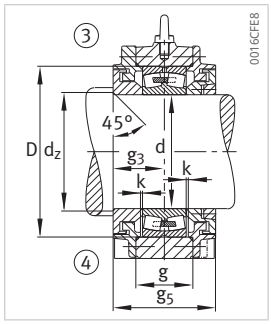




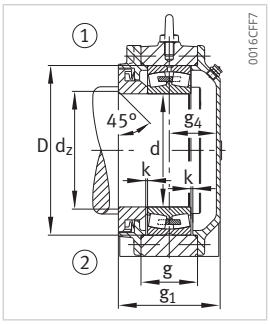
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3164-Z-T-AF-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-T-AL-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-T-BF-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-T-BL-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-Y-AF-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-Y-AL-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-Y-BF-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3164-Z-Y-BL-D	**320**	340	350	700	210	300	100	1150	940	160	52	42	M36
BND3168-Z-T-AF-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-T-AL-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-T-BF-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-T-BL-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-Y-AF-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-Y-AL-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-Y-BF-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3168-Z-Y-BL-D	**340**	360	370	745	220	360	115	1150	960	200	75	56	M48
BND3172-Z-T-AF-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-T-AL-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-T-BF-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-T-BL-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-Y-AF-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-Y-AL-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-Y-BF-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3172-Z-Y-BL-D	**360**	380	380	760	225	370	115	1200	1000	200	75	56	M48
BND3176-Z-T-AF-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-T-AL-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-T-BF-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-T-BL-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-Y-AF-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-Y-AL-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-Y-BF-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3176-Z-Y-BL-D	**380**	400	390	790	230	380	125	1200	1000	200	85	64	M56
BND3180-Z-T-AF-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-T-AL-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-T-BF-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-T-BL-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-Y-AF-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-Y-AL-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-Y-BF-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64
BND3180-Z-Y-BL-D	**400**	424	410	820	235	400	130	1280	1040	210	90	72	M64



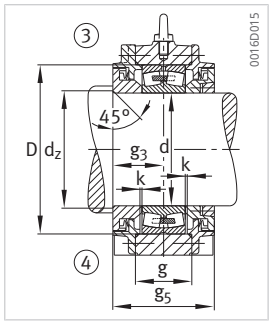
Design A
Labyrinth seal



Design B
Labyrinth seal



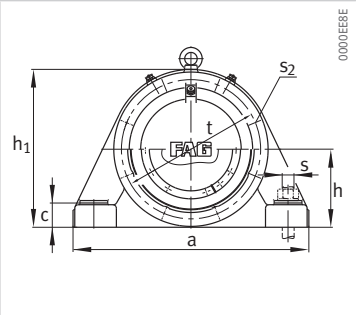
Design A
Taconite seal

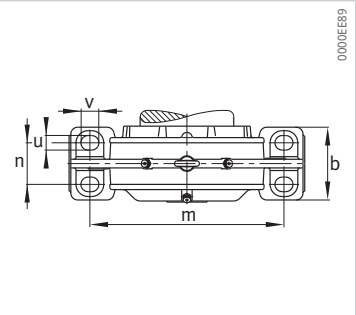


Design B
Taconite seal

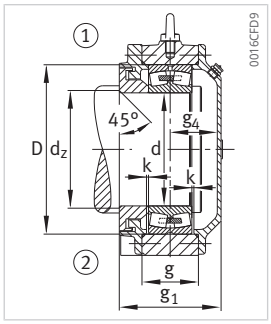
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
540	–	359	–	161	–	590	8	M24	23164	476,5
540	4	359	–	161	–	590	8	M24	23164	476,5
540	–	–	372	186	–	590	8	M24	23164	476,5
540	4	–	372	186	–	590	8	M24	23164	476,5
540	–	359	–	161	–	590	8	M24	23164	476,5
540	4	359	–	161	–	590	8	M24	23164	476,5
540	–	–	372	186	–	590	8	M24	23164	476,5
540	4	–	372	186	–	590	8	M24	23164	476,5
580	–	373	–	178	–	640	8	M24	23168	492
580	5	373	–	178	–	640	8	M24	23168	492
580	–	–	366	183	–	640	8	M24	23168	492
580	5	–	366	183	–	640	8	M24	23168	492
580	–	373	–	178	–	640	8	M24	23168	492
580	5	373	–	178	–	640	8	M24	23168	492
580	–	–	366	183	–	640	8	M24	23168	492
580	5	–	366	183	–	640	8	M24	23168	492
600	–	400	–	188	–	650	8	M24	23172	568
600	4	400	–	188	–	650	8	M24	23172	568
600	–	–	400	200	–	650	8	M24	23172	568
600	4	–	400	200	–	650	8	M24	23172	568
600	–	400	–	188	–	650	8	M24	23172	568
600	4	400	–	188	–	650	8	M24	23172	568
600	–	–	400	200	–	650	8	M24	23172	568
600	4	–	400	200	–	650	8	M24	23172	568
620	–	404	–	190	–	680	8	M30	23176	682
620	5	404	–	190	–	680	8	M30	23176	682
620	–	–	404	202	–	680	8	M30	23176	682
620	5	–	404	202	–	680	8	M30	23176	682
620	–	404	–	190	–	680	8	M30	23176	682
620	5	404	–	190	–	680	8	M30	23176	682
620	–	–	404	202	–	680	8	M30	23176	682
620	5	–	404	202	–	680	8	M30	23176	682
650	–	405	–	193	–	710	8	M30	23180	710
650	5	405	–	193	–	710	8	M30	23180	710
650	–	–	400	200	–	710	8	M30	23180	710
650	5	–	400	200	–	710	8	M30	23180	710
650	–	405	–	193	–	710	8	M30	23180	710
650	5	405	–	193	–	710	8	M30	23180	710
650	–	–	400	200	–	710	8	M30	23180	710
650	5	–	400	200	–	710	8	M30	23180	710

11.11.3 BND, for bearings with a cylindrical bore

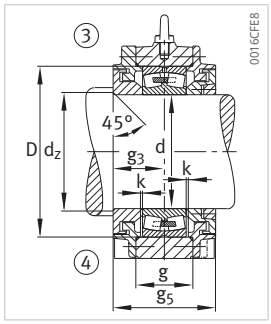




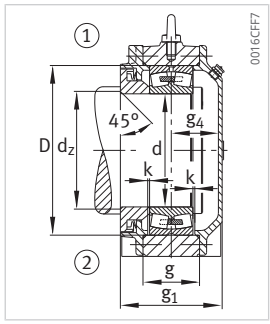
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3184-Z-T-AF-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-T-AL-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-T-BF-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-T-BL-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-Y-AF-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-Y-AL-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-Y-BF-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3184-Z-Y-BL-D	**420**	444	450	900	260	420	135	1350	1100	210	85	64	M56
BND3222-Z-T-AF-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-T-AL-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-T-BF-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-T-BL-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-Y-AF-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-Y-AL-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-Y-BF-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3222-Z-Y-BL-D	**110**	118	140	280	89,8	130	42	440	350	70	35	30	M24
BND3224-Z-T-AF-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-T-AL-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-T-BF-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-T-BL-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-Y-AF-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-Y-AL-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-Y-BF-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3224-Z-Y-BL-D	**120**	128	150	300	95	140	42	470	370	75	35	30	M24
BND3226-Z-T-AF-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-T-AL-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-T-BF-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-T-BL-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-Y-AF-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-Y-AL-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-Y-BF-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3226-Z-Y-BL-D	**130**	142	160	315	100	150	45	500	400	85	35	30	M24
BND3228-Z-T-AF-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-T-AL-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-T-BF-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-T-BL-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-Y-AF-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-Y-AL-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-Y-BF-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24
BND3228-Z-Y-BL-D	**140**	152	170	345	108	160	50	530	430	85	35	30	M24



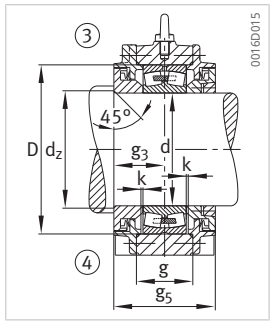
Design A
Labyrinth seal



Design B
Labyrinth seal



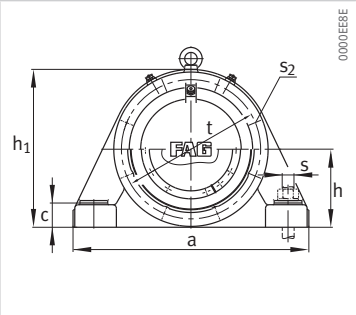
Design A
Taconite seal

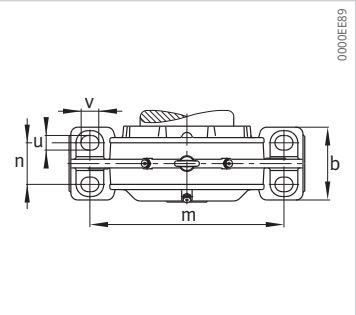


Design B
Taconite seal

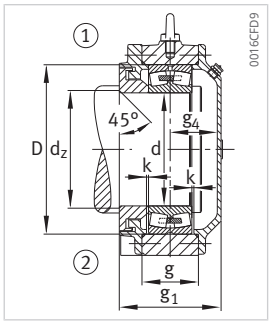
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
700	–	440	–	215	–	760	8	M30	23184	900
700	7	440	–	215	–	760	8	M30	23184	900
700	–	–	420	210	–	760	8	M30	23184	900
700	7	–	420	210	–	760	8	M30	23184	900
700	–	440	–	215	–	760	8	M30	23184	900
700	7	440	–	215	–	760	8	M30	23184	900
700	–	–	420	210	–	760	8	M30	23184	900
700	7	–	420	210	–	760	8	M30	23184	900
200	–	166	–	68	–	230	6	M12	23222	33,4
200	2,5	166	–	68	–	230	6	M12	23222	33,4
200	–	–	182	91	–	230	6	M12	23222	33,4
200	2,5	–	182	91	–	230	6	M12	23222	33,4
200	–	166	–	68	–	230	6	M12	23222	33,4
200	2,5	166	–	68	–	230	6	M12	23222	33,4
200	–	–	182	91	–	230	6	M12	23222	33,4
200	2,5	–	182	91	–	230	6	M12	23222	33,4
215	–	161	–	70	–	245	6	M12	23224	38
215	2,5	161	–	70	–	245	6	M12	23224	38
215	–	–	166	83	–	245	6	M12	23224	38
215	2,5	–	166	83	–	245	6	M12	23224	38
215	–	161	–	70	–	245	6	M12	23224	38
215	2,5	161	–	70	–	245	6	M12	23224	38
215	–	–	166	83	–	245	6	M12	23224	38
215	2,5	–	166	83	–	245	6	M12	23224	38
230	–	188	–	77	–	260	6	M12	23226	47,8
230	3	188	–	77	–	260	6	M12	23226	47,8
230	–	–	206	103	–	260	6	M12	23226	47,8
230	3	–	206	103	–	260	6	M12	23226	47,8
230	–	188	–	77	–	260	6	M12	23226	47,8
230	3	188	–	77	–	260	6	M12	23226	47,8
230	–	–	206	103	–	260	6	M12	23226	47,8
230	3	–	206	103	–	260	6	M12	23226	47,8
250	–	186	–	80	–	285	6	M16	23228	61,5
250	3	186	–	80	–	285	6	M16	23228	61,5
250	–	–	196	98	–	285	6	M16	23228	61,5
250	3	–	196	98	–	285	6	M16	23228	61,5
250	–	186	–	80	–	285	6	M16	23228	61,5
250	3	186	–	80	–	285	6	M16	23228	61,5
250	–	–	196	98	–	285	6	M16	23228	61,5
250	3	–	196	98	–	285	6	M16	23228	61,5

11.11.3 BND, for bearings with a cylindrical bore

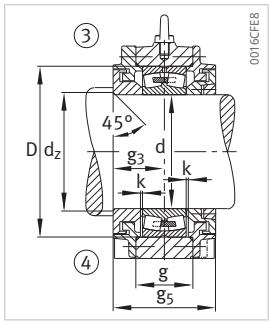




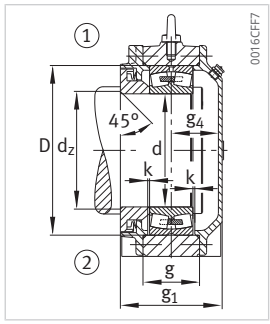
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3230-Z-T-AF-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-T-AL-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-T-BF-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-T-BL-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-Y-AF-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-Y-AL-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-Y-BF-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3230-Z-Y-BL-D	**150**	162	180	365	120	170	54	550	450	90	45	36	M30
BND3232-Z-T-AF-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-T-AL-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-T-BF-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-T-BL-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-Y-AF-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-Y-AL-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-Y-BF-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3232-Z-Y-BL-D	**160**	172	190	385	130	180	58	600	490	105	45	36	M30
BND3234-Z-T-AF-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-T-AL-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-T-BF-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-T-BL-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-Y-AF-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-Y-AL-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-Y-BF-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3234-Z-Y-BL-D	**170**	186	200	405	135	200	62	640	525	110	45	36	M30
BND3236-Z-T-AF-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-T-AL-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-T-BF-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-T-BL-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-Y-AF-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-Y-AL-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-Y-BF-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3236-Z-Y-BL-D	**180**	196	210	425	138	210	65	680	550	120	45	36	M30
BND3238-Z-T-AF-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-T-AL-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-T-BF-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-T-BL-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-Y-AF-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-Y-AL-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-Y-BF-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36
BND3238-Z-Y-BL-D	**190**	206	220	455	143	220	85	710	560	120	52	42	M36



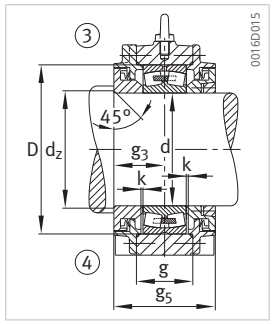
Design A
Labyrinth seal



Design B
Labyrinth seal



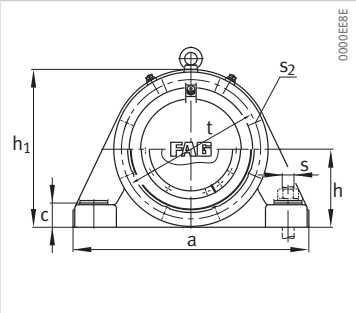
Design A
Taconite seal

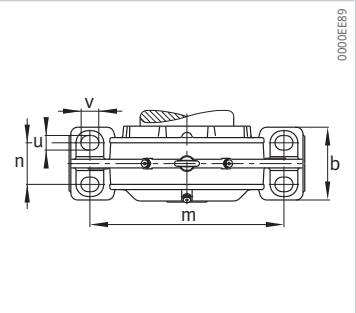


Design B
Taconite seal

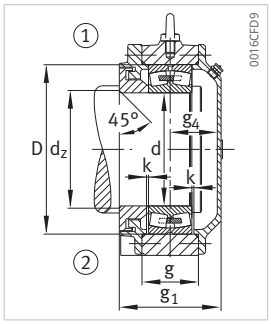
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
270	–	200	–	87	–	305	6	M16	23230	71
270	3	200	–	87	–	305	6	M16	23230	71
270	–	–	210	105	–	305	6	M16	23230	71
270	3	–	210	105	–	305	6	M16	23230	71
270	–	200	–	87	–	305	6	M16	23230	71
270	3	200	–	87	–	305	6	M16	23230	71
270	–	–	210	105	–	305	6	M16	23230	71
270	3	–	210	105	–	305	6	M16	23230	71
290	–	213	–	90	–	330	6	M16	23232	85,5
290	3	213	–	90	–	330	6	M16	23232	85,5
290	–	–	230	115	–	330	6	M16	23232	85,5
290	3	–	230	115	–	330	6	M16	23232	85,5
290	–	213	–	90	–	330	6	M16	23232	85,5
290	3	213	–	90	–	330	6	M16	23232	85,5
290	–	–	230	115	–	330	6	M16	23232	85,5
290	3	–	230	115	–	330	6	M16	23232	85,5
310	–	240	–	100	–	350	8	M16	23234	113,5
310	3	240	–	100	–	350	8	M16	23234	113,5
310	–	–	260	130	–	350	8	M16	23234	113,5
310	3	–	260	130	–	350	8	M16	23234	113,5
310	–	240	–	100	–	350	8	M16	23234	113,5
310	3	240	–	100	–	350	8	M16	23234	113,5
310	–	–	260	130	–	350	8	M16	23234	113,5
310	3	–	260	130	–	350	8	M16	23234	113,5
320	–	240	–	105	–	370	8	M16	23236	133
320	3	240	–	105	–	370	8	M16	23236	133
320	–	–	254	127	–	370	8	M16	23236	133
320	3	–	254	127	–	370	8	M16	23236	133
320	–	240	–	105	–	370	8	M16	23236	133
320	3	240	–	105	–	370	8	M16	23236	133
320	–	–	254	127	–	370	8	M16	23236	133
320	3	–	254	127	–	370	8	M16	23236	133
340	–	250	–	112	–	380	8	M16	23238	161,5
340	3	250	–	112	–	380	8	M16	23238	161,5
340	–	–	256	128	–	380	8	M16	23238	161,5
340	3	–	256	128	–	380	8	M16	23238	161,5
340	–	250	–	112	–	380	8	M16	23238	161,5
340	3	250	–	112	–	380	8	M16	23238	161,5
340	–	–	256	128	–	380	8	M16	23238	161,5
340	3	–	256	128	–	380	8	M16	23238	161,5

11.11.3 BND, for bearings with a cylindrical bore

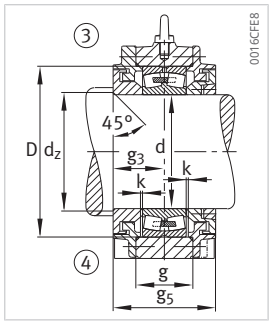




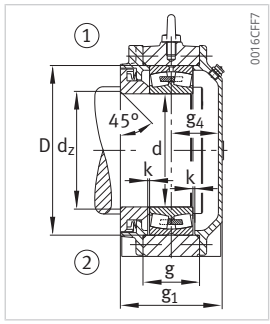
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3240-Z-T-AF-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-T-AL-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-T-BF-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-T-BL-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-Y-AF-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-Y-AL-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-Y-BF-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3240-Z-Y-BL-D	**200**	216	235	475	158	240	75	780	640	140	52	42	M36
BND3244-Z-T-AF-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-T-AL-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-T-BF-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-T-BL-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-Y-AF-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-Y-AL-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-Y-BF-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3244-Z-Y-BL-D	**220**	236	260	525	175	250	80	850	700	140	52	42	M36
BND3248-Z-T-AF-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-T-AL-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-T-BF-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-T-BL-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-Y-AF-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-Y-AL-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-Y-BF-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3248-Z-Y-BL-D	**240**	256	290	585	190	250	90	900	750	140	52	42	M36
BND3252-Z-T-AF-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-T-AL-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-T-BF-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-T-BL-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-Y-AF-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-Y-AL-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-Y-BF-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3252-Z-Y-BL-D	**260**	280	310	625	205	290	95	960	800	160	52	42	M36
BND3256-Z-T-AF-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-T-AL-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-T-BF-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-T-BL-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-Y-AF-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-Y-AL-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-Y-BF-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36
BND3256-Z-Y-BL-D	**280**	300	320	645	206	300	100	1000	840	170	52	42	M36



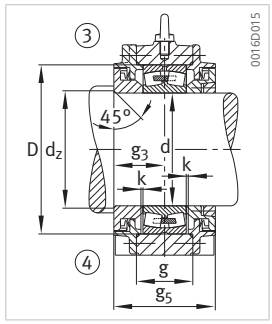
Design A
Labyrinth seal



Design B
Labyrinth seal



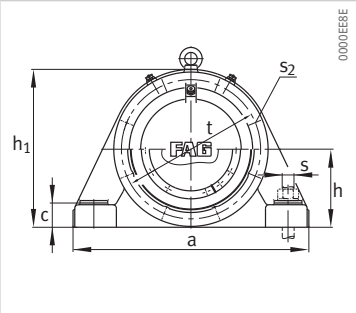
Design A
Taconite seal

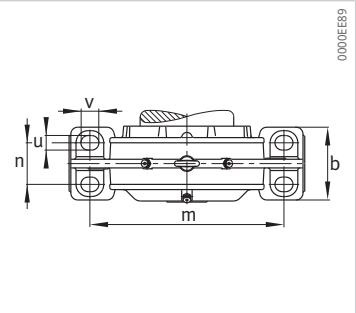


Design B
Taconite seal

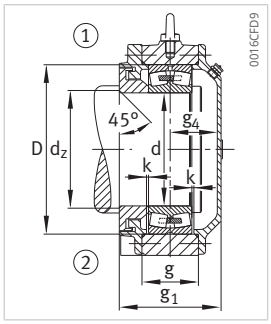
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
360	–	260	–	114	–	420	8	M16	23240	195,5
360	4	260	–	114	–	420	8	M16	23240	195,5
360	–	–	276	138	–	420	8	M16	23240	195,5
360	4	–	276	138	–	420	8	M16	23240	195,5
360	–	260	–	114	–	420	8	M16	23240	195,5
360	4	260	–	114	–	420	8	M16	23240	195,5
360	–	–	276	138	–	420	8	M16	23240	195,5
360	4	–	276	138	–	420	8	M16	23240	195,5
400	–	279	–	122	–	445	8	M20	23244	229
400	4	279	–	122	–	445	8	M20	23244	229
400	–	–	294	147	–	445	8	M20	23244	229
400	4	–	294	147	–	445	8	M20	23244	229
400	–	279	–	122	–	445	8	M20	23244	229
400	4	279	–	122	–	445	8	M20	23244	229
400	–	–	294	147	–	445	8	M20	23244	229
400	4	–	294	147	–	445	8	M20	23244	229
440	–	308	–	140	–	510	8	M20	23248	313
440	4	308	–	140	–	510	8	M20	23248	313
440	–	–	316	158	–	510	8	M20	23248	313
440	4	–	316	158	–	510	8	M20	23248	313
440	–	308	–	140	–	510	8	M20	23248	313
440	4	308	–	140	–	510	8	M20	23248	313
440	–	–	316	158	–	510	8	M20	23248	313
440	4	–	316	158	–	510	8	M20	23248	313
480	–	330	–	148	–	535	8	M20	23252	360
480	3	330	–	148	–	535	8	M20	23252	360
480	–	–	340	170	–	535	8	M20	23252	360
480	3	–	340	170	–	535	8	M20	23252	360
480	–	330	–	148	–	535	8	M20	23252	360
480	3	330	–	148	–	535	8	M20	23252	360
480	–	–	340	170	–	535	8	M20	23252	360
480	3	–	340	170	–	535	8	M20	23252	360
500	–	343	–	151	–	555	8	M24	23256	462
500	4	343	–	151	–	555	8	M24	23256	462
500	–	–	360	180	–	555	8	M24	23256	462
500	4	–	360	180	–	555	8	M24	23256	462
500	–	343	–	151	–	555	8	M24	23256	462
500	4	343	–	151	–	555	8	M24	23256	462
500	–	–	360	180	–	555	8	M24	23256	462
500	4	–	360	180	–	555	8	M24	23256	462

11.11.3 BND, for bearings with a cylindrical bore

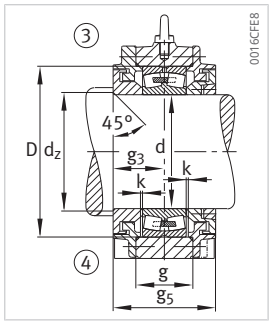




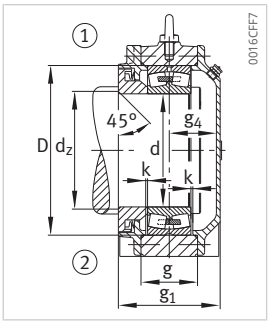
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3260-Z-T-AF-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-T-AL-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-T-BF-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-T-BL-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-Y-AF-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-Y-AL-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-Y-BF-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3260-Z-Y-BL-D	**300**	320	350	705	230	330	105	1100	920	180	75	56	M48
BND3264-Z-T-AF-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-T-AL-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-T-BF-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-T-BL-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-Y-AF-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-Y-AL-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-Y-BF-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3264-Z-Y-BL-D	**320**	340	370	745	238	360	115	1150	960	200	75	56	M48
BND3268-Z-T-AF-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-T-AL-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-T-BF-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-T-BL-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-Y-AF-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-Y-AL-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-Y-BF-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3268-Z-Y-BL-D	**340**	364	390	790	260	380	125	1200	990	200	85	64	M56
BND3272-Z-T-AF-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-T-AL-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-T-BF-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-T-BL-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-Y-AF-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-Y-AL-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-Y-BF-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3272-Z-Y-BL-D	**360**	384	410	820	267	400	130	1280	1040	210	90	72	M64
BND3276-Z-T-AF-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-T-AL-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-T-BF-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-T-BL-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-Y-AF-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-Y-AL-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-Y-BF-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64
BND3276-Z-Y-BL-D	**380**	404	425	860	295	405	135	1350	1100	225	90	72	M64



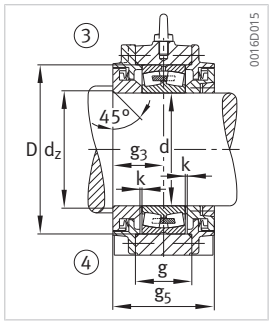
Design A
Labyrinth seal



Design B
Labyrinth seal



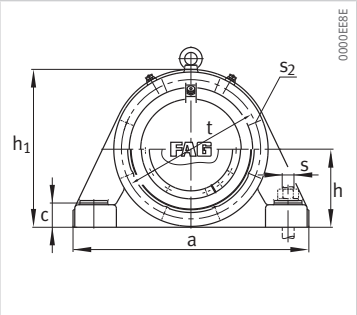
Design A
Taconite seal

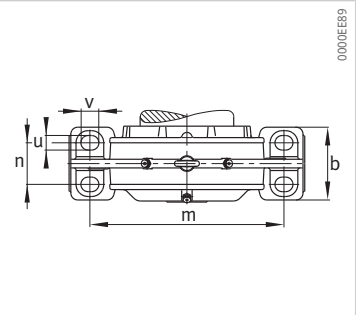


Design B
Taconite seal

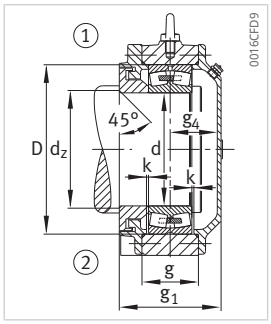
D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
540	–	369	–	175	–	600	8	M24	23260	538
540	4	369	–	175	–	600	8	M24	23260	538
540	–	–	364	182	–	600	8	M24	23260	538
540	4	–	364	182	–	600	8	M24	23260	538
540	–	369	–	175	–	600	8	M24	23260	538
540	4	369	–	175	–	600	8	M24	23260	538
540	–	–	364	182	–	600	8	M24	23260	538
540	4	–	364	182	–	600	8	M24	23260	538
580	–	391	–	187	–	640	8	M24	23264	630
580	5	391	–	187	–	640	8	M24	23264	630
580	–	–	384	192	–	640	8	M24	23264	630
580	5	–	384	192	–	640	8	M24	23264	630
580	–	391	–	187	–	640	8	M24	23264	630
580	5	391	–	187	–	640	8	M24	23264	630
580	–	–	384	192	–	640	8	M24	23264	630
580	5	–	384	192	–	640	8	M24	23264	630
620	–	434	–	205	–	680	8	M30	23268	715
620	5	434	–	205	–	680	8	M30	23268	715
620	–	–	434	217	–	680	8	M30	23268	715
620	5	–	434	217	–	680	8	M30	23268	715
620	–	434	–	205	–	680	8	M30	23268	715
620	5	434	–	205	–	680	8	M30	23268	715
620	–	–	434	217	–	680	8	M30	23268	715
620	5	–	434	217	–	680	8	M30	23268	715
650	–	437	–	209	–	710	8	M30	23272	900
650	5	437	–	209	–	710	8	M30	23272	900
650	–	–	432	216	–	710	8	M30	23272	900
650	5	–	432	216	–	710	8	M30	23272	900
650	–	437	–	209	–	710	8	M30	23272	900
650	5	437	–	209	–	710	8	M30	23272	900
650	–	–	432	216	–	710	8	M30	23272	900
650	5	–	432	216	–	710	8	M30	23272	900
680	–	489	–	244	–	745	8	M30	23276	1045
680	5	489	–	244	–	745	8	M30	23276	1045
680	–	–	464	232	–	745	8	M30	23276	1045
680	5	–	464	232	–	745	8	M30	23276	1045
680	–	489	–	244	–	745	8	M30	23276	1045
680	5	489	–	244	–	745	8	M30	23276	1045
680	–	–	464	232	–	745	8	M30	23276	1045
680	5	–	464	232	–	745	8	M30	23276	1045

11.11.3 BND, for bearings with a cylindrical bore

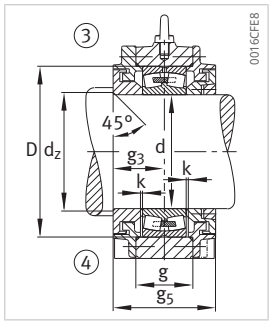




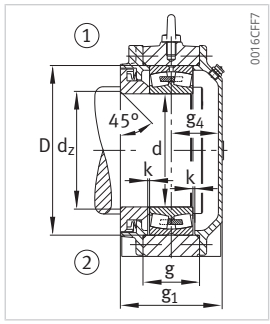
Housing	d	d_z	h	h_1	g	b	c	a	m	n	v	u	s
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	-
BND3280-Z-T-AF-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-T-AL-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-T-BF-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-T-BL-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-Y-AF-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-Y-AL-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-Y-BF-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3280-Z-Y-BL-D	**400**	424	450	900	300	450	145	1430	1160	240	90	72	M64
BND3284-Z-T-AF-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-T-AL-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-T-BF-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-T-BL-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-Y-AF-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-Y-AL-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-Y-BF-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64
BND3284-Z-Y-BL-D	**420**	450	470	950	315	470	150	1500	1220	255	90	72	M64



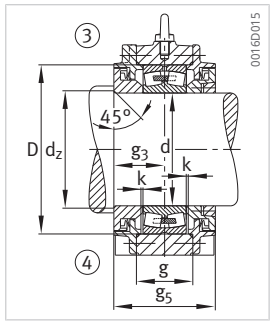
Design A
Labyrinth seal



Design B
Labyrinth seal



Design A
Taconite seal



Design B
Taconite seal

D	k	g_1	g_5	g_3	g_4	t	n	s	Bearing	m
mm	mm	mm	mm	mm	mm	mm	–	–	–	kg
720	–	504	–	237	–	790	8	M30	23280	1140
720	5	504	–	237	–	790	8	M30	23280	1140
720	–	–	504	252	–	790	8	M30	23280	1140
720	5	–	504	252	–	790	8	M30	23280	1140
720	–	504	–	237	–	790	8	M30	23280	1140
720	5	504	–	237	–	790	8	M30	23280	1140
720	–	–	504	252	–	790	8	M30	23280	1140
720	5	–	504	252	–	790	8	M30	23280	1140
760	–	510	–	240	–	835	8	M30	23284	1240
760	5	510	–	240	–	835	8	M30	23284	1240
760	–	–	510	255	–	835	8	M30	23284	1240
760	5	–	510	255	–	835	8	M30	23284	1240
760	–	510	–	240	–	835	8	M30	23284	1240
760	5	510	–	240	–	835	8	M30	23284	1240
760	–	–	510	255	–	835	8	M30	23284	1240
760	5	–	510	255	–	835	8	M30	23284	1240

12 SPA take-up housing

medias | Product catalogue |
SPA take-up bearing housing |
https://www.schaeffler.de/std/2203

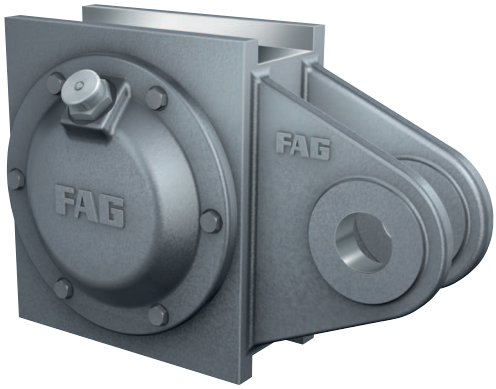
12.1 Housing design

Together with spherical roller bearings, seals and grease filling, SPA take-up housings form bearing units for very high loads.

The housings are unsplit. They were developed specifically for tensioner drum bearings in conveyor belt systems. There is a fork-shaped drawbar eye to attach to the tensioning device. The housing is guided by means of rails in the belt support structure.

Take-up housing, closed on one side, with cover

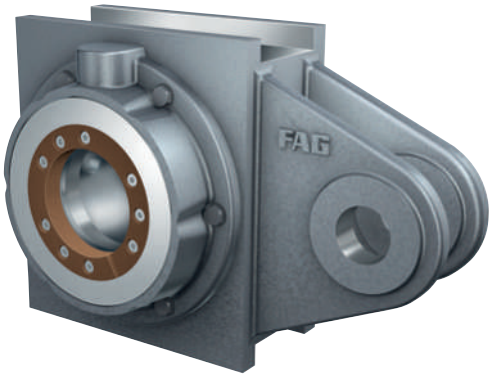
185 SPA..-AF, SPA..-AL take-up bearing housing, closed on one side, with cover



0001AD1C

Take-up bearing housing, for a continuous shaft

186 SPA..-AFSPA..-BF, SPA..-BL take-up bearing housing, for a continuous shaft



0001AD1D

12.1.1 Locating and non-locating bearings

The housings are supplied in the form of a locating or non-locating bearing design.

For the locating bearing, the bearing is clamped between the housing covers. For the non-locating bearing, the bearing can align itself axially, since the covers have shorter centring collars.

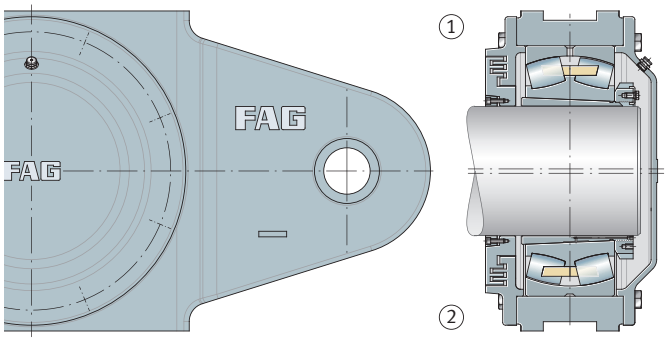
12.1.2 Housing configurations

Various housing configurations are possible for SPA take-up bearing housings, which are clearly described by the designation.

The following features can be varied in the housing configuration:

- Location of bearings with a tapered bore by means of an adapter sleeve on a shaft of constant diameter or on a stepped shaft
- Housing seal using a labyrinth seal or Taconite seal
- Continuous shaft or a housing closed on one side
- Design of the bearing as a locating or non-locating bearing

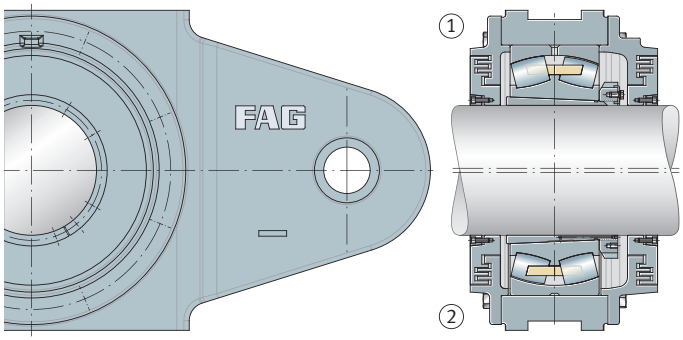
187 SPA take-up bearing housings for bearings with a tapered bore and adapter sleeve (labyrinth seal), design A



00085C25

1	SPA..-H-W-Y-AF locating bearing	2	SPA..-H-W-Y-AL non-locating bearing

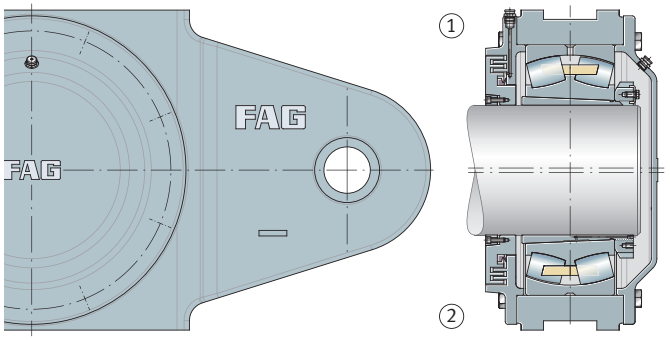
188 SPA take-up bearing housings for bearings with a tapered bore and adapter sleeve (labyrinth seal), design B



00085C29

1	SPA..-H-W-Y-BF locating bearing	2	SPA..-H-W-Y-BL non-locating bearing

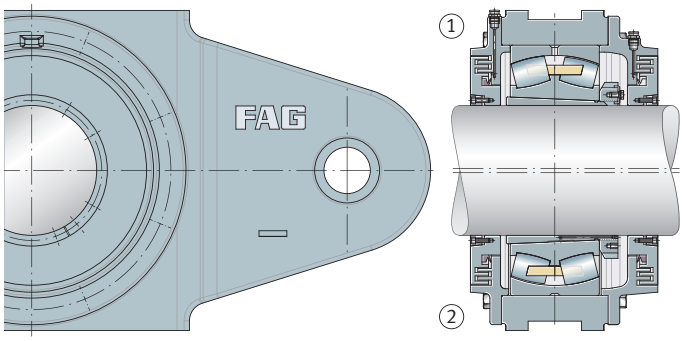
189 SPA take-up bearing housings for bearings with a tapered bore and adapter sleeve (Taconite seal), design A



00085C2D

1	SPA..-H-W-T-AF locating bearing	2	SPA..-H-W-T-AL non-locating bearing

190 SPA take-up bearing housings for bearings with a tapered bore and adapter sleeve (Taconite seal), design B



00085C31

1	SPA..-H-W-T-BF locating bearing	2	SPA..-H-W-T-BL non-locating bearing

12.2 Suitable bearings

The dimensions of SPA take-up bearing housings are tailored to 222..-K, 230..-K, 231..-K and 232..-K spherical roller bearings. The shaft diameters are 50 mm to 400 mm.

79 Suitable bearings for SPA take-up bearing housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	22211..-K	22228..-K
	23028..-K	23056..-K
	23132..-K	23176..-K
	23222..-K	23284..-K

12.3 Materials, corrosion protection

80 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Cast steel or steel	S	✓	RAL 7016, anthracite grey
Ductile cast iron GJS	D	o	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

12.4 Lubrication

SPA take-up bearing housings are designed for grease lubrication.

12.4.1 Grease types

NLGI class 3lithium soap grease types whose mineral base oil has viscosity class ISO VG 100 are suitable.

12.4.2 Grease quantities

The cavities in the bearing, the housing and the labyrinths are completely filled with grease during the initial lubrication process. The recommended grease quantities apply to this.

For relubrication, we recommend that the grease quantity is around 10 % of the grease quantity used in the initial process. If the ambient conditions result in heavy soiling of the housings, relubrication should be carried out on a daily basis using smaller quantities.

81 Recommended grease quantities for initial grease application

Bearing bore	Grease quantity for initial greasing	
	SPA22, SPA31, SPA32	SPA30
mm	g	g
180	2500	–
190	3000	–
200	3600	–
220	4200	1900
240	5000	2100
260	6000	2500
280	7000	3000
300	8000	3500
320	9000	4100
340	10500	4800
360	12000	5500
380	13000	6200
400	14500	7000
420	16000	8000

12.4.3 Relubrication

The housings have button head lubrication nipples with a standard head diameter to DIN 3404:1988 of 22 mm. The grease is fed uniformly to both rows of rollers through the circumferential slot and 3 lubrication holes in the outer ring of the spherical roller bearings.

The relubrication intervals should be tailored to the ambient conditions. The bearings should be relubricated after an interval of no more than 4 weeks.

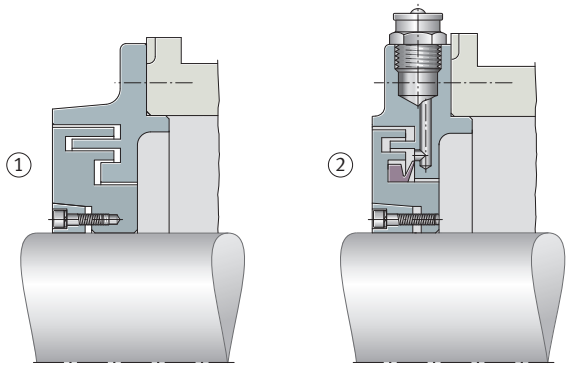
12.5 Sealing

Take-up housings in design A are intended for the bearings at shaft ends. One side is closed by a cover and the other side is sealed against the shaft. Design B is intended for a continuous shafts, in which case the housing is sealed against the shaft on both sides.

Labyrinth seals (suffix Y) or Taconite seals (suffix T) can be used. Where shafts of constant diameter are used, the seals are located on the shaft by means of slotted conical clamping rings made from laminated fabric. Labyrinth rings,Taconite seals and covers are unsplit.

The seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication.

191 Seals for SPA take-up bearing housings



1	Labyrinth seal	2	Taconite seal

12.6 Dimensions, tolerances

12.6.1 Tolerances for the bearing seat

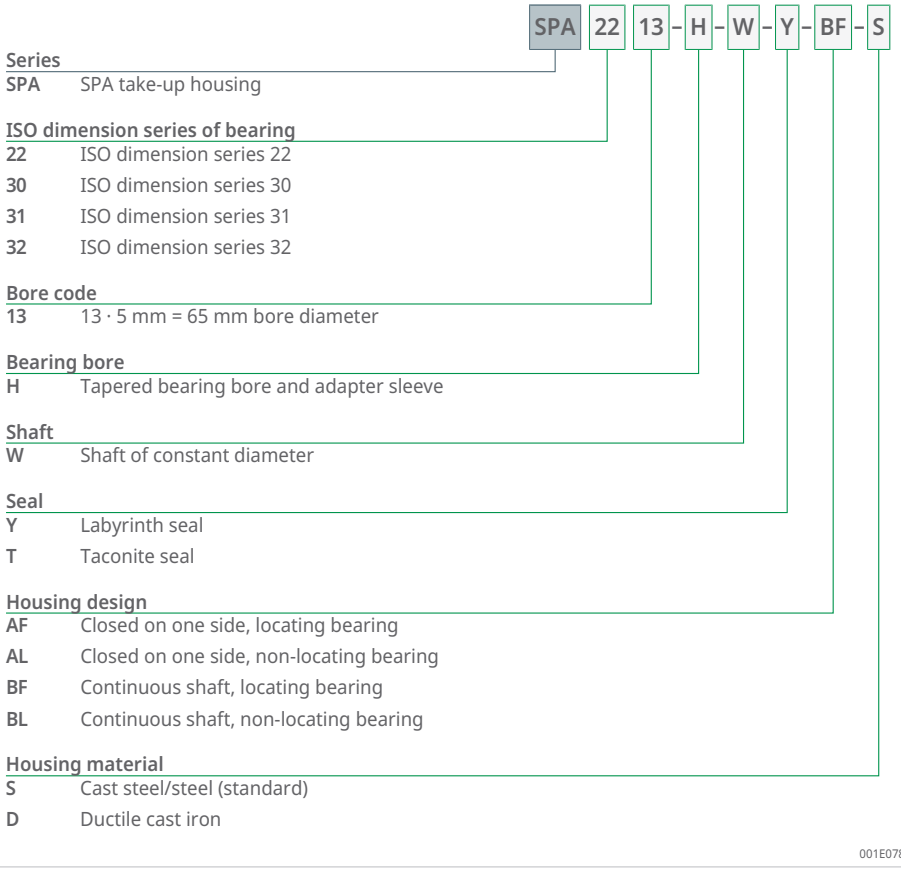
The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

12.7 Structure of the order designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

192 Structure of the designations of SPA take-up bearing housings



12.7.1 Specimen orders

For ordering an SPA take-up bearing housing, the designation contains details of the size and location method of the bearing, the shaft design, the sealing design and the design and material of the housing. The roller bearing and, if necessary, the adapter sleeve must be ordered separately.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

Example 1

Example of a housing configuration, characteristics:

- SPA take-up bearing housings made of cast steel
- For a continuous shaft with a constant diameter
- Labyrinth seal
- 23260-K-MB spherical roller bearing in the form of a locating bearing
- Secured by means of adapter sleeve

82 Order

Quantity	Description	Ordering designation, short designation
1	Take-up housing	SPA3260-H-W-Y-BF-S
1	Spherical roller bearing	23260-K-MB
1	Adapter sleeve	H3260-HG

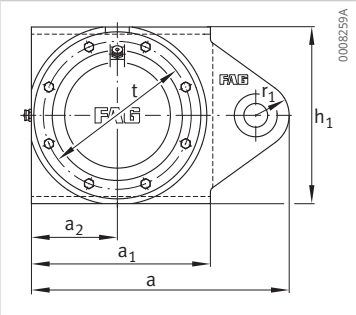
12.8 Product tables

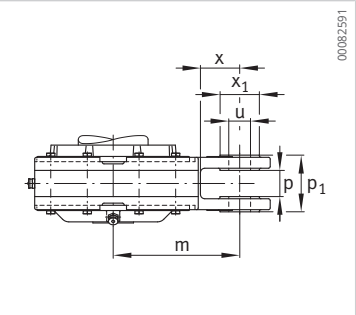
12.8.1 Explanations of the product tables

(3)	-	Locating bearing
(4)	-	Non-locating bearing
(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Total length
a_1	mm	Length of guide groove
a_2	mm	Distance to shaft axis
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
f	mm	Distance between guide groove
f_1	mm	Width of guide groove
g	mm	Housing body width
g_1	mm	Overall width
g_2	mm	Overall width
g_3	mm	Width over clamping element
g_4	mm	Spacing between the bearing centre and cover
h_1	mm	Overall height
k	mm	Axial movement distance
m	kg	Mass
m	mm	Distance between fork bore and shaft centre
n	–	Number of screw mounting holes
p	mm	Width of Inner fork
p_1	mm	Width of outer fork
r_1	mm	Radius fork
s_2	-	Fastening hole
t	mm	Pitch circle diameter of the bores
u	mm	Fork bore diameter
x	mm	Fork clearance
x_1	mm	Fork flange diameter

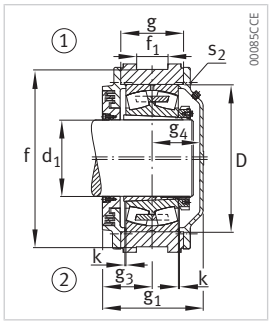
12.8.2 SPA, for bearings with a tapered bore and adapter sleeve

Unsplit

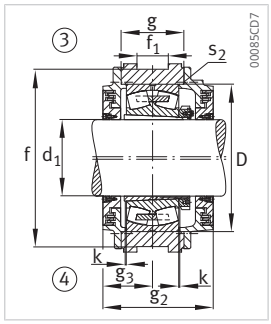




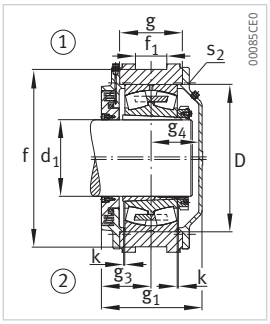
Housing	d_1	h_1	a	a_1	a_2	g_1	g_2	m	r_1	x	x_1	u	p	p_1
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
SPA2211..-S	**50**	150	210	150	75	92,5	115	105	30	30	–	20	22	42
SPA2212..-S	**55**	165	237,5	165	82,5	102,5	125	120	35	37,5	–	25	26	50
SPA2213..-S	**60**	180	265	160	80	134	144	145	40	65	60	30	30	66
SPA2216..-S	**70**	200	285	180	90	125	138	155	40	65	60	30	30	66
SPA2218..-S	**80**	230	315	210	105	119	138	170	40	65	60	30	30	66
SPA2220..-S	**90**	250	342,5	235	117,5	141	168	185	40	67,5	65	35	36	70
SPA2222..-S	**100**	270	380	265	135	149,2	177,2	200	45	70	70	40	40	76
SPA2224..-S	**110**	290	395	280	140	143	168	210	45	70	70	40	40	76
SPA2228..-S	**125**	330	445	320	160	166	196	230	55	70	70	40	40	76
SPA3028..-S	**125**	300	425	260	130	136	156	225	70	90	90	50	60	150
SPA3030..-S	**135**	320	445	280	140	149	176	235	70	95	90	50	60	150
SPA3034..-S	**150**	360	492,5	325	162,5	178	203	260	70	90	90	50	60	150
SPA3036..-S	**160**	370	540	375	180	176	201	290	70	90	90	50	60	130
SPA3038..-S	**170**	440	550	385	190	181	211	290	70	90	90	50	60	150
SPA3040..-S	**180**	440	550	385	190	201	241	290	70	90	90	50	60	150
SPA3044..-S	**200**	480	615	430	210	206	241	325	80	100	100	60	70	200
SPA3052..-S	**240**	540	910	500	245	226	261	515	150	250	140	100	123	215
SPA3056..-S	**260**	540	910	500	245	236	281	515	150	250	140	100	123	215
SPA3132..-S	**140**	370	625	370	185	216	229	350	90	160	100	60	62	132
SPA3134..-S	**150**	400	672,5	375	187,5	215	250	395	90	180	100	60	62	132
SPA3140..-S	**180**	440	650	420	210	242	278	360	80	140	100	60	70	140
SPA3144..-S	**200**	510	885	470	235	252	291	530	120	260	160	100	74	144
SPA3148..-S	**220**	540	925	520	260	276	326	515	150	250	160	100	123	215
SPA3152..-S	**240**	570	990	550	275	291	336	565	150	270	160	100	135	225
SPA3160..-S	**280**	650	1115	630	315	327	385	650	150	310	140	100	170	270
SPA3164..-S	**300**	670	1140	665	340	359	412	650	150	310	175	100	170	240
SPA3168..-S	**320**	740	1290	740	370	373	421	720	200	310	180	120	180	300
SPA3176..-S	**360**	790	1325	750	375	404	459	750	200	320	170	110	200	300
SPA3236..-S	**160**	450	600	420	210	240	274	300	90	90	120	80	72	140
SPA3240..-S	**180**	500	650	470	235	260	299	325	90	85	120	80	72	140
SPA3244..-S	**200**	540	940	520	260	279	329	530	150	250	140	100	130	200
SPA3248..-S	**220**	580	970	560	280	308	351	540	150	260	140	100	130	220
SPA3252..-S	**240**	610	1063	596	298	330	370	615	150	300	140	100	173	240
SPA3256..-S	**260**	650	1095	630	315	343	400	630	150	220	140	100	173	240
SPA3260..-S	**280**	690	1200	680	340	369	404	680	180	310	140	100	190	270
SPA3264..-S	**300**	710	1280	710	355	391	439	725	200	275	170	110	213	300
SPA3268..-S	**320**	740	1335	735	385	434	489	750	200	310	170	110	213	300
SPA3272..-S	**340**	830	1390	800	400	437	497	765	225	310	190	130	213	330
SPA3276..-S	**360**	820	1385	810	405	489	529	780	200	305	170	110	213	300
SPA3280..-S	**380**	960	1460	880	440	504	569	820	200	350	170	110	180	300
SPA3284..-S	**400**	960	1488	925	465	510	575	825	200	350	170	110	180	300



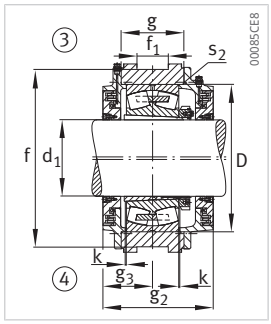
Design A,
Labyrinth seal



Design B,
Labyrinth seal



Design A,
Taconite seal



Design B,
Taconite seal

f	f_1	D	k	g	g_3	g_4	t	n	s	Bearing	Adapter sleeve	m
mm	mm	mm	mm	mm	mm	mm	mm	–	–	–	–	kg
135	34	100	2	45	52,5	35	130	6	M8	22211..-K	H311	8,5
150	35	110	2	55	57,5	40	135	6	M8	22212..-K	H312	11,5
155	53	120	2	78	72	56	135	6	M6	22213..-K	H313	15
175	53	140	2	78	69	49	155	8	M8	22216..-K	H316	24
200	53	160	2	70	60,5	52,5	185	6	M8	22218..-K	H318	21
220	40	180	2,5	65	78	57	205	6	M8	22220..-K	H320	24
240	53	200	2,5	73	82,6	60,6	230	6	M12	22222..-K	H322	34
260	53	215	2,5	77	74	61	245	6	M12	22224..-K	H3124	38
300	53	250	3	88	88	70	285	6	M16	22228..-K	H3128	48
270	45	210	3	70	68	61	240	8	M10	23028..-K	H3028	48
290	45	225	2	80	78	64	250	8	M12	23030..-K	H3030	62
330	65	260	2	102	89	81	285	8	M12	23034..-K	H3034(-HG)	87
340	60	280	3	100	88	80	310	8	M12	23036..-K	H3036(-HG)	82
400	65	290	3	105	93	80	325	8	M16	23038..-K	H3038(-HG)	125
400	65	310	3	105	105,5	87,5	340	8	M16	23040..-K	H3040(-HG)	175
440	65	340	3	115	103	95	375	8	M16	23044..-K	H3044X(-HG)	175
500	65	400	4	130	113	103	440	8	M16	23052..-K	H3052X(-HG)	214
500	65	420	4	135	118	108	460	8	M16	23056..-K	H3056(-HG)	295
340	75	270	3	116	108	100	300	8	M12	23132..-K	H3132(-HG)	115
370	65	280	3	120	115	92	330	8	M12	23134..-K	H3134(-HG)	124
410	100	340	3	135	124	108	380	8	M16	23140..-K	H3140(-HG)	165
480	105	370	4	150	134	110	420	8	M16	23144..-K	H3144X(-HG)	243
500	65	400	4	152	148	122	455	8	M20	23148..-K	H3148X(-HG)	303
540	80	440	4	175	150,5	132	490	8	M20	23152..-K	H3152X(-HG)	300
610	130	500	4	190	170	143	555	8	M24	23160..-K	H3160(-HG)	440
630	123	540	4	210	186	161	590	8	M24	23164..-K	H3164-HG	560
700	150	580	5	220	183	178	640	8	M24	23168..-K	H3168-HG	600
740	120	620	5	230	202	190	680	8	M30	23176..-K	H3176-HG	675
410	93	320	3	138	127	105	370	8	M16	23236..-K	H2336(-HG)	158
460	103	360	4	158	138	114	420	8	M16	23240..-K	H2340(-HG)	245
500	90	400	4	175	147	122	445	8	M20	23244..-K	H2344X(-HG)	168
545	95	440	4	190	158	140	510	8	M20	23248..-K	H2348X(-HG)	409
570	103	480	3	205	170	148	535	8	M20	23252..-K	H2352X(-HG)	410
610	123	500	4	206	180	151	555	8	M24	23256..-K	H2356X(-HG)	494
650	115	540	4	230	182	175	600	8	M24	23260..-K	H3260(-HG)	590
670	123	580	5	238	192	187	640	8	M24	23264..-K	H3264-HG	751
700	123	620	5	260	217	205	680	8	M30	23268..-K	H3268-HG	874
790	123	650	5	267	216	209	710	8	M30	23272..-K	H3272-HG	841
780	123	680	5	295	232	244	745	8	M30	23276..-K	H3276-HG	855
900	190	720	5	300	252	237	790	8	M30	23280..-K	H3280-HG	1520
900	190	760	5	315	255	240	835	8	M30	23284..-K	H3284-HG	1710

13 F112 flange bearing housings

medias | Product catalogue |
F112 flange bearing housings |
https://www.schaeffler.de/std/2204

13.1 Housing design

F112 flange bearing housings can be used in a wide variety of applications, for example in machine supports and other support structures. In many cases, the flange integrated in the housing is the ideal interface with the machine or system.

The bearings in these housings always act as locating bearings. Suitable bearings include 112 spherical ball bearings with a wide inner ring.

193 F112 flange bearing housings

0001ABA8

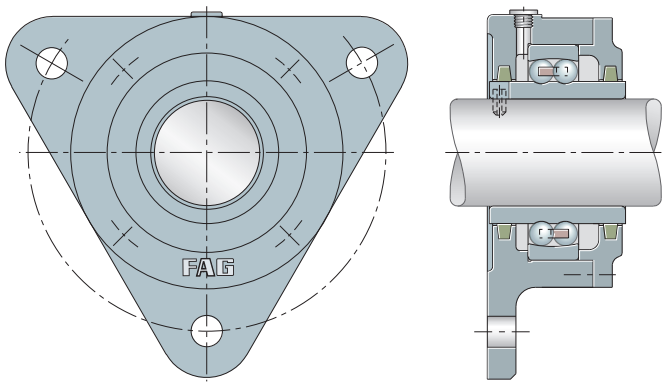
13.1.1 Locating bearing

For F112 flange bearing housings, the spherical ball bearing is clamped between the system shoulder in the housing and the cover. As a result, the bearing acts as a locating bearing.

13.1.2 Housing configuration

F112 flange bearing housings are supplied in a uniform housing configuration.

194 F112 flange bearing housings for spherical ball bearings with a wide inner ring



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13.2 Suitable bearings

The dimensions of F112 flange bearing housings are tailored to 112 spherical ball bearings with a wide inner ring. The shaft diameters are 20 mm to 60 mm.

83 Suitable bearings for F112 flange bearing housings

Bearing type	Size	
	from	to
Spherical ball bearings with a cylindrical bore and wide inner ring	11204	11212

13.3 Materials, corrosion protection

84 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron GJL	L	✓	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

13.4 Lubrication

F112 flange bearing housings are designed for grease lubrication.

13.4.1 Relubrication

There is a threaded hole M10×1 at the top of the housing. When delivered, this is sealed with a plastic stopper. A taper type or button type lubrication nipple can be screwed into this for relubrication. The lubrication nipples are not supplied with F112 flange bearing housings.

13.5 Sealing

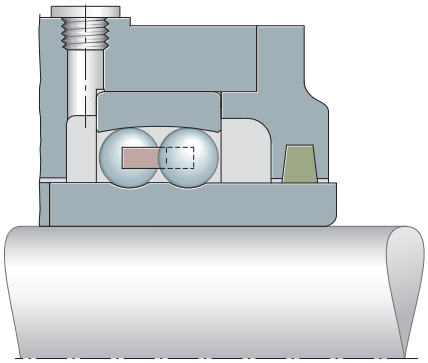
F112 flange bearing housings are sealed with felt seals.

The oil-saturated felt strips must be inserted in the relevant slots in the housing.

The felt strips are supplied by the metre and must be cut to the appropriate length on site.

The seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication.

195 Felt seal for F112 flange bearing housing



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13.6 Dimensions, tolerances

13.6.1 Tolerances for the bearing seat

The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1.

By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

13

13.7 Structure of the order designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

196 Structure of the designations of F112 flange bearing housings

001E079B

13.7.1 Specimen orders

For ordering an F112, flange bearing housing, the designation includes details of the bore diameter of the relevant self-aligning ball bearing. The roller bearing must be ordered separately.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

The felt strips for sealing the housing are supplied in the package and do not have to be ordered separately.

Example 1

Example of a housing configuration, characteristics:

- F112 flange bearing housings
- For 11204-TVH spherical ball bearings

▦85 Order

Quantity	Description	Ordering designation, short designation
1	Flange housing	F11204
1	Spherical ball bearing	11204-TVH

13.8 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

13.8.1 Flange screws

Flange screws are used to screw mount the housing flange to a mating surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

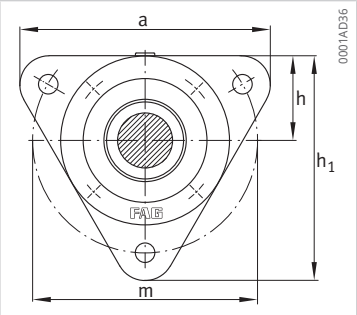
The tightening torques of the flange bolts are the same as to those of the base fastening bolts on plummer block housings.

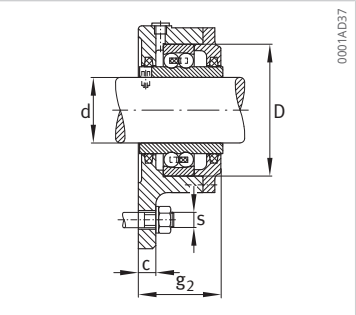
13.9 Product tables

13.9.1 Explanations of the product tables

a	mm	Base length
c	mm	Base height
d	mm	Shaft diameter
D	mm	Bearing outside diameter
g_2	mm	Overall width
h	mm	Distance to shaft axis
h_1	mm	Overall height
m	mm	Fixing hole spacing
m	kg	Mass
n	–	Number of screw mounting holes
s	inch or mm	Screw size

13.9.2 F112 flange housing for self-aligning ball bearings with a wide inner ring, triangular shape





Housing	d	h	h_1	c	a	m	s		D	g_2	m
	mm	mm	mm	mm	mm	mm	–	inch	mm	mm	kg
F11204	**20**	35	92	10	105	90	M10	$^3/_8$	47	42	0,9
F11205	**25**	38	100	10	110	96	M10	$^3/_8$	52	46	1,1
F11206	**30**	44	117	12	130	116	M10	$^3/_8$	62	49	1,5
F11207	**35**	48,5	129,5	12	145	130	M12	$^1/_2$	72	54	1,9
F11208	**40**	54	143	12	160	140	M12	$^1/_2$	80	60	2,3
F11209	**45**	60	160	15	180	160	M12	$^1/_2$	85	62,5	3,3
F11210	**50**	60	160	15	180	160	M12	$^1/_2$	90	62,5	3,6

14 F5 flange bearing housings

medias | Product catalogue |
F5 flange housings |
https://www.schaeffler.de/std/2205

14.1 Housing design

F5 flange housings can be used in a wide variety of applications, for example in general mechanical engineering. In many cases, the flange integrated in the housing is the ideal interface with the machine or system.

Depending on the size, the housing has a triangular or square shape with three or four fastening holes. Suitable bearings include spherical ball bearing, barrel roller bearings and spherical roller bearings with a tapered bore that are located on the shaft using adapter sleeves.

F5..-A flange bearing housings, closed on one side, with cover

197 F505-A to F513-A flange bearing housings, closed on one side, with cover

00193EDE

198 F515-A to F522-A flange bearing housings, closed on one side, with cover

00193EF7

F5..-B flange bearing housings, for a continuous shaft

199 F505-B to F513-B flange bearing housing, for a continuous shaft

00193EE7

200 F515-B to F522-B flange bearing housing, for a continuous shaft

00193F04

14.1.1 Locating and non-locating bearings

The bearing seats in the housing are machined so that the bearings can move and can act as non-locating bearings. Locating bearings can be achieved by inserting locating rings.

FE locating rings are used for F505 to F513 housings, FRM locating rings are used for F515 to F522 housings.

The required number of locating rings required is 1 or 2. 2 are inserted on both sides of the bearing, while a single ring is inserted on the side with the adapter sleeve nut.

Locating rings must be ordered separately.

14.1.2 Housing configurations

Various housing configurations are possible for F5 flange bearing housings, which are clearly described by the designation. For a locating bearing, the locating rings must be ordered separately.

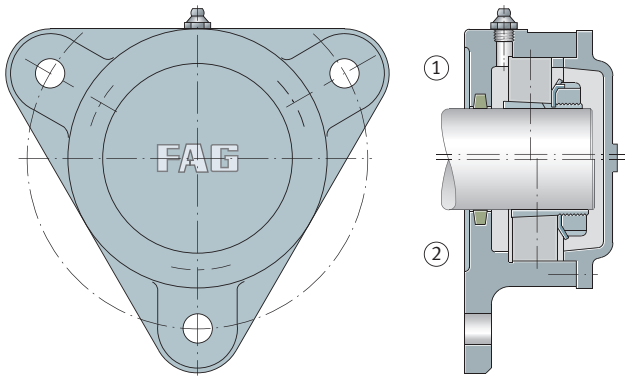
The following features can be varied in the housing configuration:

- Continuous shaft or a housing closed on one side
- Selection of the housing design depending on the bearing width
- Design of the bearing as a locating or non-locating bearing

The basic shape of the housings varies with their size:

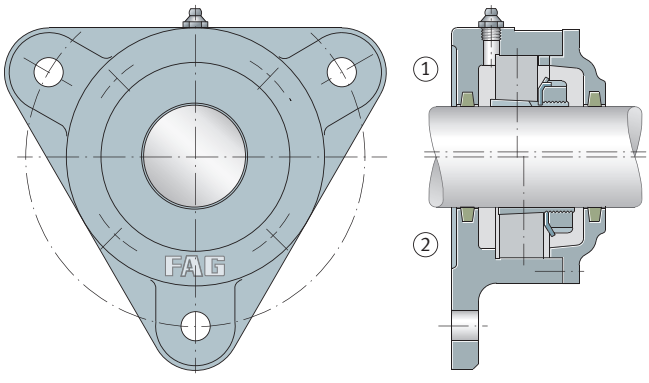
- Triangular basic design with 3 cover screws for F505, F506 and F508 flange bearing housings
- Triangular basic design with 4 cover screws for F507, F509 to F513 flange bearing housings
- Rectangular basic design with 4 cover screws for F515 to F522 flange bearing housings

201 F505, F506 and F508, for bearings with tapered bore and adapter sleeve, design A



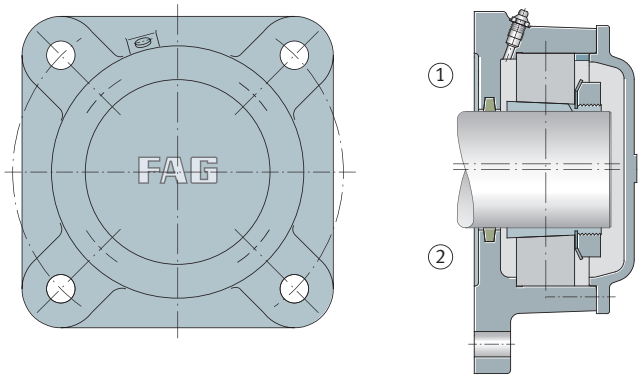
1	F5..-A, F5..-WA locating bearings	2	F5..-A, F5..-WA non-locating bearings

202 F507, F509 and F513 flange bearing housings, for bearings with a tapered bore and adapter sleeve, design B



1	F5..-B, F5..-WB locating bearings	2	F5..-B, F5..-WB non-locating bearings

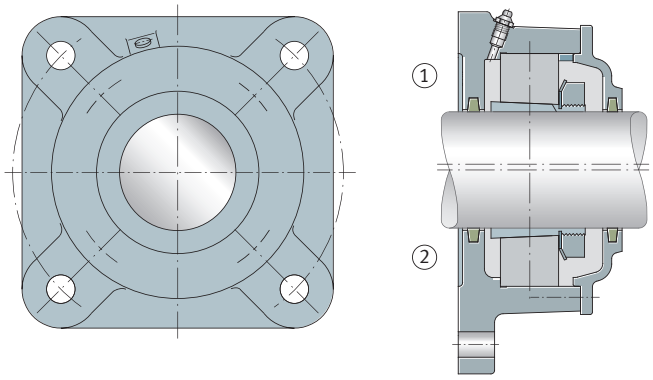
203 F515 to F522 flange bearing housings, for bearings with a tapered bore and adapter sleeve, design A



00081B32

1	F5..-A, F5..-WA locating bearings	2	F5..-A, F5..-WA non-locating bearings

204 F515 to F522 flange bearing housings, for bearings with a tapered bore and adapter sleeve, design B



00081B36

1	F5..-B, F5..-WB locating bearings	2	F5..-B, F5..-WB non-locating bearings

14

14.1.3 Accessories

Locating rings

A locating bearing can be achieved by the use of 2 locating rings.

205 FE locating rings for F505 to F513 housings

0001AD29

206 FRM locating rings for F515 to F522 housings

00019ED2

14.2 Suitable bearings

F5 flange bearing housings are designed for the installation of spherical ball bearings, barrel roller bearings and spherical roller bearings with a tapered bore, which are fastening to the shaft by adapter sleeves. The shaft diameters are 20 mm to 100 mm.

86 Suitable bearings for F5 flange bearing housings

Bearing type	Size	
	from	to
Spherical roller bearings with a tapered bore and adapter sleeve	22205..-K	22222..-K
Spherical ball bearings with a tapered bore and adapter sleeve	1205-K	1222-K
	2205-K	2222-K
Barrel roller bearings with a tapered bore and adapter sleeve	20205-K	20222-K

14.3 Materials, corrosion protection

87 Standard material and universal body coating

Material	Suffixes	Availability	Universal coating
Gray cast iron GJL	L	✓	RAL 7016, anthracite grey
Ductile cast iron GJS	D	o	RAL 7016, anthracite grey

✓ available
o available as an option

Inner and outer surfaces machined by cutting feature corrosion protection that can be easily removed. We recommend that only volatile solvents and lint-free cloths are used.

14.4 Lubrication

F5 flange bearing housings are designed for grease lubrication.

14.4.1 Relubrication

A M10×1 to DIN 71412-A conical lubricating nipple is screwed into the top of the housing for relubrication.

14.5 Sealing

F5 flange bearing housings, design A are designed for the bearings on shaft ends. One side is closed by a cover and the other side is sealed against the shaft. Design B is intended for a continuous shafts, in which case the housing is sealed against the shaft on both sides.

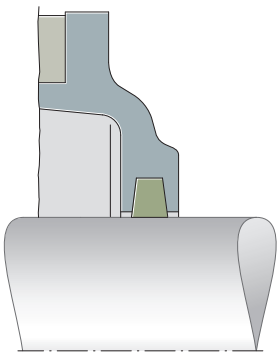
F5 flange bearing housings are sealed with felt seals.

The oil-saturated felt strips must be inserted in the relevant slots in the housing.

The felt strips are supplied by the metre and must be cut to the appropriate length on site.

The seals allow shaft misalignment of up to 0,5 ° to both sides and are suitable for grease lubrication.

207 Felt seal for F5 flange bearing housing



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14.6 Dimensions, tolerances

14.6.1 Tolerances for the bearing seat

The bearing seat in the housings is machined to the tolerance class H7 to DIN EN ISO 286-1.

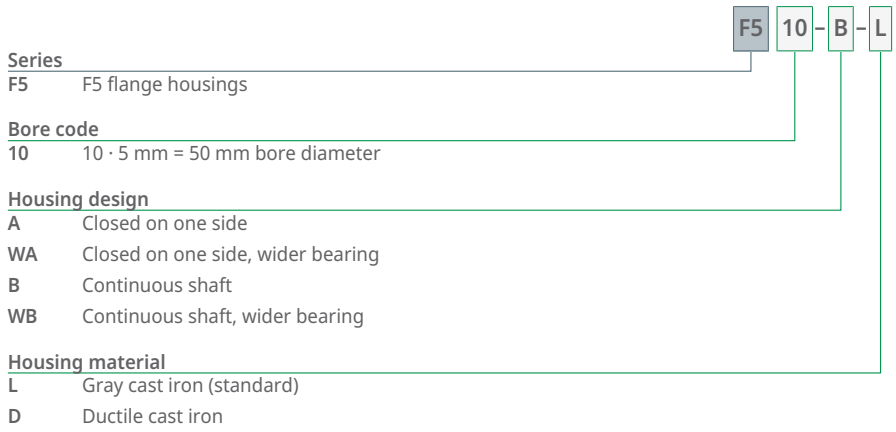
By agreement, the housings can also be supplied with bearing seats to other tolerance classes.

14.7 Structure of the order designation

The designations of the housings and associated components use a defined model. The meanings of the codes used in the designations are shown below and the full designations are also shown in the examples.

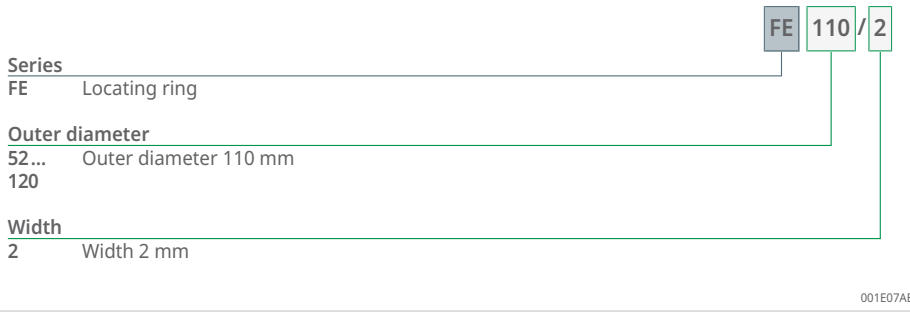
Bearing housing

208 Structure of the designations of F5 flange bearing housings

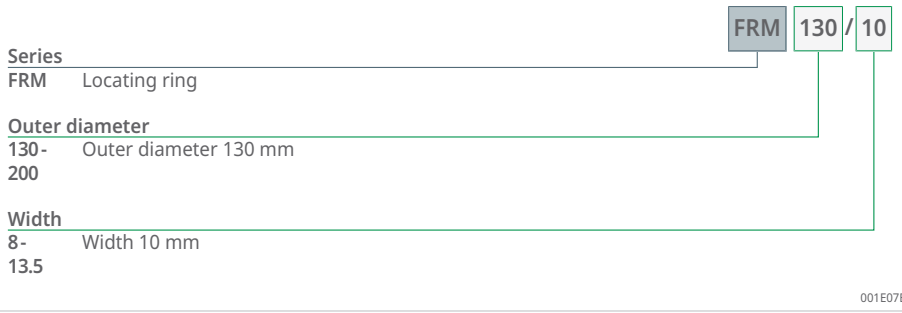


Locating rings

209 Designation structure of FE locating rings



210 Designation structure of FRM locating rings



14.7.1 Specimen orders

For ordering a F5 flange housing, the designation contains details of the size of the bearing as well as the design and material of the housing. Locating rings must be ordered separately. The roller bearing and, if necessary, the adapter sleeve must be ordered separately.

The ordering examples show the construction of an order for selected housing configurations and the appropriate bearings. For the allocation of housings, bearings and accessories for all housing sizes, see the product tables.

The felt strips for sealing the housing are supplied in the package and do not have to be ordered separately.

Example 1

Example of a housing configuration, characteristics:

- F5 flange bearing housing made of gray cast iron, closed on one side
- 2207-K-TVH-C3 spherical ball bearing in the form of a locating bearing
- Secured by means of adapter sleeve

88 Order

Quantity	Description	Ordering designation, short designation
1	Flange housing	F507-WA-L
1	Spherical ball bearing	2207-K-TVH-C3
1	Adapter sleeve	H307
1	Locating ring	FE72/2

Example 2

Example of a housing configuration, characteristics:

- F5 flange bearing housing made of gray cast iron, closed on one side
- 1220-K-M-C3 spherical ball bearing in the form of a locating bearing
- Secured by means of adapter sleeve

89 Order

Quantity	Description	Ordering designation, short designation
1	Flange housing	F520-A-L
1	Spherical ball bearing	1220-K-M-C3
1	Adapter sleeve	H220
1	Locating ring	FRM180/10
1	Locating ring	FRM180/12

Example 3

Example of a housing configuration, characteristics:

- F5 flange bearing housing made of gray cast iron
- For a continuous shaft
- 22222-E1-XL-K spherical roller bearing in the form of a non-locating bearing
- Secured by means of adapter sleeve

▦90 Order

Quantity	Description	Ordering designation, short designation
1	Flange housing	F522-WB-L
1	Spherical roller bearing	22222-E1-XL-K
1	Adapter sleeve	H322

14.8 Installation and removal

Correct assembly has a major influence on the achievable bearing service life. Careful attention must therefore be paid to the following information ➤28|2.7.

14.8.1 Flange screws

Flange screws are used to screw mount the housing flange to a mating surface. They are not supplied with the housings.

The appropriate screw size for each housing is shown in the product tables.

The relevant tightening torques are provided for screws with a metric thread to DIN 13, DIN 962:2013 and DIN ISO 965-2:1999 ➤30|2.7.2.3.

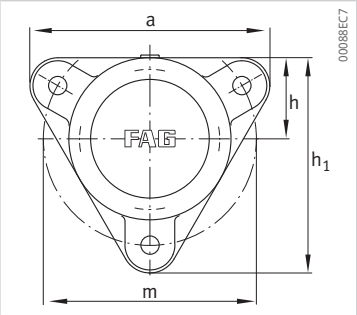
The tightening torques of the flange bolts are the same as to those of the base fastening bolts on plummer block housings.

14.9 Product tables

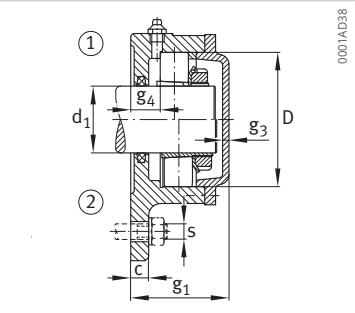
14.9.1 Explanations of the product tables

(1)	–	Locating bearing
(2)	–	Non-locating bearing
a	mm	Base length
c	mm	Base height
D	mm	Bearing outside diameter
d_1	inch or mm	Shaft diameter
g_1	mm	Overall width
g_2	mm	Overall width
g_3	mm	Housing body wall thickness
g_4	mm	Distance from flange side to bearing seat
h	mm	Distance to shaft axis
h_1	mm	Overall height
m	mm	Fixing hole spacing
m	kg	Mass
n	–	Number of screw mounting holes
s	inch or mm	Screw size

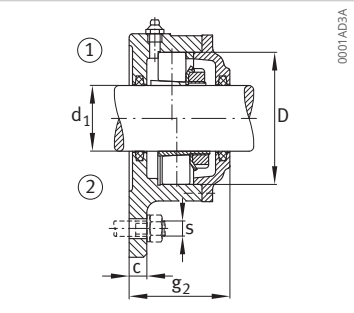
14.9.2 F5 flange housings for bearings with a tapered bore and adapter sleeve, triangular shape



Housing	d_1	h	h_1	c	a
	mm	mm	mm	mm	mm
F505-A-L	**20**	38	100	10	110
F505-B-L	**20**	38	100	10	110
F505-WA-L	**20**	38	100	10	110
F505-WB-L	**20**	38	100	10	110
F506-A-L	**25**	44	117	12	130
F506-B-L	**25**	44	117	12	130
F506-WA-L	**25**	44	117	12	130
F506-WB-L	**25**	44	117	12	130
F507-A-L	**30**	48,5	130	12	145
F507-B-L	**30**	48,5	130	12	145
F507-WA-L	**30**	48,5	130	12	145
F507-WB-L	**30**	48,5	130	12	145
F508-A-L	**35**	54	143	12	160
F508-B-L	**35**	54	143	12	160
F508-WA-L	**35**	54	143	12	160
F508-WB-L	**35**	54	143	12	160
F509-A-L	**40**	60	160	15	180
F509-B-L	**40**	60	160	15	180
F509-WA-L	**40**	60	160	15	180
F509-WB-L	**40**	60	160	15	180
F510-A-L	**45**	60	160	15	180
F510-B-L	**45**	60	160	15	180
F510-WA-L	**45**	60	160	15	180
F510-WB-L	**45**	60	160	15	180
F511-A-L	**50**	65	170	16	190
F511-B-L	**50**	65	170	16	190
F511-WA-L	**50**	65	170	16	190
F511-WB-L	**50**	65	170	16	190
F512-A-L	**55**	70	185	16	206
F512-B-L	**55**	70	185	16	206
F512-WA-L	**55**	70	185	16	206
F512-WB-L	**55**	70	185	16	206
F513-A-L	**60**	75	198	16	219
F513-B-L	**60**	75	198	16	219
F513-WA-L	**60**	75	198	16	219
F513-WB-L	**60**	75	198	16	219



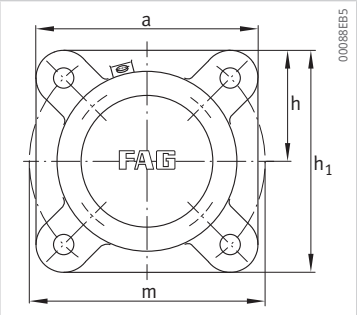
Housing design A



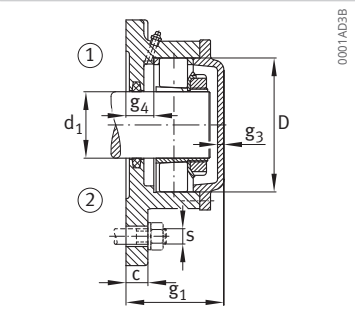
Housing design B

m	n	s		D	g_1	g_2	g_3	g_4	m
mm	–	–	inch	mm	mm	mm	mm	mm	kg
96	3	M10	$^3/_8$	52	50	–	4	16	1,2
96	3	M10	$^3/_8$	52	–	55	–	16	1,2
96	3	M10	$^3/_8$	52	50	–	4	14,5	1,2
96	3	M10	$^3/_8$	52	–	55	–	14,5	1,2
116	3	M10	$^3/_8$	62	55	–	4	18	1,6
116	3	M10	$^3/_8$	62	–	57	–	18	1,6
116	3	M10	$^3/_8$	62	55	–	4	16	1,6
116	3	M10	$^3/_8$	62	–	57	–	16	1,6
130	3	M12	$^1/_2$	72	57	–	5	17	2
130	3	M12	$^1/_2$	72	–	60	–	17	2
130	3	M12	$^1/_2$	72	63	–	5	17	2
130	3	M12	$^1/_2$	72	–	66	–	17	2
140	3	M12	$^1/_2$	80	65	–	5	22	2,5
140	3	M12	$^1/_2$	80	–	66	–	22	2,5
140	3	M12	$^1/_2$	80	70	–	5	22	2,5
140	3	M12	$^1/_2$	80	–	71	–	22	2,5
160	3	M12	$^1/_2$	85	65	–	5	22	3,6
160	3	M12	$^1/_2$	85	–	70	–	22	3,6
160	3	M12	$^1/_2$	85	69	–	5	22	3,6
160	3	M12	$^1/_2$	85	–	74	–	22	3,6
160	3	M12	$^1/_2$	90	65	–	5	20	3,8
160	3	M12	$^1/_2$	90	–	70	–	20	3,8
160	3	M12	$^1/_2$	90	71	–	5	23	3,8
160	3	M12	$^1/_2$	90	–	76	–	23	3,8
170	3	M12	$^1/_2$	100	71	–	6	23	4,1
170	3	M12	$^1/_2$	100	–	76	–	23	4,1
170	3	M12	$^1/_2$	100	77	–	6	25	4,1
170	3	M12	$^1/_2$	100	–	82	–	25	4,1
180	3	M12	$^1/_2$	110	73	–	6	24	4,6
180	3	M12	$^1/_2$	110	–	78	–	24	4,6
180	3	M12	$^1/_2$	110	79	–	6	24	4,6
180	3	M12	$^1/_2$	110	–	84	–	24	4,6
190	3	M12	$^1/_2$	120	75	–	6	24	5,4
190	3	M12	$^1/_2$	120	–	79	–	24	5,4
190	3	M12	$^1/_2$	120	83	–	6	24	5,4
190	3	M12	$^1/_2$	120	–	87	–	24	5,4

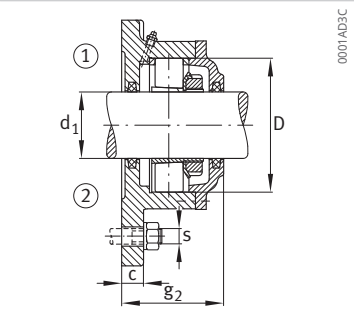
14.9.3 F5 flange housings for bearings with a tapered bore and adapter sleeve, rectangular shape



Housing	d_1	h	h_1	c	a
	mm	mm	mm	mm	mm
F515-A-L	**65**	95	190	25	190
F515-B-L	**65**	95	190	25	190
F516-A-L	**70**	98	196	25	196
F516-B-L	**70**	98	196	25	196
F517-A-L	**75**	105	210	25	210
F517-B-L	**75**	105	210	25	210
F518-A-L	**80**	105	210	25	210
F518-B-L	**80**	105	210	25	210
F520-A-L	**90**	125	250	30	250
F520-B-L	**90**	125	250	30	250
F522-A-L	**100**	135	270	30	270
F522-B-L	**100**	135	270	30	270
F522-WA-L	**100**	135	270	30	270
F522-WB-L	**100**	135	270	30	270



Housing design A



Housing design B

m	n	s		D	g_1	g_2	g_3	g_4	m
mm	–	–	inch	mm	mm	mm	mm	mm	kg
215	4	M16	$^5/_8$	130	97	–	6	30	9,5
215	4	M16	$^5/_8$	130	–	104	–	30	9,5
215	4	M16	$^5/_8$	140	101	–	6	32	10
215	4	M16	$^5/_8$	140	–	110	–	32	10
240	4	M16	$^5/_8$	150	106	–	7	31	12
240	4	M16	$^5/_8$	150	–	114	–	31	12
240	4	M16	$^5/_8$	160	110	–	7	29	13
240	4	M16	$^5/_8$	160	–	118	–	29	13
280	4	M20	$^3/_4$	180	119	–	8	29	18
280	4	M20	$^3/_4$	180	–	127	–	29	18
310	4	M20	$^3/_4$	200	128	–	8	30	22
310	4	M20	$^3/_4$	200	–	137	–	30	22
310	4	M20	$^3/_4$	200	128	–	8	30	22
310	4	M20	$^3/_4$	200	–	137	–	30	22

15 Housings for specific applications

15.1 Housings for specific industrial applications

15.1.1 Service products and special solutions

Bespoke solutions

Challenging applications in industry often place special requirements on the bearing housings. These requirements cannot always be satisfied by standard housings. In these cases, the perfect solution can be supplied in the form of special housings whose dimensions, special materials or special seals are precisely tailored to the application.

Schaeffler has many years of experience in the development of special housings. To achieve the best possible solution for bespoke cases, development work is carried out in close partnership and consultation with the customer.

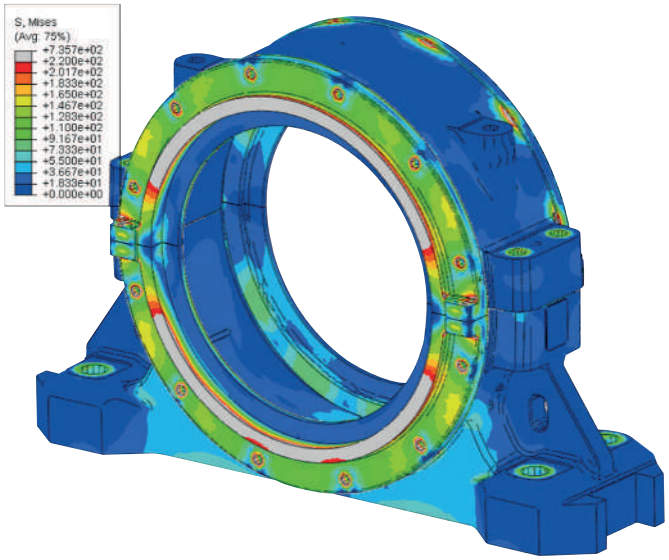
Product range

The product range for specialised industrial applications includes the following:

- Individual solutions for special requirements.
- Consideration of specific customer requirements, for example the seal, type of coating or colour.
- Calculation of housings using the FEM method.
- Supply of complete units with bearings installed in the housing.

Schaeffler also supplies the appropriate roller bearings and other accessories for individually developed housings. The industrial services experts at Schaeffler can also carry out the installation work.

211 Calculation of a housing using the FEM method



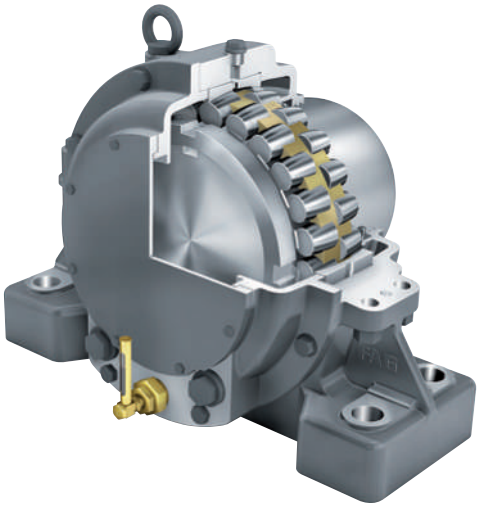
15.1.2 Application examples

15.1.2.1 Plummer block housing for the paper industry

The paper industry uses PM30 plummer block housings for drying cylinders, smoothing cylinders and Yankee cylinders. These cylinders are heated from within by steam that is fed into the cylinder through a hollow journal. The bearing housings are exposed to high ambient temperatures and high humidity. Recirculating oil lubrication is therefore necessary to dissipate the heat and allow any possible water ingress to be removed.

The bearings used are mainly spherical roller bearings, which can compensate any possible offset between the housings. The bearings are fastened either to a tapered journal or to the cylindrical shaft using a sleeve. Since substantial length increases occur due to the cylinders being heated, the housings are designed so that they allow the non-locating bearing a great deal of movement.

212 PM30 plummer block housings for the paper industry



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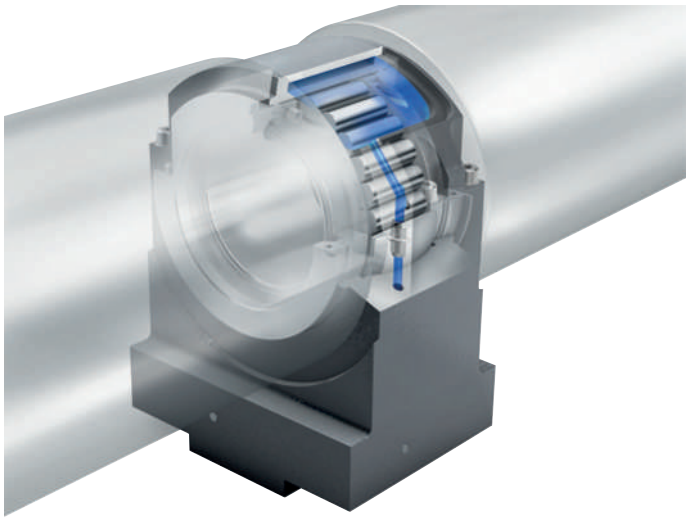
15.1.2.2 Water-cooled plummer block housings for continuous casting machines

Continuous casting is a process in the steel industry in which molten steel is converted into a solid phase. This creates long products which are then processed further in finish rolling mills.

Whilst the steel is cooling, it takes the form of an endless strand and is guided on lines of driven rollers. The strand moves through 2 opposing rows of rollers and is cooled at the same time. There are various ways of cooling the strand. Direct cooling can be carried out by aiming water straight at the strand and rollers, while indirect cooling can be achieved by cooling the rollers from the inside. Normally, these two cooling methods are combined.

The rollers are supported by means of SZA water-cooled plummer block housings developed specifically for this application. The housings have a water-cooled upper housing section. The housing is designed so that a specially split bearing is fitted with only one outer ring half. The load directions in the lines of rollers in continuous casting machines make this possible. The seals are tailored to the extreme operating conditions (high temperatures, spray water from the strand cooling system).

213 SZA water-cooled plummer block housing for continuous casting machines



15.1.2.3 Twin block housings for rotary kilns for cement production

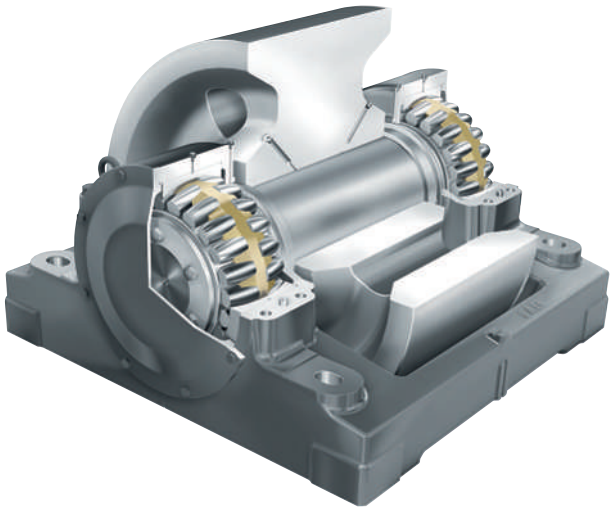
The kiln shell of a rotary kiln is supported by riding rings. Each riding ring rests on two radial support rollers. Robust roller bearings are required for the rollers to support the high radial and axial loads resulting from the movement.

The roller is rigidly connected to the shaft by an interference fit and is supported on both sides in the roller bearings. Good levering can be achieved by means of appropriate bearing spacings so that the tilting forces can be supported. Wide, low 241 spherical roller bearings with high basic load ratings are suitable for supporting radial and axial forces. They are generally lubricated with grease, which is also the case for the labyrinth seals in the housing. Arcanol LOAD400 or Arcanol LOAD1000 is used depending on the load. The housings are precisely tailored to the requirements of the application in terms of lubrication, sealing and maximum misalignment.

Schaeffler can supply complete RLR units for radial rollers, consisting of the following:

- Housing
- 2 roller bearings
- Shaft
- Roller
- Grease

214 Twin block housings for rotary kilns in cement production, RLR unit



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15.1.2.4 Converter housings for steel production

Converters are used in the steel industry to produce steel from molten pig iron. When filled, large converter vessels weigh several 100 t. The trunnion bearing used for converters is a particular challenge due to the extreme loads and ambient conditions.

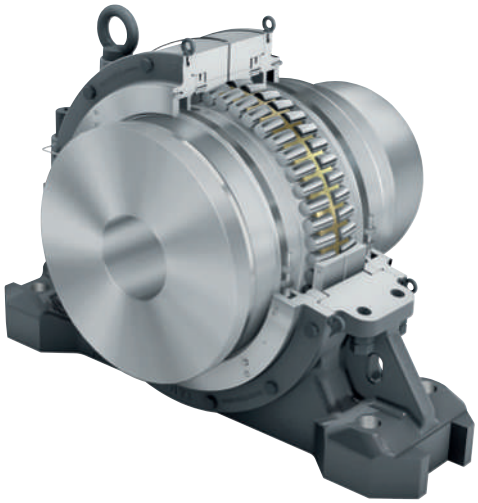
Schaeffler has developed KPG and KPGZ split plummer block housings to act as the trunnion bearings for converters and these are available as standard housings. These housings are large, solid housings for 249 spherical roller bearings with shaft diameters from 510 mm to 1320 mm. If necessary, housings for even larger shaft diameters can be designed and supplied.

Bespoke solutions have been developed for special applications on the basis of these housings. The housings can be supplemented by additional components and modified where necessary.

Complete packages comprising the following can be supplied:

- Special spherical roller bearings
- Housings with seals in various designs
- Sensors for grease condition monitoring
- Grease distributor plate with complete pipework to all lubrication points
- Sensor systems for bearing condition monitoring

215 Converter housings for steel production



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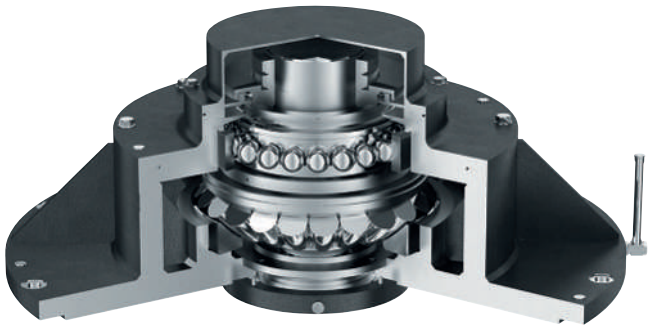
15.1.2.5 Flange bearing housing units for large electrical machinery

There are special flange housing units available for rotor bearings in large electrical machinery. One or more deep groove ball bearings, cylindrical roller bearings or angular contact ball bearings are installed in the housing depending on the application.

The flange housing units deliver solutions that are particularly cost-effective, versatile and suitable for the application. The flange housing units significantly simplify the design of the end shields, resulting in considerable reductions in the design work required on large motor housings. A wide range of sealing versions in all IP protection classes, together with various lubrication concepts ensure that they can be tailor easily to the requirements of the specific applications.

216 FKB flange bearing housing unit with oil sump lubrication



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Further information

TPI 152 | Flange housing units for large electrical machinery | https://www.schaeffler.de/std/202B

15.2 Housings for railway applications

15.2.1 Service products and special solutions

Product range and range of services

Schaeffler has many years of experience in the development of wheelset bearing housings. To achieve the best possible solution for bespoke cases, development work is carried out in close partnership and consultation with the customer.

Benefits of the wheelset bearing housing:

- Good value for money
- High quality
- Short lead times
- Long term availability guaranteed
- Proven design solutions
- Worldwide delivery capability

Product range

- Products for assembly
- Maintenance and diagnostics
- Assembly service and diagnostic service
- Technical advice on application issues
- Training tools
- Calculation programs
- Technical publications

15.2.2 Wheelset bearing housings

15.2.2.1 Housing

The housing must is a connecting part between the vehicle bogie frame and the wheelset and must therefore reliably transfer the forces. Wheelset bearing housings in locomotives, freight cars and passenger trains for local, regional traffic and high-speed transport are subject to a wide range of requirements. In terms of design, it is important to know how the forces are introduced into the housing. A housing of a suitable design will deliver good pressure distribution in the roller bearing, reducing the specific load and preventing stress peaks in the contact surfaces that transfer the load. As far as the housing is concerned, it should be the lowest possible weight whilst ensuring excellent feasibility.

Housing for freight transport with axles up to 25 t

- Approvals for all spring systems
- Welded design of sliding plates and sleeve
- Spring assembly on both sides
- One-piece housing

217 Housing for freight transport with axles up to 25 t

000B6F4B

- Guide bushes must be pressed in
- One-sided spring support
- One-sided rubber buffer
- Split housing

Housing for passenger transport

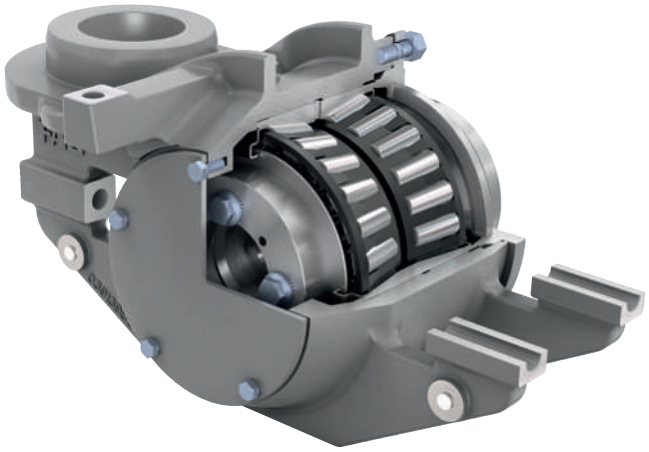
218 Housing for passenger transport, double-decker wagon

000B6F37

- Complex casting design with cast cover
- Spring assembly on both sides
- One-piece housing

Housing for locomotives

219 Housing for locomotives



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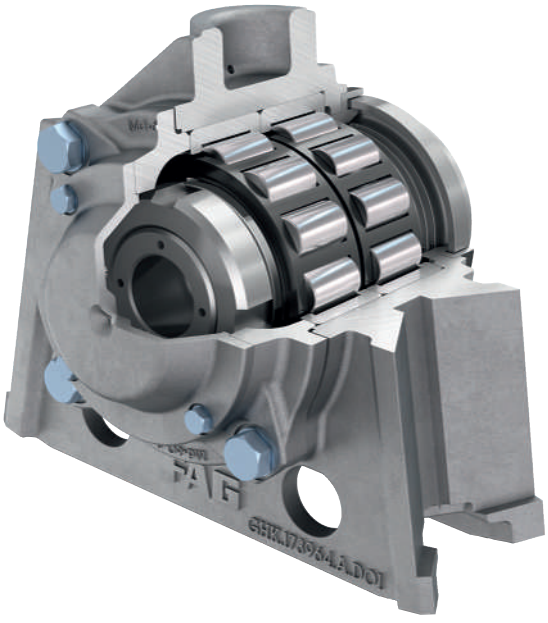
Housing for trams

- Material: aluminium
- Spring assembly on both sides
- Compact design
- One-piece and split housing design possible

220 Housing for trams



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High safety requirements apply to wheelset bearing housings. The design must be sufficient for the load in the operating conditions for long periods of several decades. A precise analysis of the loads is required to achieve this.

15.2.2.2 Materials

The standard material for wheelset bearing housings is ductile cast iron EN-GJS-400-18-LT. Other materials such as cast aluminium alloys or cast steel may be used depending on customer requirements on characteristics such as weight, strength or weldability.

15.2.2.3 Corrosion protection

All machined external surfaces of housings are coated. Inner and outer machined surfaces have anti-corrosion treatment. The coating systems of the major railway operators are registered with Schaeffler and easy to apply. If there are specific requirements relating to coatings and corrosion protection, special solutions are possible. If the customer has any special specifications, we check that they are feasible and then apply them if the result is positive.

15.2.2.4 Seals for wheelset bearing housings

To ensure that a wheelset bearing remains functional, the ingress of dirt particles or liquids must be prevent, as must the escape of lubricant. Sealing the axle makes a very important contribution to maintaining the function of the wheelset bearing. It must be taken into account that TAROL units and cylindrical roller bearing units themselves are already sealed. For the use of single bearings such as WJ, WJP cylindrical roller bearings, particular attention must be paid to sealing then against dirt and moisture ingress and to retaining grease at the roller bearing.

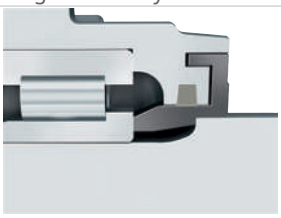
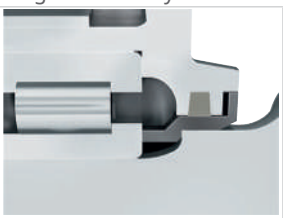
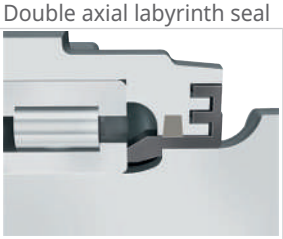
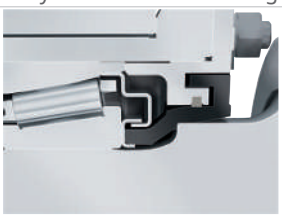
A distinction is made between contact seals and non-contact seals for these seal types. The seal type most suitable for the application is then decided based on the actual operating conditions.

For example, contact seals can very often not be used for high speeds due to the high circumferential velocities and resulting heat generation. In contrast, contact seals are necessary for trams where circumferential velocities are lower and there is a risk of track flooding.

There is a wide range of seal versions to cover the entire range of operating conditions. The housings used for wheelset bearings are designed for grease lubrication.

Other seal designs such as labyrinths made of turned parts, lamellar rings or felt ring seals require more installation space and are more expensive to buy. The use of these seal designs is required in the specifications.

91 Seals for wheelset bearing housings

Seal	Characteristics
Single axial labyrinth seal	
	• Little space required • Non-contact • Improved sealing when combined with a felt seal, but in this case it is no longer a non-contact seal
Single radial labyrinth seal	
	• Little space required • Also possible in split housings • Non-contact • Improved sealing when combined with a felt seal, but in this case it is no longer a non-contact seal
Double axial labyrinth seal	
	• Larger space required • Better labyrinth sealing action • Non-contact • Improved sealing when combined with a felt seal, but in this case it is no longer a non-contact seal
Double radial labyrinth seal	
	• Larger space required • Better labyrinth sealing action • Also possible in split housings • Non-contact • Improved sealing when combined with a felt seal, but in this case it is no longer a non-contact seal
Splash ring seal	
	• Simple design • For use only with a sealed bearing • Suitable for split and unsplit housings • Non-contact
Labyrinth seal with stuffing box packing	
	• Also suitable for high circumferential speeds • Little space required • Abrading
Combined seal consisting of labyrinth seal, felt seal and V ring seal	
	• Very good sealing to prevent the ingress of foreign matter from outside • Abrading

Schaeffler has many years of experience in the development of wheelset bearing housings. To achieve the best possible solution for each application, development work is carried out in close partnership and consultation with the customer. In addition to series housings for standard applications, for example Y25 housing for freight transport, Schaeffler develops special tailored designs for many projects to suit the specified loads and existing package space. To improve customer proximity and availability, Schaeffler products are manufactured in many regions around the world.

15.2.2.5 Further information

TPI 256 | Roller bearings in wheelset bearings |
https://www.schaeffler.de/std/202C

15.2.3 Housing unit with relubrication facility for traction motors

For rotor bearings in electric drives, Schaeffler works with customers to manufacture complete housing units with relubrication systems, which are designed for specific applications and can also be fitted with measurement systems.

221 FKB housing unit with relubrication facility for traction motors

00019990

The FKB housing units from Schaeffler offer numerous advantages in the design, manufacture, mounting and maintenance of rotor bearing arrangements. Due to the significantly simplified design of the end shields, the work involved in the development of new drive concepts is also reduced. Thanks to their highly developed design, these bearing solutions are extremely easy to maintain.

The rotor bearing arrangement must be matched to the particular operating and ambient conditions of an electric drive concept. Depending on the drive concept, widely differing bearing loads must be taken into consideration in order to achieve electric drives that are efficient, operationally reliable and cost-effective. The lubrication and sealing must be configured such that the bearings are neither undersupplied nor oversupplied with lubricant in any operating status.

Maintenance costs should be kept low. These requirements are fulfilled by the housing units for traction motors.

Depending on the size, a housing unit can be made of any of the following materials:

- ductile cast iron EN-GJS-400-15
- steel with a minimum tensile strength R_m of 400 N/mm²

The housing units can be tailored to the specific geometries of the neighbouring constructions without significant additional work since they have a bespoke design. They can be produced individually for different standard bearing types and series. As a result, there are no problems in using either standard bearings or electrically insulated bearings to prevent damage caused by electric current.

222 Cylindrical roller bearings

00019991

223 Electrically insulated bearings

00017269

224 Hybrid bearings (hybrid cylindrical roller bearings)

0001726B

Special solutions are also available to suit specific requirements, in which existing products can also be used. The housing units developed for grease lubrication have relevant relubrication systems, thus reducing the maintenance costs and effort.

Special designs with containers for collecting used grease are also possible and can be adapted to the respective customer requirements.

Further customer-specific modifications, such as sensors or measurement technology, can also be realised without major outlay.

Further information

TPI 228 | Rolling bearings in traction motors |
https://www.schaeffler.de/std/201E

16 Maintenance and service

Schaeffler supplies high quality products, services and training courses as part of its Industrial Aftermarket.

- Assembly
- Lubrication
- Condition monitoring

Schaeffler will be delighted to help you select the products, services and training courses you need wherever you are in the world.

16.1 Industrial Aftermarket

Schaeffler Industrial Aftermarket (IAM) is responsible for the spare parts and service business for end customers and sales partners in all significant industrial sectors. The Schaeffler Industrial Aftermarket service section has innovative solutions, products and services for rolling and plain bearings, to form an extensive portfolio that covers all phases in the lifecycle of the bearing and takes account of the total costs (TCO).

The aim is to help customers save on maintenance costs, improve availability and avoid unforeseen machine downtime. The Schaeffler Industrial Aftermarket Department offers every customer an individual concept solution.

Schaeffler has competence centres around the world. This makes it possible to provide customers with products, services and training quickly and professionally around the world. All service personnel undergo a comprehensive training programme and are audited at regular intervals by officially certified specialists. This ensures that services throughout the world conform to a uniformly high standard of quality.

16.2 Schaeffler Mounting Manager

medias | Engineering Tool | Mounting Manager |
https://www.schaeffler.de/std/1FE7

The Schaeffler Mounting Manager provides convenient help for the correct installation of roller bearings.

- Shows suitable mounting methods.
- Calculates the values required for mounting in relation to the reduction in radial internal clearance and drive-up distance.
- Provides information about the necessary accessories and tools.

16.3 Mounting

Schaeffler service experts offer assembly services for bearing units, comprising bearings and housings across a range of industrial sectors. They have detailed expertise and extensive experience in all industrial sectors.

Schaeffler has skilled personnel specifically trained for the Industrial Aftermarket sector at several locations around the world to provide reliable, rapid and competent support. The services are provided either at the customer's site or at Schaeffler workshops.

16.3.1 Schaeffler Mounting Service

Mounting service offerings:

- Installation and removal of bearings and bearing systems
- Troubleshooting and preparation of possible solutions
- Measurement and condition analyses
- Maintenance and inspection of bearings
- Support in achieving perfect mounting results
- Design and manufacture of special tools

225 Mounting service at a dam system

001949BA

Benefits of the mounting service:

- Worldwide portfolio of services for bearing technology
- Installation experts with comprehensive experience in highly diverse applications
- Increased equipment availability and improvements in productivity
- Improvement of assembly and dismantling processes
- Training and sensitisation of personnel for handling bearings of all types correctly

Further information

mounting-services@schaeffler.com

16.3.2 Tool rental

A wide range of special tools for correct mechanical installation and removal work and for measurement and inspection tasks can be rented from Schaeffler on a weekly basis for a charge.

- Thermal tools (for example MF-GENERATOR)
- Hydraulic tools (for example hydraulic pumps)
- Measuring and test equipment (for example enveloping circle gauges)

If one of our trained service team is asked to carry out the specific work, the equipment rental is generally free of charge.

Further information

mounting-services@schaeffler.com

16.3.3 Mechanical tools

Mechanical tools are generally used for the installation and removal of small and medium-sized bearings. The installation forces are transferred by the form fit effect. The Schaeffler range includes items such as hook and pin wrenches, socket wrenches and a wide variety of extractors.

Further information

TPI 216 | Tools for the Mechanical Mounting and Dismounting of Rolling Bearings |
https://www.schaeffler.de/std/1FDE

16.3.4 Thermal tools

Induction heating devices are used to heat roller bearings or components with a cylindrical bore for assembly and dismantling. In addition to various induction heating devices featuring mains frequency technology, Schaeffler also offers induction heating devices powered by medium frequency technology.

Adequate expansion of the housings and bearings is normally achieved at +80 °C to +100 °C. The maximum heating temperature must not be exceeded during the heating process. The temperature of roller bearings must not normally exceed +120 °C to prevent changes to the structure and hardness of the bearing. The temperature is infinitely adjustable in all heating devices.

16.3.4.1 Heating devices with medium frequency technology

Heating devices with medium frequency technology heat medium-sized to large bearings, housings and similar steel parts for installation and removal quickly and easily. The device always consists of an inductor and a generator.

The inductor can be flexible or rigid in design. The rigid design is ideal for batch applications. The flexible inductor design can be wound around the components.

Every device is designed for the specific application and is fitted with flexible or rigid inductors, depending on the workpiece. Due to its compact design, the device is also suitable for mobile use.

Benefits of the medium frequency heating device:

- Suitable for installation and removal work
- Efficiency of the generator over 90 %
- Low energy requirement
- Short heating times
- Time-controlled and temperature-controlled warm-up
- Automatic demagnetisation
- Flexible and rigid inductors possible
- Suitable for use either inside or outside the component
- Lower mains connection power than heating devices with mains frequency
- Almost silent
- Air-cooled system

Further information

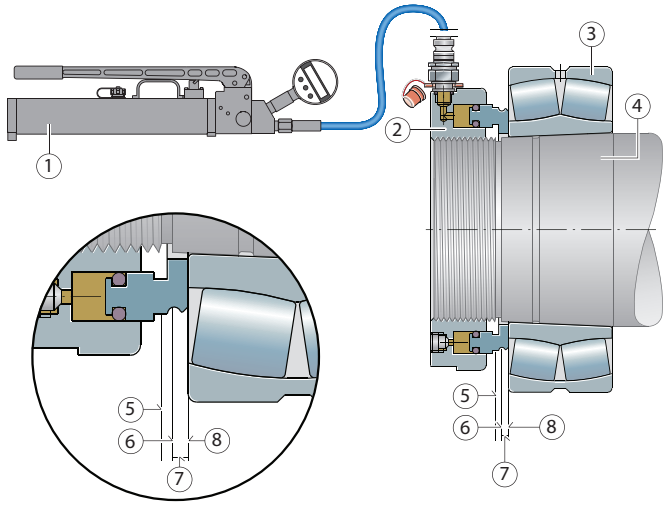
TPI 295 | Induction Heating Devices |
https://www.schaeffler.de/std/220D

16.3.5 Hydraulic tools

Hydraulic tools can be used to apply large forces. These tools are therefore ideal for the installation and removal of large bearings or components with a tapered bore.

Hydraulic nuts are used as the tool. Pressure can be generated using hand pumps or hydraulic units.

226 Assembly of a spherical roller bearing with a hydraulic pump and hydraulic nut



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1	Hydraulic pump	2	Hydraulic nut
3	Spherical roller bearing	4	Tapered seat
5	Initial position	6	Starting position
7	Sliding distance	8	Final position

16.3.5.1 Hydraulic nuts

HYDNUT hydraulic nuts are used to press parts with a tapered bore onto a tapered seat. Presses are mainly used if the sliding forces required cannot be applied using other accessories, such as shaft nuts or pressure screws.

Main applications for HYDNUT hydraulic nuts:

- Installation and removal of bearings with a tapered bore. These bearings with a tapered bore can be seated directly on a tapered shaft, an adapter sleeve or a withdrawal sleeve.
- Removal of withdrawal sleeves and adapter sleeves.

92 Hydraulic nuts

Application	Design	Designation	
		from	to
Adapter sleeves and withdrawal sleeves with metric dimensions	With metric fine pitch thread to DIN 13	HYDNUT50-E	HYDNUT200-E
	With trapezoidal thread to DIN 103		
Sleeves with Imperial dimensions	With Imperial thread to ABMA "Standards for Mounting Accessories, Section 8, Locknut Series N-00"	HYDNUT50-E-INCH	HYDNUT950-E-INCH
For high mounting forces, for example in shipbuilding	Reinforced design with smooth bore	HYDNUT100-HEAVY	HYDNUT900-HEAVY

Further information

TPI 196 | Hydraulic Nut HYDNUT |
https://www.schaeffler.de/std/1FBE

16.3.5.2 Hydraulic pumps

Hydraulic pumps are ideal for use as pressure generation devices where bearings with a tapered bore must be driven onto their tapered seat using a hydraulic nut.

First, the bearing is driven gently onto the tapered seat as far as its initial position. A suitable hydraulic nut is then screwed onto the shaft and the hydraulic pump is connected. The hydraulic pump is then operated until the pressure to reach the starting position is achieved. The bearing is then moved over the required distance and until it reaches its final position by continuing to operating the pump.

The required starting pressure and the travel distance are calculated using the Schaeffler Mounting Manager.

227 Hydraulic pumps

1	AHP hydraulic pump, manual	2	EP hydraulic pump, electric
3	UHAP hydraulic pump, UHAP hydraulic pump,		

Further information

TPI 195 | Hydraulic Pumps |
https://www.schaeffler.de/std/1F8D

medias | Engineering Tool |
Mounting Manager |
https://www.schaeffler.de/std/1FE7

16.4 Lubrication

In more than half of all cases, inadequate lubrication is the cause of unscheduled machine downtime. The service life of machine elements undergoing swivel, rotary or linear movements can be significantly extended by the use of grease types that are appropriate for the various operating and environmental conditions and be defining and observing lubrication intervals and quantities.

16.4.1 Services

Services relating to lubrication include:

- Selection of lubricants and lubrication systems
- Preparation of lubrication and maintenance schedules
- Lubrication point management
- Lubricant advice
- Lubricant studies and lubricant tests

Benefits of lubrication

- Prevention of failures
- Increase in productivity
- Reduction in the cost of lubrication

16.4.2 Lubricants

The lubricants available from Schaeffler are designed and tested for bearing.

The various grease types cover almost all applications. They have been developed by experienced application engineers and are produced by the best manufacturers on the market. Different greases are used to suit the particular application.

Arcanol roller bearing greases are subject to 100 % quality controls. The control methods used by Schaeffler are among the most demanding on the market. Arcanol roller bearing grease types therefore satisfy the very highest quality requirements.

228 Analysis of the thermal behaviour of grease types

0007FD4A

Further information

Arcanol lubricants | Safety data sheets and technical data sheets | https://www.schaeffler.de/sds

16.4.3 Lubricators

The life of lubricants is limited. In their application, they are constantly exposed to mechanical loads, ageing and contamination. For this reason, it is necessary to supply fresh lubricant at defined time intervals and thereby ensure adequate lubrication. Optimal lubrication helps to avoid consequential damage and reduces the risk of application failure.

With the automatic lubricators of the CONCEPT series, it is possible to supply almost all industrial machines and equipment precisely with oil or grease. The product range comprises simple and cost-effective single-point lubrication systems (CONCEPT1) as well as more complex lubrication systems for a larger number of lubrication points (CONCEPT2, CONCEPT4 and CONCEPT8).

OPTIME C1 is an easy to use and economical solution for automatic single point lubrication. OPTIME C1 represents a further expansion of the OPTIME system developed for the condition monitoring of rolling bearings and integrates the lubricators of series CONCEPT1.

The multi-point lubricator FAG OPTIME C4 is an important addition to the OPTIME Ecosystem. Many motor-pump and motor-fan applications can now be supplied via a central lubrication system.

229 Automatic lubricators

1	OPTIME C1	2	CONCEPT1
3	CONCEPT2	4	FAG OPTIME C4
5	CONCEPT4	6	CONCEPT8

The product range is supplemented by a wide range of accessories for automatic lubricators, manual lubricating tools, unfilled lubricant cartridges and the Schaeffler lubricant program Arcanol. These lubricants stand for very high performance in rolling bearing and linear applications.

The CONCEPT1 lubricator can be installed on the actual housing using a reducing nipple. The reducing nipple is screwed into the housing instead of a lubricating nipple.

If placement on the actual housing is not possible, the lubricators in the CONCEPT series can be connected by a lubrication line.

230 Plummer block housing with CONCEPT1 lubricator (optional OPTIME C1, CONCEPT2, CONCEPT8 or FAG OPTIME C4)

00190176

Further information

TPI 252 | Lubricators |
https://www.schaeffler.de/std/1D4E

16.5 Condition monitoring

The trouble-free, optimised operation of complex machinery and systems can only be achieved by condition-based maintenance. The preferred method of Schaeffler is vibration diagnostics.

This makes it possible to detect damage in machinery at a very early stage. This means that, for example, damaged components can be replaced as part of scheduled downtime. It also prevents unscheduled downtime.

In addition to vibration monitoring, Schaeffler can also supply a system to monitor lubrication status.

16.5.1 Schaeffler OPTIME

The OPTIME system is a complete solution for the condition monitoring and lubrication of a large number of machines. OPTIME makes automatic lubrication in maintenance even more cost-effective, as it prevents unplanned downtimes caused by excessive or inadequate lubrication. OPTIME also eliminates the need for regular inspections to check the lubricators. The OPTIME system warns the user if a lubricator is malfunctioning or when the CONCEPT1 lubricant cartridge needs replacing.

Special attention was paid to the following points during the development of the OPTIME system:

- Simple commissioning
- Easy expansion
- Versatile application options

Benefits

- Comprehensive, automated lubrication
- Wireless IoT solution that can be extended at any time
- Several 100 units can be integrated easily in a single day
- Up to 50 % lower costs compared to manual lubrication

The work required by the user is kept to a minimum in every process step. One component of the OPTIME system is the OPTIME C1 wireless lubricators, which are used with a CONCEPT1 cartridge and combine with the gateway to form a mesh network. OPTIME sensors can be added to expand the mesh network. Another component of the OPTIME system is the associated cloud services, which allow the condition of the lubricators and the condition of the machines to be checked using a suitable device with an app or dashboard.

231 OPTIME concept with OPTIME C1, OPTIME sensors, gateway and digital service

001C5563

1	Lubricators	2	Sensors
3	Gateway	4	Dashboard
5	Mobile App	6	Condition monitoring for electric motors

Further information

CSS 0179 | No Surprises with the Lubricators |
https://www.schaeffler.de/std/1FF6

TPI 271 | Lubricator | OPTIME C1 |
https://www.schaeffler.de/std/1FC2

medias | OPTIME C1
https://www.schaeffler.de/std/2039

16.5.2 ProLink Condition Monitoring System (CMS)

The ProLink Condition Monitoring System (CMS) from Schaeffler is a multi-channel system for monitoring complex machines or entire systems.

ProLink CMS has a modular design and consists of a main processor module for signal processing and up to 4 vibration modules for signal acquisition of machine vibrations.

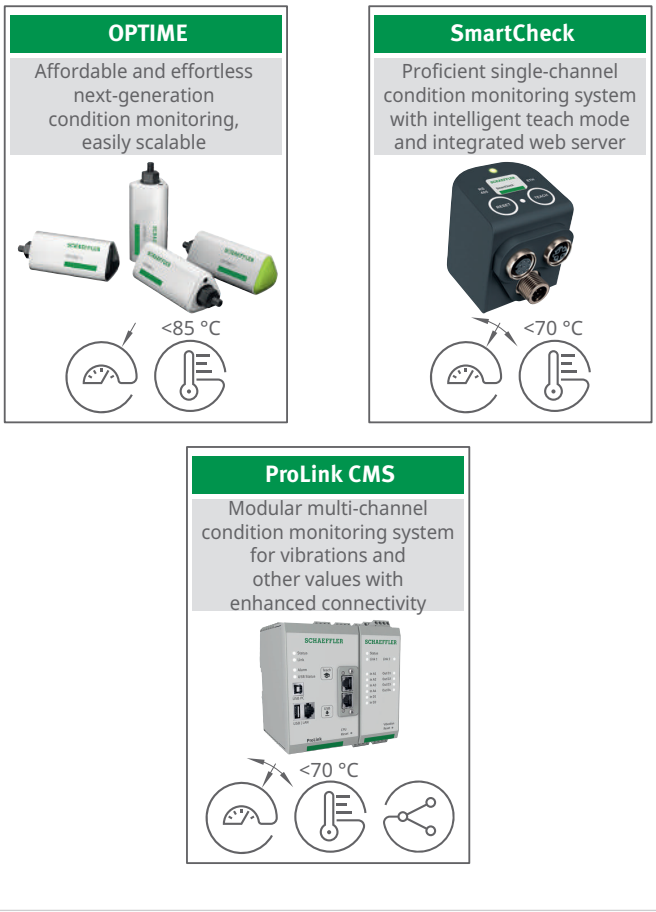
Machine vibrations are recorded using up to 16 acceleration sensors, digitalised in the vibration modules and sent to the processor module for evaluation.

232 Schaeffler ProLink CMS

Thanks to its separate sensor technology, ProLink CMS can be used wherever the use of OPTIME sensors or SmartCheck is not possible on account of the environment. As with SmartCheck, a measurement task is also automatically pre-installed for each sensor in ProLink CMS. The learning module makes it easy to get started in condition monitoring.

As integration into the customer infrastructure is particularly useful for a multichannel system, optional functionalities such as OPC UA or email are also available in ProLink CMS.

233 Overview of Schaeffler OPTIME, SmartCheck and ProLink CMS



16

Further information

OPL | ProLink CMS - Systematic Condition Monitoring |
https://www.schaeffler.de/std/1FF4

GTS 0129 | Integrated condition monitoring system for a water treatment facility |
https://www.schaeffler.de/std/1FF3

16.5.3 Schaeffler SmartCheck

SmartCheck is a compact online measuring system for permanent machine monitoring and is suitable for the early detection of roller bearing damage, imbalances and misalignment, for example.

234 Schaeffler SmartCheck

0018F062

Despite its compact size, SmartCheck is a complete monitoring device and includes an acceleration sensor and full electronic evaluation system. The device is installed on the unit it is to monitor. Simple units such as pumps, motors or fans can be monitored without any additional configuration using the pre-installed measurement task and learning mode. This measurement task can be adjusted and expanded at any time to accommodate changing requirements.

SmartCheck makes the following possible:

- Correlation of machine parameters with process parameters
- Condition-based maintenance
- Increased system availability

The optional interface which complies with the Open Platform Communication Unified Architecture OPC US) standard allows all measurement data and alarm information to be transferred or integrated into the customer infrastructure with ease. For example, the alarm system can be displayed in the visualisation system or work orders can be triggered in a maintenance planning system.

Optionally, a cyclical or alarm-controlled status email can be sent to the maintenance engineer. Measurement data can be added to the status email, which can then be sent straight to Schaeffler Monitoring Services GmbH for detailed analysis. The characteristic value set generated from the measured values enables the very precise monitoring of the electrical machine.

Further information

medias | Lifetime solutions
https://www.schaeffler.de/std/2037

16.5.4 Services and training

In addition to innovative solutions and products relating to roller and plain bearings, Schaeffler offers a wide range of customer-specific services for system maintenance and quality assurance.

Services

Services include the following:

- Routine measurements and routine inspections
- Installation and setup of measurement equipment and systems for condition monitoring
- Troubleshooting and fault resolution on complex industrial systems
- Comprehensive full-service packages
- Global support through state-of-the-art technologies, such as remote diagnostics or augmented reality

Training

The range of training courses available from Schaeffler covers a wide range of product training and analysis training courses that use first-hand practical experience to strengthen your internal skills.

Our training courses start with basic knowledge and an overview of the products that can be used. More advanced training courses cover deeper theoretical expertise and are supplemented by practical exercises. They will also prepare you to have your knowledge of condition monitoring tested and certified at various levels in our certification courses which comply with DIN ISO 18436-2.

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